

MASTER QUESTION COMPENDIUM

ELECTROCHEMISTRY

EXHAUSTIVE PROBLEM CORPUS & COMPLETE UNIFIED QUESTION BANK

JEE Main • JEE Advanced • Chemistry Olympiad (INChO/IChO)

Complete Cross-Book Problem Bank Synthesizing All 8 Classical Authorities:

Neeraj Kumar • Narendra Avasthi • GRB (O.P. Tandon) • Pearson • Cengage Theory •
Cengage DPP • Essential Physical Chemistry • Peter Atkins

Features full categorization into Core Subtopics with co-located redundant variants, followed by the exhaustive book-by-book exercise question modules (over 1,750+ individual questions).

Exhaustive Master Problem Edition

September 27, 2026

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Part I

Core Topic Clusters with Co-Located Variants

Chapter 1

Master Problem Bank: Electrolytic Conduction & Ionic Migration

Part I Organization: Electrolytic Conduction, Kohlrausch & Transport Numbers

This part compiles all questions and quantitative problems on electrolytic conduction, resistance, conductivity, molar conductivity, Kohlrausch's law, transport numbers, and conductometric titrations from all 8 books. Identical and redundant variants from other books are co-located beneath each primary problem with full source citations.

1.1 Subtopic 1.1: Resistance, Resistivity, Conductivity (κ) & Cell Constant (G^*)

Q.1.1 [Primary Source: Neeraj Kumar Part 2 Ex-I Q1 / Pearson Ex-I Q106]
Which of the following aqueous solutions of NaCl exhibits the highest electrical resistance?

- | | |
|-----------------|----------------|
| (a) 1 N NaCl | (c) 2 N NaCl |
| (b) 0.05 N NaCl | (d) 0.1 N NaCl |

Cross-Book Redundant / Identical Variants:

- [Variant v1: GRB Objective Set-1 Q8, p. 66] Which of the following solutions of KCl has the lowest electrical conductance? (a) 0.1 M (b) 1.0 M (c) 0.01 M (d) 0.001 M. [*Identical concept testing inverse relation $R \propto 1/\kappa$*].
- [Variant v2: Narendra Avasthi Level-1 Q12, p. 8] Resistance of an electrolyte solution increases when: (a) concentration increases (b) concentration decreases (c) temperature increases (d) electrodes are moved closer.
- [Variant v3: Cengage DPP 3.1 Q2, p. 1] The resistance of a 0.05 M solution of an electrolyte is $100\ \Omega$. The conductivity of the solution is $0.5\ \text{S m}^{-1}$. The cell constant of the conductivity cell is: (a) $50\ \text{m}^{-1}$ (b) $20\ \text{m}^{-1}$ (c) $5\ \text{m}^{-1}$ (d) $200\ \text{m}^{-1}$.

Q.1.2 [Primary Source: GRB Practice Problem 1, p. 46 / Essential 24.7]

The resistance of a 0.1 M solution of an electrolyte is $100\ \Omega$. If the electrodes of the conductivity cell are 2.0 cm apart and each has an area of cross-section 4.0 cm^2 , calculate:

- (a) Cell constant G^* (in cm^{-1})
- (b) Specific conductivity κ (in S cm^{-1})
- (c) Molar conductivity Λ_m (in $\text{S cm}^2\text{ mol}^{-1}$)

Cross-Book Redundant / Identical Variants:

- **[Variant v1: Neeraj Kumar Part 2 Ex-I Q4, p. 35]** A conductivity cell has two platinum electrodes separated by 1.5 cm with plate area 4.5 cm^2 . The cell constant is: (a) 0.33 cm^{-1} (b) 3.0 cm^{-1} (c) 0.5 cm^{-1} (d) 2.0 cm^{-1} .
- **[Variant v2: Essential Physical Chemistry Solved Ex 2, p. 864]** A conductance cell whose electrodes are 1.25 cm apart and have area 2.5 cm^2 was filled with a 0.05 M solution of a salt. The resistance was found to be $52.5\ \Omega$. Calculate the cell constant and specific conductance.
- **[Variant v3: Pearson Ex-I Q4, p. 2]** If the distance between two planar electrodes is doubled and their surface area is halved, the cell constant becomes: (a) unchanged (b) doubled (c) four times (d) one-fourth.

Q.1.3 [Primary Source: Narendra Avasthi Level-1 Q15, p. 8 / CNG-TH p. 51]

The resistance of a conductivity cell filled with 0.02 M KCl solution at 25°C is $175\ \Omega$. The specific conductivity of 0.02 M KCl is $0.002768\text{ S cm}^{-1}$. When the same cell is filled with 0.001 M AgNO_3 solution, the resistance is $1050\ \Omega$. The cell constant and the specific conductivity of the AgNO_3 solution are:

- (a) $G^* = 0.4844\text{ cm}^{-1}, \kappa = 4.613 \times 10^{-4}\text{ S cm}^{-1}$
- (b) $G^* = 0.2422\text{ cm}^{-1}, \kappa = 2.306 \times 10^{-4}\text{ S cm}^{-1}$
- (c) $G^* = 0.9688\text{ cm}^{-1}, \kappa = 9.226 \times 10^{-4}\text{ S cm}^{-1}$
- (d) $G^* = 0.4844\text{ cm}^{-1}, \kappa = 9.226 \times 10^{-4}\text{ S cm}^{-1}$

Cross-Book Redundant / Identical Variants:

- **[Variant v1: GRB Solved Example 4, p. 13]** A conductivity cell with 0.01 M KCl gives $R = 150\ \Omega$ ($\kappa = 1.41 \times 10^{-3}\text{ S cm}^{-1}$). When filled with 0.01 M HCl, resistance is $40\ \Omega$. Determine the cell constant and molar conductivity of HCl.
- **[Variant v2: Neeraj Kumar Part 2 Ex-I Q8, p. 35]** Specific conductance of 0.1 M KCl is 0.0129 S cm^{-1} . Resistance with this solution is $100\ \Omega$. What is the cell constant?
- **[Variant v3: Essential 24.8 Solved Ex 5, p. 865]** The specific conductivity of $N/50$ KCl solution at 25°C is $0.002765\text{ mho cm}^{-1}$. If the resistance of a cell containing this solution is $400\ \Omega$, what is the cell constant?

1.2 Subtopic 1.2: Molar Conductivity (Λ_m) & Equivalent Conductivity (Λ_{eq})

Q.1.4 [Primary Source: Neeraj Kumar Part 2 Ex-I Q18 / NA Level-1 Q24]

For an electrolyte of the general stoichiometric formula $A_x B_y$, what is the exact algebraic relationship between its molar conductivity Λ_m and its equivalent conductivity Λ_{eq} ?

- (a) $\Lambda_{eq} = (x + y)\Lambda_m$ (c) $\Lambda_m = \frac{\Lambda_{eq}}{xz_+}$
 (b) $\Lambda_m = (xz_+)\Lambda_{eq} = (yz_-)\Lambda_{eq}$ (d) $\Lambda_{eq} = (xz_+ \cdot yz_-)\Lambda_m$

Cross-Book Redundant / Identical Variants:

- [Variant v1: GRB Objective Set-1 Q22, p. 67] For aluminium sulphate $Al_2(SO_4)_3$, the ratio Λ_m/Λ_{eq} is: (a) 6 (b) 1/6 (c) 5 (d) 3.
- [Variant v2: Pearson Ex-I Q22, p. 3] If Λ_m of $Al_2(SO_4)_3$ is $x \text{ S cm}^2 \text{ mol}^{-1}$, then Λ_{eq} is: (a) $x/6$ (b) $6x$ (c) $x/3$ (d) $x/2$.
- [Variant v3: Cengage DPP 3.1 Q8, p. 1] The molar conductivity of a 0.05 M $BaCl_2$ solution is $240 \text{ S cm}^2 \text{ mol}^{-1}$. The equivalent conductivity is: (a) 480 (b) 120 (c) 240 (d) $60 \text{ S cm}^2 \text{ equiv}^{-1}$.

Q.1.5 [Primary Source: GRB Practice Problem 18, p. 48 / Essential 24.11]

The conductivity of a 0.01 M solution of magnesium sulphate ($MgSO_4$) is $1.26 \times 10^{-3} \text{ S cm}^{-1}$. Calculate:

- (a) Molar conductivity Λ_m
 (b) Equivalent conductivity Λ_{eq}

Cross-Book Redundant / Identical Variants:

- [Variant v1: Narendra Avasthi Level-1 Q28, p. 9] If the conductivity of 0.001 M $CaCl_2$ is $2.6 \times 10^{-4} \text{ S cm}^{-1}$, find Λ_m and Λ_{eq} .
- [Variant v2: Essential 24.11 Ex 9, p. 868] The specific conductance of 0.01 N solution of potassium chloride is $0.00141 \Omega^{-1} \text{ cm}^{-1}$. What is its equivalent conductivity?

1.3 Subtopic 1.3: Debye–Hückel–Onsager (DHO) Theory & High-Field Effects

Q.1.6 [Primary Source: Atkins Topic 6C / Neeraj Kumar Part 2 Ex-II Sec F Q3]

Derive the Debye–Hückel–Onsager equation for a univalent electrolyte in solution:

$$\Lambda_m = \Lambda_m^\circ - (A + B\Lambda_m^\circ)\sqrt{c}$$

Differentiate physically between:

- (a) The relaxation (asymmetry) retarding force
- (b) The electrophoretic retarding drag

Under what conditions do the Wien effect and the Debye–Falkenhagen effect occur?

Cross-Book Redundant / Identical Variants:

- **[Variant v1: Narendra Avasthi Level-3 Passage 1, p. 37]** In high alternating frequency ($> 10^7$ Hz), the Debye–Falkenhagen effect increases molar conductivity because: (a) solvent viscosity drops (b) ionic atmosphere remains spherically symmetric (c) ion pairs dissociate completely (d) temperature rises.
- **[Variant v2: Pearson Ex-II Sec A Q42, p. 19]** At electric fields $> 10^5$ V cm $^{-1}$, the Wien effect causes Λ_m of strong electrolytes to approach Λ_m° because the ion speed exceeds the relaxation time of the atmosphere.
- **[Variant v3: Cengage Theory p. 58]** For a 1 : 1 strong electrolyte, the slope of Λ_m vs $\sqrt{c}$ depends on solvent dielectric constant ϵ_r , absolute temperature T , and viscosity η .

1.4 Subtopic 1.4: Ostwald Dilution Law & Degree of Dissociation (α)

Q.1.7 [Primary Source: GRB Practice Problem 32, p. 49 / Essential 24.14]

The molar conductivity of a 0.01 M solution of acetic acid is 16.5 S cm 2 mol $^{-1}$ at 25°C. If $\Lambda_m^\circ(\text{CH}_3\text{COOH}) = 390.5$ S cm 2 mol $^{-1}$, calculate:

- (a) Degree of dissociation α
- (b) Dissociation constant K_a of acetic acid

Cross-Book Redundant / Identical Variants:

- **[Variant v1: Neeraj Kumar Part 2 Ex-I Q35, p. 37]** The molar conductivity of 0.025 M methanoic acid is 46.1 S cm 2 mol $^{-1}$. Given $\Lambda_m^\circ(\text{HCOOH}) = 405.5$ S cm 2 mol $^{-1}$, calculate its degree of dissociation α and K_a .
- **[Variant v2: Narendra Avasthi Level-1 Q40, p. 10]** For a weak monoprotic acid HA, $\Lambda_m = 20$ S cm 2 mol $^{-1}$ at 0.04 M. If $\Lambda_m^\circ = 400$ S cm 2 mol $^{-1}$, the dissociation constant K_a is: (a) 1.05×10^{-4} (b) 2.10×10^{-4} (c) 4.20×10^{-5} (d) 1.05×10^{-5} .
- **[Variant v3: Cengage DPP 3.1 Q16, p. 2]** According to Ostwald's dilution law, a plot of $1/\Lambda_m$ versus $C\Lambda_m$ gives: (a) slope = $1/(K_a(\Lambda_m^\circ)^2)$ and intercept = $1/\Lambda_m^\circ$ (b) slope = K_a and intercept = Λ_m° (c) a parabolic curve (d) horizontal line.

1.5 Subtopic 1.5: Kohlrausch's Law of Independent Migration of Ions

Q.1.8 [Primary Source: Neeraj Kumar Part 2 Ex-I Q45 / NA Level-1 Q52]

The limiting molar conductivities at 25°C are: $\Lambda_m^\circ(\text{NaCl}) = 126.4 \text{ S cm}^2 \text{ mol}^{-1}$, $\Lambda_m^\circ(\text{HCl}) = 426.2 \text{ S cm}^2 \text{ mol}^{-1}$, $\Lambda_m^\circ(\text{CH}_3\text{COONa}) = 91.0 \text{ S cm}^2 \text{ mol}^{-1}$. Calculate the limiting molar conductivity of acetic acid $\Lambda_m^\circ(\text{CH}_3\text{COOH})$.

Cross-Book Redundant / Identical Variants:

- [Variant v1: GRB Solved Example 18, p. 17] Calculate Λ_m° for acetic acid given $\Lambda_m^\circ(\text{HCl}) = 425.9$, $\Lambda_m^\circ(\text{NaCl}) = 126.4$, $\Lambda_m^\circ(\text{CH}_3\text{COONa}) = 91.0 \text{ S cm}^2 \text{ mol}^{-1}$.
- [Variant v2: Essential 24.15 Solved Ex 14, p. 875] From the following molar conductivities at infinite dilution: $\Lambda_m^\circ(\text{Ba}(\text{OH})_2) = 457.6$, $\Lambda_m^\circ(\text{BaCl}_2) = 240.6$, $\Lambda_m^\circ(\text{NH}_4\text{Cl}) = 129.8 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$, calculate Λ_m° for ammonium hydroxide (NH_4OH).
- [Variant v3: Pearson Ex-I Q55, p. 5] The limiting equivalent conductance of K_2SO_4 , $\text{Al}_2(\text{SO}_4)_3$, and KCl are x, y, z . The limiting equivalent conductance of alum $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ is: (a) $x + y$ (b) $(x + y)/2$ (c) $x + y + z$ (d) $2x + 2y$.

Q.1.9 [Primary Source: GRB Practice Problem 45, p. 50 / NA Level-2 Q14]

The specific conductivity of a saturated aqueous solution of silver chloride (AgCl) at 25°C is $3.41 \times 10^{-6} \text{ S cm}^{-1}$ and that of water used is $1.60 \times 10^{-6} \text{ S cm}^{-1}$. If $\lambda^\circ(\text{Ag}^+) = 61.9 \text{ S cm}^2 \text{ mol}^{-1}$ and $\lambda^\circ(\text{Cl}^-) = 76.3 \text{ S cm}^2 \text{ mol}^{-1}$, calculate:

- Solubility of AgCl in mol L^{-1} and in g L^{-1}
- Solubility product K_{sp} of AgCl at 25°C

Cross-Book Redundant / Identical Variants:

- [Variant v1: Neeraj Kumar Part 2 Ex-II Sec A Q12, p. 41] Specific conductance of saturated BaSO_4 is $4.58 \times 10^{-6} \text{ S cm}^{-1}$ and that of pure water is $1.52 \times 10^{-6} \text{ S cm}^{-1}$. If $\Lambda_m^\circ(\text{BaSO}_4) = 286 \text{ S cm}^2 \text{ mol}^{-1}$, find K_{sp} of BaSO_4 .
- [Variant v2: Essential 24.15 Solved Ex 18, p. 877] The specific conductivity of a saturated solution of PbSO_4 is $4.2 \times 10^{-5} \text{ S cm}^{-1}$ and that of water is $1.5 \times 10^{-6} \text{ S cm}^{-1}$. Given $\Lambda_m^\circ(\text{PbSO}_4) = 152 \text{ S cm}^2 \text{ mol}^{-1}$, calculate the solubility and K_{sp} .

1.6 Subtopic 1.6: Ionic Mobility, Hittorf Rule & Moving Boundary Method

Q.1.10 [Primary Source: Atkins Topic 6D / GRB 12.9 Solved Ex 25, p. 18]

The limiting molar ionic conductivities of K^+ and Cl^- at $25^\circ C$ are $73.5 \text{ S cm}^2 \text{ mol}^{-1}$ and $76.3 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. Calculate:

- The absolute ionic mobility of K^+ and Cl^- in $\text{cm}^2 \text{ V}^{-1} \text{ s}^{-1}$
- The drift velocity of K^+ ion in a potential gradient of 5.0 V cm^{-1}
- The transport numbers t_{K^+} and t_{Cl^-} in infinitely dilute KCl solution

Cross-Book Redundant / Identical Variants:

- [Variant v1: Essential 24.18 Solved Ex 20, p. 881] In a Hittorf cell with AgNO_3 solution and silver electrodes, before electrolysis the cathode compartment contained 0.227 g of AgNO_3 . After passing a current, the cathode compartment contained 0.205 g of AgNO_3 while 0.032 g of silver was deposited in a silver coulometer. Calculate the transport number of Ag^+ and NO_3^- .
- [Variant v2: Neeraj Kumar Part 2 Ex-II Sec F Q4, p. 51] In a moving boundary experiment using 0.1 M HCl with CdCl_2 as follower electrolyte, a current of 11.5 mA moved the boundary through 7.5 cm in 18.2 min in a tube of diameter 0.4 cm. Calculate t_{H^+} .
- [Variant v3: Narendra Avasthi Level-2 Q24, p. 26] Explain why the transport number of an ion is not an invariant property of the ion, but depends on the co-ion and concentration.

1.7 Subtopic 1.7: Conductometric Titrations

Q.1.11 [Primary Source: Neeraj Kumar Part 2 Ex-II Sec C Passage 2 / GRB p. 20]

Sketch and explain the exact shape of the conductometric titration curve for each of the following systems (titrant added from burette to analyte in beaker):

- Strong acid vs. strong base: 25 mL 0.1 M HCl titrated with 0.1 M NaOH
- Weak acid vs. strong base: 25 mL 0.1 M CH₃COOH titrated with 0.1 M NaOH
- Strong acid vs. weak base: 25 mL 0.1 M HCl titrated with 0.1 M NH₄OH
- Mixture of strong and weak acid vs. strong base: (HCl + CH₃COOH) titrated with NaOH
- Precipitation titration: 25 mL 0.1 M AgNO₃ titrated with 0.1 M KCl

Cross-Book Redundant / Identical Variants:

- **[Variant v1: Pearson Ex-II Sec A Q48, p. 20]** In the titration of a mixture of HCl and CH₃COOH with NaOH, how many distinct breaks are observed in the conductance curve? (a) 1 (b) 2 (c) 3 (d) no sharp break.
- **[Variant v2: Narendra Avasthi Level-3 Passage 4, p. 38]** During the conductometric titration of CH₃COOH with NaOH, why does conductance initially decrease slightly before rising steadily?
- **[Variant v3: Cengage Theory p. 74]** In which conductometric titration is the equivalence point characterized by a sharp minimum?

Chapter 2

Master Problem Bank: Galvanic Cells, Electrodes & Electrochemical Series

Part II Organization: Galvanic Cells, Reference Electrodes & EC Series

This part compiles all questions and quantitative problems on electrode potentials, standard hydrogen electrode (SHE), calomel, silver-silver chloride, Daniell cell construction, salt bridge mechanics, and the electrochemical series from all 8 books. Identical and redundant variants from other books are co-located beneath each primary problem with full source citations.

2.1 Subtopic 2.1: Electrode Potential & Electrical Double Layer

Q.2.1 [Primary Source: Neeraj Kumar Part 1 Ex-I Q1 / GRB 12.11]

A metal rod is dipped in a solution of its own ions. Its single electrode potential is independent of:

- (a) Temperature of the solution
- (b) Concentration of the metal ions in solution
- (c) Surface area of the metal rod exposed to the solution
- (d) Nature of the metal and solvent

Cross-Book Redundant / Identical Variants:

- [Variant v1: Pearson Ex-I Q1, p. 1] The electrode potential of a metal does not depend on: (a) area of electrode (b) concentration of ions (c) temperature (d) pressure in case of gas electrodes.
- [Variant v2: Narendra Avasthi Level-1 Q76, p. 12] An intensive property among the following is: (a) electrode potential E (b) resistance R (c) Gibbs free energy change ΔG (d) heat capacity.
- [Variant v3: Cengage Theory p. 3] Explain the origin of the potential difference at a metal-solution interface using the concept of Helmholtz electrical double layer and Stern's compact/diffuse model.

Q.2.2**[Primary Source: Atkins Topic 6C.1 / GRB 12.16]**

Why is the potential of the Standard Hydrogen Electrode (SHE) defined as zero at all temperatures, whereas the standard reduction potential of the saturated calomel electrode (SCE) has a non-zero temperature coefficient ($\partial E/\partial T \approx -0.65 \text{ mV K}^{-1}$)?

Cross-Book Redundant / Identical Variants:

- **[Variant v1: Neeraj Kumar Part 1 Ex-I Q7, p. 1]** A standard hydrogen electrode has zero electrode potential because: (a) hydrogen is easiest to oxidize (b) this electrode potential is adopted as zero by international thermodynamic convention (c) hydrogen has one electron (d) hydrogen is the lightest element.
- **[Variant v2: Narendra Avasthi Level-1 Q82, p. 13]** The calomel electrode is reversible with respect to: (a) Hg_2^{2+} (b) Hg^{2+} (c) Cl^- (d) H^+ .
- **[Variant v3: Pearson Ex-I Q15, p. 2]** What is the composition and standard potential of the saturated calomel electrode at 25°C ? (a) $+0.2422 \text{ V}$ (b) $+0.2800 \text{ V}$ (c) $+0.3338 \text{ V}$ (d) 0.0000 V .

2.2 Subtopic 2.2: Daniell Cell & Salt Bridge Dynamics**Q.2.3****[Primary Source: GRB 12.13 Solved Ex 33 / NA Level-1 Q88]**

For the Daniell cell $\text{Zn(s)} \mid \text{ZnSO}_4(\text{aq}, 1 \text{ M}) \parallel \text{CuSO}_4(\text{aq}, 1 \text{ M}) \mid \text{Cu(s)}$:

- Write the anode half-reaction, cathode half-reaction, and overall cell reaction.
- What is the function of the salt bridge, and why does current stop flowing if the salt bridge is removed?
- Why cannot KCl be used in a salt bridge if the cell contains Ag^+ or Pb^{2+} ions?

Cross-Book Redundant / Identical Variants:

- **[Variant v1: Neeraj Kumar Part 1 Ex-I Q22, p. 2]** A salt bridge contains KNO_3 in agar-agar gel because: (a) K^+ and NO_3^- have almost equal ionic velocities ($t_+ \approx t_- \approx 0.5$) (b) agar-agar conducts current (c) it prevents mixing of solutions (d) all of these.
- **[Variant v2: Pearson Ex-I Q12, p. 2]** What happens when a Daniell cell operates for a long period? (a) mass of Zn rod increases (b) concentration of Cu^{2+} increases (c) mass of Cu rod increases and $[\text{Zn}^{2+}]$ increases (d) cell potential increases.
- **[Variant v3: Cengage DPP 3.2 Q5, p. 3]** In which of the following electrochemical cells is a salt bridge NOT required? (a) Lead-acid storage battery (b) Daniell cell (c) Standard cadmium cell (d) Concentration cell without transference.

2.3 Subtopic 2.3: Electrochemical Series & Feasibility of Reactions

Q.2.4 [Primary Source: Neeraj Kumar Part 1 Ex-I Q3 / PRSN Ex-I Q8]

The standard reduction potentials are given as: $E^\circ(\text{Mg}^{2+}/\text{Mg}) = -2.37 \text{ V}$, $E^\circ(\text{Al}^{3+}/\text{Al}) = -1.66 \text{ V}$, $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$, $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V}$, $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$, $E^\circ(\text{Ag}^+/\text{Ag}) = +0.80 \text{ V}$. Which of the following processes will occur spontaneously under standard conditions?

- (a) Storing aqueous CuSO_4 in a zinc container
- (b) Stirring an $\text{Al}(\text{NO}_3)_3$ solution with a copper spoon
- (c) Precipitation of silver from AgNO_3 solution upon adding copper turnings
- (d) Evolution of H_2 gas upon adding copper to dilute HCl

Cross-Book Redundant / Identical Variants:

- [Variant v1: GRB Objective Set-1 Q45, p. 70] Which metal will displace hydrogen from dilute H_2SO_4 ? (a) Ag (b) Cu (c) Fe (d) Au.
- [Variant v2: Narendra Avasthi Level-1 Q92, p. 14] Given $E^\circ(\text{Cr}^{3+}/\text{Cr}) = -0.74 \text{ V}$, $E^\circ(\text{MnO}_4^-/\text{Mn}^{2+}) = +1.51 \text{ V}$, $E^\circ(\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}) = +1.33 \text{ V}$, $E^\circ(\text{Cl}/\text{Cl}^-) = +1.36 \text{ V}$. The strongest oxidizing agent is: (a) MnO_4^- (b) Cl^- (c) Cr^{3+} (d) Mn^{2+} .
- [Variant v3: Cengage Theory Concept App 1 Q4, p. 14] Can an aqueous solution of 1 M FeSO_4 be stored in a nickel vessel? Given $E^\circ(\text{Ni}^{2+}/\text{Ni}) = -0.25 \text{ V}$, $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V}$.

Chapter 3

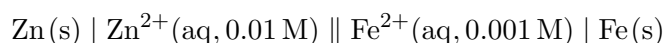
Master Problem Bank: Thermodynamics of Cells & Nernst Equation

Part III Organization: Nernst Equation & Cell Thermodynamics

This part compiles all questions and multi-step quantitative problems on the Nernst equation, equilibrium constants, cell Gibbs energy, enthalpy, entropy, temperature coefficients, and reversible cell heat from all 8 books. Identical and redundant variants from other books are co-located beneath each primary problem with full source citations.

3.1 Subtopic 3.1: Nernst Equation Calculations

Q.3.1 [Primary Source: Neeraj Kumar Part 1 Ex-I Q42 / NA Level-1 Q105]
Calculate the EMF of the cell at 25°C:



Given: $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.763 \text{ V}$ and $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.440 \text{ V}$.

- | | |
|---------------|---------------|
| (a) +0.3525 V | (c) +0.3230 V |
| (b) +0.2935 V | (d) -0.3230 V |

Cross-Book Redundant / Identical Variants:

- [Variant v1: Pearson Ex-I Q45, p. 4] For the cell $\text{Cd} \mid \text{Cd}^{2+}(0.01 \text{ M}) \parallel \text{Ag}^+(0.5 \text{ M}) \mid \text{Ag}$, given $E^\circ(\text{Cd}^{2+}/\text{Cd}) = -0.40 \text{ V}$ and $E^\circ(\text{Ag}^+/\text{Ag}) = +0.80 \text{ V}$, calculate E_{cell} at 298 K.
- [Variant v2: GRB Practice Problem 12, p. 47] Calculate the potential of a Daniell cell when $[\text{Zn}^{2+}] = 0.5 \text{ M}$ and $[\text{Cu}^{2+}] = 0.01 \text{ M}$. Standard potential $E_{\text{cell}}^\circ = 1.10 \text{ V}$.
- [Variant v3: Cengage DPP 3.3 Q3, p. 5] The potential of a hydrogen electrode at 298 K in contact with a solution of $\text{pH} = 3$ and $P_{\text{H}_2} = 1 \text{ atm}$ is: (a) -0.177 V (b) $+0.177 \text{ V}$ (c) -0.059 V (d) 0.000 V .

Q.3.2 [Primary Source: Narendra Avasthi Level-2 Q38 / PRSN Ex-II Q18]

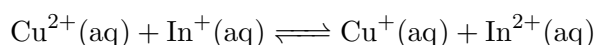
For the cell $\text{Pt} | \text{H}_2(1 \text{ bar}) | \text{HA}(0.1 \text{ M}) || \text{Ag}^+(0.1 \text{ M}) | \text{Ag}$, the cell EMF is measured to be 0.985 V at 298 K. Given $E^\circ(\text{Ag}^+/\text{Ag}) = +0.800 \text{ V}$, calculate the acid dissociation constant K_a of the weak monobasic acid HA.

Cross-Book Redundant / Identical Variants:

- **[Variant v1: Neeraj Kumar Part 1 Ex-II Sec A Q24, p. 14]** The EMF of the cell $\text{Pt}, \text{H}_2(1 \text{ atm}) | \text{Buffer} (\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}) || \text{KCl}(\text{sat}) | \text{Hg}_2\text{Cl}_2, \text{Hg}$ is 0.518 V. If $E_{\text{SCE}} = 0.242 \text{ V}$, calculate the pH of the buffer.
- **[Variant v2: GRB Practice Problem 24, p. 48]** A hydrogen electrode dipping into a solution of an organic acid of unknown strength gives an EMF of 0.236 V when coupled with a standard calomel electrode. Calculate the pH and $[\text{H}^+]$ of the solution.

3.2 Subtopic 3.2: Equilibrium Constant (K_{eq}) from E°_{cell} **Q.3.3 [Primary Source: GRB Practice Problem 34, p. 49 / NA Level-1 Q118]**

Find the equilibrium constant K_{eq} at 298 K for the reaction:



Given: $E^\circ(\text{Cu}^{2+}/\text{Cu}^+) = +0.15 \text{ V}$ and $E^\circ(\text{In}^{2+}/\text{In}^+) = -0.42 \text{ V}$.

- | | |
|-----------------------|---------------------------|
| (a) 1.0×10^5 | (c) 2.1×10^{19} |
| (b) 4.4×10^9 | (d) 8.8×10^{-10} |

Cross-Book Redundant / Identical Variants:

- **[Variant v1: Neeraj Kumar Part 1 Ex-I Q65, p. 6]** Calculate the equilibrium constant for the disproportionation reaction $2 \text{Cu}^+(\text{aq}) \rightleftharpoons \text{Cu}^{2+}(\text{aq}) + \text{Cu}(\text{s})$ at 25°C . Given $E^\circ(\text{Cu}^+/\text{Cu}) = +0.52 \text{ V}$ and $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$.
- **[Variant v2: Pearson Ex-I Q60, p. 6]** For the cell reaction $\text{Sn}(\text{s}) + 2 \text{Fe}^{3+}(\text{aq}) \longrightarrow \text{Sn}^{2+}(\text{aq}) + 2 \text{Fe}^{2+}(\text{aq})$, $E^\circ_{\text{cell}} = 0.61 \text{ V}$. The equilibrium constant K is approximately: (a) 10^{10} (b) 10^{20} (c) 10^{30} (d) 10^5 .
- **[Variant v3: Cengage DPP 3.3 Q7, p. 5]** The relationship between ΔG° and E°_{cell} implies that for an electrochemical cell to be in chemical equilibrium, the required condition is: (a) $E^\circ_{\text{cell}} = 0$ (b) $E_{\text{cell}} = 0$ (c) $\Delta G^\circ = 0$ (d) $Q = 1$.

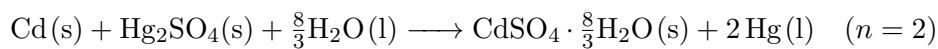
3.3 Subtopic 3.3: Cell Thermodynamics: ΔG , ΔS , ΔH & Temperature Coefficient

Q.3.4 [Primary Source: Atkins Topic 6C.4 / Neeraj Kumar Part 1 Ex-II Sec A Q32]

The EMF of the standard Weston cadmium cell is expressed as a function of temperature:

$$E_t = 1.01845 - 4.05 \times 10^{-5}(t - 20) - 9.5 \times 10^{-7}(t - 20)^2 \quad (\text{in Volts})$$

where t is the temperature in $^{\circ}\text{C}$. For the cell reaction:



Calculate at $t = 25^{\circ}\text{C}$:

- The temperature coefficient $\left(\frac{\partial E}{\partial T}\right)_P$
- The Gibbs free energy change ΔG in kJ mol^{-1}
- The entropy change ΔS in $\text{J K}^{-1} \text{mol}^{-1}$
- The reaction enthalpy ΔH in kJ mol^{-1}
- The reversible heat exchange q_{rev} with the surroundings

Cross-Book Redundant / Identical Variants:

- [Variant v1: GRB Practice Problem 28, p. 48] The EMF of a galvanic cell is 1.05 V at 298 K and its temperature coefficient is $-4.0 \times 10^{-4} \text{ V K}^{-1}$. If $n = 2$, calculate ΔH for the cell reaction.
- [Variant v2: Narendra Avasthi Level-2 Q48, p. 28] A cell has $E = 1.20 \text{ V}$ and $\Delta H = -200 \text{ kJ mol}^{-1}$ for a 2-electron process. The temperature coefficient $(\partial E / \partial T)_P$ is: (a) $+5.4 \times 10^{-4} \text{ V K}^{-1}$ (b) $-5.4 \times 10^{-4} \text{ V K}^{-1}$ (c) $+2.7 \times 10^{-4} \text{ V K}^{-1}$ (d) zero.
- [Variant v3: Cengage DPP 3.4 Q4, p. 7] If a cell absorbs heat from its surroundings during reversible operation at constant T and P , then: (a) $(\partial E / \partial T)_P > 0$ (b) $(\partial E / \partial T)_P < 0$ (c) $\Delta H > \Delta G$ (d) Both (a) and (c).

Chapter 4

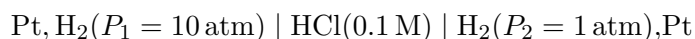
Master Problem Bank: Concentration Cells & Liquid Junction Potential

Part IV Organization: Concentration Cells & Liquid Junction Potential

This part compiles all questions and quantitative problems on electrode concentration cells (gas & amalgam), electrolyte concentration cells without transference, concentration cells with transference, and the liquid junction potential (E_{lj}) from all 8 books. Identical and redundant variants from other books are co-located beneath each primary problem with full source citations.

4.1 Subtopic 4.1: Electrode Concentration Cells (Gas & Amalgam)

Q.4.1 [Primary Source: Neeraj Kumar Part 1 Ex-I Q95 / GRB 12.24]
Calculate the potential at 25°C of the gas concentration cell:



- (a) +0.0296 V (b) -0.0296 V (c) +0.0591 V (d) 0.000 V

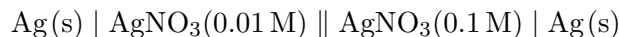
Cross-Book Redundant / Identical Variants:

- [Variant v1: Pearson Ex-I Q90, p. 8] For the gas cell $\text{Pt}, \text{Cl}_2(P_1) \mid \text{NaCl}(1 \text{ M}) \mid \text{Cl}_2(P_2), \text{Pt}$, the cell reaction is spontaneous if: (a) $P_1 > P_2$ (b) $P_2 > P_1$ (c) $P_1 = P_2$ (d) independent of pressure.
- [Variant v2: Narendra Avasthi Level-2 Q55, p. 30] For a zinc amalgam concentration cell $\text{Zn}(\text{Hg}, a_1) \mid \text{ZnSO}_4(\text{aq}) \mid \text{Zn}(\text{Hg}, a_2)$, the EMF is given by: (a) $\frac{RT}{2F} \ln \frac{a_1}{a_2}$ (b) $\frac{RT}{2F} \ln \frac{a_2}{a_1}$ (c) $\frac{RT}{F} \ln \frac{a_1}{a_2}$ (d) zero.
- [Variant v3: Cengage Theory p. 30] In an amalgam concentration cell, why is $E_{\text{cell}}^\circ = 0.00 \text{ V}$ identically?

4.2 Subtopic 4.2: Electrolyte Concentration Cells Without Transference

Q.4.2 [Primary Source: GRB Practice Problem 38, p. 49 / PRSN Ex-II Q22]

The EMF of the concentration cell without transference:



is 0.0575 V at 25°C.

- Calculate the theoretical EMF assuming complete ionization and unit activity coefficients.
- If the mean activity coefficient of 0.01 M AgNO_3 is $\gamma_1 = 0.89$, determine the mean activity coefficient γ_2 of the 0.1 M solution.

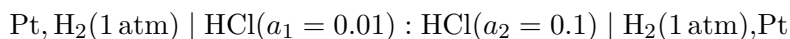
Cross-Book Redundant / Identical Variants:

- [Variant v1: Neeraj Kumar Part 1 Ex-I Q98, p. 9] The EMF of the cell $\text{Zn} \mid \text{Zn}^{2+}(0.001 \text{ M}) \parallel \text{Zn}^{2+}(0.1 \text{ M}) \mid \text{Zn}$ at 298 K is: (a) 0.0591 V (b) 0.0296 V (c) 0.1182 V (d) zero.
- [Variant v2: Narendra Avasthi Level-1 Q135, p. 19] If the concentration of both half-cells in an electrolyte concentration cell is doubled, the cell EMF will: (a) double (b) halve (c) remain unchanged (d) become zero.
- [Variant v3: Cengage DPP 3.3 Q18, p. 6] For a concentration cell reversible with respect to anions, $\text{Ag, AgCl(s)} \mid \text{KCl}(a_1) \parallel \text{KCl}(a_2) \mid \text{AgCl(s), Ag}$, the cell EMF is positive when: (a) $a_1 > a_2$ (b) $a_2 > a_1$ (c) $a_1 = a_2$ (d) always negative.

4.3 Subtopic 4.3: Concentration Cells With Transference & Liquid Junction Potential (E_{lj})

Q.4.3 [Primary Source: Atkins Topic 6C.2 / Neeraj Kumar Part 1 Ex-II Sec C Passage 1]

For the cell with direct liquid junction:



The EMF with transference is measured to be $E_{wt} = 0.0955 \text{ V}$ at 25°C .

- Derive the expression $E_{wt} = 2t_- \frac{RT}{F} \ln \frac{a_2}{a_1}$.
- Calculate the transport number of the chloride ion t_{Cl^-} and the proton t_{H^+} .
- Calculate the liquid junction potential E_{lj} at the interface.

Cross-Book Redundant / Identical Variants:

- [Variant v1: Pearson Ex-II Sec A Q52, p. 20]** The liquid junction potential for a 1 : 1 electrolyte boundary between activities a_1 and a_2 is given by: (a) $(t_+ - t_-) \frac{RT}{F} \ln(a_2/a_1)$ (b) $(2t_- - 1) \frac{RT}{F} \ln(a_2/a_1)$ (c) $(1 - 2t_+) \frac{RT}{F} \ln(a_2/a_1)$ (d) All of these are equivalent.
- [Variant v2: GRB 12.26 Solved Ex 42, p. 33]** Explain why the liquid junction potential is virtually eliminated when a saturated KCl bridge is inserted between two solutions.
- [Variant v3: Narendra Avasthi Level-3 Passage 2, p. 37]** In which case is the liquid junction potential negative for $a_2 > a_1$? (a) $t_+ > t_-$ (b) $t_- > t_+$ (c) $t_+ = t_-$ (d) always positive.

Chapter 5

Master Problem Bank: Advanced Equilibrium Applications of EMF

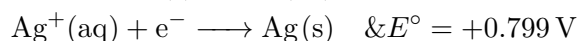
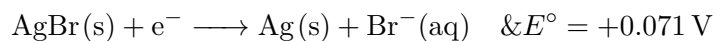
Part V Organization: Equilibrium Applications, pH, K_{sp} , Latimer & Frost

This part compiles all questions and multi-step analytical problems on solubility products (K_{sp}), potentiometric pH determination (SHE, quinhydrone, glass electrode), complex ion formation constants, and Latimer/Frost oxidation state diagrams from all 8 books. Identical and redundant variants from other books are co-located beneath each primary problem with full source citations.

5.1 Subtopic 5.1: Determination of Solubility Product (K_{sp})

Q.5.1 [Primary Source: Neeraj Kumar Part 1 Ex-II Sec F Q8 / NA Level-1 Q128]

Calculate the solubility product K_{sp} of silver bromide (AgBr) at 25°C from the following standard reduction potentials:



(a) 5.0×10^{-13}

(b) 2.5×10^{-12}

(c) 1.0×10^{-10}

(d) 7.2×10^{-15}

Cross-Book Redundant / Identical Variants:

- [Variant v1: GRB Practice Problem 50, p. 50] The EMF of the cell $\text{Ag} \mid \text{AgI(s)}, 0.05 \text{ M KI} \parallel 0.05 \text{ M AgNO}_3 \mid \text{Ag}$ is 0.788 V at 25°C. Calculate the solubility product of AgI.
- [Variant v2: Pearson Ex-II Sec E Q18, p. 38] For the cell $\text{Pb} \mid \text{PbSO}_4(\text{s}), \text{SO}_4^{2-}(0.1 \text{ M}) \parallel \text{Pb}^{2+}(0.1 \text{ M}) \mid \text{Pb}$, the EMF is 0.236 V at 298 K. Calculate $K_{sp}(\text{PbSO}_4)$.
- [Variant v3: Cengage Theory Concept App 5 Q2, p. 43] How can the solubility product of calomel (Hg_2Cl_2) be determined from standard electrode potentials?

5.2 Subtopic 5.2: Potentiometric pH Determination (SHE, Quinhydrone & Glass)

Q.5.2 [Primary Source: GRB 12.28 Solved Ex 48 / NA Level-2 Q36]

A quinhydrone electrode is set up in an unknown acidic buffer solution at 25°C and coupled with a saturated calomel electrode ($E_{\text{SCE}} = +0.242 \text{ V}$). The cell EMF is measured to be 0.180 V with the quinhydrone electrode as the positive pole. Given $E^\circ(\text{Q}, 2\text{H}^+/\text{H}_2\text{Q}) = +0.699 \text{ V}$:

- Calculate the pH of the buffer solution.
- Why cannot the quinhydrone electrode be used accurately in solutions of $\text{pH} > 8.5$?

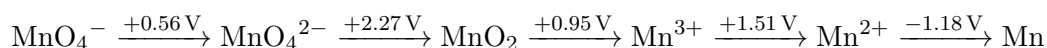
Cross-Book Redundant / Identical Variants:

- [Variant v1: Neeraj Kumar Part 1 Ex-II Sec A Q28, p. 14] The EMF of a cell consisting of a hydrogen electrode dipping in a solution of unknown pH coupled with SCE is 0.450 V. The pH is: (a) 3.52 (b) 7.61 (c) 4.25 (d) 5.80.
- [Variant v2: Pearson Ex-II Sec A Q58, p. 21] In alkaline solution ($\text{pH} > 9$), hydroquinone undergoes: (a) oxidation by dissolved oxygen (b) ionization of phenolic $-\text{OH}$ groups ($K_a \approx 10^{-10}$) disrupting the equimolar $[\text{Q}] = [\text{H}_2\text{Q}]$ ratio (c) dimerization (d) precipitation.
- [Variant v3: Cengage Theory p. 40] Write the potential equation for a glass electrode and explain how it determines pH across the full range $1 \leq \text{pH} \leq 12$.

5.3 Subtopic 5.3: Latimer & Frost Oxidation State Diagrams

Q.5.3 [Primary Source: Atkins Focus 6 / Neeraj Kumar Part 1 Ex-II Sec B Q14]

The Latimer diagram for manganese species in 1 M acid solution ($\text{pH} = 0$) is:



- Calculate the standard potential E° for the direct 5-electron reduction $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$.
- Will manganate ion (MnO_4^{2-}) undergo spontaneous disproportionation in acid solution? Justify thermodynamically by calculating E_{disp}° .
- Will Mn^{3+} undergo spontaneous disproportionation into MnO_2 and Mn^{2+} ?

Cross-Book Redundant / Identical Variants:

- [Variant v1: Pearson Ex-II Sec A Q68, p. 22] On a Frost diagram (NE° vs N), a species that lies above the line connecting its two neighboring oxidation states is: (a) stable to disproportionation (b) prone to spontaneous disproportionation (c) a reducing agent only (d) in its ground electronic state.
- [Variant v2: Narendra Avasthi Level-3 Passage 5, p. 39] Given the standard reduction potentials: $\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$ ($E^\circ = 0.77 \text{ V}$) and $\text{Fe}^{2+} + 2\text{e}^- \longrightarrow \text{Fe}$ ($E^\circ = -0.44 \text{ V}$), calculate E° for $\text{Fe}^{3+} + 3\text{e}^- \longrightarrow \text{Fe}$.
- [Variant v3: Cengage Theory p. 46] In a Latimer diagram, comproportionation occurs spontaneously between two oxidation states when: $E_{\text{right}}^\circ < E_{\text{left}}^\circ$ for the intermediate state.

Chapter 6

Master Problem Bank: Electrolysis, Faraday's Laws & Overvoltage

Part VI Organization: Electrolysis, Faraday's Laws & Kinetics

This part compiles all questions and quantitative problems on preferential discharge, overpotential (η) on various electrode materials, Faraday's 1st and 2nd laws, current efficiency, and industrial electrolytic cells from all 8 books. Identical and redundant variants from other books are co-located beneath each primary problem with full source citations.

6.1 Subtopic 6.1: Preferential Discharge Theory & Overpotential (η)

Q.6.1 [Primary Source: Neeraj Kumar Part 2 Ex-II Sec B Q4 / GRB 12.3]

Predict the primary product liberated at the cathode and at the anode during the electrolysis of an aqueous 1 M solution of NaCl under the following two distinct experimental conditions:

- (a) Using inert platinum (Pt) electrodes at low current density
- (b) Using a liquid mercury (Hg) cathode and graphite anode (Castner–Kellner cell)

Explain the physical role of hydrogen overvoltage on mercury ($\eta_{\text{H}_2/\text{Hg}} \approx 1.5 \text{ V}$) in determining the cathodic product.

Cross-Book Redundant / Identical Variants:

- [Variant v1: Pearson Ex-II Sec A Q56, p. 21] During the electrolysis of aqueous NaCl with mercury cathode, sodium is discharged instead of hydrogen because: (a) $E^\circ(\text{Na}^+/\text{Na}) > E^\circ(\text{H}^+/\text{H}_2)$ (b) hydrogen overvoltage on mercury is exceptionally high (c) sodium forms amalgam with mercury lowering its discharge potential (d) both (b) and (c).
- [Variant v2: Narendra Avasthi Level-1 Q142, p. 20] During electrolysis of aqueous CuSO_4 solution using copper electrodes, the process taking place at the anode is: (a) evolution of O_2 (b) evolution of SO_2 (c) dissolution of copper into Cu^{2+} (d) deposition of sulfate.
- [Variant v3: Cengage Theory p. 78] Why does chlorine gas evolve at the anode during electrolysis of concentrated aqueous NaCl even though thermodynamic standard reduction potential of oxygen ($E^\circ = 1.23 \text{ V}$) is lower than that of chlorine ($E^\circ = 1.36 \text{ V}$)?

6.2 Subtopic 6.2: Faraday's 1st and 2nd Laws & Current Efficiency

Q.6.2 [Primary Source: GRB Practice Problem 64, p. 52 / NA Level-1 Q58]

A constant current of 3.0 A is passed for 2.0 hours through three electrolytic cells connected in series containing solutions of: (A) $\text{AgNO}_3(\text{aq})$, (B) $\text{CuSO}_4(\text{aq})$, (C) $\text{CrCl}_3(\text{aq})$. Assuming 100% current efficiency, calculate the mass deposited at the cathode in each cell:

- (a) Mass of silver (Atomic mass = 108.0)
- (b) Mass of copper (Atomic mass = 63.5)
- (c) Mass of chromium (Atomic mass = 52.0)

Cross-Book Redundant / Identical Variants:

- [Variant v1: Neeraj Kumar Part 2 Ex-I Q68, p. 38] The same quantity of electrical charge that deposited 1.08 g of silver from AgNO_3 will deposit what mass of aluminium from molten AlCl_3 ? (a) 0.09 g (b) 0.27 g (c) 0.81 g (d) 1.08 g.
- [Variant v2: Essential 24.4 Solved Ex 3, p. 863] What current strength in amperes is required to deposit 5.00 g of gold from a solution of HAuCl_4 in 1.0 hour? (At. mass of Au = 197.0, $n = 3$).
- [Variant v3: Pearson Ex-I Q82, p. 7] In an electroplating process, a current of 10.0 A deposits 12.0 g of zinc in 1.0 hour. The current efficiency (η_{curr}) is: (a) 98.5% (b) 82.4% (c) 65.2% (d) 50.0%.

Q.6.3 [Primary Source: Narendra Avasthi Level-2 Q10, p. 25 / PRSN Ex-II Q8]

An aqueous solution of 0.1 M CuSO_4 (1.0 L) is electrolyzed between inert Pt electrodes with a steady current of 9.65 A for 1000 seconds.

- (a) Calculate the mass of copper deposited at the cathode.
- (b) Calculate the total volume of oxygen gas evolved at the anode measured at STP (0°C , 1 atm).
- (c) Calculate the final pH of the solution after electrolysis (assume volume remains constant at 1.0 L).

Cross-Book Redundant / Identical Variants:

- [Variant v1: Cengage DPP 3.2 Q19, p. 4] A current strength of 96.5 A is passed for 10 seconds through 1 L of 0.1 M aqueous CuSO_4 . The pH of the resulting solution is: (a) 1.0 (b) 2.0 (c) 3.0 (d) 7.0.
- [Variant v2: Neeraj Kumar Part 2 Ex-II Sec A Q34, p. 44] When 100 mL of 0.2 M AgNO_3 is electrolyzed until half the silver is deposited, how many moles of H^+ ions are produced in the electrolyte?
- [Variant v3: GRB Practice Problem 68, p. 52] Electrolysis of dilute H_2SO_4 for 1.0 hour produces 112 mL of O_2 at STP at the anode. What volume of H_2 is evolved at the cathode at the same temperature and pressure?

Chapter 7

Master Problem Bank: Commercial Cells, Fuel Cells & Corrosion

Part VII Organization: Commercial Batteries, Fuel Cells & Corrosion

This part compiles all questions and multi-step quantitative problems on primary dry cells, secondary storage batteries (lead-acid, nicad, lithium-ion), hydrogen-oxygen fuel cells, and electrochemical corrosion mechanisms from all 8 books. Identical and redundant variants from other books are co-located beneath each primary problem with full source citations.

7.1 Subtopic 7.1: Secondary Batteries (Lead-Acid Storage Accumulator)

Q.7.1 [Primary Source: Neeraj Kumar Part 1 Ex-II Sec C Passage 3 / GRB 12.31]

For the lead-acid storage accumulator during discharge:

- Write the balanced half-reactions at the negative electrode (Pb) and positive electrode (PbO₂), and the overall net cell reaction.
- A lead storage battery contains 3.0 L of 38% (*w/w*) H₂SO₄ (density = 1.28 g cm⁻³). After discharging at a steady current of 4.0 A for 50.0 hours, calculate the mass of H₂SO₄ consumed and the mass of PbSO₄ formed on the plates.
- Explain why the electrolyte density falls to $\approx 1.15 \text{ g cm}^{-3}$ upon complete discharge, and how this property is utilized to assess the state of charge.

Cross-Book Redundant / Identical Variants:

- [Variant v1: Narendra Avasthi Level-3 Passage 4, p. 38] During recharging of a lead-acid cell, the reaction occurring at the electrode connected to the positive terminal of the charger is: (a) $\text{PbSO}_4 \longrightarrow \text{Pb}$ (b) $\text{PbSO}_4 \longrightarrow \text{PbO}_2$ (c) $\text{Pb} \longrightarrow \text{PbSO}_4$ (d) $\text{H}_2\text{O} \longrightarrow \text{H}_2$.
- [Variant v2: Pearson Ex-II Sec D Q8, p. 34] A commercial lead storage battery delivers 100 A · h of charge. The theoretical minimum mass of PbO₂ consumed is: (a) 446 g (b) 223 g (c) 892 g (d) 112 g.
- [Variant v3: Cengage Theory p. 95] What is the nominal single-cell voltage of a fully charged lead-acid cell, and how many cells are placed in series to construct a standard 12 V automobile battery?

7.2 Subtopic 7.2: Primary Batteries & Secondary Alkaline Cells

Q.7.2 [Primary Source: GRB 12.30 Objective Q125 / NA Level-1 Q180]

In a standard Leclanché dry cell:

- What constitutes the anode and cathode materials?
- What is the chemical function of MnO_2 in the electrolyte paste?
- What is the role of ZnCl_2 in preventing gas pressure buildup from released NH_3 ?

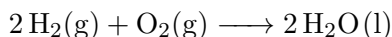
Cross-Book Redundant / Identical Variants:

- [Variant v1: Pearson Ex-I Q102, p. 11] In the alkaline dry cell, the electrolyte paste is: (a) $\text{NH}_4\text{Cl} + \text{ZnCl}_2$ (b) KOH (c) H_2SO_4 (d) NaOH .
- [Variant v2: Neeraj Kumar Part 1 Ex-II Sec D Q4, p. 28] In a nickel-cadmium (Nicaid) rechargeable cell, the reaction at the cathode during discharge is: (a) reduction of $\text{NiO}(\text{OH})$ to $\text{Ni}(\text{OH})_2$ (b) oxidation of Cd (c) reduction of $\text{Cd}(\text{OH})_2$ (d) oxidation of Ni .
- [Variant v3: Cengage Theory p. 93] Why does a mercury button cell maintain a constant operating voltage ($\approx 1.35 \text{ V}$) throughout its lifetime, whereas a Leclanch dry cell voltage drops continuously?

7.3 Subtopic 7.3: Fuel Cells & Thermodynamic Efficiency (η)

Q.7.3 [Primary Source: Atkins Topic 6C.4 / Neeraj Kumar Part 1 Ex-II Sec C Passage 5]

For the alkaline hydrogen–oxygen fuel cell operating at 298 K:



Given: $\Delta G^\circ = -474.4 \text{ kJ}$ and $\Delta H^\circ = -571.6 \text{ kJ}$ for the reaction as written ($n = 4$).

- Calculate the standard theoretical cell potential E°_{cell} .
- Calculate the maximum theoretical thermodynamic efficiency $\eta_{\text{th}} = \frac{\Delta G^\circ}{\Delta H^\circ} \times 100\%$.
- Explain why fuel cell efficiencies ($\sim 70\text{--}80\%$) exceed the Carnot efficiency limit of conventional thermal power plants.

Cross-Book Redundant / Identical Variants:

- [Variant v1: GRB 12.32 Solved Ex 52, p. 42] In the Apollo space mission, the alkaline $\text{H}_2\text{--O}_2$ fuel cell produced both electrical power and drinking water for astronauts. Write the electrode reactions in concentrated hot KOH electrolyte.
- [Variant v2: Narendra Avasthi Level-3 Passage 6, p. 40] If a fuel cell reaction has a positive temperature coefficient $(\partial E / \partial T)_P > 0$, its thermodynamic efficiency η : (a) exceeds 100% (b) is less than 100% (c) equals zero (d) cannot be defined.
- [Variant v3: Cengage Theory p. 100] Contrast the alkaline fuel cell with the Proton Exchange Membrane Fuel Cell (PEMFC) in terms of electrolyte and operating temperature.

7.4 Subtopic 7.4: Corrosion Mechanism, Rusting & Protection

Q.7.4

[Primary Source: GRB 12.33 / NA Level-1 Q195]

Explain the electrochemical mechanism of rusting of iron in the presence of moist air containing dissolved CO_2 :

- Write the anode and cathode half-reactions that form the localized galvanic micro-cells.
- What is the chemical formula of rust?
- Distinguish between sacrificial protection (galvanization with zinc) and cathodic protection using impressed direct current. Why is zinc preferred over tin for protecting iron sheets that are exposed to surface scratches?

Cross-Book Redundant / Identical Variants:

- [Variant v1: Pearson Ex-II Sec D Q18, p. 35] Rusting of iron is accelerated by: (a) presence of electrolytes (salts like NaCl) (b) acidic environment ($\text{pH} < 7$) (c) contact with less electropositive metals (e.g. Cu, Sn) (d) all of the above.
- [Variant v2: Neeraj Kumar Part 1 Ex-II Sec D Q8, p. 28] In cathodic protection of underground pipelines, blocks of which active metal are connected to the steel pipe? (a) Magnesium or Zinc (b) Copper (c) Silver (d) Lead.
- [Variant v3: Cengage Theory p. 104] Explain why coating iron with tin (Sn) accelerates corrosion if the tin coating is scratched, whereas coating with zinc (Zn) continues to protect the underlying iron even when scratched.

Part II

Exhaustive Source Book Question Banks

Chapter 8

Neeraj Kumar [NRJ]: Complete Exercise Question Bank

Problem 1

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q1]

A metal rod is dipped in a solution of its ions. Its electrode potential is independent of

- (a) temperature of the solution
- (b) concentration of the solution
- (c) area of the metal exposed
- (d) nature of the metal

Problem 2

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q2]

Indicator electrode is

- (a) SHE
- (b) Calomel electrode
- (c) Ag/AgCl electrode
- (d) Quinhydrone electrode

Problem 3

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q3]

The position of some metals in the electrochemical series in decreasing electropositive character is given as: $\text{Mg} > \text{Al} > \text{Zn} > \text{Cu} > \text{Ag}$. What will happen if a copper spoon is used to stir a solution of aluminium nitrate?

- (a) The spoon will get coated with aluminium.
- (b) An alloy of copper and aluminium is formed.
- (c) The solution becomes blue.
- (d) No chemical change will take place.

Problem 4

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q4]

Four colourless salt solutions are placed in separate test tubes and a strip of copper is placed in each. Which solution nally turns blue?

- (a) AgNO_3 (b) $\text{Pb}(\text{NO}_3)_2$
- (c) $\text{Zn}(\text{NO}_3)_2$ (d) $\text{Cd}(\text{NO}_3)_2$

Problem 5

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q5]

The metal that cannot be obtained on reduction of its oxide by aluminium is

- (a) K (b) Mn
- (c) Cr (d) Fe

Problem 6

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q6]

Beryllium is placed above magnesium in the group II. Beryllium dust, therefore, when added to MgCl_2 solution will

- (a) have no effect
- (b) precipitate Mg metal
- (c) precipitate MgO
- (d) lead to dissolution of Be metal

Problem 7

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q7]

A standard hydrogen electrode has zero electrode potential because

- (a) hydrogen is easiest to oxidize
- (b) this electrode potential is assumed to be zero
- (c) hydrogen atom has only one electron
- (d) hydrogen is the lightest element

Problem 8

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q8]

The standard reduction potential values of three metallic cations, X, Y and Z are +0.52, -3.03 and -1.18 V, respectively. The order of reducing power of the corresponding metals is

- (a) $Y > Z > X$
- (b) $X > Y > Z$
- (c) $Z > Y > X$
- (d) $Z > X > Y$

EXERCISE I (JEE MAIN) Electrochemistry CHAPTER 8
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Problem 9

[Neeraj Kumar [NRJ] EXERCISE I Q9]

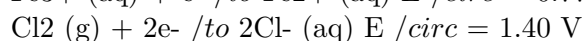
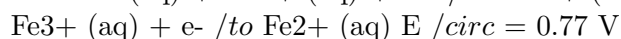
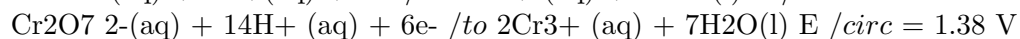
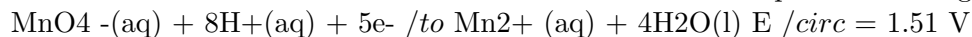
A gas 'X' at 1 atm is bubbled through a solution containing a mixture of 1 M Y^- and 1 M Z^- at 25°C . If the reduction potential of $\text{Z}^- > \text{Y}^- > \text{X}$, then,

- (a) Y^- will oxidize X and not Z^-
- (b) Y^- will oxidize Z^- and not X
- (c) Y^- will oxidize both X and Z^-
- (d) Y^- will reduce both X and Z^-

Problem 10

[Neeraj Kumar [NRJ] EXERCISE I Q10]

Standard electrode potential data are useful for understanding the suitability of an oxidant in a redox titration. Some half-cell reactions and their standard potentials are given below:



Identify the only incorrect statement regarding the quantitative estimation of aqueous $\text{Fe}(\text{NO}_3)_2$

- (a) MnO_4^- can be used in aqueous HCl
- (b) $\text{Cr}_2\text{O}_7^{2-}$ can be used in aqueous HCl
- (c) MnO_4^- can be used in aqueous H_2SO_4
- (d) $\text{Cr}_2\text{O}_7^{2-}$ can be used in aqueous H_2SO_4

Problem 11

[Neeraj Kumar [NRJ] EXERCISE I Q11]

The decreasing order of standard electrode potential of Mg, K, Ba and Ca, is

- (a) K, Ca, Ba, Mg
- (b) Ba, Ca, K, Mg
- (c) Ca, Mg, K, Ba
- (d) Mg, Ca, Ba, K

Problem 12

[Neeraj Kumar [NRJ] EXERCISE I Q12]

A metal having negative reduction potential, when dipped in the solution of its own ions, has a tendency to

- (a) remain as metal atoms
- (b) become electrically positive
- (c) become electrically negative
- (d) be deposited from the solution

Problem 13

[Neeraj Kumar [NRJ] EXERCISE I Q13]

The calomel electrode is reversible with respect to

- (a) Hg_2^{2+} (b) H^+
- (c) Hg_2^{2+} (d) Cl^-

Problem 14

[Neeraj Kumar [NRJ] EXERCISE I Q14]

Which one of the following does not get oxidized by bromine water?

- (a) Fe^{2+} to Fe^{3+} (b) Cu^+ to Cu^{2+}
- (c) Mn^{2+} to MnO_4^- (d) Sn^{2+} to Sn^{4+}

Problem 15

[Neeraj Kumar [NRJ] EXERCISE I Q15]

In salt bridge, normally KCl is used because

- (a) it is a strong electrolyte.
- (b) it is good conductor of electricity.
- (c) K^+ and Cl^- ions have nearly same ionic mobility.
- (d) it is an ionic compound.

Problem 16

[Neeraj Kumar [NRJ] EXERCISE I Q16]

By how much would the oxidizing power of $\text{MnO}_4^-/\text{Mn}^{2+}$ couple change if the H^+ ions concentration is decreased 100 times at 25°C ?

- (a) increases by 189 mV
- (b) decreases by 189 mV
- (c) will increase by 19 mV
- (d) will decrease by 19 mV

Problem 17

[Neeraj Kumar [NRJ] EXERCISE I Q17]

The solution of CuSO_4 , in which copper rod is immersed, is diluted to 10 times. The reduction electrode potential

- (a) increases by 0.0295 V
- (b) decreases by 0.0295 V
- (c) increases by 0.059 V
- (d) decreases by 0.059 V

Problem 18

[Neeraj Kumar [NRJ] EXERCISE I Q18]

The standard reduction potential of oxygen in acidic solution is +1.23 V. What is the standard reduction potential of oxygen in basic solution?

- (a) +0.404 V (b) -0.404 V
- (c) +2.056 V (d) -2.056 V

Problem 19

[Neeraj Kumar [NRJ] EXERCISE I Q19]

The standard reduction potentials of $\text{Cu}^{2+} | \text{Cu}$ and $\text{Cu}^{2+} | \text{Cu}^+$ are 0.337 V and 0.153 V, respectively. The standard electrode potential of $\text{Cu}^+ | \text{Cu}$ half-cell is

- (a) 0.184 V (b) 0.827 V
(c) 0.521 V (d) 0.490 V

Problem 20

[Neeraj Kumar [NRJ] EXERCISE I Q20]

The electrode potential of hydrogen electrode in neutral solution and 298 K is

- (a) -0.413 V (b) zero
(c) -0.826 V (d) +0.413 V

Problem 21

[Neeraj Kumar [NRJ] EXERCISE I Q21]

Electrode potential will be more for hydrogen electrode at pH (at the same temperature)

- (a) 4 (b) 3 (c) 2 (d) 5

Problem 22

[Neeraj Kumar [NRJ] EXERCISE I Q22]

Saturated solution of KNO_3 is used to make 'salt bridge' because

- (a) velocity of K^+ is greater than that of NO_3^-
(b) velocity of NO_3^- is greater than that of K^+
(c) velocities of both K^+ and NO_3^- are nearly the same
(d) KNO_3 is highly soluble in water

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Problem 23

[Neeraj Kumar [NRJ] EXERCISE I Q23]

The standard reduction potentials of $\text{Pt} | \text{Cr}_2\text{O}_7^{2-}, \text{Cr}^{3+}$; $\text{Pt} | \text{MnO}_4^-, \text{Mn}^{2+}$; $\text{Pt} | \text{Ce}^{4+}, \text{Ce}^{3+}$ in the presence of acid are 1.33 V, 1.51 V and 1.61 V, respectively, at 25°C. The decreasing order of oxidizing power is

- (a) $\text{Cr}_2\text{O}_7^{2-} > \text{MnO}_4^- > \text{Ce}^{4+}$
(b) $\text{MnO}_4^- > \text{Cr}_2\text{O}_7^{2-} > \text{Ce}^{4+}$
(c) $\text{Ce}^{4+} > \text{MnO}_4^- > \text{Cr}_2\text{O}_7^{2-}$
(d) $\text{MnO}_4^- > \text{Ce}^{4+} > \text{Cr}_2\text{O}_7^{2-}$

Problem 24

[Neeraj Kumar [NRJ] EXERCISE I Q24]

The standard reduction potentials at 25°C of $\text{Li}^+ | \text{Li}$, $\text{Ba}^{2+} | \text{Ba}$, $\text{Na}^+ | \text{Na}$ and $\text{Mg}^{2+} | \text{Mg}$ are -3.05, -2.73, -2.71 and -2.37 V, respectively. Which is the strongest reducing agent?

- (a) Li (b) Ba
(c) Na (d) Mg

Problem 25

[Neeraj Kumar [NRJ] EXERCISE I Q25]

Some standard electrode potentials are given:

$\text{Fe}^{2+} + 2e^- \rightarrow \text{Fe}$; $E^\circ = -0.440 \text{ V}$

$\text{Fe}^{3+} + 3e^- \rightarrow \text{Fe}$; $E^\circ = -0.036 \text{ V}$

The standard electrode potential for: $\text{Fe}^{3+} + e^- \rightarrow \text{Fe}^{2+}$, is

- (a) -0.476 V (b) -0.404 V
(c) +0.988 V (d) +0.772 V

Problem 26

[Neeraj Kumar [NRJ] EXERCISE I Q26]

The dissociation constant for CH_3COOH is 1.8×10^{-5} at 298 K. The electrode potential for the half-cell: $\text{Pt} \mid \text{H}_2 (1 \text{ bar}) \mid 0.5 \text{ M} - \text{CH}_3\text{COOH}$, at 298 K is ($\log 2 = 0.3$; $\log 3 = 0.48$; $2.303 \text{ RT/F} = 0.06$)

- (a) -0.3024 V (b) -0.1512 V
(c) $+0.3024 \text{ V}$ (d) $+0.1512 \text{ V}$

Problem 27

[Neeraj Kumar [NRJ] EXERCISE I Q27]

At 25°C , the solubility product of CuCl is 2.0×10^{-7} and $E^\circ_{\text{Cl}^-/\text{Cl}_2}$ is 1.36 V . The value of $E^\circ_{\text{Cu}^{2+}/\text{Cu}}$ is ($\log 2 = 0.3$; $2.303 \text{ RT/F} = 0.06$)

- (a) $+1.08 \text{ V}$ (b) $+0.53 \text{ V}$
(c) $+1.682 \text{ V}$ (d) $+0.878 \text{ V}$

Problem 28

[Neeraj Kumar [NRJ] EXERCISE I Q28]

At 25°C , the solubility product of $\text{Pb}(\text{OH})_2 (\text{s})$ is 1.2×10^{-15} [Given: $E^\circ_{\text{Pb}^{2+}/\text{Pb}} = -0.13 \text{ V}$; $E^\circ_{\text{H}_2\text{O}/\text{OH}^-} = 1.23 \text{ V}$; $2.303 \text{ RT/F} = 0.06$]

- (a) 1.36×10^{-5} (b) 1.0×10^{-7}
(c) 1.0×10^{-14} (d) 1.25×10^{-15}

Problem 29

[Neeraj Kumar [NRJ] EXERCISE I Q29]

The standard potentials of $\text{MnO}_4^- \mid \text{Mn}^{2+}$ and $\text{MnO}_2 \mid \text{Mn}^{2+}$ electrodes in acid solution are 1.51 V and 1.23 V , respectively. Standard electrode potential for the electrode, $\text{MnO}_4^- \mid \text{MnO}_2$ in acid solution is

- (a) $+1.697 \text{ V}$ (b) $+5.09 \text{ V}$
(c) $+0.28 \text{ V}$ (d) $+1.37 \text{ V}$

Problem 30

[Neeraj Kumar [NRJ] EXERCISE I Q30]

The following reactions represent the reduction of IO_3^- ion into I^- ion in acidic and basic medium. Predict in which medium IO_3^- ion will act as a better oxidizing agent?

$\text{IO}_3^- + 6\text{H}^+ + 6\text{e}^- \rightarrow \text{I}^- + 3\text{H}_2\text{O}$; $E^\circ = +0.907 \text{ V}$

$\text{IO}_3^- + 3\text{H}_2\text{O} + 6\text{e}^- \rightarrow \text{I}^- + 6\text{OH}^-$; $E^\circ = +0.260 \text{ V}$

- (a) Acid medium (b) Basic medium
(c) Equally in both (d) Not predictable Galvanic Cell

Problem 31

[Neeraj Kumar [NRJ] EXERCISE I Q31]

The correct cell diagram for the following reaction and E° for the cell is

$2\text{AgBr}(\text{s}) + \text{H}_2(\text{g}) \rightarrow 2\text{Ag}(\text{s}) + 2\text{H}^+ + 2\text{Br}^-$

$E^\circ_{\text{AgBr}/\text{Ag}} = +0.10 \text{ V}$.

- (a) $(\text{Pt}) \mid \text{H}_2 \mid \text{H}^+ \parallel \text{Br}^- \mid \text{AgBr} \mid \text{Br}_2 (\text{Pt})$; $E^\circ = 0.10 \text{ V}$
(b) $(\text{Pt}) \mid \text{H}_2 \mid \text{H}^+ \parallel \text{Br}^- \mid \text{AgBr} \mid \text{Br}_2 (\text{Pt})$; $E^\circ = -0.10 \text{ V}$
(c) $(\text{Pt}) \mid \text{Br}_2 \mid \text{AgBr} \mid \text{Br}^- \parallel \text{H}^+ \mid \text{H}_2 (\text{Pt})$; $E^\circ = 0.10 \text{ V}$
(d) $(\text{Pt}) \mid \text{Br}_2 \mid \text{AgBr} \mid \text{Br}^- \parallel \text{H}^+ \mid \text{H}_2 (\text{Pt})$; $E^\circ = -0.10 \text{ V}$

Problem 32

[Neeraj Kumar [NRJ] EXERCISE I Q32]

In an experimental set-up for the measurement of EMF of a half-cell using a reference electrode and a salt bridge, when the salt bridge is removed, the voltage

- (a) remains the same
- (b) increases to maximum
- (c) decreases half the value
- (d) drops to zero

Problem 33

[Neeraj Kumar [NRJ] EXERCISE I Q33]

After some time, the voltage of an electrochemical cell becomes zero. This is because

- (a) their electrode potential becomes zero.
- (b) their reduction potential become equal but have opposite sign.
- (c) their reduction potential become equal and have the same sign.
- (d) the ions of the electrolyte in the salt bridge stop moving.

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Problem 34

[Neeraj Kumar [NRJ] EXERCISE I Q34]

The cell reaction for the given cell is spontaneous if Pt, Cl₂ (P₁ atm) | Cl⁻ | Cl₂ (P₂ atm), Pt

- (a) P₁ > P₂ (b) P₁ < P₂
- (c) P₁ = P₂ (d) P₁ = 1 atm

Problem 35

[Neeraj Kumar [NRJ] EXERCISE I Q35]

Which one of the following statements is incorrect regarding an electrochemical cell?

- (a) The electrode on which oxidation takes place is called anode.
- (b) Anode is the negative pole.
- (c) The direction of the current is same as that of the direction of ow of electrons.
- (d) The ow of current is partly due to ow of electrons and partly due to ow of ions.

Problem 36

[Neeraj Kumar [NRJ] EXERCISE I Q36]

When an electric current is drawn from a galvanic cell

- (a) EMF suddenly increases.
- (b) EMF gradually increases and attains a maximum value.
- (c) EMF decreases and nally falls to zero.
- (d) EMF must remain constant.

Problem 37

[Neeraj Kumar [NRJ] EXERCISE I Q37]

Identi cation of anode and cathode in an electrochemical cell is made by the use of

- (a) Galvanometer (b) Salt bridge
- (c) Voltmeter (d) Potentiometer

Problem 38

[Neeraj Kumar [NRJ] EXERCISE I Q38]

For the cell Zn | Zn²⁺ || Cu²⁺ | Cu, if the concentration of both, Zn²⁺ and Cu²⁺ ions are doubled, the EMF of the cell

- (a) doubles (b) reduces to half
- (c) remains same (d) becomes zero

Problem 39

[Neeraj Kumar [NRJ] EXERCISE I Q39]

The standard EMF of a galvanic cell can be calculated from

- (a) the size of the electrode
- (b) the pH of the solution
- (c) the amount of metal in the anode
- (d) the E° values of the half-cells

Problem 40

[Neeraj Kumar [NRJ] EXERCISE I Q40]

The value of equilibrium constant for a feasible cell reaction must be

- (a) <1 (b) Zero
- (c) $= 1$ (d) >1

Problem 41

[Neeraj Kumar [NRJ] EXERCISE I Q41]

If the cell reaction is spontaneous then

- (a) $\Delta G^\circ = -ve$ (b) $E^\circ = -ve$
- (c) $E^\circ = +ve$ (d) $\Delta G^\circ = -ve$

Problem 42

[Neeraj Kumar [NRJ] EXERCISE I Q42]

Which of the following pair of metals, when coupled, will give maximum EMF for a voltaic cell?

- (a) Fe and Cu (b) Pb and Au
- (c) Cu and Au (d) Ca and Cu

Problem 43

[Neeraj Kumar [NRJ] EXERCISE I Q43]

When a lead storage battery is charged

- (a) PbO_2 dissolves.
- (b) the lead electrode becomes coated with lead sulphate.
- (c) sulphuric acid is regenerated.
- (d) the amount of acid decreases.

Problem 44

[Neeraj Kumar [NRJ] EXERCISE I Q44]

Use of lithium metal as an electrode in high energy density batteries is due to

- (a) lithium is the lightest element.
- (b) lithium has the highest oxidation potential.
- (c) lithium is quite reactive.
- (d) lithium does not corrode readily.

Problem 45

[Neeraj Kumar [NRJ] EXERCISE I Q45]

A depolarizer used in dry cell batteries is

- (a) NH_4Cl (b) MnO_2
- (c) KOH (d) Na_3PO_4

Problem 46

[Neeraj Kumar [NRJ] EXERCISE I Q46]

Which is correct about fuel cells?

- (a) Cell continuously run as long as fuels are supplied.
- (b) These are more efficient and free from pollution.
- (c) These are used to provide power and drinking water to astronauts in space programme.
- (d) All of these

Problem 47

[Neeraj Kumar [NRJ] EXERCISE I Q47]

When a lead storage battery is discharged

- (a) SO₂ is evolved
- (b) lead sulphate is consumed
- (c) lead is formed
- (d) sulphuric acid is consumed

Problem 48

[Neeraj Kumar [NRJ] EXERCISE I Q48]

For a cell reaction involving a two- electron change, the standard EMF of the cell is found to be 0.295 V at 25°C. The equilibrium constant of the reaction at 25°C will be

- (a) 1×10^{10} (b) 1×10^{-10}
- (c) 29.5×10^{-2} (d) 2×10^{10}

Problem 49

[Neeraj Kumar [NRJ] EXERCISE I Q49]

The ECell for Ag(s) | AgI (satd) || Ag+ (0.10 M) | Ag (s) is +0.413 V. What is the value of K_{sp} of AgI?

- (a) 1.0×10^{-8} (b) 1.0×10^{-7}
- (c) 1.0×10^{-14} (d) 1.0×10^{-16}

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[Neeraj Kumar [NRJ] EXERCISE I Q50]

Assuming that hydrogen behaves as an ideal gas, what is the EMF of the cell at 25°C if P₁ = 600 mm and P₂ = 420 mm: Pt | H₂ (P₁) | HCl | H₂ (P₂) | Pt? [Given: $2.303 RT/F = 0.06$, $\log 7 = 0.85$]

- (a) -0.0045 V (b) -0.0 V
- (c) +0.0045 V (d) +0.0015 V

Problem 51

[Neeraj Kumar [NRJ] EXERCISE I Q51]

For the electrochemical cell, M | M⁺ || X⁻ | X, E_{M⁺ / M} = 0.44 V and E_{X / X⁻} = 0.33 V. From this data, one can deduce that

- (a) M + X⁻ → M⁺ + X is the spontaneous reaction
- (b) M⁺ + X⁻ → M + X is the spontaneous reaction
- (c) E_{cell} = 0.77 V
- (d) E_{cell} = - 0.77 V

Problem 52

[Neeraj Kumar [NRJ] EXERCISE I Q52]

The EMF of the cell: Zn | Zn²⁺ (0.01 M) || Fe²⁺ (0.001 M) | Fe at 298 K is 0.2905 V, then the value of equilibrium constant for the cell reaction is

- (a) $e^{0.3200295}$. .
- (b) $10^{0.3200295}$. .
- (c) $10^{0.2600295}$. .
- (d) $10^{0.3200591}$. .

Problem 53

[Neeraj Kumar [NRJ] EXERCISE I Q53]

For the cell reaction: 4Br⁻ + O₂ + 4H⁺ → 2Br₂ + 2H₂O; E_{cell} = 0.18 V. The value of (log K_C) at 298 K is [$2.303 RT/F = 0.06$]

- (a) 12 (b) 6
- (c) 18 (d) 3

Problem 54

[Neeraj Kumar [NRJ] EXERCISE I Q54]

The standard EMF for the cell reaction: $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$ is 1.10 volts at 25°C. The EMF of the cell reaction when 0.1 M Cu^{2+} and 0.1 M Zn^{2+} solutions are used at 25°C is

- (a) 1.10 V (b) 1.041 V
(c) -1.10 V (d) -1.041 V

Problem 55

[Neeraj Kumar [NRJ] EXERCISE I Q55]

For the cell: $\text{Ni} \mid \text{Ni}^{2+} \parallel \text{Cu}^{2+} \mid \text{Cu}$; $E^\circ = 0.77 \text{ V}$. By which of the following activity, E_{cell} will increase?

- (a) On decreasing $[\text{Ni}^{2+}]$
(b) On decreasing $[\text{Cu}^{2+}]$
(c) On increasing mass of Ni electrode
(d) On increasing mass of Cu electrode

Problem 56

[Neeraj Kumar [NRJ] EXERCISE I Q56]

The standard EMF of a Daniel cell at 298 K is E_1 . When the concentration of ZnSO_4 is 1.0 M and that of CuSO_4 is 0.01 M, the EMF becomes E_2 at 298 K. The correct relationship between E_1 and E_2 is

- (a) $E_1 = E_2$ (b) $E_2 = 0$
(c) $E_1 > E_2$ (d) $E_1 < E_2$

Problem 57

[Neeraj Kumar [NRJ] EXERCISE I Q57]

For a reaction $\text{A(s)} + 2\text{B}^+ \rightarrow \text{A}^{2+} + \text{B(s)}$; K_C has been found to be 1012. The EMF of the cell is

- (a) 0.354 V (b) 0.708 V
(c) 0.534 V (d) 0.453 V

Problem 58

[Neeraj Kumar [NRJ] EXERCISE I Q58]

An electrochemical cell is set up as follows:

$\text{Pt} \mid \text{H}_2(\text{g}) (1 \text{ atm}) \mid 0.001 \text{ M HCl} \parallel 0.1 \text{ M HA} \mid \text{H}_2(\text{g}) (1 \text{ atm}) \mid \text{Pt}$

EMF of this cell is zero because (pK_a of $\text{HA} = 5$)

- (a) molar concentrations of acids are different
(b) Temperature is constant
(c) pH of two solutions are same
(d) both are standard hydrogen electrodes

Problem 59

[Neeraj Kumar [NRJ] EXERCISE I Q59]

The reaction: $\text{H}_2(\text{g}) + 2\text{AgCl(s)} \rightarrow 2\text{H}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) + 2\text{Ag(s)}$ occurs in the galvanic cell

- (a) $\text{Ag} \mid \text{AgCl(s)} \mid \text{KCl(aq)} \mid \text{AgNO}_3(\text{aq}) \mid \text{Ag}$
(b) $\text{Pt} \mid \text{H}_2(\text{g}) \mid \text{HCl(aq)} \mid \text{AgNO}_3(\text{aq}) \mid \text{Ag}$
(c) $\text{Pt} \mid \text{H}_2(\text{g}) \mid \text{HCl(aq)} \mid \text{AgCl(s)} \mid \text{Ag}$
(d) $\text{Pt} \mid \text{H}_2(\text{g}) \mid \text{KCl(aq)} \mid \text{AgCl(s)} \mid \text{Ag}$

Problem 60

[Neeraj Kumar [NRJ] EXERCISE I Q60]

The standard reduction potentials in acidic conditions are 0.77 V and 0.53 V, respectively, for $\text{Fe}^{3+} | \text{Fe}^{2+}$ and $\text{I}_3^- | \text{I}^-$ couples. The equilibrium constant for the reaction: $2\text{Fe}^{3+} + 3\text{I}^- \rightarrow 2\text{Fe}^{2+} + \text{I}_3^-$, is $(2.303 RT/F = 0.06)$

- (a) 2 /times 108 (b) 108
(c) 104 (d) 10-8

Problem 61

[Neeraj Kumar [NRJ] EXERCISE I Q61]

The following electrochemical cell has been set up: $\text{Pt(s)} | \text{Fe}^{3+}, \text{Fe}^{2+} (a = 1) || \text{Ce}^{4+}, \text{Ce}^{3+} (a = 1) | \text{Pt(s)}$; $E_{\text{Fe}^{3+}|\text{Fe}^{2+}} = 0.77 \text{ V}$; $E_{\text{Ce}^{4+}|\text{Ce}^{3+}} = 1.61 \text{ V}$. If an ammeter is connected between the two platinum electrodes, predict the direction of flow of current. Will the current increase or decrease with time?

- (a) Ce electrode to Fe electrode, decrease
(b) Ce electrode to Fe electrode, increase
(c) Fe electrode to Ce electrode, decrease
(d) Fe electrode to Ce electrode, increase

Problem 62

[Neeraj Kumar [NRJ] EXERCISE I Q62]

For the reaction: $\text{H}_2 (1 \text{ bar}) + 2\text{AgCl(s)} \rightarrow 2\text{Ag(s)} + 2\text{H}^+ (0.1 \text{ M}) + 2\text{Cl}^- (0.1 \text{ M})$; $\Delta G^\circ = -48,250 \text{ J}$ at 25°C . The EMF of cell in which the given reaction takes place is

- (a) 0.25 V (b) 0.37 V
(c) 0.13 V (d) 0.049 V

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Problem 63

[Neeraj Kumar [NRJ] EXERCISE I Q63]

A cell contains two hydrogen electrodes. The negative electrode is in contact with a solution of 10^{-6} M hydrogen ions. The EMF of the cell is 0.118 V at 25°C . The concentration of hydrogen ions at the positive electrode is

- (a) 10^{-6} M (b) 10^{-3} M
(c) 10^{-4} M (d) 10^{-5} M

Problem 64

[Neeraj Kumar [NRJ] EXERCISE I Q64]

The standard potentials of OCl^-/Cl^- and Cl^-/Cl_2 are 0.94 V and -1.36 V, respectively. The E value of OCl^-/Cl_2 will be

- (a) 3.24 V (b) -0.42 V
(c) -2.30 V (d) 0.52 V

Problem 65

[Neeraj Kumar [NRJ] EXERCISE I Q65]

From the following E° values for the half-cells:

- (i) $\text{D} \rightarrow \text{D}^{2+} + 2\text{e}^-$; $E^\circ = -1.5 \text{ V}$
(ii) $\text{B}^+ + \text{e}^- \rightarrow \text{B}$; $E^\circ = -0.5 \text{ V}$
(iii) $\text{A}_3^- \rightarrow \text{A}_2^- + \text{e}^-$; $E^\circ = 1.5 \text{ V}$
(iv) $\text{C}_2^+ + \text{e}^- \rightarrow \text{C}^+$; $E^\circ = +0.5 \text{ V}$

Which combination of two half-cells would result in a cell with largest potential?

- (a) i and iii (b) i and iv
(c) iii and iv (d) ii and iv

Problem 66

[Neeraj Kumar [NRJ] EXERCISE I Q66]

Which of the following statements does not differentiate between electrochemical cell and electrolytic cell?

- (a) Spontaneous or non-spontaneous nature of the chemical process.
- (b) Chemical reactions occurring at the electrodes
- (c) Positive and negative nature of anode.
- (d) EMF measurement.

Problem 67

[Neeraj Kumar [NRJ] EXERCISE I Q67]

The electrode through which electrons enter the electrolytic solution is

- (a) cathode
- (b) anode
- (c) may be anode or cathode
- (d) both, anode and cathode

Problem 68

[Neeraj Kumar [NRJ] EXERCISE I Q68]

Which process occurs in the electrolysis of an aqueous solution of nickel chloride at nickel anode?

- (a) $\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^-$ (b) $\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}$
- (c) $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ (d) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$

Problem 69

[Neeraj Kumar [NRJ] EXERCISE I Q69]

If mercury is used as a cathode during the electrolysis of an aqueous NaCl solution, the ions discharged at cathode are

- (a) H^+ (b) Na^+
- (c) OH^- (d) Cl^-

Problem 70

[Neeraj Kumar [NRJ] EXERCISE I Q70]

Electrochemical equivalent is more for

- (a) Hydrogen (b) Silver
- (c) Copper (d) Zinc

Problem 71

[Neeraj Kumar [NRJ] EXERCISE I Q71]

The electrolytic bath used in gold plating of copper articles contains

- (a) Molten gold
- (b) Copper sulphate(aq)
- (c) $\text{AuCl}_3(\text{aq})$
- (d) $\text{AuCl}_3 + \text{NaCN}(\text{aq})$

Problem 72

[Neeraj Kumar [NRJ] EXERCISE I Q72]

Copper can be deposited from acidified copper sulphate and alkaline cuprous cyanide. If the same current is passed for a definite time

- (a) the amount of copper deposited from acidic copper sulphate will be higher
- (b) the amount of copper deposited from alkaline cuprous cyanide will be higher
- (c) the same amount of copper will be deposited
- (d) copper will not deposit in either case

Problem 73

[Neeraj Kumar [NRJ] EXERCISE I Q73]

In the electrolytic cell, flow of electrons is from

- (a) Cathode to anode in solution
- (b) Cathode to anode through external supply
- (c) Cathode to anode through internal supply
- (d) Anode to cathode through internal supply

Problem 74

[Neeraj Kumar [NRJ] EXERCISE I Q74]

Electrolytic cell is used to convert

- (a) Chemical energy to electrical energy
- (b) Electrical energy to chemical energy
- (c) Chemical energy to mechanical energy
- (d) Electrical energy to mechanical energy

Problem 75

[Neeraj Kumar [NRJ] EXERCISE I Q75]

Faraday's law of electrolysis fails when

- (a) temperature is increased
- (b) inert electrodes are used
- (c) a mixture of electrolytes is used
- (d) in none of these cases

Problem 76

[Neeraj Kumar [NRJ] EXERCISE I Q76]

Using same quantity of current, which among Na, Mg and Al is deposited more (by mass) during electrolysis of their molten salts?

- (a) Na (b) Mg
- (c) Al (d) All in same

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Problem 77

[Neeraj Kumar [NRJ] EXERCISE I Q77]

A certain current liberated 0.50 g of hydrogen in 2 h. How many grams of copper can be liberated by the same current owing for the same time in a copper sulphate solution? ($\text{Cu} = 63.5$)

- (a) 12.7 g (b) 15.88 g
- (c) 31.75 g (d) 63.5 g

Problem 78

[Neeraj Kumar [NRJ] EXERCISE I Q78]

A current of 3.7 A is passed for 6 h between nickel electrodes in 0.50 L of 2 M solution of $\text{Ni}(\text{NO}_3)_2$. The molarity of Ni^{2+} at the end of electrolysis is

- (a) 1.172 M (b) 0.172 M
- (c) 0.586 M (d) 2 M

Problem 79

[Neeraj Kumar [NRJ] EXERCISE I Q79]

The current efficiency of an electrodeposition of copper metal in which 9.8 g of copper is deposited by a current of 3 A for 10000 s, from aqueous copper sulphate solution, is about

- (a) 60% (b) 99%
- (c) 92% (d) 75%

Problem 80

[Neeraj Kumar [NRJ] EXERCISE I Q80]

On passing electricity through dilute H_2SO_4 solution, the mass of substances liberated at the cathode and anode are in the ratio of

- (a) 1:8 (b) 8:1
(c) 1:32 (d) 1:16

Problem 81

[Neeraj Kumar [NRJ] EXERCISE I Q81]

The electrochemical equivalents of two substances are E_1 and E_2 . The current that must pass to deposit the same amount at the cathodes in the same time must be in the ratio of

- (a) $E_1:E_2$ (b) $E_2:E_1$
(c) $(E_1 - E_2):E_2$ (d) $E_1:(E_2 - E_1)$

Problem 82

[Neeraj Kumar [NRJ] EXERCISE I Q82]

The same quantity of electricity is passed through one molar solution of H_2SO_4 and one molar solution of HCl . The amount of hydrogen evolved from H_2SO_4 as compared to that from HCl is

- (a) the same
(b) twice as such
(c) one half as such
(d) dependent on size of electrode

Problem 83

[Neeraj Kumar [NRJ] EXERCISE I Q83]

In the electrolysis of acidified AgNO_3 solution using Pt-electrodes, the anode reaction is

- (a) $2\text{NO}_3^- \rightarrow 2\text{NO}_2 + \text{O}_2 + 2\text{e}^-$
(b) $\text{NO}_3^- \rightarrow \text{NO} + \frac{1}{2}\text{O}_2 + \text{e}^-$
(c) $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + \text{O}_2 + 4\text{e}^-$
(d) $\text{Pt} \rightarrow \text{Pt}^{3+} + 3\text{e}^-$

Problem 84

[Neeraj Kumar [NRJ] EXERCISE I Q84]

Two platinum electrodes were immersed in a solution of CuSO_4 and electric current was passed through the solution. After some time, it was found that colour of CuSO_4 disappeared with the evolution of gas at the electrode. The colourless solution contains

- (a) platinum sulphate
(b) copper sulphate
(c) copper hydroxide
(d) sulphuric acid

Problem 85

[Neeraj Kumar [NRJ] EXERCISE I Q85]

A solution containing 1.0 M each of $\text{Cu}(\text{NO}_3)_2$, $\text{Mg}(\text{NO}_3)_2$, AgNO_3 , $\text{Hg}(\text{NO}_3)_2$ is being electrolysed using inert electrodes. The values of standard electrode potential are: $\text{Ag}^+ | \text{Ag} = 0.80 \text{ V}$, $\text{Hg}_2^{2+} | \text{Hg} = 0.79 \text{ V}$, $\text{Cu}^{2+} | \text{Cu} = 0.34 \text{ V}$, $\text{Mg}^{2+} | \text{Mg} = -2.37 \text{ V}$. With increasing voltage, the sequence of deposit of metals on the cathode will be

- (a) Ag, Hg, Cu, Mg
(b) Mg, Cu, Hg, Ag
(c) Ag, Mg, Cu
(d) Cu, Hg, Ag

Problem 86

[Neeraj Kumar [NRJ] EXERCISE I Q86]

During the electrolysis of an aqueous salt solution, the pH in the space near one of the electrode was increased and the other one was decreased. The salt solution was

- (a) NaCl (very dilute) (b) ZnCl_2
(c) NaCl (Conc.) (d) $\text{Cu}(\text{NO}_3)_2$

Problem 87

[Neeraj Kumar [NRJ] EXERCISE I Q87]

Two electrolytic cells, one containing acidified ferrous chloride and another acidified ferric chloride, are connected in series. The mass ratio of iron deposited at cathodes in the two cells will be

- (a) 3:1 (b) 2:3
(c) 1:1 (d) 3:2

Problem 88

[Neeraj Kumar [NRJ] EXERCISE I Q88]

A galvanic cell is set up from a zinc bar weighing 100 g and 1.0 L of 1.0 M copper sulphate solution. How long would the cell run if it is assumed to deliver a steady current of 1.0 A? ($\text{Zn} = 65.4$)

- (a) 53.6 h (b) 26.8 h
(c) 81.97 h (d) 40.99 h

Problem 89

[Neeraj Kumar [NRJ] EXERCISE I Q89]

An ion is reduced to the element when it absorbs 6 times 10^{20} electrons. The number of equivalents of the ion is

- (a) 0.10 (b) 0.01
(c) 0.001 (d) 0.0001

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Problem 90

[Neeraj Kumar [NRJ] EXERCISE I Q90]

In the lead storage battery, the anode reaction is $\text{Pb(s)} + \text{HSO}_4^- + \text{H}_2\text{O} \rightarrow \text{PbSO}_4(\text{s}) + \text{H}_3\text{O}^+ + 2\text{e}^-$. How many grams of Pb will be used up to deliver 1 A for 100 h? ($\text{Pb} = 208$)

- (a) 776 g (b) 388 g
(c) 194 g (d) 0.1 g

Problem 91

[Neeraj Kumar [NRJ] EXERCISE I Q91]

The copper anode of a cell containing silver nitrate solution weighs 60.0 g. After passing current for some time, it is found that 3.24 g of silver is deposited on the platinum cathode. What is the molar weight of the anode? ($\text{Ag} = 108$, $\text{Cu} = 64$)

- (a) 0.96 g (b) 60 g
(c) 59.04 g (d) 60.96 g

Problem 92

[Neeraj Kumar [NRJ] EXERCISE I Q92]

The quantity of electricity required for the reduction of 1 mole of Fe_2O_3 to Fe is

- (a) 1F (b) 0.33F
(c) 3F (d) 6F

Problem 93

[Neeraj Kumar [NRJ] EXERCISE I Q93]

Three faradays of electricity is passed through molten Al_2O_3 , aqueous solutions of CuSO_4 and molten NaCl . The amounts of Al, Cu and Na deposited at the cathodes will be in the molar ratio of

- (a) 1:2:3 (b) 3:2:1
(c) 1:1.5:3 (d) 6:3:2

Problem 94

[Neeraj Kumar [NRJ] EXERCISE I Q94]

Electrolysis of a solution of HSO_4^- ions produces $\text{S}_2\text{O}_8^{2-}$. Assuming 75% current efficiency, what current should be employed to achieve a production rate of 1 mole of $\text{S}_2\text{O}_8^{2-}$ per hour?

- (a) 71.5 A (b) 35.7 A
(c) 53.0 A (d) 143 A

Problem 95

[Neeraj Kumar [NRJ] EXERCISE I Q95]

The number of Faradays required to produce 1 g-atom of Mg from MgCl_2 is

- (a) 1 (b) 2
(c) 0.5 (d) 4

Problem 96

[Neeraj Kumar [NRJ] EXERCISE I Q96]

Passage of 96,500 coulomb of electricity liberates _____ L of O_2 at 273°C and 2 atm during electrolysis.

- (a) 5.6 (b) 16.8
(c) 22.4 (d) 11.2

Problem 97

[Neeraj Kumar [NRJ] EXERCISE I Q97]

An electrolytic cell contains a solution of Ag_2SO_4 and platinum electrodes. A current is passed until 1.6 g of O_2 has been liberated at anode. The amount of silver deposited at cathode would be

- (a) 108.0 g (b) 1.6 g
(c) 10.8 g (d) 21.6 g

Problem 98

[Neeraj Kumar [NRJ] EXERCISE I Q98]

When 12,000 coulombs of electricity is passed through the electrolyte, 3.0 g of a metal of atomic mass 96.5 g/mol is deposited. The electro-valency of the metal cation in the electrolyte is

- (a) +4 (b) +3
(c) +2 (d) -4

Problem 99

[Neeraj Kumar [NRJ] EXERCISE I Q99]

How many electrons flow when a current of 5 A is passed through a solution for 200 s?

- (a) 6.022×10^{23} (b) 6.24×10^{21}
(c) 6.024×10^{21} (d) 6.022×10^{20}

Problem 100

[Neeraj Kumar [NRJ] EXERCISE I Q100]

The current of 9.65 A flowing for 10 min deposits 3.0 g of a metal. The equivalent weight of the metal is

- (a) 10 (b) 30
(c) 50 (d) 96.5

Problem 101**[Neeraj Kumar [NRJ] EXERCISE I Q101]**

If a current of 1.0 A is drawn from the Daniel cell for 96.5 min, the cathode will gain in weight by ($\text{Cu} = 63.5$, $\text{Zn} = 65.4$)
 (a) 1.905 g (b) 1.962 g
 (c) 3.81 g (d) 3.924 g

Problem 102**[Neeraj Kumar [NRJ] EXERCISE I Q102]**

The current required to produce oxygen at the rate of 2.8 ml (0/°C, 1 atm) per second during electrolysis of acidulated water is
 (a) 48.25 A/s (b) 24.12 A/s
 (c) 96.5 A/s (d) 0.0048 A/s

Problem 103**[Neeraj Kumar [NRJ] EXERCISE I Q103]**

Sodium amalgam is prepared by electrolysis of aqueous NaCl using 10 g mercury as cathode. How many Faraday of electricity is required to prepare 18.7 g of amalgam if efficiency of 50% is maintained?
 (a) 0.1 F (b) 0.2 F
 (c) 0.05 F (d) 0.16 F

Problem 104**[Neeraj Kumar [NRJ] EXERCISE I Q104]**

Element A (atomic mass = 112) and element B (atomic mass = 27) form chlorides. Solutions of these chlorides are electrolysed separately and it is found that when the same quantity of electricity is passed, 5.6 g of A was deposited while only 0.9 g of B was deposited. The valency of B is 3. The valency of A is
 (a) 1 (b) 2
 (c) 3 (d) 4
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Problem 105**[Neeraj Kumar [NRJ] EXERCISE I Q105]**

The same current was passed successively through solution of zinc-ammonium sulphate and nickel-ammonium sulphate rendered alkaline with ammonia. The weights of zinc and nickel deposited in a certain time were found to be 22.89 g and 20.55 g, respectively. Given that the chemical equivalent weight of zinc is 32.7, what is the chemical equivalent weight of nickel?
 (a) 58.71 (b) 29.36
 (c) 14.39 (d) 36.42 Conductance

Problem 106**[Neeraj Kumar [NRJ] EXERCISE I Q106]**

Which of the following solutions have highest resistance?
 (a) 1N - NaCl (b) 0.05N - NaCl
 (c) 2N - NaCl (d) 0.1N - NaCl

Problem 107**[Neeraj Kumar [NRJ] EXERCISE I Q107]**

Variation of molar conductance of an electrolytic solution with temperature is that it
 (a) increases with increase of temperature
 (b) decreases with increase of temperature
 (c) first increases then decreases
 (d) is not affected by temperature

Problem 108**[Neeraj Kumar [NRJ] EXERCISE I Q108]**

Which pure substance will not conduct electricity?

- (a) Molten NaCl (b) Molten KOH
(c) Liquid HCl (d) Liquid Hg

Problem 109**[Neeraj Kumar [NRJ] EXERCISE I Q109]**

The correct order of molar conductance at infinite dilution of LiCl, NaCl and KCl is

- (a) $\text{LiCl} > \text{NaCl} > \text{KCl}$ (b) $\text{KCl} > \text{NaCl} > \text{LiCl}$
(c) $\text{NaCl} > \text{KCl} > \text{LiCl}$ (d) $\text{LiCl} > \text{KCl} > \text{NaCl}$

Problem 110**[Neeraj Kumar [NRJ] EXERCISE I Q110]**

The molar conductance of a strong electrolyte at infinite dilution

- (a) tends to a finite value, which is above that at higher concentration
(b) tends to a finite value, which is below that at higher concentration
(c) tends to zero
(d) tends to a finite value, which is equal to that at high concentration

Problem 111**[Neeraj Kumar [NRJ] EXERCISE I Q111]**

The best conductor of electricity is a 0.1 M solution of

- (a) Boric acid (b) Sulphuric acid
(c) Acetic acid (d) Propanoic acid

Problem 112**[Neeraj Kumar [NRJ] EXERCISE I Q112]**

The specific conductance of AgCl solution in water was determined to be $1.8 \times 10^{-6} \text{ } \Omega^{-1} \text{ cm}^{-1}$ at 298 K. The molar conductances at infinite dilution, of Ag^+ and Cl^- are 67.9 and $82.1 \text{ } \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$, respectively. What is the solubility of AgCl in water?

- (a) $1.2 \times 10^{-8} \text{ M}$ (b) $1.44 \times 10^{-10} \text{ M}$
(c) $1.2 \times 10^{-5} \text{ M}$ (d) $1.44 \times 10^{-16} \text{ M}$

Problem 113**[Neeraj Kumar [NRJ] EXERCISE I Q113]**

Equivalence conductance at infinite dilution of NH_4Cl , NaOH and NaCl are 129.8, 217.4 and $108.9 \text{ } \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$, respectively. If the equivalent conductance of 0.01 N solution of NH_4OH is $9.532 \text{ } \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$, then the degree of dissociation of NH_4OH at this temperature is

- (a) 0.04/(b) 2.1/
(c) 4.0/(d) 44.7/

Problem 114**[Neeraj Kumar [NRJ] EXERCISE I Q114]**

The resistance of 1 M CH_3COOH solution is $250 \text{ } \Omega$, when measured in a cell of cell constant 125 m^{-1} . The molar conductivity, in $\Omega^{-1} \text{ m}^2 \text{ mol}^{-1}$ is

- (a) 5.0×10^{-4} (b) 500
(c) 2×10^{-3} (d) 200

Problem 115**[Neeraj Kumar [NRJ] EXERCISE I Q115]**

How does the electrical conductivity of 20 ml of 0.2 M – MgSO₄ change when 0.5 M – Ba(OH)₂ solution is gradually added in it, to excess?

- (a) decreases continuously
- (b) increases continuously
- (c) increases and then decreases
- (d) decreases and then increases

Problem 116**[Neeraj Kumar [NRJ] EXERCISE I Q116]**

The equivalent conductivity (in $\Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$) of 1.0 M – H₂SO₄ solution of specific conductance $2.6 \times 10^{-1} \text{ cm}^{-1}$, is

- (a) 1.3×10^2 (b) 6.5×10^1
- (c) 1.3×10^{-1} (d) 2.6×10^2

Problem 117**[Neeraj Kumar [NRJ] EXERCISE I Q117]**

The molar conductance of a 0.01 M solution of acetic acid was found to be $16.30 \Omega^{-1} \text{ cm}^{-1} \text{ mol}^{-1}$ at 25°C. The ionic conductances of hydrogen and acetate ions at infinite dilution are 349.8 and 40.9 $\Omega^{-1} \text{ cm}^{-1} \text{ mol}^{-1}$, respectively, at the same temperature. What percentage of acetic acid is dissociated at this concentration?

- (a) 0.04172/(b) 4.172/
- (c) 41.72/(d) 0.4172/

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Problem 118**[Neeraj Kumar [NRJ] EXERCISE I Q118]**

The distance between two electrodes of a cell is 2.5 cm and area of each electrode is 5 cm². The cell constant is

- (a) 0.5 m⁻¹ (b) 12.5 cm³
- (c) 2.0 cm (d) 50 m⁻¹

Problem 119**[Neeraj Kumar [NRJ] EXERCISE I Q119]**

The molar conductivity of NH₄Cl, OH⁻ and Cl⁻ at infinite dilution is 150, 200 and 75 $\Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$, respectively. If the molar conductivity of a 0.01 M-NH₄OH solution is 22 $\Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$, then its degree of dissociation is

- (a) 0.146 (b) 0.063
- (c) 0.080 (d) 0.293

Problem 120**[Neeraj Kumar [NRJ] EXERCISE I Q120]**

Calculate the ionic product of water at 25°C from the following data:

Conductivity of water = $5.5 \times 10^{-6} \text{ mho m}^{-1}$

$\lambda_{\text{H}^+}^{\circ} = 0.035 \text{ mho m}^2 \text{ mol}^{-1}$

$\lambda_{\text{OH}^-}^{\circ} = 0.020 \text{ mho m}^2 \text{ mol}^{-1}$

- (a) $2 \times 10^{-14} \text{ M}^2$ (b) $1 \times 10^{-7} \text{ M}$
- (c) $1 \times 10^{-8} \text{ M}^2$ (d) $1 \times 10^{-14} \text{ M}^2$

Problem 121**[Neeraj Kumar [NRJ] EXERCISE I Q121]**

Calculate K_a of acetic acid if its 0.05 M solution has molar conductivity of $7.814 \times 10^{-4} \text{ } \Omega^{-1} \text{ m}^2 \text{ mol}^{-1}$ at 25°C . Given: Λ° for $\text{CH}_3\text{COOH} = 3.907 \times 10^{-2} \text{ } \Omega^{-1} \text{ m}^2 \text{ mol}^{-1}$.

- (a) 2×10^{-5} (b) 1.8×10^{-5}
(c) 4×10^{-4} (d) 0.02

Problem 122**[Neeraj Kumar [NRJ] EXERCISE I Q122]**

Equal volumes of 0.015 M – CH_3COOH and 0.015 M – NaOH solutions are mixed together. What would be the molar conductivity of mixture if conductivity of CH_3COONa is $6.3 \times 10^{-4} \text{ S cm}^2 \text{ mol}^{-1}$?

- (a) $0.84 \text{ S cm}^2 \text{ mol}^{-1}$ (b) $8.4 \text{ S cm}^2 \text{ mol}^{-1}$
(c) $84 \text{ S cm}^2 \text{ mol}^{-1}$ (d) $42 \text{ S cm}^2 \text{ mol}^{-1}$

Problem 123**[Neeraj Kumar [NRJ] EXERCISE I Q123]**

Calculate Λ° (in $\Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$) for SrCl_2 at 25°C , from the following data: Conc. 0.25 M 1.0 M Λ° (in $\Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$) 260 250

- (a) 270 (b) 265
(c) 240 (d) 275

Problem 124**[Neeraj Kumar [NRJ] EXERCISE I Q124]**

The resistance of a solution A is $50 \text{ } \Omega$ and that of solution B is $100 \text{ } \Omega$, both solutions being taken in the same conductivity cell. If equal volumes of solution A and B are mixed, what will be the resistance of the mixture using the same cell? Assume that there is no increase in the degree of dissociation of A and B on mixing.

- (a) $150 \text{ } \Omega$ (b) $75 \text{ } \Omega$
(c) $33.33 \text{ } \Omega$ (d) $66.67 \text{ } \Omega$

Problem 125**[Neeraj Kumar [NRJ] EXERCISE I Q125]**

In a conductivity cell, the two platinum electrodes, each of area 10 cm^2 are held 1.5 cm apart. The cell contained 0.05 N solution of a salt. If the two electrodes are just half dipped into the solution which has a resistance of $50 \text{ } \Omega$, the equivalent conductance of the salt solution, in $\Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$, is

- (a) 120 (b) 60
(c) 240 (d) 3000

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Problem 126**[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q1]**

A student made the following observations in the laboratory:

- (I) Clean copper metal did not react with 1 M – $\text{Pb}(\text{NO}_3)_2$ solution.
(II) Clean lead metal dissolves in 1 M – AgNO_3 solution and crystals of Ag metal appeared.
(III) Clean silver metal did not react with 1 M – $\text{Cu}(\text{NO}_3)_2$ solution.

The order of decreasing reducing character of the three metals is

- (a) $\text{Cu} > \text{Pb} > \text{Ag}$ (b) $\text{Cu} > \text{Ag} > \text{Pb}$
(c) $\text{Pb} > \text{Cu} > \text{Ag}$ (d) $\text{Pb} > \text{Ag} > \text{Cu}$

Problem 127 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q2]

We have an oxidation-reduction system: $[\text{Fe}(\text{CN})_6]^{3-} + e^- \rightleftharpoons [\text{Fe}(\text{CN})_6]^{4-}$; $E = +0.36 \text{ V}$. The ratio of concentrations of oxidized and reduced form at which the potential of the system becomes 0.24 V , is [Given: $2.303 \text{ RT/F} = 0.06$]

- (a) 2:1 (b) 1:2
(c) 1:20 (d) 1:100

Problem 128 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q3]

The standard reduction potential for the process: $[\text{Co}(\text{H}_2\text{O})_6]^{3+} + e^- \rightarrow [\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is 1.8 V . The standard reduction potential for the process: $[\text{Co}(\text{NH}_3)_6]^{3+} + e^- \rightarrow [\text{Co}(\text{NH}_3)_6]^{2+}$ is 0.1 V . Which of the complex ion, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ or $[\text{Co}(\text{NH}_3)_6]^{2+}$ can be oxidized to the corresponding cobalt (III) complex, by oxygen, in basic medium, under standard condition? [Given: $E^\circ_{\text{O}_2/\text{OH}^-} = 0.4 \text{ V}$]

- (a) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ (b) $[\text{Co}(\text{NH}_3)_6]^{2+}$
(c) both (d) none of these

Problem 129 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q4]

The standard reduction potential for the reactions: $\text{Ag}^+ + e^- \rightarrow \text{Ag}$ and $\text{Ag}(\text{NH}_3)_2^+ + e^- \rightarrow \text{Ag} + 2\text{NH}_3$ are $+0.79 \text{ V}$ and $+0.37 \text{ V}$, respectively. From these values and the Nernst equation, what should be K_f for the $\text{Ag}(\text{NH}_3)_2^+$ ion? [Given: $2.303 \text{ RT/F} = 0.06$]

- (a) 1.0×10^{-7} (b) 1.0×10^7
(c) 2.15×10^{19} (d) 4.64×10^{-20}

Problem 130 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q5]

The overall formation constant for the reaction of 6 mole of CN^- with cobalt (II) is 1×10^{19} . What is the formation constant for the reaction of 6 moles of CN^- with cobalt (III)? Given that $[\text{Co}(\text{CN})_6]^{3-} + e^- \rightarrow [\text{Co}(\text{CN})_6]^{4-}$; $E^\circ = -0.83 \text{ V}$
 $[\text{Co}^{3+}] + e^- \rightarrow [\text{Co}^{2+}]$; $E^\circ = +1.81 \text{ V}$
and $2.303 \text{ RT/F} = 0.06$.

- (a) 1.0×10^{63} (b) 1.0×10^{25}
(c) 1.0×10^{-25} (d) 1.0×10^{-63}

Problem 131 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q6]

For the process: $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$; $\log[\text{Cu}^{2+}]$ vs. E_{red} graph is shown in the figure, where $OA = 0.34 \text{ V}$. The electrode potential of the half-cell of $\text{Cu}|\text{Cu}^{2+} (0.1 \text{ M})$ will be [$2.303 \text{ RT/F} = 0.06$] E_{red} $\log[\text{Cu}^{2+}]$

- (a) -0.31 V (b) $+0.31 \text{ V}$
(c) -0.37 V (d) $+0.37 \text{ V}$

Problem 132 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q7]

Tell which of the following statements accurately describes the effect of adding CN^- to the cathode of a cell with a cell reaction: $\text{Cd} + 2\text{Ag}^+ \rightarrow 2\text{Ag} + \text{Cd}^{2+}$, $E^\circ = 1.2 \text{ V}$

- (a) E increases because $\text{Cd}(\text{CN})_4^{2-}$ forms
(b) E decreases because $\text{Cd}(\text{CN})_4^{2-}$ forms
(c) E increases because $\text{Ag}(\text{CN})_2^-$ forms
(d) E decreases because $\text{Ag}(\text{CN})_2^-$ forms

Problem 133 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q8]

For an electrochemical cell $\text{Zn} | \text{Zn}^{2+} (\text{C1 M}) || \text{Cu}^{2+} (\text{C2 M}) | \text{Cu}$, the decrease in free energy at a given temperature is a function of

- (a) $\ln \text{C1}$ (b) $\ln \text{C2}$
 (c) $\ln \text{C2}/\text{C1}$ (d) $\ln \text{C1}/\text{C2}$

Problem 134 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q9]

Select the correct option if it is known that $K_{\text{sp}}(\text{AgCl}) > K_{\text{sp}}(\text{AgBr}) > K_{\text{sp}}(\text{AgI})$

- (a) $\text{E}(\text{AgI} | \text{Ag}) > \text{E}(\text{AgBr} | \text{Ag}) > \text{E}(\text{AgCl} | \text{Ag})$
 (b) $\text{E}(\text{AgI} | \text{Ag}) < \text{E}(\text{AgBr} | \text{Ag}) < \text{E}(\text{AgCl} | \text{Ag})$
 (c) $\text{E}(\text{AgI} | \text{Ag}) < \text{E}(\text{AgCl} | \text{Ag}) < \text{E}(\text{AgBr} | \text{Ag})$
 (d) $\text{E}(\text{AgI} | \text{Ag}) = \text{E}(\text{AgBr} | \text{Ag}) = \text{E}(\text{AgCl} | \text{Ag})$ EXERCISE II (JEE ADVANCED)

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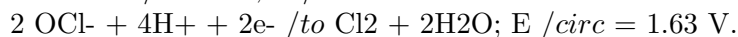
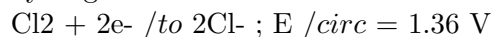
Problem 135 [Neeraj Kumar [NRJ] EXERCISE II Q10]

The EMF of a galvanic cell composed of two hydrogen electrodes is 272 mV. What is the pH of the solution in which the anode is immersed if the cathode is in contact with a solution of $\text{pH} = 3$?

- (a) 3 (b) 6.7
 (c) 7.6 (d) 1.6

Problem 136 [Neeraj Kumar [NRJ] EXERCISE II Q11]

At what pH does the potential (EMF) for the disproportionation of chlorine change from a negative value to a positive value, assuming 1.0 bar pressure and 1.0 M concentration for all species except hydrogen ion? Given:



$$[2.303 \text{ RT}/\text{F} = 0.06]$$

- (a) 4.5 (b) 1.5
 (c) 2.25 (d) 9.0

Problem 137 [Neeraj Kumar [NRJ] EXERCISE II Q12]

The EMF of the cell: $\text{Hg(l)} | \text{Hg}_2\text{Cl}_2(\text{s}), \text{KCl sol. (1.0N)} | \text{Quinohydrone} | \text{Pt}$, is 0.210 V at 298 K. What is the pH of the quinohydrone solution, the potential of the normal calomel electrode is 0.279 V and E° for the quinohydrone electrode is 0.699 V, both at the same temperature. $[2.303 \text{ RT}/\text{F} = 0.06]$

- (a) 3.5 (b) 7.0
 (c) 1.85 (d) -3.5

Problem 138 [Neeraj Kumar [NRJ] EXERCISE II Q13]

The potential (EMF) of a cell consisting of an anode of silver in 0.10 M AgNO_3 solution and a cathode of Pt immersed in a solution of 1.6 M $\text{Cr}_2\text{O}_7^{2-}$, 0.4 M Cr^{3+} and 0.1 M H^+ is (E $^\circ(\text{Ag}^+ | \text{Ag}) = 0.80 \text{ V}$ and $E^\circ(\text{Cr}_2\text{O}_7^{2-} | \text{Cr}^{3+}) = 1.33 \text{ V}$) $[2.303 \text{ RT}/\text{F} = 0.06]$

- (a) 0.46 V (b) 0.60 V
 (c) 0.53 V (d) -0.17 V

Problem 139**[Neeraj Kumar [NRJ] EXERCISE II Q14]**

The EMF of the cell: $\text{Zn-Hg (C1M)} \mid \text{Zn}^{2+}(\text{aq}) \mid \text{Zn-Hg (C2M)}$ at 25°C , if the concentrations of the zinc amalgams are: $\text{C1} = 10 \text{ g per } 100 \text{ g of mercury}$, $\text{C2} = 1 \text{ g per } 100 \text{ g mercury}$, is
 (a) 0.059 V (b) 0.0295 V
 (c) 0.59 V (d) 0.295 V

Problem 140**[Neeraj Kumar [NRJ] EXERCISE II Q15]**

What is the equilibrium constant of the reaction: $2\text{Fe}^{3+} + \text{Au} \rightarrow 2\text{Fe}^{2+} + \text{Au}^{3+}$? Given $E_{\text{Au}^{3+}/\text{Au}} = 1.68 \text{ V}$, $E_{\text{Au}^{3+}/\text{Au}^{2+}} = 1.50 \text{ V}$, $E_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.75 \text{ V}$ and $2.303 \text{ RT/F} = 0.06$.
 (a) 1×10^{22} (b) 1×10^{-22}
 (c) 1×10^{-11} (d) 1×10^{-72}

Problem 141**[Neeraj Kumar [NRJ] EXERCISE II Q16]**

What is the EMF of the cell: $\text{Pt, H}_2 (1 \text{ atm}) \mid \text{CH}_3\text{COOH} (0.1 \text{ M}) \parallel (0.01 \text{ M}) \text{NH}_4\text{OH} \mid \text{H}_2 (1 \text{ atm}), \text{Pt}$? Given: K_a for $\text{CH}_3\text{COOH} = 1.8 \times 10^{-5}$, K_b for $\text{NH}_4\text{OH} = 1.8 \times 10^{-5}$, $2.303 \text{ RT/F} = 0.06$, $\log 1.8 = 0.25$
 (a) 0.465 V (b) -0.465 V
 (c) -0.2325 V (d) -0.93 V

Problem 142**[Neeraj Kumar [NRJ] EXERCISE II Q17]**

Two electrochemical cells are assembled in which the following reactions occur:

$\text{V}^{2+} + \text{VO}^{2+} + 2\text{H}^+ \rightarrow 2\text{V}^{3+} + \text{H}_2\text{O}$; $E_{\text{Cell}} = 0.616 \text{ V}$

$\text{V}^{3+} + \text{Ag}^+ + \text{H}_2\text{O} \rightarrow \text{VO}^{2+} + \text{Ag(s)} + 2\text{H}^+$; $E_{\text{Cell}} = 0.439 \text{ V}$

If $E_{\text{Ag}^+/\text{Ag}} = 0.799 \text{ V}$, what is $E_{\text{V}^{3+}/\text{V}^{2+}}$?

- (a) -0.256 V (b) +0.256 V
 (c) +1.854 V (d) -1.854 V

Problem 143**[Neeraj Kumar [NRJ] EXERCISE II Q18]**

The EMF of cell: $\text{H}_2(\text{g}) \mid \text{Buffer} \parallel \text{Normal calomel electrode}$, is 0.70 V at 25°C , when barometric pressure is 760 mm. What is the pH of the buffer solution? $E_{\text{Calomel}} = 0.28 \text{ V}$. [$2.303 \text{ RT/F} = 0.06$]

- (a) 3.5
 (b) 7.0
 (c) tending to zero
 (d) tending to 14.0

Problem 144**[Neeraj Kumar [NRJ] EXERCISE II Q19]**

What is the solubility product of a saturated solution of Ag_2CrO_4 in water at 298 K if the EMF of the cell: $\text{Ag} \mid \text{Ag}^+ (\text{satd. Ag}_2\text{CrO}_4) \parallel \text{Ag}^+ (0.1 \text{ M}) \mid \text{Ag}$ is 0.162 V at 298 K? [$2.303 \text{ RT/F} = 0.06$, $\log 2 = 0.3$]

- (a) 2.0×10^{-4} (b) 3.2×10^{-11}
 (c) 8.0×10^{-12} (d) 4.0×10^{-12}

Problem 145**[Neeraj Kumar [NRJ] EXERCISE II Q20]**

A $\text{Tl}^+|\text{Tl}$ couple was prepared by saturating 0.10 M KBr with TlBr and allowing Tl^+ ions form the insoluble bromide to equilibrate. This couple was observed to have a potential -0.444 V with respect to $\text{Pb}^{2+}|\text{Pb}$ couple in which Pb^{2+} was 0.10 M. What is the K_{sp} of TlBr . [Given: $E^\circ_{\text{Pb}^{2+}|\text{Pb}} = -0.126 \text{ V}$, $E^\circ_{\text{Tl}^+|\text{Tl}} = -0.336 \text{ V}$, $\log 2.5 = 0.4$, $2.303 \text{ RT/F} = 0.06$]

- (a) 4.0×10^{-6} (b) 2.5×10^{-4}
 (c) 4.0×10^{-5} (d) 6.3×10^{-3}

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Problem 146**[Neeraj Kumar [NRJ] EXERCISE II Q21]**

The EMFs of the cell obtained by combining separately Zn and Cu electrodes of a Daniel cell with normal calomel electrodes are 1.083 V and - 0.018 V, respectively, at 25 °C. If the potential of normal calomel electrode is - 0.28 V, the EMF of the Daniel cell is

- (a) 1.065 V (b) 1.101 V
 (c) 0.803 V (d) 0.262 V

Problem 147**[Neeraj Kumar [NRJ] EXERCISE II Q22]**

A hydrogen electrode placed in a buffer solution of CH_3COONa and CH_3COOH in the ratio's $x:y$ and $y:x$ has electrode potential values E_1 and E_2 volts, respectively, at 25°C. The pK_a value of acetic acid is

- (a) $(E_1 + E_2)/0.118$ (b) $(E_2 - E_1)/0.118$
 (c) $-(E_1 + E_2)/0.118$ (d) $(E_1 - E_2)/0.118$

Problem 148**[Neeraj Kumar [NRJ] EXERCISE II Q23]**

The standard reduction potential at 25°C of the reaction $2\text{H}_2\text{O} + 2e^- \rightarrow \text{H}_2 + 2\text{OH}^-$ is -0.84 V. Calculate equilibrium constant for the reaction: $2\text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{OH}^-$ at 25°C. ($2.303 \text{ RT/F} = 0.06$)

- (a) 10^{-14} (b) 10^{14}
 (c) 10^7 (d) 10^{-7}

Problem 149**[Neeraj Kumar [NRJ] EXERCISE II Q24]**

The voltage of the cell given below is -0.61 V.

$\text{Pt} | \text{H}_2 (1 \text{ bar}) | \text{NaHSO}_3 (0.4 \text{ M}), \text{Na}_2\text{SO}_3 (6.4 \times 10^{-2} \text{ M}) || \text{Zn}^{2+} (0.4 \text{ M}) | \text{Zn}$

If $E^\circ_{\text{Zn}^{2+}|\text{Zn}} = -0.76 \text{ V}$, Calculate K_{a2} of H_2SO_4 . ($2.303 \text{ RT/F} = 0.06$)

- (a) 3.2×10^{-4} (b) 3.2×10^{-2}
 (c) 3.2×10^{-3} (d) 6.4×10^{-7}

Problem 150**[Neeraj Kumar [NRJ] EXERCISE II Q25]**

Estimate the cell potential of a Daniel cell having 1.0 M Zn^{2+} and originally having 1.0 M Cu^{2+} after sufficient ammonia has been added to the cathode compartment to make the NH_3 concentration 2.0 M. Given: $E^\circ_{\text{Zn}^{2+}|\text{Zn}} = -0.76 \text{ V}$, $E^\circ_{\text{Cu}^{2+}|\text{Cu}} = +0.34 \text{ V}$, K_f for $\text{Cu}(\text{NH}_3)_4^{2+} = 1 \times 10^{12}$ [$2.303 \text{ RT/F} = 0.06$].

- (a) 1.10 V (b) 0.704 V
 (c) 0.396 V (d) 1.496 V

Problem 151**[Neeraj Kumar [NRJ] EXERCISE II Q26]**

The minimum mass of NaOH required to be added in RHS to consume all the H^+ present in RHS of the cell of EMF, +0.70 V at 25°C, before its use.

$Zn | Zn^{2+} (0.01 M) || HCl (V = 500 \text{ ml}) | H_2 (1 \text{ bar}) | Pt$

Given: $E^\circ_{Zn^{2+}/Zn} = -0.76 \text{ V}$; $2.303 RT/F = 0.06$

- (a) 2.0 g (b) 4.0 g
(c) 0.2 g (d) 0.4 g

Problem 152**[Neeraj Kumar [NRJ] EXERCISE II Q27]**

The EMF of the cell: $Ag, AgCl \text{ in } 0.1 M - KCl || \text{satd. } NH_4NO_3 || 0.1 M - AgNO_3, Ag$ is 0.42 V at 25°C. 0.1 M - KCl is 50/AgNO₃ is 40/of AgCl is (2.303 RT/F = 0.06)

- (a) 1.0 $\times 10^{-10}$ (b) 2.0 $\times 10^{-9}$
(c) 1.0 $\times 10^{-9}$ (d) 2.0 $\times 10^{-10}$

Problem 153**[Neeraj Kumar [NRJ] EXERCISE II Q28]**

On the basis of information available from the reaction

$4Al + 3O_2 \rightarrow 2Al_2O_3$; $\Delta G = -965 \text{ kJ/mol of } O_2$

The minimum EMF required to carry out electrolysis of Al_2O_3 is

- (a) 0.833 V (b) 2.5 V
(c) 5.0 V (d) 1.67 V

Problem 154**[Neeraj Kumar [NRJ] EXERCISE II Q29]**

The theoretical efficiency of a hypothetical cell is about 84/

$A(s) + B^{2+}(aq) \rightarrow A^{2+}(aq) + B(s)$; $\Delta H = -285 \text{ kJ}$,

then, the standard EMF of the cell is

- (a) 1.10 V (b) 1.24 V
(c) 2.48 V (d) 2.20 V

Problem 155**[Neeraj Kumar [NRJ] EXERCISE II Q30]**

To a beaker containing 0.1 M - HCl, a little pure solid AgCl is added. Both a silver and a hydrogen electrode ($P_{H_2} = 1.0 \text{ bar}$) are then placed in the solution. What is the approximate value of EMF measured between the electrodes at 25°C? (Given: $E^\circ_{Ag^+/Ag} = 0.80 \text{ V}$ and $K_{sp}(AgCl) = 10^{-10}$) [$2.303 RT/F = 0.06$]

- (a) 0.92 V (b) 0.32 V
(c) 1.28 V (d) 0.56 V

Problem 156**[Neeraj Kumar [NRJ] EXERCISE II Q31]**

Although aluminium is above hydrogen in the electrochemical series, it is stable in air and water. Why?

- (a) Aluminium is non-reactive metal.
(b) A layer of oxide is formed at metal surface, which prevent further reaction.
(c) The reaction is thermodynamically unfavourable, i.e., ΔG is not negative.
(d) Kinetically, the reaction has very low activation energy and hence do not occur.

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Problem 157**[Neeraj Kumar [NRJ] EXERCISE II Q32]**

When silver chloride is dissolved in a large excess of ammonia, practically all silver ion can be assumed to exist in form of a single ionic species $[\text{Ag}_x(\text{NH}_3)_y]^{x+}$. Compute the values of x and y using the two following cells:

Cell I: $\text{Ag} \mid 4.0 \times 10^{-4} \text{ M} - \text{AgCl}, 1 \text{ M} - \text{NH}_3 \parallel 4 \times 10^{-2} \text{ M} - \text{AgCl}, 1 \text{ M} - \text{NH}_3 \mid \text{Ag}$;
 $E_{\text{cell}} = 0.118 \text{ V}$ at 298 K.

Cell II: $\text{Ag} \mid 3 \times 10^{-3} \text{ M} - \text{AgCl}, 1 \text{ M} - \text{NH}_3 \parallel 3 \times 10^{-3} \text{ M} - \text{AgCl}, 0.1 \text{ M} - \text{NH}_3 \mid \text{Ag}$;
 $E_{\text{cell}} = 0.118 \text{ V}$ at 298 K.

(a) $x = 1, y = 2$ (b) $x = 1, y = 4$

(c) $x = 2, y = 4$ (d) $x = 1, y = 6$

Problem 158**[Neeraj Kumar [NRJ] EXERCISE II Q33]**

When metallic copper is shaken with a solution of a copper salt, the reaction $\text{Cu} + \text{Cu}^{2+} \rightleftharpoons 2\text{Cu}^+$ proceeds. When equilibrium is established at 298 K, $[\text{Cu}^{2+}]/[\text{Cu}^+]^2 = 1.667 \times 10^6 \text{ M}^{-1}$. If the standard potential of the $\text{Cu}^{2+} \mid \text{Cu}$ half-cell is $+0.3376 \text{ V}$, what is the standard potential of $\text{Cu}^+ \mid \text{Cu}$ half-cell? (Given: $2.303 \text{ RT}/F = 0.06$, $\log 2 = 0.3$, $\log 3 = 0.48$)

(a) -0.3732 V (b) 0.6752 V

(c) 0.5242 V (d) 0.151 V

Problem 159**[Neeraj Kumar [NRJ] EXERCISE II Q34]**

Zinc granules are added in excess to a 500 ml of 1.0 M nickel nitrate solution at 25°C until the equilibrium is reached. If the standard reduction potential of $\text{Zn}^{2+} \mid \text{Zn}$ and $\text{Ni}^{2+} \mid \text{Ni}$ are -0.75 V and -0.24 V , respectively, the concentration of Ni^{2+} in solution at equilibrium is ($2.303 \text{ RT}/F = 0.06$)

(a) $1.0 \times 10^{-17} \text{ M}$ (b) $1.0 \times 10^{17} \text{ M}$

(c) $5 \times 10^{-17} \text{ M}$ (d) $2 \times 10^{-17} \text{ M}$

Problem 160**[Neeraj Kumar [NRJ] EXERCISE II Q35]**

The EMF for the cell: $\text{Ag(s)} \mid \text{AgCl(s)} \mid \text{KCl}(0.2 \text{ M}) \parallel \text{KBr}(0.001 \text{ M}) \mid \text{AgBr(s)} \mid \text{Ag(s)}$ at 25°C is ($K_{\text{sp}}(\text{AgCl}) = 2.0 \times 10^{-10}$; $K_{\text{sp}}(\text{AgBr}) = 4.0 \times 10^{-13}$, $2.303 \text{ RT}/F = 0.06$, $\log 2 = 0.3$)

(a) 0.024 V (b) -0.024 V

(c) -0.24 V (d) -0.012 V

Problem 161**[Neeraj Kumar [NRJ] EXERCISE II Q36]**

Two students use same stock solution of ZnSO_4 but different solutions of CuSO_4 . The EMF of one cell is 0.03 V higher than the other. The concentration of CuSO_4 in the cell with higher EMF value is 0.5 M . The concentration of CuSO_4 in the other cell is ($2.303 \text{ RT}/F = 0.06$)

(a) 0.05 M (b) 5.0 M

(c) 0.5 M (d) 0.005 M

Problem 162**[Neeraj Kumar [NRJ] EXERCISE II Q37]**

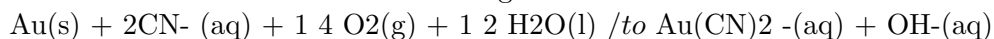
Two weak acid solutions HA_1 and HA_2 each with the same concentration and having pK_a values 3 and 5 are placed in contact with hydrogen electrodes (1 atm, 25°C) and are interconnected through a salt bridge. EMF of the cell is

(a) 0.0295 V (b) 0.118 V

(c) 0.0885 V (d) 0.059 V

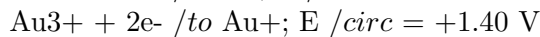
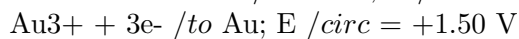
Problem 163**[Neeraj Kumar [NRJ] EXERCISE II Q38]**

Consider the reaction of extraction of gold from its ore:



Use the following data to calculate ΔG° for the above reaction.

$$K_f [\text{Au}(\text{CN})_2^-] = X$$



$$(a) -RT \ln X + 1.29 F \quad (b) -RT \ln X - 1.29 F$$

$$(c) +RT \ln X + 2.11 F \quad (d) -RT \ln X - 2.11 F$$

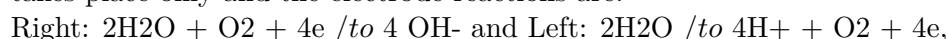
Problem 164**[Neeraj Kumar [NRJ] EXERCISE II Q39]**

F₂ gas cannot be obtained by the electrolysis of any aqueous fluoride salt because

- (a) F₂ is the strongest oxidizing agent
- (b) F₂ easily combines with water
- (c) F₂ readily combines with the electrodes
- (d) F⁻ can never be oxidized

Problem 165**[Neeraj Kumar [NRJ] EXERCISE II Q40]**

A volume of 100 ml of a buffer of 1 M – NH₃ and 1 M – NH₄⁺ is placed in two half-cells connected by a salt bridge. A current of 1.5 A is passed through the cell for 20 min. If electrolysis of water takes place only and the electrode reactions are:



then, the pH of the

- (a) right electrode will increase
- (b) left electrode will increase
- (c) both electrode will increase
- (d) both electrode will decrease

Problem 166**[Neeraj Kumar [NRJ] EXERCISE II Q41]**

When molten ICl₃ is electrolysed using platinum electrodes

- (a) I₂ is evolved at cathode and Cl₂ at anode
- (b) Cl₂ is evolved at cathode and I₂ at anode
- (c) I₂ is evolved at cathode and both I₂ and Cl₂ at anode
- (d) electrolysis does not take place

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Problem 167**[Neeraj Kumar [NRJ] EXERCISE II Q42]**

When electric current is passed through a cell having an electrolyte, positive ions move towards the cathode and negative ions towards the anode. If the cathode is pulled out of the solution, then the

- (a) positive and negative ions will move towards the anode
- (b) positive ions will start moving towards the anode, the negative ions will stop moving
- (c) negative ions will continue to move towards the anode, the positive ions will start moving randomly
- (d) positive and negative ions will start moving randomly

Problem 168**[Neeraj Kumar [NRJ] EXERCISE II Q43]**

When 10-6 M – HCl is electrolysed

- (a) O₂ is produced at the anode
- (b) H₂ is produced at the anode
- (c) Cl₂ is produced at the anode
- (d) Cl₂ and O₂ are produced at the anode

Problem 169**[Neeraj Kumar [NRJ] EXERCISE II Q44]**

A dilute aqueous solution of Na₂SO₄ is electrolysed using platinum electrodes. The products at the anode and cathode are, respectively,

- (a) O₂, H₂ (b) S₂O₈²⁻, Na
- (c) O₂, Na (d) S₂O₈²⁻, H₂

Problem 170**[Neeraj Kumar [NRJ] EXERCISE II Q45]**

In the electrolysis of a fused salt, the mass of substance deposited on an electrode will not depend on

- (a) temperature of bath
- (b) current intensity
- (c) time of electrolysis
- (d) electrochemical equivalent of ions

Problem 171**[Neeraj Kumar [NRJ] EXERCISE II Q46]**

When an electric current is passed through an aqueous solution of the following, the concentrations of cation as well as anion will not change for (assume constant volume of the electrolytic solution)

- (a) CsCl (b) KNO₃
- (c) AgNO₃ (d) HCl

Problem 172**[Neeraj Kumar [NRJ] EXERCISE II Q47]**

Which of the following substances: Na, Hg, S, Pt and graphite can be used as electrodes in electrolytic cells having aqueous solution?

- (a) Na, Pt and graphite (b) Na and Hg
- (c) Hg, Pt and graphite (d) Na and S

Problem 173**[Neeraj Kumar [NRJ] EXERCISE II Q48]**

An ammeter and a copper voltameter are connected in series in an electric circuit through which a constant direct current flows. The ammeter shows 0.5 A. If 0.635 g of Cu is deposited in 0.965 h, what is the percentage error of ammeter? (Cu = 63.5)

- (a) 5 / (b) 10 /
- (c) 9 / (d) 90 /

Problem 174**[Neeraj Kumar [NRJ] EXERCISE II Q49]**

The same current is passed through acidulated water and stannous chloride solution. What volume of dry detonating gas at 0/*circ*C and 1 atm is evolved from water, when 1.20 g of tin is deposited from the other solution? (Sn = 120)

- (a) 112 ml (b) 336 ml
- (c) 224 ml (d) 672 ml

Problem 175**[Neeraj Kumar [NRJ] EXERCISE II Q50]**

The preparation of LiOH by the electrolysis of a 35% current efficiency of 80% was formed by the passage of 2.5 A for 4825 s?

- (a) 1.92 g (b) 2.40 g
(c) 0.96 g (d) 0.672 g

Problem 176**[Neeraj Kumar [NRJ] EXERCISE II Q51]**

Assuming that copper contains only iron, silver and gold as impurities. After passage of 12.4 A for 4825 s, the mass of anode decreased by 20.00 g and the cathode increased by 19.05 g. The percentages of iron and copper in the original sample are, respectively, (Cu = 63.5, Fe = 56)

- (a) 4.75/(b) 2.8/
(c) 95.25/(d) 95.25/

Problem 177**[Neeraj Kumar [NRJ] EXERCISE II Q52]**

After electrolysis of an aqueous sodium chloride solution, it was found that the solution is being neutralized by 60 ml N – HCl solution. During the same period of electrolysis, 3.18 g of copper was deposited in a copper voltameter in series. What is the percentage of the theoretical yield of sodium hydroxide obtained? (Cu = 63.6)

- (a) 40/(b) 60/
(c) 30/(d) 80/

Problem 178**[Neeraj Kumar [NRJ] EXERCISE II Q53]**

A lead storage battery has initially 200 g of lead and 200 g of PbO₂ plus excess H₂SO₄. Theoretically, how long could this cell deliver a current of 10.0 A, without recharging, if it were possible to operate it so that the reaction goes to completion? (Pb = 208)

- (a) 16083.33 s (b) 8041.67 s
(c) 44.68 s (d) 18557.7 s

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Problem 179**[Neeraj Kumar [NRJ] EXERCISE II Q54]**

A volume of 100 ml of 0.6 N-CuSO₄ solution is electrolysed between two platinum electrodes till the concentration in the residual liquid is 0.1 N, when a steady current of 5.0 A is used. How long should the current be passed to get the above change?

- (a) 965 s (b) 96500 s
(c) 1930 s (d) 482.5 s

Problem 180**[Neeraj Kumar [NRJ] EXERCISE II Q55]**

A constant current flowed for 2 h through a potassium iodide solution, oxidizing the iodide ion to iodine. At the end of the experiment, the iodine was titrated with 72 ml of 1.0 M-Na₂S₂O₃ solution. What was the average rate of current flow, in ampere?

- (a) 0.965 A (b) 1.93 A
(c) 0.483 A (d) 0.0965 A

Problem 181**[Neeraj Kumar [NRJ] EXERCISE II Q56]**

Anthracene, C₁₄H₁₀, can be oxidized to anthraquinone, C₁₄H₈O₂. What weight of anthraquinone can be produced by the passage of a current of 1 A of 40 min if the current efficiency is 96.5/

- (a) 0.862 g (b) 0.832 g
(c) 0.624 g (d) 0.738 g

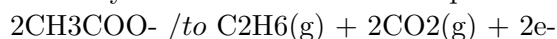
Problem 182**[Neeraj Kumar [NRJ] EXERCISE II Q57]**

Most of the copper used to make wire has been electrically refined by depositing it from copper salts solution (divalent) on to a cathode. What is the cost of electrical energy required per kg of copper if the cost of electricity is Rs 4.00 per kWh and the cell operates at 0.33 V? The electrochemical equivalent of copper is 0.00033 g/coulomb.

- (a) Rs 11.11 (b) Rs 5.55
(c) Rs 2.22 (d) Rs 1.11

Problem 183**[Neeraj Kumar [NRJ] EXERCISE II Q58]**

Electrolysis of an acetate solution produces ethane according to the reaction:



What total volume of ethane and CO_2 would be produced at 0°C and 1 atm, if a current of 0.5 A is passed through the solution for 482.5 min? Assume current efficiency 80%

- (a) 1.344 L (b) 2.688 L
(c) 4.032 L (d) 1.792 L

Problem 184**[Neeraj Kumar [NRJ] EXERCISE II Q59]**

To perform an analysis of a mixture of metal ions by electro-deposition, the second metal to be deposited must not be plating out until the concentration ratio of the second to the first is about 106. What must be the minimum difference in standard potential of the two metals which form dipositive ions in order for such an analysis to be feasible?

- (a) 0.177 V (b) 0.354 V
(c) 0.708 V (d) 0.088 V

Problem 185**[Neeraj Kumar [NRJ] EXERCISE II Q60]**

During the electrolysis of 0.1 M CuSO_4 solution using copper electrodes, a depletion of Cu^{2+} occurs near the cathode with a corresponding excess near the anode, owing to inefficient stirring of the solution. If the local concentration of Cu^{2+} near the anode and cathode are, respectively, 0.12 M and 0.08 M, the back EMF developed at 298 K is ($\log 1.5 = 0.18$, $2.303 RT/F = 0.06$)

- (a) 0.33 V (b) 5.4 mV
(c) 2.7 mV (d) 10.8 mV

Problem 186**[Neeraj Kumar [NRJ] EXERCISE II Q61]**

The following galvanic cell:



was operated as an electrolytic cell as Cu as the anode and Zn as the cathode. A current of 0.4825 A was passed for 10 h and then the cell was allowed to function as galvanic cell. What would be the cell EMF of the cell at 25°C? Assume that the only electrode reactions occurring were those involving $\text{Cu} | \text{Cu}^{2+}$ and $\text{Zn} | \text{Zn}^{2+}$. Given: $E^\circ_{\text{Cu}^{2+}|\text{Cu}} = +0.34 \text{ V}$, $E^\circ_{\text{Zn}^{2+}|\text{Zn}} = -0.76 \text{ V}$, $2.303 RT/F = 0.06$, $\log 1.9 = 0.28$)

- (a) 1.10 V (b) 1.0616 V
(c) 1.1084 V (d) 1.1768 V

Problem 187**[Neeraj Kumar [NRJ] EXERCISE II Q62]**

At the Nangal fertilizer plant in Punjab, hydrogen is produced by the electrolysis of water. The hydrogen is used for the production of ammonia and nitric acid (by the oxidation of ammonia). If the average production of ammonium nitrate is 1152 kg/day, the daily consumption of electricity (in A /day) is

- (a) 96,500 (b) 48,250
(c) 32,166.67 (d) 24,125

Problem 188**[Neeraj Kumar [NRJ] EXERCISE II Q63]**

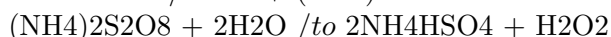
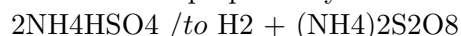
Lactic acid, $\text{HC}_3\text{H}_5\text{O}_3$, produced in 1 g sample of muscle tissue was titrated using phenolphthalein as indicator against OH^- ions which were obtained by the electrolysis of water. As soon as OH^- ions are produced, they react with lactic acid and at complete neutralization, immediately a pink colour is noticed. If electrolysis was made for 1158 s using 50.0 mA current to reach the end point, what was the percentage of lactic acid in muscle tissue?

- (a) 5.4/(b) 2.7/
(c) 10.8/(d) 0.054/

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Problem 189**[Neeraj Kumar [NRJ] EXERCISE II Q64]**

H_2O_2 can be prepared by successive reactions:

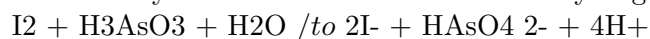


The first reaction is an electrolytic reaction and second is steam distillation. What amount of current would have to be used in first reaction to produce enough intermediate to yield 102 g pure H_2O_2 per hour. Assume current efficiency 50/

- (a) 643.33 A (b) 321.67 A
(c) 160.83 A (d) 1286.67 A

Problem 190**[Neeraj Kumar [NRJ] EXERCISE II Q65]**

In an analytical determination of arsenic, a solution containing arsenious acid, H_3AsO_3 , KI and a small amount of starch is electrolysed. The electrolysis produces free I_2 from I^- ions and the I_2 immediately oxidizes the arsenious acid to hydrogen arsenate ion, HAsO_4^{2-} .



When the oxidation of arsenic is complete, the free iodine combines with the starch to give a deep blue colour. If during a particular run, it takes 96.5 s for a current of 1.68 mA to give an end point (indicated by the blue colour), how many g of arsenic are present in the solution. ($\text{As} = 75$)

- (a) 6.3 /times 10^{-4} g (b) 1.26 /times 10^{-4} g
(c) 3.15 /times 10^{-5} g (d) 6.3 /times 10^{-5} g

Problem 191**[Neeraj Kumar [NRJ] EXERCISE II Q66]**

Electrolysis of a solution of MnSO_4 in aqueous sulphuric acid is a method for the preparation of MnO_2 as per the reaction: $\text{Mn}^{2+}(\text{aq}) + 2\text{H}_2\text{O} \xrightarrow{\text{to}} \text{MnO}_2(\text{s}) + 2\text{H}^+(\text{aq}) + \text{H}_2(\text{g})$. Passing a current of 19.3 A for 2 h gives only 52.2 g of MnO_2 . The current efficiency is ($\text{Mn} = 55$)

- (a) 8.33/(b) 83.33/
(c) 41.67/(d) 100/

Problem 192**[Neeraj Kumar [NRJ] EXERCISE II Q67]**

Copper sulphate solution (250 ml) was electrolysed using a platinum anode and a copper cathode. A constant current of 2 mA was passed for 19.3 min. It was found that after electrolysis the absorbance of the solution was reduced to 50/value. Calculate the concentration of copper sulphate in the solution to begin with.

- (a) 9.6×10^{-5} M (b) 4.8×10^{-5} M
(c) 2.4×10^{-5} M (d) 1.2×10^{-5} M

Problem 193**[Neeraj Kumar [NRJ] EXERCISE II Q68]**

Calculate the quantity of electricity that would be required to reduce 12.3 g of nitrobenzene to aniline if the current efficiency for the process is 50/how much energy will be consumed?

- (a) 0.3 F, 86.85 kJ (b) 0.6 F, 173.7 kJ
(c) 1.2 F, 347.4 kJ (d) 0.6 F, 173.7 kJ

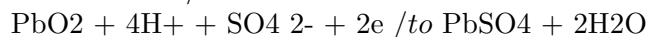
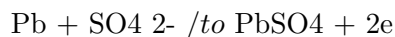
Problem 194**[Neeraj Kumar [NRJ] EXERCISE II Q69]**

Perdisulphuric acid, $\text{H}_2\text{S}_2\text{O}_8$ can be prepared by electrolytic oxidation of H_2SO_4 , oxygen and hydrogen gases are by products. In such an electrolysis, 9.08 L of H_2 and 2.27 L of O_2 were generated at STP. What is the mass of $\text{H}_2\text{S}_2\text{O}_8$ formed?

- (a) 0 (b) 77.6 g
(c) 38.8 g (d) 19.4 g

Problem 195**[Neeraj Kumar [NRJ] EXERCISE II Q70]**

During the discharge of a lead storage battery, the density of sulphuric acid fell from 1.5 to 1.1 g/ml. Sulphuric acid of density 1.5 g/ml is 40% by weight, and that of density 1.1 g/ml is 10% H_2SO_4 by weight. The battery holds 3.6 L of the acid and the volume remained practically constant during the discharge. Calculate the number of ampere-hours which the battery should have been used. The electrode reactions are:



- (a) 965 (b) 482.5
(c) 1930 (d) 241.25

Problem 196**[Neeraj Kumar [NRJ] EXERCISE II Q71]**

A cell whose resistance, when filled with 0.1 M – KCl is 200 Ω , is measured to be 6400 Ω , when filled with 0.003 M – NaCl solution. What is the molar conductance of NaCl solution, in $\Omega^{-1} \text{cm}^2 \text{mol}^{-1}$ if the molar conductance of 0.1 M – KCl is 120 $\Omega^{-1} \text{cm}^2 \text{mol}^{-1}$?

- (a) 41.67 (b) 250
(c) 125 (d) 375

Problem 197**[Neeraj Kumar [NRJ] EXERCISE II Q72]**

At 18°C, the ionic mobilities of NH_4^+ and CrO_4^{2-} ions are 6.6×10^{-8} and $5.4 \times 10^{-8} \text{ m}^2 \text{volt}^{-1} \text{s}^{-1}$ at infinite dilution. What is the molar conductance of ammonium chromate solution in $\Omega^{-1} \text{m}^2 \text{mol}^{-1}$?

- (a) 0.01795 (b) 0.01158
(c) 1.2×10^{-8} (d) 1.86×10^{-7}

Problem 198**[Neeraj Kumar [NRJ] EXERCISE II Q73]**

At 25°C, the molar conductance at infinite dilution for HCl solution is $4.25 \times 10^{-2} \text{ m}^2 \text{ mol}^{-1}$, while its specific conductance is $382.5 \times 10^{-1} \text{ m}^{-1}$. If the degree of dissociation is 90/

- (a) 0.9 M (b) 1.0 M
(c) 0.1 M (d) 1.1 M

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Problem 199**[Neeraj Kumar [NRJ] EXERCISE II Q74]**

A big irregular shaped vessel contained water, conductivity of which was $2.56 \times 10^{-3} \text{ S} \cdot \text{m}^{-1}$. 585 g of NaCl was then added to the water and conductivity after the addition of NaCl, was found to be $3.06 \times 10^{-3} \text{ S} \cdot \text{m}^{-1}$. The molar conductivity of NaCl at this concentration is $1.5 \times 10^{-2} \text{ S} \cdot \text{m}^2 \text{ mol}^{-1}$. The capacity of vessel if it is filled with water, is

- (a) $3 \times 10^4 \text{ L}$ (b) 30 L
(c) $3 \times 10^8 \text{ L}$ (d) $3 \times 10^5 \text{ L}$

Problem 200**[Neeraj Kumar [NRJ] EXERCISE II Q75]**

The molar conductivity of 0.10 M solution of MgCl_2 is $100 \text{ mho cm}^2 \text{ mol}^{-1}$, at 25°C. A cell with electrodes that are 1.50 cm^2 in surface area and 0.50 cm apart is filled with 0.10 M MgCl_2 solution. How much current will flow when the potential difference between the electrodes is 5 volts?

- (a) 0.03 A (b) 3.0 A
(c) 0.15 A (d) 15 A

Problem 201**[Neeraj Kumar [NRJ] EXERCISE II Q76]**

For Na^+ , the value of symbol λ° is $50.0 \text{ cm}^2 \text{ mol}^{-1}$. The speed of Na^+ ion in the solution, if in the cell, electrodes are 5 cm apart and to which a potential of 19.3 volt is applied is

- (a) $2 \times 10^{-3} \text{ cm/s}$ (b) $1 \times 10^{-3} \text{ cm/s}$
(c) $2 \times 10^{-4} \text{ cm/s}$ (d) $2 \times 10^{-2} \text{ cm/s}$

Problem 202**[Neeraj Kumar [NRJ] EXERCISE II Q77]**

The conductivity of a saturated solution of AgCl at 298 K was found to be $3.40 \times 10^{-6} \text{ cm}^{-1}$; the conductivity of water used to make up the solution was $1.60 \times 10^{-6} \text{ cm}^{-1}$. Determine the solubility of AgCl in water in mole per litre at 298 K. The equivalent conductivity of AgCl at infinite dilution is $150.0 \text{ cm}^2 \text{ eq}^{-1}$.

- (a) 1.44×10^{-10} (b) 1.2×10^{-5}
(c) 3.33×10^{-5} (d) 1.2×10^{-8}

Problem 203**[Neeraj Kumar [NRJ] EXERCISE II Q78]**

Resistance of 0.2 M solution of an electrolyte is $50 \text{ } \Omega$. The specific conductance of the solution is $1.4 \text{ S} \cdot \text{m}^{-1}$. The resistance of 0.5 M solution of the same electrolyte is $280 \text{ } \Omega$. The molar conductivity of 0.5 M solution of the electrolyte (in $\text{S} \cdot \text{m}^2 \text{ mol}^{-1}$) is

- (a) 5×10^{-3} (b) 5×10^3
(c) 5×10^2 (d) 5×10^{-4}

Problem 204**[Neeraj Kumar [NRJ] EXERCISE II Q79]**

The conductivity of a saturated solution containing AgA ($K_{sp} = 3 \times 10^{-14}$) and AgB ($K_{sp} = 1 \times 10^{-14}$) is $3.75 \times 10^{-8} \Omega^{-1} \text{cm}^{-1}$. If the limiting molar conductivity of Ag^+ and A^- ion is 60 and $80 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$, respectively, the limiting molar conductivity of B^- (in $\Omega^{-1} \text{cm}^2 \text{mol}^{-1}$) is

- (a) 135 (b) 67.5
(c) 270 (d) 190

Problem 205**[Neeraj Kumar [NRJ] EXERCISE II Q80]**

The conductivity of saturated solution of $\text{Ba}_3(\text{PO}_4)_2$ is $1.2 \times 10^{-5} \Omega^{-1} \text{cm}^{-1}$. The limiting equivalent conductivities of BaCl_2 , K_3PO_4 and KCl are 160, 140 and $100 \Omega^{-1} \text{cm}^2 \text{eq}^{-1}$, respectively. The solubility product of $\text{Ba}_3(\text{PO}_4)_2$ is

- (a) 10^{-5} (b) 1.08×10^{-23}
(c) 1.08×10^{-25} (d) 1.08×10^{-27} Section B (One or More than one Correct)

Problem 206**[Neeraj Kumar [NRJ] EXERCISE II Q1]**

Which of the following statement(s) differentiate between electrochemical cell and electrolytic cell?

- (a) Spontaneous or non-spontaneous nature of the chemical process.
(b) Chemical reactions occurring at the electrodes.
(c) Positive and negative nature of anode.
(d) Dependence on Faraday's law.

Problem 207**[Neeraj Kumar [NRJ] EXERCISE II Q2]**

Pick up the false statement(s):

- (a) The net chemical change in a galvanic cell reaction is always redox reactions.
(b) In a galvanic cell made of copper and cadmium electrodes, cadmium electrode may act as anode.
(c) Standard potential increases with increasing concentration of the electrolyte.
(d) Calomel electrode is a reference electrode having 0.00 volt potential.

Problem 208**[Neeraj Kumar [NRJ] EXERCISE II Q3]**

Consider the cell: $\text{Ag(s)}, \text{AgCl(s)} \mid \text{KCl (0.1 M)} \mid \text{Hg}_2\text{Cl}_2(\text{s}), \text{Hg(l)}$. The cell potential

- (a) increases on increasing concentration of Cl^- ions.
(b) decreases on decreasing concentration of Cl^- ions.
(c) is independent of concentration of Cl^- ions.
(d) is independent of amounts of AgCl and Hg_2Cl_2 .

Problem 209**[Neeraj Kumar [NRJ] EXERCISE II Q4]**

The passage of electricity in the Daniel cell when Zn and Cu electrodes are connected

- (a) from Cu to Zn inside the cell
(b) from Cu to Zn outside the cell
(c) from Zn to Cu inside the cell
(d) from Zn to Cu outside the cell

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Problem 210**[Neeraj Kumar [NRJ] EXERCISE II Q5]**

From an electrolyte, one mole of electron will deposit at cathode

- (a) 63.5 g of Cu (b) 24 g of Mg
(c) 11.5 g of Na (d) 9.0 g of Al

Problem 211

[Neeraj Kumar [NRJ] EXERCISE II Q6]

Which of the following statements is/are correct?

- (a) The conductivity of molten NaCl is due to movement of Na⁺ and Cl⁻ ions.
- (b) Solid NaCl is good conductor of electricity.
- (c) Molten sodium is a good conductor because of mobile electrons.
- (d) Resistivity is reciprocal of molar conductivity of electrolyte.

Problem 212

[Neeraj Kumar [NRJ] EXERCISE II Q7]

Ionic conductance at infinite dilution of Al³⁺ and SO₄²⁻ ions are 60 and 80 Ω⁻¹ cm² eq⁻¹, respectively. The correct detail(s) regarding Al₂(SO₄)₃ is/are

- (a) the molar conductance is 140 Ω⁻¹ cm² mol⁻¹.
- (b) the equivalent conductance is 140 Ω⁻¹ cm² eq⁻¹.
- (c) the molar conductance is 840 Ω⁻¹ cm² mol⁻¹.
- (d) the molar conductance is 23.33 Ω⁻¹ cm² mol⁻¹.

Problem 213

[Neeraj Kumar [NRJ] EXERCISE II Q8]

The function(s) of salt bridge in a cell is/ are

- (a) It maintains standard electrode potential of cell constant which depends on several factors.
- (b) It completes the electrical circuit.
- (c) It separates both the solutions from each other.
- (d) It maintains the electrical neutrality of both electrolytic solutions.

Problem 214

[Neeraj Kumar [NRJ] EXERCISE II Q9]

The reactions taking place in the dry cell are:

Anode: Zn → Zn²⁺ + 2e⁻Cathode: 2MnO₂ + 2NH₄⁺ + 2e⁻ → Mn₂O₃ + 2NH₃ + H₂O

The minimum mass of reactants, if a dry cell is to generate 0.25 A for 9.65 h, are (Mn = 55, Zn = 65.4) (neglect any other chemical reactions occurring in the cell)

- (a) 2.943 g Zn (b) 7.83 g MnO₂
- (c) 1.62 g NH₄⁺ (d) 3.915 g MnO₂

Problem 215

[Neeraj Kumar [NRJ] EXERCISE II Q10]

EMF of the cell: Cd(s) | CdCl₂·5H₂O (sat.) | AgCl(s) | Ag(s) is +0.70 V at 0°C and +0.60 V at 50°C. If ΔH° and ΔS° are temperature independent, then the correct information(s) regarding the cell reaction is/are

- (a) ΔG° = -115.8 kJ at 50°C
- (b) ΔG° = 135.1 kJ at 0°C
- (c) ΔS° = -386 J/K
- (d) ΔH° = -221.178 kJ

Problem 216

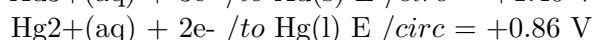
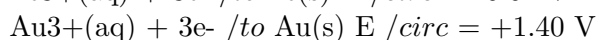
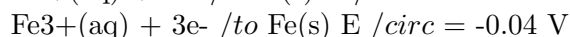
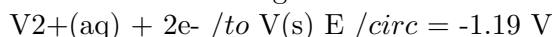
[Neeraj Kumar [NRJ] EXERCISE II Q11]

Two litre solution of a buffer mixture containing 1.0 M – NaH₂PO₄ and 1.0 M – Na₂HPO₄ is placed in two compartments (one litre in each) of an electrolytic cell. The platinum electrodes are inserted in each compartment and 1.25 A current is passed for 965 min. Assuming electrolysis of only water at each compartment, what will be pH in each compartment after passage of above charge? (pK_a for H₂PO₄⁻ = 2.15, log 7 = 0.85)

- (a) Anode: 3.00 (b) Cathode: 3.00
- (c) Anode: 1.30 (d) Cathode: 1.30

Problem 217**[Neeraj Kumar [NRJ] EXERCISE II Q12]**

For the reduction of NO_3^- ion in an aqueous solution, E° is +0.96 V. Values of E° for some metal ions are given below:



The pair(s) of metal that is(are) oxidized by NO_3^- in aqueous solution is(are)

- (a) V and Hg (b) Hg and Fe
(c) Fe and Au (d) Fe and V

Problem 218**[Neeraj Kumar [NRJ] EXERCISE II Q13]**

Among the following, the intensive property is (properties are)

- (a) Molar conductivity (b) Electromotive force
(c) Resistance (d) Heat capacity

Problem 219**[Neeraj Kumar [NRJ] EXERCISE II Q14]**

In a galvanic cell, the salt bridge

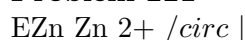
- (a) does not participate chemically in the cell reaction.
(b) stops the diffusion of ions from one electrode to another.
(c) is necessary for the occurrence of the cell reaction.
(d) ensures mixing of the two electrolytic solutions.

Problem 220**[Neeraj Kumar [NRJ] EXERCISE II Q15]**

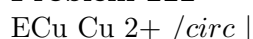
A lead storage cell is discharged which causes the H_2SO_4 electrolyte to change from a concentration of 40% by weight. The original volume of solution was 1 L. Identify the correct statement(s):

- (a) The overall cell reaction is: $\text{Pb}(\text{s}) + \text{PbO}_2(\text{s}) + 2\text{H}_2\text{SO}_4(\text{aq}) \rightarrow 2\text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$.
(b) A total of 2.0 moles of H_2SO_4 is reacted.
(c) The total charge released from anode of the cell is 1.93×10^5 coulomb.
(d) The mass of electrolytic solution has decreased.

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Problem 221**[Neeraj Kumar [NRJ] EXERCISE II Q1]**

- (a) +0.24 V (b) -0.76 V
(c) +1.76 V (d) -0.34 V

Problem 222**[Neeraj Kumar [NRJ] EXERCISE II Q2]**

- (a) -0.76 V (b) +1.34 V
(c) +1.76 V (d) +0.34 V

Problem 223

[Neeraj Kumar [NRJ] EXERCISE II Q3]

Zn-Cu Cell

- (a) +1.10 V
 (b) -1.10 V
 (c) 0.0 V
 (d) Indeterminate Comprehension II The standard reduction potential of the $\text{Ag}^+|\text{Ag}$ electrode at 298 K is 0.80 V. The solubility product of AgI is 6.4×10^{-17} at 298 K. ($2.303 RT/F = 0.06$, $\log 2 = 0.3$)

Problem 224

[Neeraj Kumar [NRJ] EXERCISE II Q4]

The potential of $\text{Ag}^+|\text{Ag}$ electrode in a saturated solution of AgI at 298 K is

- (a) -0.314 V (b) +0.314 V
 (c) -0.172 V (d) +0.172 V

Problem 225

[Neeraj Kumar [NRJ] EXERCISE II Q5]

The standard reduction potential of $\text{I}^-|\text{AgI}|\text{Ag}$ electrode at 298 K is

- (a) -0.314 V (b) +0.314 V
 (c) -0.172 V (d) +0.172 V

Problem 226

[Neeraj Kumar [NRJ] EXERCISE II Q6]

The potential of $\text{I}^- (0.04 \text{ M})|\text{AgI}|\text{Ag}$ electrode at 198 K is

- (a) -0.088 V
 (b) +0.088 V
 (c) -0.172 V
 (d) +0.172 V Comprehension III The Edison storage cell is represented as $\text{Fe(s)}|\text{FeO(s)}|\text{KOH(aq)}|\text{Ni}_2\text{O}_3\text{(s)}|\text{NiO(s)}|\text{Ni(s)}$. The half-cell reactions are $\text{Ni}_2\text{O}_3\text{(s)} + \text{H}_2\text{O(l)} + 2\text{e}^- \rightarrow 2\text{NiO(s)} + 2\text{OH}^-$; $E^\circ = +0.40 \text{ V}$ $\text{FeO(s)} + \text{H}_2\text{O(l)} + 2\text{e}^- \rightarrow \text{Fe(s)} + 2\text{OH}^-$; $E^\circ = -0.87 \text{ V}$

Problem 227

[Neeraj Kumar [NRJ] EXERCISE II Q7]

What is the cell reaction?

- (a) $\text{Ni}_2\text{O}_3\text{(s)} + \text{Fe(s)} \rightarrow 2\text{NiO(s)} + \text{FeO(s)}$
 (b) $2\text{NiO(s)} + \text{FeO(s)} \rightarrow \text{Ni}_2\text{O}_3\text{(s)} + \text{Fe(s)}$
 (c) $\text{Ni}_2\text{O}_3\text{(s)} + \text{FeO(s)} \rightarrow 2\text{NiO(s)} + \text{Fe(s)} + \text{O}_2$
 (d) None of these

Problem 228

[Neeraj Kumar [NRJ] EXERCISE II Q8]

What is the standard cell EMF?

- (a) 1.27 V
 (b) 0.47 V
 (c) -1.27 V
 (d) -0.47 V

Problem 229

[Neeraj Kumar [NRJ] EXERCISE II Q9]

How does cell EMF depend on increasing the concentration of KOH ?

- (a) increases
 (b) decreases
 (c) remains unaffected
 (d) none of these

Problem 230**[Neeraj Kumar [NRJ] EXERCISE II Q10]**

What is the maximum amount of electrical energy that can be obtained from one mole of Ni_2O_3 ?

- (a) 2.54 J (b) 245.11 kJ
(c) 122.56 kJ (d) 61.28 kJ

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Problem 231**[Neeraj Kumar [NRJ] EXERCISE II Q11]**

In which dish, the solution is acidic?

- (a) Dish A (b) Dish B
(c) Both (d) None

Problem 232**[Neeraj Kumar [NRJ] EXERCISE II Q12]**

How many moles of electrons pass through the circuit when 0.60 mole of Hg_2^{2+} and 0.30 mole of HNO_2 are produced in the cell that contains 0.50 mole of Hg_2^{2+} and 0.40 mole of NO_3^- at the beginning of the reaction?

- (a) 0.30 (b) 0.60
(c) 0.15 (d) 1.20

Problem 233**[Neeraj Kumar [NRJ] EXERCISE II Q13]**

How long will it take to produce 0.10 mole of HNO_2 by this reaction if a current of 10 A passes through the cell?

- (a) 965 s (b) 193 s

(c) 1930 s (d) 482.5 s

Comprehension V A fuel cell is the device to convert the energy of a fuel into electrical energy without the use of heat engine, where the fuel is burnt directly. Such conversions are possible because the combustion reactions are essentially redox reactions and highly exothermic as well as highly exergonic. Electrical energy can be obtained indefinitely from a fuel cell as long as the outside supply of fuel is maintained. In hydrogen-oxygen fuel cell, the following reactions take place:

Anode reaction: $2\text{H}_2(\text{g}) + 4\text{OH}^-(\text{aq}) \rightarrow 4\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$

Cathode reaction: $\text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^- \rightarrow 4\text{OH}^-(\text{aq})$

Overall reaction: $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$

The overall reaction has a value of $\Delta H^\circ = -285.8 \text{ kJ}$ and $\Delta G^\circ = -237.39 \text{ kJ}$ at 25°C per mole of $\text{H}_2\text{O}(\text{l})$.

Problem 234**[Neeraj Kumar [NRJ] EXERCISE II Q14]**

What is the standard EMF of the cell?

- (a) 0.615 V (b) 1.23 V
(c) 2.46 V (d) 0.74 V

Problem 235**[Neeraj Kumar [NRJ] EXERCISE II Q15]**

What volume of gaseous H_2 (at STP), when combined with excess O_2 in the fuel cell, is needed to produce 23.739 kJ of useful work under standard conditions?

- (a) 2.27 L (b) 4.54 L
(c) 1.13 L (d) 2.24 L

Problem 236

[Neeraj Kumar [NRJ] EXERCISE II Q16]

Suppose the concentration of hydroxide ion in the cell is doubled at 298 K. The cell voltage will be

- (a) reduced by half
- (b) increased by a factor of 2
- (c) increased by a factor of 4
- (d) unchanged

Problem 237

[Neeraj Kumar [NRJ] EXERCISE II Q17]

What is the approximate value of ΔS for the fuel cell reaction at 25°C?

- (a) - 0.1624 JK⁻¹ (b) -162.4 JK⁻¹
- (c) +162.4 JK⁻¹ (d) +0.1624 JK⁻¹

Problem 238

[Neeraj Kumar [NRJ] EXERCISE II Q18]

The theoretical efficiency of the fuel cell is given by

- (a) 83.06% (b) 100%
- (c) 67.53% (d) 97.88%

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Problem 239

[Neeraj Kumar [NRJ] EXERCISE II Q19]

Which of the following reactions may occur at the anode of the iron hull?

- (a) $\text{Fe}^{2+}(\text{aq}) \rightarrow \text{Fe}^{3+}(\text{aq}) + \text{e}^-$
- (b) $\text{Fe}(\text{s}) \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$
- (c) $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$
- (d) $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{OH}^-(\text{aq}) + \text{H}_2(\text{g}) + 2\text{e}^-$

Problem 240

[Neeraj Kumar [NRJ] EXERCISE II Q20]

What is the standard voltage generated by the corrosion reaction on the iron hull of a ship?

- (a) -1.676 V (b) - 0.782 V
- (c) 1.782 V (d) 1.676 V

Problem 241

[Neeraj Kumar [NRJ] EXERCISE II Q21]

Which of the following materials could not serve as a sacrificial anode for lead?

- (a) Copper (b) Iron
- (c) Magnesium (d) Manganese

Problem 242

[Neeraj Kumar [NRJ] EXERCISE II Q22]

An ammeter reading shows a current of 0.50 A running through the iron hull. How many grams of Fe(s) are lost in 1.0 h?

- (a) 0.261 g (b) 0.522 g
- (c) 1.044 g (d) 0.783 g

Problem 243**[Neeraj Kumar [NRJ] EXERCISE II Q23]**

A ship that has sunk to the bottom of the ocean may still exhibit corrosion. One difference in the corrosion reaction between ship above water and one that has sunk of the bottom of the ocean may be that

- (a) the oxidation reaction of the sunken ship does not include H₂O as a reactant.
- (b) the oxidation reaction of the sunken ship does not include O₂ as a reactant.
- (c) the reduction reaction of the sunken ship does not include H₂O as a reactant.
- (d) the reduction reaction of the sunken ship does not include O₂ as a reactant.

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Problem 244**[Neeraj Kumar [NRJ] EXERCISE II Q24]**

How much of nickel is plated on the cathode per hour?

- (a) 16.43 g (b) 32.85 g
- (c) 19.7 g (d) 9.85 g

Problem 245**[Neeraj Kumar [NRJ] EXERCISE II Q25]**

What is the thickness of the plating if the cathode consists of a sheet of metal 4.0 cm², which is to be coated on both sides?

- (a) 1.38 mm (b) 2.76 mm
- (c) 0.69 mm (d) 23.0 mm

Problem 246**[Neeraj Kumar [NRJ] EXERCISE II Q26]**

What volume of H₂ at 0/*circ*C and 1 atm is formed per hour?

- (a) 6.27 L (b) 3.76 L
- (c) 2.5 L (d) 5.01 L

Problem 247**[Neeraj Kumar [NRJ] EXERCISE II Q27]**

At the end of the electrolysis, how many grams of the gaseous product appear at the anode?

- (a) 4.48 g (b) 1.79 g
 - (c) 2.69 g (d) 7.46 g
- Comprehension VIII Tollen's reagent is used for the detection of aldehyde when a solution of AgNO₃ is added to glucose with NH₄OH then gluconic acid is formed.

$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$; $E^\circ_{\text{red}} = 0.8 \text{ V}$

$\text{C}_6\text{H}_{12}\text{O}_6 + \text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_7 \text{ (Gluconic acid)} + 2\text{H}^+ + 2\text{e}^-$; $E^\circ_{\text{oxd}} = -0.05 \text{ V}$

$\text{Ag}(\text{NH}_3)_2^+ + \text{e}^- \rightarrow \text{Ag(s)} + 2 \text{NH}_3$; $E^\circ_{\text{red}} = 0.337 \text{ V}$ [Use $2.303 \text{ RT/F} = 0.0592$ and $\text{F/RT} = 38.92$ at 298 K]

Problem 248**[Neeraj Kumar [NRJ] EXERCISE II Q28]**

$2\text{Ag}^+ + \text{C}_6\text{H}_{12}\text{O}_6 + \text{H}_2\text{O} \rightarrow 2\text{Ag(s)} + \text{C}_6\text{H}_{12}\text{O}_7 + 2\text{H}^+$. Find $\ln K$ of this reaction.

- (a) 66.13 (b) 58.38
- (c) 28.30 (d) 46.29

Problem 249**[Neeraj Kumar [NRJ] EXERCISE II Q29]**

When ammonia is added to the solution, pH is raised to 11. Which half-cell reaction is affected by pH and by how much?

- (a) E_{oxd} will increase by a factor of 0.65 from E°_{oxd}
- (b) E_{oxd} will decrease by a factor of 0.65 from E°_{oxd}
- (c) E_{red} will increase by a factor of 0.65 from E°_{red}
- (d) E_{red} will decrease by a factor of 0.65 from E°_{red}

Problem 250

[Neeraj Kumar [NRJ] EXERCISE II Q30]

Ammonia is always added in this reaction. Which of the following must be incorrect?

- (a) NH_3 combines with Ag^+ to form a complex
- (b) $\text{Ag}(\text{NH}_3)_2^+$ is a stronger oxidizing agent than Ag^+
- (c) In absence of ammonia, silver salt of gluconic acid is formed
- (d) NH_3 has affected the standard reduction potential of glucose-gluconic acid electrode

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Problem 251

[Neeraj Kumar [NRJ] EXERCISE II Q31]

The overall cell reaction is

- (a) $\text{H}_2(\text{g}) + 2\text{AgCl}(\text{s}) \rightarrow 2\text{HCl}(\text{aq}) + 2\text{Ag}(\text{s})$
- (b) $\text{H}_2(\text{g}) + 2\text{Ag}^+(\text{aq}) \rightarrow 2\text{H}^+(\text{aq}) + 2\text{Ag}(\text{s})$
- (c) $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$
- (d) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{aq})$

Problem 252

[Neeraj Kumar [NRJ] EXERCISE II Q32]

ΔH for the cell reaction per mol of AgCl is

- (a) -99,974 J (b) -49,987 J
- (c) +99,974 J (d) +49,987 J

Problem 253

[Neeraj Kumar [NRJ] EXERCISE II Q33]

ΔS for the cell reaction per mol of AgCl is

- (a) -193 J/K (b) -96.5 J/K
- (c) +96.5 J/K (d) +193 J/K

Problem 254

[Neeraj Kumar [NRJ] EXERCISE II Q34]

The solubility of AgCl in water at 25°C is ($2.303 RT/F = 0.058$)

- (a) 10^{-10} M (b) 10^{-12} M
 - (c) 10^{-5} M (d) 10^{-6} M
- Comprehension X For the reaction: $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$. Given
 Species ΔG° of (kJ / mol) at 25°C $\text{Ag}^+(\text{aq})$ +77 $\text{Cl}^-(\text{aq})$ -129 $\text{AgCl}(\text{s})$ -109
 $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$; $E^\circ = 0.80$ V
 $\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$; $E^\circ = -0.76$ V

Problem 255

[Neeraj Kumar [NRJ] EXERCISE II Q35]

The cell representations of the above reaction and E_{cell} are

- (a) $\text{Ag}(\text{s}) \mid \text{AgCl}(\text{s}) \mid \text{Cl}^-(\text{aq}) \parallel \text{Ag}^+(\text{aq}) \mid \text{Ag}(\text{s})$; $E_{\text{cell}} = 0.59$ V
- (b) $\text{Ag}(\text{s}) \mid \text{Ag}^+(\text{aq}) \parallel \text{Cl}^-(\text{aq}) \mid \text{AgCl}(\text{s}) \mid \text{Ag}(\text{s})$; $E_{\text{cell}} = 0.59$ V
- (c) $\text{Ag}(\text{s}) \mid \text{Cl}^-(\text{aq}) \parallel \text{Ag}^+(\text{aq}) \mid \text{Ag}$; $E_{\text{cell}} = 0.59$ V
- (d) $\text{Ag}(\text{s}) \mid \text{Ag}^+(\text{aq}) \parallel \text{Cl}^-(\text{aq}) \mid \text{AgCl}(\text{s}) \mid \text{Ag}(\text{s})$; $E_{\text{cell}} = 1.18$ V

Problem 256

[Neeraj Kumar [NRJ] EXERCISE II Q36]

The solubility product of AgCl at 298 K is

- (a) 2×10^{-10} (b) 10^{-10}
- (c) 10^{-10} (d) 10^{-5}

Problem 257**[Neeraj Kumar [NRJ] EXERCISE II Q37]**

A quantity of 6.539 $\times 10^{-2}$ g of metallic zinc is added to 100 ml of saturated solution of AgCl. The value of $\log \left[\frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2} \right]$ is ($E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76$ V, $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80$ V)

- (a) 5.288 (b) 52.88
(c) 528.8 (d) 26.44

Problem 258**[Neeraj Kumar [NRJ] EXERCISE II Q38]**

How many moles of Ag will be precipitated in the above reaction?

- (a) 10^{-3} (b) 2×10^{-3}
(c) 10^{-6} (d) 10^{-5}

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Problem 259**[Neeraj Kumar [NRJ] EXERCISE II Q39]**

Cell constant of the conductivity cell is

- (a) 4 cm⁻¹ (b) 0.4 cm⁻¹
(c) 40 cm⁻¹ (d) 0.04 cm⁻¹

Problem 260**[Neeraj Kumar [NRJ] EXERCISE II Q40]**

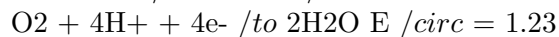
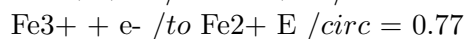
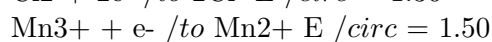
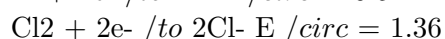
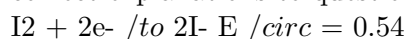
Conductivity of water (in $\Omega^{-1}\text{cm}^{-1}$) is

- (a) 0.4 (b) 0.04
(c) 0.0004 (d) 0.00004

Problem 261**[Neeraj Kumar [NRJ] EXERCISE II Q41]**

Volume (in L) of water in the pool is

- (a) 1,25,000 (b) 12,500
(c) 1250 (d) 125
- Comprehension XII Redox reactions play a pivotal role in chemistry and biology. The values of standard redox potential (E°) of two half-cell reactions decide which way the reaction is expected to proceed. A simple example is Daniel cell in which zinc goes into solution and copper gets deposited. Given below are a set of half-cell reactions (acidic medium) along with their E° (V with respect to normal hydrogen electrode) values. Using this data, obtain the correct explanations to questions below:

**Problem 262****[Neeraj Kumar [NRJ] EXERCISE II Q42]**

Among the following, identify the correct statement?

- (a) Chloride ion is oxidized by O_2
(b) Fe^{2+} is oxidized by iodine
(c) Iodide ion is oxidized by chlorine
(d) Mn^{2+} is oxidized by chlorine

Problem 263**[Neeraj Kumar [NRJ] EXERCISE II Q43]**

While Fe^{3+} is stable, Mn^{3+} is not stable in acid solution because

- (a) O_2 oxidizes Mn^{2+} to Mn^{3+}
- (b) O_2 oxidizes both Mn^{2+} to Mn^{3+} and Fe^{2+} to Fe^{3+}
- (c) Fe^{3+} oxidizes H_2O to O_2

(d) Mn^{3+} oxidizes H_2O to O_2 Comprehension XIII The concentration of potassium ions inside a biological cell is at least 20 times higher than the outside. The resulting potential difference across the cell is important in several processes such as transmission of nerve impulses and maintaining the ion balance. A simple model for such a concentration cell involving a metal 'M' is $\text{M}(\text{s}) \mid \text{M}^+(\text{aq}; 0.05 \text{ molar}) \parallel \text{M}^+(\text{aq}; 1 \text{ molar}) \mid \text{M}(\text{s})$ For the above electrolytic cell, the magnitude of the cell potential, $|\text{E}_{\text{cell}}| = 70 \text{ mV}$.

Problem 264**[Neeraj Kumar [NRJ] EXERCISE II Q44]**

For the above cell

- (a) $\text{E}_{\text{cell}} < 0$; $\Delta G > 0$ (b) $\text{E}_{\text{cell}} > 0$; $\Delta G < 0$
- (c) $\text{E}_{\text{cell}} < 0$; $\Delta G > 0$ (d) $\text{E}_{\text{cell}} > 0$; $\Delta G < 0$

Problem 265**[Neeraj Kumar [NRJ] EXERCISE II Q45]**

If the 0.05 molar solution of M^+ is replaced by a 0.0025 molar M^+ solution, then the magnitude of the cell potential would be

- (a) 35 mV (b) 70 mV
- (c) 140 mV (d) 700 mV

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Problem 266**[Neeraj Kumar [NRJ] EXERCISE II Q46]**

The value of ΔG (kJ mol^{-1}) for the given cell is (take $1 \text{ F} = 96500 \text{ C mol}^{-1}$)

- (a) -5.7 (b) 5.7
- (c) 11.4 (d) -11.4

Problem 267**[Neeraj Kumar [NRJ] EXERCISE II Q47]**

The solubility product (K_{sp} ; $\text{mol}^3 \text{ dm}^{-3}$) of MX_2 at 298 K based on the information available for the given concentration cell is (take $2.303 \times R \times 298 / \text{F} = 0.059$)

- (a) 1×10^{-15} (b) 4×10^{-15}

(c) 1×10^{-12} (d) 4×10^{-12} Comprehension XV Conductance measurements are frequently employed to find the end points of acid-base and other titrations. The principle involved is that electrical conductance of a solution depends upon the number and mobility of ions. In the conductometric titrations, the conductance of the resulting solution is measured at different stages on adding some volume of the standard solution and then a graph is plotted from the experimental observations to get the end point. Conductometric titrations have several advantages. Coloured solutions, titration of weak acid and weak base, etc. cannot be titrated by normal methods, but can be titrated successfully by this method. Further, no special care is needed in titration because the end point is determined graphically.

Problem 268**[Neeraj Kumar [NRJ] EXERCISE II Q48]**

Which of the following graph truly represents the titration of HCl solution against NaOH solution?

- (a) Cond. VNaOH
- (b) Cond. VNaOH
- (c) Cond. VNaOH
- (d) Cond. VNaOH

Problem 269**[Neeraj Kumar [NRJ] EXERCISE II Q49]**

Which of the following graph truly represents the titration of CH_3COOH solution against NaOH solution?

- (a) Cond. VNaOH
- (b) Cond. VNaOH
- (c) Cond. VNaOH
- (d) Cond. VNaOH

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Problem 270**[Neeraj Kumar [NRJ] EXERCISE II Q50]**

Which of the following graph truly represents the titration of HCl solution against NH_4OH solution?

- (a) Cond. VNH_4OH
- (b) Cond. VNH_4OH
- (c) Cond. VNH_4OH
- (d) Cond. VNH_4OH

Problem 271**[Neeraj Kumar [NRJ] EXERCISE II Q51]**

Which of the following graph truly represents the titration of a solution containing a mixture of HCl and CH_3COOH against NaOH solution?

- (a) Cond. VNaOH
- (b) Cond. VNaOH
- (c) Cond. VNaOH
- (d) Cond. VNaOH

Problem 272**[Neeraj Kumar [NRJ] EXERCISE II Q52]**

Which of the following graph truly represents the titration of AgNO_3 solution against KCl solution?

- (a) Cond. VKCl
- (b) Cond. VKCl
- (c) Cond. VKCl
- (d) Cond. VKCl

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Problem 273**[Neeraj Kumar [NRJ] EXERCISE II Q53]**

Which of the following graph truly represents the titration of CH_3COOH solution against NH_4OH solution?

- (a) Cond. VNH_4OH
- (b) Cond. VNH_4OH
- (c) Cond. VNH_4OH

(d) Cond. VNH_4OH Section D (Assertion - Reason) The following questions consist of two statements. Mark (a) If both statements are CORRECT, and Statement II is the CORRECT explanation of Statement I. (b) If both statements are CORRECT, and Statement II is NOT the CORRECT explanation of Statement I. (c) If Statement I is CORRECT, but Statement II is INCORRECT. (d) If Statement I is INCORRECT, but Statement II is CORRECT.

Problem 274**[Neeraj Kumar [NRJ] EXERCISE II Q1]**

Statement I: Electrolysis of $\text{CuCl}_2(\text{aq})$ gives 1 mole of Cu and 1 mole of Cl_2 by the passage of suitable charge.

Statement II: Equal equivalents of Cu and Cl_2 are formed during the passage of same charge.

Problem 275

[Neeraj Kumar [NRJ] EXERCISE II Q2]

Statement I: Zinc metal can displace Ag metal from a solution containing the complex $[\text{Ag}(\text{CN})_2]^-$.Statement II: $E_{\text{Zn}/\text{Zn}^{2+}}$ is greater than $E_{\text{Ag}/\text{Ag}^+}$.**Problem 276**

[Neeraj Kumar [NRJ] EXERCISE II Q3]

Statement I: In electrolysis, the quantity of electricity needed for depositing 1 mole of silver is different from that required for 1 mole copper.

Statement II: The atomic masses of silver and copper are different.

Problem 277

[Neeraj Kumar [NRJ] EXERCISE II Q4]

Statement I: In an electrochemical cell, anode and cathode are, respectively, negative and positive electrode.

Statement II: At anode, oxidation takes place and at cathode reduction takes place.

Problem 278

[Neeraj Kumar [NRJ] EXERCISE II Q5]

Statement I: A metal having negative reduction potential when dipped in the solution of its own ions has a tendency to pass into the solution.

Statement II: Metals having negative reduction potential have large hydration energy.

Problem 279

[Neeraj Kumar [NRJ] EXERCISE II Q6]

Statement I: Specific conductance decreases with dilution while molar conductance increases.

Statement II: On dilution, number of ions per unit volume decreases but total number of ions increases considerably.

Problem 280

[Neeraj Kumar [NRJ] EXERCISE II Q7]

Statement I: When acidified zinc sulphate solution is electrolysed between zinc electrodes, it is zinc that is deposited at the cathode and evolution of hydrogen gas does not take place.

Statement II: The electrode potential of zinc becomes less negative than hydrogen as the overvoltage for the hydrogen evolution on zinc is quite large.

Problem 281

[Neeraj Kumar [NRJ] EXERCISE II Q8]

Statement I: The conductivity of solutions of different electrolytes in the same solvent and at a given temperature is same.

Statement II: The conductivity depends on the charge and size of the ions, the concentrations of ions and ease with which the ions move under potential gradient.

Problem 282

[Neeraj Kumar [NRJ] EXERCISE II Q9]

Statement I: Sodium ions are discharged in preference to hydrogen ions at a mercury cathode.

Statement II: Na^+ is a strong reducing agent in comparison to H^+ ion.Visit www.jeebooks.in for more**Problem 283**

[Neeraj Kumar [NRJ] EXERCISE II Q10]

Statement I: The conductivity of metals decreases while that of electrolytic solution increases with increase in temperature.

Statement II: Electrons in metals are very tightly held by the nucleus and are not free to move.

Problem 284**[Neeraj Kumar [NRJ] EXERCISE II Q11]**

Statement I: An electrochemical cell can be set up only when the redox reaction is spontaneous.

Statement II: A reaction is spontaneous if free energy change at constant temperature and pressure is negative.

Problem 285**[Neeraj Kumar [NRJ] EXERCISE II Q12]**Statement I: In the Daniel cell, if concentrations of Cu^{2+} and Zn^{2+} ions are doubled, the EMF of cell does not change.

Statement II: If the concentration of ions in contact with the metal is doubled, the electrode potential will be doubled.

Problem 286**[Neeraj Kumar [NRJ] EXERCISE II Q13]**Statement I: KCl, NaCl, NH_4Cl , etc., cannot be used in the salt bridge of a cell containing silver ion.Statement II: A salt bridge contains concentrated solution of an inert electrolyte like KCl, KNO_3 , K_2SO_4 , etc., in agar-agar.**Problem 287****[Neeraj Kumar [NRJ] EXERCISE II Q14]**

Statement I: During electrolysis of aqueous sodium acetate solution, the molar ratio of gases formed at cathode and anode is 1:3.

Statement II: Acetate ion discharges at anode and H^+ ion, at cathode.**Problem 288****[Neeraj Kumar [NRJ] EXERCISE II Q15]**

Statement I: Electrode potential of any electrode will change on changing any of its intensive properties.

Statement II: Any intensive property of system changes on changing any of its properties, whether intensive or extensive. Section E (Column Match)

Problem 289**[Neeraj Kumar [NRJ] EXERCISE II Q1]**

Match Column I with Column II Column I Column II (Electrolysis product using inert electrodes)

(A) Dilute solution of HCl (P) O_2 evolved at anode (B) Dilute solution of NaCl (Q) H_2 evolved at cathode (C) Concentrated solution of NaCl (R) Cl_2 evolved at anode (D) AgNO_3 solution (S) Ag deposited at cathode**Problem 290****[Neeraj Kumar [NRJ] EXERCISE II Q2]**

An aqueous solution of 'X' is added slowly to an aqueous solution of 'Y' as shown in List I. The variation in conductivity of these reactions is given in List II. Match List I with List II. List I (X + Y) List II (A) $(\text{C}_2\text{H}_5)_3\text{N} + \text{CH}_3\text{COOH}$ (P) Conductivity decreases and then increases (B) $\text{KI}(0.1 \text{ M}) + \text{AgNO}_3(0.01 \text{ M})$ (Q) Conductivity decreases and then does not change much (C) $\text{CH}_3\text{COOH} + \text{KOH}$ (R) Conductivity increases and then does not change much (D) $\text{NaOH} + \text{HI}$ (S) Conductivity does not change much and then increases

Problem 291

[Neeraj Kumar [NRJ] EXERCISE II Q3]

The standard reduction potential data at 25°C is given below:

$$E^\circ(\text{Fe}^{3+}, \text{Fe}^{2+}) = +0.77 \text{ V}$$

$$E^\circ(\text{Fe}^{2+}, \text{Fe}) = -0.44 \text{ V}$$

$$E^\circ(\text{Cu}^{2+}, \text{Cu}) = +0.34 \text{ V}$$

$$E^\circ(\text{Cu}^+, \text{Cu}) = +0.52 \text{ V}$$

$$E^\circ[\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}] = +1.23 \text{ V}$$

$$E^\circ[\text{O}_2(\text{g}) + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-] = +0.40 \text{ V}$$

$$E^\circ(\text{Cr}^{3+}, \text{Cr}) = -0.74 \text{ V}$$

$$E^\circ(\text{Cr}^{2+}, \text{Cr}) = -0.91 \text{ V}$$

Match E° of the redox pair in List I with the values given in List II. List I List II (A) $E^\circ(\text{Fe}^{3+}, \text{Fe})$ (P) -0.70 V (B) $E^\circ(4\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{OH}^-)$ (Q) -0.4 V (C) $E^\circ(\text{Cu}^{2+} + \text{Cu} \rightarrow 2\text{Cu}^+)$ (R) -0.04 V (D) $E^\circ(\text{Cr}^{3+}, \text{Cr}^{2+})$ (S) -0.83 V

Problem 292

[Neeraj Kumar [NRJ] EXERCISE II Q4]

Match the following Column I Column II (A) Concentration cell (P) $\text{Ag} | \text{AgCl} | \text{Cl}^- || \text{Ag}^+ | \text{Ag}$ (B) Spontaneous cell reaction (Q) $\text{Ag} | \text{AgCl} | \text{Cl}^- || \text{Br}^- | \text{AgBr} | \text{Ag}$ (C) Non-spontaneous cell reaction (R) $\text{Ag} | \text{Ag}^+ (0.1 \text{ M}) || \text{Ag}^+ (1.0 \text{ M}) | \text{Ag}$ (S) $\text{Ag} | \text{AgCl} | \text{Cl}^- (0.1 \text{ M}) || \text{Cl}^- (1.0 \text{ M}) | \text{AgCl} | \text{Ag}$

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Problem 293

[Neeraj Kumar [NRJ] EXERCISE II Q1]

Percentage of aniline hydrochloride hydrolysed in its M/40 solution at 25°C is (Given: $E^\circ(\text{H}^+ | \text{H}_2) = -0.18 \text{ V}$, $2.303 \text{ RT/F} = 0.06$)

Problem 294

[Neeraj Kumar [NRJ] EXERCISE II Q2]

The standard reduction potential for $\text{Cu}^{2+} | \text{Cu}$ electrode is +0.34 V. The solubility product of $\text{Cu}(\text{OH})_2$ is 1.0×10^{-19} . The pH of solution at which the reduction potential for the above electrode becomes 0.31 V, is ($2.303 \text{ RT/F} = 0.06$)

Problem 295

[Neeraj Kumar [NRJ] EXERCISE II Q3]

The reduction potential at 25°C for $\text{Fe}^{3+} | \text{Fe}^{2+}$ electrode is +0.718 V. If $E^\circ(\text{Fe}^{2+} | \text{Fe}) = -0.44 \text{ V}$ and $E^\circ(\text{Fe}^{3+} | \text{Fe}) = -0.04 \text{ V}$, the ratio of molar concentrations of Fe^{2+} to Fe^{3+} ions in solution is ($2.303 \text{ RT/F} = 0.06$, $\log 5 = 0.7$)

Problem 296

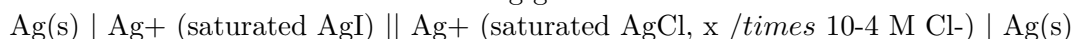
[Neeraj Kumar [NRJ] EXERCISE II Q4]

If $[\text{Fe}^{3+}]$ at equilibrium, when potassium iodide is added to a solution of Fe^{3+} initially at 0.50 M until $[\text{I}^-] = 1.0 \text{ M}$, is $x \times 10^{-5} \text{ M}$, the value of x is (Given $E^\circ(\text{Fe}^{3+} | \text{Fe}^{2+}) = +0.77 \text{ V}$, $E^\circ(\text{I}_2 | \text{I}^-) = +0.53 \text{ V}$, $2.303 \text{ RT/F} = 0.06$)

Problem 297

[Neeraj Kumar [NRJ] EXERCISE II Q5]

If it is desired to construct the following galvanic cell:



to have $E_{\text{Cell}} = 0.102 \text{ V}$, what should be the value of x to get $[\text{Cl}^-]$, which must be present in the cathodic half-cell to achieve the desired EMF. Given K_{sp} of AgCl and AgI are 1.8×10^{-10} and 8.1×10^{-17} , respectively. ($2.303 \text{ RT/F} = 0.06$, $\log 2 = 0.3$)

Problem 298

[Neeraj Kumar [NRJ] EXERCISE II Q6]

An alloy of lead (valency = 2)-thallium (valency = 1) containing 70% be electroplated onto a cathode from a perchloric acid solution. How many hours (approx.) would be required to deposit 5.0 g of this alloy at a current of 1.10 A? (Pb = 208, Tl = 204)

Problem 299

[Neeraj Kumar [NRJ] EXERCISE II Q7]

Iridium was plated from a solution containing IrCl_6 for 2.0 h with a current of 0.075 A. The Iridium deposited on the cathode weighed 0.36 g. If the oxidation state of Ir in IrCl_6 is x , then the value of $(x + y)$ is (Ir = 192)

Problem 300

[Neeraj Kumar [NRJ] EXERCISE II Q8]

The electrolysis of cold sodium chloride solution produces sodium hypochlorite by reacting NaOH and Cl_2 thoroughly. How long (in days) will a cell operate to produce 10 L of 7.45% solution of NaClO if the cell current is 2.5 A? Assume that the density of solution is 1.0 g/ml.

Problem 301

[Neeraj Kumar [NRJ] EXERCISE II Q9]

A volume of 500 ml of 0.1 M – CuSO_4 solution is electrolysed for 5 min at a current of 0.161 A. If Cu is produced at one electrode and oxygen at the other, the approximate pH of the final solution is

Problem 302

[Neeraj Kumar [NRJ] EXERCISE II Q10]

A test for complete removal of Cu^{2+} ions from a solution of Cu^{2+} is to add $\text{NH}_3(\text{aq})$. A blue colour signifies the formation of complex $[\text{Cu}(\text{NH}_3)_4]^{2+}$ having $K_f = 1.1 \times 10^{13}$ and thus confirms the presence of Cu^{2+} in solution. 250 ml of 0.1 M – CuSO_4 is electrolysed by passing a current of 5 A for 1351 s. After passage of this charge, sufficient quantity of NH_3 is added to electrolysed solution maintaining $[\text{NH}_3] = 0.10$ M. If $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is detectable up to its concentration as low as 1×10^{-5} M, would a blue colour be shown by the electrolysed solution on addition of NH_3 . Mark '1', if the answer is 'yes' and mark '2', if the answer is 'no'.

Problem 303

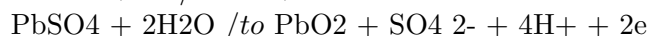
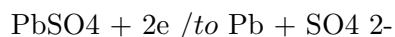
[Neeraj Kumar [NRJ] EXERCISE II Q11]

By passing a certain amount of charge through NaCl solution, 9.08 L of chlorine gas were liberated at STP. When the same amount of charge is passed through a nitrate solution of metal M, 52.8 g of the metal was deposited. If the specific heat of metal is 0.032 Cal/°C-g, the valency of metal is

Problem 304

[Neeraj Kumar [NRJ] EXERCISE II Q12]

The electrode reactions for charging of a lead storage battery are:



The electrolyte in the battery is an aqueous solution of sulphuric acid. Before charging, the specific gravity of the liquid was found to be 1.10 (16/ H_2SO_4 by wt.). After charging for 9659 h, the specific gravity of the liquid was found to be 1.42 (40/2 L of the liquid and the volume remains constant during charge, the average current (in A) used for charging the battery is

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Problem 305

[Neeraj Kumar [NRJ] EXERCISE II Q13]

A dilute solution of KCl was placed between two platinum electrodes, 12 cm apart, across which a potential of 1.93 volts was applied. How far (in cm) would the K^+ ion move in 20 h at 25°C? Ionic conductance of K^+ ion at infinite dilution at 25°C is 7.5×10^{-3} mho $\text{m}^2 \text{mol}^{-1}$.

Problem 306**[Neeraj Kumar [NRJ] EXERCISE II Q14]**

In the refining of silver by electrolytic method, what will be the final mass (in gm) of 72.8 g silver anode (60/1 h? ($\text{Ag} = 108$)

Problem 307**[Neeraj Kumar [NRJ] EXERCISE II Q15]**

The pH of 100 L of concentrated KCl solution after the electrolysis for 10 s using 9.65 A at 298 K is Four-digit Integer Type

Problem 308**[Neeraj Kumar [NRJ] EXERCISE II Q1]**

The specific conductivity of a saturated solution of AgCl is $2.80 \times 10^{-4} \text{ mho m}^{-1}$ at 25°C . If $\lambda_{\text{Ag}^+} = 6.19 \times 10^{-3} \text{ mho m}^2 \text{ mol}^{-1}$ and $\lambda_{\text{Cl}^-} = 7.81 \times 10^{-3} \text{ mho m}^2 \text{ mol}^{-1}$, the solubility of silver chloride (in order of 10^{-5} g l^{-1}) at 25°C , is

Problem 309**[Neeraj Kumar [NRJ] EXERCISE II Q2]**

The electrode potential (in millivolts) of $2\text{Ag(s)} + \text{S}^{2-}(\text{aq}) \rightarrow \text{Ag}_2\text{S(s)} + 2\text{e}^-$ in a solution buffered at $\text{pH} = 3$ and which is also saturated with $0.1 \text{ M} - \text{H}_2\text{S}$, is (Given: for H_2S , $K_{a1} = 10^{-8}$; $K_{a2} = 10^{-13}$, K_{sp} of $\text{Ag}_2\text{S} = 4 \times 10^{-48}$, $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}$, $\log 2 = 0.3$, $2.303 \text{ RT/F} = 0.06$)

Problem 310**[Neeraj Kumar [NRJ] EXERCISE II Q3]**

An excess of liquid mercury is added to an acidified solution of $10^{-3} \text{ M} - \text{Fe}^{3+}$. It is found that 10/The value of $E_{\text{Hg}^{2+}/\text{Hg}}$ (in mV), assuming that the only reaction that occurs is: $2\text{Hg} + 2\text{Fe}^{3+} \rightarrow \text{Hg}_2^{2+} + 2\text{Fe}^{2+}$, is (Given: $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.7724 \text{ V}$, $\log 2 = 0.3$, $\log 3 = 0.48$, $2.303 \text{ RT/F} = 0.06$)

Problem 311**[Neeraj Kumar [NRJ] EXERCISE II Q4]**

Determine potential (in mV) of the cell: $\text{Pt} | \text{Fe}^{2+}, \text{Fe}^{3+} || \text{Cr}_2\text{O}_7^{2-}, \text{Cr}^{3+}, \text{H}^+ | \text{Pt}$ in which $[\text{Fe}^{2+}] = 0.75 \text{ M}$, $[\text{Fe}^{3+}] = 0.75 \text{ M}$, $[\text{Cr}_2\text{O}_7^{2-}] = 2 \text{ M}$, $[\text{Cr}^{3+}] = 4 \text{ M}$ and $[\text{H}^+] = 1 \text{ M}$. Given: $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$; $E^\circ = 0.77 \text{ V}$

$14 \text{H}^+ + 6\text{e}^- + \text{Cr}_2\text{O}_7^{2-} \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$; $E^\circ = 1.35 \text{ V}$

$2.303 \text{ RT/F} = 0.06$, $\log 2 = 0.3$, $\log 3 = 0.48$

Problem 312**[Neeraj Kumar [NRJ] EXERCISE II Q5]**

An alloy weighing 2.70 mg of Pb-Ag was dissolved in desired amount of HNO_3 and volume was made 250 ml. A silver electrode was dipped in solution and E_{cell} of the cell:

$\text{Pt}, \text{H}_2(1 \text{ bar}) | \text{H}^+(1 \text{ M}) || \text{Ag}^+ | \text{Ag}$

was 0.50 V at 298 K. The percentage of lead in alloy is (Given: $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}$, $\text{Ag} = 108$, $2.303 \text{ RT/F} = 0.06$)

Problem 313**[Neeraj Kumar [NRJ] EXERCISE II Q6]**

Given $E^\circ = 0.08 \text{ V}$ for $\text{Fe}^{3+}(\text{cyt b}) | \text{Fe}^{2+}(\text{cyt b})$ couple and $E^\circ = 0.20 \text{ V}$ for $\text{Fe}^{3+}(\text{cyt c1}) | \text{Fe}^{2+}(\text{cyt c1})$ couple, where E° represents the standard state reduction potentials at $\text{pH} = 7.0$ at 25°C and cyt is an abbreviation for cytochromes. The value of K_{eq} for the reaction $\text{Fe}^{3+}(\text{cyt c1}) + \text{Fe}^{2+}(\text{cyt b}) \rightarrow \text{Fe}^{2+}(\text{cyt c1}) + \text{Fe}^{3+}(\text{cyt b})$ is ($2.303 \text{ RT/F} = 0.06$)

Problem 314**[Neeraj Kumar [NRJ] EXERCISE II Q7]**

Calculate potential (in mV) of the cell: $\text{Cu} \mid \text{Mn(s)} \mid \text{MnCl}_2(0.001 \text{ M}), \text{HCl}(0.01 \text{ M}) \mid \text{O}_2(0.25 \text{ bar}) \mid \text{Pt} \mid \text{Cu}$. Given, $E = -1.185 \text{ V}$ for the $\text{Mn}^{2+} \mid \text{Mn}$ couple and 1.229 V for the $\text{O}_2 \mid \text{H}_2\text{O}, \text{H}^+$ couple. ($2.303 \text{ RT/F} = 0.06$, $\log 2 = 0.3$)

Problem 315**[Neeraj Kumar [NRJ] EXERCISE II Q8]**

The temperature coefficient of the cell: $\text{Zn} \mid \text{ZnCl}_2 \mid \text{AgCl(s)} \mid \text{Ag}$ is $+0.001 \text{ V/K}$. The entropy change (in J/K) accompanying this reaction, at 298 K is

Problem 316**[Neeraj Kumar [NRJ] EXERCISE II Q9]**

For the reaction: $4\text{Al(s)} + 3\text{O}_2(\text{g}) + 6\text{H}_2\text{O(l)} + 4\text{OH}^-(\text{aq}) \rightarrow 4\text{Al(OH)}_4^-(\text{aq})$; $E_{\text{cell}} = 2.5 \text{ V}$. If ΔG° for $\text{H}_2\text{O(l)}$ and $\text{OH}^-(\text{aq})$ are -280.0 and -156.25 kJ/mol , respectively, the magnitude of ΔG° for $\text{Al(OH)}_4^-(\text{aq})$ (in kJ/mol) is

Problem 317**[Neeraj Kumar [NRJ] EXERCISE II Q10]**

In a Zn-MnO_2 Cell, the anode is made up of Zn and cathode of carbon rod surrounded by a mixture of MnO_2 , Carbon, NH_4Cl and ZnCl_2 in aqueous base. If 8.7 g MnO_2 is present in cathodic compartment, how many days the dry cell will continue to give a current of $3.99 \times 10^{-3} \text{ A}$?

Problem 318**[Neeraj Kumar [NRJ] EXERCISE II Q11]**

An amount of 19 g of molten SnCl_2 is electrolysed for some time. Inert electrodes are used. 1.19 g of tin is deposited at the cathode. No substance is lost during electrolysis. If the ratio of the masses of SnCl_2 and SnCl_4 after electrolysis is $x : 261$, the value of x is ($\text{Sn} = 119$)
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Problem 319**[Neeraj Kumar [NRJ] Exercise I Q12]**

An object whose surface area is 80 cm^2 is to be plated with an even layer of gold $8.0 \times 10^{-4} \text{ cm}$ thick. The density of gold is 19.7 g/cm^3 . The object is placed in a solution of $\text{Au(NO}_3)_3$ and a current of 2.4 A is applied. The time (in seconds) required for the electroplating to be completed, assuming that the layer of gold builds up evenly, is ($\text{Au} = 197$)

Problem 320**[Neeraj Kumar [NRJ] Exercise I Q13]**

We have taken a saturated solution of AgBr . K_{sp} of AgBr is 12×10^{-14} . If 10^{-7} moles of AgNO_3 is added to 1 litre of this solution, and conductivity (specific conductance) of this solution in terms of $10^{-7} \text{ S cm}^{-1}$ units. Given $\lambda^\circ(\text{Ag}^+) = 6 \times 10^{-3} \text{ S m}^2 \text{ mol}^{-1}$, $\lambda^\circ(\text{Br}^-) = 8 \times 10^{-3} \text{ S m}^2 \text{ mol}^{-1}$, $\lambda^\circ(\text{NO}_3^-) = 7 \times 10^{-3} \text{ S cm}^2 \text{ mol}^{-1}$.

Problem 321**[Neeraj Kumar [NRJ] Exercise I Q14]**

On passing electricity through nitrobenzene solution, it is converted into azobenzene. The mass of azobenzene (in mg) produced, if the same quantity of electricity produces oxygen just sufficient to burn 96 g of fullerene (C_{60}), is

Problem 322

[Neeraj Kumar [NRJ] Exercise I Q15]

The conductivity of saturated solution of sparingly soluble salt, $\text{Ba}_3(\text{PO}_4)_2$, is $1.2 \times 10^{-5} \text{ } \Omega^{-1}\text{cm}^{-1}$. The limiting equivalent conductances of BaCl_2 , K_3PO_4 and KCl are 160, 140 and $100 \text{ } \Omega^{-1}\text{cm}^2 \text{ eq}^{-1}$, respectively. The K_{sp} of $\text{Ba}_3(\text{PO}_4)_2$ (in the order of 10^{-25}) is

Problem 323

[Neeraj Kumar [NRJ] Exercise I Q30]

(a) Galvanic Cell

Problem 324

[Neeraj Kumar [NRJ] Exercise I Q65]

(a) Electrolysis

Problem 325

[Neeraj Kumar [NRJ] Exercise I Q99]

(b) 100. (c) 101. (a) 102. (a) 103. (b) 104. (b) 105. (b) Conductance

Problem 326

[Neeraj Kumar [NRJ] Exercise I Q106]

(b) 107. (a) 108. (c) 109. (b) 110. (a) 111. (b) 112. (c) 113. (c) 114. (a) 115. (d)

Problem 327

[Neeraj Kumar [NRJ] Exercise I Q116]

(a) 117. (b) 118. (d) 119. (c) 120. (d) 121. (a) 122. (c) 123. (a) 124. (d) 125. (a) ANSWER KEYS
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Problem 328

[Neeraj Kumar [NRJ] Exercise II Q80]

(b) Section B (One or More than one Correct)

Problem 329

[Neeraj Kumar [NRJ] Exercise II Q15]

(a), (b), (c), (d) Section C Comprehension I

Problem 330

[Neeraj Kumar [NRJ] Exercise II Q3]

(a) Comprehension II

Problem 331

[Neeraj Kumar [NRJ] Exercise II Q6]

(a) Comprehension III

Problem 332

[Neeraj Kumar [NRJ] Exercise II Q10]

(b) Comprehension IV

Problem 333

[Neeraj Kumar [NRJ] Exercise II Q13]

(c) Comprehension V

Problem 334

[Neeraj Kumar [NRJ] Exercise II Q18]

(a) Comprehension VI

Problem 335

(d) Comprehension VII

[Neeraj Kumar [NRJ] Exercise II Q23]

Problem 336

(a) Comprehension VIII

[Neeraj Kumar [NRJ] Exercise II Q27]

Problem 337

(d) Comprehension IX

[Neeraj Kumar [NRJ] Exercise II Q30]

Problem 338

(c) Comprehension X

[Neeraj Kumar [NRJ] Exercise II Q34]

Problem 339

(c) Comprehension XI

[Neeraj Kumar [NRJ] Exercise II Q38]

Problem 340

(a) Comprehension XII

[Neeraj Kumar [NRJ] Exercise II Q41]

Problem 341

(d) Comprehension XIII

[Neeraj Kumar [NRJ] Exercise II Q43]

Problem 342

(c) Comprehension XIV

[Neeraj Kumar [NRJ] Exercise II Q45]

Problem 343

(b) Comprehension XV

[Neeraj Kumar [NRJ] Exercise II Q47]

Problem 344

(b) ANSWER KEYS

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[Neeraj Kumar [NRJ] Exercise II Q53]

Problem 345

(c) Section E (Column Match)

[Neeraj Kumar [NRJ] Exercise II Q15]

Problem 346

A /to P, Q; B /to P, Q; C /to Q, R; D /to P, S

[Neeraj Kumar [NRJ] Exercise II Q1]

Problem 347

A /to R; B /to S; C /to Q; D /to P

[Neeraj Kumar [NRJ] Exercise II Q2]

Problem 348

A /to R; B /to S; C /to P; D /to Q

[Neeraj Kumar [NRJ] Exercise II Q3]

Problem 349

[Neeraj Kumar [NRJ] Exercise II Q4]

A /to R, S; B /to P, R; C /to Q, S Section F (Subjective) Single-digit Integer Type

Problem 350

[Neeraj Kumar [NRJ] Exercise II Q15]

(9) Four-digit Integer Type

Problem 351

[Neeraj Kumar [NRJ] Exercise II Q15]

(0108) ANSWER KEYS

Visit www.jeebooks.in for more**Problem 352**

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q1]

Potential of electrode is independent from the area of electrode immersed in solution.

Problem 353

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q3]

Copper cannot reduce Al^{3+} .**Problem 354**

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q4]

Cu will reduce Ag^+ and itself becomes Cu^{2+} .**Problem 355**

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q5]

K is more active metal than Al.

Problem 356

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q6]

Mg is better reducing agent.

Problem 357

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q8]

Lower the reduction potential, strong is the reducing agent.

Problem 358

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q9]

 $Y^- + X \rightarrow Y^{2-} + X^+$ is possible.**Problem 359**

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q10]

 MnO_4^- - Will also oxidize HCl.**Problem 360**

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q11]

Electrochemical series

Problem 361

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q12]

 $M(s); E^\circ_{M^{n+}/M} = -0.76 V$ $2H^+(aq) + 2e^- \rightarrow H_2(g); E^\circ_{H^+/H_2} = 0 V$

Hence, reverse reaction is spontaneous.

Problem 362 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q13]
 $\text{Hg} \text{Cl}_2 \text{aq} \text{Cl}_2 \text{aq} \rightleftharpoons \text{Hg}^{2+} \text{aq} + 2 \text{Cl}^- \text{aq}$
 $E^\circ = 0.92 \text{ V}$
 $E = 0.92 - \frac{0.059}{2} \log \frac{[\text{Hg}^{2+}]}{[\text{Cl}_2]}$

Problem 363 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q14]
 $E^\circ_{\text{MnO}_4^-/\text{Mn}^{2+}} = 1.51 \text{ V}$, $E^\circ_{\text{Br}_2/\text{Br}^-} = 1.07 \text{ V}$
 $E^\circ_{\text{cell}} = 1.51 - 1.07 = 0.44 \text{ V}$

Problem 364 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q15]
 Electrolyte used must have nearly same ionic mobility.

Problem 365 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q16]
 $\text{MnO}_4^- + 8 \text{H}^+ + 5 \text{e}^- \rightarrow \text{Mn}^{2+} + 4 \text{H}_2\text{O}$
 $E^\circ = 1.51 \text{ V}$
 $E = 1.51 - \frac{0.059}{5} \log \frac{[\text{Mn}^{2+}]}{[\text{MnO}_4^-][\text{H}^+]^8}$
 $= -0.1888 \text{ V}$

Problem 366 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q17]
 $\text{Cu}^{2+} + \text{e}^- \rightarrow \text{Cu}^+$
 $E^\circ = 0.52 \text{ V}$
 $E = 0.52 - \frac{0.059}{1} \log [\text{Cu}^+]$
 $= E^\circ - 0.0295 \text{ V}$

Problem 367 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q18]
 $\text{O}_2 + 4 \text{H}^+ + 4 \text{e}^- \rightarrow 2 \text{H}_2\text{O}$
 $E^\circ = 1.23 \text{ V}$
 $E = 1.23 - \frac{0.059}{4} \log \frac{1}{[\text{H}^+]^4}$
 $= -0.059 \text{ V}$

Problem 368 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q19]
 $\text{Cu}^{2+} + \text{e}^- \rightarrow \text{Cu}^+$
 $E^\circ = 0.52 \text{ V}$
 $E = 0.52 - \frac{0.059}{1} \log [\text{Cu}^+]$
 $= 0.52 - 0.059 \log 2 = 0.48 \text{ V}$

Problem 369 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q20]
 $2 \text{H}^+ + 2 \text{e}^- \rightarrow \text{H}_2$
 $E^\circ = 0 \text{ V}$
 $E = 0 - \frac{0.059}{2} \log \frac{1}{[\text{H}^+]^2}$
 $= -0.0413 \text{ V}$

Problem 370 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q22]
 Ionic mobility of ions in salt bridge should be same.

Problem 371 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q23]
 Higher reduction potential, stronger is the oxidizing agent.

Problem 372

[Neera] Kumar [NRJ] EXERCISE I (JEE MAIN) Q24]

Lower the reduction potential, stronger is the reducing agent.

Problem 373

[Neera] Kumar [NRJ] EXERCISE I (JEE MAIN) Q25]

E E E V Fe Fe Fe Fe Fe Fe /circ = /times /circ - /times /circ - = + + + + + 3 2 3 2 3 2 3 2 0
772 / / / .

Problem 374

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q26]

H g H e 2 2 2 () (aq)

$$+ + - \text{E E E (H) P H /CH COOH H /H H /H}^2 \text{H}^2 \text{3}^2 + 2^2 = = /circ -$$
 $+ + 0.062 \cdot \log$
$$= -$$
$$/times () = -$$

/times /times - 0 0 06 2 0 06 2 1 8 10 0 5 1 2 5 2 . log . log . . Ka C PH

$$= +0.1512 \text{ V}$$

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Problem 375

[Neeraj] Kumar [NRJ] EXERCISE I (JEE MAIN) Q27]

E E K Cl /CuCl/Cu Cu Cu sp + /circ = /circ - - / . log 0 06 1 1

or, $1\ 28\ 0\ 06\ 1\ 1\ 2\ 10\ 7\ \dots \log / = /circ -$

$$/times - E_{Cu} Cu +$$
$$/implies E V Cu Cu + /circ = / .1682$$

Problem 376

[Neera] Kumar [NRJ] EXERCISE I (JEE MAIN) Q28]

E E K OH Pb(OH) Pb Pb Pb sp P /circ = /circ -

$$-// // . \log 2 \ 2 \ 0 \ 06 \ 2 \ 1$$

or, $(-0.55) = (-0.13) - 0.0621 \cdot \log$

$$K_{sp}$$
$$K_{sp} = 1.0 \times 10^{-14}$$

Problem 377

[Neera] Kumar [NRJ] EXERCISE I (JEE MAIN) Q29]

E E E MnO MnO MnO Mn MnO Mn 2 4 4 2 2 2 5 2 5 2 - - + + = /times - /times - / / /

$$= /times - /times = 5\ 1\ 51\ 2\ 1\ 23\ 3\ 1\ 697\ \dots\ V$$

Problem 378

[Neera] Kumar [NRJ] EXERCISE I (JEE MAIN) Q30]

For stronger oxidizing agent, reduction potential should be high. Galvanic Cell

Problem 379

[Neera] Kumar [NRJ] EXERCISE I (JEE MAIN) Q31]

Reduction occur at right electrode.

Problem 380

[Neera] Kumar [NRJ] EXERCISE I (JEE MAIN) Q32]

Circuit becomes incomplete.

Problem 381

[Neera] Kumar [NRJ] EXERCISE I (JEE MAIN) Q33]

$$\text{ECell} = (\text{ERP})_{\text{R}} - (\text{ERP})_{\text{L}} = 0$$

[Neera] Kumar [NRJ] EXERCISE I (JEE MAIN) Q34]

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q38]

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q39]

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q42]

[Neera] Kumar [NRJ] EXERCISE I (JEE MAIN) Q43]

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q45]

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q47]

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q48]

[Neera] Kumar [NRJ] EXERCISE I (JEE MAIN) Q49]

[Neera] Kumar [NRJ] EXERCISE I (JEE MAIN) Q50]

Problem 392 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q51]
 $E_{\text{cell}} / \text{circ} = - = 0.44033011 \dots \text{V}$

Problem 393 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q52]
 Cell reaction: $\text{Zn} + \text{Fe}^{2+} \rightarrow \text{Zn}^{2+} + \text{Fe}$
 $E_{\text{cell}} / \text{circ} = -$
 $[] + + 0.059222 \dots \log []$
 or, $0.290500592005920010001 \dots \log \dots =$
 $-$
 K_{eq}
 $K_{\text{eq}} =$
 $10^{0.320295} \dots$
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Problem 394 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q53]
 $E_{\text{K cell eq}} / \text{circ} =$
 $0.06 \dots \log n / \text{implies } 0.18064 \dots \log =$
 K_{eq}
 $\log K_{\text{c}} = 12$

Problem 395 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q54]
 $E_{\text{RT F Zn Cu cell}} / \text{circ} = -$
 $+ + 222 \ln [] []$

Problem 396 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q55]
 $E_{\text{RT F Ni Cu cell}} / \text{circ} = -$
 $+ + 222 \ln [] []$

Problem 397 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q56]
 $E_{\text{E } 0.059 \text{ E } 21} = -$
 $= -12100010059 \log \dots$

Problem 398 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q57]
 $E_{0.059 \text{ V cell}} / \text{circ} =$
 $= 210035412 \log \dots$

Problem 399 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q58]
 $[] \cdot \text{HA} + \text{H}^+ + \text{K}^+ + \text{M}^+ + \text{a} + - - =$
 $= / \text{times} = 10^{0.11053}$

Problem 400 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q60]
 $E_{\text{n K cell eq}} / \text{circ} =$
 $0.06 \dots \log$
 $/ \text{implies } (\dots) \cdot \log 0.770530062 - =$
 K_{eq}
 $K_{\text{eq}} = 108$

Problem 401 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q61]
Current flows from right electrode to left electrode in external circuit.

Problem 402 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q62]
 $\Delta G_{\text{cell}} = -nFE_{\text{cell}} \implies -48250 = -2 \times 96500 \times E_{\text{cell}}$
 $E_{\text{cell}} = 0.25 \text{ V}$
 Now, $E_{\text{cell}} = E_{\text{Cl}^+/\text{Cl}_2} - E_{\text{H}^+/\text{H}_2}$
 $+ 0.0592 \log [\text{H}^+]$
 $= -0.0592 \log (0.1) = 0.0592 \text{ V}$

Problem 403 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q63]
 Net cell reaction: $\text{H}_2 + \text{C} \rightleftharpoons \text{H}^+ + \text{C}^{2-}$ (10⁻⁶ M)
 $E_{\text{cell}} = -0.0591 \log K \implies C = 10^{-4} \text{ M}$

Problem 404 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q64]
 $E_{\text{cell}} = E_{\text{OCl}^-/\text{Cl}_2} - E_{\text{Cl}^-/\text{Cl}_2} = 1.21 \text{ V} - 0.94 \text{ V} = 0.27 \text{ V}$

Problem 405 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q65]
 For (i) and (iii); $E_{\text{cell}} = 3.0 \text{ V}$. Electrolysis

Problem 406 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q66]
 Oxidation occurs at anode and reduction at cathode, in both.

Problem 407 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q69]
 Due to high over-voltage potential, reduction of H_2O becomes less favorable.

Problem 408 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q70]
 Electrochemical equivalent, $Z = \frac{E}{F}$ and E is maximum for silver.

Problem 409 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q72]
 $W = \frac{Q}{F} \times n$
 For smaller n , w will be higher.

Problem 410 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q73]
 Cathode to anode in internal supply. $+ - e^- e^-$

Problem 411 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q76]
 Equivalent weight is maximum for Na.

Problem 412 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q77]
 $n_{\text{eq}} \text{ of Cu} = \frac{0.516352 \text{ g}}{15.875 \text{ g}} = \frac{1}{2}$
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Problem 413 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q78]
 Ni will oxidize and Ni^{2+} will reduce resulting no change in Ni^{2+} Concentration.

Problem 414 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q79]
 $W_{\text{QF}} = \frac{9.863523 \times 10000}{96500} = \frac{10.1}{10.2} = 0.99$

Problem 415 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q80]
 Cathode = H_2 , anode = O_2 and $W_{\text{H}_2} = 1.8 \text{ g}$

Problem 416 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q81]
 $W = Z \cdot i \cdot t$ implies $Z = \frac{W}{i \cdot t}$
 $\frac{1.12221 \text{ g}}{1 \text{ A} \cdot 2 \text{ hr}} = \frac{Z}{1 \text{ A} \cdot 2 \text{ hr}}$

Problem 417 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q83]
 Informative 8.4. At cathode: $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu(s)}$
 At anode: $2\text{H}_2\text{O(l)} \rightarrow \text{O}_2\text{(g)} + 4\text{H}^+\text{(aq)} + 4e^-$
 Hence, solution becomes rich in H_2SO_4 .

Problem 418 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q85]
 Reduction occurs first for the species having higher reduction potential.

Problem 419 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q86]
 For very dilute NaCl solution,
 Cathode: $2\text{H}_2\text{O(l)} + 2e^- \rightarrow \text{H}_2\text{(g)} + 2\text{OH}^-\text{(aq)}$; pH
 Anode: $2\text{H}_2\text{O(l)} \rightarrow \text{O}_2\text{(g)} + 4\text{H}^+\text{(aq)} + 4e^-$; pH

Problem 420 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q87]
 $W_{\text{EAA}} = \frac{1.212232 \text{ g}}{10.1} = \frac{1.212232 \text{ g}}{10.1}$

Problem 421 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q88]
 $n_{\text{eq}} \text{ of Zn} = \frac{100.65423058 \text{ g}}{65.38 \text{ g}} = 1.54$
 $n_{\text{eq}} \text{ of CuSO}_4 = 1.0 \times 2 = 2.0 \text{ (L.R.)}$
 $Q_{\text{F}} = \frac{2.1096500 \text{ C}}{96500 \text{ C}} = 2.18 \times 10^{-5} \text{ F}$
 $t = \frac{193000 \text{ sec}}{3600 \text{ hr}} = 53.6 \text{ hr}$

Problem 422 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q89]
 $n_{\text{eq}} = \frac{6.1061020233 \text{ g}}{63.55 \text{ g}} = 0.096$

Problem 423 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q90]
 $neq Q F = \frac{208.21}{100} \times \frac{3600}{96500} \times \frac{100}{100} = \frac{100}{100} \times \frac{100}{100}$
 $\implies w \approx 388 \text{ gm}$

Problem 424 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q91]
 $neq \text{Ag deposited} = neq \text{Cu dissolved}$
 or, $3.241081642 \times \frac{1}{100} = \frac{1}{100} \times w \implies w = 0.96 \text{ gm}$
 Final mass of Cu anode = $60 - 0.96 = 59.04 \text{ gm}$

Problem 425 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q92]
 $neq \text{Fe}_2\text{O}_3 \text{ reduced} = n \times \frac{1}{100} \times \frac{3600}{96500} \times \frac{100}{100} = \frac{1}{100} \times \frac{100}{100} \times \frac{100}{100}$
 Charge needed = $6 F$

Problem 426 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q93]
 Number of moles, $n = \frac{neq}{n\text{-factor}}$
 $n_{\text{Al}} : n_{\text{Cu}} : n_{\text{Na}} = 3 : 3 : 3 = 1 : 1 : 1$

Problem 427 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q94]
 $2.224282 \text{ HSO}_3\text{O}_2 \text{ e}^- \rightarrow \text{SO}_4^{2-} + 2\text{H}^+$
 Now, $neq Q F = \frac{1}{100} \times \frac{3600}{96500} \times \frac{100}{100} = \frac{1}{100} \times \frac{100}{100} \times \frac{100}{100}$
 $i = 71.48 \text{ A}$

Problem 428 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q95]
 Number of Faradays required = $neq = 1 \times \frac{1}{100} \times \frac{3600}{96500} \times \frac{100}{100} = \frac{1}{100} \times \frac{100}{100} \times \frac{100}{100}$

Problem 429 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q96]
 $neq = n \times \frac{1}{100} \times \frac{3600}{96500} \times \frac{100}{100} = \frac{1}{100} \times \frac{100}{100} \times \frac{100}{100}$
 or, $1.008215464 \times \frac{1}{100} = \frac{1}{100} \times \frac{3600}{96500} \times \frac{100}{100} \implies V = 5.6 \text{ L}$

Problem 430 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q97]
 $neq \text{Ag} = neq \text{O}_2 \implies w_{\text{O}_2} = \frac{108}{16} \times \frac{1}{100} \times \frac{3600}{96500} \times \frac{100}{100} = \frac{1}{100} \times \frac{100}{100} \times \frac{100}{100}$
 $\implies w = 21.6 \text{ gm}$

Problem 431 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q98]
 $neq Q F = \frac{3.965}{12000} \times \frac{3600}{96500} \times \frac{100}{100} = \frac{1}{100} \times \frac{100}{100} \times \frac{100}{100}$
 $\implies n = 4$

Problem 432 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q99]
 Number of electrons
 $= Q e = \frac{5.200160210624}{100} \times \frac{3600}{96500} \times \frac{100}{100} = \frac{1}{100} \times \frac{100}{100} \times \frac{100}{100}$

Problem 433 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q100]
 $neq Q F = \frac{3.0965}{1060} \times \frac{3600}{96500} \times \frac{100}{100} = \frac{1}{100} \times \frac{100}{100} \times \frac{100}{100}$
 $\implies E = 50$

Problem 434

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q101]

Cu will deposit at cathode.

$$n_{\text{eq}} Q F = \frac{63.5}{1000} \times \frac{1}{2} \times \frac{96500}{1000} = \frac{w}{1.905} \quad w = 1.905 \text{ gm}$$

Problem 435

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q102]

$$n_{\text{eq}} Q F = \frac{22400}{1000} \times \frac{1}{4} \times \frac{96500}{1000} = \frac{i}{48.25} \quad i = 48.25 \text{ A/sec}$$

Problem 436

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q103]

Mass of Na deposited

$$= \frac{23}{1000} \times \frac{1}{1} \times \frac{18710}{1000} \times \frac{623}{1000} = 2.623 \text{ gm}$$

$$\text{Now, } n_{\text{eq}} Q F = \frac{23}{1000} \times \frac{1}{1} \times \frac{623}{1000} = \frac{Q}{0.2} \quad Q \approx 0.2 F$$

Problem 437

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q104]

$$n_{\text{eq}} A = n_{\text{eq}} B$$

$$\text{or, } \frac{56}{112} \times \frac{1}{2} = \frac{927}{3} \times \frac{1}{VA} \quad VA = 2$$

Valency of A, VA = 2

Problem 438

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q105]

$$n_{\text{eq}} \text{Zn} = n_{\text{eq}} \text{Ni} \quad \frac{22}{1000} \times \frac{1}{2} \times \frac{8932}{1000} = \frac{2055}{1000} \times \frac{1}{n_{\text{eq}}} \quad n_{\text{eq}} = 29.36$$

$$E_{\text{Ni}} = 29.36$$

Visit www.jeebooks.in for more**Problem 439**

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q106]

On dilution, conductance decreases and hence resistance increase.

Problem 440

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q108]

HCl is molecular.

Problem 441

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q109]

Radius of aq.ion : $K^+ < Na^+ < Li^+$ **Problem 442**

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q110]

On dilution, m increases.

Problem 443

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q111]

H₂SO₄ is strong electrolyte**Problem 444**

[Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q112]

$$\text{For AgCl, } \frac{1}{\lambda} = \frac{1}{\lambda_{\text{Ag}^+}} + \frac{1}{\lambda_{\text{Cl}^-}} \quad m_{\text{Ag}^+} = 108, m_{\text{Cl}^-} = 35.5$$

$$= \frac{67.9}{1000} + \frac{82.1}{1000} = \frac{150}{1000} \quad \lambda = 150 \text{ cm}^2 \text{ mol}^{-1}$$

$$\text{Now, } \lambda = \frac{m C}{k} \quad \text{implies } \frac{150}{1000} = \frac{1.8 \times 10^{-6}}{k} \quad k = 1.2 \times 10^{-8} \text{ mol cm}^{-3}$$

$$S = 1.2 \times 10^{-8} \text{ mol cm}^{-3} = 1.2 \times 10^{-5} \text{ mol e}^{-1}$$

Problem 445 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q113]
 $\frac{1}{\text{circ}} = \frac{1}{\text{circ}} + \frac{1}{\text{circ}} - \frac{1}{\text{circ}}$ eq(NH OH) eq(NH Cl) eq(NaOH) eq(NaCl) 4 4
 $= 129.8 + 217.4 - 108.9 = 238.3 \text{ } \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
 Now, $\frac{1}{\alpha} = \frac{1}{\text{circ}} \frac{1}{\lambda} \frac{1}{\lambda} \text{ eq eq } 9 \text{ } 532 \text{ } 238 \text{ } 3 \text{ } 0 \text{ } 04 \text{ } 4 \text{ } . \text{ } . \text{ } . \text{ } /$

Problem 446 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q114]
 $= \frac{1}{\text{times}} \frac{1}{\text{times}} = \frac{1}{\text{times}} - \text{m C } \frac{1}{\kappa} 1 \text{ } 250 \text{ } 125 \text{ } 1 \text{ } 10 \text{ } 5 \text{ } 10 \text{ } 3 \text{ } 4 \text{ } \Omega^{-1} \text{ m}^2 \text{ mol}^{-1}$

Problem 447 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q115]
 $\text{MgSO}_4(\text{aq}) + \text{Ba}(\text{OH})_2 (\text{added})$
 $\text{Mg}(\text{OH})_2(\text{s}) + \text{BaSO}_4(\text{s})$
 First number of ions decreases and then increases.

Problem 448 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q116]
 $= \frac{1}{\text{times}} \frac{1}{\text{times}} \frac{1}{\text{times}} = \frac{1}{\text{times}} - \text{eq C } \frac{1}{\kappa} 2 \text{ } 6 \text{ } 10 \text{ } 1 \text{ } 2 \text{ } 10 \text{ } 1 \text{ } 3 \text{ } 10 \text{ } 1 \text{ } 3 \text{ } 2 \text{ } . \text{ } (\text{ }) \text{ } . \text{ } \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$

Problem 449 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q117]
 $\frac{1}{\alpha} = \frac{1}{\text{circ}} = + = \text{m m or } 16 \text{ } 30 \text{ } 349 \text{ } 8 \text{ } 40 \text{ } 9 \text{ } 0 \text{ } 04172 \text{ } 4 \text{ } 172 \text{ } . \text{ } (\text{ } . \text{ }) \text{ } . \text{ } /$

Problem 450 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q118]
 Cell constant, $C \text{ A cm m} = = = - - 1 \text{ } 2 \text{ } 5 \text{ } 5 \text{ } 0 \text{ } 5 \text{ } 50 \text{ } 1 \text{ } 1 \text{ } . \text{ } .$

Problem 451 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q119]
 $\frac{1}{\text{circ}} = \frac{1}{\text{circ}} - \frac{1}{\text{circ}} + \frac{1}{\text{circ}} - \text{m(m(m(Cl m(NH OH) NH Cl)) OH) } 4 \text{ } 4$
 $= 150 - 75 + 200 = 275 \text{ } \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
 Now, $\frac{1}{\alpha} = \frac{1}{\text{circ}} = \text{m m } 22 \text{ } 275 \text{ } 0 \text{ } 08 \text{ } .$

Problem 452 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q120]
 $\frac{1}{\text{circ}} = \frac{1}{\text{circ}} + \frac{1}{\text{circ}} = + + - \text{m(m(m(OH . . H O) H)) } 2 \text{ } 0 \text{ } 035 \text{ } 0 \text{ } 020 \text{ } \frac{1}{\lambda} \frac{1}{\lambda}$
 $= 0.055 \text{ mho m}^2 \text{ mol}^{-1}$
 Now, $= \text{m C } \frac{1}{\kappa} \text{ } \text{implies } 0 \text{ } 055 \text{ } 5 \text{ } 5 \text{ } 10 \text{ } 6 \text{ } . \text{ } . = \frac{1}{\text{times}} - C$
 $C = 10^{-4} \text{ mol m}^{-3} = 10^{-7} \text{ M } \text{implies } K_w = C^2 = 10^{-14} \text{ M}^2$

Problem 453 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q121]
 $\frac{1}{\alpha} = \frac{1}{\text{circ}} = \frac{1}{\text{times}} \frac{1}{\text{times}} = - - \text{m m } 7 \text{ } 814 \text{ } 10 \text{ } 3 \text{ } 907 \text{ } 10 \text{ } 0 \text{ } 02 \text{ } 4 \text{ } 2 \text{ } . \text{ } . \text{ } .$
 Now, $K \text{ C C a} =$
 $-$
 $= \frac{1}{\text{times}} = \frac{1}{\text{times}} - \frac{1}{\alpha} \frac{1}{\alpha} \frac{1}{\alpha} 2 \text{ } 2 \text{ } 2 \text{ } 5 \text{ } 1 \text{ } 0 \text{ } 02 \text{ } 0 \text{ } 05 \text{ } 2 \text{ } 10$
 $(\text{ } . \text{ }) \text{ } .$

Problem 454 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q122]
 $n \text{ C m} = = \frac{1}{\text{times}} \frac{1}{\text{times}} = - - \frac{1}{\kappa} 6 \text{ } 3 \text{ } 10 \text{ } 0 \text{ } 015 \text{ } 2 \text{ } 10 \text{ } 84 \text{ } 4 \text{ } 3 \text{ } . \text{ } . \text{ } \text{ S cm}^2 \text{ mol}^{-1}$

Problem 455 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q123]

 $\kappa = \frac{1}{\rho}$
 $\kappa = \frac{1}{\rho}$ (for strong electrolyte)

or, $\frac{1}{\rho} = \frac{1}{\kappa} \times \frac{1}{\text{m A}}$

(1)

 $\frac{1}{\rho} = \frac{1}{\kappa} \times \frac{1}{\text{m A}}$

(2)

 $\frac{1}{\rho} = \frac{1}{\kappa} \times \frac{1}{\text{m A}}$
Problem 456 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q124]

 $\frac{1}{\rho} = \frac{1}{\kappa} \times \frac{1}{\text{m A}}$ and $\frac{1}{\rho} = \frac{1}{\kappa} \times \frac{1}{\text{m A}}$
 $\frac{1}{\rho} = \frac{1}{\kappa} \times \frac{1}{\text{m A}}$

Now, $\frac{1}{\rho} = \frac{1}{\kappa} \times \frac{1}{\text{m A}}$

=

 $\frac{1}{\rho} = \frac{1}{\kappa} \times \frac{1}{\text{m A}}$
 $\frac{1}{\rho} = \frac{1}{\kappa} \times \frac{1}{\text{m A}}$
Problem 457 [Neeraj Kumar [NRJ] EXERCISE I (JEE MAIN) Q125]

Conductivity of solution will remain unchanged but the cell constant becomes double.

 $\frac{1}{\rho} = \frac{1}{\kappa} \times \frac{1}{\text{m A}}$
 $\frac{1}{\rho} = \frac{1}{\kappa} \times \frac{1}{\text{m A}}$
 $\frac{1}{\rho} = \frac{1}{\kappa} \times \frac{1}{\text{m A}}$

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Problem 458 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q1]

(I) Cu cannot reduce Pb

(II) Pb can reduce Ag

(III) Ag cannot reduce Cu.

Hence, reducing power: $\text{Pb} > \text{Cu} > \text{Ag}$
Problem 459 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q2]

 $E_{\text{cell}} = \frac{1}{n} \times \frac{RT}{F} \ln \frac{[\text{O}]}{[\text{R}]}$
 $E_{\text{cell}} = \frac{1}{n} \times \frac{RT}{F} \ln \frac{[\text{O}]}{[\text{R}]}$
 $E_{\text{cell}} = \frac{1}{n} \times \frac{RT}{F} \ln \frac{[\text{O}]}{[\text{R}]}$
 $E_{\text{cell}} = \frac{1}{n} \times \frac{RT}{F} \ln \frac{[\text{O}]}{[\text{R}]}$
Problem 460 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q3]

For the complex ion to get oxidised, its reduction potential should be low.

Problem 461 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q4]

 $\text{Ag} + \text{NH}_3 \rightleftharpoons \text{Ag}(\text{NH}_3)_2^+$

; . . . V

Now, $E_{\text{cell}} = \frac{1}{n} \times \frac{RT}{F} \ln \frac{[\text{O}]}{[\text{R}]}$
 $E_{\text{cell}} = \frac{1}{n} \times \frac{RT}{F} \ln \frac{[\text{O}]}{[\text{R}]}$

Kf

 $E_{\text{cell}} = \frac{1}{n} \times \frac{RT}{F} \ln \frac{[\text{O}]}{[\text{R}]}$

Problem 462 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q5]

Given: $\text{Co}^{3+} + e^- \rightleftharpoons \text{Co}^{2+}$; $E^\circ = 1.81 \text{ V}$;

$\Delta G^\circ = -nFE^\circ = -1 \times 96485 \times 1.81 = -1.75 \times 10^5 \text{ J mol}^{-1}$

$\text{Co(CN)}_6^{3-} + e^- \rightleftharpoons \text{Co(CN)}_6^{4-}$;

$E^\circ = -0.83 \text{ V}$;

$\Delta G^\circ = -nFE^\circ = -1 \times 96485 \times (-0.83) = 8.01 \times 10^4 \text{ J mol}^{-1}$

$\text{Co(CN)}_6^{3-} + e^- \rightleftharpoons \text{Co(CN)}_6^{4-}$;

$K_f = 10^{19}$;

$\Delta G^\circ = -RT \ln K_f$

Required $\text{Co(CN)}_6^{3-} + e^- \rightleftharpoons \text{Co(CN)}_6^{4-}$;

$K_f = ?$;

$\Delta G^\circ = -RT \ln K_f$

Now, $\Delta G^\circ = \Delta G^\circ_{\text{Co}^{3+}/\text{Co}^{2+}} + \Delta G^\circ_{\text{Co(CN)}_6^{3-}/\text{Co(CN)}_6^{4-}}$

or, -

$-1.75 \times 10^5 = -RT \ln K_f + RT \ln 10^{19}$

or, $RT \ln K_f$

$= -1.75 \times 10^5 + RT \ln 10^{19}$

$\implies \log K_f = \frac{-1.75 \times 10^5 + RT \ln 10^{19}}{RT} = -10.63$

$K_f = 10^{10.63}$

Problem 463 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q6]

$\text{E}^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34 \text{ V}$, $\text{E}^\circ_{\text{Cu}^{+}/\text{Cu}} = 0.52 \text{ V}$, $\text{E}^\circ_{\text{Cu}^{2+}/\text{Cu}^{+}} = 0.16 \text{ V}$

$\text{E}^\circ_{\text{Cu}^{2+}/\text{Cu}} = \frac{2 \times 0.34 + 1 \times 0.52}{2 + 1} = 0.31 \text{ V}$

$\text{E}^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34 + 0.03 \log(0.1) = 0.31 \text{ V}$

$\text{E}^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.31 \text{ V}$

Problem 464 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q7]

$\text{E}^\circ_{\text{RT F Cd Ag cell}} = \frac{RT}{F} \ln K_f$

$+ 2.22 \ln [] []$

As CN^- will form complex with Ag^+ ion in the cathodic compartment, E_{cell} will decrease.

Problem 465 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q8]

$-\Delta G^\circ =$

$= -nFE^\circ$

$= -nFE^\circ + nFE^\circ_{\text{Zn}^{2+}/\text{Zn}} - nFE^\circ_{\text{Cu}^{2+}/\text{Cu}}$

Problem 466 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q9]

$\text{E}^\circ_{\text{RT F K X Ag X|Ag Ag|Ag sp}} = \frac{RT}{F} \ln K_f$

$- + \ln 1$

Problem 467 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q10]

Cell reaction: H^+ (cathode, $\text{pH} = 3$) H^+ (anode, $\text{pH} = ?$)

$\text{E}^\circ_{\text{H anode H P P cell anode H catho}} = -$

$= - + + 0.0591 \log [] []_{\text{cathode}} \cdot \text{de H}$

or, $0.272 = 0.0593 \log []_{\text{anode H}} \implies \text{pH} = 7.6$

Problem 468 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q11]
 $\text{Cl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{Cl}^- + \text{ClO}^- + 2\text{H}^+$;
 $E^\circ_{\text{cell}} = 1.36 \text{ V}$
Now, $E_{\text{cell}} = 0.06 \text{ V}$
or, $0.027 = -\frac{0.06}{0.059} = -\text{pH}$

Problem 469 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q12]
 $E_{\text{cell}}(\text{quinohydrone}|\text{calomel}) = 0.210 \text{ V}$
 $E_{\text{cell}}(\text{quinohydrone}|\text{calomel}) = 0.279 \text{ V}$
 $\therefore \text{quinohydrone} = 0.489 \text{ V}$ EXERCISE II (JEE ADVANCED)
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Problem 470 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q13]
Anode: $\text{Ag(s)} \rightarrow \text{Ag}^+(\text{aq}) + \text{e}^-$
Cathode:
 $\text{Cr}^{3+}(\text{aq}) + \text{H}_2\text{O(l)} \rightarrow \text{CrO}_4^{2-}(\text{aq}) + 7\text{H}^+(\text{aq}) + 3\text{e}^-$
Net:
 $6\text{Ag(s)} + \text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O(l)} \rightarrow 6\text{Ag}^+(\text{aq}) + \text{CrO}_4^{2-}(\text{aq}) + 14\text{H}^+(\text{aq})$
 $E_{\text{cell}} = 1.33 \text{ V} - 0.80 \text{ V} = 0.53 \text{ V}$
 $\log K = \frac{nE_{\text{cell}}}{0.059} = \frac{6 \times 0.53}{0.059} = 54.24$
 $K = 10^{54.24}$

Problem 471 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q14]
Net cell reaction: $\text{Zn} + \text{Hg}^{2+}(\text{C1M}) \rightarrow \text{Zn}^{2+}(\text{C2M}) + \text{Hg}$
 $E_{\text{cell}} = 0.059 \text{ V}$
 $\log K = \frac{nE_{\text{cell}}}{0.059} = \frac{2 \times 0.059}{0.059} = 2$
 $K = 10^2$

Problem 472 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q15]
 $E_{\text{cell}} = 0.06 \text{ V}$
 $\log K = \frac{nE_{\text{cell}}}{0.059} = \frac{2 \times 0.06}{0.059} = 2.03$
 $K = 10^{2.03}$

Problem 473 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q16]
 $\text{H}^+(\text{C1M}) + \text{H}_2\text{O(l)} \rightleftharpoons \text{H}_3\text{O}^+(\text{C2M})$
 $K = 10^{-14}$
 $\text{H}^+(\text{C1M}) + \text{H}_2\text{O(l)} \rightleftharpoons \text{H}_3\text{O}^+(\text{C2M})$
 $K = 10^{-14}$
Net cell reaction, assuming as concentration cell:
 $\text{H}^+(\text{C2M}) + \text{H}_2\text{O(l)} \rightleftharpoons \text{H}^+(\text{C1M}) + \text{H}_2\text{O(l)}$
 $E_{\text{cell}} = 0$
 $\log K = \frac{nE_{\text{cell}}}{0.059} = 0$
 $K = 1$

Problem 474 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q17]
 $E_{\text{V V V}} / \text{circ} = / \text{times} + / \text{times} - / \text{times} = + + 2 \ 3 \ 1 \ 0 \ 616 \ 1 \ 0 \ 439 \ 1 \ 0 \ 799 \ 0 \ 256 \mid$
 $E_{\text{V V V}} / \text{circ} = - + + 3 \ 2 \ 0 \ 256 \mid .$

Problem 475 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q18]
 $E_{\text{E n H P cell cell H}} = / \text{circ} -$
 $+ 0 \ 06 \ 2 \ 2 . \log []$
 or, $0 \ 70 \ 0 \ 28 \ 0 \ 0 \ 06 \ 2 \ 1 \ 2 . (.) . \log [] = - -$
 $+ \text{H} / \text{implies PH} = 7.0$

Problem 476 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q19]
 Net cell reaction:
 $\text{Ag C M) Ag C K M sp} + + = / \text{to}$
 $=$
 $(. / 1 \ 2 \ 1 \ 3 \ 0 \ 1 \ 2 \ 4$
 Now, $E_{\text{C C cell}} = -$
 $0 \ 0 \ 06 \ 1 \ 2 \ 1 . \log$
 $/ \text{implies } 0.162 = -$
 $0 \ 06 \ 2 \ 0 \ 1 \ 1 \ 3 . \log (.) . / K_{\text{sp}}$
 $K_{\text{sp}} = 4 / \text{times } 10^{-12}$

Problem 477 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q20]
 $E_{\text{E V Tl Tl Pb Pb}} + + - = - \mid \mid . 2 \ 0 \ 444$
 or, $E_{\text{E Pb Tl Tl Tl Pb Pb}} + + - (.) -$
 $= - + + \mid \mid . \log [] [] . 2 \ 0 \ 06 \ 2 \ 0 \ 444 \ 2 \ 2$
 or, $[(.) (.)] . \log . . . - - -$
 $= - 0 \ 336 \ 0 \ 126 \ 0 \ 03 \ 0 \ 1 \ 0 \ 1 \ 0 \ 444 \ 2 \ K_{\text{sp}}$
 $K_{\text{sp}} = 4 / \text{times } 10^{-6}$

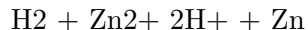
Problem 478 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q21]
 $E_{\text{cell}} = E_{\text{Cu}} - E_{\text{Zn}} = (E_{\text{Cu}} - E_{\text{calomel}}) - (E_{\text{Zn}} - E_{\text{calomel}})$
 From question $E_{\text{calomel}} - E_{\text{Zn}} = 1.083 \text{ V}$
 and $E_{\text{calomel}} - E_{\text{Cu}} = - 0.018 \text{ V}$

Problem 479 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q22]
 The potential of hydrogen electrode at $\text{H}_2(1 \text{ bar})$ may be expressed as $E = - 0.059 \text{ PH}$
 Now, $E_{\text{PKa } 1 \ 0 \ 059} = - +$
 $. \log x y$
 and $E_{\text{PKa } 2 \ 0 \ 059} = - +$
 $. \log y x$
 $\text{P E E) Ka } 1 \ 2 = - + (.0 \ 118$
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Problem 480 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q23]
 For the given reaction, $E_{\text{E E E cell given hydrogen}} / \text{circ} = / \text{circ} - / \text{circ}$
 $(-0.84) - 0 =$
 $0 \ 06 \ 1 . \log K_{\text{eq}} / \text{implies } K_{\text{eq}} = 10^{-14}$

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q24]

Net cell reaction may be written as



E E n H P cell cell H = /circ -

$$+ + 0.06222 \cdot \log [\text{Zn}]$$

or, $(\cdot) (\cdot) \cdot \log [\cdot] \cdot = \cdot$

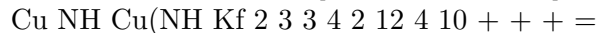
/times + 0 61 0 76 0 06 2 1 0 4 2 H

$$[\text{H}^+] = 2 \times 10^{-3} \text{ M}$$

Now, $K_H SO_H SO_{a2} 3 2 3 3 2 2 10 6 4 10 0 4 = = /times /times /times + - - - [] [] () (.)$.

$$= 3.2 \text{ } /times 10^{-4}$$

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q25]

 $)$,

1.0 M excess

100/

Equ. x 2.0 M 1.0 M

10 1 0 2 0 12 4 = /times . (.) x /implies x = = /times - - 10 16 6 25 10 12 14 .

Now, cell reaction: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$

E E Zn Cu cell cell = /circ -

$$+ + 0 \text{ } 06 \text{ } 2 \text{ } 2 \text{ } 2 . \log [] []$$

$$\equiv - -$$

$$/times = - \left[. \left(. \right) \right] . \log . . . 0 \ 34 \ 0 \ 76 \ 0 \ 06 \ 2 \ 1 \ 0 \ 6 \ 25 \ 10 \ 0 \ 704 \ 14 \ \text{V}$$

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q26]

Net cell reaction: $\text{Zn} + 2\text{H}^+ \rightarrow \text{Zn}^{2+} + \text{H}_2$

E E Zn P H cell cell H = /circ -

$$+ + 0.06222 \cdot \log [] []$$

or, $0.70 \ 0.076 \ 0.062 \ 0.011 \ 2. \cdot \left[\begin{pmatrix} . \\ . \end{pmatrix} \right] \cdot \log \cdot \left[\begin{pmatrix} . \\ . \end{pmatrix} \right] = -$

$$/times + \mathbb{H}$$

/implies [H+] = 0.01 M

Moles of HCl in RHS = $500 \times 0.01 \times 1000 \times 5 \times 10^{-3} \times \text{times} = \text{times} \times$.

Mass of NaOH needed = $5 \times 10^{-3} \times 40 = 0.2$ gm

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q27]

On assuming concentration cell, the net cell reaction is $\text{Ag}^+ (\text{C1M, Right}) \rightarrow \text{Ag}^+ (\text{C2M, left})$

Now, C M 1 0 1 40 100 0 04 = /times = . .

and C K Cl K K M sp sp sp 2 0 1 50 100 0 05 = = /times = - [] . .

Now, E C C cell = - 0 0 06 1 2 1 . log

or, $0.42 \pm 0.06 \pm 0.05 \pm 0.04 \dots \log / \dots = -K_{sp} / \text{implies } K_{sp} = 2 \times 10^{-10}$

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q28]

$$\Delta G_{\text{cell}} = -nFE_{\text{cell}} \implies -965 \times 3 \times 103 = -12 \times 96500 \times E_{\text{cell}}$$

$$E_{\text{cell}} = 2.5 \text{ V}$$

Problem 486 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q29]

Theoretical efficiency = $-\Delta G^\circ / \Delta G$ $\implies 0.84 \times 285 = -\Delta G^\circ / \Delta G$
 $\Delta G^\circ / \Delta G = -0.84 \times 285 \text{ KJ} = -nF E^\circ_{\text{cell}}$
 or, $0.84 \times 285 \times 10^3 = 2 \times 96500 \times E^\circ_{\text{cell}}$
 $\implies E^\circ_{\text{cell}} = 1.24 \text{ V}$

Problem 487 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q30]

Net cell reaction: $\text{Ag(s)} + \text{H}^+ + \text{Cl}^- \rightarrow \text{AgCl(s)} + \frac{1}{2} \text{H}_2(\text{g})$ But for E_{cell} calculation, reaction may be written as



Now, $E^\circ_{\text{Ag}^+/\text{Ag}} - E^\circ_{\text{H}^+/\text{H}_2} = E^\circ_{\text{cell}}$

$$+0.061 \text{ V} - 0 \text{ V} = E^\circ_{\text{cell}}$$

$E^\circ_{\text{cell}} = 0.061 \text{ V}$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln \frac{[\text{Ag}^+]}{[\text{H}^+]^{1/2} [\text{Cl}^-]}$$

It means that actual reaction is in reverse direction and $E_{\text{cell}} = 0.32 \text{ V}$

Problem 488 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q32]

)

$aM^a + bM^b \rightleftharpoons cM^c$

Eqn. $aM^a + bM^b \rightleftharpoons cM^c$

$K = \frac{[\text{Ag(NH}_3\text{)}^+]}{[\text{Ag}^+][\text{NH}_3]}$

$$+ \ln \frac{[\text{Ag}^+]}{[\text{Ag(NH}_3\text{)}^+]} = - \ln K$$

$a \ln \frac{[\text{Ag}^+]}{[\text{Ag(NH}_3\text{)}^+]} = - \ln K$

Cell reaction: $\text{Ag}^+ (\text{Right}) \rightleftharpoons \text{Ag}^+ (\text{Left})$

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Problem 489 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q33]

$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q$

$$= 0.061 \text{ V} - \frac{0.059}{2} \log \frac{[\text{Cu}^+][\text{Cu}^{2+}]}{[\text{Cu}^+][\text{Cu}^{2+}]}$$

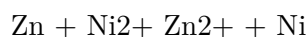
Now, $E_{\text{cell}} = 0.061 \text{ V} - \frac{0.059}{2} \log \frac{[\text{Cu}^+][\text{Cu}^{2+}]}{[\text{Cu}^+][\text{Cu}^{2+}]}$

(1)

$$\text{and } E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q = 0.061 \text{ V} - \frac{0.059}{2} \log \frac{[\text{Cu}^+][\text{Cu}^{2+}]}{[\text{Cu}^+][\text{Cu}^{2+}]} = 0.061 \text{ V} - \frac{0.059}{2} \log \frac{[\text{Cu}^+][\text{Cu}^{2+}]}{[\text{Cu}^+][\text{Cu}^{2+}]}$$

$$\text{or, } E_{\text{cell}} = 0.061 \text{ V} - \frac{0.059}{2} \log \frac{[\text{Cu}^+][\text{Cu}^{2+}]}{[\text{Cu}^+][\text{Cu}^{2+}]} = 0.061 \text{ V} - \frac{0.059}{2} \log \frac{[\text{Cu}^+][\text{Cu}^{2+}]}{[\text{Cu}^+][\text{Cu}^{2+}]}$$

From (1) and (2), $E_{\text{cell}} = 0.05242 \text{ V}$

Problem 490 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q34]

$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q$

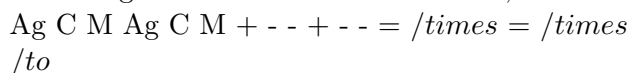
$$\implies (-0.24) - (-0.75) = 0.03 \log K_{\text{eq}}$$

$K_{\text{eq}} = 10^{17} \implies$ It means that Ni^{2+} will react almost completely and $[\text{Zn}^{2+}] \approx 1.0 \text{ M}$

Now, $10^{-17} = \frac{[\text{Ni}]}{[\text{Ni}^{2+}]}$

Problem 491 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q35]

Assuming the cell as concentration cell, the cell reaction may be written as



$$= \quad / \text{times} = 1 \quad 13 \quad 10 \quad 2 \quad 10 \quad 9 \quad 4 \quad 10 \quad 0 \quad 001 \quad 4 \quad 10 \quad 2 \quad 10 \quad 0 \quad 2 \quad 10 \quad . \quad .$$

Now, $E_{\text{cell}} = -$

$$/ \text{times} = - - - 0 \quad 0 \quad 06 \quad 1 \quad 10 \quad 4 \quad 10 \quad 0 \quad 024 \quad 9 \quad 10 \quad . \quad \log .$$

Problem 492 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q36]

$$0 \quad 03 \quad 0 \quad 06 \quad 2 \quad 0 \quad 3 \quad 0 \quad 5 \quad 2 \quad 2 \quad . \quad . \quad \log [] [] . \quad \log . [] \text{ lower} =$$

=

+ + + Cu Cu Cu higher lower

$$[\text{Cu}^{2+}]_{\text{lower}} = 0.05 \text{ M}$$

Problem 493 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q37]

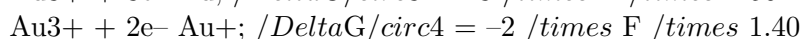
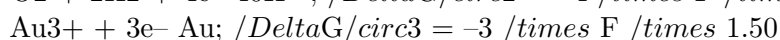
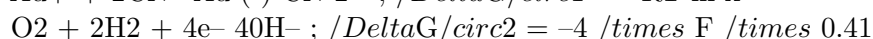
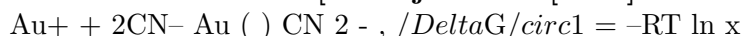
Cell reaction: $\text{H}^+(\text{C}_1, \text{HA}_1) \mid \text{H}^+(\text{C}_2, \text{HA}_2)$

$$E_{\text{cell}} = -$$

=

$$0 \quad 0 \quad 059 \quad 1 \quad 0 \quad 059 \quad 2 \quad 1 \quad 2 \quad 1 \quad . \quad \log . \quad \log$$

$$= - = 0 \quad 059 \quad 2 \quad 0 \quad 059 \quad 1 \quad 2 \quad . \quad () . \quad V \quad P \quad P \quad K \quad K \quad a \quad a$$

Problem 494 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q38]

$$\text{From } \Delta G^\circ_1 + \Delta G^\circ_2 - \Delta G^\circ_3 + \Delta G^\circ_4 = - + G \quad G \quad G \quad G \quad G$$

RT

$$F \text{ required } 1 \quad 2 \quad 3 \quad 4 \quad 1 \quad 4 \quad 1 \quad 29 \quad , \quad \ln . \quad x$$

Problem 495 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q40]

PH of right electrode will increase due to formation of OH^- ion.

Problem 496 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q43]

At low $[\text{Cl}^-]$, O_2 becomes anode product

Problem 497 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q46]

In the electrolysis of aq. KNO_3 , neither K^+ are NO_3^- participate in electrode reaction.

Problem 498 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q47]

Na will react with water. S will not conduct electricity.

Problem 499 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q48]

$$w \quad E \quad Q \quad F = \quad / \text{implies} \quad 0 \quad 635 \quad 63 \quad 5 \quad 2 \quad 0 \quad 965 \quad 3600 \quad 96500 \quad . \quad . \quad . \quad / \text{times} = \quad / \text{times} \quad / \text{times} \quad i$$

$$/ \text{implies} \quad i = 2 \quad 3 \quad 6 \quad . \quad A$$

$$/ . \quad . \quad . \quad / \text{error} = - = 2 \quad 3 \quad 6 \quad 0 \quad 5 \quad 2 \quad 3 \quad 6 \quad 10$$

Problem 500 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q49]

Detonating gas is mixture of H_2 and O_2

$n_{eq} Sn = n_{eq} H_2 = n_{eq} O_2 \implies 1 \times 2 \times 120 = 2 \times 4 \times 2 \times n \implies n = 15$
 n and $n_{H_2} = 15$ and $n_{O_2} = 15$

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Problem 501 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q50]

$n_{eq} \text{ of LiOH} = Q/F \implies w = 24 \times 1.25 \times 4825 / 96500 \implies w = 12.48$

Problem 502 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q51]

$n_{eq} \text{ Cu deposited at cathode} = Q/F$

or, $w = 63.5 \times 2 \times 12.4 \times 4825 / 96500 \implies w = 19.685 \text{ gm}$

But the increase in mass of cathode is only 19.05 gm It represents that 20 gm of sample contains only 19.05 gm Cu.

$20/100 = 19.05/25 \implies$

Now, $Q/F \times n_{Cu} = n_{Fe} \times e_{eq} \implies$ (oxidised at anode)

or, $12.4 \times 4825 / 96500 \times 2 = 19.05 / 56 \times 2 \implies w = 0.56 \text{ gm}$

Percentage of Fe = $0.56/20 \times 100 = 2.8\%$

Problem 503 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q52]

Theoretical n_{eq} of NaOH formed = $n_{eq} Cu$

$= 3 \times 18 \times 63.6 \times 2 \times 0.1 \implies$

Actual n_{eq} of NaOH formed = $60 \times 1 \times 1000 \times 0.06 \implies$

Percentage yield = $0.06/0.1 \times 100 = 60\%$

Problem 504 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q53]

Cell reaction during discharge:

$Pb + PbO_2 + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O$

$n_{Pb} \text{ taken} = 200/208$ and $n_{PbO} \text{ taken} = 200/240 \implies$

Hence, PbO_2 is L.R.

Now, $n_{eq} PbO_2 \times Q/F = 200/240 \times 2 \times 10 \times 96500 \implies$

$\implies t = 16083.33 \text{ sec}$

Problem 505 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q55]

Q/F of $I \times Na \times S \times O \times e_{eq} = 2 \times 2 \times 3 \times n \implies$

or, $i \times t / 96500 = 2 \times 3600 \times 96500 / 72 \times 10 \times 1000 \implies i = 0.965 \text{ A}$

Problem 506 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q56]

$C_{14}H_{10} + 2H_2O \rightarrow C_{14}H_8O_2 + 6H^+ + 6e^-$

$n_{eq} C_{14}H_8O_2 = Q/F \implies w = 208 \times 6 \times 1 \times 40 \times 60 \times 0.965 / 96500 \implies$

$w = 0.832 \text{ gm}$

Problem 507 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q57]

Number of coulombs required = $1000 \times 0.00033 \times 1000 \text{ per Kg Cu}$

Energy required = $1000 \times 0.00033 \times 0.33 \times 106 \text{ J}$

Cost of electricity

$= 4 \times 10 \times 3600 \times 10 \times 1.11 \times 3.6 \text{ Rupee}$

Problem 508 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q58]

neq CH_3COO^- oxidised = Q F

or, $n \times \text{times} = \frac{\text{times}}{\text{times}} \times \frac{\text{times}}{\text{times}} \times \frac{1}{0.54825600896500} \dots \implies n = 0.12$

Moles of $(\text{C}_2\text{H}_6 + \text{CO}_2)$ produced

= $\frac{32012018}{\text{times}} = \dots$

and total volume = $0.18 \times 22.4 = 4.032 \text{ L}$

Problem 509 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q59]

$\Delta G^\circ =$

= $E^\circ \times 0.0592 \times 10 \times 0.1776 \times \log \dots$

Problem 510 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q60]

Back EMF =

= $\frac{\text{times}}{\text{times}} - 0.06201200854103 \times \log \dots \times V$

Problem 511 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q61]

Equivalent of charge used

= $\frac{0.482510360096500018 \dots \times \text{times}}{\text{times}} = F$

Cell reaction during charge:

$\text{Cu} \mid \text{Zn} \mid \text{Cu} \mid \text{Zn} + \frac{\text{to}}{2}$

$+ + + 22$

$\frac{1001100001}{\text{times}} = \dots \text{ mole}$ $\frac{1001100001}{\text{times}} = \dots \text{ mole}$

= $0.2 \text{ eq} = 0.2 \text{ eq}$

Final $0.2 - 0.18 = 0.2 + 0.18$

= $0.02 \text{ eq} = 0.38 \text{ eq}$

= $0.01 \text{ mole} = 0.19 \text{ mole}$

Final $[\dots] \dots \text{Zn} \mid \text{M} \mid \frac{20011001000001}{\text{times}} + = \frac{\text{times}}{\text{times}} =$

and $[\dots] \dots \text{Cu} \mid \text{M} \mid \frac{20191001000019}{\text{times}} + + \frac{\text{times}}{\text{times}} =$

Now, cell reaction as galvanic cell:

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Problem 512 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q62]

Reactions involved are

$2\text{H}_2\text{O} \xrightarrow{\text{electrolysis}}$

$\frac{\text{to}}{2}$

+

$\text{N}_2 + 3\text{H}_2 \xrightarrow{2\text{NH}_3}$

$\text{NH}_3 + 2\text{O}_2 \xrightarrow{\text{HNO}_3 + \text{H}_2\text{O}}$

$\text{NH}_3 + \text{HNO}_3 \xrightarrow{\text{NH}_4\text{NO}_3}$

For 1 mole NH_4NO_3 , 3 moles of H_2O should be electrolyzed. Hence for 1152 Kg NH_4NO_3 , moles of H_2 needed = $\frac{\text{times}}{\text{times}} \times \frac{\text{times}}{\text{times}} = \frac{\text{times}}{\text{times}} \times \frac{3115210804321034}{\text{times}} \dots$

Now, $\text{neqH} \times Q \times F \times 2 = \frac{\text{times}}{\text{times}} \times \frac{4.32 \times \text{times}}{104 \times \text{times}} \times 2 = i \times \frac{\text{times}}{\text{times}} \times \frac{\text{times}}{\text{times}} \times \frac{24360096500}{\text{times}}$

$i = 96500 \text{ A/day}$

Problem 513 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q63]

neq $\text{HC}_3\text{H}_5\text{O}_3 = n\text{OH} \times Q \times F =$

or, $w \times \frac{90150101158965003}{\text{times}} = \frac{\text{times}}{\text{times}} \times \frac{\text{times}}{\text{times}} \times \frac{\text{times}}{\text{times}} \implies w = 0.054 \text{ gm}$

$\frac{110054 \dots}{\text{times}} =$

Problem 514 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q64]

no. of H₂O formed = $\frac{2 \times 4 \times 2 \times 8}{2} = 32$

and no. of NH₃ = $\frac{4 \times 2 \times 8}{2} = 16$

$\therefore$ mass of H₂O = $\frac{32 \times 18}{1000} = 0.576$ g

mass of NH₃ = $\frac{16 \times 17}{1000} = 0.272$ g

+

mass of H₂O = 0.576 g

Problem 515 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q65]

no. of As₂O₃ = 3 and no. of H₃AsO₃ = 3

or, $\frac{3 \times 198}{1000} = \frac{3 \times 100}{1000} \times \frac{w}{1000} \therefore w = 6.3$ g

Problem 516 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q66]

no. of e⁻ = $\frac{52 \times 2 \times 87 \times 2 \times 19 \times 3 \times 3600 \times 96500}{1000} = 1.8333 \times 10^6$

$\therefore$ $\frac{1.8333 \times 10^6}{2} = 916650$ e⁻

Problem 517 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q67]

no. of Cu²⁺ reduced = $\frac{Q}{2F} = \frac{2 \times 10 \times 19 \times 3 \times 60 \times 96500 \times 3}{2 \times 96500} = 360$

$\therefore$ $\frac{360 \times 160}{1000} = 57.6$ g

or, $\frac{360 \times 160}{1000} = 57.6$ g

Problem 518 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q68]

no. of e⁻ = $\frac{12 \times 3 \times 123 \times 6 \times 1000}{1000} = 22320$ e⁻

and Energy consumed = $\frac{22320 \times 3.0}{1000} = 66.96$ kJ

Problem 519 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q69]

no. of H₂ (at cathode) = no. of O₂ + no. of H₂S₂O₈ (at anode)

or, $\frac{9 \times 0.8 \times 22 \times 7 \times 2 \times 27 \times 22 \times 7 \times 4 \times 194 \times 2 \times 1000}{1000} = \frac{9 \times 0.8 \times 22 \times 7 \times 2 \times 27 \times 22 \times 7 \times 4 \times 194 \times 2 \times 1000}{1000} + \frac{9 \times 0.8 \times 22 \times 7 \times 2 \times 27 \times 22 \times 7 \times 4 \times 194 \times 2 \times 1000}{1000} \times \frac{w}{1000}$

$\therefore w = 38.8$ g

Problem 520 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q70]

Initial mass of H₂SO₄,

$w_1 = \frac{3600 \times 1.5 \times 100 \times 2160}{1000} = 11664$ g

Final mass of H₂SO₄,

$w_2 = \frac{3600 \times 1.1 \times 100 \times 396}{1000} = 14793.6$ g

Moles of H₂SO₄ consumed = $\frac{w_1 - w_2}{98} = \frac{11664 - 14793.6}{98} = -134.65$

Now, no. of H₂SO₄ = $\frac{Q}{2F} = \frac{18 \times 10^6}{2 \times 96500} = 9233.27$

Number of ampere-hr = 482.5

Problem 521 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q71]

$\frac{m}{m} \text{NaCl KCl NaCl NaCl KCl NaCl KCl} / \text{C} / \text{C} \text{ R G C R G} () () () (/ \kappa / \kappa 1 1 1$
 $=$
 $1 \text{ C R C}) \text{ R C}) \text{ KCl KCl NaCl} (($
 or, $= \frac{1}{\text{times}} \frac{1}{\text{times}} m \text{NaCl} (. . 120 200 0 1 6400 0 003$
 $\implies m(\text{NaCl}) = 125 \text{ } \Omega^{-1} \text{ cm}^{-1} \text{ mol}^{-1} 2 2 4 2 2 8 2 \text{ SO S O e } - - - / \text{to} +$
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Problem 522 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q72]

$\frac{1}{\text{circ}} = \frac{1}{\text{times}} \frac{1}{\text{circ}} + + - m m \text{NH CrO NH CrO} [()] () () 4 2 4 4 4 2 2 \frac{1}{\lambda} \frac{1}{\lambda}$
 $= (2 \frac{1}{\text{times}} 6.6 \frac{1}{\text{times}} 10^{-8} + 5.4 \frac{1}{\text{times}} 10^{-8}) \frac{1}{\text{times}} 96500$
 $= 0.01795 \text{ } \Omega^{-1} \text{ m}^2 \text{ mol}^{-1}$

Problem 523 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q73]

$\frac{1}{\text{circ}} = m m \text{C} / \alpha / \kappa$
 or, $0.9 \frac{1}{\text{times}} 4.25 \frac{1}{\text{times}} 10^{-2} = 382 5 3 . \text{C} 10 \frac{1}{\text{times}} \implies \text{C} = 0.1 \text{ M}$

Problem 524 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q74]

$= m \text{K C} \implies 1.5 \frac{1}{\text{times}} 10^{-2} = 3 06 10 2 56 10 3 3 . . \frac{1}{\text{times}} - \frac{1}{\text{times}} - - \text{C}$
 $\text{C mol m mol l V} = \frac{1}{\text{times}}$
 $= - - - 1 30 1 30 10 585 58 5 3 3 1 / .$
 $\text{V} = 3 \frac{1}{\text{times}} 105 \text{ L}$

Problem 525 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q75]

$=$
 $=$
 $m \text{C G G C} / \kappa \implies 100 1 0 5 1 5 0 1 10 3 = \frac{1}{\text{times}} \frac{1}{\text{times}} - \text{R} . . .$
 $\implies \text{R} / \Omega = 100 3$
 Now, $\text{V} = \text{IR} \implies \text{I V R A} = = = 5 100 3 0 15 / .$

Problem 526 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q76]

Ionic mobility, $\frac{1}{\mu} \frac{1}{\lambda} \frac{1}{\text{circ}} = = \frac{1}{\text{circ}} \text{ speed of ion Pot. gradient F m}$
 or, speed $19 3 5 50 96500 .$
 $= \implies \text{speed} = 2 \frac{1}{\text{times}} 10^{-3} \text{ cm/s}$

Problem 527 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q77]

$\text{eq} = \text{C} / \kappa \implies 150 3 4 10 1 6 10 5 6 6 \frac{1}{\text{times}} \frac{1}{\text{times}} - \frac{1}{\text{times}} - - . .$
 $\text{S} = 1.2 \frac{1}{\text{times}} 10^{-8} \text{ mol cm}^{-3} = 1.2 \frac{1}{\text{times}} 10^{-5} \text{ M}$

Problem 528 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q78]

$\text{K} = \text{G G}^* \implies \frac{1}{\kappa} 1 4 1 280 1 50 . / / =$
 $\implies \frac{1}{\kappa} = 0.25 \text{ s m}^{-1} \text{ for } 0.5 \text{ M}$
 Now, $= = \frac{1}{\text{times}} = \frac{1}{\text{times}} - - m \text{C s m mol} / \kappa 0 25 0 5 10 5 10 3 4 2 1 . .$

Problem 529

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q79]

Ag A S Ag A () ()M

$$+ + - + \text{ x y xM}$$

Ag B S Ag B M () ()M

$$+ + - + x y y$$

Now, $(x + y) x = 3$ /times 10-14 and

$$(x + y) y = 1 \text{ /times } 10^{-14}$$

[Ag]₇, [A]₇ M; [B]₇ M. + - - - - = + = /times = = /times = = /times x y x y 2 10 1 5 10 0 5 10 7

Now, $/kappasolution = /times = /times /times /times + /times /times /times + /times - - -$
 $- 3\ 75\ 10\ 2\ 10\ 10\ 60\ 1\ 5\ 10\ 10\ 80\ 0\ 5\ 10\ 8\ 7\ 3\ 7\ 3\ .\ .\ . - - /circ /times\ 7\ 3\ 10 /lambda\ B$
 $/lambda B /circ - - = 135 /Omega\ cm\ mol\ 1\ 2\ 1$

Problem 530

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q80]

$$/circ = /circ + /circ - /circ \text{ eq eq eq eq [Be (Po)] [BeCl] [K Po] [K] } 3 \ 4 \ 2 \ 2 \ 3 \ 4 \ Cl$$
$$= + - = - - 160 \ 140 \ 100 \ 200 \ 1 \ 2 \ 1 \ / \Omega \text{ cm eq}$$

Now, $\text{neg} = \text{/implies} = \text{/times} - \text{/kappa}$ C C 200 1 2 10 5 .

/implies C = 6 /times 10-8 eq cm-3

$$= 6 \text{ } /times 10^{-5} \text{ N}$$
$$= 10^{-5} \text{ M}$$

Now, Ksp = $\frac{[times]}{[times]}$ = $\frac{108}{108} \cdot \frac{10}{1} \cdot \frac{08}{10} \cdot \frac{5}{5} \cdot \frac{5}{5} \cdot \frac{23}{S} ()$.

Section B (One or More than one Correct)

Problem 531

[Neeraj] Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q1]

Net cell reaction is spontaneous in electrochemical cell but non-spontaneous in electrolytic cell.
Cathode is +ve in electrochemical cell but -ve in electrolytic cell.

Problem 532

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q2]

For the cell: $\text{Ag(s)} \mid \text{AgCl(s)} \mid \text{Cl}^- \parallel \text{Ag}^+ \mid \text{Ag(s)}$, the net cell reaction is $\text{Ag}^+ + \text{Cl}^- \rightarrow \text{AgCl(s)}$.

Problem 533

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q3]

Left electrode: $\text{Ag(s)} + \text{Cl}^- (\text{aq}) \rightarrow \text{AgCl (s)} + \text{e}^-$ *times 2* Right electrode : $\text{Hg}_2\text{Cl}_2(\text{s}) + 2\text{e}^- \rightarrow 2\text{Hg(l)} + 2\text{Cl}^- (\text{aq})$ Net reaction: $2\text{Ag(s)} + \text{Hg}_2\text{Cl}_2(\text{s}) \rightarrow 2\text{AgCl(s)} + 2\text{Hg (l)}$

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Problem 534

[Neeraj] Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q5]

$\text{Cu} \text{ Mg} \text{ Na} \text{ Al} = \text{ } / \text{times} = \text{ } / \text{times} = \text{ } / \text{times} = \text{ } 63 \text{ } 5 \text{ } 63 \text{ } 5 \text{ } 2 \text{ } 2 \text{ } 24 \text{ } 24 \text{ } 2 \text{ } 2$
 $11 \text{ } 5 \text{ } 23 \text{ } 1 \text{ } 0 \text{ } 5 \text{ } \dots ; \dots ; 9 \text{ } 27 \text{ } 3 \text{ } 1 \text{ } / \text{times} =$

Problem 535

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q7]

$$/circeq = 60 + 80 = 140 \text{ } /Omega-1 \text{ cm}^2 \text{ eq}^{-1}$$
$$/circm = 140 /times 6 = 840 /Omega-1 \text{ cm}^2 \text{ eq}^{-1}$$

Problem 536

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q8]

Salt bridge does not change standard potential of any electrode.

Problem 537 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q9]

Moles of electron involved

$$= 0.25965 \times 3600 \times 96500 \times 0.09 \dots \text{times} =$$

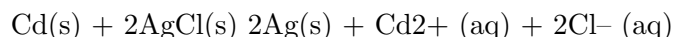
$$\text{Mass of Zn involved} = 0.0926542 \times 943 \dots \text{times} = \text{gm}$$

$$\text{Mass of MnO}_2 \text{ involved} = 0.09 \times 87 = 7.83 \text{ gm}$$

$$\text{Mass of NH}_4^+ \text{ involved} = 0.09 \times 18 = 1.62 \text{ gm}$$

Problem 538 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q10]

Net cell reaction is



$$\Delta G^\circ = -nFE^\circ = -2 \times 96500 \times 0.6115800 \text{ C nFE J}$$

$$\Delta G^\circ = -2 \times 96500 \times 0.7135100 \text{ C nFE J}$$

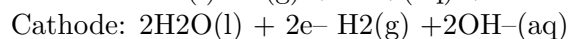
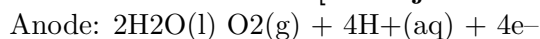
$$\Delta G^\circ =$$

$$= -S T T n F E^\circ \text{ J/K}$$

$$= -2 \times 96500 \times 0.60750386 \dots$$

$$\Delta H^\circ = \Delta G^\circ + T \Delta S^\circ$$

$$= (-135100) + 273 \times (-386) = -240478 \text{ J}$$

Problem 539 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q11]

$$\text{neq H}^+ \text{ produced} = \text{neq OH}^- \text{ produced} = Q/F$$

$$\text{or, } n \times \text{H}^+ \text{ or OH}^- \text{ produced} = \text{times} = 1.12596560 \times 96500 \times 0.75 \dots$$



$$1.0 \text{ M } 0.75 \text{ M } 1.0 \text{ M}$$

$$\text{Final } 0.25 \text{ M } 1.75 \text{ M}$$

$$P \text{ HPO}_4^{2-} \text{ H}_2\text{PO}_4^- \text{ K}^+ = + = + = - - a \log \left[\frac{[\text{H}_2\text{PO}_4^-]}{[\text{HPO}_4^{2-}]} \right] \dots 4.2024215025075130$$



$$1.0 \text{ M } 0.75 \text{ M } 1.0 \text{ M}$$

$$\text{Final } 0.25 \text{ M } 0.75 \text{ M } 1.75 \text{ M}$$

$$P \text{ HPO}_4^{2-} \text{ H}_2\text{PO}_4^- \text{ K}^+ = + = + = - - a \log \left[\frac{[\text{H}_2\text{PO}_4^-]}{[\text{HPO}_4^{2-}]} \right] \dots 4.202421517502530$$

Problem 540 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q12]V, Fe and Hg will be oxidised by NO₃⁻**Problem 541** [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q13]

Resistance and heat capacity depends on quantity.

Problem 542 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q15]

Net cell reaction of discharge is



x mole x mole

Initial mass of H₂SO₄,

$$w_1 = 1000 \times 1.26 \times 40 \times 100 \times 504 = \text{gm}$$

Final mass of H₂SO₄,

$$w_2 = (1260 - 98x + 18x) \times 28 \times 100$$

$$= (352.8 - 22.4x) \text{ gm}$$

From reaction, $504 - 98x = 352.8 - 22.4x$ $\implies x = 2$ Section C (Comprehensions) Comprehension I

Problem 543 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q1]
 $E_{\text{cell}} = E_{\text{H}^+/\text{H}_2} - E_{\text{Zn}^{2+}/\text{Zn}}$ because $E_{\text{H}^+/\text{H}_2}$ was higher
 or, $0.76 = 1.00 - E_{\text{Zn}^{2+}/\text{Zn}}$ $\implies E_{\text{Zn}^{2+}/\text{Zn}} = 0.24 \text{ V}$

Problem 544 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q2]
 $E_{\text{cell}} = E_{\text{Cu}^{2+}/\text{Cu}} - E_{\text{H}^+/\text{H}_2}$ because $E_{\text{Cu}^{2+}/\text{Cu}}$ was higher
 or, $0.34 = E_{\text{Cu}^{2+}/\text{Cu}} - 1.00 \implies E_{\text{Cu}^{2+}/\text{Cu}} = 1.34 \text{ V}$

Problem 545 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q3]
 $E_{\text{cell}} = E_{\text{Cu}^{2+}/\text{Cu}} - E_{\text{Zn}^{2+}/\text{Zn}} = 1.34 - 0.24 = 1.10 \text{ V}$
 Visit www.jeebooks.in for more

Problem 546 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q4]
 $E_{\text{cell}} = E_{\text{Ag}^+/\text{Ag}} - E_{\text{K}^+/\text{K}}$
 $= -0.06 + \frac{0.059}{1} \log \frac{1}{0.06} = +0.314 \text{ V}$

Problem 547 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q5]
 $E_{\text{cell}} = E_{\text{Ag}^+/\text{Ag}} - E_{\text{K}^+/\text{K}}$
 $= -0.06 + \frac{0.059}{1} \log \frac{1}{0.06} = 0.172 \text{ V}$

Problem 548 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q6]
 $E_{\text{cell}} = E_{\text{Ag}^+/\text{Ag}} - E_{\text{I}^-/\text{I}_2}$
 $= 0.06 - \frac{0.059}{2} \log \frac{1}{0.06} = -0.172 - 0.06 \log 0.04 = -0.088 \text{ V}$ Comprehension III

Problem 549 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q7]
 $\Delta G^\circ = -nFE^\circ$
 $\Delta G^\circ = -2 \times 96500 \times 1.27 = -245110 \text{ J}$ Comprehension IV

Problem 550 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q8]
 $E_{\text{cell}} = (0.40) - (-0.87) = 1.27 \text{ V}$

Problem 551 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q9]
 Net cell reaction is independent from OH^- (aq)

Problem 552 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q10]
 $-\Delta G^\circ = nFE_{\text{cell}} = 2 \times 96500 \times 1.27 = 245110 \text{ J}$ Comprehension IV

Problem 553 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q11]
 H_3O^+ is only needed in balancing cathode reaction:
 $\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{NO} + 2\text{H}_2\text{O}$
 $\text{H}_2\text{O} + \text{e}^- \rightarrow \text{OH}^-$

Problem 554 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q12]
 Moles of electron needed = 2 /times moles of HNO₂ formed

Problem 555 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q13]
 $\text{neq HNO}_3 = 2 \times 10^5 \times 1930 \times 10^{-6} = 386 \text{ g}$ Comprehension V

Problem 556 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q14]
 $\Delta G^\circ = -nFE^\circ = -103 \times 96500 \times 1.23 = -123 \text{ kJ mol}^{-1}$

Problem 557 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q15]
 Moles of H₂ needed = $\frac{237.39}{22.7} = 10.46$
 Volume of H₂ needed = $10.46 \times 22.7 = 237.39 \text{ L}$

Problem 558 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q16]
 E_{cell} is independent from [OH⁻]

Problem 559 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q17]
 $\Delta G^\circ = -nFE^\circ = -103 \times 96500 \times 1.23 = -123 \text{ kJ mol}^{-1}$

Problem 560 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q18]
 $\Delta G^\circ = -nFE^\circ = -103 \times 96500 \times 1.23 = -123 \text{ kJ mol}^{-1}$

Problem 561 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q19]
 Refer theory given in passage.

Problem 562 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q20]
 $E^\circ_{\text{Fe}^{2+}/\text{Fe}} = -0.44 \text{ V}$, $E^\circ_{\text{H}^+/\text{H}_2} = 0 \text{ V}$

Problem 563 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q21]
 $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ V}$, $E^\circ_{\text{Pb}^{2+}/\text{Pb}} = -0.13 \text{ V}$ and hence Cu will not oxidise easily.

Problem 564 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q22]
 $\text{neq Fe} = \frac{56}{55.85} \times \frac{10}{100} = 0.102 \text{ g}$

Problem 565 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q23]
 Theoretical Comprehension VII

Problem 566 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q24]
 $n_{\text{eq Ni}} Q F = \frac{w}{\text{implies } w} \frac{58.7}{2} \frac{15}{3600} \frac{0.6}{96500} \dots \text{implies } w = 9.85 \text{ gm}$

Problem 567 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q25]
 $V = A \frac{1}{\text{times } t} \text{implies } 9.85 \frac{8.9}{4} \frac{0.2}{2} \dots \text{implies } t = 0.138 \text{ cm}$
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Problem 568 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q26]
 $n_{\text{eq H}} Q F = \frac{VH_2}{2} \frac{22.4}{2} \frac{15}{3600} \frac{0.4}{96500} \dots \text{implies } VLH_2 = 25 \text{ cm}^3$

Problem 569 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q27]
 Anode produced is only O₂ gas.
 $n_{\text{eq O}} Q F = \frac{w}{\text{implies } w} \frac{8}{15} \frac{3600}{96500} \text{implies } w = 4.477 \text{ gm}$ Comprehension VIII

Problem 570 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q28]
 $E_{\text{cell}} = 0.8 - 0.05 = 0.75 \text{ V}$
 Now, $E_{\text{RT}} n F K_{\text{cell}} = \ln \text{implies } 0.75 \frac{1}{2} \frac{38.92}{92} \dots \ln K = 58.38$

Problem 571 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q29]
 The oxidation reaction of glucose contains H⁺ and
 $E_{\text{H}} = \frac{0.0592}{2} \log \dots \text{implies } E - E_{\text{H}} = 0.0592 \text{ V}$

Problem 572 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q30]
 Standard potential is independent from ammonia concentration. Comprehension IX

Problem 573 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q31]
 Left electrode: $\text{H}_2(\text{g}) \rightarrow 2\text{H}^+(\text{aq}) + 2\text{e}^-$
 Right electrode: $2\text{AgCl}(\text{s}) + 2\text{e}^- \rightarrow 2\text{Ag}(\text{s}) + 2\text{Cl}^-(\text{aq})$
 Net reaction: $\text{H}_2(\text{g}) + 2\text{AgCl}(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{Cl}^-(\text{aq})$

Problem 574 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q32]
 $\Delta G_{\text{cell}} = -nFE_{\text{cell}} = -2 \times 96500 \times 0.23 = -44390 \text{ J at } 15^\circ\text{C}$
 Now, $\Delta H_{\text{cell}} = \Delta G_{\text{cell}} + T \Delta S_{\text{cell}} = (-44390) + 288 \times (-193) = -99974 \text{ J} = -49987 \text{ J/mole AgCl}$

Problem 575 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q33]
 $\Delta S_{\text{cell}} = -193 \text{ J/K} = -96.5 \text{ J/K per mole AgCl}$

Problem 576 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q34]

$$\Delta G^\circ_{298} = \Delta H^\circ - T \Delta S^\circ = (-49987) - 298(-96.5)$$

$$\text{or, } -1 \times 96500 E^\circ = -21230$$

$$\implies E^\circ = 0.22 \text{ V} = E^\circ_{\text{Cl AgCl/Ag}}$$

$$\text{or, } E^\circ_{\text{K Cl AgCl/Ag Ag Ag sp}} = E^\circ + \frac{RT}{nF} \log 0.0581$$

$$\text{or, } 0.220800581 \dots \log = +K_{\text{sp}} \implies K_{\text{sp}} = 1 \times 10^{-10}$$

Hence, Solubility, $S = K_{\text{sp}} = 1 \times 10^{-5}$ Comprehension X

Problem 577 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q35]

Add Ag(s) in both sides of given reaction to get cell reaction. Now, for E°_{cell}

$$\Delta G^\circ = -nFE^\circ \implies [-109] - [77 + (-129)] \times 103$$

$$= -1 \times 96500 E^\circ_{\text{cell}}$$

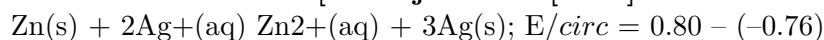
$$E^\circ_{\text{cell}} = 0.59 \text{ V}$$

Problem 578 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q36]

$$E^\circ_{\text{K cell eq}} = 0.059 \log n \implies 0.059 \log 1 =$$

$$K_{\text{sp}}$$

$$K_{\text{sp}} = 10^{-10}$$

Problem 579 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q37]

Now, $E^\circ_{\text{K eq}} =$

$$0.059 \log n$$

$$\implies 1.56059222 \dots \log [\text{Zn}^{2+}] [\text{Ag}] =$$

$$+ + \text{Zn Ag}$$

$$\log [\text{Zn}^{2+}] [\text{Ag}] \cdot \text{Zn Ag}^{2.5288} + + =$$

Problem 580 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q38]

$$\text{Moles of Zn added} = 6.5391065391023 \dots \times 10^{-3} =$$

$$\text{Moles of Ag}^+ \text{ present} = 10.10001001056 \dots \times 10^{-3} = (\text{L.R.})$$

$$\text{Moles of Ag precipitated} = 10^{-6}$$

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Problem 581 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q39]

For KCl: $\gamma_{\pm} =$

$$\frac{m}{m^\circ} \gamma_{\pm} \implies \gamma^\circ = \frac{m}{m^\circ}$$

$$\text{or, } \gamma^\circ = \frac{m}{m^\circ} \gamma = 200 \times (0.02 \times 10^{-3}) \times 100 = 0.4 \text{ cm}^{-1}$$

Problem 582 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q40]

$$\frac{1}{\rho} = \frac{V_m}{M} \implies \rho = \frac{M}{V_m}$$

$$= \frac{M}{V_m} = \frac{M}{\Omega} \implies \Omega = \frac{M}{\rho}$$

$$= 1.1000004410511 \dots$$

Problem 583 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q41]For NaCl: $\kappa =$ $\kappa = \frac{1000 \times C}{\rho}$ or, $125 \times 1000 \times 1 = \frac{1000 \times 0.4585}{\rho}$ $\rho = 3.668 \text{ g cm}^{-3}$ $\rho = 1.25 \text{ g cm}^{-3}$ $\rho = 1.25 \times 10^3 \text{ kg m}^{-3}$ $\rho = 1.25 \times 10^3 \text{ kg m}^{-3}$ Comprehension XII**Problem 584** [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q42] $\frac{E_{\text{cell}}}{nF} > \frac{E_{\text{cell}}}{nF} - \frac{RT}{nF} \ln \frac{1}{I}$ **Problem 585** [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q43] $E_{\text{cell}} = E_{\text{MnO}_4^-/\text{Mn}^{2+}} - E_{\text{H}^+/\text{H}_2} = 1.51 \text{ V} - 0 = 1.51 \text{ V}$ Comprehension XIII**Problem 586** [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q44]Net reaction: $\text{M}^{2+} + \text{M}^{3+} \rightleftharpoons \text{M}^{3+} + \text{M}^{2+}$

Higher Conc. Lower Conc.

 $E_{\text{cell}} > 0, \Delta G_{\text{cell}} < 0$ **Problem 587** [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q45] $70 \text{ mV} = \frac{RT}{F} \ln \frac{1}{0.05}$

(1)

 $E_{\text{cell}} = \frac{RT}{F} \ln \frac{1}{0.05}$ $E_{\text{cell}} = 0.059 \text{ V}$ $\ln \frac{1}{0.05} = 2.9957$

(2)

and $E_{\text{cell}} = 0$. Hence, $E_{\text{req}} = 140 \text{ mV}$ Comprehension XIV**Problem 588** [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q46] $\Delta G_{\text{cell}} = -nFE_{\text{cell}} = -2 \times 96500 \times 0.059 = -11387 \text{ J}$ **Problem 589** [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q47] $E_{\text{cell}} = E_{\text{Left}} - E_{\text{Right}} = 0.059 \text{ V} - 0.059 \text{ V} = 0$ $0.059 \text{ V} = 0.059 \text{ V} \cdot \log \frac{1}{K_{\text{sp}}}$ or, $0.059 \text{ V} = 0.059 \text{ V} \cdot \log \frac{1}{K_{\text{sp}}}$ $K_{\text{sp}} = 4 \times 10^{-15}$ $K_{\text{sp}} = 4 \times 10^{-15}$ Comprehension XV**Problem 590** [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q48] $\text{H}^+ + \text{Cl}^- + (\text{NaOH}) \rightarrow \text{Na}^+ + \text{Cl}^- + \text{H}_2\text{O}$ Conductance first decreases and H^+ ions are replaced by Na^+ ions. After equivalent point, conductance increases due to increase in number of ions.

Problem 591 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q49]
 $\text{CH}_3\text{COOH} + (\text{NaOH added}) \text{CH}_3\text{COO}^- + \text{Na}^+ + \text{H}_2\text{O}$

As number of ions increases, conductance increases. slight decrease initially was due to some dissociated CH_3COOH . After equivalence point, in place of CH_3COO^- ion, number of OH^- ions increases and hence slope becomes greater.

Problem 592 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q50]
 $\text{H}^+ + \text{Cl}^- + (\text{NH}_4\text{OH added})$
 $\text{NH}_4^+ \text{Cl}^- + \text{H}_2\text{O} + \text{OH}^-$

As H^+ ions are replaced by NH_4^+ ions, conductance decreases. After equivalent point, it become almost constant as the dissociation of added NH_4OH will be suppressed in presence of NH_4^+ ions.

Problem 593 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q51]
 HCl will neutralize first followed by CH_3COOH .

Problem 594 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q52]
 Ionic mobilities of Ag^+ and K^+ ions do not differ largely

Problem 595 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q53]
 First number of ions increases and then remain almost constant. added
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Problem 596 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q1]
 $\text{CuCl}_2 \rightarrow \text{Cu}^{2+} + 2\text{Cl}^-$

Problem 597 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q2]
 $E^\circ_{\text{Zn}^{2+}/\text{Zn}} < E^\circ_{\text{Ag}^+/\text{Ag}} < E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}}$

Problem 598 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q3]
 Reason is different charges on ions.

Problem 599 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q5]
 Negative reduction potential means greater tendency to get oxidised.

Problem 600 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q6]
 Number of ions increases considerably only for weak electrolytes.

Problem 601 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q9]
 It is due to very high over voltage potential of hydrogen at mercury cathode.

Problem 602 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q12]
 $E = E^\circ - \frac{RT}{nF} \ln Q$
 $E = E^\circ - \frac{RT}{nF} \log Q$

Problem 603 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q13]
 Cl^- will combine with Ag^+ to precipitate AgCl .

Problem 604 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q14]
 At Cathode:
 $2\text{H}^+(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$
 At Anode:
 $2\text{C}_2\text{H}_3\text{COO}^-(\text{aq}) \rightarrow \text{C}_2\text{H}_4(\text{g}) + 2\text{CO}_2(\text{g}) + 2\text{e}^-$

Problem 605 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q15]
 Theoretical Section E (Column Match)

Problem 606 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q3]
 $\text{E} = \text{E}^\circ - \frac{0.059}{n} \log \frac{[\text{Fe}^{2+}][\text{Fe}^{3+}]}{[\text{Fe}^{2+}][\text{Fe}^{3+}]}$
 $\text{E} = \text{E}^\circ - \frac{0.059}{n} \log \frac{[\text{Fe}^{2+}][\text{Fe}^{3+}]}{[\text{Fe}^{2+}][\text{Fe}^{3+}]}$
 $\text{E} = \text{E}^\circ - \frac{0.059}{n} \log \frac{[\text{Fe}^{2+}][\text{Fe}^{3+}]}{[\text{Fe}^{2+}][\text{Fe}^{3+}]}$
 $\text{E} = \text{E}^\circ - \frac{0.059}{n} \log \frac{[\text{Fe}^{2+}][\text{Fe}^{3+}]}{[\text{Fe}^{2+}][\text{Fe}^{3+}]}$

Problem 607 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q4]
 For concentration cell, both half cell must have same concentration.
 P. $\text{Ag} | \text{AgCl} | \text{Ag}^+(\text{aq})$ spontaneous, $K = 10^4$
 Q. $\text{Ag} | \text{Br}^- | \text{AgBr} | \text{Br}^-(\text{aq})$ Non-Spontaneous eq. $K = 10^{-4}$
 R. $\text{Ag} | \text{Ag}^+(\text{aq})$ spontaneous, $K = 10^4$
 S. $\text{Cl}^- | \text{Cl}_2(\text{g}) | \text{Cl}^-(\text{aq})$ Non spontaneous, $K = 10^{-4}$
 Section F (Subjective) Single-digit Integer Type

Problem 608 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q1]
 $\text{E} = \text{E}^\circ - \frac{0.059}{n} \log \frac{[\text{H}^+]}{[\text{H}^+]}$
 $\text{E} = \text{E}^\circ - \frac{0.059}{n} \log \frac{[\text{H}^+]}{[\text{H}^+]}$
 $\text{E} = \text{E}^\circ - \frac{0.059}{n} \log \frac{[\text{H}^+]}{[\text{H}^+]}$
 $\text{E} = \text{E}^\circ - \frac{0.059}{n} \log \frac{[\text{H}^+]}{[\text{H}^+]}$
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Problem 609 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q2]

$E = -$

$\frac{1}{2} \log [\text{Cu}]$

$0.31 - 0.34 - 0.06 - 0.1 - 0.1 - 0.2 - 0.2 \dots \log [\text{Cu}] [\text{Cu}] \cdot M = -$

$\log [\text{OH}^-] = + +$

$\log [\text{OH}^-] = + + + = \log K_{sp} \text{H Cu p } 2.19 - 9.10 - 0.1 - 10.5$

Problem 610 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q3]

$E = E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} - \frac{0.059}{3} \log \frac{[\text{Fe}^{2+}]^3}{[\text{Fe}^{3+}]}$

Now, $E = E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} - \frac{0.059}{3} \log \frac{[\text{Fe}^{2+}]^3}{[\text{Fe}^{3+}]}$

or, $0.78 - 0.06 - 0.05 - 0.2 - 0.2 - 0.2 \dots \log [\text{Fe}^{2+}] [\text{Fe}^{2+}] [\text{Fe}^{2+}] = - \log [\text{Fe}^{3+}]$

Problem 611 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q4]

$2\text{Fe}^{3+} + 2\text{I}^- \rightarrow 2\text{Fe}^{2+} + \text{I}_2$;

0.5 M excess

100%

Equ. CM 1.0 M 0.5 M

Ecell V; $E^\circ = 0.77 - 0.53 = 0.24 \text{ V}$

$K_{eq} = 10^8$

Now, $10.5 - 1.0 - 5.10 - 8.2 - 2.2 - 5 = \log K_{eq} = \log K_{eq} = \log K_{eq} = \log K_{eq}$

Problem 612 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q5]

Assuming the cell as concentration cell, net cell reaction is

$\text{Ag} + \text{Ag}^+ \rightarrow \text{Ag} + \text{Ag}^+$

$E = 0$

$E = \frac{0.059}{1} \log \frac{[\text{Ag}^+]_1}{[\text{Ag}^+]_2}$

and $E_{\text{cell}} = -0.06 - 0.1 - 0.1 - 0.1 \dots \log$

or, 0.102

$-0.06 \log \frac{[\text{Cl}^-]_1}{[\text{Cl}^-]_2} = 0.102$

$\log \frac{[\text{Cl}^-]_1}{[\text{Cl}^-]_2} = -1.7$

Problem 613 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q6]

$n_{\text{eq}} \text{Pb} = n_{\text{eq}} \text{Tl} = 1$

$\log \frac{[\text{Pb}^{2+}]}{[\text{Tl}^+]} = \log \frac{[\text{Pb}^{2+}]}{[\text{Tl}^+]} = \log \frac{[\text{Pb}^{2+}]}{[\text{Tl}^+]} = \log \frac{[\text{Pb}^{2+}]}{[\text{Tl}^+]}$

$\dots t$

$t = 3597.4 \text{ sec} ; 1 \text{ hr}$

Problem 614 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q7]

$n_{\text{Q}} = n_{\text{F}} = 1$

Now, $x + 6(-1) = y$

Problem 615 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q8]

$2\text{NaCl} + 2\text{H}_2\text{O} \xrightarrow{\text{electrolysis}} 2\text{NaOH} + \text{H}_2 + \text{Cl}_2$

to

+

$2\text{NaOH} + \text{Cl}_2 \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$

+

$n\text{NaClO formed} = n\text{Cl}_2 \text{ produced from electrolysis}$

or, $\frac{10}{100} = \frac{1}{74.5} \times \frac{100}{2} \times \frac{1}{2} \times \frac{1}{96500} \times t$ (n-factor of Cl_2 in electrolysis)

$t = 7.72 \times 10^5 \text{ sec} = 8.93 \text{ days} \approx 9 \text{ days}$

Problem 616 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q9]

Cathode: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$

Anode $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$

Moles of e^- used = $Q/F = \frac{0.1615 \times 60 \times 96500}{96500} = 0.969$

Eq. of Cu^{2+} present = $\frac{500}{1000} = 0.5$ excess

Moles of H^+ produced = Moles of $\text{e}^- = 0.969$

$[\text{H}^+]_{\text{final}} = \frac{0.969}{500} = 0.001938 \text{ M} \Rightarrow \text{pH} = 3.0$

Problem 617 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q10]

cathode: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$

$\frac{250}{1000} = \frac{1}{1000} \times \frac{1}{2} \times \frac{1}{96500} \times t$

$t = 0.025 \text{ mole} = 0.07 \text{ mole}$

As Cu^+ will not remain stably in solution, no complex formation.

Problem 618 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q11]

neq metal = neq Cl_2 $\Rightarrow \frac{52.89}{100} = \frac{1}{2} \times \frac{1}{96500} \times t \times F$

At. wt. (approx) = $\frac{64}{100} = 0.64$

valency At wt Eq.wt = $\frac{200}{66} = 3$

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Problem 619 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q13]

Initial mass of H_2SO_4 ,

$w_1 = 2000 \times \frac{16}{100} = 320 \text{ gm}$

Final mass of H_2SO_4 ,

$w_2 = 2000 \times \frac{14}{100} = 280 \text{ gm}$

Now, neq H_2SO_4 produced = Q/F

or, $\frac{1136 - 352}{98} = \frac{1}{96500} \times t \times F$

$t = 2 \text{ A}$

Problem 620 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q13]

Ionic mobility, $\frac{u}{\lambda} = \frac{v}{E} = \frac{d}{dt} = \frac{d}{F} \times \frac{1}{d}$ Speed of ion Potential gradient

or, $\frac{7.5}{10} = \frac{1}{96500} \times \frac{1}{0.12} \times \frac{1}{3} \times \frac{1}{d}$ distance 20 3600

$\Rightarrow d = 0.09 \text{ m} = 9 \text{ cm}$

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q14]

$$\text{neq Ag oxidised} = \text{Q F} \text{ /implies w } 108 \text{ 1 9 65 1 3600 96500 /times} = \text{/times /times} .$$
$$w = 38.88 \text{ gm}$$

Mass of anode dissolved = 38 88 100 60 64 8 . . /times /times gm

Final mass of anode = $72.8 - 64.8 = 8$ gm

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q15]

Cathode: $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$

Anode: $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$

$$\text{neq OH}^- \text{ produced} = Q F$$

/implies nOH-/times = /times = - 1 9 65 10 96500 10 3 .

[OH-] na1 = 10 100 10 3 5 - - = M /*implies* PH = 9.0 Four digit integer type

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q1]

$$/Lambda/circ = /circ + /circ - m m m (AgCl) (AgCl) (Cl) /lambda /lambda$$
$$= \text{times} + \text{times} = \text{times} \text{ --- } 6 \ 19 \ 10 \ 7 \ 81 \ 10 \ 14 \ 00 \ 10 \ 3 \ 3 \ 3 \ 1 \ 2 \ 1 \ . \ . \ . \ / \Omega \text{ m mol}$$

Now, $/Lambdam \text{ C S} = /implies /times = /times - - /kappa \text{ 14 10 2 8 10 3 4}$.

/implies S = 0.02 mol m⁻³

$$= 2 \text{ } /times \text{ } 10^{-5} \text{ M}$$
$$= 287 \text{ } /times 10^{-5} \text{ g/L}$$

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q2]

[S] K [H S] [] . () 2 1 2 2 2 8 13 3 16 10 10 0 1 10 10 - + - - - - =

$= \text{times} \text{ times} = \text{a a K H M}$

$$[\text{Ag}^+] [\text{S}^{2-}] + \dots = \text{times} = \text{times} \text{ K M sp } 2 \ 48 \ 16 \ 16 \ 4 \ 10 \ 10 \ 2 \ 10$$

Now, $E_{\text{Ag}^+/\text{Ag}} = -0.8006 \text{ V} - \frac{0.0591}{1} \log = -0.142 \text{ V}$

Required potential = 142 mV

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q3]

$$2\text{Hg}(1) + 2\text{Fe}^{3+} \rightleftharpoons \text{Hg}^{2+} + 2\text{Fe}^{2+}$$

excess 10⁻³ M

Equ. 10-3 - x x 2 x

= /times - 10 100 10 3M

$$x = 9 \times 10^{-4}$$

Now, $\text{Keg} = \text{times} - \text{times} / \text{times} / \text{times} / \text{times} = \text{times} - - - - - \text{x x x 2 10 9 10 2 9 10 1}$

10 9 10 2 2 3 2 4 4 2 4 2 3 4 () () ()

Now, $E_{\text{cell}} = E_{\text{Fe}^{2+}/\text{Fe}} - E_{\text{Hg}^{2+}/\text{Hg}} = 0.062 \text{ V}$. $\log$

or, 0 7724 0 06 2 9 10 2 3 4 . . log - /circ =

$$/times - E_{Hg} / Hg^{2+}$$
$$E(V_{\text{Hg}}/ \text{Hg}^{2+})_{\text{circ}} = 0.815 \text{ V}.$$

[Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q4]

The cell reaction is

6 14 6 2 7 3 3 Fe Cr O H Fe Cr H O 2+ 2 7 2 2 + + /to + + - + + +

$$E_{n \text{ cell cell}} = /circ -$$
$$+++++0.063632262728 \cdot \log [\text{Fe}] [\text{Cr}] [\text{Fe}] [\text{Cr O}] [\text{H}]$$
$$= - \frac{1}{\text{times}} \frac{1}{\text{times}} \frac{1}{\text{times}} \left(\frac{1}{\text{times}} \right) \cdot \log \left(\frac{1}{\text{times}} \right) \left(\frac{1}{\text{times}} \right) \left(\frac{1}{\text{times}} \right) 1.3507700660754075216268$$
$$= 0.571 \text{ V}$$

Problem 627 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q5]Cell reaction: $\text{H}_2(\text{g}) + 2\text{Ag}^+ \rightarrow 2\text{H}^+ + 2\text{Ag}(\text{s})$ E E P cell cell $\text{H}_2 = \frac{1}{2} \log \frac{[\text{H}^+]}{[\text{Ag}^+]}$ $+ + 0.06222 \cdot \log [\text{H}^+][\text{Ag}^+]$ Visit www.jeebooks.in for more**Problem 628 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q6]**E K cell eq $\frac{1}{n} \log K = 0.06 \cdot \log n \Rightarrow (0.2 - 0.08) = 0.06 \cdot \log K$ $\Rightarrow \log K = 1 \Rightarrow K = 100$ **Problem 629 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q7]**Left electrode: $\text{Mn}(\text{s}) \rightarrow \text{Mn}^{2+} + 2\text{e}^-$ Right electrode: $2\text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^-$ Net cell reaction: $2\text{Mn}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{Mn}^{2+} + \text{O}_2(\text{g}) + 4\text{H}^+$ Now, E E Mn P cell cell $2\text{H}^+ + \text{O}_2 = \frac{1}{2} \log \frac{[\text{H}^+]^4}{[\text{Mn}^{2+}]^2}$ $+ + 0.0641242 \cdot \log \frac{[\text{H}^+]^4}{[\text{Mn}^{2+}]^2}$ $= - \frac{1}{2} \log \frac{[\text{H}^+]^4}{[\text{Mn}^{2+}]^2} \Rightarrow \log \frac{[\text{H}^+]^4}{[\text{Mn}^{2+}]^2} = 1.229118500640001001025124$ $= 2.633 \text{ V}$ **Problem 630 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q8]** $\Delta S_{\text{m}} = \frac{1}{T} \left(\frac{\partial \Delta G}{\partial T} \right)_P$ $= \frac{1}{T} \left(\frac{\partial \Delta G}{\partial T} \right)_P = 296500.000193 \text{ J/K}$ **Problem 631 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q9]** $\Delta_r G^\circ = -nFE^\circ = -nFE^\circ_{\text{Products}} + nFE^\circ_{\text{Reactants}}$ or, $\Delta_r G^\circ = -nFE^\circ_{\text{Products}} + nFE^\circ_{\text{Reactants}}$ $-129600.25100044030628044 \cdot \log \frac{[\text{OH}^-]}{[\text{Al}(\text{OH})_4^-]} = -156.25$ $\Delta_r G^\circ = -156.25 \text{ kJ/mol (OH)}_4^-$ **Problem 632 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q10]** $\ln K = \frac{nFE^\circ}{RT} \Rightarrow \ln K = \frac{2 \cdot 96500 \cdot 3}{8.314 \cdot 298} = 2.418$ $t = 2.418 \cdot 106 \text{ sec} \approx 28 \text{ days}$ **Problem 633 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q11]** $\text{SnCl}_2 + 2\text{Cl}^- \rightarrow \text{SnCl}_4$ Cathode: $\text{Sn}^{2+} + 2\text{e}^- \rightarrow \text{Sn}$ Anode: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ $\text{Cl}_2 + \text{SnCl}_2 \rightarrow \text{SnCl}_4$ Moles of SnCl_2 taken = $19.19001 = 19.19$ Moles of Sn produced = $19.119001 = 19.12$ $=$ moles of Cl_2 produced $=$ moles of SnCl_4 formedand moles of SnCl_4 left = $0.1 - (0.01 + 0.01) = 0.08$ m m SnCl $\text{SnCl}_2 + 0.08190012611520261 = 0.08190012611520261$

Problem 634 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q12]
 $neq\ Au = Q\ F / implies / times / times / times / times = / times - 80\ 8\ 0\ 10\ 19\ 7\ 3\ 197\ 2\ 4\ 96\ 500$
 $4 \dots t$
 $t = 772\ sec$

Problem 635 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q13]
 $AgBr(s) \rightleftharpoons Ag^+ + Br^-$
 $(10^{-7} + x)M \times M$
 Now, $(10^{-7} + x) \times x = 12 / times 10^{-14} / implies x = 3 / times 10^{-7}$
 Final solution: $[Ag^+] = 4 / times 10^{-7} M$, $[Br^-] = 3 / times 10^{-7} M$; $[] NO\ M\ 3\ 7\ 10\ -\ - =$
 Now, $/kappa solution = /lambda / circm (Ag^+) / times [Ag^+] + /lambda / circm (Br^-) / times [Br^-]$
 $+ /lambda / circm () [] NO\ NO\ 3\ 3\ -\ - / times$
 $= 6 / times 10^{-3} / times (4 / times 10^{-7} / times 103) + 8 / times 10^{-3} / times (3 / times 10^{-7} / times 103) + 7 / times 10^{-3} (10^{-7} / times 10^{-3})$

Problem 636 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q14]
 $C_6O + 6O_2 \rightarrow 6CO_2$
 Moles of O_2 needed = $60\ 60\ 96\ 60\ 12\ 8\ 60 / times = / times / times = nC$
 $neq\ O_2 = neq\ azobenzene / implies 8 / times 4 = w\ 182\ 8 / times$
 $/ implies w = 728\ gm + 8H^+ + 8e^- + 4H_2O\ N\ N\ NO_2\ 2$

Problem 637 [Neeraj Kumar [NRJ] EXERCISE II (JEE ADVANCED) Q15]
 $/Lambda / circ eq [Ba (PO_4)_3] = 160 + 140 - 100$
 $= 200\ Ohm^{-1}\ cm^2\ eq^{-1}$
 Now, $/circ eq = eq = /kappa\ C / implies 200\ 1\ 2\ 10\ 5\ 5 = / times - .$
 $S = 6 / times 10^{-8}\ eq/cm^3 = 6 / times 10^{-5}\ N = 10^{-5}\ M$
 $K_{sp} = 108\ S_5 = 108 / times 10^{-25}\ M^5$
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Chapter 9

GRB (O.P. Tandon): Complete Practice and Objective Bank

Problem 1

[GRB (O.P. Tandon) p.48 Q2]

$0.77 + 0.0591 \log \frac{[Zn^{2+}]}{[Fe^{3+}]^2}$ (95) $3 \times 10^{-5} \text{ g Zn} \times 25 \times 2 = 0.77 + 0.0226 \log \frac{[Zn^{2+}]}{[Fe^{3+}]^2}$
 volt - $\frac{0.77}{0.0591} = \log \frac{[Zn^{2+}]}{[Fe^{3+}]^2}$ $\log \frac{[Zn^{2+}]}{[Fe^{3+}]^2} = \frac{0.77}{0.0591} = 13.0289$
 $\frac{[Zn^{2+}]}{[Fe^{3+}]^2} = 10^{13.0289} = 1.07 \times 10^{13}$
 to Cu $2+$ in equilibrium? $\frac{[Cu^{2+}]}{[Fe^{3+}]^2} = 1.07 \times 10^{13}$
 $\frac{[Cu^{2+}]}{[Fe^{3+}]^2} = 1.07 \times 10^{13}$ $\frac{[Cu^{2+}]}{[Fe^{3+}]^2} = 1.07 \times 10^{13}$
 Solution: The reaction is, $Zn + CuSO_4 \rightleftharpoons Cu + ZnSO_4$ or $Zn + Cu^{2+} \rightleftharpoons Cu + Zn^{2+}$ Eo. 0.0591 V
 $[Zn^{2+}] = 1.0 \times 10^{-3} \text{ M}$ cell -2- og $[Cu^{2+}]$ At equilibrium, E cell 0 Eo 0.0591 V, $[Zn^{2+}] = 1.0 \times 10^{-3} \text{ M}$
 $-2-$ og $[Cu^{2+}]$ 2 + 2 Eo log $[Zn^{2+}]$: X cell $[Cu^{2+}]$ 0.0591 or (E) 0.76 + 0.34 V
 1.10 V) : 2 X 1.10 : 37.225 0.0591 (Zn^{2+}) 1.679 X 1037 : 1 $[Cu^{2+}]$ Example 22. An excess of liquid mercury is added to an acidified solution of $1.0 \times 10^{-3} \text{ M Fe}^{3+}$. It is found that 5/Fe $3+$ remains at equilibrium at 25°C Calculate Eo 2+Hg. Hg g assuming that the only reaction that occurs is $2Hg + 2Fe^{3+} \rightleftharpoons 2Hg^{2+} + 2Fe^{2+}$ (Given, Eo Fe^{3+}/Fe^{2+} : 0.77 volt). e Fe Solution: . 2Hg At equilibrium, excess At equilibrium, Ecell + $2Fe^{3+} \rightleftharpoons 2Hg^{2+} + 2Fe^{2+}$ $10^{-3} \times 5. 10^{-3} \times 95 100 \times 100$ o (IIT 1995) + $2Fe^{2+} + 10^{-3} \times 95 100$ c;;/textasciitilde()/textasciitilde()E-sample 23. Prove that for two half reactions having potentials E1 and E2 which are combined to yield a third half reaction, having a potential E3, Solution: or or :;I;/textasciitilde()l+n'!!l. . n3 !!G3 !!G1 + !!G2 -n3FE3 :;: - n1 FE) - n2FE2 n3E3 == n1 E1 + n2E2 + .>Example 24. What is the standard potential of the Tl^{3+}/Tl electrode? $Tl^{3+} + 2e^- \rightleftharpoons Tl$; Eo: 1.26 volt $Tl + e^- \rightleftharpoons Tl^+$; Eo - 0.336 volt Solution: $Tl^{3+} + 2e^- \rightleftharpoons Tl$; nFE o 2x 1.26 x F: 2.52F $Tl + e^- \rightleftharpoons Tl^+$; nFE o : Ix (- 0.336) x F: - 0.336F . Adding 2.52F - 0.336F 2.184 nF 3 : 0.728 volt Example 25. Calculate the minimum mass of NaOH required to be added in RHS to consume all the H⁺ present in RHS of the cell of emf + 0.701 volt at 25°C before its use. Also . report the emf of the cell after addition of NaOH 0.760 V Solution: The cell reaction is, $Zn + 2H^+ \rightleftharpoons Zn^{2+} + H_2$

Problem 2

[GRB (O.P. Tandon) p.51 Q1]

Electrochemistry is a branch of physical chemistry which deals with the relationship between chemical energy and electrical energy and how one can be converted into another. The subject is divided into two categories: (i) use of electrical energy to produce chemical changes (electrolysis) and (ii) conversion of chemical energy into electrical energy (production of electricity by spontaneous redox reactions).

Problem 3**[GRB (O.P. Tandon) p.51 Q2]**

Conductors are the substances which allow the passage of electric current through them. Those which do not allow the flow of electric current through them are called insulators. Conductors are of two types: (a) Metallic or electronic conductors are those which remain unchanged as current flows through them. These conductors transfer electric current by transfer of electrons without transfer of matter. Metals such as Cu, Ag, Al, Pt, etc., and non-metals like carbon (graphite) and various alloys belong to this class. (b) Electrolytic conductors are the conductors like aqueous solutions of acids, bases and salts or substance in molten state which allow the flow of electric current with chemical decomposition. These conductors are termed as electrolytes. The substances whose aqueous solutions do not conduct electric current are called non-electrolytes. To pass the current through an electrolytic conductor; two metallic rods or plates are required. These are termed electrodes. The electrode which is connected with the positive terminal of battery or through which electric current enters the solution is termed anode and the electrode which is connected with negative terminal of battery or through which electric current leaves the solution is termed cathode. Actually the anode is the electrode

Problem 4**[GRB (O.P. Tandon) p.52 Q4]**

Preferential discharge theory: If more than one type of coating an inferior metal with a superior metal by electrolysis, of ion is, attracted towards a particular electrode, then the ion

Problem 5**[GRB (O.P. Tandon) p.52 Q6]**

Arrhenius theory of electrolytic dissociation: It was discharged is the one which requires least energy. put forward by Arrhenius in 1884 to explain the properties of The, decreasing order' of the discharge potential or the electrolytic solutions. The main points of the theory are: increasmg ruder of deposition fol the cations and anions is as (i) AD eleetfelyte, when dissolved in water, breaks up into foHows: tWQ types Qf cllarged particles called iQns. The iQn carrying Cations': K^+ , Na^+ , Ca^{2+} , Mg^{2+} , Al^{3+} , Zn^{2+} , H^+ , Cu^{2+} , Ag^+ , Au^{3+} /bullet positive charge is called cation and the ion carrying negative 2 ' , 'charge is called anion. Anions : SO_4^{2-} , NO_3^- , OH^- , Cr^{3-} , Br^- , r^- , " AB/textasciitilde() A^+ (cation) + B^- (anion) ,S.FanidaY'slaws 6felectrolysis: These present the.. . . rellltionship,between the ,qmmntity of electric charge and the . (19 T/textasciitilde()e proces/textasciitilde() o/textasciitilde()spl/textasciitilde()ttmg of the m?leculs of the elec-trolyte 'amount of the substance deposlted-artlie electrode.these were _____,_____J11tO...lons.- 1 and callecLu/textasciitilde()JlIsatio/textasciitilde().Jhe.../textasciitilde()acti/textasciitilde()D-of.thejotaJ number of given by Faraday in 1834. ' , " /textasciitilde()o/textasciitilde()ecules present m solut1On as 10ns IS known as degree of " , (i) FirstJaw:,. ')When an electric curre/textasciitilde()t is /textasciitilde()assed thr/textasciitilde()ugh an ION:::tion/textasciitilde() , , electrolyte; Ihearnountof substancedeoslted Isproport1Onal to s de oted by(x . ,the quantity (f electric charge which flows' through the a = Number of molecules dissociated into ions electrQlyte.' Total number of molecules dissolved Matheniati/textasciitilde()ally, or WocQ $W = ZQ = Z \times I \times t$ where, Zis a constant, known as electrochemical equivalent, I is ' the c/textasciitilde()nt in amperes and t is the time in seconds. When $Q = 1$ coulomb or,oile ampere of current is passed for 1 seCbnd, W /textasciitilde() Z. 'Zisthts-themassofthesubstance deposited 'by one" coulomb: (ii)Second' law: Wheri /bullet the same quantity of current, is passed, through ,different elec-trolytes, the masse/textasciitilde() . of different substances deposited on the electrodes will be in the ratio of their equivalent masses. F.orexample,ifsame quantity of current is ' passeq through ,copper sulphate and silverttitrate solutions, then 'M;/textasciitilde()ssofcoppei deposited Equi/textasciitilde()areritm/textasciitilde()s /textasciitilde()f copper 'Massof silver deposited 'Equivalentmass of silver , ' One grrurtequivalent of /textasciitilde()y substance is deposited by passing same aniOUllt Of charge, i.e., 96500 coulomb (1 faraday). ' . E:::dZ x 96500 , E ' or:,. Z;::-' 96500 Th h ' db/cdot nF, , e C arge carne y anIon = , coulomb , , 6.02 X 10²³ When; n = '1, ' F' 96500 The fundamental unit of charge = ' ;:: -/textasciitilde()- , 6.02 x 10²³ 6.02 x 10²³ The degree of ionisation depends upon: (a) Nature of solute (b) Nature of solvent (c) Dilution and temperature. ' (iii) Ions present in solution constantly reunite to form neutral molecules; thus, there is a state of dynamic equilibrium between ionised and unionised molecules. AB/textasciitilde() $A^+ + B^- \rightleftharpoons [A + B]$, , , ' ;:: K (ionisation constant) $[AB]$, (iv) The ions are discharged always in equivalent aniounts when current is passed through electrolytic solution no matter what their relative speeds are. (v) The electrolytic solution is always neutral in nature as the total' charge on cations is equal to the total charge on anions. However, it is not necessary that, the number of cations and anions should be always equal. (vi) The properties of electrolytes in solution an: , the properties of ions present in the solution . (vii) The ions act as molecules for colligative properties. (viii) The conductivity of the solution depends onthe nature and number of ions. ' ' Evidences in favour of Arrhenius theory: (i) Ohm's law applicability: No part of the currenit is used in splitting up the molecules into ions. (ii) X-ray studies have shown that ions are present in solid electrolytes. The ionic compounds in molten state behave as good conductors. (iii) Ionic reactions are observed in the case of electrolytes. (iv) Constant value of heat of

Problem 6**[GRB (O.P. Tandon) p.53 Q9]**

Galvanic or Voltaic or Electrochemical cell: It is the

Problem 7**[GRB (O.P. Tandon) p.53 Q7]**

Conductivity is the property of the conductor which device in which chemical energy is converted into electrical facilitates the flow of electricity through it. It is equal to the energy. In this cell a redox reaction is carried out in an indirect reciprocal of resistance. manner and the decrease in free energy during the chemical C d' . I 1 process appears as electrical energy. . on uctlvty = Resistance = R The common galvanic cell, dry cell and the lead' storage It is expressed as $\kappa = \frac{1}{R}$ battery are the devices for converting chemical energy into electrical energy. Daniell cell is the typical example of galvanic Specific conductivity is defined as the reciprocal of specific resistance; i.e., it is the conductance of 1 cm³ of a conductor. It is cell. It consists of two half-cells, one containing zinc electrode dipping in solution of zinc sulphate (1 M) and another containing represented by the symbol K. copper electrode dipping in solution of copper sulphate (1 M). Ki I x conductivity Zinc electrode acts as anode. At this electrode oxidation occurs. $\text{Zn} \rightarrow \text{Zn}^{2+} + 2e^-$. COReelectrode acts as cathode: At this h $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$. TS", .../C...Jlu...1 2.-'-+:-t:+/textasciitilde()2:ee:....-.....=7.=/cdot /textasciitilde()Cl...llU _____. .../Wh'/Lue/textasciitilde()n:l-L.tbwe/textasciitilde()/cdot+t/w"uo- —Were, t ,*lstaRce uetween I ectr04lS anu tl -area 0 cross-section of electrode. half-cells are connected by a salt bridge and the metal electrodes' In the case of electrolytic solutions, specific conductance is are joined externally, the electrons from zinc electrode. (-ve the conductance of a solution of definite dilution enclosed in a electrode) move towards copper electrode (+ve electrode): i.e., cell having two electrodes, of unit. area separated by one current flows from cathode to anode. The cell reaction is: centimetre. The units of spe.cifi<;;. con.ductance are $\Omega^{-1} \text{cm}^{-1}$. Zn-+ - /textasciitilde()l'tn...:2+!:!... —li'<- ICPlu't—=—'— The ratio of II a is called cell constant. Equivalent conductance: Conductivity of a solution containing 19 equivalent of the electrolyte. It is denoted by 'N. If the concentration of solution is C g equivalent per litre/textasciitilde() then A K 1000 C or A=i(xV where, V is the volume in mL containing Ig equivalent of the electrolyte. The unit of equivalent conductance is $\Omega^{-1} \text{cm}^2 \text{equiv}^{-1}$. Molar conductance: Conductance of a solution containing 1 g mole of an electrolyte. If V is volume in mL containing 19 mole, then - Molar conductance, $\Lambda_m = \frac{1000}{V} \kappa$)1 /textasciitilde() K X V If C is the concentration of the solution in g me le per litre, then 1000)1=KX C . Its unit is $\Omega^{-1} \text{cm}^2 \text{mol}^{-1}$. . Molar conductance EqUIvalent conductance = . n Molecular where, n Equivalentmass

Problem 8**[GRB (O.P. Tandon) p.53 Q8]**

Kohlrausch's law: At infinite dilution, each ion makes a definite contribution towards equivalent conductance of the electrolyte. The value of equivalent conductance at infinite dilution for any electrolyte is the sum of conductances at infinite dilution of its constituent ions. Thus, $\Lambda_m^\infty = \Lambda_a^\infty + \Lambda_c^\infty$ Λ_a^∞ and Λ_c^∞ are called the ionic conductances of anion and cation at infinite dilution respectively. It can be used to determine degree of dissociation. S/textasciitilde()lt-bridge allows the flow of current by completing the circuit and lTlaintains . electrical neutrality .. It also prevents liquid-HqtiidjunctionpoiintiaJ.- -" – A galvanic cell is represented in the following manner: LHS Anode (-ve) $\text{Zn} | \text{Zn}^{2+} (1 \text{ M})$ Oxidation half-cell $\text{Zn} \rightarrow \text{Zn}^{2+} + 2e^-$ Salt-bridge RHS Cathode (+ve) $\text{Cu} | \text{Cu}^{2+} (1 \text{ M})$ I ClI Reduction half-cell $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$, 10.. Electrftdepotential/textasciitilde()- When—ametal—is—placoo—in—'a"— solution of its ions, the metal either acquires a positive charge or negative charge with respect to solution. On account of this a definite potential is developed betwe.en. the metal and the solution. This potential difference is termed electrode potential. The magnitude of petential depends on the nature of el(ectrode, concentration of ions and temperature. It is a ,measure of the tendency of the metal to lose or gain electrons or a measure of the relative tendency to undergo. oxidation or reductio.n. Depending on. the nature of the metal electrode, the electrode potential is of two types: (i) Oxidation potential: When electrode is negatively charged with respect to solution, i.e., oxidation occurs and the' electrode acts as anode $\text{M} \rightarrow \text{M}^{n+} + ne^-$ (ii) Reduction potential: When electrode is positively charged with respect to solution, i.e., reduction occurs and the, electrode acts as cathode, $\text{M}^{n+} + ne^- \rightarrow \text{M}$

Problem 9**[GRB (O.P. Tandon) p.54 Q12]**

Emf of a cell: The difference in potentials of two half-cells is known as the emf of the cell
 $E_{\text{cell}} = \text{Red. pot. cathode} - \text{Red. pot. anode}$
 $E_{\text{cell}} = \text{Red. pot. cathode} + \text{Oxid. pot. anode}$
 Oxid. potential anode = Oxid. potential cathode

Problem 10**[GRB (O.P. Tandon) p.54 Q13]**

Some more reference electrodes: Since, a hydrogen electrode is difficult to prepare and maintain, it is usually replaced by other reference electrodes which are known as secondary reference electrodes. These are convenient to handle and prepared easily. Calomel electrode and silver-silver chloride are used as reference electrodes. The potentials of these electrodes are fixed on hydrogen scale but their values depend on strong electrolyte concentration.

Problem 11**[GRB (O.P. Tandon) p.54 Q14]**

Reversible and irreversible cells: A cell is said to be reversible if the following two conditions are fulfilled: (i) The chemical reaction of the cell stops when an exactly equal external emf is applied. (ii) The chemical reaction of the cell is reversed and the current flows in opposite direction when the external emf is slightly higher than that of the cell. Any other cell which does not obey the above two conditions is termed as irreversible. Daniell cell is reversible but Zn | H₂SO₄ (aq) | Ag cell is irreversible in nature.

Problem 12**[GRB (O.P. Tandon) p.54 Q15]**

Prediction for occurrence of a redox reaction: Any redox reaction would occur spontaneously if the free energy change (ΔG) is negative. $\Delta G = -nFE_{\text{cell}}$ where, 'n' is the number of electrons involved, F is the value of Faraday and E_{cell} is the cell emf. ΔG can be negative if E_{cell} is positive. When E_{cell} is positive, the cell reaction will be spontaneous and serves as a source of electrical energy.

Example: Consider the following cell: $\text{Zn} | \text{Zn}^{2+} (1\text{M}) || \text{Cu}^{2+} (1\text{M}) | \text{Cu}$

The cell potential depends on the concentration of ions in solution in contact with the electrode. Increase in concentration of cations results in an increase of reduction potential of an electrode. Consider $\text{M}^{n+} + ne^- \rightleftharpoons \text{M}$

Reduction potential: $E_{\text{red}} = E_{\text{red}}^{\circ} - \frac{0.0591}{n} \log \frac{[\text{M}]}{[\text{M}^{n+}]}$

Oxidation potential: $E_{\text{ox}} = E_{\text{ox}}^{\circ} + \frac{0.0591}{n} \log \frac{[\text{M}]}{[\text{M}^{n+}]}$

Cell potential: $E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$

Example: $\text{Zn} | \text{Zn}^{2+} (1\text{M}) || \text{Cu}^{2+} (1\text{M}) | \text{Cu}$

$E_{\text{cell}} = E_{\text{Cu}^{2+}/\text{Cu}} - E_{\text{Zn}^{2+}/\text{Zn}}$

$E_{\text{cell}} = 0.34 - (-0.76) = 1.10\text{V}$

Problem 13**[GRB (O.P. Tandon) p.54 Q17]**

Electrochemical series: When the electrodes (metals and non-metals) in contact with their ions are arranged on the basis of values of their standard reduction potentials or standard oxidation potentials, the resulting series is called the electrochemical or activity series of the elements.

Problem 14**[GRB (O.P. Tandon) p.55 Q18]**

Relation between emf and equilibrium constant: $\Delta G^{\circ} = -2.303RT \log K$

Also, $\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ}$

So, $-nFE_{\text{cell}}^{\circ} = -2.303RT \log K$

$E_{\text{cell}}^{\circ} = \frac{2.303RT}{nF} \log K$

At $T = 298\text{K}$, $E_{\text{cell}}^{\circ} = \frac{0.0591}{n} \log K$

Problem 15**[GRB (O.P. Tandon) p.55 Q19]**

Primary cells are those which have definite life and become dead over a time. These cannot be recharged; Examples are dry cell, mercury cell, etc. Secondary cells are the cells which can be recharged by passing direct current through them. Lead storage battery and nickel-cadmium cell, are examples of secondary cells. Lead storage battery consists of spongy lead as anode, a grid of lead packed with PbO_2 as cathode and an aqueous solution of H_2SO_4 (38% electrolyte). $Pb + PbO_2 + 4H^+ + 2SO_4^{2-} \rightarrow 2PbSO_4 + 2H_2O$ Voltage 2.0 volt. The concentration of H_2SO_4 decreases during discharging. $2PbSO_4 + 2H_2O \rightarrow Pb + PbO_2 + 4H^+ + 2SO_4^{2-}$. Name of Anode Cathode Electrolyte used Dry cell Zn Graphite $MnO_2 + C$ (around cathode) $NH_4Cl + ZnCl_2$ (around anode) Mercury cell Zn Graphite $HgO + KOH$ (moist) Lead storage $Pb/PbO_2/H_2SO_4$ (38% battery mass) Lithium battery Li Metal sulphide $LiClO_4$ in organic solvent Ni-Cad cell $Cd/NiO/KOH$ solution Fuel cell Porous carbon Porous carbon Cone. $NaOH$ or H_2O_2 (with catalyst) KOH (aq.).

Problem 16**[GRB (O.P. Tandon) p.55 Q20]**

Fuel cell: The cells which convert chemical energy of fuels directly into electrical energy are known as fuel cells. One of such cells is fuel cell (H_2/O_2) in which electrical energy is generated by the use of H_2 and O_2 . In fuel cell $NaOH$ is used as an electrolyte and hydrogen gas is diffused at the anode made of porous carbon. At the cathode oxygen is diffused through a porous carbon. The following chemical reactions occur: $2H_2 \rightarrow 2H_2O + 2e^-$ (i) At anode, $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ (ii) At cathode, Net cell reaction $2H_2 + O_2 \rightarrow 2H_2O + \text{energy}$ Fuel cells are quite efficient, and free from pollution. These are used in spacecrafts.

Problem 17**[GRB (O.P. Tandon) p.55 Q21]**

Concentration cells: If two plates of the same metal are dipped separately into two solutions of the same electrolyte and are connected with a salt bridge, the whole arrangement is found to act as galvanic cell. Such cells are known as concentration cells. These are of two types: (i) Electrode concentration cells, e.g., $Pt, H_2(p_1) | H^+(aq) | H_2(p_2), Pt$. If $p_1 > p_2$ oxidation occurs at LHS electrode and reduction occurs at RHS electrode; $E = \frac{0.0591}{n} \log \frac{p_1}{p_2}$ (ii) Electrolyte concentration cells, e.g., $M | M^{n+}(C_1) || M^{n+}(C_2) | M$. If $C_2 > C_1$, $E = \frac{0.0591}{n} \log \frac{C_2}{C_1}$.

Problem 18**[GRB (O.P. Tandon) p.56 Q1]**

Match the List-I with List-II and List-III: List-I List-II List-III (Quantity) (Symbol) (Unit) (a) Conductivity (P) $\Omega^{-1} m^{-1}$ (b) Cell constant (q) K (v) cm^{-1} (c) Molar conductance (r) $\Omega^{-1} cm^2 mol^{-1}$ (d) Equivalent (s) $l A$ (x) $\Omega^{-1} cm^2 eq^{-1}$ conductance

Problem 19**[GRB (O.P. Tandon) p.56 Q2]**

Match the salts in the Column-I with their Use in Column-II: Column-I (a) KCl (b) Agar-agar (c) $0.1 N$ KCl (d) Quinhydrone Column-II (q) Calomel electrode (r) Used in ice cream (s) Redox electrode (t) Match the Column-I with Column-II: (a) (1) (Degree of ionisation) (P) α (b) t^+ (Transport number) (c) Fraction of a mole undergoing ionization (d) $NaCl$ (q) $NmIA$ (r) $U^+ + U^-$ (s) $NIAO$ (t) $A_m = A_e$

Problem 20**[GRB (O.P. Tandon) p.56 Q4]**

Match the physical quantities in the List-I with their units in List-II: List-I List-II (a) Resistance (P) Ω (b) Resistivity (q) Ωm (c) Conductivity (r) $\Omega^{-1} m^{-1}$ (d) Specific conductance (s) $\Omega^{-1} m^{-1}$

Problem 21**[GRB (O.P. Tandon) p.56 Q5]**

Match the Column-I with Column-II: Column-I . Column-II (a) Specific conductance, K (P) Λ° ; Λ° , (b) Molar conductance, Λ_m (q) Decreases with dilution (c) Resistance of electrolyte (r) Increases with dilution solution (s) Degree of ionization of weak electrolyte, α (t) Increases with increase in the distance between parallel plates

Problem 22**[GRB (O.P. Tandon) p.56 Q6]**

Match Column-I with Column-II: Column-I (Combination of half-cell reactions) Column-II (potential of overall process) (a) $6\text{Fe} + \text{Br}_2 \rightarrow 6\text{Fe}^{2+} + 2\text{Br}^-$ (P) $E^\circ = 0.56 \text{ V}$ (Q) $E^\circ = -0.61 \text{ V}$ (R) $2\text{Fe} + \text{Br}_2 \rightarrow 2\text{Fe}^{3+} + 2\text{Br}^-$ (S) $E^\circ = 0.45 \text{ V}$ (T) $\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{SO}_3 + \text{H}_2\text{O}$ (U) $E^\circ = 0.17 \text{ V}$ (V) $E^\circ = ?$ (c) $\text{ClO}_2 + 6\text{H}^+ + 6\text{e}^- \rightarrow \text{Cr} + 3\text{H}_2\text{O}$ (W) $E^\circ = +1.45 \text{ V}$ (X) $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$ (Y) $2\text{Cr}^{3+} + 7\text{H}_2\text{O} \rightarrow \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^-$ (Z) $E^\circ = -0.535 \text{ V}$ (r) $E^\circ = +0.36 \text{ V}$ (d) $\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn(s)}$ (0) $E^\circ = -0.76 \text{ V}$ (s) $E^\circ = +1.47 \text{ V}$ (Ag) $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag(s)}$ (0) $E^\circ = +0.80 \text{ V}$ (Zn) $\text{Zn(s)} + 2\text{Ag}^+ \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{Ag(s)}$ (0) $E^\circ = ?$

Problem 23**[GRB (O.P. Tandon) p.56 Q7]**

Match the Column-I with Column-II: Column-I . (a) Concentration cell (b) Edison cell (c) Mercury cell (d) Dry cell Column-II (P) Fe is oxidised by Ni^{2+} (q) Zinc anode (r) HgO cathode (s) $\text{O} = \text{O}$

Problem 24**[GRB (O.P. Tandon) p.56 Q8]**

Match the Column-I with Column-II: Column-I (Electrode) (a) Calomel (b) Glass (c) Hydrogen (d) Quinhydrone Column-II (Type) (P) Reference (q) Redox (r) Membrane (s) Gas I

Problem 25**[GRB (O.P. Tandon) p.57 Q9]**

Match the Column-I with Column-II: Column-I (a) Kohlrausch law (b) Λ_m (c) K (κ) (d) Ci Column-II (P) $\Lambda_m / \Lambda^\circ$ (Q) $J = \frac{1}{R} \frac{d\Lambda_m}{dC}$ (r) $\Lambda^\circ = \frac{1}{C} \lim_{C \rightarrow 0} \Lambda_m$ (S) $K \times 1000 \text{ M}$

Problem 26**[GRB (O.P. Tandon) p.57 Q10]**

Match the Column-I with Column-II: Column-I (a) $E^\circ = 0$ (b) $E^\circ = 0$ (c) $\text{LiG} = \text{O}$ (d) 1 Faraday Column-II (p) Cell is discharged (q) $Q = K$ (r) 96500 coulomb (8) 1 mol electrons (t) Concentration cell $\left[\frac{4}{\text{textasciitilde{}}}\right]$.

Problem 27**[GRB (O.P. Tandon) p.57 Q1]**

(a-q, u), (b-s, v), (c-p, w), (d/textasciitilde{r}, x)

Problem 28**[GRB (O.P. Tandon) p.57 Q2]**

(a-q), (b-p, r), (c-p), (d-s)

Problem 29**[GRB (O.P. Tandon) p.57 Q3]**

(a-q, s), (b/textasciitilde{p}, r), (c-q, s), (d-t) .4. (a-p, q), (b-r), (c/textasciitilde{s}), (d-s)

Problem 30**[GRB (O.P. Tandon) p.57 Q5]**

(a-q), (b-r), (c-q, s), Cd-p, r)

Problem 31

[GRB (O.P. Tandon) p.57 Q6]

(a-q), (b-r), (c-s), (d-p)

Problem 32

[GRB (O.P. Tandon) p.57 Q7]

(a-s), (b-p), (c-q, r), (d-r)

Problem 33

[GRB (O.P. Tandon) p.57 Q8]

(a-p), (b-r), (c-s), (d-q)

Problem 34

[GRB (O.P. Tandon) p.57 Q9]

(a-r), (b-s), (c-q), (d-p)

Problem 35

[GRB (O.P. Tandon) p.57 Q10]

(a-t), (b-p, q), (c-p, q), (d-r, s)

Problem 36

[GRB (O.P. Tandon) p.58 Q1]

How many coulombs are required for the following reductions? (i) 1 mole of Ag^+ ions to Ag (ii) 1 mole of Cu^{2+} ions to Cu (iii) 1 mole of MnO_4^- ions to Mn^{2+} - [Ans. (i) 96500 coulomb (ii) 193000 coulomb (iii) 96500 coulomb] .

Problem 37

[GRB (O.P. Tandon) p.58 Q2]

How many faradays are needed to reduce 2 gram-mole of Cu^{2+} to Cu metal? . [Ans. 4 faraday]

Problem 38

[GRB (O.P. Tandon) p.58 Q3]

How many faradays are released when 12.7 g of copper metal is changed into copper ions? . n. A lead storage battery is discharged in which the following electrode reaction takes place: $\text{Pb(s)} + \text{PbO}_2(\text{s}) + 2\text{H}_2\text{SO}_4(\text{l}) \rightarrow 2\text{PbSO}_4 + 2\text{H}_2\text{O}$ The original volume in electrolyte is 1 litre. During discharge the concentration of H_2SO_4 changes from 34.6% (density 1.261 g/mL) to 27% . [Ans. 1.254 faraday]

Problem 39

[GRB (O.P. Tandon) p.58 Q12]

The chemical reaction given below: $\text{C}_{12}(\text{g}) + \text{S}_{02}(\text{g}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{C}_{\text{aq}} + 3\text{H}^+(\text{aq})$

1. [Ans. 1.254 faraday] (i) Give the half-cell reactions. (ii) Calculate the number of coulombs required to deposit 40.5 g of Al when the electrode reaction is: $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ assuming the cell becomes non-operative when 90% of initial Cl_2 has been used up? [Ans. 40.5 faraday]

Problem 40

[GRB (O.P. Tandon) p.58 Q4]

Calculate the number of coulombs required to deposit 40.5 g of Al when the electrode reaction is: $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ assuming the cell becomes non-operative when 90% of initial Cl_2 has been used up? [Ans. 40.5 faraday]

Problem 41**[GRB (O.P. Tandon) p.58 Q5]**

Calculate the number of coulombs required to deposit 50 g of silver at cathode from silver nitrate solution.

Problem 42**[GRB (O.P. Tandon) p.58 Q13]**

A current of 80 microampere is passed through a solution of (Atomic mass of silver = 108) AgNO_3 for 32 minutes using platinum electrodes. A uniform [Ans. 44675.9 coulomb] single atom thick layer is deposited covering 86/

Problem 43**[GRB (O.P. Tandon) p.58 Q6]**

How much electric charge is required to produce 20.0 g of cathode surface. If total surface area of cathode is 601.7 cm^2 , calcium from molten CaCl_2 ? ' calculate the area covered by one Ag atom. [Ans. 96500 coulomb] [Ans. $5.4 \times 10^{-16} \text{ cm}^2$]

Problem 44**[GRB (O.P. Tandon) p.58 Q7]**

If 3 faradays of electricity are passed through an iron (II)

Problem 45**[GRB (O.P. Tandon) p.58 Q14]**

Anthracene can be electrolytically oxidised to anthraquinone. bromide solution, how many grams Of iron metal will be /circ deposited? (At. mass of iron = 56) II [Ans. 84 g]

Problem 46**[GRB (O.P. Tandon) p.58 Q8]**

A certain quantity of electricity deposits 0.54 g of Ag from silver nitrate solution. What volume of hydrogen will be liberated by the same quantity of electricity at 27/circC and 750 mm of Hg pressure? Mass of [Hint: Mass of Ag 1 x 0.54 Mass of H_2 0.005 g 108 /textasciitilde()olumeof H_2 0.005 x 0.0821 x 300 x 760 2 x 750 0.06234 litre] (V =/textasciitilde(). RT) M P

Problem 47**[GRB (O.P. Tandon) p.58 Q9]**

In the electrolysis of an aqueous cupric bromide solution, how many grams of bromine are finned on passing a current of 1 ampere for 16 minutes and 5 seconds? Write the anode and the cathode half reactions. [Ans. 0.80 g; cathode reaction $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$; anode reaction $2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{e}^-$]

Problem 48**[GRB (O.P. Tandon) p.58 Q10]**

A current of 1.5 amperes is passed through a/cdotsolution of a salt of a bivalent metal for 30 minutes. Increase in mass of cathode is 0.8898 g. Find the atomic mass of the metal. [f/ 115. 63,6] 3[0] $\text{C}_{14}\text{H}_{10}$ /textasciitilde() Anthracene II /circ Anthraquinone What mass of anthraquinone can be produced by the passage of 1 ampere current for an hour at 100/[Ans. 1.2932 g]

Problem 49**[GRB (O.P. Tandon) p.58 Q15]**

Dal lake has water 8.2×10^{12} litre approximately. A power reactor produces electricity at the rate of 1.5×10^6 coulomb per second at an appropriate voltage. How much time would it take to electro lyse the lake? [Ans. 1.9 million year]

Problem 50**[GRB (O.P. Tandon) p.58 Q16]**

How many grams of H_2 and O_2 are produced during the electrolysis of water by a 1.30 ampere current for 5 hours? What volumes of dry gases are produced at NTP? [Ans. 0.242 g H_2 ; 1.94 g O_2 ; 2.72 litre H_2 ; 3.6 litre O_2 ,]

Problem 51**[GRB (O.P. Tandon) p.59 Q17]**

How many coulombs of electricity would be required to reduce the iron in 36.0 g of potassium hexacyano ferrate(III), $K_3Fe(CN)_6$ to metallic iron?

Problem 52**[GRB (O.P. Tandon) p.59 Q26]**

Three electrolytic cells A, B and C containing electrolytes zinc sulphate, silver nitrate and copper sulphate respectively, were connected in series. A steady current of 1.50 ampere was passed through them until 1.45 g of silver was deposited at the cathode of cell B. How long did the current flow? What masses of copper and zinc were deposited? 1 mole 3×96500 C (At. mass: Cu 63.5, Zn = 65.4, Ag = 107.8)

Problem 53**[GRB (O.P. Tandon) p.59 Q18]**

What current strength in ampere will be required to liberate 5 g of iodine from potassium iodide solution in 30 minutes? [Ans. 865 second; 0.427 g Cu; 0.44 g Zn]

Problem 54**[GRB (O.P. Tandon) p.59 Q27]**

An electric current passing for 6 minutes through a dilute H_2SO_4 solution gave 40 mL of the electrolytic gas ($H_2 + O_2$)

Problem 55**[GRB (O.P. Tandon) p.59 Q19]**

The mass of copper deposited from a solution of copper sulphate by a uniform current of 0.25 ampere flowing for one hour is 0.295 g. Find the equivalent mass of copper. (1 faraday [Hint: 96500 coulomb give 11.2 litre of H_2 and 5.6 litre of O_2 = 96500 coulomb] . i.e. 16.8 litre of electrolytic gas at NTP.) [Ans. 31.75]

Problem 56**[GRB (O.P. Tandon) p.59 Q28]**

The same quantity of electricity that liberated 2.158 g silver

Problem 57**[GRB (O.P. Tandon) p.59 Q20]**

The solution of a salt of a metal of atomic mass 112 was passed through a solution of a gold salt and 1.314 g of metal was deposited. Find the valency of the metal. The equivalent mass of silver is 107.9. The mass of the metal deposited was 0.788 g. Find the valency of the metal. What is the oxidation state of gold in this gold salt? (At. mass of gold = 197) [Ans. 2] [Eq. mass Oxidation state 3]

Problem 58**[GRB (O.P. Tandon) p.59 Q21]**

If a monovalent metal ion carries 1.6×10^{-19} coulomb of charge. Calculate the volume of Cl_2 at NTP produced during electricity, what is the amount of electricity carried by one electrolysis of MgCl_2 which produces 6.50 g Mg. gram molecular-mass of the metal ions? - CAt. lllasJ'. QfMg=..)/.pmJ)._____ [Ans. 9.6336×10^4 coulomb] [w; 5.99litre]

Problem 59**[GRB (O.P. Tandon) p.59 Q22]**

Calculate approximately how much current is necessary to :m. How long does it take to deposit 100 g of Al from an produce oxygen gas at the rate of 1 mL per second? electrolytic cell containing Al_2O_3 using a current of 125 [Ans. 11.23 ampere] ampere? [Hint: 1 g-eqivalent of oxygen or 22400 mL is liberated by [Al/textasciitilde(). 8577.8 second] 96500 coulomb.] 4 '11, 109 fairly concentrated solution of CuSO_4 is electrolysed

Problem 60**[GRB (O.P. Tandon) p.59 Q23]**

0.5 faraday of electricity was required to deposit all the copper using 1.01 faraday of electricity. Calculate the mass of the in 500 mL of a copper sulphate solution. What is the normality resulting solution. of the copper sulphate solution? . [AII,< 9.6025 g] [Ans .. Normality = IN] J), The density of copper is 8.94 g mL⁻¹. Find out the number of

Problem 61**[GRB (O.P. Tandon) p.59 Q24]**

Copper is deposited at an electrode according to the reaction coulombs needed to plate an area 10 cm² to a thickness 2×10^{-2} cm using CuSO_4 solution as electrolyte. (At. mass of Cu = 63.5) When a current of 1.34 ampere was passed through a solution of copper sulphate for 10 hours, 15.885 g of copper was deposited. (i) How many coulombs of electricity were passed through the solution? (ii) What mass of Ag (At. mass 108) could be plated on a spoon the solution? (iii) How many moles of copper were deposited? (iv) How many coulombs are required for the deposition of 1 mole of copper? (v) If a current of 0.3 ampere is drawn from a Daniell cell for 1 hour, what would be the change in mass of electrodes? (vi) If the charge on an electron is 1.6×10^{-19} coulomb mass of Cu = 63.5 and Zn = 65.31) calculate the value of Avogadro's number. (vii) [Ans; (i) 482.40 coulomb (ii) 0.25 mol (iii) 2×96500 coulomb (iv) 0.356 g Cu deposited, 0.366 g of Zn dissolves] An element A (At. mass 112) and an element B (At. mass 27) (iv) 6.03×10^{23} form chlorides. Solutions of these chlorides are electrolysed Calculate the mass and volume at NTP of hydrogen and separately and it is found that when the same quantity of chlorine that will be formed by passing 10,000 coulomb of electricity is passed, 5.6 g of A was deposited while in the charge through an aqueous solution of potassium chloride. other cell only 0.9 g of B was deposited. What is the valency The cell reaction is: of A if the valency of B is 3? (Dhanbad 1991) $2\text{KCl} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{O}_2 + \text{H}_2$ [2] [Ans. 0.1036 g H₂, volume 1.063 litre; 3.678 g Cl₂, volume 1.1603 litre] electrolytic production of 245 g NaClO₄ from NaClO₃ The anode efficiency for the desired reaction is 60% [Ji. 6.43 X 10⁵ coulomb]

Problem 62**[GRB (O.P. Tandon) p.60 Q37]**

Calculate the mass of HgCl_2 which can be prepared by the reduction of mercury(II) ions in the presence of chloride ions by the passage of 5.0 ampere current for 3.0 hours. [ADS. 132.34 g] [Hint: $2\text{Hg}^{2+} + 2\text{Cl}^- + 2e^- \rightarrow \text{Hg}_2\text{Cl}_2$] 2 F 1 mole

[GRB (O.P. Tandon) p.60 Q38]

How long will it take for a uniform current of 6.0 ampere to deposit 78,0 g gold from a solution of AuCl₃? What mass of chlorine gas will be formed simultaneously at the anode in the electrolytic cell? [ADS: 19104 second; chlorine liberated = 42.17 g] [HiDt: AuCl₃ + 3e⁻ → Au + 3Cl⁻ (Cathode) 2Cl⁻ → Cl₂ + 2e⁻ (Anode)] I mole Z F

[GRB (O.P. Tandon) p.60 Q43]

The resistance of 0.01 N solution at 25 DC is 200 Ω . Cell constant of the conductivity cell is unity. Calculate the equivalent conductance of the solution. (ADS, 500 Ω -I cm² eq⁻¹)

[GRB (O.P. Tandon) p.60 Q44]

A conductivity cell was filled with 0.01 M solution of KCl which was known to have a specific conductivity of $0.1413 \text{ } \Omega^{-1} \text{ m}^{-1}$ at 298 K. Its measured resistance at 298 K was $94.3 \text{ } \Omega$. When the cell was filled with 0.02 M AgNO₃ solution, its resistance was $50.3 \text{ } \Omega$. Calculate (i) cell constant and (ii) the specific conductance of AgNO₃ solution. [ADs. 13.32 m^{-1} ; $2.648 \times 10^{-1} \text{ } \Omega^{-1} \text{ m}^{-1}$]

[GRB (O.P. Tandon) p.60 Q45]

A conductance cell was calibrated by filling it with a 0.02 M solution of potassium chloride. (specific conductance = 0.2768 $\Omega^{-1}\text{m}^{-1}$) and measuring the resistance at 298 K which was found to be 457.3 Ω . The cell was then filled

[GRB (O.P. Tandon) p.60 Q39]

How long will it take 5 ampere of current to deposit 2 g of
with a calcium chloride solution containing 0.555 g of CaCl_2 ...
/textasciitilde()/textasciitilde()/textasciitilde()/textasciitilde()/textasciitilde()/textasciitilde()/textasciitilde()/textasciitilde()
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itilde()/textasciitilde() _____/textasciitilde() _____/textasciitilde() _____
/textasciitilde()/cdot/textasciitilde()C/textasciitilde()oEPp/textasciitilde()e/textasciitilde()rc.lfr/textasciitilde()o
,—____—.!per litre. The measured resistance was 1050 Ω . Calculate (Given, CE of copper
= 32, F = 96500 coulomb) the molar conductivity of Cl_2 solution. (Ranchi 1996) [ADS. 0.0241
mho $\text{m}^2 \text{mol}^{-1}$] [ADS. 20.1 minute] [Hint: -Determine cell constant with KCl data. its value is

[GRB (O.P. Tandon) p.60 Q40]

The amount of lactic acid, HC_3O_3 produced in a sample of 126.6 m-I . muscle tissue was analysed by reaction with hydroxide ion. $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$ in the sample mixture by Sp. _____ condn of CaCl_2 soln. = Cell const. $\times$ Resistance _____ .. electrolysis. The cathode reaction was, $\text{H}^+ + \text{e}^- = \text{H}_2$ $\times$ - = 0.1206 / Ω - rri- , $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ (g) + 20W(aq.) Hydroxide ion reacts with lactic acid as soon as it is produced. The end point of the reaction is detected with an acid-base indicator. It required 115 seconds for a current of 15.6 mA to reach end point. How many grams of lactic acid (a monoprotic acid) were present in the sample? [ADS. 1.674×10^{-3} g] [Hint: No. of moles of lactic acid = No. of moles of OH^- used ;, No. of faraday used in electrolysis $N = \frac{Q}{F}$ $\text{Co} / \text{circled} / \text{dot} \text{d} / \text{dot} \text{I} \times \text{t} 15.6 \times 10^{-3} \times 115$ urn er 0 lara ay use = - = . . 96500 96500 = 1.86×10^{-5} Mass of lactic acid = Number of moles $\times$ Molecular mass = $1.86 \times 10^{-5} \times 90 = 1.674 \times 10^{-3}$ g]

Problem 69**[GRB (O.P. Tandon) p.60 Q41]**

In what direction, can the reaction: $2\text{NaCl} + \text{Fe}_2(\text{SO}_4)_3 \rightarrow 2\text{FeSO}_4 + \text{Cl}_2 + \text{Na}_2\text{SO}_4$ proceed spontaneously? (Given: $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.77\text{V}$; $E^\circ_{\text{I}^-/\text{Cl}_2} = 1.36\text{V}$) (BCECE 2005) [ADS. Forward]

Problem 70**[GRB (O.P. Tandon) p.60 Q42]**

How many faradays of electricity will be required to completely electrolyse one mole of molten Al_2O_3 to produce Al metal and O_2 gas? IJEE (West Bengal) 2005 [ADS. 3 F] Sp. cond. Molar conductance = $\dots$ Conc. $1050\text{ g dm}^{-3} = 0.555\text{ mol dm}^{-3} = 5\text{ mol m}^{-3}$] 111.0

Problem 71**[GRB (O.P. Tandon) p.60 Q46]**

The molar conductivities at infinite dilution of KCl , KNO_3 and AgNO_3 at 298 K are $0.01499\text{ mho m}^2\text{ mol}^{-1}$, $0.01250\text{ mho m}^2\text{ mol}^{-1}$ and $0.01334\text{ mho m}^2\text{ mol}^{-1}$ respectively. What is the molar conductivity of AgCl at infinite dilution at this temperature? [ADS. $0.01383\text{ mho m}^2\text{ mol}^{-1}$] 47, The electrodes in a conductivity cell have area $1.2 \times 10^{-4}\text{ m}^2$ and they are fixed $3 \times 10^{-3}\text{ m}$ apart. A solution containing 200 g equivalent. of the electrolyte per m^3 of the solution has a resistance of $60\text{ }\Omega$ at 298 K. Calculate the equivalent conductivity of the solution. [ADS. $2.09 \times 10^{-2}\text{ mho m}^2\text{ eq}^{-1}$]

Problem 72**[GRB (O.P. Tandon) p.60 Q48]**

The molar conductivities of NH_4^+ ion and Cl^- ion are $73.5\text{ mho cm}^2\text{ mol}^{-1}$ and $76.2\text{ mho cm}^2\text{ mol}^{-1}$ respectively. The specific conductivity of $0.1\text{ M NH}_4\text{Cl}$ is $1.288 \times 10^{-2}\text{ mho cm}^{-1}$. What is the dissociation constant of NH_4Cl ? [Ans. 0.86]

Problem 73**[GRB (O.P. Tandon) p.60 Q49]**

The specific conductivity of a saturated solution of silver chloride at 18°C is $1.24 \times 10^{-6}\text{ mho cm}^{-1}$ after subtracting that of water. Ionic conductances at infinite dilution of Ag^+ and Cl^- ions at this temperature are 53.8 and 65.3 respectively. Calculate the solubility of silver chloride in gram per litre. [ADS. $1.494 \times 10^{-3}\text{ g per litre}$] 1

Problem 74**[GRB (O.P. Tandon) p.61 Q50]**

Given: $\Lambda^\circ_{\text{Mg}^{2+}} = 53.06\text{ cm}^2\text{ mol}^{-1}$, $\Lambda^\circ_{\text{UO}_2^{2+}} = 80\text{ cm}^2\text{ mol}^{-1}$, $\Lambda^\circ_{\text{Al}^{3+}} = 63\text{ cm}^2\text{ mol}^{-1}$. Calculate the values of $\Lambda^\circ_{\text{Al}_2(\text{SO}_4)_3}$ and $\Lambda^\circ_{\text{MgSO}_4}$. [Ans. $858\text{ cm}^2\text{ mol}^{-1}$, $266.12\text{ cm}^2\text{ mol}^{-1}$] (i) Iron reduces Zn^{2+} ions (ii) Iron reduces Ni^{2+} ions. Given: $E^\circ_{\text{Fe}^{2+}/\text{Fe}} = -0.76\text{V}$; $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.44\text{V}$ and $E^\circ_{\text{Ni}^{2+}/\text{Ni}} = -0.25\text{V}$ [Ans. Iron reduces Ni^{2+} ions as the E°_{cell} in this case has positive value.] 57 Can a solution of 1 M CuSO_4 be stored in a vessel made of nickel metal? Given: $E^\circ_{\text{Ni}^{2+}/\text{Ni}} = -0.25\text{V}$ [Ans. No]

Problem 75**[GRB (O.P. Tandon) p.61 Q51]**

Hydrofluoric acid is a weak acid. At 25°C; the molar conductivity of 0.002 M HF is $176.2\text{ cm}^2\text{ mol}^{-1}$. If its $\Lambda^\circ = 405.2\text{ cm}^2\text{ mol}^{-1}$, calculate its degree of dissociation and equilibrium constant at the given concentration. Ans. $\alpha = 0.34$ No; The reaction takes place between CuSO_4 and nickel [Ans. $\alpha = 0.435$, $K = 6.7 \times 10^{-4}$] 5

[GRB (O.P. Tandon) p.61 Q52]

For mercury at O/*circ*C specific conductance,

[GRB (O.P. Tandon) p.61 Q58]

An answer whether, under standard conditions, the following reactions are possible or not: _____

*K/textasciitilde()+1.Q/textasciitilde()/textasciitilde()/textasciitilde()/textasciitilde()m/textasciitilde()./textasciitilde()
 -(/textasciitilde()I/textasciitilde())/textasciitilde()Wrr/cdot 1/textasciitilde()
 itilde()II/textasciitilde()c/textasciitilde()o/textasciitilde()p=p/textasciitilde()cr/textasciitilde()r/textasciitilde()e,
 E/textasciitilde()g+/Ag _____/textasciitilde()—/textasciitilde()—
 _____/textasciitilde()—/textasciitilde()—/textasciitilde()—/textasciitilde()—
 /textasciitilde()—/textasciitilde()/cdot,_____,(a) If the resistance of a cell containing
 mercury is 0.243 Ω, $E_{\text{cell}} = 0.799$ volt; $E_{\text{Cu}^{2+}/\text{Cu}} = -0.337$ volt. $\text{Cu}^{2+} + \text{Fe} \rightarrow \text{Cu} + \text{Fe}^{2+}$, what is
 the cell constant of the cell? (ii) Will Fe^{3+} be reduced to Fe^{2+} by Sn^{2+} ion? Given:
 (b) If the same cell is filled with KCl solution at 25°C, the $\text{Fe}^{3+} + \text{Fe}^{2+} = 0.771$ volt;
 resistance of the cell is $3.966 \times 10^4 \Omega$. What is the conductivity of KCl solution?
 $\text{Sn}^{2+} + \text{Sn}^{4+} = -0.250$ volt _____ J; (c) If the average _____/textasciitilde()ss-
 s/textasciitilde()ctio!llLJ!/textasciitilde()iL9L.JhtL_____/textasciitilde()ll/textasciitilde()j/textasciitilde()L
 —(iii)—Would you use a silver spoon to stir a solution...? 0.9643 mm^2 , what is the effective
 distance between the $\text{Cu}(\text{NO}_3)_2$ electrodes? [Ans. (i) Yes, $E_{\text{cell}} \text{ Cu}^{2+} + \text{Fe} \rightarrow \text{Cu} + \text{Fe}^{2+}$ is positive
 (+ 0.462 [Ans. (a) $2.58476 \times 10^5 \text{ S/m}$; (b) 6.517 S/m ; (c) 0.2492 m] volt)

[GRB (O.P. Tandon) p.61 Q53]

The mobility' of the NH_4^+ ions is $7.623 \times 10^{-8} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$. (ii) Yes, E_{cell} Pt | Sn^{4+} , Sn^{2+} || Fe^{3+} , Fe^{2+} | Pt is positive (+0.621 volt) . " Calculate: ΔG ; (iii) Yes, the reaction does not occur.] (a) the molar conductivity of NH_4^+ ion;

[GRB (O.P. Tandon) p.61 Q59]

The electrode potentials of two half reactions are as follows: (b) velocity of the ion if 15 volt are applied across the electrodes 25 cm apart. [Ans. (a) $73.55 \times 10^{-4} \text{ S m}^2/\text{mol}$; (b) 4.574 I-lrnl] 54; . Construct electric cells for the following reactions: (a) $\text{Fe} + \text{Cu}^{2+} \longrightarrow \text{Cu} + \text{Fe}^{2+}$ (b) $\text{Cu} + 2\text{Ag}^{+} \longrightarrow 2\text{Ag} + \text{Cu}^{2+}$ (c) $6\text{Fe}^{2+} + \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^{+} \longrightarrow 6\text{Fe}^{3+} + 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ (d) $\text{Cd} + 12\text{I}^{-} \longrightarrow \text{Cd}^{2+} + 2\text{I}_2$ (e) $2\text{Fe}^{3+} + 2\text{Cl}^{-} \longrightarrow 2\text{Fe}^{2+} + \text{Cl}_2$ [Ans. (a) $\text{Fe} | \text{Fe}^{2+} || \text{Cu}^{2+} | \text{Cu}$ (b) $\text{Cu} | \text{Cu}^{2+} || \text{Ag}^{+} | \text{Ag}$ (c) $\text{Pt} | \text{Fe}^{2+}, \text{Fe}^{3+} || \text{H}^{+}, \text{Cr}_2\text{O}_7^{2-} | \text{Cr}^{3+} | \text{Pt}$ (d) $\text{Cd} | \text{Cd}^{2+} || \text{I}^{-} | \text{I}_2 | \text{Pt}$ (e) $\text{Pt} | \text{Cl}_2, \text{Cl}^{-} || \text{Fe}^{3+}, \text{Fe}^{2+} | \text{Pt}$] 5/. Predict whether the following reaction will occur spontaneously or not: $\text{CO}_2 + \text{Sn} \longrightarrow \text{Co} + \text{Sn}^{2+}$ $E^\circ = 0.277 \text{ V}$ and $E^\circ = 0.136 \text{ V}$ $\text{Co} | \text{Co}^{2+} || \text{Sn}^{2+} | \text{Sn}$ [Ans. No; E_{cell} is negative = -0.141 V]

[GRB (O.P. Tandon) p.61 Q56]

An iron wire is immersed in a solution containing ZnSO_4 and NiSO_4 . Predict giving reasons which of the following reactions is likely to proceed?

$\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$, $E^\circ = +0.76$ volt

$\text{Ce}^{4+} + \text{e}^- \longrightarrow \text{Ce}^{3+}$; $E^\circ = +1.60$ volt

Giving reason, describe if Ce^{3+} can be oxidised by Fe^{3+} . [Hint: $\text{Ce}^{3+} + \text{Fe}^{3+} \longrightarrow \text{Ce}^{4+} + \text{Fe}^{2+}$ $\text{Ce}^{3+} \longrightarrow \text{Ce}^{4+} + \text{e}^-$ (oxidation); $E^\circ = -1.60$ volt $\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$ (reduction); $E^\circ = +0.76$ volt Adding both, E° of the reaction = -0.84 volt The oxidation of Ce^{3+} by Fe^{3+} is not possible.]

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[GRB (O.P. Tandon) p.63 Q78]

At equimolar concentrations of Fe^{2+} and Fe^{3+} , what must $[\text{Ag}^+]$ be so that the voltage of the galvanic cell made from $\text{K}^+/\text{I}^-/\text{gandP-e3+1'Fe2+el-ect1'ooe/textasciitilde()requals zero}$? The Teacfioffs $\text{Fe}^{2+} + \text{Ag}^+ \rightleftharpoons \text{Fe}^{3+} + \text{Ag}$. Detennine $t/\text{textasciitilde()e, /textasciitilde()equilibrium} \dots -0^{\circ}\text{C}/\text{textasciitilde()}/\text{textasciitilde()tant at } 25^{\circ}\text{C}$ for the reaction. (Given: $E_{\text{Ag}^+/\text{Ag}} = 0.799$ volt and $E_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.771$ volt) [Hint: $= -0.771 + 0.799 = 0.028$ volt At $t/\text{textasciitilde()quilibrium}$, $E_{\text{cell}} = 0$ - EO _____ $0.0591 \text{ V} - \text{cell I}^{\circ} \log]\text{Ag}^+]/\text{circ } 1 = E_{\text{cell}} - 0.0591 \log -]\text{Ag}^+] = 0.34 \text{ V}$ at $25^{\circ}\text{C} = 298 \text{ K} = nE_{\text{cell}}/0.0591 \text{ V} = 3.0] [\text{Ag}^+]$

[GRB (O.P. Tandon) p.63 Q79]

Using Nernst equation for the cell reaction, $\text{Pb} + \text{Sn}^{2+} \rightleftharpoons \text{Pb}^{2+} + \text{Sn}$; $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{2} \log \frac{[\text{Pb}^{2+}]}{[\text{Sn}^{2+}]}$,
Calculate the ratio for which $E_{\text{cell}} = 0$. (Given: $E_{\text{Pb}^{2+}/\text{Pb}}^{\circ} = 0.13$ volt and $E_{\text{Sn}^{2+}/\text{Sn}}^{\circ} = 0.14$ volt) , $[\text{Pb}^{2+}]$ (Ans. $[\text{Pb}^{2+}] = 0.5 [\text{Sn}^{2+}]$)

[GRB (O.P. Tandon) p.63 Q80]

, Determine the potential of a Daniell cell, initially containing 1.00 L each of 1.0 M copper(II) ion and 1.0 M zinc(II) ion, after passage of 105 coulomb charge. The E° of the Daniell cell is 1.10 volt. [Ans. 1.09 volt] This electrode is now coupled with $\text{el}_2 + 2\text{e}^- \longrightarrow 2\text{Cl}^-$ electrode.]

[GRB (O.P. Tandon) p.63 Q82]

Given: EO 0.15 volt . _____ Cu/textasciitilde()_____+_____e _____ u"LCU;_____. _____ ...
 _____E/textasciitilde() ::,_____05 volL. "n"/textasciitilde()/textasciitilde() _____ n
 Calculate potential for ClI 2+ + 2e- /textasciitilde() Cu. [Ans. 0.325 volt]

[GRB (O.P. Tandon) p.63 Q83]

What is the standard electrode potential for the electrode MnO₄⁻/MnO₂ in solution?

[GRB (O.P. Tandon) p.63 Q84]

(Given: $E^\circ_{\text{O}_2/\text{H}_2\text{O}} = 1.51$ volt, $E^\circ_{\text{Mn}^{4+}/\text{Mn}^{2+}} = 1.7$ volt) [Hint: 1.23 volt) $\text{MnO}_2 + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$; $E^\circ_{\text{O}_2/\text{H}_2\text{O}} = 1.23$ volt $\text{MnO}_2 + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{MnO} + 2\text{H}_2\text{O}$; $E^\circ_{\text{O}_2/\text{H}_2\text{O}} = ?$ $E^\circ_{\text{O}_2/\text{H}_2\text{O}} = 5 \times 1.51 - 2 \times 1.23 = 7.55 - 2.46 = 5.09$ volt] 333 What ratio of Pb^{2+} to Sn^{2+} concentration is needed to reverse the following cell reaction? $\text{Sn(s)} + \text{Pb}^{2+}(\text{aq.}) \rightarrow \text{Sn}^{2+}(\text{aq.}) + \text{Pb(s)}$ $E^\circ_{\text{O}_2/\text{H}_2\text{O}} = -0.136$ volt and $E^\circ_{\text{Sn}^{2+}/\text{Sn}} = 0.126$ volt [Ans.

[GRB (O.P. Tandon) p.63 Q85]

for the cell $\text{Mg} \mid \text{Mg}^{2+} \parallel \text{Ag}^+ \mid \text{Ag}$, calculate the equilibrium constant at 25°C and the maximum work that can be obtained during the operation of the cell. (Given: $E^\circ_{\text{Mg}^{2+}/\text{Mg}} = -2.37$ volt and 0.80 volt) $\text{Mg}/\text{l g}^{-1}$. [Ans. $K = 2.86 \times 10^{107}$; $W_{\text{max}} = 6.118 \times 10^5 \text{ J}$].

[GRB (O.P. Tandon) p.64 Q86]

Determine the potential for the cell: Pt | Fe²⁺, Fe³⁺ || Cr²⁺, Cr³⁺ | Pt in which [Fe²⁺] and [Fe³⁺] are 0.5M and 0.75M respectively and [Cr²⁺] and [Cr³⁺] are 2M, 4M and 1M respectively. Given: $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.770 \text{ volt}$, $14\text{H}^+ + 6\text{e}^- + \text{Cr}_2\text{O}_7^{2-} = 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$; $E^\circ = 1.35 \text{ volt}$ [Ans. 0.56 volt]

Problem 105**[GRB (O.P. Tandon) p.64 Q87]**

For the measurement of the solubility product of AgCl the following cell is constructed: Ag | AgCl | KCl (0.1 M) || AgNO₃ (0.1 M) | Ag. The emf of the cell is 0.45 volt. In the cell, KCl is dissociated to the extent of 83/86. [Ans. 1.735×10^{-10}] [Hint: $E_{\text{cell}} = E^{\circ}_{\text{cell}} - 0.0591 \log K_{\text{sp}}$]

Problem 106**[GRB (O.P. Tandon) p.64 Q88]**

Excess of AgCl is added to 0.1 M solution of KBr at 298 K. Calculate the equilibrium concentrations of Br⁻ and Cl⁻ ions. $E^{\circ}_{\text{AgCl/Ag}} = 0.222$ volt, $E^{\circ}_{\text{AgBr/Ag}} = 0.095$ volt. [Ans. $[\text{Br}^-] = 0.09929$ mol L⁻¹, $[\text{Cl}^-] = 0.00071$ mol L⁻¹] [Hint: $\text{AgCl} + \text{Br}^- \rightleftharpoons \text{AgBr} + \text{Cl}^-$ At equilibrium, $(0.1 - x) \times \text{Apply } 1 \text{ K.E. cell} = -0.0591 \log (0.1 - x) = 0.0591$]

Problem 107**[GRB (O.P. Tandon) p.64 Q89]**

The emf of the cell Cu | CuSO₄ || CuSO₄ | Cu is -0.03 V at $a = 0.1 \times 25^{\circ}\text{C}$. Calculate the activity of copper sulphate solution in the right hand side electrode. [Ans. 0.009616 mol L⁻¹] [Hint: Apply $E = E^{\circ} - 0.0591 \log Q$]

Problem 108**[GRB (O.P. Tandon) p.64 Q90]**

The cell, Pt | H₂ (1 atm) || H⁺ (pH = x) | normal calomel electrode, has an emf of 0.67 volt at 25°C . Calculate the pH of the solution. The oxidation potential of calomel electrode on the hydrogen scale is -0.28 volt. [Ans. pH = 6.61]

Problem 109**[GRB (O.P. Tandon) p.64 Q91]**

The emf of the cell, Ag | AgI, KI (0.05 M) || AgNO₃ (0.05 M) | Ag is 0.78 volt. Calculate the solubility product of AgI. [Ans. 1.101×10^{-16}]

Problem 110**[GRB (O.P. Tandon) p.64 Q92]**

The standard reduction potential for Cu²⁺ / Cu is +0.34 V. Calculate the reduction potential at pH = 14 for the above couple. K_{sp} of Cu(OH)₂ is 1.0×10^{-19} . [Ans. -0.22 V] (TIT 1996)

Problem 111**[GRB (O.P. Tandon) p.64 Q93]**

Calculate the potential of a cell in which H⁺ electrode is immersed in a solution of pH 3.5 and in a solution of pH 10.7. [Ans. 0.4255 volt]

Problem 112**[GRB (O.P. Tandon) p.64 Q94]**

Calculate the emf of the cell, Cl₂ (1 atm) | NaCl (aq.) | Cl₂ (0.1 atm) [Ans. -0.0295 volt]

Problem 113**[GRB (O.P. Tandon) p.64 Q95]**

The standard free energy change for the reaction, $\text{H}_2(\text{g}) + 2\text{AgCl}(\text{s}) \rightleftharpoons 2\text{Ag}(\text{s}) + 2\text{H}^+(\text{aq.}) + 2\text{Cl}^-(\text{aq.})$ is -10.26 kcal mol⁻¹ at 25°C . A cell using the above reaction is operated at 25°C under $P_{\text{H}_2} = 1$ atm, $[\text{H}^+] = 1$ and $[\text{Cl}^-] = 0.1$. Calculate the emf of the cell. [Ans. 0.340 volt]

Problem 114**[GRB (O.P. Tandon) p.64 Q96]**

For the reaction, $\text{Fe}^{3+} + 3\text{e}^- \rightarrow \text{Fe}$. E° is - 0.036 volt and the standard electrode potential for $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$ is - 0.771 volt. Calculate E° for $\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$. [Ans. - 0.4395 volt]

Problem 115**[GRB (O.P. Tandon) p.64 Q97]**

The standard reduction potential at 25°C of the reaction $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$ is - 0.8277 volt. Calculate the equilibrium constant for the reaction, $2\text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{OH}^-$ at 25°C [Ans. 1.014×10^{-14}] [Hint: $\text{H}_2 + 2\text{e}^- \rightarrow 2\text{H}$ (Cathode); $E^\circ = -0.8277$ volt $\text{H}_2 + 2\text{H}^+ \rightarrow 2\text{H}_2$ (Anode); $E^\circ = 0$ for the cell = - 0.8277 volt Apply now $E^\circ = 0.0591 \log K$, $n = 1$].

Problem 116**[GRB (O.P. Tandon) p.64 Q98]**

Calculate the emf of the following cell: $\text{Pt}(\text{H}_2, 1 \text{ atm}) | \text{CH}_3\text{CH}_2\text{COOH} (0.15 \text{ M}) || 0.01 \text{ M NH}_4\text{OH} | \text{H}_2 (1 \text{ atm}) | \text{Pt}$ for $\text{CH}_3\text{CH}_2\text{COOH} = 1.4 \times 10^{-5}$ K_b for $\text{NH}_4\text{OH} = 1.8 \times 10^{-5}$ [Ans. - 0.4603 volt] [Hint: $\text{CH}_3\text{CH}_2\text{COOH} \rightleftharpoons \text{CH}_3\text{CH}_2\text{COO}^- + \text{H}^+$ $K_a = \frac{0.15 \times 1.4 \times 10^{-5}}{1.449 \times 10^{-3}}$ [OH] in $\text{NH}_4\text{OH} = \frac{0.01 \times 1.8 \times 10^{-5}}{0.4242 \times 10^{-14}}$ $[\text{H}^+] = 2.3573 \times 10^{-14}$, 0.4242×10^{-14} $[\text{H}^+] = 0.0591 \log \frac{[\text{H}^+]}{[\text{H}_2]^{1/2}}$ $E_{\text{cell}} = 0.0591 \log \frac{[\text{H}_2]^{1/2}}{[\text{H}^+]}$]

Problem 117**[GRB (O.P. Tandon) p.64 Q99]**

Calculate equilibrium constant for $\text{I}_2 + 2\text{I}^- \rightleftharpoons \text{I}_3^-$ at 298 K from the following information: $\text{I}_2(\text{aq.}) + 2\text{e}^- \rightarrow 2\text{I}^-$; $E^\circ = 0.6197$ volt $\text{I}_3^- + 2\text{e}^- \rightarrow 3\text{I}^-$; [Ans. 706.9] [Hint: $\text{I}_2 + 2\text{e}^- \rightarrow 2\text{I}^-$; $E^\circ = 0.5355$ volt $\text{I}_3^- + 2\text{e}^- \rightarrow 3\text{I}^-$; $E^\circ = -0.5355$ volt $E^\circ = 0.6197 - 0.5355 = 0.0842$ volt]

Problem 118**[GRB (O.P. Tandon) p.65 Q100]**

101. K antilo' [$nE^\circ = \text{antilo}$ $[2 \times 0.0842 \times 706.9] \text{ g}$ 0.0591 g 0.0591 A lead storage battery has initially 200 g of lead and 200 g of PbO_2 , plus excess H_2SO_4 , How long could this cell deliver a current of 10 amp, without recharging, if it was possible to operate it so that the reaction goes to completion? [Ans. 4.48 hour] $\text{Zn}(\text{s}) + 2\text{AgCl}(\text{s}) \rightarrow \text{ZnCl}_2 (0.555 \text{ M}) + 2\text{Ag}(\text{s})$ $E^\circ = 1.015$ volt ($dE = -4.02 \times 10^{-4}$ volt per degree. dT IP Find ΔG , ΔH , ΔS).

Problem 119**[GRB (O.P. Tandon) p.65 Q105]**

What will be the value of Λ° for 0.001 M aqueous NH_3 solution? $K_b = 1.6 \times 10^{-5}$ and $\Lambda^\circ = 2.38 \times 10^{-2} \text{ } \Omega^{-1} \text{ m}^2 \text{ mol}^{-1}$ [Ans. $2.998 \times 10^{-3} \text{ } \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$] [Hint: $K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]} = \frac{0.126 \times 1.6 \times 10^{-5}}{0.001 \times a^2}$ $a = 0.126$ (Λ° , = molar conductance at concentration 'c' and Λ° : molar conductance at infinite dilution) $\frac{\Lambda^\circ}{\Lambda} = \frac{1000}{c}$ $\Lambda^\circ = \frac{1000}{0.001} \times \frac{\Lambda}{1000} = 1000 \times \frac{0.126}{1000} = 0.126$ $\Lambda^\circ = 1.95 \times 10^{-3} \text{ } \Omega^{-1} \text{ m}^2 \text{ mol}^{-1}$ $\Delta G^\circ = -nFE^\circ = -2 \times 96500 \times 1.015 = -195895 \text{ J}$, $\Delta H^\circ = -195.895 \text{ kJ}$. $\Delta S^\circ = \frac{\Delta H^\circ - \Delta G^\circ}{T} = \frac{-195.895 \times 10^3 - (-195895)}{273} = -0.07758 \text{ kJ} = -77.58 \text{ J} = -18.55 \text{ cal}$]

Problem 120**[GRB (O.P. Tandon) p.65 Q102]**

Calculate the equilibrium constant for the reaction, $2\text{Fe}^{3+} + 3\text{r} \rightarrow 2\text{Fe}^{2+} + 13$

Problem 121**[GRB (O.P. Tandon) p.65 Q103]**

The standard reduction potentials in acidic conditions are 0.77 and 0.54 V respectively for $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$ and $\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{Mn}^{2+} + 2\text{H}_2\text{O}$. Electrolysis of a solution of MnSO_4 in aqueous sulphuric acid is a method for the preparation of MnO_2 as the per reaction, $\text{Mn}^{2+}(\text{aq.}) + 2\text{H}_2\text{O} \rightarrow \text{MnO}_2 + 2\text{H}^+(\text{aq.}) + \text{H}_2$. Passing a current of 27 A for 24 hours gives one kg of MnO_2 . What is the value of current efficiency? Write the reactions taking place at the cathode and at the anode. IIT (May) 1997 [Hint: Apply $E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} = 0.77 - 0.54 = 0.23$ V. $Q = I \times t = 27 \times 24 \times 60 \times 60 = 2.29 \times 10^6$ C. $n = 2$ for MnO_2 . $Q = n \times F \times \text{moles of } \text{MnO}_2 = 2 \times 96500 \times \text{moles of } \text{MnO}_2$. $\text{moles of } \text{MnO}_2 = \frac{1000}{86.94} = 11.5$. $2.29 \times 10^6 = 2 \times 96500 \times 11.5 \times \text{efficiency}$. $\text{efficiency} = \frac{2.29 \times 10^6}{2 \times 96500 \times 11.5} = 104.8\%$ Reactions: Anode: $\text{Mn}^{2+} \rightarrow \text{MnO}_2 + 2\text{H}^+ + 2\text{e}^-$ Cathode: $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$]

Problem 122**[GRB (O.P. Tandon) p.65 Q104]**

Calculate the number of kWh of electricity is necessary to produce 1 metric ton (1000 kg) of aluminium by Hall process in a cell operating at 15 V [Ans. 4.47×10^4 kWh]

Problem 123**[GRB (O.P. Tandon) p.65 Q106]**

107. A weak monobasic acid is 5% solution. The limiting molar conductivity at infinite dilution is $4.00 \times 10^{-2} \text{ } \Omega^{-1} \text{ m}^2 \text{ mol}^{-1}$. Calculate the conductivity of a 0.05 molar solution of the acid. [Ans. $8.92 \times 10^{-4} \text{ } \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$] [Hint: Dissociation constant of the acid $K_a = \frac{C\alpha^2}{1-\alpha}$. $2.5 \times 10^{-5} = \frac{0.05 \times \alpha^2}{1-\alpha}$. $\alpha = 0.0223$. We know that, $\Lambda = \Lambda^\infty \alpha$. $\Lambda = 4.00 \times 10^{-2} \times 0.0223 = 8.92 \times 10^{-4} \text{ } \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$] 1.05 g of a lead ore containing impurity of Ag was dissolved in quantity of HNO_3 and the volume was made 350 mL. A Ag electrode was dipped in the solution and Ecell of, Pt(H_2) | H^+ (1 M) || Ag^+ | Ag was 0.503 V at 298 K. Calculate E° . 0.80 V $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$ [Ans. 0.0339 V] [Hint: Anode: $\text{H}_2(\text{g}) \rightarrow 2\text{H}^+ (1 \text{ M}) + 2\text{e}^-$ Cathode: $\text{Ag}^+ (x) + \text{e}^- \rightarrow \text{Ag}(\text{s})$ $E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} = 0.80 - 0 = 0.80$ volt $Q = E_{\text{cell}} \times n \times F = 0.80 \times 1 \times 96500 = 77200$ C. $Q = n \times F \times \text{moles of } \text{Ag}^+ = 1 \times 96500 \times \text{moles of } \text{Ag}^+$. $\text{moles of } \text{Ag}^+ = \frac{77200}{96500} = 0.80$ g of Ag = 0.80 g. $1.05 - 0.80 = 0.25$ g of impurity]

Problem 124**[GRB (O.P. Tandon) p.66 Q108]**

Calculate E° of the following half-cell reaction at 298 K: $\text{Ag}(\text{NH}_3)_2^+ + \text{e}^- \rightarrow \text{Ag} + 2\text{NH}_3$ [Ans. 0.373 volt] [ADS. 0.373 volt] $E^\circ = 0.80 \text{ V} - 0.427 \text{ V} = 0.373 \text{ V}$ Set-1: Questions with single correct answer

Problem 125**[GRB (O.P. Tandon) p.66 Q1]**

An electrolyte is a substance which: (a) conducts electricity (b) decomposes on heating (c) is acidic in nature (d) when dissolved in water, dissociates into ions

Problem 126**[GRB (O.P. Tandon) p.66 Q2]**

The theory of ionisation was presented by: (a) Faraday (b) Arrhenius (c) Ostwald (d) Rutherford

Problem 127**[GRB (O.P. Tandon) p.66 Q3]**

Dissociation of an electrolyte in water into negative and positive ions is called: (a) ionisation (b) electrolysis (c) decomposition (d) hydrolysis

Problem 128**[GRB (O.P. Tandon) p.66 Q4]**

Degree of ionisation is equal to: (a) total number of moles of the electrolyte present in solution (b) total number of moles of the electrolyte dissociated into ions (c) number of moles dissociated / total number of moles dissolved (d) total number of moles dissolved / number of moles dissociated

Problem 129**[GRB (O.P. Tandon) p.66 Q5]**

Conductivity of aqueous solution of an electrolyte depends on: (a) molecular mass of the electrolyte (b) boiling point of solvent (c) degree of ionisation (d) volume of the solvent

Problem 130**[GRB (O.P. Tandon) p.66 Q6]**

Degree of ionisation does not depend on: (a) nature of the solvent (b) nature of the electrolyte (c) dilution (d) molecular mass of the electrolyte

Problem 131**[GRB (O.P. Tandon) p.66 Q7]**

Substances which give good conducting aqueous solution are called: (a) weak electrolytes (b) strong electrolytes (c) non-electrolytes (d) catalysts

Problem 132**[GRB (O.P. Tandon) p.66 Q8]**

The number of ions given by one molecule of $K_4Fe(CN)_6$ after complete dissociation is: **[Hint: Anode EO 0.80 volt $Q = [Ag + HNH_3] 2 [Ag(NH_3)_2^+ Ag(s) + Ag^+ + 2NH_3$ 6×10^{-8} (n I) $E^\circ_{text{asciitilde{ell}}(v - 0,80)$ At equilibrium, $E = 0$.. $E = E^\circ - \frac{0.0591}{n} \log Q$ (a) 5 (b) 11 (c) 2 (d) 10]**

Problem 133**[GRB (O.P. Tandon) p.66 Q9]**

The amount of electricity required to produce one mole of copper from copper sulphate solution will be .

Problem 134**[GRB (O.P. Tandon) p.66 Q10]**

(a) 1 faraday (c) 2 faraday **[Hint: $Cu^{2+} + 2e^-$ (VITEEE 2008) (b) 2.33 faraday (d) 1.33 faraday Cu (s) 1 mole copper requires 2 mole electron, i.e. 2 faraday charge.]** The process in which chemical change occurs on passing electricity is termed: (a) ionisation (b) neutralisation (c) electrolysis (d) hydrolysis

Problem 135**[GRB (O.P. Tandon) p.66 Q11]**

Which of the following condition is correct for operation of electrolytic cell? (a) $b.G = 0, E = 0$ (c) $b.G > 0, E < 0$ (b) $b.G < 0, E > 0$ (d) $b.G > 0, E > 0$

Problem 136**[GRB (O.P. Tandon) p.66 Q12]**

Which one is the correct equation that represents the first law of electrolysis? (a) $mZ = et$ (b) $m = eZt$ (c) $me = Zt$ (d) $e = mZt$ **13..** When one coulomb of electricity is passed through an electrolytic solution, the mass deposited on the electrode is equal to: (a) equivalent weight (b) molecular weight (c) electrochemical equivalent (d) one gram

Problem 137**[GRB (O.P. Tandon) p.66 Q14]**

One faraday is equal to: (a) 9650 coulomb (c) 19640 coulomb (b) 10,000 coulomb (d) 96500 coulomb

Problem 138**[GRB (O.P. Tandon) p.66 Q15]**

When one faraday of electric current is passed, the mass deposited is equal to: (a) one gram equivalent (b) one gram mole (c) electrochemical equivalent (d) half gram equivalent

Problem 139**[GRB (O.P. Tandon) p.66 Q16]**

On passing one faraday of electricity through a dilute solution of an acid, the volume of hydrogen obtained at NTP is: (a) 22400 mL (b) 1120 mL (c) 2240 mL (d) 11200 mL

Problem 140**[GRB (O.P. Tandon) p.67 Q17]**

w g of copper is deposited in a copper voltan1.eter when an electric current of 2 ampere is passed for 2 hours. If one ampere of electric current is passed for 4 hours in the same voltameter, copper deposited will be: (a) w (b) w/2 (c) w/4 (d) 2w

Problem 141**[GRB (O.P. Tandon) p.67 Q18]**

Copper sulphate solution is electrolysed between two platinum electrodes. A current is passed until 1.6 g of oxygen is liberated at anode. Thl/textasciitilde() amount of copper deposited at the cathode during the same period is: (a) 6.36 g (b) 63.6 g (c) 12.7 g (d) 3.2 g

Problem 142**[GRB (O.P. Tandon) p.67 Q19]**

When electricity is passed through a solution of AlCl_3 13.5 g

Problem 143**[GRB (O.P. Tandon) p.67 Q28]**

The extent of ionisation increases: (a) with increase in concentration of the electrolyte (b) on decreasing temperature of solution (c) on addition of excess of water (d) on stirring the solution vigorously

Problem 144**[GRB (O.P. Tandon) p.67 Q29]**

Molten sodium chloride conducts electricity due to the presence of: (a) free electrons (b) free ions (c) free molecules (d) free atoms of Na and Cl

Problem 145**[GRB (O.P. Tandon) p.67 Q30]**

When NaCl is dissolved in water, the sodium ion is: (a) oxidised (b) reduced (c) hydrolysed (d) hydrated of Al is discharged. The amount of charge passed is:

Problem 146**[GRB (O.P. Tandon) p.67 Q31]**

A solution of sodium sulphate in water is electrolysed using /textasciitilde() -' -.....(a)1.5 .. r- lo) — o/textasciitilde() — r — (c) — t:U/textasciitilde() /textasciitilde() (G) — r.OF — - m — ptatinum- etelectrod'es — :-The proauctsafCathooe and' anode are 20.' When the same electric current is passed through the solution of different electrolytes in series, the amounts of elements deposited on the electrodes are in the ratio oftheir: (a) atomic numbers (b) atomic masses . ____ . ____ /textasciitilde() ____ /textasciitilde() ____ M ____ /cdot ____ /cdot/textasciitilde() ____ ____ . ____ . ____ ... ____ .. /textasciitilde() ____ /textasciitilde() ____, ____ . ____ .. (c) specific gravities (d) equivalent masses

Problem 147**[GRB (O.P. Tandon) p.67 Q21]**

Faraday's laws of electrolysis are related to: ____ (a) atomic number of the cation (b) atomic number of the anion (c), equivalent mass of the electrolyte (d) speed of the cation

Problem 148**[GRB (O.P. Tandon) p.67 Q22]**

The specific conductance of a 0.01 M solution of KCl is 0.0014 $\Omega^{-1}\text{cm}^{-1}$ at 25°C. Its equivalent conductance is: (a) 14 (b) 140 (c) 1.4 (d) 0.14

Problem 149**[GRB (O.P. Tandon) p.67 Q23]**

The equivalent conductivity of 0.1 N CH_3COOH at 25°C is 80 and at infinite dilution 400 $\Omega^{-1}\text{cm}^2\text{mol}^{-1}$. The degree of dissociation of CH_3COOH is: (a) 1 (b) 0.2 (c) 0.1 (d) 0.5

Problem 150**[GRB (O.P. Tandon) p.67 Q24]**

The number of electrons involved when one faraday of electricity is passed through an electrolytic solution is: (a) 96500 (b) 8×10^6 (c) 12×10^{16} (d) 6×10^{23}

Problem 151**[GRB (O.P. Tandon) p.67 Q25]**

One faraday of charge was passed through the electrolytic cells placed in series containing solutions of Ag^+ , Ni^{2+} and Cr^{3+} respectively. The amount of Ag (At. mass 108), Ni (At. mass 59) and Cr (At. mass 52) deposited will be: Ag Ni Cr (a) 108g 29.5g 17.5g (b) 108 g 59.0 g 52.0 g (c) 108g 108.0g 108.0g (d) 108g 117.5g 166.0g

Problem 152**[GRB (O.P. Tandon) p.67 Q26]**

One faraday of electricity will liberate one gram mole of the metal from the solution of: (a) BaCl_2 (b) CuSO_4 (c) AlCl_3 (d) NaCl

Problem 153**[GRB (O.P. Tandon) p.67 Q27]**

Strong electrolytes are those which: (a) dissolve readily in water (b) conduct electricity (c) dissociate into ions even at high concentration (d) dissociate into ions at high dilution respectively: (a) H_2O_2 (c) Na_2O (b) H_2 (d) SO_2

Problem 154**[GRB (O.P. Tandon) p.67 Q32]**

The electric conduction of a salt solution in water depends on: (a) shape of molecules (b) size of its molecules (c) size of solvent molecules (d) extent of its ionisation

Problem 155**[GRB (O.P. Tandon) p.67 Q33]**

In electroplating, the article to be electroplated is connected as: (a) cathode (b) anode (c) anode (d) cathode

Problem 156**[GRB (O.P. Tandon) p.67 Q34]**

The amount of electricity that can deposit 108 g of silver from silver nitrate solution is: (AFMC 1993) (a) 1 ampere (b) 1 coulomb (c) 1 faraday (d) 2 ampere

Problem 157**[GRB (O.P. Tandon) p.67 Q35]**

A certain current liberated 0.504 g of hydrogen in 2 hours. How many grams of copper can be liberated by the same current flowing for the same time in CuSO_4 solution? (a) 12.7g (b) 15.9g (c) 31.8g (d) 63.5g

Problem 158**[GRB (O.P. Tandon) p.67 Q36]**

If the specific resistance of a solution of concentration C g equivalent litre⁻¹ is R , then its equivalent conductance is: (a) $100R$ (b) RC (c) 1000 (d) $C \times 1000 RC \times 1000R$

Problem 159**[GRB (O.P. Tandon) p.67 Q37]**

If the specific conductance and conductance of a solution are same, then its cell constant is equal to: (a) 1 (b) 0 (c) 10 (d) 100

Problem 160**[GRB (O.P. Tandon) p.67 Q38]**

On increasing the dilution, the specific conductance: (Jiwaji 1990) (a) increases (b) decreases (c) remains constant (d) none of these

Problem 161**[GRB (O.P. Tandon) p.67 Q39]**

The distance between two electrodes of a cell is 2.5 cm and area of each electrode is 5 cm² /bullet The cell constant is: (Jabalpur 1990) (a) 2 (b) 12.5 (c) 7.5 (d) 0.5

Problem 162**[GRB (O.P. Tandon) p.67 Q40]**

At 25°C , the molar conductances at infinite dilution for the strong electrolytes NaOH, NaCl and BaCl₂ are 248×10^{-4} , 126×10^{-4} and 280×10^{-4} S m² mol⁻¹ respectively. $\Lambda^\infty(\text{Ba(OH)}_2)$ in S m² mol⁻¹: (EAMCET 20(9)) (a) 52.4×10^{-4} (b) 524×10^{-4} (c) 402×10^{-4} (d) 262×10^{-4} [Hint : $\Lambda^\infty(\text{Ba(OH)}_2) = \Lambda^\infty(\text{BaCl}_2) + 2\Lambda^\infty(\text{NaOH}) - 2\Lambda^\infty(\text{NaCl}) = 280 \times 10^{-4} + 2 \times 248 \times 10^{-4} - 2 \times 126 \times 10^{-4} = 524 \times 10^{-4}$ S m² mol⁻¹]

Problem 163**[GRB (O.P. Tandon) p.68 Q41]**

The electrochemical cell stops working after some time because: (VITEEE 2008) (a) electrode potential of both the electrodes becomes zero (b) electrode potential of both the electrodes becomes equal (c) one of the electrodes is eaten away (d) the cell reaction gets reversed

Problem 164**[GRB (O.P. Tandon) p.68 Q42]**

Which reaction will take place at cathode when fused calcium chloride is electrolysed? (a) $\text{Ca}^{2+} + 2e^- \rightarrow \text{Ca}$ (b) $\text{Cl}^- \rightarrow \text{Cl} + e^-$

Problem 165**[GRB (O.P. Tandon) p.68 Q43]**

In electrolysis oxidation takes place at: (a) both the electrodes (b) cathode (c) anode (d) in the solution

Problem 166**[GRB (O.P. Tandon) p.68 Q44]**

The equation representing the process by which standard reduction potential of zinc can be defined is: (a) $\text{Zn}^{2+}(\text{s}) + 2e^- \rightarrow \text{Zn}(\text{s})$ (b) $\text{Zn}^{2+}(\text{aq}) + 2e^- \rightarrow \text{Zn}(\text{s})$ (c) $\text{Zn}^{2+}(\text{g}) + 2e^- \rightarrow \text{Zn}(\text{s})$ (d) $\text{Zn}^{2+}(\text{aq.}) + 2e^- \rightarrow \text{Zn}(\text{s})$

Problem 167**[GRB (O.P. Tandon) p.68 Q45]**

The measured potential for, $\text{Mg}^{2+} + 2e^- \rightarrow \text{Mg}(\text{s})$ does not depend upon: (a) raising the temperature (b) increasing the concentration of Mg^{2+} ions (c) making the magnesium plate bigger (d) purity of magnesium plate

[GRB (O.P. Tandon) p.68 Q46]

All cells do not contain: (a) an anode (b) a cathode (c) ions (d) a porous partition

[GRB (O.P. Tandon) p.68 Q47]

When lead accumulator is charged, it is: (a) an electrolytic cell (b) a galvanic cell (c) a Daniell cell (d) none of these

[GRB (O.P. Tandon) p.68 Q48]

The standard electrode potentials for the reactions, $\text{Ag}^+(\text{aq.}) + \text{e}^- \rightarrow \text{Ag(s)}$ $\text{Sn}^{2+}(\text{aq.}) + 2\text{e}^- \rightarrow \text{Sn(s)}$ at 25°C are 0.80 volt and - 0.14 volt respectively. The emf of the cell, $\text{Sn} | \text{Sn}^{2+}(1 \text{ M}) || \text{Ag}^+(1 \text{ M}) | \text{Ag}$ is: (a) 0.66 volt (b) 0.80 volt (c) 1.08 volt (d) 0.94 volt

[GRB (O.P. Tandon) p.68 Q49]

The cathodic reaction in electrolysis of dilute H_2SO_4 with platinum electrode is: (a) oxidation (b) reduction (c) both oxidation and reduction (d) neutralization

[GRB (O.P. Tandon) p.68 Q50]

Strongest reducing agent is: (a) K (b) Mg (c) Al (d) I

[GRB (O.P. Tandon) p.68 Q51]

The metal oxide which decomposes on heating, is: (a) ZnO (b) HgO (c) Al_2O_3 (d) CuO

[GRB (O.P. Tandon) p.68 Q52]

Thereaction./textasciitilde()H2(g) + AgCl(s) = H + (aq.) + Cnaq.) + Ag(s) occurs in the galvanic cell: (AFMC 2009) (a) I AgCl(s) I KCl (soln.) II AgNO3 (soln.) lAg (b) Pt I H2(g) I HCl (soln.) II AgNO3 (soln.) I Ag (c) 'Pt I H2(g) I HCl (soln.) II AgCl(s) I Ag (d) Pt I H2(g) I KCl (soln.) II AgCl(s) I Ag

[GRB (O.P. Tandon) p.68 Q53]

The standard oxidation potentials, EO_j, for the half reactions are as, Zn /textasciitilde() Zn²⁺ + 2e⁻; EO = + 0.76 volt Pe /textasciitilde() Pe²⁺ + 2e⁻; EO = + 0.41 volt lneemfofthecen/textasciitilde()/textasciitilde()+-+Zn /textasciitilde() Zn²⁺+;-j:-Pe IS: ICET (Karnataka) 2009] (a) + 0.35 volt (b) - 0.35 volt (c) +1.17 volt (d) -1.17 volt

[GRB (O.P. Tandon) p.68 Q54]

The standard reduction-potentials (at 25°C) of the following half reactions are given against each; . . . 55:

Problem 177**[GRB (O.P. Tandon) p.68 Q58]**

59. $\text{Zn}^{2+}(\text{aq.}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$; $\text{Cr}^{3+}(\text{aq.}) + 3\text{e}^- \rightarrow \text{Cr}(\text{s})$; $2\text{W} + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$; $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$; Which is the strongest reducing agent? (a) Zn (b) Cr (c) $\text{H}_2(\text{g})$ (d) $\text{Fe}^{2+}(\text{aq.})$ Hydrogen gas will not reduce heated: - 0.762 - 0.740 0.00 0.77 (VITEEE 2007) (a) cupric oxide (b) ferric oxide (c) stannic oxide (d) aluminium oxide A solution containing one mole per litre of each $\text{Cu}(\text{NO}_3)_2$; AgNO_3 ; $\text{Hg}_2(\text{NO}_3)_2$ and $\text{Mg}(\text{NO}_3)_2$ is being electrolysed by using inert electrodes. The values of standard electrode potentials (reduction potentials) are $\text{Ag}/\text{Ag}^+ 0.80$ volt, $2\text{H}^+/\text{H}_2 = 0.79$ volt, $\text{Cu}/\text{Cu}^{2+} = 0.24$ volt, $\text{Mg}/\text{Mg}^{2+} = -2.37$ volt. With increasing voltage, the sequence of deposition of metals on the cathode will be: [PMT (Kerala) 2004] (a) Ag, Hg, Cu (b) Cu, Hg, Ag (c) Ag, Hg, Cu, Mg (d) Mg, Cu, Hg, Ag Four colourless salt solutions are placed in separate test tubes and a strip of copper is dipped in each. Which solution finally turns blue? (a) $\text{Pb}(\text{NO}_3)_2$ (b) AgNO_3 (c) $\text{Zn}(\text{NO}_3)_2$ (d) $\text{Cd}(\text{NO}_3)_2$ Red hot carbon will remove oxygen from the oxide XO and YO but not from ZO. Y will remove oxygen from XO. Use this evidence to deduce the order of activity of the three metals X, Y and Z putting the most active first: (a) XYZ (b) ZYT (c) YXZ (d) ZXY Which of the following metals does not give the following reaction? $\text{M} + \text{Water} \rightarrow \text{Oxide or hydroxide} + \text{H}_2$ (a) Iron (b) Sodium (c) Mercury (d) Magnesium

Problem 178**[GRB (O.P. Tandon) p.69 Q60]**

Which is the best reducing agent? (a) F (b) Cl^- (c) Br^- (d) I^-

Problem 179**[GRB (O.P. Tandon) p.69 Q61]**

If a spoon of copper metal is placed in a solution of ferrous sulphate: (a) Cu will precipitate out (b) iron will precipitate (c) Cu and Fe will precipitate (d) no reaction will take place

Problem 180**[GRB (O.P. Tandon) p.69 Q62]**

Among Na, Hg, S, Pt and graphite, which can be used as electrodes in electrolytic cells having aqueous solutions? (a) Hg and Pt (b) Hg, Pt and graphite and S

Problem 181**[GRB (O.P. Tandon) p.69 Q63]**

The most reactive metal among the following is: (a) Al (b) Ni (c) Pb (d) Cu

Problem 182**[GRB (O.P. Tandon) p.69 Q64]**

Which of the following metals is most readily corroded in moist air? (a) Copper (b) Iron (c) Silver (d) Nickel

Problem 183**[GRB (O.P. Tandon) p.69 Q65]**

Which one will liberate Br_2 from KBr? (a) HI (b) I_2 (c) Cl_2 (d) SO_2

Problem 184**[GRB (O.P. Tandon) p.69 Q66]**

Which one of the following is not the correct representation? (a) Red. pot. of cathode + Oxid. pot. of anode (b) $E_{\text{cell}} :: \text{Red. pot. of cathode} - \text{Oxid. pot. of anode}$ (c) $E_{\text{cell}} :: \text{Red. pot. of cathode} + \text{Red. pot. of anode}$ (d) $E_{\text{cell}} :: \text{Oxid. pot. of cathode} + \text{Oxid. pot. of anode}$

[GRB (O.P. Tandon) p.69 Q67]

Which of the following represents the potential of silver wire dipped into 0.1 M AgNO₃ solution at 25°C? (a) (E; 0.059) (b) (E; 0.059) (c) (E; -0.059) (d) 0.059

[GRB (O.P. Tandon) p.69 Q68]

If the solution of the CuSO_4 in which copper rod is immersed is diluted to 10 times, the electrode potential: (a) increases by 0.030 volt (b) decreases by 0.030 volt (c) increases by 0.059 volt (d) decreases by 0.0059 volt

[GRB (O.P. Tandon) p.69 Q69]

A solution of Cu(II) sulphate is reacted with KCl and KI. In which case will the Cu²⁺ be reduced to Cu⁺? (a) In both the cases (b) When reacted with KCl (c) When reacted with KI (d) In both the case/textasciitilde() but in presence of H⁺

[GRB (O.P. Tandon) p.69 Q70]

From the electrochemical series, it can be concluded that: (a) Zn^{2+} will liberate H_2 from 1 M HCl . (b) Ag metal reacts spontaneously with Zn^{2+} (c) Zn metal will liberate H_2 from 1 M HCl (d) Ag metal will liberate H_2 from 1 M HCl

[GRB (O.P. Tandon) p.69 Q71]

The potential of a hydrogen electrode at pH = 1 is: (a) 0.059 volt (b) 0 volt (c) 0.059 volt (d) 0.59 volt

[GRB (O.P. Tandon) p.69 Q72]

10. Which is not true for a standard hydrogen electrode? (a) The hydrogen ion concentration is 1 M (b) Temperature is 25°C (c) Pressure of hydrogen is 1 atmosphere (d) It contains a metallic conductor which does not absorb H₂

[GRB (O.P. Tandon) p.69 Q73]

The electrode potential becomes equal to standard electrode potential when reactants and products ratio is: (a) equal to I (b) greater than I (c) less than I (d) none of these

[GRB (O.P. Tandon) p.69 Q74]

For the half-cell reaction, $\text{Au}^{3+} + 3\text{e}^- \rightarrow \text{Au}$ the value of n used in Nernst equation is: (a) 3 (b) 2 (c) 1 (d) 3×96500

When a piece of sodium metal is put in water, a reaction takes place to yield hydrogen gas because: (a) sodium loses electrons (b) sodium acts as an oxidising agent (c) _____ water loses electron (d) water acts as a reducing agent

[GRB (O.P. Tandon) p.69 Q76]

Which one is the wrong statement about electrochemical series? (a) Active metals have negative reduction potentials (b) Active nonmetals have positive reduction potentials (c) Metals above hydrogen liberate hydrogen from acids (d) Metals below hydrogen are strong reducing agents

Problem 194**[GRB (O.P. Tandon) p.69 Q77]**

The reduction potential values are given below: $3+ \text{Al} \rightarrow \text{Al} = -1.67$ volt, $\text{Mg} + 2\text{H}^+ \rightarrow \text{Mg} + \text{H}_2 = 2.34$ volt, $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu} = +0.34$ volt, $\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn} = -0.53$ volt. Which one is the best reducing agent? (a) Al (b) Mg (c) Cu (d) Zn

Problem 195**[GRB (O.P. Tandon) p.69 Q78]**

From the values given in question No. 77, which one is the best oxidising agent? (a) Al (b) Mg (c) Zn (d) Cu

Problem 196**[GRB (O.P. Tandon) p.69 Q79]**

When iron is rusted, it is: (a) reduced (b) oxidised (c) evaporated (d) decomposed

Problem 197**[GRB (O.P. Tandon) p.69 Q80]**

Galvanization of iron denotes coating with: (a) Cu (b) Sn (c) Zn (d) Al

Problem 198**[GRB (O.P. Tandon) p.69 Q81]**

The standard electrode potentials of four elements A, B, C and D are -3.05, 1.66, -0.40 and 0.80 volt. The highest chemical activity will be shown by: (a) A (b) B (c) C (d) D

Problem 199**[GRB (O.P. Tandon) p.69 Q82]**

Which of the following methods does not liberate hydrogen? (a) $\text{Zn} + \text{H}_2\text{SO}_4$ (dil.) (b) $\text{Mg} + \text{H}_2\text{SO}_4$ (dil.) (c) $\text{Cu} + \text{H}_2\text{SO}_4$ (dil.) (d) $\text{Zn} + \text{HCl}$ (dil.)

Problem 200**[GRB (O.P. Tandon) p.69 Q83]**

A depolarizer used in dry cell is: (a) ammonium chloride (b) manganese dioxide (c) potassium oxide (d) sodium phosphate. The oxide which can be reduced by hydrogen is: (a) Na_2O (b) CaO (c) K_2O (d) CuO

Problem 201**[GRB (O.P. Tandon) p.69 Q85]**

The reference electrode is made from which of the following? (a) ZnCl_2 (b) CuSO_4 (c) Hg_2Cl_2 (d) HgCl_2

Problem 202**[GRB (O.P. Tandon) p.70 Q86]**

Given, standard electrode potentials; $\text{Fe}^{3+} + 3\text{e}^- \rightarrow \text{Fe}$; $E^\circ = -0.036$ volt $\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$; $E^\circ = -0.440$ volt The standard electrode potential E° for $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$ is: (a) 0.476 volt (b) 0.404 volt (c) 0.440 volt (d) 0.772 volt [Hint: Apply $\Delta G^\circ = -nFE^\circ$ $\text{Fe}^{3+} + 2\text{e}^- \rightarrow \text{Fe}$ $\Delta G^\circ = 3 \times F \times (-0.036) = -0.108F$... (i) $\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$ $\Delta G^\circ = 2 \times F \times (-0.440) = -0.88F$... (ii) Subtracting eq. (i) from eq. (ii), $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$ $\Delta G^\circ = -0.772F$ $E^\circ = -\Delta G^\circ / nF = 0.772$ volt] Ca and Li -metals may be arranged in the decreasing order of their standard electrode potentials as: (a) K, Ca and Li (b) Li, K and Ca (c) Li, Ca and K (d) Ca, Li and K

Problem 203**[GRB (O.P. Tandon) p.70 Q88]**

In a galvanic cell energy changes occur like: (a) chemical energy $\rightarrow$ electrical energy (b) electrical energy $\rightarrow$ chemical energy (c) chemical energy $\rightarrow$ internal energy (d) internal energy $\rightarrow$ electrical energy

Problem 204**[GRB (O.P. Tandon) p.70 Q89]**

The reaction is spontaneous if the cell potential is: (a) positive (b) negative (c) zero (d) infinite

Problem 205**[GRB (O.P. Tandon) p.70 Q90]**

Is the reaction, $2\text{Al} + 3\text{Fe}^{2+} \rightarrow 2\text{Al}^{3+} + 2\text{Fe}$ possible? (a) No, because standard oxidation potential of $\text{Al} < \text{Fe}$ (b) Yes, because standard oxidation potential of $\text{Al} > \text{Fe}$ (c) Cannot be predicted (d) Yes, because aluminium is a strong oxidising agent

Problem 206**[GRB (O.P. Tandon) p.70 Q91]**

In an experimental set-up for the measurement of emf of a half-cell using a reference electrode and a salt bridge, when the salt bridge is removed, the voltage: (a) does not change (b) increases to maximum (c) decreases half the value (d) drops to zero

Problem 207**[GRB (O.P. Tandon) p.70 Q92]**

More electronegative elements have: (a) negative reduction potential (b) tendency to lose electrons (c) positive reduction potential (d) positive oxidation potential

Problem 208**[GRB (O.P. Tandon) p.70 Q93]**

Which metal will be deposited in the galvanic cell? $\text{Cu} \mid \text{Cu}^{2+} \parallel \text{Ag} \mid \text{Ag}$ (a) Cu (b) Ag (c) Both (d) None of these

Problem 209**[GRB (O.P. Tandon) p.70 Q94]**

In the cell $\text{Zn} \mid \text{Zn}^{2+} \parallel \text{Cu}^{2+} \mid \text{Cu}$, the negative electrode is: (a) Cu (b) Cu^{2+} (c) Zn (d) Zn^{2+}

Problem 210**[GRB (O.P. Tandon) p.70 Q95]**

Which of the following gains electrons more easily? (a) Na^+ (b) Zn^{2+} (c) Al^{3+} (d) H^+

Problem 211**[GRB (O.P. Tandon) p.70 Q96]**

The potential of a hydrogen electrode at pH 10 is: (a) 0.51 volt (b) 0 volt (c) -0.591 volt (d) 0.059 volt

Problem 212**[GRB (O.P. Tandon) p.70 Q97]**

The emf of the cell, $\text{Ni} \mid \text{Ni}^{2+} (1.0 \text{ M}) \parallel \text{Ag}^+ (1.0 \text{ M}) \mid \text{Ag}$ (0.25 volt, E° for $\text{Ag}^+/\text{Ag} = 0.80$ volt) is given by: (a) $-0.25 + 0.80 = 0.55$ volt (b) $-0.25 - (+0.80) = -1.05$ volt (c) $0 + 0.80 - (-0.25) = 1.05$ volt (d) $-0.80 - (-0.25) = -0.55$ volt

For the reaction of the cell, $\text{Zn} \mid \text{Zn}^{2+} (1.0 \text{ M}) \parallel \text{Cd}^{2+} (1.0 \text{ M}) \mid \text{Cd}$ is: (a) $\text{Cd} \rightarrow \text{Cd}^{2+} + 2\text{e}^-$ (b) $\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$ (c) $\text{Cd} + \text{Zn}^{2+} \rightarrow \text{Cd}^{2+} + \text{Zn}$ (d) $\text{Zn} + \text{Cd}^{2+} \rightarrow \text{Cd} + \text{Zn}^{2+}$

Problem 213**[GRB (O.P. Tandon) p.70 Q99]**

The position of some metals in the electrochemical series in decreasing electropositive character is given as $Mg > Al > Zn > Cu > Ag$. What will happen if a copper spoon is used to stir a solution of aluminium nitrate? (a) The spoon will get coated with aluminium (b) An alloy of aluminium and copper is formed (c) The solution becomes blue (d) There is no reaction

Problem 214**[GRB (O.P. Tandon) p.70 Q100]**

The half-cell reaction is the one that: (a) takes place at one electrode (b) consumes half a unit of electricity (c) involves half a mole of electrolyte (d) goes halfway to completion Which Nernst equation is true to find out the potential of non-standard electrochemical cell from the following? .
 $Fe(s) \rightleftharpoons Fe^{2+}(aq) + 2e^{-}$ (Pt) ICE: (Gujarat) 2008J . 0.0592 V (a) $E_{cell} = E^{\circ}_{cell} - \frac{0.0592}{n} \log \frac{[Fe^{2+}]}{[Fe]}$ (b) $E = E^{\circ} - 0.0592 \log \frac{[Fe^{2+}]}{[Fe]}$ (c) $E_{cell} = E^{\circ}_{cell} - \frac{0.0592}{n} \log \frac{[Fe^{2+}]}{[Fe]}$ (d) $E_{cell} = E^{\circ}_{cell} - \frac{0.0592}{n} \log [Fe]$

Problem 215**[GRB (O.P. Tandon) p.70 Q102]**

During the electrolysis of fused NaCl, which reaction occurs at anode? (a) Chloride ions are oxidised (b) Sodium ions are oxidised (c) Chloride ions are reduced (d) Sodium ions are reduced

Problem 216**[GRB (O.P. Tandon) p.70 Q103]**

$E^{\circ}(Ni^{2+}/Ni) = -0.25$ volt, $E^{\circ}(Au^{3+}/Au) = 1.50$ volt. The emf of the voltaic cell,

Problem 217**[GRB (O.P. Tandon) p.71 Q104]**

Which of the following does not occur at cathode? (a) $Ag^{+} + e^{-} \rightarrow Ag$ (b) $Fe^{2+} + 2e^{-} \rightarrow Fe$ (c) $Cu^{2+} + 2e^{-} \rightarrow Cu$ (d) $2H^{+} + 2e^{-} \rightarrow H_2$

Problem 218**[GRB (O.P. Tandon) p.71 Q105]**

The strong oxidising agent has: (a) high value of reduction potential (b) high value of oxidation potential (c) low value of reduction potential to lose electrons

Problem 219**[GRB (O.P. Tandon) p.71 Q106]**

The passage of electricity in the Daniell cell when Zn and electrodes are connected is from: (a) Cu to Zn in the cell (b) Cu to Zn outside the cell (c) Zn to Cu in the cell (d) Zn to Cu outside the cell

Problem 220**[GRB (O.P. Tandon) p.71 Q107]**

Which of the following can be used as an electrode? (a) A glass rod (b) A wooden stick (c) A nail (d) A soda straw

Problem 221**[GRB (O.P. Tandon) p.71 Q108]**

Which ions are reduced at platinum electrode prior to: (a) Zn^{2+} (b) Cu^{2+} (c) Ag^{+} (d) H^{+}

Problem 222**[GRB (O.P. Tandon) p.71 Q109]**

Which of the following statements is wrong? (a) F₂ is the strongest oxidising agent as its reduction potential is high (b) Li is the weakest reducing agent as its reduction potential is low (c) Li is the strongest reducing agent as its oxidation potential is high (d) F⁻ ion does not show reducing property

Problem 223**[GRB (O.P. Tandon) p.71 Q110]**

Out of Cu, Ag, Fe and Zn, the metal which can displace all others from their salt solution is: (a) Ag (b) Cu (c) Fe (d) Zn. When the electric current is passed through a cell having an electrolyte, the positive ions move towards cathode and negative ions towards the anode. If the cathode is pulled out of the solution: (a) the positive and negative ions will move towards anode (b) the positive ions will start moving towards the anode while negative ions will stop moving (c) the negative ions will continue to move towards anode while positive ions will stop moving (d) the positive and negative ions will start moving randomly

Problem 224**[GRB (O.P. Tandon) p.71 Q112]**

The oxidation potentials of Zn, Cu, Ag, H₂ and Ni are 0.76, -0.34, -0.80, 0 and 0.25 volt respectively. Which of the following reactions will provide maximum voltage? (a) Zn + Cu²⁺ → Zn²⁺ + Cu (b) Zn + 2Ag⁺ → Zn²⁺ + 2Ag (c) H₂ + Cu²⁺ → 2H⁺ + Cu (d) H₂ + Ni²⁺ → 2H⁺ + Ni [PET (KeraJa) 2007] (e) Zn(s) + 2H⁺(aq.) → Zn²⁺(aq.) + H₂(g)

Problem 225**[GRB (O.P. Tandon) p.71 Q113]**

Which one of the following will increase the voltage of the cell? Sn + 2Ag⁺ → Sn²⁺ + 2Ag (a) Increase in the size of silver rod (b) Increase in the concentration of Sn²⁺ ions (c) Increase in the concentration of Ag⁺ ions (d) None of the above

Problem 226**[GRB (O.P. Tandon) p.71 Q114]**

A chemist wants to produce Cl₂(g) from molten NaCl. How many grams could be produced if he uses a steady current of 2 ampere for 2.5 minutes? (a) 3.55 g (b) 1.775 g (c) 0.110 g (d) 0.1775 g

Problem 227**[GRB (O.P. Tandon) p.71 Q115]**

In the electrolysis of CuCl₂ solution, the mass of cathode increases by 0.224 g. What volume of Cl₂ gas is liberated? (a) 0.224 litre (b) 1.12 litre (c) 0.05 mole (d) 0.05 mole of Cu²⁺ passed into the solution. (e) 0.05 mole of Cl₂ gas is liberated

Problem 228**[GRB (O.P. Tandon) p.71 Q116]**

Consider the reaction, Cl₂(g) + 2Br⁻(aq.) → 2Cl⁻(aq.) + Br₂(l). The emf of the cell, when [Cl⁻] = [Br⁻] = 0.1 M and Cl₂ gas is at 1 atm pressure, will be: (E⁰ for the above reaction is 0.29 volt). (a) 0.54 volt (b) 0.35 volt (c) 0.24 volt (d) -0.29 volt. If A₊, A₋ and A refer to equivalent conductance of a cation, an anion and equivalent conductance of the salt at infinite dilution, then according to Kohlrausch's law: (a) A₊ = A₋ + A (b) A₊ = A₋ - A (c) A₊ = A₋ + A (d) A₊ = A₋ - A

Problem 229**[GRB (O.P. Tandon) p.71 Q118]**

How much silver will be obtained by that quantity of current which displaces 5.6 litre of H₂? [PMT (Pb.) 1993] (a) 54 g (b) 13.5 g (c) 20 g (d) 108 g

Problem 230**[GRB (O.P. Tandon) p.71 Q119]**

The specific conductance of a salt of 0.01 M concentration is 1.061×10^{-4} . Molar conductance of the same solution will be: (a) 1.061×10^{-4} (b) 1.061 (c) 10.61 IPMT (Pb.) 1993 (d) 106.1

Problem 231**[GRB (O.P. Tandon) p.71 Q120]**

What is the number of coulombs required for the conversion of one mole of MnO_2 to one mole of Mn^{2+} ? [MBBS (Orissa) 1993] (a) 5×96500 (b) 3×96500 (c) 96500 (d) 9650

Problem 232**[GRB (O.P. Tandon) p.71 Q121]**

Which of the following solutions of NaCl will have the highest specific conductance? (a) 0.001 N (b) 0.1 N (c) 0.01 N (d) 1.0 N

Problem 233**[GRB (O.P. Tandon) p.71 Q122]**

The equivalent conductance of a 1 N solution of an electrolyte is nearly: (a) 103 times its specific conductance (b) 10^{-3} times its specific conductance (c) 100 times its specific conductance (d) the same as its specific conductance

Problem 234**[GRB (O.P. Tandon) p.71 Q123]**

Zinc is coated over iron to prevent rusting of iron because: (a) it is cheaper than iron

Problem 235**[GRB (O.P. Tandon) p.72 Q124]**

Free energy change (ΔG) is related to the emf of the cell (E) (a) $E = -\Delta G/nF$ (c) $\Delta G = RT \cdot -F \log E/n$ (b) $\Delta G = -nFE$ (d) $\Delta G = nF \cdot \log E/RT$ 125 .. The standard reduction potentials at 25°C of Li^+/Li , Ba^{2+}/Ba , Na^+/Na and Mg^{2+}/Mg are -3.05, 2.73, 2.71 and -2.37 volt respectively. Which one of the following is the strongest oxidising agent? (a) Na^+ (c) Ba^{2+} (b) Li^+ (d) Mg^{2+}

Problem 236**[GRB (O.P. Tandon) p.72 Q126]**

Three faradays of electricity was passed through an aqueous solution of iron(II) bromide. The mass of iron metal (At. mass 56) deposited at the cathode is: (EAMCET 1991) (a) 56 g (b) 84 g (c) 112 g (d) 168 g 127 /bullet. The standard electrode potentials of Zn, Ag and Cu are -0.76, 0.80 and 0.34 volt respectively; then: (a) Ag can oxidise Zn and Cu (b) Ag can reduce Zn^{2+} and Cu^{2+} (c) Zn can reduce Ag^+ and Cu^{2+} (d) Cu can oxidise Zn and Ag

Problem 237**[GRB (O.P. Tandon) p.72 Q128]**

The standard emf for the cell reaction, $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$ is 1.10 volt at 25°C. The emf for the cell reaction when 0.1 M Cu^{2+} and 0.1 M Zn^{2+} solutions are used at 25°C is: (a) 1.10 volt (c) 1.10 volt (b) 0.110 volt (d) 0.110 volt (MLNR 1994)

Problem 238**[GRB (O.P. Tandon) p.72 Q129]**

Three mole of electrons are passed through three solutions in succession containing AgNO_3 , CuSO_4 and AuCl_3 respectively. The ratio of amounts of cations reduced at cathode will be: (a) 1 : 3 (b) 2 : 1 : 3 (c) 3, 2 : 1 (d) 6. 3 : 2

Problem 239**[GRB (O.P. Tandon) p.72 Q130]**

In the electrolysis of an aqueous solution of NaOH, 2.8 litre of oxygen gas at NTP was liberated at anode. How much of hydrogen gas was liberated at cathode? (a) 2.8 litre (b) 5.6 litre (c) 11.1-litre (d) 22.4 litre

Problem 240**[GRB (O.P. Tandon) p.72 Q131]**

Two half-cells have potentials 0.44 and 0.799 volt respectively. These two are coupled to make a galvanic cell. Which of the following will be true? (a) Electrode of half-cell potential- 0.44 V will act as anode (b) Electrode of half-cell potential - 0.44 V will act as cathode (c) Electrode of half-cell potential 0.799 V will act as anode (d) Electrode of half-cell potential - 0.44 V will act as a/cdot positive terminal

Problem 241**[GRB (O.P. Tandon) p.72 Q132]**

When a lead storage battery is charged: (a) PbO₂ dissolves (b) the lead electrode becomes coated with lead sulphate/cdot (c) sulphuric acid is regenerated (d) the amount of acid decreases

Problem 242**[GRB (O.P. Tandon) p.72 Q137]**

An example of a simple fuel cell is: (a) lead storage battery (b) H₂ - O₂ cell (c) Daniell cell (d) Leclanche cell For the cell reaction, $\text{Mg(s)} + 2\text{Ag}^+(\text{aq.}) \rightarrow \text{Mg}^{2+}(\text{aq.}) + 2\text{Ag(s)}$. E° is +3.17 V at 298 K. The value of E° and Q at + 2+ /textasciitilde() Ag and Mg concentrations of 0.001 M and 0.02 M respectively are: (ISAT 2010) (a) 3.04 V, -605.8 kJ mol⁻¹, 20000 (b) 3.04 V, 611.8 kJ mol⁻¹, 20000 (c) 3.13 V, -604 kJ mol⁻¹, 20 3.04 -611.8 20000 $e^\circ = +3.17$ V, $n = 2$ $Q = \frac{[Mg^{2+}]}{[Ag^+]^2} = \frac{0.02}{(0.001)^2} = 20000$ $\Delta G^\circ = -nFE^\circ = -2 \times 96500 \times 3.17 = -611.8 \text{ kJ}$ $0.059 E = e^\circ - \log Q$ $0.059 \times 3.17 = 3.04 - \log Q$ Which of the following statements is correct? (a) The temperature coefficient of electrolytic conductance is positive (b) The temperature coefficient of electrolytic resistance is negative (c) The resistance of an electrolyte decreases with decreasing temperature (d) The resistance of electrolytic conductors is independent of temperature Kohlrausch's law states that at: ICB1',E (PMT) 2008] (a) infinite dilution, each ion makes definite contribution to conductance of an electrolyte, whatever be the nature of the other ion of the electrolyte (b) infinite dilution, each ion makes definite contribution to the equivalent conductance of an electrolyte, whatever be the nature of the other ion of the electrolyte (c) finite dilution, each ion makes definite contribution to the equivalent conductance of an electrolyte, whatever be the nature of the other ion of the electrolyte (d) infinite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte depending on the nature of the other ion of the electrolyte Which one of the following conditions will increase the voltage of the cell represented by the equation? $\text{Cu(s)} + (\text{aq.}) \rightarrow \text{Cu}^{2+}(\text{aq.}) + 2\text{Ag(s)}$ (KCET 2006) (a) Increase in the dimension of Cu electrode (b) Increase in the dimension of Ag electrode (c) Increase in the concentration of Cl^- (d) Increase in the concentration of Ag^+ ion [Hint: Cell voltage will increase, either by increasing the concentration of Cu^{2+} ion or by decreasing the concentration of Ag^+]

Problem 243**[GRB (O.P. Tandon) p.73 Q138]**

Which of the following reactions occurs at cathode during potential of $Z > Y > X$, then: (IIT 1999) charging of storage battery? (MGIMS 2007) (a) Y will oxidise X and not Z (a) $\text{Pb}^{2+} + 2e^- \rightarrow \text{Pb}$ (b) Y will oxidise Z and not X (b) $\text{Pb} \rightarrow \text{Pb}^{2+} + 2e^-$ (c) Y will oxidise both Z and X (c) $\text{Pb}^{2+} + \text{SO}_4^{2-} \rightarrow \text{PbSO}_4$ (d) Y will reduce both X and Z

Problem 262**[GRB (O.P. Tandon) p.74 Q164]**

The emf of the cell $\text{Ni}|\text{Ni}^{2+}(-1.0\text{M})||\text{Au}^{3+}(\text{LOM})|\text{Au}$ is – (b) Halving the $[\text{Cu}^{2+}]$ for $\text{Ni}^{2+}|\text{Ni} = -0.25\text{ V}$; E° for $\text{Au}^{3+}|\text{Au} = 1.5\text{ V}$: (c) Doubling the size of Cu electrode (d) Halving the size of Ag electrode As a lead storage battery is charged: IPMT (M.P) 1993, 2000] (a) lead dioxide dissolves (b) sulphuric acid is regenerated (c) lead electrode becomes coated with lead sulphate (d) the concentration of sulphuric acid decreases In the electrochemical reaction, $2\text{Fe}^{3+} + \text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{Fe}^{2+}$ increasing the concentration of Fe^{2+} : ICEE (Kenia) 20001 (a) increases the cell emf (b) increases the current flow (c) decreases the cell emf (d) alters the pH of the solution In the cell reaction, $\text{Cu(s)} + 2\text{Ag}^+(\text{aq.}) \rightarrow \text{Cu}^{2+}(\text{aq.}) + 2\text{Ag(s)}$. $E^\circ_{\text{cell}} = 0.46\text{ V}$. By doubling the concentration of Cu^{2+} , E_{cell} is: [MEE (Kerala) 20001 (a) doubled (b) halved (c) increased but less than double (d) decreased by a small fraction The conductivity of 0.01 mol dm^{-3} aqueous acetic acid at 300 K is $19.5 \times 10^{-5}\text{ }\Omega^{-1}\text{ cm}^{-1}$ and the limiting molar conductivity of acetic acid at the same temperature is $390\text{ }\Omega^{-1}\text{ cm}^2\text{ mol}^{-1}$. The degree of dissociation of acetic acid is: (a) 0.5 (b) 0.05 [IAS (Pre.) 1995] (c) 5×10^{-3} (d) 5×10^{-7} The ionization constant of a weak electrolyte is 25×10^{-6} while the equivalent conductance of its 0.01 M solution is $19.6\text{ S cm}^2\text{ eq}^{-1}$. The equivalent conductance of the electrolyte at infinite dilution (in $\text{S cm}^2\text{ eq}^{-1}$) will be: (a) 250 (c) 392 (b) 196 (d) 384 [IAS (Pre.) 1998] ICET (Pb.) 2000; PMT (MP) 1999 (a) 0.25 V (b) $+1.75\text{ V}$ (c) -1.25 V (d) -1.75 V

Problem 263**[GRB (O.P. Tandon) p.74 Q165]**

For the reduction of silver ions with copper metal; the standard cell potential was found to be $+0.46\text{ V}$ at 25°C . The value of

Problem 264**[GRB (O.P. Tandon) p.74 Q166]**

167. standard Gibbs energy, ΔG° will be : (F, 96500 C mol^{-1}) (a) -89 kJ (b) -89 J (c) -44.5 kJ (d) -98 kJ (AIPMT 2010) In electrolysis of NaCl when Pt electrode is taken then H_2 is liberated at cathode while with Hg cathode it forms sodium amalgam: [CBSE (PMT) 20021 (a) Hg is more inert than Pt (b) more voltage is required to reduce at Hg than at Pt (c) Na is dissolved in Hg while it does not dissolve in Pt (d) concentration of F^- ions is larger when Pt electrode is taken Zn gives H_2 gas with H_2SO_4 and HCl but not with HNO_3 because: [CBSE (PMT) 20021 (a) Zn acts as oxidising agent when reacts with HNO_3 (b) HNO_3 is weaker acid than H_2SO_4 and HCl (c) in electrochemical series, Zn is above hydrogen (d) NO_3^- is reduced in preference to hydronium ion, H_3O^+

Problem 265**[GRB (O.P. Tandon) p.74 Q168]**

A current is passed through two voltmeters connected in series. The first voltmeter contains $\text{XSO}_4(\text{aq.})$ while the second voltmeter contains $\text{YSO}_4(\text{aq.})$. The relative atomic masses of X and Y are in the ratio 2: 1. The ratio of the mass of X liberated to the mass of Y liberated is: (a) 1: 1 (b) 1: 2 (c) 2: 1 (d) none of these

Problem 266**[GRB (O.P. Tandon) p.74 Q169]**

Which of the following reactions is possible at anode? (AIEEE 2002) (a) $2\text{Cr}^{3+} + 7\text{H}_2\text{O} \rightarrow \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+$ (b) $\text{F}_2 \rightarrow 2\text{F}^-$ (c) $\text{O}^{2-} + 2\text{H}^+ \rightarrow \text{H}_2\text{O}$ (d) None of the above

Problem 267**[GRB (O.P. Tandon) p.75 Q171]**

Conductivity (Unit Siemen's 'S') is directly proportional to area of the vessel and the concentration of the solution in it and is inversely proportional to the length of the vessel, then the $\frac{\kappa}{c}$ is proportional to $\frac{1}{l}$
 (a) $\frac{1}{l}$ (b) $\frac{1}{l^2}$ (c) $\frac{1}{l^3}$ (d) $\frac{1}{l^4}$

Problem 268**[GRB (O.P. Tandon) p.75 Q172]**

$\text{Zn(s)} + \text{Cu}^{2+}(\text{aq.}) \rightarrow \text{Cu(s)} + \text{Zn}^{2+}(\text{aq.})$ $E^\circ = 1.10$ volt. Ecell = 1.1591 volt when: $\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$ is (a) 0.01 (b) 0.1 (c) 1 (d) 10

Problem 269**[GRB (O.P. Tandon) p.75 Q173]**

In which of the following cells will the emf be independent of the activity of the chloride ions?

Problem 270**[GRB (O.P. Tandon) p.75 Q174]**

(a) $\text{Zn} | \text{ZnCl}_2(\text{aq.}) || \text{Pt} | \text{Cl}_2(\text{g})$ (b) $\text{Zn} | \text{ZnCl}_2(\text{aq.}) || \text{KCl}(\text{aq.}) | \text{AgCl}(\text{s}), \text{Ag}(\text{s})$ (c) $\text{Ag}, \text{AgCl}(\text{s}) | \text{KCl}(\text{aq.}) | \text{Pt} | \text{Cl}_2(\text{g})$ (d) $\text{Hg}, \text{Hg}_2\text{Cl}_2(\text{s}) | \text{KCl}(\text{aq.}) || \text{AgNO}_3(\text{aq.}) | \text{Ag}(\text{s})$
 Standard electrode potential data are useful for understanding the suitability of an oxidant in a redox titration. Some half-cell reactions and their standard potentials are given below:
 $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O(l)}$; $E^\circ = 1.51$ volt
 $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O(l)}$; $E^\circ = 1.33$ volt
 $\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq.})$; $E^\circ = 1.36$ volt
 Identify the only incorrect statement regarding the quantitative estimation of aqueous $\text{Fe}(\text{NO}_3)_3$: (a) MnO_4^- can be used in aqueous HCl (b) $\text{Cr}_2\text{O}_7^{2-}$ can be used in aqueous HCl (c) MnO_4^- can be used in aqueous H_2SO_4 (d) $\text{Cr}_2\text{O}_7^{2-}$ can be used in aqueous H_2SO_4

Problem 271**[GRB (O.P. Tandon) p.75 Q175]**

The standard reduction potential E° , for the half reaction are: $\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn(s)}$; $E^\circ = 0.76$ V $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu(s)}$; $E^\circ = 0.34$ V The emf for the cell reaction, $\text{Zn(s)} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu(s)}$ is: (a) 0.42 V (b) - 0.42 V (c) - 1.1 V (d) + 1.1 V

Problem 272**[GRB (O.P. Tandon) p.75 Q176]**

In a galvanic cell, the electrons flow from: (a) anode to cathode through the solution (b) cathode to anode through the solution (c) anode to cathode through the external circuit (d) cathode to anode through the external circuit

Problem 273**[GRB (O.P. Tandon) p.75 Q177]**

The standard emf of a galvanic cell involving cell reaction with $n = 2$ is found to be 0.295 V at 25°C. The equilibrium constant of the reaction would be: Given: $F = 96500$ C mol⁻¹; $R = 8.314$ JK⁻¹ mol⁻¹ (a) 2×10^{11} (b) 4×10^{12} (c) 1×10^2 (d) 1×10^1

Problem 274**[GRB (O.P. Tandon) p.75 Q178]**

The emf of the cell, $\text{Zn} | \text{Zn}^{2+} (0.01 \text{ M}) || \text{Fe}^{2+} (0.001 \text{ M}) | \text{Fe}$ at 298 K is 0.2905 V then the value of equilibrium constant for the cell reaction is: [IIT (S) 2004] (a) 0.3210.0295 (b) $10/\text{circ}.3210.0295$ (c) $10/\text{circ}/\text{cdot}261$ 0.0295 (d) $10/\text{circ}.3210.0591$

Problem 275**[GRB (O.P. Tandon) p.75 Q179]**

The standard emf of the following electrodes are;

Problem 276**[GRB (O.P. Tandon) p.75 Q180]**

181. $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = +0.77 \text{ V}$; $E^\circ_{\text{Sn}^{2+}/\text{Sn}} = -0.14 \text{ V}$ under standard conditions, the potential for the reaction, $\text{Sn(s)} + 2\text{Fe}^{3+}(\text{aq.}) \rightarrow 2\text{Fe}^{2+}(\text{aq.}) + \text{Sn}^{2+}(\text{aq.})$ is: (AIEEE 2004) (a) 1.68 V (b) 1.40 V (c) 0.91 V (d) 0.63 V The highest electrical conductivity of the following aqueous solutions is of: (AIEEE 2005) (a) 0.1 M acetic acid (b) 0.1 M chloroacetic acid (c) 0.1 M fluoroacetic acid (d) 0.1 M difluoroacetic acid Aluminium oxide may be electrolysed at 1000°C to furnish aluminium metal (Atomic mass = 27 amu; 1 faraday = 96500 coulomb). The cathode reaction is: $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ To prepare 5.12 kg of aluminium metal by this method we would require: (AIEEE 2005) . (a) $5049 \times 10^7 \text{ C}$ of electricity (b) $1.83 \times 10^7 \text{ C}$ of electricity (c) $5.049 \times 10^4 \text{ C}$ of electricity (d) $5.49 \times 10^{10} \text{ C}$ of electricity

Problem 277**[GRB (O.P. Tandon) p.75 Q182]**

During the process of electrolytic refining of copper, some metals present as impurity settle as 'anode mud'. These are: (AIEEE 2005) (a) Sn and Ag (b) Pb and Zn (c) Ag and Au (d) Fe and Ni

Problem 278**[GRB (O.P. Tandon) p.75 Q183]**

When an acid cell is charged, then: (AFMC 2005) (a) voltage of the cell increases (b) electrolyte of the cell dilutes

Problem 279**[GRB (O.P. Tandon) p.76 Q184]**

How many coulombs of electricity are required for the reduction of 1 mole of MnO_4^- to Mn^{2+} ? [IPMT (Kerala) 2005] (a) 96500 C (b) $4.93 \times 10^5 \text{ C}$ (c) $4.83 \times 10^5 \text{ C}$ (d) $9.65 \times 10^6 \text{ C}$ (e) $5.62 \times 10^5 \text{ C}$.

Problem 280**[GRB (O.P. Tandon) p.76 Q185]**

The standard electrode potential of Ag^+/Ag is +0.80 V and of Cu^{2+}/Cu is +0.34 V. These electrodes are connected through a salt bridge and if: . (PET (Kerala) 2005] (a) copper electrode acts as cathode, then E_{cell} is +0.46 volt (b) silver electrode acts as anode, then E_{cell} is +0.46 volt (c) copper electrode acts as anode, then E_{cell} is +0.46 volt (d) silver electrode acts as cathode, then E_{cell} is -0.34 volt (e) silver electrode acts as anode, then E_{cell} is +0.46 volt

Problem 281**[GRB (O.P. Tandon) p.76 Q186]**

The half-cell reaction for the corrosion, $\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$; $E^\circ = 1.23 \text{ V}$. $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$; $E^\circ = -0.44 \text{ V}$ Find the ΔG° (in kJ) for the overall reaction: (IIT (S) 2005) (a) 76 kJ (b) 322 kJ . (c) -161 kJ (d) 152 kJ [Hint: Applying $\text{Fe(s)} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$; $\Delta G^\circ = 2F \times 0.44$ and $\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O(l)}$; $\Delta G^\circ = -2F \times 1.23$]
 $\Delta G^\circ = (-2F \times 0.44) + (-2F \times 1.23) = (-2 \times 96500 \times 0.44) + (-2 \times 96500 \times 1.23)$
 $= -322310 \text{ J} = -322 \text{ kJ}$

Problem 282**[GRB (O.P. Tandon) p.76 Q187]**

What is the cell reaction occurring in Daniell cell (galvanic cell)? [CET (J and K) 2006] (a) $\text{Cu(s)} + \text{ZnSO}_4(\text{aq.}) \rightarrow \text{CuSO}_4(\text{aq.}) + \text{Zn(s)}$ (b) $\text{Zn(s)} + \text{CuSO}_4(\text{aq.}) \rightarrow \text{Cu(s)} + \text{ZnSO}_4(\text{aq.})$ (c) $\text{Ni(s)} + \text{ZnSO}_4(\text{aq.}) \rightarrow \text{NiSO}_4(\text{aq.}) + \text{Zn(s)}$ (d) $2\text{Na(s)} + \text{CdSO}_4(\text{aq.}) \rightarrow \text{Na}_2\text{SO}_4(\text{aq.}) + \text{Cd(s)}$ 188. What are the units of equivalent conductivity of a solution? .

Problem 283**[GRB (O.P. Tandon) p.76 Q189]**

[CET (J and K) 2006] (a) $\text{mho cm}^2 \text{equiv}^{-1}$ (b) $\text{ohm cm}^{-1} \text{equiv}^{-1}$ (c) $\text{Inho cm}^{-2} \text{g equiv}^{-1}$ (d) $\text{mho cm}^2 \text{g equiv}^{-1}$ The amount of copper deposited by one faraday current will be maximum in an acidic solution of one litre of: (a) 1M CuCl_2 (c) 5M CuSO_4 (e) 10M CuPz [PMT (KeraJa) 2006] (b) 2M $\text{Cu(NO}_3)_2$ (d) 5M $\text{Cu}_3(\text{PO}_4)_2$ [Hint: Greater is the equivalent mass, more is the amount deposited by 1 F charge.]

Problem 284**[GRB (O.P. Tandon) p.76 Q190]**

The reduction potential values of M, N and O are + 2.46 V, - 1.13 V, 3.13 V respectively. Which of the following orders is correct regarding their reducing property? . (JEE (Orissa) 2006) (a) $O > N > M$ (b) $O > M > N$ (c) $M > N > O$ (d) $M > O > N$

Problem 285**[GRB (O.P. Tandon) p.76 Q191]**

The molar conductivities of NaOAc and NaCl at infinite dilution in water at 25°C are 91 and $426.2 \text{ S cm}^2 \text{mol}^{-1}$ respectively. To calculate, Λ° of NaOAc the additional value required is: (AIEEE 2006) (a) Λ° of H_2O (b) Λ° of Cl^- (c) Λ° of NaOH (d) Λ° of NaCl

Problem 286**[GRB (O.P. Tandon) p.76 Q192]**

The equivalent conductances at infinite dilution of HCl and NaCl are 426.15 and $126.15 \text{ mho cm}^2 \text{equiv}^{-1}$ respectively. It can be said that the mobility of: . (CET (Gujarat) 2006) (a) H^+ ions is much more than that of Cl^- ions (b) Cl^- ions is much more than that of H^+ ions (c) H^+ ions is much more than that of Na^+ ions (d) Na^+ ions is much more than that of H^+ ions

Problem 287**[GRB (O.P. Tandon) p.76 Q193]**

The tendencies of the electrodes made up of Cu, Zn and Ag to release electrons when dipped in their respective salt solutions decrease in the order: (VITEEE 2006) (a) $\text{Zn} > \text{Ag} > \text{Cu}$ (b) $\text{Cu} > \text{Zn} > \text{Ag}$ (c) $\text{Zn} > \text{Cu} > \text{Ag}$ (d) $\text{Ag} > \text{Cu} > \text{Zn}$

Problem 288**[GRB (O.P. Tandon) p.76 Q194]**

The electrode reaction that takes place at the anode of $\text{CH}_4 - \text{O}_2$ fuel cell is: (VITEEE 2006) . (a) $2\text{O}_2 + 8\text{H}^+ + 8\text{e}^- \rightarrow 4\text{H}_2\text{O}$ (b) $\text{CH}_4 + 2\text{H}_2\text{O} \rightarrow \text{CO}_2 + 8\text{H}^+ + 8\text{e}^-$ (c) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$ (d) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$

Problem 289**[GRB (O.P. Tandon) p.76 Q196]**

Zn(s) | Zn²⁺ (1M) || Cu²⁺ (1M) | Cu(s) ($E^\circ = +1.10$ V) was allowed to be completely discharged at 298K. The relative concentration of Zn²⁺ to Cu²⁺ is: [Cu²⁺] (AIEEE 2007) (a) 9.65×10^4 (b) $\text{antilog}(24.08)$ (c) 37.3 (d) 173. [Hint: $E = 0 - 0.059 \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$]
The efficiency of a cell is given by: (CaSE (Med.) 2007) (a) 11G (b) AG / 11S (c) 11G (d) 11G

Problem 290**[GRB (O.P. Tandon) p.77 Q197]**

An electric current of 1 amp is passed through acidulated water for 160 minutes and 50 seconds. What is the volume of the hydrogen liberated at the anode (as reduced to NTP)? (SCRA2007) (a) 1.12 litre (b) 2.24 litre (c) 11.2 litre (d) 22.4 litre [Hint: $V = \frac{I \times t}{96500} \times 22.4$]
= 11.2 litre

Problem 291**[GRB (O.P. Tandon) p.77 Q198]**

The resistance of a solution is found to be $2.5 \times 10^3 \Omega$. The equivalent conductance of the solution is (cell constant = 1.25 cm⁻¹): [PMT (Kerala) 2007] (a) $2.5 \Omega^{-1} \text{cm}^2 \text{equiv}^{-1}$ (b) $5 \Omega^{-1} \text{cm}^2 \text{equiv}^{-1}$

Problem 292**[GRB (O.P. Tandon) p.77 Q205]**

Electrolysis of dilute aqueous NaCl solution was carried out by passing 10 milli ampere current. The time required to liberate 0.01 mole of H₂ gas at the cathode is: (1 faraday = 96500 C mol⁻¹) (IIT 2008) (a) 9.65×10^4 sec (b) 19.3×10^4 sec (c) 28.95×10^4 sec (d) 38.6×10^4 sec [Hint: Mass of 0.01 mol H₂ = 0.02 g. $W = \frac{Z \times I \times t}{96500}$]
 $0.02 = \frac{1 \times 10 \times t}{96500}$ $t = 19.3 \times 10^4$ sec (c) $2.5 \Omega^{-1} \text{cm}^2 \text{equiv}^{-1}$ (e) $1.25 \Omega^{-1} \text{cm}^2 \text{equiv}^{-1}$ (d) $5 \Omega^{-1} \text{cm}^2 \text{equiv}^{-1}$
The emf of a cell containing sodium/copper electrodes is 3.05 V, if the electrode potential of copper electrode is

Problem 293**[GRB (O.P. Tandon) p.77 Q199]**

For strong electrolytes, the plot of molar conductance versus $\sqrt{C}$ is: (VITEEE 2007) (a) linear (b) sinusoidal (c) sinusoidal (d) circular

Problem 294**[GRB (O.P. Tandon) p.77 Q200]**

How long (in hours) must a current of 5 ampere be maintained to electroplate 60 g of calcium from molten CaCl₂? (VITEEE 2007) (a) 27 hours (b) 8.3 hours (c) 11 hours (d) 16 hours

Problem 295**[GRB (O.P. Tandon) p.77 Q201]**

Emf of hydrogen electrode in term of pH is (at 1 atm pressure):

Problem 296**[GRB (O.P. Tandon) p.77 Q204]**

(MHT/CET 2007) RT (a) $E_H = \frac{RT}{2F} \ln \frac{[Fe^{2+}]}{[Fe^{3+}]}$ (c) $E_H = 2.303RT \log \frac{[Fe^{2+}]}{[Fe^{3+}]}$ (b) $E = RT \ln \frac{[Fe^{2+}]}{[Fe^{3+}]}$ (d) $E_H = -0.0591 \text{ pH}$ The rusting of iron is catalysed by which of the following? (a) Fe (c) Zn (MGIMS 2007) (d) H^+ On the basis of E° values, the strongest oxidising agent is: $[Fe(CN)_6]^{3-}$ (c) $[Fe(CN)_6]^{4-}$ (d) Fe^{2+} Given $E^\circ([Fe(CN)_6]^{3-}/[Fe(CN)_6]^{4-}) = -0.72 \text{ V}$, $E^\circ(Fe^{2+}/Fe) = -0.42 \text{ V}$. The potential for the cell, $Cr|Cr^{3+}(0.1 \text{ M})||Fe^{2+}(0.1 \text{ M})|Fe$ is: (AIEEE 2008) (a) -0.26 V (b) 0.26 V (c) 0.339 V (d) -0.339 V [Hint: $E_{\text{cell}} = -0.42 - (-0.72) = +0.30 \text{ V}$ $2Cr(s) + 3Fe^{2+}(0.1 \text{ M}) \rightarrow 2Cr^{3+}(0.1 \text{ M}) + 3Fe(s)$ $Q = \frac{[Cr^{3+}]^2}{[Fe^{2+}]^3} = \frac{[0.1]^2}{[0.1]^3} = 10$ According to Nernst equation, $0.059 E = 0 - \frac{0.059}{3} \log 10$ $E = -0.019 \text{ V}$ (": $n = 6$) + 0.34 V , the electrode potential of sodium is: reamed (Karnataka) 2008 (a) -2.7 V (b) $+2.7 \text{ V}$ (c) -3.7 V (d) $+3.7 \text{ V}$

Problem 297**[GRB (O.P. Tandon) p.77 Q207]**

What is the number of moles of oxygen gas evolved at the anode during the electrolysis of 180 g of water? (SeRA 2009) (a) 2.5 (b) 5.0 (c) 7.5 (d) 10.0 The Gibbs energy for the decomposition of Al_2O_3 at 500°C is as follows: $2Al_2O_3 \rightarrow 4Al + 3O_2$; $\Delta_r G^\circ = +966 \text{ kJ mol}^{-1}$ The potential difference needed for the electrolytic reduction of Al_2O_3 at 500°C is at least: (AIEEE 2010) (a) 2.5 V (b) 5.0 V (c) 4.5 V (d) 3.0 V [Hint: In the given reaction, $3[2Al^{3+} + 3e^- \rightarrow 2Al]$ ($n = 4$) $\Delta_r G^\circ = -nFE$ $E = \frac{966 \times 1000}{4 \times 96500} = 2.5 \text{ V}$] Set-2: The questions given below may have more than one correct answers

Problem 298**[GRB (O.P. Tandon) p.77 Q208]**

What is the difference between galvanic cell and electrolytic cell? (a) In galvanic cell, electrical energy is produced while in electrolytic cell electrical energy is consumed (b) In galvanic cell, anode is (-)ve while in electrolytic cell anode is (+)ve (c) In galvanic cell, cathode is (+)ve while in electrolytic cell anode is (-)ve (d) All are correct

Problem 299**[GRB (O.P. Tandon) p.77 Q1]**

What is the difference between galvanic cell and electrolytic cell? (a) In galvanic cell, electrical energy is produced while in electrolytic cell electrical energy is consumed (b) In galvanic cell, anode is (-)ve while in electrolytic cell anode is (+)ve (c) In galvanic cell, cathode is (+)ve while in electrolytic cell anode is (-)ve (d) All are correct

Problem 300**[GRB (O.P. Tandon) p.77 Q2]**

Ag | Ag⁺, KI || AgI | Ag emf is E , then K_{sp} of AgI is given as: $nF \ln \frac{K_{sp}}{[Ag^+][I^-]}$ (a) $K_{sp} = 2.303RT \log \frac{E}{E^\circ}$ (b) $\ln K_{sp} = \frac{nF}{RT} (E - E^\circ)$ (c) $\ln K_{sp} = -\frac{nF}{RT} E^\circ$ (d) $10 K_{sp} = \frac{nF}{RT} E^\circ$ $E = 2.303RT \log \frac{K_{sp}}{[Ag^+][I^-]}$

Problem 301**[GRB (O.P. Tandon) p.77 Q3]**

A hydrogen electrode is placed in a buffer solution of acetic acid and sodium acetate in the ratio $x:y$. The pH of the solution is $\frac{1}{2} \log \frac{[CH_3COO^-]}{[CH_3COOH]}$ (a) $\frac{1}{2} \log \frac{[CH_3COO^-]}{[CH_3COOH]}$ (b) $\frac{1}{2} \log \frac{[CH_3COOH]}{[CH_3COO^-]}$ (c) $\frac{1}{2} \log \frac{[CH_3COO^-]}{[CH_3COOH]}$ (d) $\frac{1}{2} \log \frac{[CH_3COOH]}{[CH_3COO^-]}$

Problem 302**[GRB (O.P. Tandon) p.78 Q5]**

The main factors which affect corrosion are: (a) position of metal in electrochemical series (b) presence of CO_2 in water (c) presence of impurities in metal (d) presence of protective coating

Problem 303**[GRB (O.P. Tandon) p.78 Q6]**

Which is correct about silver plating? (a) Anode-pure Ag (b) Cathode—object to be electroplated (c) Electrolyte- $Na[Ag(CN)_2]$ (d) Ag^+ concentration should be maintained at 10^{-3} M (e) Ag^+ concentration should be maintained at 10^{-3} M

Problem 304**[GRB (O.P. Tandon) p.78 Q7]**

Lead storage battery contains: (a) Pb rod as anode (b) Pb rod as cathode (c) Pb plates coated with PbO₂ act as cathode (d) electrolyte is, H₂SO₄.

Problem 305**[GRB (O.P. Tandon) p.78 Q8]**

During the electrolysis of AgNO₃ (using Pt electrodes) concentration around cathode as well as anode falls from 4M to 3M. What will happen if this happened with Ag electrodes? (a) Result will remain same (b) Concentration around cathode will, fall from 4M to 3M but around anode will increase from 4M to 5M (c) Reverse of statement (b) (d) Concentration increases from 4M to 5M on both the electrodes

Problem 306**[GRB (O.P. Tandon) p.78 Q9]**

Emf of the cell Pt: H₂ (1 atm) | H⁺(aq.) || AgCl | Ag is 0.27 V and 0.26 V at 25°C and 35°C. Heat of reaction occurring inside the cell at 25°C is: (a) - 54.5 kJ (c) - 26.05 kJ

Problem 307**[GRB (O.P. Tandon) p.78 Q10]**

Given that, (b) 26.05 kJ (d) + 54.5 kJ $\text{Ni}^{2+}/\text{Ni} = 0.25 \text{ V}$, $\text{Cu}^{2+}/\text{Cu} = 0.34 \text{ V}$, $\text{Ag}^+/\text{Ag} = 0.80 \text{ V}$ and $\text{Zn}^{2+}/\text{Zn} = -0.76 \text{ V}$ which of the following reactions under standard condition will not take place in the specified direction? (a) $\text{Ni}^{2+}(\text{aq.}) + \text{Cu}(\text{s}) \rightarrow \text{Ni}(\text{s}) + \text{Cu}^{2+}(\text{aq.})$ (b) $\text{Cu}(\text{s}) + 2\text{Ag}^+(\text{aq.}) \rightarrow \text{Cu}^{2+}(\text{aq.}) + 2\text{Ag}(\text{s})$ (c) $\text{Cu}(\text{s}) + 2\text{W}(\text{aq.}) \rightarrow \text{Cu}^{2+}(\text{aq.}) + \text{H}_2(\text{g})$ (d) $\text{Zn}(\text{s}) + 2\text{W}(\text{aq.}) \rightarrow \text{Zn}^{2+}(\text{aq.}) + 3\text{H}_2(\text{g})$ 11. Which of the following statements is/are correct? (a) One faraday is the charge carried by one mole of electrons (b) If same quantity of electricity flows through the solutions of 0.1 M AgNO₃ and 0.1 M CuSO₄ solutions, same weight of silver and copper will be deposited (c) Electrochemical equivalent has the units of grams per coulomb (d) Passage of one faraday of electricity produces one gram equivalent of the substance at the electrode

Problem 308**[GRB (O.P. Tandon) p.78 Q12]**

Which of the following statements is/are not correct? (a) Zn-Cu cell is called Daniell cell (b) Rust is Fe₂O₃. (c) Saline water slows down rusting (d) Pure metals undergo corrosion faster than impure metals

Problem 309**[GRB (O.P. Tandon) p.78 Q13]**

In electrolysis of very dilute NaOH solution using platinum electrodes: (a) H₂ is evolved at cathode (b) H₂ is produced at anode (c) Na is obtained at cathode (d) O₂ is produced at anode 14. We observe blue colour if: (a) Cu electrode is placed in the AgNO₃ solution (b) Cu electrode is placed in the ZnSO₄ solution (c) Cu electrode is placed in dil. H₂SO₄ (d) Cu electrode is placed in dil. H₂SO₄

Problem 310**[GRB (O.P. Tandon) p.78 Q15]**

In which of the following cell (s); $E_{\text{cell}} = E^\circ_{\text{cell}}$? (a) $\text{Cu}(\text{s}) | \text{Cu}^{2+}(0.01 \text{ M}) || \text{Ag}^+(0.1 \text{ M}) | \text{Ag}(\text{s})$ (b) $\text{Pt}(\text{H}_2) | \text{pH} = 1 || \text{Zn}^{2+}(0.01 \text{ M}) | \text{Zn}(\text{s})$ (c) $\text{Pt}(\text{H}_2) | \text{pH} = 1 || \text{Zn}^{2+}(1 \text{ M}) | \text{Zn}(\text{s})$ (d) $\text{Pt}(\text{H}_2) | \text{pH} = 0.01 \text{ M} || \text{Zn}^{2+}(0.01 \text{ M}) | \text{Zn}(\text{s})$

Problem 311**[GRB (O.P. Tandon) p.78 Q16]**

Rusting on the surface of iron involves: (a) $\text{Fe(s)} \longrightarrow \text{Fe}^{2+}(\text{aq.}) + 2\text{e}^-$ (at anodic site)
 (b) $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq.}) + 4\text{e}^- \longrightarrow 2\text{H}_2\text{O(l)}$ (at cathodic) (c) $4\text{Fe}^{2+}(\text{aq.}) + \text{O}_2(\text{g}) + 4\text{H}_2\text{O(l)} \longrightarrow 4\text{Fe}(\text{OH})_2(\text{s}) + 4\text{H}^+(\text{aq.})$
 (d) $\text{Fe}_2\text{O}_3(\text{s}) + \text{xH}_2\text{O(l)} \longrightarrow \text{Fe}(\text{OH})_3(\text{s}) + \text{xH}^+(\text{aq.})$

Problem 312**[GRB (O.P. Tandon) p.78 Q17]**

Fuel cell involves following reaction(s): (a) $\text{O}_2(\text{g}) + 2\text{H}_2\text{O(l)} + 4\text{e}^- \longrightarrow 4\text{OH}^-(\text{aq.})$ (at cathode) (b) $\text{O}_2(\text{g}) + 2\text{H}_2\text{O(l)} + 4\text{e}^- \longrightarrow 4\text{OH}^-(\text{aq.})$ (at anode) (c) $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{H}_2\text{O(l)}$ (at anode) (d) $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{H}_2\text{O(l)}$ (at cathode) .

Problem 313**[GRB (O.P. Tandon) p.78 Q18]**

In the following question, more than one of the answers given may be correct. Select the correct answers and mark it according to the code: Codes'; [BHU (Mains) 2008] (a) 1, 2 and 3 are correct (b) 1 and 2 are correct (c) 2 and 4 are correct (d) 1 and 3 are correct [Hint: In a cell $\text{Zn(s)} | \text{Zn}^{2+}(\text{aq.}) || \text{H}^+(\text{aq.}) | \text{H}_2(\text{Pt})$; the addition of H_2SO_4 to the cathode compartment, will:

Problem 314**[GRB (O.P. Tandon) p.78 Q3]**

shift equilibrium to left

Problem 315**[GRB (O.P. Tandon) p.78 Q4]**

shift equilibrium to right]

Problem 316**[GRB (O.P. Tandon) p.79 Q1]**

(A) When acidified zinc sulphate solution is electrolysed between zinc electrodes, it is zinc that is deposited at the cathode and hydrogen evolution does not take place; . (R) The electrode potential of zinc is more negative than hydrogen as the over voltage for the hydrogen evolution on zinc is quite large.

Problem 317**[GRB (O.P. Tandon) p.79 Q2]**

(A) In electrolysis, the quantity of electricity needed for depositing 1 mole of silver is different from that required for 1 mole of copper. (R) The molecular weights of silver and copper are different. (AIIMS 1996)

Problem 318**[GRB (O.P. Tandon) p.79 Q3]**

(A) Equivalent conductance of all electrolytes decreases with increasing concentration. (R) Lesser number of ions are available per gram equivalent at higher concentration. . (AIIMS .1999)

Problem 319**[GRB (O.P. Tandon) p.79 Q4]**

(A) Zinc displaces copper from copper sulphate solution. (R) The E° of Zn is -0.76 volt and that of copper is +0.34 volt. (AIIMS 1999)

Problem 320**[GRB (O.P. Tandon) p.79 Q5]**

(A) An electrochemical cell can be set-up only if the redox reaction is spontaneous. (R) A reaction is spontaneous iff free energy change is negative.

Problem 321**[GRB (O.P. Tandon) p.79 Q6]**

(A) If an aqueous solution of NaCl is electrolysed, the product obtained at the cathode is H₂ gas and not Na. (R) Gases are liberated faster than the metals.

Problem 322**[GRB (O.P. Tandon) p.79 Q7]**

(A) Specific conductance decreases with dilution whereas equivalent conductance increases. (R) On dilution, number of ions per cc decreases but total number of ions increases considerably. (A) The cell constant of a cell depends upon the nature of the material of the electrodes. (R) The observed conductance of a solution depends upon the nature of the material of the electrodes.

Problem 323**[GRB (O.P. Tandon) p.79 Q9]**

(A) The ratio of specific conductivity to the observed conductance does not depend upon the concentration of the solution taken in the conductivity cell. (R) Specific conductivity decreases with dilution whereas observed conductance increases with dilution.

Problem 324**[GRB (O.P. Tandon) p.79 Q10]**

(A) Molar conductivity of a weak electrolyte at infinite dilution cannot be determined experimentally. (R) Kohlrausch's law helps to find the molar conductivity of a weak electrolyte at infinite dilution.

Problem 325**[GRB (O.P. Tandon) p.79 Q11]**

(A) One coulomb of electric charge deposits weight equal to the electrochemical equivalent of the substance. (R) One faraday deposits one mole of the substance. (A) If standard reduction potential for the reaction, $\text{Ag}^+ + e^- \rightarrow \text{Ag}$ is 0.80 volt, then for the reaction, $2\text{Ag}^+ + 2e^- \rightarrow 2\text{Ag}$, it will be 1.60 volt. (R) If concentration of Ag^+ ions is doubled, the electrode potential is also doubled.

Problem 326**[GRB (O.P. Tandon) p.79 Q13]**

(A) Gold is deposited in a vessel made of copper, iron, nickel, chromium, zinc or tin. (R) Gold is very precious metal.

Problem 327**[GRB (O.P. Tandon) p.79 Q14]**

(A) In the Daniell cell, if concentrations of Cu^{2+} and Zn^{2+} ions are doubled, the emf of the cell will be doubled. (R) If the concentration of ions in contact with the metals is doubled, the electrode potential is doubled.

Problem 328**[GRB (O.P. Tandon) p.79 Q15]**

(A) H₂ + O₂ fuel cell gives a constant voltage throughout its life. (R) In this fuel cell, H₂ reacts with OH⁻ ions, yet the overall concentration of OH⁻ ions does not change.

Problem 329**[GRB (O.P. Tandon) p.79 Q16]**

(A) Presence of CO₂ in the air accelerates corrosion. (R) CO₂ is a poisonous gas.

Problem 330**[GRB (O.P. Tandon) p.79 Q17]**

(A) For the Daniell cell, $\text{Zn} \mid \text{Zn}^{2+} \parallel \text{Cu}^{2+} \mid \text{Cu}$ with $E_{\text{cell}} = 1.1$ volt, the application of opposite potential greater than 1.1 V results into flow of electrons from cathode to anode. (R) Zn is deposited at anode and Cu is dissolved at cathode. (AIIMS 2006)

Problem 331**[GRB (O.P. Tandon) p.79 Q18]**

(A) A current of 96.5 ampere is passed into aqueous AgNO_3 solution for 100 second. The weight of silver deposited is 10.8 g. (Atomic weight of Ag = 108) (R) The mass of a substance deposited during the electrolysis of an electrolyte is inversely proportional to the quantity of electricity passing through the electrolyte. [EAMCET (Engg.) 2006]

Problem 332**[GRB (O.P. Tandon) p.79 Q19]**

(A) According to Kohlrausch's Law, the molar conductance of a strong electrolyte at infinite dilution is sum of molar conductivities of its ions. (R) The current carried by cation and anion is always equal. (AIIMS 2007)

Problem 333**[GRB (O.P. Tandon) p.79 Q20]**

(A) The cell potential of mercury cell is 1.35 V, which remains constant. (R) In mercury cell, the electrolyte is a paste of KOH and ZnO. (AIIMS 2008)

Problem 334**[GRB (O.P. Tandon) p.80 Q71]**

(e) . 72. (d) . 73. (a)

Problem 335**[GRB (O.P. Tandon) p.80 Q112]**

(b) ____ JJJ. ____ J<:L ____ 1 14 ____ (<:L ____ JI/textasciitilde()./cdot ____ 7J9) ____ 11/textasciitilde() ____ (Q) ____ 117 /bullet. ____ (liL ____ 118. ____ /textasciitilde()/textasciitilde()) ____ , ____ 112!.. ____ Jf) ____ 12fi. ____ jliL ____ . /bullet

Problem 336**[GRB (O.P. Tandon) p.80 Q14]**

(a, e) .', 6 /bullet. (c).

Problem 337**[GRB (O.P. Tandon) p.81 Q1]**

When aluminium oxide (Al_2O_3) is electrolysed for the production of aluminium metal. For a given quantity of electricity, the number of moles of aluminium obtained if the volume of O_2 gas obtained is 201.6 litre measured at NTP, is: (a:) 9 (b) 6 (c) 12 (d) 4.5 201.6 [Hint: Number of equivalents of oxygen 36 5.6 (Equivalent volume of oxygen == 5.6 litre at NTP) $\therefore$ Number of equivalents of Al = 36]

Problem 338**[GRB (O.P. Tandon) p.81 Q2]**

A smuggler could not carry gold by depositing iron on the gold surface since: (a) gold is denser (b) iron rusts (c) gold has higher reduction potential than iron . . d) gold has lower reduction potential than iron

Problem 339**[GRB (O.P. Tandon) p.81 Q3]**

Oil electrolysis, which of the following does not give out oxygen? (a) Acidic water using Pt electrode (b) Fused NaOH using Pt electrode (c) Dilute H_2SO_4 using Pt electrode (d) Dilute H_2SO_4 using Cu electrode

Problem 348**[GRB (O.P. Tandon) p.82 Q13]**

How many coulombs are required for the oxidation of 1 mole

Problem 349**[GRB (O.P. Tandon) p.82 Q14]**

of H_2O_2 to O_2 ? (a) $9.65 \times 10^4 \text{ C}$ (c) $1.93 \times 10^5 \text{ C}$ (b) 93000 C (d) $19.3 \times 10^2 \text{ C}$ [Hint: $\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$]
 [Hint: $1 \text{ mol H}_2\text{O}_2 \rightarrow 2 \text{ mol e}^- \Rightarrow 2 \times 96500 \text{ C} = 193000 \text{ C} = 1.93 \times 10^5 \text{ C}$]. W " ItE 96500 E W x 965JO $= 22.2 \times 96500 = 59.5$ It 2 x 5 x 3600 . . Atomic.
 mass EqUivalent mass = $\frac{\text{Atomic mass}}{\text{Oxidation state}}$ $59.5 = 177 \text{ n .}$;

Problem 350**[GRB (O.P. Tandon) p.82 Q15]**

When water is electrolysed, hydrogen and oxygen gases are produced. If 1.008 g of H_2 is liberated at cathode, what mass of O_2 is formed at the anode? (a) 32g (b) 16g (c) 8g (d) 4g [Hint: $\text{W}_1 \text{ W}_2 = 1.008 \times 8 = 8 \text{ g}$ where, E_1 and E_2 are equivalent mass of hydrogen and oxygen respectively.]

Problem 351**[GRB (O.P. Tandon) p.82 Q16]**

The cell potential (E) and the free energy change (ΔG) accompanying an electrochemical reaction are related by: (a) $\Delta G = nF \log E$ (b) $\Delta G = nFE$ (c) $-\Delta G = nFE$ (d) $-\Delta G = nE \cdot \log E$

Problem 352**[GRB (O.P. Tandon) p.82 Q11]**

The units of conductivity are: (a) siemens-1 cm-1 (b) siemens cm (c) siemens cm-1 (d) siemens cm-2 mol-1
 HI- The calomel electrode used as a reference electrode contains: (a) PbO_2 - PbSO_4 mixture (b) HgCl_2 (c) Hg_2Cl_2 (d) ZnCl_2 . 19.
 KCl is used in salt bridge because: (a) it forms a good jelly with agar-agar (b) it is a strong electrolyte (c) it is a good conductor of electricity (d) the transference number of K^+ and Cl^- ions are almost equal [Hint: Only those electrolytes are used in salt bridge whose ions have same transference number in agar-agar gel.]

Problem 353**[GRB (O.P. Tandon) p.82 Q20]**

The increase in the equivalent conductance of a salt solution on dilution is due to increase in the: (a) attraction between the ions (b) degree of ionization of the salt (c) molecular attraction (d) association of the salt

Problem 354**[GRB (O.P. Tandon) p.82 Q21]**

When 96500 coulombs of electricity are passed through nickel sulphate solution, the amount of nickel deposited will be: (a) 1 mol (b) 0.5 mol (c) 0.1 mol (d) 2 mol [Hint: 1 faraday deposits 1 equivalent of nickel. 1 equivalent of $\text{Ni} = 1/2$ mole of nickel.]

Problem 355**[GRB (O.P. Tandon) p.82 Q22]**

$\text{Al}^{3+}(\text{aq.}) + 3\text{e}^- \rightarrow \text{Al}(\text{s})$; $E^\circ = -1.66 \text{ V}$ $\text{Cu}^{2+}(\text{aq.}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$; $E^\circ = +0.34 \text{ V}$ potentials? (a) 1.32 V (b) 2.00 V [Hint; The cell will be: (c) 2.30 V (d) 4.34 V
 $\text{Al}(\text{s}) | \text{Al}^{3+}(\text{aq.}) || \text{Cu}^{2+}(\text{aq.}) | \text{Cu}(\text{s})$ $E_{\text{cell}} = E_{\text{athode}} - E_{\text{athode}} = E_{\text{Cu}^{2+}/\text{Cu}} - E_{\text{Al}^{3+}/\text{Al}} = +0.34 - (-1.66) = +2.00 \text{ V}$]

Problem 356**[GRB (O.P. Tandon) p.82 Q23]**

For which of these oxidation/reduction pairs will the reduction potential vary with pH? I. AmO^+/AmO II. $\text{AmO}^+/\text{Am}^{4+}$ III. $\text{Am}^{4+}/\text{Am}^{2+}$ (a) I only (b) II only (c) I and II only (d) I, II and III (Hint: $4\text{H}^+ + \text{AmO}^+ + 2\text{e}^- \rightarrow \text{Am}^{4+} + 2\text{H}_2\text{O}$ It includes H^+ ions; hence the electrode potential depends on pH.)

Problem 357**[GRB (O.P. Tandon) p.82 Q24]**

$2\text{Ag} + (\text{aq.}) + \text{Cu(s)} \rightarrow \text{Cu}^{2+} (\text{aq.}) + 2\text{Ag(s)}$ The standard potential E° for this reaction is 0.46 V. Which change will increase the potential the most? (a) Doubling the $[\text{Ag}^+]$ (b) Halving the $[\text{Cu}^{2+}]$ (c) Doubling the size of the Cu(s) electrode (d) Decreasing the size of the Ag electrode by one-half

Problem 358**[GRB (O.P. Tandon) p.82 Q25]**

$10\text{Cl}^- (\text{aq.}) + 2\text{MnO}_4^{2-} (\text{aq.}) + 16\text{H}^+ (\text{aq.}) \rightarrow 5\text{Cl}_2 (\text{g}) + 2\text{Mn}^{2+} (\text{aq.}) + 8\text{H}_2\text{O(l)}$. The value of E° for this reaction is 0.15 V. What is the value of the equilibrium constant (K) for this reaction? (a) 2.4×10^{25} (b) 4.9×10^{12} (c) 1.2×10^5 (d) 3.4×10^2 Hint: $K = \text{antilog} \left[\frac{nE^\circ}{0.059} \right]$ U. What takes place when zinc metal is added to an aqueous solution containing magnesium nitrate and silver nitrate? */bullet*

Problem 359**[GRB (O.P. Tandon) p.83 Q4]**

No reaction takes place (a) I and 2 only (b) I and 3 only (c) 1, 2 and 3 only (d) 4 only

Problem 360**[GRB (O.P. Tandon) p.83 Q27]**

In the galvanizing process, iron is coated with zinc. The resulting chemical protection is most similar to that provided when: (a) a magnesium bar is connected to an iron pipe (b) an iron can is plated with tin (c) copper pipes are connected using lead solder (d) a copper pipe is covered with epoxy paint

Problem 361**[GRB (O.P. Tandon) p.83 Q28]**

What is the sign of ΔG° and the value of K for an

Problem 362**[GRB (O.P. Tandon) p.83 Q33]**

electrochemical cell for which 0.80 V? (c) (d) $K > 1$ < 1 < 1 'Consider a voltaic cell based on these half-cells: $\text{Ag}^+ (\text{aq.}) + \text{e}^- \rightarrow \text{Ag(s)}$; $E^\circ = 0.80\text{V}$ $\text{Cd(s)} \rightarrow \text{Cd}^{2+} (\text{aq.}) + 2\text{e}^-$; $E^\circ = 0.40\text{V}$ Identify the anode and give the voltage of this cell under standard conditions. (a) Ag ; $E^\circ = 0.40\text{V}$ (c) Cd ; $E^\circ = 1.20\text{V}$ (b) Ag ; $E^\circ = 2.00\text{V}$ (d) Cd ; $E^\circ = 2.00\text{V}$ [Hint: Anode has lower standard reduction potential; thus Cd will be considered as anode. $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = 0.80 - 0.40 = 1.20\text{V}$] If the E° for a given reaction has a negative value, which gives the correct relationships for the values of ΔG° and K ? (a) $\Delta G^\circ > 0$; $K < 1$ (b) $\Delta G^\circ > 0$; $K > 1$ (c) $\Delta G^\circ < 0$; $K > 1$ (d) $\Delta G^\circ < 0$; $K < 1$ Which of the following solutions is used as an anti-rusting solution? (a) Na_2SO_4 (b) Na_3PO_4 (c) Na_3BO_3 (d) Na_2S The pressure of hydrogen gas is increased from 1 atm to 100 atm. Keeping the H^+ (1 M) constant, the voltage of the hydrogen half-cell at 25°C will be: (a) 0.059 V (b) 0.59 V (c) 0.0295 V (d) 0.118 V Efficiency of the following cell is 84% $\text{A(s)} + (\text{aq.}) \rightarrow \text{A}^{2+} (\text{aq.}) + \text{B(s)}$; $\Delta G^\circ = -285\text{kJ}$ Then the standard electrode potential of the cell will be: (a) 1.20 V (b) 2.40 V (c) 1.10 V (d) 1.24 V $\Delta G^\circ = -nFE^\circ$ [Hint: Efficiency $\Delta G^\circ = -285 \times 1000 \text{ J} = -285000 \text{ J}$ $-285000 = -n \times 96500 \times E^\circ$ $E^\circ = 1.24\text{V}$]

Problem 363

[GRB (O.P. Tandon) p.83 Q34]

A chemical reaction will be spontaneous if: (a) is positive (b) ΔG is negative (c) reaction quotient, $Q < K$ (d) E_{cell} is negative

Problem 364

[GRB (O.P. Tandon) p.83 Q35]

Match the List-I with List-II: List-I (Electrode) List-II (Type)

Problem 365

[GRB (O.P. Tandon) p.83 Q1]

Calomel (A) Reference

Problem 366

[GRB (O.P. Tandon) p.83 Q3]

Hydrogen (C) Membrane

Problem 367

[GRB (O.P. Tandon) p.83 Q4]

Quinhydrone (D) Gas Codes: (a) I-A 2-C 3-D 4-B (b) I-B 2-A 3-D 4-C (c) I-C 2-B 3-A 4-D (d) I-D 2-A 3-C 4-B mho and its equivalent conductance is 1.53 mho cm² eq⁻¹. The K_{sp} for BaSO₄ will be: (a) 4×10^{-12} M (b) 4×10^{-6} M - 4×10^{-6} M²

Problem 368

[GRB (O.P. Tandon) p.83 Q37]

The standard reduction potential of hydrogen is zero because: (a) it is assumed (b) hydrogen is easiest to oxidise (c) hydrogen has single electron (d) hydrogen is electronegative In the following three questions, three statements I, II and III are given. Mark: (a) if all the statements are correct (b) if II and III are correct- (c) if I and III are correct (d) if only II is correct IBHU (Mains) 2007

Problem 369

[GRB (O.P. Tandon) p.83 Q38]

I. Conductance of electrolyte solution increases with temperature. II. Resistivity is reciprocal of molar conductivity of electrolyte. III. Cell constant has unit cm⁻¹. (a) (b) (c) (d)

Problem 370

[GRB (O.P. Tandon) p.83 Q39]

1. The conductivity of molten NaCl is due to movement of Na⁺ and Cl⁻ ions. II. Solid NaCl is also conductor of electricity. III. Molten sodium is a good conductor because of mobile electrons. (a) (b) (c) (d)

Problem 371

[GRB (O.P. Tandon) p.83 Q40]

1. Cathode is -ve terminal both in electrochemical and electrolytic cells. II. Reduction occurs at cathode both in galvanic as well as electrolytic cells. III. Chemical change in electrolytic cell is non-spontaneous. (a) (b) (c) (d)

Problem 372

[GRB (O.P. Tandon) p.84 Q41]

In an experiment, 0.04 F was passed through 400 mL of a 1M solution of NaCl. What would be the pH of the solution after the electrolysis? [PMT (Kerala) 2007] (a) 8 (b) 10 (c) 13 (d) 6 [Hint: Electrolysis of aq. NaCl gives hydrogen gas at cathode and oxygen gas at anode, the electrolyte solution contains NaOH after electrolysis. Number of equivalents of NaOH formed = 0.04 Normality, $N \times V = 0.04 \times 1000 = 0.1$ eq. $\therefore$ 400 mL of 0.1M NaOH. $\therefore$ pOH = 1. pH = 13]

Problem 373**[GRB (O.P. Tandon) p.84 Q42]**

An alloy of Pb and Ag weighing 1.08 g was dissolved in dilute acid and the volume made to 100 mL. A silver electrode was used and the potential was 0.80 V, what is the percentage of Ag in the alloy? (At 25°C, $R/F = 0.06$) (PET (Kerala) 2001) (a) 25 (b) 2.50 (c) 10 (d) 1 (e) 50
 Hint: Overall cell reaction is: $\text{H}_2(\text{g}) + 2\text{Ag}^+(\text{aq.}) \rightleftharpoons 2\text{H}^+(\text{aq.}) + 2\text{Ag}(\text{s})$
 $E^\circ = 0.06 \times 2.303 \log \frac{[\text{H}^+]^2}{[\text{Ag}^+]^2}$
 $0.80 = 0.06 \times 2.303 \log \frac{[\text{H}^+]^2}{[\text{Ag}^+]^2}$
 $2 \log [\text{Ag}^+] = 0.05 \text{ M}$
 Number of moles of Ag in 100 mL = 0.005
 Mass of silver = 0.005 x 108 g = 0.54 g
 Percentage of Ag in alloy = $\frac{0.54}{1.08} \times 100 = 50\%$

Problem 374**[GRB (O.P. Tandon) p.84 Q43]**

Select the correct statements about dry cell: (a) It is also called Leclanche cell. (b) It is also called Daniell cell (c) Electrolyte used is moist paste of NH_4Cl and ZnCl_2 (d) Cathodic process is: $2\text{MnO}_2(\text{s}) + 2\text{NH}_4^+(\text{aq.}) + 2\text{e}^- \rightarrow \text{Mn}_2\text{O}_3(\text{s}) + 2\text{NH}_3(\text{g})$

Problem 375**[GRB (O.P. Tandon) p.84 Q44]**

Given the standard oxidation potentials, $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$ $E^\circ = +0.4 \text{ V}$, $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$ $E^\circ = -0.8 \text{ V}$, $\text{Fe} \rightarrow \text{Fe}^{3+} + 3\text{e}^-$ $E^\circ = -0.9 \text{ V}$. It is easier to oxidise Fe^{2+} to Fe^{3+} in: (a) acid medium (b) alkaline medium (c) neutral medium (d) both in acidic and alkaline mediums
 Hint: E° for $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$ is positive in alkaline solution, therefore, it is easier to oxidise to Fe^{3+} in alkaline medium.]

Problem 376**[GRB (O.P. Tandon) p.84 Q45]**

Dipping iron article into a strongly alkaline solution of sodium phosphate: (VIJEE 2008) (a) does not affect the article. (b) forms $\text{Fe}_2\text{O}_3 \cdot \text{H}_2\text{O}$ on the surface (c) forms iron phosphate film (d) forms ferric hydroxide

Problem 377**[GRB (O.P. Tandon) p.84 Q46]**

For the redox process, $\text{Zn}(\text{s}) + \text{Cu}^{2+}(\text{aq.}) \rightleftharpoons \text{Zn}^{2+}(\text{aq.}) + \text{Cu}(\text{s})$, $E^\circ_{\text{cell}} = 1.10 \text{ V}$ which graph correctly represents E_{cell} (Y-axis) as a function of $[\text{Zn}^{2+}]$ (X-axis)?
 (a) $E_{\text{cell}} = 1.10 \text{ V} - \frac{0.06}{2} \log [\text{Zn}^{2+}]$ (b) $E_{\text{cell}} = 1.10 \text{ V} + \frac{0.06}{2} \log [\text{Zn}^{2+}]$ (c) $E_{\text{cell}} = 1.10 \text{ V} - \frac{0.06}{2} \log \frac{1}{[\text{Zn}^{2+}]}$ (d) $E_{\text{cell}} = 1.10 \text{ V} + \frac{0.06}{2} \log \frac{1}{[\text{Zn}^{2+}]}$
 A fuel cell involves combustion of the butane at 1 atm and 298 K: $\text{C}_4\text{H}_{10}(\text{g}) + \frac{13}{2}\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 5\text{H}_2\text{O}(\text{l})$. $\Delta_f G^\circ = -2746 \text{ kJ/mol}$ what is E° of a cell? (a) +4.74 V (b) +0.547 V (c) +1.09 V (d) +4.37 V [Hint: In the reaction $\text{C}_4\text{H}_{10}(\text{g}) + \frac{13}{2}\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 5\text{H}_2\text{O}(\text{l})$ Change in oxidation number of carbon = $4 \times (4 - 0) = +16$. Number of electrons involved in cell process will be 26. $E^\circ = \frac{-\Delta_f G^\circ}{nF} = \frac{26 \times 96500}{26 \times 96500} = 1.09 \text{ V}$]

Problem 378**[GRB (O.P. Tandon) p.84 Q48]**

Molar conductance Λ_m is plotted against $\sqrt{C}$ (mol litre⁻¹) for three electrolytes (NaCl , HCl , NH_4OH) (3)

[GRB (O.P. Tandon) p.85 Q49]

In the concentration cell $\text{HCl NaCl NaCl HCl} \parallel 3 \text{ NH}_4\text{OH NH}_4\text{OH} \parallel \text{HCl NaCl Pt(H)}_2 \parallel \text{HCl NaCl Pt(H)}_2$ (O.M) iiHA (1M) I (H⁺) Pt .2 NaA (1M) NaA (1M) 2 (PKa ofHA = 4) Cell potential will be: (a) 0.03 V (b) 0.06 V (c) - 0.06 V (d) 0.03 V pH Anode 3 pH Cathode 4 (fromeq. I) Eccll - 0.06 V] 50;-In the following-proceess'ofdisproportionation:' $\text{E}^\circ_{\text{ClO}_3/\text{ClO}_2} = 1.04 \text{ V}$ $\text{E}^\circ_{\text{ClO}_2/\text{ClO}} = 0.36 \text{ V}$ Chlorate ClO_3^- ; + ClO_2 ; + 0.33 V Per chlorate ion ion [4' : /textasciitilde() - 1- (c)

[GRB (O.P. Tandon) p.85 Q50]

(d) , (c) (c) (b) (a) (d) (b) Initial concentration of chlorate ion was 0.1 M, The equilibrium concentration of per chlorate ion will be : (a) 0,19 V (b) 0J M (c) 0,024 M (d) 0.019 M [Hint: $\text{ClO}_2(\text{aq.}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{ClO}_2^-(\text{aq.}) + 2\text{H}^+(\text{aq.}) + 2\text{e}^-$ $2\text{H}^+ + \text{ClO}_2^-(\text{aq.}) + 2\text{e}^- \rightleftharpoons \text{ClO}_2(\text{aq.}) + \text{H}_2\text{O}(\text{l})$ $\text{ZClO}_2(\text{aq.}) / \text{ClO}_2(\text{aq.}) + \text{ClO}_2^-(\text{aq.}) = 0.33 - 0.36 = -0.03 \text{ V}$ $E - E^\circ = 0.059 \text{ V} \log Q$ - $\log n$ At equilibrium, $E = 0$, $n = 2$, $Q = K$ $0.03 - 0.059 \log K = 2 \text{ClO}_2 / \text{ClO}_2(\text{aq.}) + \text{ClO}_2^-(\text{aq.})$ $0.1 \text{ o o } 0.1$ $2x \text{ x x } K = x/x =$ $-(\text{OX} : - 2\text{X}) / 10 - \dots$ $x = 0.019$]

[GRB (O.P. Tandon) p.86 Q1]

How many grams of water will be electrolysed by 96500 coulomb charge?

[GRB (O.P. Tandon) p.86 Q2]

Number of faradays required to convert 1 mole of Cr^{2+} into

[GRB (O.P. Tandon) p.86 Q4]

The ratio Of(/textasciitilde): J for Ca3 (P04 h will be equal to :

[GRB (O.P. Tandon) p.86 Q6]

If an aqueous solution of NiCl_2 is electrolysed using platinum electrode by a current of 5 amp, then what volume of Cl_2 gas in litres at STP will be produced? 7, Charge of 6.24×10^{18} electrons will be (in coulomb):

[GRB (O.P. Tandon) p.86 Q8]

A current of 2 amp-when passed for 5 hour through a molten salt deposits 22.2 g of metal of atomic mass 177. The positive oxidation state of the metal in the metal salt is :

[GRB (O.P. Tandon) p.86 Q9]

. Cr(s) | Cr³⁺ || Fe²⁺ | Fe(s) In above cell, the value of n in the Nernst equation: i.e., $E = E^\circ - \frac{0.05910}{n} \log Q$ will be: n

[GRB (O.P. Tandon) p.86 Q10]

In the Nernst Equation, Q will be equal to the equilibrium constant Kc, when the cell potential E is equal to :

Problem 388**[GRB (O.P. Tandon) p.86 Q5]**

12 (s) / r (0.1 M) half-cell is connected to W (aq) / (H₂ 1 atm) || a / textasciitilde() If / textasciitilde() c (: lLa! (U: tsc / textasciitilde() R1! 9tenti / textasciitilde() lJ found to J, le 0.7714 V J (0.535 V, the pH of W / Hi half-cell will be : .. -.-.... —.. - / textasciitilde() / textasciitilde() - / textasciitilde() ——— -.-. —.-.-.-, -.- / textasciitilde() ., .. -.-.- / textasciitilde(), .. -.- / textasciitilde() —

Problem 389**[GRB (O.P. Tandon) p.87 Q1]**

Equivalent mass of sulphuric acid in lead storage battery is: (a) 49 (b) 98 (c) 24.5 (d) none of these

Problem 390**[GRB (O.P. Tandon) p.87 Q2]**

Normalities of sulphuric acid before and after discharge are: (a) 5.15, 2.32 (b) 2.32; 5.15 (c) 5.15, 5.15 (d) 2.32, 2.32 (c) 265.04 amp-hrs (d) 26.504 amp-hrs

Problem 391**[GRB (O.P. Tandon) p.87 Q4]**

The amount of charge which the battery must have been used is: (a) 9.88 F (b) 8.98 F (c) 8.89 F (d) 7.88 F 50 / cdot / cdot Which oHhe / cdot fo U (') wing-takes / cdot place in discharge-process at / cdot anode? (a) PbO₂ + 4H⁺ + SO / textasciitilde() - + 2e⁻ PbSO₄ + 2H₂O. (b) PbSO₄ + 2H₂O / textasciitilde() PbO₂ + 4W + SO / textasciitilde() - + 2e⁻ (c) Pb + SO / textasciitilde() - / textasciitilde() PbSO₄ + 2e⁻ (d) PbSO₄ + 2e / textasciitilde() / textasciitilde() Pb + SO!- /bullet Passage2 Electrolysis is the process in which electrical energy is converted to chemical energy. In electrolytic cell, oxidation takes place at anode and reduction at cathode. Electrode process depends on the electrode taken for electrolysis. Amount of substance liberated at an electrode is directly proportional to the amount of charge passed through it. The mass of substance liberated at electrode is calculated using the following relation: $m = \frac{ItE}{96500}$ Here, E represents the equivalent mass and 96500 C is called the Faraday constant. Faraday (96500 C) is the charge of 1 mole electron, i.e., 6.023 × 10²³ electrons; it is used to liberate one gram equivalent of the substance. Answer the following questions:

Problem 392**[GRB (O.P. Tandon) p.87 Q1]**

Th / textasciitilde() platinum electrodes were immersed in a solution of cupric sulphate (CuSO₄) and electric current is passed through the solution. After sometime, it was observed that the colour of copper sulphate disappeared with evolution of a gas at the electrode. The colourless solution contains: . (a) platinum sulphate (b) copper nitrate (c) copper sulphate (d) sulphuric acid

Problem 393**[GRB (O.P. Tandon) p.87 Q2]**

The passage of current liberates H₂ at cathode and Cl₂ at anode. The solution is: (a) copper chloride in water (b) NaCl in water (c) mercuric chloride in water (d) AuCl₃ in water

Problem 394**[GRB (O.P. Tandon) p.87 Q3]**

On -electrolysis of dilute sulphuric acid using platinum electrodes, the product obtained at the anode will be: (a) hydrogen (b) oxygen (c) hydrogen sulphide (d) sulphur oxide

Problem 395**[GRB (O.P. Tandon) p.87 Q4]**

How many faradays are required to reduce 1 mol BrO₃ to Br⁻? 003 005 / textasciitilde() 6 004

Problem 396**[GRB (O.P. Tandon) p.87 Q5]**

Calculate the volume of gas liberated at the anode at STP during the electrolysis of a CuSO_4 solution by a current of 1 A passed for 16 minutes and 5 seconds: (a) 224 mL (b) 56 mL (c) 112 mL (d) 448 mL

Problem 397**[GRB (O.P. Tandon) p.87 Q6]**

The quantity of electricity required to liberate 112 cc hydrogen at STP is _____ (Comed (Karnataka) / (a) 965 C (b) 9650 C (c) 96500 C (d) 4825 C)

Passage 3 The concentration of potassium ions inside a biological cell is at least twenty times higher than the outside. The resulting potential difference across the cell is important in several processes such as transmission of nerve impulses and maintaining the ion balance. A simple model for such a concentration cell involving a metal M is: $\text{M(s)} | \text{M}^+(\text{aq}, 0.05 \text{ molar}) || \text{M}^+(\text{aq}, 1 \text{ molar}) | \text{M(s)}$ For the above electrolytic cell, the magnitude of the cell potential is 70 mV. Answer the following questions:

Problem 398**[GRB (O.P. Tandon) p.87 Q1]**

For the above cell (a) $E_{\text{cell}} < 0$; $\Delta G > 0$ (b) $E_{\text{cell}} > 0$; $\Delta G < 0$ (c) $E_{\text{cell}} < 0$; $\Delta G > 0$ (d) $E_{\text{cell}} > 0$; $\Delta G < 0$

Problem 399**[GRB (O.P. Tandon) p.87 Q2]**

If the 0.05 molar solution of M^+ is replaced by a 0.0025 molar M^+ solution, then the magnitude of the cell potential would be: (a) 35 mV (c) 140 mV (b) 70 mV (d) 700 mV (IIT 2010 !Hint: 1 (b) Electrolyte concentration cell will be spontaneous when the concentration in cathodic half cell is greater than that of anodic half cell. Thus, the given cell is spontaneous hence $\Delta G < 0$, $E_{\text{cell}} > 0$

Problem 400**[GRB (O.P. Tandon) p.87 Q2]**

(c) $E_{\text{cell}} = 0$, for every concentration cell $E = 0.059 \log \frac{[\text{M}^+]_{\text{anode}}}{[\text{M}^+]_{\text{cathode}}} = 0.059 \log \frac{0.05}{0.0025} = +153 \text{ mV}$ It is close to 140 mV.]

Problem 401**[GRB (O.P. Tandon) p.88 Q1]**

On the basis of information available for the reaction: $4 \text{Al} + 3 \text{O}_2 \rightarrow 2 \text{Al}_2\text{O}_3$; $\Delta G = -827 \text{ kJ}$ per mol of O_2 consumed, the minimum emf required to carry out an electrolysis of Al_2O_3 is: (Given: $1 \text{ F} = 96500 \text{ C}$) (a) 2.14 V (b) 4.28 V (c) 6.42 V (d) 8.56 V [Hint: $\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^-$ i. e., 4 mol Al $\rightarrow$ 12 mol e^- $\therefore n = 4$ $\therefore \Delta G = -nFE$ $-827 \times 1000 = -4 \times 96500 \times E$ $E = 2.14 \text{ V}$]

Problem 402**[GRB (O.P. Tandon) p.88 Q2]**

The equilibrium constant K_c will be equal to Q , when: (a) $E = E_0$ (b) $RT \ln F = 1$ (c) $E = 0$ (d) $E_0 = 1$

Problem 403**[GRB (O.P. Tandon) p.88 Q3]**

The nature of graph of E_{cell} against $\log K_c$ is: (a) straight line (b) parabola (c) hyperbola (d) elliptical curve

Problem 404**[GRB (O.P. Tandon) p.88 Q4]**

The equilibrium constant K_c for the reaction: $\text{Cu(s)} + 2\text{Ag}^+(\text{aq.}) \rightleftharpoons \text{Cu}^{2+}(\text{aq.}) + 2\text{Ag(s)}$ ($E^\circ_{\text{cell}} = 0.46 \text{ V}$) will be: (a) $\text{antilog } 15.6$ (c) $\text{antilog } 1.5$ (b) $\text{antilog } 2.5$ (d) $\text{antilog } 12.2$

Problem 405**[GRB (O.P. Tandon) p.88 Q5]**

In an electrochemical cell, $\text{Zn(s)} \mid \text{Zn}^{2+} (1 \text{ M aq.}) \parallel \text{Cu}^{2+} (1 \text{ M aq.}) \mid \text{Cu(s)}$ is 1.10 volt at 25°C . The equilibrium constant for the cell reaction: will be: (a) 10^{-37} (b) 10^{37} (c) 10^{-39} (d) 10^{39}
 concentration: $\Lambda_m = \Lambda^\circ - b\sqrt{c}$ where, Λ° = molar conductance at infinite dilution, b = constant
 The degrees of dissociation of weak electrolytes are calculated as:
 Answer the following questions:

Problem 406**[GRB (O.P. Tandon) p.88 Q1]**

Which of the following decreases on dilution of electrolyte solution? . (a) Equivalent conductance . (b) Molar conductance (c) Specific conductance . (d) Conductance

Problem 407**[GRB (O.P. Tandon) p.88 Q2]**

The correct order of equivalent conductances at infinite dilution of LiCl , NaCl and KCl is: (a) $\text{LiCl} > \text{NaCl} > \text{KCl}$ (b) $\text{KCl} > \text{NaCl} > \text{LiCl}$ (c) $\text{NaCl} > \text{KCl} > \text{LiCl}$ (d) $\text{LiCl} > \text{KCl} > \text{NaCl}$

Problem 408**[GRB (O.P. Tandon) p.88 Q3]**

For which of the following electrolytic solutions Λ_m and Λ° are equal? . (a) BaCl_2 (b) KCl (c) $\text{Al}_2(\text{SO}_4)_3$ (d) CaCl_2

Problem 409**[GRB (O.P. Tandon) p.88 Q4]**

The conductance of a solution of an electrolyte is equal to that of its specific conductance. The cell constant of the conductivity cell is equal to: . (a) resistance (b) faraday (c) zero (d) unity

Problem 410**[GRB (O.P. Tandon) p.88 Q5]**

Which of the following equality holds good for the strong electrolytes? (a) $\Lambda = \Lambda^\circ$ (b) $\Lambda = \Lambda^\circ$ (c) $\Lambda = \Lambda^\circ$ (d) $\Lambda = \Lambda^\circ$
 Passage 6 (b) $\Lambda = \Lambda^\circ$ (d) $\Lambda = \Lambda^\circ$
 At infinite dilution, when the dissociation of electrolyte is complete, each ion makes a definite contribution towards the molar conductance of electrolyte, irrespective of the nature of the other ion with which it is associated.

Problem 411**[GRB (O.P. Tandon) p.89 Q2]**

Which of the following oxides will be thermally most stable? (a) ZnO (b) MgO (c) Cu_2O (d) Ag_2O

Problem 412**[GRB (O.P. Tandon) p.89 Q3]**

Which of the following reactions is not correct? (a) $2\text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2\text{SO}_4 + \text{H}_2$ (b) $\text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{FeSO}_4 + \text{H}_2$ (c) $\text{Mg} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2$

Problem 417**[GRB (O.P. Tandon) p.89 Q5]**

The Unit of molar conductance of an electrolyte solution will be: (a) $\Omega^{-1} \text{cm}^2 \text{mol}^{-1}$ (b) $\text{S cm}^2 \text{mol}^{-1}$ (c) $\text{S cm}^2 \text{mol}^{-1}$ (d) $\Omega^{-1} \text{cm}^2 \text{mol}^{-1}$. The potential associated with each electrode is known as electrode potential. If the concentration of each species taking part in the electrode reaction is unity (if any appears in the electrode reaction, it is confined to 1 atmospheric pressure) and further the reaction is carried out at 298 K, then the potential of each electrode is said to be the standard electrode potential. By convention, the standard electrode potential of hydrogen electrode is 0.0 volt. The electrode potential value for each electrode process is a measure of relative tendency of the active species in the process to remain in the oxidised/reduced form. A negative E° means that the redox couple is a stronger reducing agent than the H^+ / H_2 couple. A positive E° means that the redox couple is a weaker reducing agent than the H^+ / H_2 couple. The metal with greater positive value of standard reduction potential forms the oxide of greater thermal stability. Answer the following questions:

1. Given the standard reduction potentials, $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) = +0.77 \text{ V}$, $E^\circ(\text{Ag}^+/\text{Ag}) = +0.80 \text{ V}$, $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V}$, $E^\circ(\text{Hg}^{2+}/\text{Hg}) = +0.79 \text{ V}$ (a) Mg (b) Zn (c) Sn (d) Ag

Chemical reactions involve interaction of atoms and molecules. A large number of atoms/molecules (approximately 6.023×10^{23}) are present in a few grams of any chemical compound varying with their atomic/molecular masses. To handle such large numbers conveniently, the mole concept was introduced. This concept has implications in diverse areas such as analytical chemistry, biochemistry, electrochemistry and radiochemistry. The following example illustrates a typical case, involving chemical/electrochemical reaction, which requires a clear understanding of the mole concept. A 4.0 molar aqueous solution of NaCl is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes (Atomic mass: Na = 23; Hg = 200; 1 Faraday = 96500 coulombs). (IIT 2007) Answer the following questions:

Problem 418**[GRB (O.P. Tandon) p.89 Q1]**

The total number of moles of chlorine gas evolved is: (a) 0.5 (b) 1.0 (c) 2.0 (d) 3.0 [Hint: Number of moles of NaCl = $MV = 4 \times 500 = 2$. $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ 2 mol Cl^- ions give 1 mol Cl_2]

Problem 419**[GRB (O.P. Tandon) p.89 Q2]**

If the cathode is a Hg electrode, the maximum weight (in g) of amalgam formed from this solution is: (a) 200 (b) 225 (c) 400 (d) 446 [Hint: Electrolysis gives 2 mol sodium at cathode. Thus, amalgam (Na / Hg) will contain 2 mol of each sodium and Hg. Mass of amalgam = $2 \times 200 + 2 \times 23 = 446$]

Problem 420**[GRB (O.P. Tandon) p.89 Q3]**

The total charge (in coulombs) required for complete electrolysis is: (a) 24125 (b) 48250 (c) 96500 (d) 193000 [Hint: 2 mol electrons will be required, therefore, required charge will be 2 faraday or 193000 coulombs.]

Problem 421**[GRB (O.P. Tandon) p.90 Q5]**

(a, b, c) Passage 7.

Problem 422**[GRB (O.P. Tandon) p.90 Q3]**

(d) ASSIGNMENT NO. 12 'SECTION 1' Straight Objective Type Questions This section contains 10 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which only one is correct.

Problem 423**[GRB (O.P. Tandon) p.90 Q1]**

The specific conductance (K) of an electrolyte of 0.1N concentration is related to equivalent conductance (Λ_e) by the following formula: [ICET (J and K) 2007], (a) $\Lambda_e = K$ (b) $\Lambda_e = 10K$ (c) $\Lambda_e = 100K$ (d) $\Lambda_e = 10000K$

Problem 424**[GRB (O.P. Tandon) p.90 Q2]**

The standard electrode potentials (E°) of A, B and C are +0.68 V, 2.54 V, 0.50 V respectively. The order of their reducing power is: (MHT-CET 2007) (a) $A > B > C$ (b) $A > C > B$ (c) $C > B > A$ (d) $B > C > A$

Problem 425**[GRB (O.P. Tandon) p.90 Q3]**

In the electrochemical reaction, $Zn + 2Fe^{2+} \rightarrow Zn^{2+} + 2Fe$, increasing the concentration of Fe^{2+} : (IJEE (W8) 2007) (a) increasing the cell emf (b) increasing the current flow (c) decrease the cell emf (d) alter the pH of the solution

Problem 426**[GRB (O.P. Tandon) p.90 Q4]**

Fully charged lead storage battery contains 1.5 L of 5 M H_2SO_4 . If 2.5 amp of current is taken from the cell for 965 minutes, then what will be the molarity of remaining H_2SO_4 ? Assume that volume of battery fluid to be constant: (a) 4 M (b) 3.5 M (c) 2M (d) 4.25 M

Problem 427**[GRB (O.P. Tandon) p.90 Q5]**

In a hydrogen-oxygen cell, 67.2 litre of H_2 at STP is used in 15 minutes. What is the average current produced? (a) 549.4 amp (b) 643.33 amp (c) 965 amp (d) 129.8 amp

Problem 428**[GRB (O.P. Tandon) p.90 Q6]**

The cell reaction involving quinhydrone electrode is: $OH^- + 2H^+ \rightleftharpoons H_2O$ OR $2H^+ + 2e^- \rightleftharpoons H_2$ What will be the electrode potential at pH: 3? (a) 1.48 V (b) 1.20 V (c) 1.10 V (d) 1.30 V
 Hint: $E^\circ_{H^+/H_2} = 0.0591 \log [H^+]^2 = 1.30 + 0.0591 \times 3 = 1.48 \text{ V}$

Problem 429**[GRB (O.P. Tandon) p.90 Q7]**

The standard reduction potential E° for Cr^{3+}/Cr and Fe^{2+}/Fe are 0.86 V and -1.10 V respectively. The E° value of Cr^{3+}/Fe^{2+} will be: (a) +1.96 V (b) -1.96 V (c) +0.24 V (d) 0.24 V
 Hint: $OCr^{3+} + H_2 + e^- \rightarrow Cr + 2H^+$ $E^\circ = 0.86 \text{ V}$ $Cl_2 + e^- \rightarrow Cl^-$ $E^\circ = -1.10 \text{ V}$, $2' = -0.24 \text{ V}$

Problem 430**[GRB (O.P. Tandon) p.90 Q8]**

The standard reduction potential for the following two reactions, are given: $AgCl + e^- \rightarrow Ag(s) + Cl^-$ (aq.); $E^\circ = 0.22 \text{ V}$... (i) $Ag^+ (aq.) + e^- \rightarrow Ag(s)$, $E^\circ = 0.80 \text{ V}$... (ii) The solubility product of $AgCl$ under standard condition will be: (a) $1.613 \times 10^{-5} \text{ M}^2$ (b) $1.535 \times 10^{-8} \text{ M}^2$ (c) $3.213 \times 10^{-10} \text{ M}^2$ (d) $1.535 \times 10^{-10} \text{ M}^2$

Problem 431**[GRB (O.P. Tandon) p.91 Q9]**

The equilibrium constant for the reaction, question has following 4 choices (a), (b), (c) and (d), out of . $\text{Cu(s)} + \text{Cu}^{2+}(\text{aq.}) \rightleftharpoons 2\text{Cu}^+(\text{aq.})$ which only one is correct.' . . (a) Statement-1 is true; statement-2 is true; statement-2 is a $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34 \text{ V}$. $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.15 \text{ V}$ correct explanation for statement-I. (Given: $\log 3.72 \times 10^{-5} = 0.571$) (b) Statement-1 is true; statement-2 is true; statement-2 is not (a) 3.72×10^{-6} .. (b) 3.72×10^{-5} . Is correct explanation for statement-I. (c) 3.72×10^{-7} (d) 3.72×10^{-8} (c) Statement-1 is true; statement-2 is false. . . ' 022 ://1,' (d) Statement-I is false; statement-2 is true.

Problem 432**[GRB (O.P. Tandon) p.91 Q10]**

Cations absorb 6.023×10^{23} electrons for the reduction. How

Problem 433**[GRB (O.P. Tandon) p.91 Q16]**

StatementM .. During the electrolysis of water, two faraday of SECTION/II Multiple Answers TYPE Objective Questions

Problem 434**[GRB (O.P. Tandon) p.91 Q11]**

. In an electrolytic cell: (a) anode is positively charged (b) cathode is negatively charged (c) oxidation takes place at anode . (d) reduction takes place at cathode

Problem 435**[GRB (O.P. Tandon) p.91 Q12]**

One gram equivalent of a substance is liberated at an electrode by: (a) 6.023×10^{23} electrons (b) 96500 C (c) 1 amp current for 1 second (d) 1 amp current for 96500 sec

Problem 436**[GRB (O.P. Tandon) p.91 Q13]**

If 9 gm. H_2O is electrolysed completely with the current of 50/(a) 96500 charge is required (b) $2 \times 96500 \text{ C}$ charge is required (c) 5.6 L of O_2 at STP will be formed (d) 11.2 L of O_2 at STP will be formed

Problem 437**[GRB (O.P. Tandon) p.91 Q14]**

A galvanic cell involves the following reaction: $\text{Zn(s)} + 2\text{Ag}^+(\text{aq.}) \rightleftharpoons \text{Zn}^{2+}(\text{aq.}) + 2\text{Ag(s)}$ Select the correct statements among the following: (a) Zinc is negatively charged (b) The given redox process is spontaneous (c) $\text{Ag}^+ + e^- \rightarrow \text{Ag}$, takes place at anode (d) $\text{Zn(s)} \rightarrow \text{Zn}^{2+} + 2e^-$, takes place at cathode

Problem 438**[GRB (O.P. Tandon) p.91 Q15]**

Given that, $E^\circ_{\text{Ni}^{2+}/\text{Ni}} = -0.25 \text{ V}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ V}$; $E^\circ_{\text{Ag}^+/\text{Ag}} = +0.80 \text{ V}$; $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$ Which of the following redox processes will not take place in specified direction? (a) $\text{Ni}^{2+}(\text{aq.}) + \text{Cu(s)} \rightarrow \text{Ni(s)} + \text{Cu}^{2+}(\text{aq.})$ (b) $\text{Cu(s)} + 2\text{Ag}^+(\text{aq.}) \rightarrow \text{Cu}^{2+}(\text{aq.}) + 2\text{Ag(s)}$ Because Statement-2: In the electrolysis of water, two faraday of charge will produce half mole of H_2 gas and one fourth mole of O_2 gas. 17.-Statement-1: Aqueous solution of CuSO_4 —complete electrolysis using platinum electrode. Because Statement-2: CuSO_4 is converted to Cu(OH)_2 on electrolysis. 18; Statement-I: Sodium ions are discharged at a mercury cathode in preference to hydrogen ion. Because Statement-2: Na^+ is stronger reducing agent than H^+ .

Problem 439**[GRB (O.P. Tandon) p.91 Q19]**

Statement-I: KCl and NH₄Cl cannot be used in salt bridge of a cell containing Ag⁺, Hg²⁺ and Tl⁺ ions. /cdotBecause Statement-2: Cell will be destroyed due to precipitation of metal chlorides.

Problem 440**[GRB (O.P. Tandon) p.91 Q20]**

Statement-I: The voltage of mercury cell remains constant for its life time. . Because Statement-2: Overall cell reaction does not involve any ion.

Problem 441**[GRB (O.P. Tandon) p.91 Q21]**

Statement-I: In alkaline version of dry cell, /textasciitilde()Cl is replaced by KOH. Because Statement-2: Zinc container does not undergo corrosion ill alkaline medium. /cdot/cdot/cdotSECTION-IV Matrix-Matching Type Questions , This section contains 3 questions;. Each question contains statement given in two columns which have to be matched. Statements (a, b, c and d) in Coiumn-I have to be matched with statements (p, q, r and s) in Column-II. The answers to these questions have to be appropriately bubbled as illustrated in the following examples: If the correct matches are (a-p,s); (b-q,r); (c-P,q) and (d-s); then correct bubbled 4 x 4 matrix should be as follows:

Problem 442**[GRB (O.P. Tandon) p.92 Q22]**

Match the Column-I with Column-II: Column-I Column-II (a) Nickel-Cadmium cell (p) Used in auto vehicles (b) Lithium battery (q) Secondary cell

Problem 443**[GRB (O.P. Tandon) p.92 Q23]**

Match the Column-I with Col limn-II: Column-I Column-TI (a) Specific conductance, K (p) A/textasciitilde(), I A/textasciitilde(), (b) Molar conductance, Am (q) Decreases with diluti:on (c) Resistance of electrolyte (r) Increases with dilution solution, R (d) Degree of ionization of (s) Increases with increase in weak electrolyte, a the distance between parallel plates

Problem 444**[GRB (O.P. Tandon) p.92 Q24]**

Match the Column-I with Column-II: Column-I (a) Concentration cell (b) Edison cell (c) Mercury cell (d) Dry cell

Problem 445**[GRB (O.P. Tandon) p.92 Q23]**

(a-q) (b-r) (c-q,s) (d-p,r) Column-II (P) Fe is oxidised by Ni20 3 (q) Zinc anode (r) HgO cathode (s) EO 0

Problem 446**[GRB (O.P. Tandon) p.92 Q24]**

(a-s) (b-p) (c-q,r) (d-q) SEcnON ... V Linked Comprehension Type Questions Redox reactions play a pivotal role in chemistry and biology. The values of standard redox potential (EO) of two half-cell reactions decide which way the reaction is expected to proceed. A simple example is a Daniell cell in which zinc goes into solution and copper gets deposited. Given below are a set of half-cell reactions (acidic medium) along with their EO (V with respect to normal hydrogen electrode) values.

$\text{I}_2 + 2\text{e}^- \rightarrow 2\text{I}^-$ EO 0.54
 $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$ EO 1.36
 $\text{Mn}^{3+} + \text{e}^- \rightarrow \text{Mn}^{2+}$ EO 1.50
 $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$ EO 0.77
 $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$ E () 1.23 (IIT 2007)

Answer the following questions: (b) Fe^{2+} is oxidised by iodine. (c) Iodide ion is oxidised by chlorine (d) Mn^{2+} is oxidised by chlorine [Hint: Species with greater reduction potential, oxidises other with lower reduction potential.]

Problem 447**[GRB (O.P. Tandon) p.92 Q26]**

While Fe^{3+} is stable, Mn^{3+} is not stable in acid solution because: (a) O_2 oxidises Mn^{1+} to Mn^{3+} (b) O_2 oxidises both Mn^{2+} and Fe^{2+} (c) Fe^{3+} oxidises H_2O to O_2 (d) Mn^{3+} oxidises H_2O to O_2 [Hint: Mn^{3+} oxidises H_2O to O_2 because the standard reduction potential of ($\text{Mn}^{3+} / \text{Mn}^{2+}$) is greater than that of ($\text{O}_2 / \text{H}_2\text{O}$).]

Problem 448**[GRB (O.P. Tandon) p.92 Q22]**

(a-q) (b-q) (c-r,s) (d-p,q)

Chapter 10

Narendra Avasthi: Complete Graded Problem Bank (Levels 1, 2 and 3)

Problem 1

NaCl(Molten)withPtelectrode $\text{Cl}_2(\text{g})$ $\text{Na}(\text{l})$

[Narendra Avasthi Q1]

Problem 2

NaCl(aq)withPtelectrode $\text{Cl}_2(\text{g})$ $\text{H}(\text{g})$

[Narendra Avasthi Q2]

Problem 3

$\text{Na}_2\text{SO}_4(\text{aq})$ with Pt electrode $\text{O}_2(\text{g})$ $\text{H}_2(\text{g})$

[Narendra Avasthi Q3]

Problem 4

$\text{NaNO}_3(\text{aq})$ withPtelectrode $\text{O}_2(\text{g})$ $\text{H}_2(\text{g})$

[Narendra Avasthi Q4]

Problem 5

$\text{AgNO}_3(\text{aq})$ withPtelectrode $\text{O}_2(\text{g})$ $\text{Ag}(\text{s})$

[Narendra Avasthi Q5]

Problem 6

$[\text{CuSO}_4(\text{aq})]$ with inert electrode $\text{O}_2(\text{g})$ $\text{Cu}(\text{s})$

[Narendra Avasthi Q6]

Problem 7**[Narendra Avasthi Q7]**

$\text{CuSO}_4(\text{aq})$ with copper electrode Cu dissolve $\text{Cu}(\text{s})$ Galvanic Cell This cell converts chemical energy into electrical energy. Anode of Zn Salt Cathode of Cu Bridge (-ve electrode) (+ve electrode) Cotton (Anodic cell) plug (Cathodic cell) Galvanic cell is made up of two half cells i.e., anodic and cathodic. Oxidation takes place at anode and reduction at cathode. It is also known as voltaic cell. It may be represented as shown in Fig. Zinc rod immersed in ZnSO_4 behaves as anode and copper rod immersed in CuSO_4 behaves as cathode. Anode half cell reaction: $\text{Zn}(\text{s}) \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$ (oxidation) Cathode half cell reaction: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ (reduction) Overall process: $\text{Zn}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Cu}(\text{s}) + \text{Zn}^{2+}(\text{aq})$ In galvanic cell like Daniel cell; electrons flow from anode (zinc rod) to the cathode (copper rod) through external circuit; zinc dissolves as Zn^{2+} ; Cu^{2+} ion in the cathode cell picks up two electrons and becomes deposited at cathode. Salt Bridge It is U-shaped tube contains saturated solution of inert electrolyte like KCl , KNO_3 , NH_4Cl and NH_4NO_3 , etc. in agar-agar gel. Cation and anion of inert electrolyte have same mobility. Ions of inert electrolyte do not mix with electrolytic solution present in the half cell. Ions of inert electrolyte do not participate in electrochemical change. — PAGE 3 — 390 PROBLEMS IN CHEMISTRY Function of Salt Bridge It completes the electrical circuit. It maintains the two half cell electrically neutral by the flow of ions. Representation of a cell (IUPAC conventions): Let us illustrate the convention taking the example of Daniel cell. $\therefore \text{Zn}(\text{s}) | \text{ZnSO}_4(\text{aq}) || \text{CuSO}_4(\text{aq}) | \text{Cu}(\text{s})$ (i) Two half cells are separated by double vertical lines and it indicates salt bridge or any type of porous partition. (ii) EMF (electromotive force) may be written on the right hand side of the cell. (iv) Single vertical lines indicate the phase separation between electrode and electrolyte solution. (s) n(b) + nl(bo) + uz () uz (-) (+) Anode Cathode $E_{\text{cell}} = E_{\text{OP}}(\text{LHs}) + E_{\text{RR}}(\text{RHs})$ $E_{\text{cell}} = E_{\text{OP}}(\text{LH}) + E_{\text{RP}}(\text{RHs}) - E_{\text{RP}}(\text{LHS})$ Nernst's Equation $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log Q$ At 298 K temp. $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log Q$; where Q is reaction quotient. cell n Cell Thermodynamics (i) $\Delta G = -nFE_{\text{cell}}$ (1). (i) Equilibrium constant (K), $\log K = \frac{nE_{\text{cell}}^{\circ}}{0.0591}$ (iv) Temp. coefficient of cell: $\left(\frac{\partial E_{\text{cell}}}{\partial T}\right)_P$ (v) Enthalpy of reaction inside the cell, $\Delta H = nFE_{\text{cell}} + nT \left(\frac{\partial E_{\text{cell}}}{\partial T}\right)_P$ Concentration Cell Anode and cathode both are formed by same substance so $E_{\text{cell}} = 0$, such type of cell is known as concentration cell. Concentration gradient may arise either in electrode material or in electrolyte. Thus there are two types of concentration cell. Electrode gas conc. cell: $\text{Pt}, \text{H}_2(\text{g}) | \text{H}^+(\text{aq}) | \text{H}_2(\text{g}), \text{Pt}$ B atm P2 atm $0.0591 \log \frac{P_2}{P_1}$ — PAGE 4 — ELECTROCHEMISTRY Electrolyte conc. cell: $\text{Zn}(\text{s}) | \text{ZnSO}_4(\text{C}_1) || \text{ZnSO}_4(\text{C}_2) | \text{Zn}(\text{s})$ $0.0591 \log \frac{C_2}{C_1}$ Different Type, of Electrode S.No. Name of Electrode Anode Cathode

Problem 8**[Narendra Avasthi Q1]**

Hydrogen electrode $\text{Pt}(\text{s}) | \text{H}_2(\text{g}) | \text{H}^+(\text{aq})$ $\text{H}^+(\text{aq}) | \text{H}_2(\text{g}) / \text{Pt}(\text{s})$ $0.0591 \log \frac{P_2}{P_1} + \frac{0.0591}{2} \log \frac{P_2}{P_1}$ $2 \log \frac{P_2}{P_1}$

Problem 9**[Narendra Avasthi Q2]**

Metal-metal ion electrode $\text{M}(\text{s}) | \text{M}^{n+}(\text{aq})$ $\text{M}^{n+}(\text{aq}) | \text{M}(\text{s})$ $0.0591 \log [\text{M}^{n+}]$ $0.0591 \log \frac{1}{[\text{M}^{n+}]}$

Problem 10**[Narendra Avasthi Q3]**

Calomel electrode $\text{Hg}(\text{l}), \text{Hg}_2\text{Cl}_2(\text{s}) / \text{Cl}^-(\text{aq})$ $\text{Cl}^-(\text{aq}) | \text{Hg}_2\text{Cl}_2(\text{s}), \text{Hg}(\text{l})$ $0.0591 \log \frac{1}{[\text{Cl}^-]}$ $0.0591 \log \frac{1}{[\text{Cl}^-]}$

Problem 11**[Narendra Avasthi Q4]**

Redox electrode $\text{Pt}(\text{s}) / \text{Fe}^{2+}(\text{aq}), \text{Fe}^{3+}(\text{aq})$ $\text{Fe}^{3+}(\text{aq}), \text{Fe}^{2+}(\text{aq}) | \text{Fe}(\text{s})$ $0.0591 \log \frac{[\text{Fe}^{3+}]}{[\text{Fe}^{2+}]}$ $0.0591 \log \frac{[\text{Fe}^{3+}]}{[\text{Fe}^{2+}]}$

Problem 12**[Narendra Avasthi Q5]**

Metal insoluble salt anion $\text{Ag (s)} | \text{AgCl(s)} | \text{Cl}^-(\text{aq}) | \text{Cl}^-(\text{aq}) | \text{AgCl(s)} | \text{Ag (s)}$ electrode 0.0591 1

0.0591 $\log \log[\text{Cl}^-] - 1 [\text{Cl}^-] - 1$ Batteries Electrochemical cells can be used as batteries. Batteries are of two kinds: Primary battery: Where the reaction occurs only once and can not be reused once it becomes dead over the course of time. For examples, dry cell (Leclanche cell), Mercury cell. Secondary battery: Which can be recharged by passing current through it in the opposite direction so that it can be used again. For example, lead-acid battery, Nickel-cadmium cell. Fuel Cells methane, methanol, etc.) directly into electrical energy. Fuel cells are more efficient thermodynamically and more of the energy of the reaction can be made available for useful work provided that the supply of reactants is maintained. Methanol, ethanol, hydrazine, formaldehyde, carbon monoxide can be used as fuels in fuel cells apart from hydrogen. — PAGE 5 — 0 PROBLEMS IN CHEMISTRY

Corrosion When the metal exposed to some environment it gets converted to its oxides. The oxidative deterioration of metal is known as corrosion. Example: Rusting of iron, tarnishing of silver, development of green coating on copper and bronze etc. Oxidation $\text{Fe(s)} \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$ Reduction $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O(l)}$. Atmospheric oxygen oxidised $\text{Fe}^{2+} + 2\text{Fe}^{2+}(\text{aq}) + 2\text{H}_2\text{O(l)} + \text{O}_2(\text{g}) \rightarrow \text{Fe}_2\text{O}_3(\text{s}) + 4\text{H}^+(\text{aq})$ 2 Hydrated feric oxide ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O(s)}$) is known as rust. Factors which enhance corrosion : i) Presence of impurities in the metal. (i) Presence of moisture. (i) Presence of electrolyte. Prevention of corrosion: (i) Barrier protection by oil/grease layer, paints or electroplating. (i) Sacrificial protection by coating the metal with more electropositive metal. Conductance Conductance (G): It is defined as the reciprocal of the electrical resistance i.e., $G = 1/R$. It measures the ease with which the current flows through a conductor. Unit : Siemen, S or Ω^{-1} . Specific resistance or resistivity (ρ): The resistivity or specific resistance is defined as the resistance in Ω of a conductor having length equal to 1 cm and area of cross-section equal to 1 cm^2 . or $R = \rho \frac{l}{A}$ where ρ = specific resistance, l = length of conductor, A = area of cross-section of the conductor. Unit: $\Omega \text{ cm}$. W Specific conductance or conductivity (k): It is defined as the reciprocal of specific resistance. i.e., $k = 1/\rho$. $R = \rho \frac{l}{A}$ $k = \frac{G \times A}{l}$ where G^* = cell constant A When $l = 1 \text{ cm}$ and $A = 1 \text{ cm}^2$; $k = G$ Thus conductivity is the conductance of one centimeter cube or conductance of one cm cube of the solution of an electrolyte. Unit : $\Omega^{-1} \text{ cm}^{-1}$, Scm^{-1} . Molar conductivity (Λ_m): The conducting power of all the ions produced by dissolving 1 mole of an electrolyte in solution. — PAGE 6 — 393 ELECTROCHEMISTRY

Mathematically, $\Lambda_m = \frac{k \times 1000}{M}$ (M = molarity) M Variation of conductivity and molar conductivity with concentration: For weak and strong electrolytes Conductivity decreases with concentration. Molar conductivity increases with decrease in concentration. 01 $\frac{1}{\text{cm}}$ Strong electrolyte (NaCl) Weak electrolyte $\frac{1}{C}$ Kohlrausch's Law. According to this law, at infinite dilution, when the dissociation is complete, each ion makes a definite contribution towards molar conductivity of the electrolyte irrespective of the nature of the other ions present. The molar conductivity of an electrolyte at infinite dilution is the sum of the ionic conductivities of the cations and the anions each multiplied by the number of ions present in one formula unit of the electrolyte e.g., AB . The equivalent conductivity of an electrolyte at infinite dilution is the sum of the equivalent conductivities of the cations and anions. $(\Lambda_m^\infty) = x\lambda^\infty(\text{A}^+) + y\lambda^\infty(\text{B}^-)$ $N_q. (\Lambda_m^\infty) = 2g.(\text{A}^+) + 2g.(\text{B}^-)$ Application

(a) Determination of equivalent/molar conductivities of weak electrolytes at infinite dilution Λ_m^∞ (b) Determination of degree of dissociation (α) and equilibrium constant (K) of an electrolyte at a given dilution Molar conductance at concentration C $M/\alpha = \text{Molar conductance at infinite dilution}$ WV $\text{HAH} + A$ Initial conc. c . Conc. at equilibrium $C - \frac{C}{\alpha}$ $C - \frac{C}{\alpha}$ — PAGE 7 — PROBLEMS IN CHEMISTRY

$C - \frac{C}{\alpha}$ Λ_m^∞ (c) Determination of the solubility of a sparingly soluble salt Since the solution is saturated at infinite dilution $A = A$ and molarity = solubility. $\frac{1}{\text{cm}^3}$ Solubility (S) = $k \times 1000$ For sparingly soluble salt AxB , : Solubility product $K_{sp} = x^s \cdot y^s$ $s \times + y$ Conductometric Titration the titration (for burette) is called conductometric titration: Titration of HCl(aq) with NaOH(aq) (Strong base. Vs Strong acid) $\text{HCl(aq)} + \text{NaOH(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$ Endpoint Volume of NaOH added Titration of $\text{CH}_3\text{COOH(aq)}$ with NaOH(aq) (Weak acid Vs Strong base) $\text{CH}_3\text{COOH(aq)} + \text{NaOH(aq)} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O(l)}$ Conductance Endpoint Volume of NaOH added — PAGE 8 — ELECTROCHEMISTRY 395

Level 1. A cell reaction would be spontaneous if the cell potential and ΔG are respectively: (a) positive and negative (b) negative, negative (c) zero, zero (d) positive, zero 2. Which of the following statement is correct? (a) Cathode is -ve terminal in both, galvanic and electrolytic cells (b) Anode is +ve terminal in both, galvanic and electrolytic cells (c) Cathode and anode are -ve terminal in electrolytic and galvanic cell. (d) Cathode and anode are +ve terminal in electrolytic and galvanic cell. 3. Electrolytes when dissolved in water dissociate into ions because: (a) They are unstable. (b) The water dissolves it. (c) The force of repulsion increases

Problem 13**[Narendra Avasthi Q5]**

The amount of an ion liberated on an electrode during electrolysis does not depend upon: (a) Conductance of the solution (b) Current strength (c) Time (d) Electrochemical equivalent of the element

Problem 14**[Narendra Avasthi Q6]**

How many electrons are there in one coulomb of electricity? (a) 6.023×10^{23} (b) 1.64×10^{-24} (c) 6.24×10^{18} (d) 6.24×10^{-24} 7. How many coulombs are provided by a current of 0.010 mA in the calculator battery that can operate for 1000 hours? (a) 1.0 (b) 10 (c) 0.010 (d) 36 8. How many minutes are required to deliver 3.21×10^3 coulombs using a current of 500 A used in the commercial production of chlorine? (a) 8.3 (b) 5.3×10^4 (c) 6420 (d) 107 9. Passage of a current for 548 seconds through a silver coulometer results in the deposition of 0.746 g of silver. What is the current (in A)? (a) 1.22 (b) 1.16 (c) 1.07 (d) 1.00 10. Electrolysis can be used to determine atomic masses. A current of 0.550 A deposits 0.55 g of a certain metal in 100 minutes. Calculate the atomic mass of the metal if eq. wt. = mole. wt. / 3 (a) 100 (b) 45.0 (c) 48.25 (d) 144.75 11. Beryllium occurs naturally in the form of beryl. The metal is produced from its ore by electrolysis after the ore has been converted to the oxide and then to the chloride. How many grams of Be(s) is deposited from a BeCl_2 solution by a current of 5.0 A that flows for 1.0 h? (Atomic weight: Be = 9) (a) 0.840 (b) 1.68 (c) 1.42 (d) 1.08 — PAGE 9 — 396 PROBLEMS IN CHEMISTRY 12. How many minutes will it take to plate out 5.0 g of Cr from a $\text{Cr}_2(\text{SO}_4)_3$ solution using a current of 1.50 A? (Atomic weight : Cr = 52.0) (a) 254 (b) 30 (c) 152 (d) 103 13. Calculate the current (in mA) required to deposit 0.195 g of platinum metal in 5.0 hours from a solution of PtCl_2 : (Atomic weight: Pt = 195) (a) 310 (b) 31 (c) 21.44 (d) 5.36

Problem 15**[Narendra Avasthi Q14]**

How many Faradays are required to reduce 0.25 g of Nb (V) to the metal? (Atomic weight : Nb = 93g) (a) 2.7×10^{-3} (b) 1.3×10^{-2} (c) 2.7×10^{-2} (d) 7.8×10^{-3}

Problem 16**[Narendra Avasthi Q15]**

One gm metal M^{3+} was discharged by the passage of 1.81×10^{23} electrons. What is the atomic weight of metal? (a) 33.35 (b) 133.4 (c) 66.7 (d) None of these 16. Total charge required for the oxidation of two moles MnO_4^- into MnO_2 in presence of alkaline medium is : (a) 5F (b) 10 F (c) 20 F (d) None of these 17. The electrolytic decomposition of dilute sulphuric acid with platinum electrode, cathodic reaction is : (a) Reduction of H^+ (b) Oxidation of SO_2 (c) Reduction SO_2 (d) Oxidation of H_2O 18. Which one of the following metals can not be obtained on electrolysis of aqueous solution of its salts ? (a) Mg (b) Ag (c) Cu (d) Cr 19. A solution of potassium sulphate in water is electrolysed using inert electrodes. The products at the cathode and anode are respectively. (a) H_2, O_2 (b) O_2, H_2 (c) O_2, Na (d) None of these 20. The passage of current through a solution of certain electrolyte results in the evolution of $\text{H}_2(\text{g})$ at cathode and $\text{Cl}_2(\text{g})$ at anode. The electrolytic solution is : (a) Water (b) aq. H_2SO_4 (c) aq. NaCl (d) aq. CuCl_2

Problem 17**[Narendra Avasthi Q21]**

When an aqueous solution of H_2SO_4 is electrolysed, the product at anode is : (a) H^+ (b) OH^- (c) SO_2 (d) O_2

Problem 18**[Narendra Avasthi Q22]**

An aqueous solution of Na_2SO_4 in water is electrolysed using Pt electrodes. The products at the cathode and anode are respectively: (a) H_2, SO_2 (b) O_2, NaOH (c) H_2, O_2 (d) O_2, SO_2 23. The electrolysis of a solution resulted in the formation of $\text{H}_2(\text{g})$ at the cathode and $\text{O}_2(\text{g})$ at the anode. The solution is : (a) $\text{AgCl}(\text{aq})$ (b) $\text{H}_2\text{SO}_4(\text{aq})$ (c) highly concentrated $\text{NaCl}(\text{aq})$ solution (d) $\text{CuCl}_2(\text{aq})$ 24. If mercury is used as cathode in the electrolysis of aqueous NaCl solution, the ion discharged at cathode is : (a) H^+ (b) Na^+ (c) OH^- (d) Cl^-

Problem 19**[Narendra Avasthi Q25]**

A dilute aqueous solution of CuSO_4 is electrolyzed using platinum electrodes. The products at the anode and cathode are : (a) O_2, H_2 (b) H_2, O_2 (c) O_2, Cu (d) $\text{S}_2\text{O}_8^{2-}, \text{H}_2$ — PAGE 10 — 397 ELECTROCHEMISTRY

Problem 20**[Narendra Avasthi Q26]**

What products are formed during the electrolysis of concentrated aqueous solution of sodium chloride? (I) $\text{Cl}_2(\text{g})$ at anode (II) NaOH as electrolyte (III) $\text{H}_2(\text{g})$ at cathode (a) I only (b) I and II only (c) I and III only (d) I, II and III

Problem 21**[Narendra Avasthi Q27]**

Which of the following aqueous solution produces metal after electrolysis ? (a) $\text{K}_2\text{Cr}_2\text{O}_7$ (b) KMnO_4 (c) CH_3COONa (d) CuCl_2

Problem 22**[Narendra Avasthi Q28]**

How much time is required for complete decomposition of 4 moles of water using 4 ampere? (a) $3.86 \times 10^5 \text{ sec}$ (b) $1.93 \times 10^5 \text{ sec}$ (c) 96500 sec (d) 48250 sec

Problem 23**[Narendra Avasthi Q29]**

An aqueous solution containing 1 M each of $\text{Au}^{3+}, \text{Cu}^{2+}, \text{Ag}^+, \text{Li}^+$ is being electrolysed by using inert electrodes. The value of standard potentials are: $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}, E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34 \text{ V}$. With increasing voltage, the sequence of deposition of metals on the cathode will be: (a) $\text{Li}, \text{Cu}, \text{Ag}, \text{Au}$ (b) $\text{Cu}, \text{Ag}, \text{Au}$ (c) $\text{Au}, \text{Ag}, \text{Cu}$ (d) $\text{Au}, \text{Ag}, \text{Cu}, \text{Li}$ 30. If 0.50 L of a 0.60 M SnSO_4 solution is electrolyzed for a period of 30.0 min using a current of 4.60 A. If inert electrodes are used, what is the final concentration of Sn^{2+} remaining in the solution? [at. wt. of $\text{Sn} = 119$] (a) 0.342 M (b) 0.544 M (c) 0.389 M (d) 0.514 M 31. A 100.0 mL dilute solution of Ag^+ is electrolyzed for 15.0 minutes with a current of 1.25 mA and the silver is removed completely. What was the initial $[\text{Ag}^+]$? (a) $2.32 \times 10^{-1} \text{ M}$ (b) $2.32 \times 10^{-4} \text{ M}$ (c) $2.32 \times 10^{-3} \text{ M}$ (d) $1.16 \times 10^{-4} \text{ M}$

Problem 24**[Narendra Avasthi Q32]**

A 250.0 mL sample of a 0.20 M Cr^{3+} is electrolyzed with a current of 96.5 A. If the remaining $[\text{Cr}^{3+}]$ is 0.1 M the duration of process is : (a) 25 sec (b) 225 sec (c) 150 sec (d) 75 sec 33. The element indium is to be obtained by electrolysis of a molten halide of the element. Passage of a current of 3.20 A for a period of 40.0 min results in formation of 3.05 g of In . What is the oxidation state of indium in the halide melt? (Atomic weight: $\text{In} = 114.8$) (a) 3 (b) 2 (c) 5 (d) 1

Problem 25**[Narendra Avasthi Q34]**

An electrolysis of a oxytungsten complex ion using 1.10 A for 40 min produces 0.838 g of tungsten. What is the charge on tungsten in the material? (Atomic weight: $W = 184$) (a) 6 (b) 2 (c) 4 (d) 1

35. In the electrolysis of aqueous NaCl, what volume of $Cl_2(g)$ is produced in the time that it takes to liberate 5.0 liter of $H_2(g)$? Assume that both gases are measured at STP (a) 5.0 (b) 2.50 (c) 7.50 (d) 10.0

36. How many grams of Cr are deposited in the electrolysis of solution of $Cr(NO_3)_3$ in the same time that it takes to deposit 0.54g of Ag in a silver coulometer arranged in series with the $Cr(NO_3)_3$ cell? (Atomic weight: $Cr = 52.0$; $Ag = 108$) (a) 0.0866 (b) 0.0288 (c) 0.173 (d) 0.220

37. In the electrolysis of a $CuSO_4$ solution, how many grams of Cu are plated out on the cathode in the time that it takes to liberate 5.6 litre of $O_2(g)$, measured at STP at the anode? (a) 31.75 (b) 14.2 (c) 4.32 (d) None of these

38. Ammonium perchlorate, NH_4ClO_4 , used in the solid fuel in the booster rockets on the space shuttle, is prepared from sodium perchlorate, $NaClO_4$, which is produced commercially by the electrolysis of a hot, stirred solution of sodium chloride. How many faradays are required to produce 1.0 kg of sodium perchlorate? $NaCl + 4H_2O \rightarrow NaClO_4 + 4H_2$ (a) 40.3 (b) 18.3 (c) 31.6 (d) 65.3

Problem 26**[Narendra Avasthi Q39]**

In the commercial preparation of aluminum, aluminum oxide (Al_2O_3) is electrolyzed at $1000^\circ C$. How many coulombs of electricity are required to give 54 kg of aluminum? Assume following reaction takes place at cathode: $Al^{3+} + 3e^- \rightarrow Al$ (a) 17.3×10^8 (b) 3.21×10^7 (c) 1.82×10^4 (d) 57.9×10^7

40. When molten lithium chloride ($LiCl$) is electrolyzed, lithium metal is formed at the cathode. If current efficiency is 75%, charge pass through the cell? (Atomic weight : $Li = 7$) (a) 0.105 (b) 0.120 (c) 0.28 (d) 0.240

41. Sodium metal is produced commercially by the electrolysis of molten sodium chloride and chlorine is produced as a by product. How many litres of chlorine at 1.8 atm and $27^\circ C$ will be produced if a current of 1.0×10^3 A is passed through NaCl (l) for 9.65 h? (a) 2463 (b) 460 (c) 1800 (d) 1231.6

42. $H_2(g)$ and $O_2(g)$, can be produced by the electrolysis of water. What total volume (in L) of O_2 and H_2 are produced at STP when a current of 30 A is passed through a K_2SO_4 (aq) solution for 193 min.? (a) 20.16 (b) 40.32 (c) 60.48 (d) 80.64

Problem 27**[Narendra Avasthi Q43]**

The cost of 2 Rs/kWh of operating an electric motor for 10 hours takes 10 amp at 110V is: (a) 79200 Rs (b) 22000 Rs (c) 220 Rs (d) 22 Rs

Problem 28**[Narendra Avasthi Q44]**

A 1 M solution of H_2SO_4 is electrolyzed. Select right statement with products at anode and cathode respectively: Given: $2SO_2 \rightarrow S_2O_2 + 2e^-$; $E^\circ = -2.01$ V $H_2O(l) \rightarrow 2H^+(aq) + 1/2 O_2(g) + 2e^-$; $E^\circ = -1.23$ V (a) concentration of H_2SO_4 remain constant; H_2, O_2 (b) concentration of HSO_4^- increases; O_2, H_2 (c) concentration of H_2SO_4 decreases; O_2, H_2 (d) concentration of H_2SO_4 remains constant; S_2O, H_2

45. Cadmium amalgam is prepared by electrolysis of a solution of $CdCl_2$ using a mercury cathode. How long should a current of 4 A be passed in order to prepare 10 g amalgam on cathode of 4.5 g Hg? (atomic wt. of $Cd = 112$) (a) 400 sec (b) 215.40 sec (c) 861.6 sec (d) 430.8 sec

Problem 29**[Narendra Avasthi Q46]**

Use of electrolysis is in : (a) Electrorefining (b) Electroplating (c) Both (a) and (b) (d) None of these

Problem 30

[Narendra Avasthi Q47]

When a solution of AgNO_3 (1 M) is electrolyzed using platinum anode and copper cathode. What are the products obtained at two electrodes? $E^\circ = +0.0\text{V}$; $\text{H}^+|\text{H}_2$ — PAGE 12 — ELECTRO-CHEMISTRY 399 E°/circ (a) $\text{Cu} \rightarrow \text{Cu}^{2+}$ at anode; $\text{Ag}^+ \rightarrow \text{Ag}$ at cathode (b) $\text{H}_2\text{O} \rightarrow \text{O}_2$ at anode; $\text{Cu}^{2+} \rightarrow \text{Cu}$ at cathode (c) $\text{H}_2\text{O} \rightarrow \text{O}_2$ at anode; $\text{Ag}^+ \rightarrow \text{Ag}$ at cathode (d) $\text{NO}_3^- \rightarrow \text{NO}_2$ at anode; $\text{Ag}^+ \rightarrow \text{Ag}$ at cathode

Problem 31

[Narendra Avasthi Q48]

Which of the following statements is correct about Galvanic cell ? (a) It converts chemical energy into electrical energy. (b) It converts electrical energy into chemical energy. (c) It converts metal from its free state to the combined state. (d) It converts electrolyte into individual ions. 49. E°/circ for $\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$ is 1.36 V; E°/circ for $\text{Cl}^-(\text{g}) \rightarrow 1/2\text{Cl}_2(\text{g}) + \text{e}^-$ is : (a) 1.36 V (b) -1.36 V (c) -0.68 V (d) 0.68 V

Problem 32

[Narendra Avasthi Q50]

When two half-cells of electrode potential of E_1 and E_2 are combined to form a cell of electrode potential E_3 , then (when n_1, n_2 and n_3 are no. of electrons exchanged in first, second and combined half-cells) : $E_3 = E_2 - E_1$ (b) $E_3 = n_3 E_1 - E_2 n_2$ (c) $E_3 =$ (d) $E = E_1 + E_2$ n 51. The function of a salt bridge is to : (a) maintain electrical neutrality of both half cells (b) increase the cell potential at the positive electrode (c) decrease the cell potential at the negative electrode (d) eliminate the impurities present in the electrolyte

Problem 33

[Narendra Avasthi Q52]

Saturated solution of KNO_3 with agar-agar is used to make 'salt bridge' because : (a) size of K^+ is greater than that of NO_3^- (b) velocity of NO_3^- is greater than that of K^+ (c) velocities of K^+ and NO_3^- are nearly the same (d) both velocity and sizes of K^+ and NO_3^- ions are same 53. A salt bridge contains : (a) A saturated solution of KCl and agar-agar (b) A saturated solution of KNO_3 and agar-agar (c) A saturated solution of NH_4NO_3 and agar-agar (d) All of these

Problem 34

[Narendra Avasthi Q54]

The nature of curve of E_{cell} vs. $\log K$ is : (a) straight line (b) parabola (c) hyperbola (d) elliptical curve

Problem 35

[Narendra Avasthi Q55]

Consider the following equations for a cell reaction $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$; $E^\circ/\text{circ} = x\text{V}$, $K_{\text{eq}} = K_1$ $2\text{A} + 2\text{B} \rightarrow 2\text{C} + 2\text{D}$; $E^\circ/\text{circ} = y\text{V}$, $K_{\text{eq}} = K_2$ — PAGE 13 — 400 PROBLEMS IN CHEMISTRY then: (a) $x = y, K_1 = K_2$ (b) $x = 2y, K_1 = 2K_2$ (c) $x = y, K_1 = K_2$ (d) $x^2 = y, K_1 = K_2$

Problem 36

[Narendra Avasthi Q56]

Which graph correctly correlates E_{cell} as a function of concentrations for the cell $\text{Zn}(\text{s}) + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{Ag}(\text{s})$, $E_{\text{cell}} = 1.56\text{V}$ Y-axis : E_{cell} , X-axis : $\log_{10} [\text{Zn}^{2+}] [\text{Ag}^+]$ 1.56V .56V (a) (b) 0 1.0 1.0 0 1.0 1.56V (c) (d) 1.56V -1.0 0 1.0 -1.0 0 1.0.

Problem 37

[Narendra Avasthi Q57]

The Nernst equation $E = E^\circ/\text{circ} - \frac{RT}{nF} \ln Q$ indicates that the Q will be equal to equilibrium constant K when: (a) $E = E^\circ/\text{circ}$ (b) $\frac{RT}{nF} = 1$ (c) $E = \text{zero}$ (d) $E^\circ/\text{circ} = 1$

Problem 38

[Narendra Avasthi Q58]

The cell reaction $2\text{Ag}^+(\text{aq}) + \text{H}_2(\text{g}) \rightarrow 2\text{H}^+(\text{aq}) + 2\text{Ag}(\text{s})$, is best represented by :
 (a) $\text{Ag}(\text{s})|\text{Ag}^+(\text{aq})|\text{H}^+(\text{aq})|\text{H}_2(\text{g})|\text{Pt}(\text{s})$ (b) $\text{Pt}(\text{s})|\text{H}_2(\text{g})|\text{H}^+(\text{aq})|\text{Ag}^+(\text{aq})|\text{Ag}(\text{s})$ (c) $\text{Ag}(\text{s})|\text{Ag}^+(\text{aq})|\text{H}_2(\text{g})|\text{H}^+(\text{aq})|\text{Pt}(\text{s})$ (d) $\text{Ag}^+(\text{aq})|\text{Ag}(\text{s})|\text{H}_2(\text{g})|\text{H}^+(\text{aq})|\text{Pt}(\text{s})$
 The cell reaction $\text{Cu}(\text{s}) + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{Ag}(\text{s})$ is best represented by:
 (a) $\text{Cu}(\text{s})|\text{Cu}^{2+}(\text{aq})||\text{Ag}^+(\text{aq})|\text{Ag}(\text{s})$ (b) $\text{Ag}(\text{s})|\text{Ag}^+(\text{aq})||\text{Cu}^{2+}(\text{aq})|\text{Cu}(\text{s})$ (c) $\text{Cu}(\text{s})|\text{Cu}^{2+}(\text{aq})||\text{Ag}^+(\text{aq})|\text{Ag}(\text{s})$ (d) $\text{Ag}(\text{s})|\text{Ag}^+(\text{aq})||\text{Cu}^{2+}(\text{aq})|\text{Cu}(\text{s})$

Problem 39

[Narendra Avasthi Q60]

The cell reaction $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{Fe}^{2+}(\text{aq}) \rightarrow 6\text{Fe}^{3+}(\text{aq}) + 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\text{l})$, is best represented by :
 (a) $\text{Pt}(\text{s})|\text{Fe}^{2+}(\text{aq}), \text{Fe}^{3+}(\text{aq})||\text{Cr}_2\text{O}_7^{2-}(\text{aq}), \text{Cr}^{3+}(\text{aq})|\text{Pt}(\text{s})$ (b) $\text{Pt}(\text{s})|\text{Cr}_2\text{O}_7^{2-}(\text{aq}), \text{Cr}^{3+}(\text{aq})||\text{Fe}^{3+}(\text{aq}), \text{Fe}^{2+}(\text{aq})|\text{Pt}(\text{s})$ (c) $\text{Fe}^{2+}(\text{aq})|\text{Fe}^{3+}(\text{aq})||\text{Cr}_2\text{O}_7^{2-}(\text{aq})|\text{Cr}^{3+}(\text{aq})$ (d) $\text{Cr}_2\text{O}_7^{2-}(\text{aq})|\text{Cr}^{3+}(\text{aq})||\text{Fe}^{3+}(\text{aq})|\text{Fe}^{2+}(\text{aq})$

Problem 40

[Narendra Avasthi Q61]

Select the correct cell reaction of the cell $\text{Ag}(\text{s})|\text{Ag}^+(\text{aq})||\text{Cu}^{2+}(\text{aq})|\text{Cu}(\text{s})$:
 (a) $2\text{Ag}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \rightarrow 2\text{Ag}^+(\text{aq}) + \text{Cu}(\text{s})$ (b) $\text{Cu}(\text{s}) + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{Ag}(\text{s})$
 (c) $2\text{Ag}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Cu}(\text{s}) + 2\text{Ag}^+(\text{aq})$ (d) $\text{Cu}^{2+}(\text{aq}) + 2\text{Ag}^+(\text{aq}) \rightarrow 2\text{Ag}(\text{s}) + \text{Cu}(\text{s})$
 Select the correct cell reaction of the cell $\text{Pt}(\text{s})|\text{Cl}_2(\text{g})|\text{Cl}^-(\text{aq})||\text{Ag}^+(\text{aq})|\text{Ag}(\text{s})$:
 (a) $\text{Cl}_2(\text{g}) + \text{Ag}^+(\text{aq}) \rightarrow \text{Ag}(\text{s}) + 2\text{Cl}^-(\text{aq})$ (b) $\text{Cl}_2(\text{g}) + \text{Ag}(\text{s}) \rightarrow 2\text{Cl}^-(\text{aq}) + \text{Ag}^+(\text{aq})$ (c) $2\text{Cl}^-(\text{aq}) + 2\text{Ag}^+(\text{aq}) \rightarrow 2\text{Ag}(\text{s}) + \text{Cl}_2(\text{g})$ (d) $\text{AgCl}(\text{s}) \rightarrow \text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq})$

Problem 41

[Narendra Avasthi Q63]

Standard electrode potential of SHE at 298K is: (a) 0.05V (b) 0.10 V (c) 0.50 V (d) 0.00v

Problem 42

[Narendra Avasthi Q64]

The e.m.f. of the following galvanic cells :
 (a) $\text{Zn}|\text{Zn}^{2+}(1\text{ M})||\text{Cu}^{2+}(1\text{ M})|\text{Cu}$ (b) $\text{Zn}|\text{Zn}^{2+}(0.1\text{ M})||\text{Cu}^{2+}(1\text{ M})|\text{Cu}$ (c) $\text{Zn}|\text{Zn}^{2+}(1\text{ M})||\text{Cu}^{2+}(0.1\text{ M})|\text{Cu}$ (d) $\text{Zn}|\text{Zn}^{2+}(0.1\text{ M})||\text{Cu}^{2+}(0.1\text{ M})|\text{Cu}$ are represented by E_1, E_2, E_3 and E_4 respectively. Which of the following statement is true?
 (a) $E_1 > E_2 > E_3 > E_4$ (b) $E_3 > E_2 > E_1 > E_4$ (c) $E_1 > E_3 = E_4 > E_2$ (d) $E_2 > E_3 = E_4 > E_1$
 Based on the cell notation for a spontaneous reaction, at the anode: $\text{Ag}(\text{s})|\text{AgCl}(\text{s})|\text{Cl}^-(\text{aq})||\text{Br}_2(\text{l})|\text{C}(\text{s})$
 (a) AgCl becomes reduced (b) Ag becomes oxidized (c) Br_2 becomes oxidized (d) Br_2 becomes reduced
 Given the listed standard electrode potentials, what is E° for the cell: $4\text{BiO}^+(\text{aq}) + 3\text{N}_2\text{H}_4(\text{aq}) \rightarrow 4\text{Bi}(\text{s}) + 3\text{N}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l}) + 7\text{H}^+(\text{aq})$
 $\text{BiO}^+(\text{aq}) + 2\text{H}^+(\text{aq}) + 3\text{e}^- \rightarrow \text{Bi}(\text{s}) + \text{H}_2\text{O}(\text{l}), E^\circ = +0.32\text{V}$
 $\text{N}_2\text{H}_4(\text{aq}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow \text{N}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l}), E^\circ = -0.23\text{V}$
 (a) +0.55 (b) +0.34 (c) +1.88 (d) +0.09
 What is the standard electrode potential for the reduction of HClO ? $\text{HClO}(\text{aq}) + \text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 Given: $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{e}^- \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\text{l}), E^\circ = 1.80\text{V}$
 $\text{HClO}(\text{aq}) + \text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}), E^\circ = 1.80\text{V}$
 (a) 1.39 (b) 1.54 (c) 1.22 (d) 0.90
 The E° for the following cell is +0.34 V. $\text{In}(\text{s})|\text{In}(\text{OH})_3(\text{aq})||\text{SbO}_2(\text{aq})|\text{Sb}(\text{s})$. Using $E^\circ = -1.0\text{V}$ for the $\text{In}(\text{OH})_3/\text{In}$ couple, calculate E° for the SbO_2/Sb half-reaction:
 (a) -1.34 (b) +0.66 (c) +0.82 (d) -0.66
 From the following half-cell reactions and their potentials, what is the smallest possible standard e.m.f. for spontaneous reactions?
 $\text{PO}_4^{3-}(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{HPO}_4^{2-}(\text{aq}) + 2\text{OH}^-(\text{aq}), E^\circ = -1.05\text{V}$
 $\text{PbO}_2(\text{s}) + \text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{PbO}(\text{s}) + 2\text{OH}^-(\text{aq}), E^\circ = +0.28\text{V}$
 $\text{IO}_3^-(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^- \rightarrow \text{IO}^-(\text{aq}) + 4\text{OH}^-(\text{aq}), E^\circ = +0.56\text{V}$
 (a) +0.00 (b) +0.74 (c) +0.56 (d) +0.28

Problem 43

[Narendra Avasthi Q71]

Which substance is the best oxidizing agent in Q. no. 47? (a) IO_3^- (b) IO^- (c) PbO (d) PO_3

Problem 44

[Narendra Avasthi Q72]

Consider the following half-cell reactions and associated standard half-cell potentials and determine the maximum voltage that can be obtained by combination resulting in spontaneous processes:
 $\text{AuBr (aq)} + 3\text{e}^- \rightarrow \text{Au(s)} + 4\text{Br(aq)}; E^\circ = -0.86 \text{ V}$
 $\text{Eu}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Eu}^{2+}(\text{aq}); E^\circ = -0.43 \text{ V}$
 $\text{Sn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Sn(s)}; E^\circ = -0.14 \text{ V}$
 $\text{IO}^-(\text{aq}) + \text{H}_2\text{O(l)} + 2\text{e}^- \rightarrow \text{I}^-(\text{aq}) + 2\text{OH}^-(\text{aq}); E^\circ = +0.49 \text{ V}$
 (a) +0.72 (b) +1.54 (c) +1.00 (d) +1.35

Problem 45

[Narendra Avasthi Q73]

The position of some metals in the electrochemical series in decreasing electropositive character is $\text{Mg} > \text{Al} > \text{Zn} > \text{Cu} > \text{Ag}$. What will happen if copper spoon is used to stir a solution of aluminium nitrate? (a) The spoon gets coated with aluminium. (b) An alloy of aluminium and copper is formed. (c) No reaction occurs (d) The solution starts turning blue 74. Zn can displace: (a) Mg from its aqueous solution (b) Cu from its aqueous solution (c) Na from its aqueous solution (d) Al from its aqueous solution

Problem 46

[Narendra Avasthi Q75]

Based on the following information arrange four metals A, B, C and D in order of decreasing ability to act as reducing agents : (1) Only A, B and C react with 1 M HCl to give $\text{H}_2(\text{g})$ (II) When C is added to solutions of the other metal ions, metallic B and D are formed (III) Metal C does not reduce A^{n+} . (a) $\text{C} > \text{A} > \text{B} > \text{D}$ (b) $\text{C} > \text{A} > \text{D} > \text{B}$ (c) $\text{A} > \text{C} > \text{D} > \text{B}$ (d) $\text{A} > \text{C} > \text{B} > \text{D}$ 76. When an aqueous solution of CuSO_4 is stirred with a silver spoon then: (a) Cu^+ will be formed (b) Ag^+ will be formed (c) Cu^{2+} will be deposited (d) None of these 77. Based on the following information arrange four metals, A, B, C and D in order of increasing ability to act as reducing agents : (1) Only C reacts with 1 M HCl to give $\text{H}_2(\text{g})$ (II) When A is added to solution of the other metal ions, metallic D is formed but not B or C (a) $\text{D} < \text{A} < \text{C} < \text{B}$ (b) $\text{A} < \text{D} < \text{C} < \text{B}$ (c) $\text{B} < \text{D} < \text{A} < \text{C}$ (d) $\text{D} < \text{A} < \text{B} < \text{C}$

Problem 47

[Narendra Avasthi Q78]

In the reaction : $4\text{Fe(s)} + 3\text{O}_2(\text{g}) \rightarrow 4\text{Fe}^{3+}(\text{aq}) + 6\text{O}^{2-}(\text{aq})$ — PAGE 16 — ELECTROCHEMISTRY 403 which of the following statement is incorrect? (a) A redox reaction (b) Fe is reducing agent (c) O_2 is an oxidizing agent (d) Fe is reduced to Fe^{3+} 79. Which of the following is displaced by Fe? (a) Ag (b) Zn (c) Na (d) All of these

Problem 48

[Narendra Avasthi Q80]

The standard potential at 25°C for the following half reactions is given : $\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg}; E^\circ = -2.37 \text{ V}$ When Zinc dust is added to the solution of MgCl_2 . (a) ZnCl_2 is formed (b) Mg is precipitated (c) Zn dissolved in the solution (d) No reaction takes place 81. The element which can displace three other halogens from their compound is: (a) F (b) Cl (c) Br (d) I

Problem 49

[Narendra Avasthi Q82]

Using the standard half-cell potential listed, calculate the equilibrium constant for the reaction: $\text{Co(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Co}^{2+}(\text{aq}) + \text{H}_2(\text{g})$ at 298 K $\text{Co}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Co(s)}; E^\circ = -0.277 \text{ V}$ (a) 2.3×10^9 (b) 4.8×10^4 (c) 4.8×10^7 (d) 4.8×10^{11} $\text{H}_2(\text{g}) + 2\text{AgCl(s)} \rightarrow 2\text{Ag(s)} + 2\text{HCl(aq)}$ (a) 2.8×10^7 (b) 5.2×10^8 (c) 5.2×10^6 (d) 5.2×10^3

Problem 50

[Narendra Avasthi Q84]

Electrode potential of the half cell $\text{Pt(s)} | \text{Hg(l)} | \text{Hg}_2\text{Cl}_2(\text{s}) | \text{Cl}^-(\text{aq})$ can be increased by : (a) Increasing $[\text{Cl}^-]$ (b) Decreasing $[\text{Cl}^-]$ (c) Increasing $\text{Hg}_2\text{Cl}_2(\text{s})$ (d) Decreasing Hg(l)

Problem 51

[Narendra Avasthi Q85]

The equilibrium constant for the following general reaction is 1030. Calculate E° for the cell at 298 K. $2X_2(s) + 3Y_2(aq) \rightarrow 2X_2(aq) + 3Y(s)$ (a) +0.105 V (b) +0.2955 V (c) 0.0985 V (d) -0.2955 V

86. A solution containing H^+ and D^+ ions is in equilibrium with a mixture of H_2 and D_2 gases at 25°C. If the partial pressures of both gases are 1.0 atm, find the ratio of $[D^+]/[H^+]$: = -0.003 V (a) 1.23 (b) 1.12 (c) 0.11 (d) 1.0

Problem 52

[Narendra Avasthi Q87]

The E° at 25°C for the following reaction is 0.55 V. Calculate the ΔG° in kJ: $4BiO^+(aq) + 3N_2H_4(l) \rightarrow 4Bi(s) + 3N_2(g) + 4H_2O(l) + 7H^+(aq)$ (a) -637 (b) -424 (c) -106 (d) -318.5

88. Use the following E° for the electrode potentials, calculate ΔG° in kJ for the indicated reaction: $(b) D + H_8 + (b) O_u W + (b) D + S(1) O_2 H + (b) D + u W + (b) + S MnO_4(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$; $E^\circ = +1.51$ V — PAGE 17 — 404 $Ce^{4+}(aq) + e^- \rightarrow Ce^{3+}(aq)$ $E^\circ = +1.61$ V (a) -9.65 (b) -24.3 (c) -48.25 (d) -35.2

89. Consider an electrochemical cell in which the following reaction occurs and predict which changes will decrease the cell voltage: $Fe^{2+}(aq) + Ag^+(aq) \rightarrow Ag(s) + Fe^{3+}(aq)$ (1) decrease the $[Ag^+]$ (I) increase in $[Fe]$ (II) increase the amount of Ag (a) 1 (b) II and III (c) II (d) I and II

90. Consider the following equation for an electrochemical cell reaction. Which of the following changes in condition will increase the cell voltage? $H_2(g) + PbCl_2(s) \rightarrow Pb(s) + 2HCl(aq)$ (1) dissolve concentrated HCl in the cell solution (II) increase the pressure of $H_2(g)$ (II) increase the amount of Pb(s) (a) III (b) I and II (c) II and III (d) 11

Problem 53

[Narendra Avasthi Q91]

The standard electrode potential for the following reaction is +1.33 V. What is the potential at pH=2.0? $Cr_2O_7^{2-}(aq, 1M) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq, 1M) + 7H_2O(l)$ (a) +1.820 V (b) +1.990 V (c) +1.608 V (d) +1.0542 V

Problem 54

[Narendra Avasthi Q92]

The standard electrode potential for the following reaction is -0.57 V. What is potential at pH=12.0? $TeO_2(aq, 1M) + 3H_2O(l) + 4e^- \rightarrow Te(s) + 6OH^-(aq)$ (a) -0.17 V (b) +0.21 V (c) -0.39 V (d) +1.95 V

Problem 55

[Narendra Avasthi Q93]

$Co|Co_2(C_2) || Co_2(C_2) |Co$; for this cell, ΔG° is negative if: (a) $C_2 > C$ (b) $C > C_2$ (c) $C = C_2$ (d) unpredictable

Problem 56

[Narendra Avasthi Q94]

What will be the emf for the given cell? $Pt|H(g, P)|H^+(aq)||H(g, P)|Pt$ (a) (c) (d) None of these

F P2 2F " P2 F" R 95. For the electrochemical cell $Pt(s)|H_2(g)|H^+(1M) || Cu^{2+}(1M) |Cu(s)$, which one of the following statements is true? (a) H_2 is liberated at anode and Cu is deposited at cathode. (b) H_2 is liberated at cathode and Cu is deposited at anode. (c) Oxidation occurs at cathode. (d) Reduction occurs at anode.

Problem 57

[Narendra Avasthi Q96]

In a concentration cell the same reactants are present in both the anode and the cathode compartments, but at different concentrations. Calculate the emf of a cell containing 0.040 M Cr^{3+} in one compartment and 1.0 M Cr^{3+} in the other if Cr electrodes are used in both. (a) 0.028 V (b) 0.249 V (c) 0.083 V (d) 0.125 V

Problem 58

[Narendra Avasthi Q97]

A 1.0 M solution of Cd^{2+} is added to excess iron and the system is allowed to reach equilibrium: $\text{Cd}^{2+}(\text{aq}) + \text{Fe}(\text{s}) \rightleftharpoons \text{Cd}(\text{s}) + \text{Fe}^{2+}(\text{aq})$; $E^\circ = 0.037 \text{ V}$ — PAGE 18 — 405 T (a) 0.195 (b) 0.097 (c) 0.053 (d) 0.145 98. The measured voltage for the reaction with the indicated concentrations is 1.50 V. Calculate E for $\text{Cr}(\text{s}) + 3\text{Ag}^+(\text{aq}, 0.10 \text{ M}) \rightleftharpoons \text{Ag}(\text{s}) + \text{Cr}^{3+}(\text{aq}, 0.30 \text{ M})$ (a) 1.35 (b) 1.40 (c) 1.65 (d) 1.55

Problem 59

[Narendra Avasthi Q99]

Calculate the standard voltage that can be obtained from an ethane oxygen fuel cell at 25°C . $\text{C}_2\text{H}_6(\text{g}) + 7/2\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l})$; $\Delta G^\circ = -1467 \text{ kJ}$ (a) +0.91 (b) +0.54 (c) +0.72 (d) +1.08 /cdot to be 0.7714 V If $E^\circ = 0.535 \text{ V}$, find the pH of H^+ half-cell. (a) 1 (b) 3 (c) 5 (d) 7 101. What is the SRP for $\text{S}_2^{2-}|\text{CuS}|\text{Cu}$ half-cell? Given: $K_{\text{sp}}(\text{CuS}) = 10^{-35}$ and (a) 1.034 V (b) 1.0 V (c) -0.694 V (d) 0.694 V 102. Given the following standard electrode potentials, the K_s for PbBr_2 is : $\text{PbBr}_2(\text{s}) + 2\text{e}^- \rightleftharpoons \text{Pb}(\text{s}) + 2\text{Br}^-(\text{aq})$; $E^\circ = -0.248 \text{ V}$ $\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Pb}(\text{s})$; $E^\circ = -0.126 \text{ V}$ (a) 7.4×10^{-5} (b) 4.9×10^{-14} (c) 5.2×10^{-6} (d) 2.3×10^{-13} 103. The standard free energy change for the following reaction is -210 kJ. What is the standard cell potential? $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ (a) +0.752 (b) +1.09 (c) +0.420 (d) +0.640

Problem 60

[Narendra Avasthi Q104]

At equilibrium : (a) $E_{\text{cell}} = 0$, $\Delta G = 0$ (b) $E_{\text{cell}} = 0$, $\Delta G \neq 0$ (c) both are correct (d) none is correct 105. The E° for the following reaction at the indicated concentrations is 1.50 V. Calculate the ΔG in kJ at 25°C : $\text{Cr}(\text{s}) + 3\text{Ag}^+(\text{aq}, 0.1 \text{ M}) \rightleftharpoons \text{Ag}(\text{s}) + \text{Cr}^{2+}(\text{aq}, 0.1 \text{ M})$ (a) -140.94 (b) -295 (c) -212 (d) -422.83 106. If E° is 1.69 V and $E^\circ_{\text{Au}^+/\text{Au}} = 1.9 \text{ V}$ (a) 0.19 V (b) 2.945 V (c) 1.255 V (d) None of these 107. Consider the following standard electrode potentials and calculate the equilibrium constant at 25°C for the indicated disproportionation reaction: $3\text{Mn}^{2+}(\text{aq}) \rightleftharpoons 2\text{Mn}(\text{s}) + \text{Mn}^{2+}(\text{aq})$ $\text{Mn}^{2+}(\text{aq}) + \text{e}^- \rightleftharpoons \text{Mn}^{2+}(\text{aq})$; $E^\circ = 1.51 \text{ V}$ $\text{Mn}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Mn}(\text{s})$; $E^\circ = -1.185 \text{ V}$ (a) 1.2×10^{-43} (b) 2.4×10^{-73} (c) 6.3×10^{-92} (d) 1.5×10^{-62} — PAGE 19 — 406 PROBLEMS IN CHEMISTRY 108. A galvanic cell is composed of two hydrogen electrodes, one of which is a standard one. In which of the following solutions should the other electrode be immersed to get maximum e.m.f. : (a) 0.1 M HCl (b) 0.1 M H_2SO_4 (c) 0.1 M NH_4OH (d) 0.01 M HCOOH (a) $\text{C} < \text{C}_2$ (b) $\text{C} > \text{C}_2$ (c) $\text{C}_2 > \text{C}$ (d) Both (a) and (c) 110. By how much is the oxidizing power of $\text{Cr}_2\text{O}_7^{2-}|\text{Cr}^{3+}$ couple decreased if the H^+ concentration is decreased from 1 M to 10^{-3} M at 25°C ? (a) 0.001 V (b) 0.207 V (c) 0.441 V (d) 0.414 V 111. The temperature coefficient of a cell whose operation is based on the reaction $\text{Pb}(\text{s}) + \text{HgCl}_2(\text{aq}) \rightleftharpoons \text{PbCl}_2(\text{aq}) + \text{Hg}(\text{l})$ is : $E = 1.5 \times 10^{-4} \text{ V/K}$ at 298 K dT The change in entropy (in J/K mol) during the operation is : (a) 8627 (b) 57.9 (c) 28.95 (d) 14.475 112. Thermodynamic efficiency of a cell is given by : (a) H (b) nFE (c) nEF (d) $nFE_0 \Delta G / \Delta H$

Problem 61

[Narendra Avasthi Q113]

Calculate the value of equilibrium constant (K) for the reaction : $\text{Zn}^{2+}(\text{aq}) + 4\text{OH}^-(\text{aq}) \rightleftharpoons \text{Zn}(\text{OH})_2(\text{aq})$ Given : $\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Zn}(\text{s})$; $E^\circ = -0.76 \text{ V}$ $\text{Zn}(\text{OH})_2(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Zn}(\text{s}) + 4\text{OH}^-(\text{aq})$; $E^\circ = -1.36 \text{ V}$ $F = 96485 \text{ C/mol}$ (a) 1010 (b) 2×10^{10} (c) 1020 (d) none of these 114. Which of the following statement is false for fuel cells? (a) They are more efficient (b) They are free from pollution (c) They run till reactants are active (d) Fuel burned with O_2 115. When a lead storage battery is charged it acts as: (a) a fuel cell (b) an electrolytic cell (c) a galvanic cell (d) a concentration cell 116. The metal that forms a self-protecting film of oxide to prevent corrosion is : (a) Na (b) Al (c) Cu (d) Au 117. Rusting of iron is catalyzed by which of the following? (a) Fe (b) Zn (c) O_2 (d) H^+ 118. Which of the following is a highly corrosive salt ? (a) Hg_2Cl_2 (b) HgCl_2 (c) FeCl_2 (d) PbCl_2

Problem 62**[Narendra Avasthi Q119]**

The Zn acts as sacrificial or cathodic protection to prevent rusting of iron because : (a) Eop of Zn < Eop of Fe (b) Eop of Zn > Eop of Fe (c) Eop of Zn = Eop of Fe (d) Zn is cheaper than iron — PAGE 20 — 407 120. In electrochemical corrosion of metals, the metal undergoing corrosion: (a) acts as anode (b) acts as cathode (c) undergoes reduction (d) None

Problem 63**[Narendra Avasthi Q121]**

When an acid cell is charged, then : (a) Voltage of cell increases (b) Resistance of cell increases (c) Electrolyte of cell dilutes (d) None of these 122. Electrolytic conduction is due to the movement of: (a) electrons (b) ions (c) atoms (d) electrons as well as ions 1.23. Molten sodium chloride conducts electricity due to the presence of: (a) Free electron (b) Free ions (c) Free molecules (d) Atoms of sodium and chlorine 124. Pure water does not conduct electricity because it : (a) is neutral (b) is readily decomposed (c) is almost totally unionized (d) has a low boiling point

Problem 64**[Narendra Avasthi Q125]**

The relation among conductance (G), specific conductance (k) and cell constant (l/A) is : (a) $G = K \cdot l/A$ (b) $G = K \cdot A$ (c) $GK = l/A$ (d) $G = KAl/A$

Problem 65**[Narendra Avasthi Q126]**

If x is specific resistance (in S-cm) of the electrolyte solution and y is the molarity of the solution, then Λ_m (in $S\ cm^2\ mol^{-1}$) is given by : (a) $1000x$ (b) 1000 (c) 1000 (d) xy y x Kx 1000 127. Equivalent conductance can be expressed in terms of specific conductance (k) and (a) $k \times N$ (b) $K/\text{times}1000$ (c) $K \times N$ (d) $K \times N \times 1000$ N 1000

Problem 66**[Narendra Avasthi Q128]**

Resistance of a decimolar solution between two electrodes 0.02 meter apart and $0.0004\ m^2$ in - area was found to be $50\ \Omega$. Specific conductance (k) is: (a) $0.1\ S\ m^{-1}$ (b) $1\ S\ m^{-1}$ (c) $10\ S\ m^{-1}$ (d) $4/\text{times}10^{-4}\ S\ m^{-1}$ 129. Resistance of 0.1 M KCl solution in a conductance cell is $300\ \Omega$ and conductivity is $0.013\ S\ cm^{-1}$. The value of cell constant is : (a) $3.9\ cm^{-1}$ (b) $39\ m^{-1}$ (c) $3.9\ m^{-1}$ (d) None of these 130. Ionisation constant of a weak acid (HA) in terms of Λ and Λ_m is : $C\Lambda_m$ CA^2m (a) $K_a =$ (b) $K_a = (\nu - \nu)V / (\alpha V - \nu) C(\Lambda_m)^2$ (c) $K_a =$ (d) None of these $A(\Lambda_m - \Lambda_m)$ 131. When a concentrated solution of an electrolyte is diluted? (a) its specific conductance increases (b) its equivalent conductance decreases — PAGE 21 — 408 (c) its specific conductance decreases and equivalent conductance increases (d) both specific and equivalent conductance increase 132. Molar conductivity of a solution of an electrolyte AB is $150\ S\ cm^2\ mol^{-1}$. If it ionises as $AB_3 \rightarrow A^{3+} + 3B^-$, its equivalent conductivity will be : (a) 150 (in $S\ cm^2\ eq^{-1}$) (b) 75 (in $S\ cm^2\ eq^{-1}$) (c) 50 (in $S\ cm^2\ eq^{-1}$) (d) 80 (in $S\ cm^2\ eq^{-1}$) 133. Equivalent conductivity of $Fe_2(SO_4)_3$ is related to molar conductivity by the expression: $\nu = b\nu$ (e) (b) $A_{eq} = \Lambda_m/3$ $\nu = b\nu$ () (d) $A_{eq} = \Lambda_m/6$

Problem 67**[Narendra Avasthi Q134]**

The limiting equivalent conductivity of NaCl, KCl and KBr are 126.5, 150.0 and $151.5\ S\ cm^2\ eq$, respectively. The limiting equivalent ionic conductance for Br^- is $78\ S\ cm^2\ eq^{-1}$. The limiting equivalent ionic conductance for Na^+ ions would be : (a) 128 (b) 125 (c) 49 (d) 50 135. The specific conductance of a saturated solution of silver bromide is $k\ S\ cm^{-1}$. The limiting ionic conductivity of Ag^+ and Br^- ions are x and y, respectively. The solubility of silver bromide in $g\ L^{-1}$ is : (molar mass of $AgBr = 188$) $x(\text{g}) / \text{times}188$ (c) $K/\text{times}1000/\text{times}188$ (d) $x+y$ 1000 $x-y$ $x+y$ $x+y$ K 188

Problem 68**[Narendra Avasthi Q136]**

The resistance of 0.1 N solution of formic acid is 200 Ω and cell constant is 2.0 cm⁻¹. The equivalent conductivity (in S cm² eq⁻¹) of 0.1 N formic acid is : (a) 100 (b) 10 (c) 1 (d) none of these

Problem 69**[Narendra Avasthi Q137]**

A conductance cell was filled with a 0.02 M KCl solution which has a specific conductance of 2.768 $\times 10^{-3}$ Ω^{-1} cm⁻¹. If its resistance is 82.4 Ω at 25°C, the cell constant is : (a) 0.2182 cm⁻¹ (b) 0.2281 cm⁻¹ (c) 0.2821 cm⁻¹ (d) 0.2381 cm⁻¹

Problem 70**[Narendra Avasthi Q138]**

The ionic conductivity of Ba²⁺ and Cl⁻ at infinite dilution are 127 and 76 Ω^{-1} cm² eq⁻¹ respectively. The equivalent conductivity of BaCl₂ at infinity dilution (in Ω^{-1} cm² eq⁻¹) would be : (a) 203 (b) 279 (c) 101.5 (d) 139.5. Unit of ionic mobility is : (a) m V⁻¹ s⁻¹ (b) m² v⁻¹ s⁻¹ (c) m² v⁻¹ s⁻¹ (d) m² V s⁻¹

Problem 71**[Narendra Avasthi Q140]**

Λ°_{AgCl} can be obtained : (a) by extrapolation of the graph Λ and $\sqrt{C}$ to zero concentration (b) by known values of Λ° of AgNO₃, HCl and HNO₃ (c) both (a) and (b) (d) None of these

Problem 72**[Narendra Avasthi Q141]**

The conductance of a salt solution (AB) measured by two parallel electrodes of area 100 cm² separated by 10 cm was found to be 0.0001 Ω^{-1} . If volume enclosed between two electrode contain 0.1 mole of salt, what is the molar conductivity (S cm² mol⁻¹) of salt at same concentration: (a) 10 (b) 0.1 (c) 1 (d) none of these — PAGE 22 — ELECTROCHEMISTRY 409-442. The conductivity of a strong electrolyte: (a) Increases on dilution (b) Decreases on dilution (c) Does not change with dilution (d) Depends upon density of electrolytes 143. The increase in equivalent conductance of a weak electrolyte with dilution is due to: (a) Increase in degree of dissociation and decrease in ionic mobility (b) Decrease in degree of dissociation and decrease in ionic mobility (c) Increase in degree of dissociation and increase in ionic mobility (d) Decrease in degree of dissociation and increase in ionic mobility 144. Strong electrolytes are those which: (a) Conduct electricity (b) Dissolve readily in water (c) Dissociate into ions at high dilution (d) Completely dissociate into ions 145. The electric conduction of a salt solution in water depends on the: (a) Size of its molecules (b) Shape of its molecules (c) Size of solvent molecules (d) Extent of its ionization 146. A graph was plotted between molar conductivity of various electrolytes (NaCl, HCl and NH₄OH) and $\sqrt{C}$ (in mol L⁻¹). Correct setting: conductivity Molar $\sqrt{C}$ (a) I(NaCl), II(HCl), III(NH₄OH) (b) I(HCl), II(NaCl), II(NH₄OH) (c) I(NH₄OH), I(NaCl), III(HCl) (d) I(NH₄OH), II(HCl), II(NaCl) 147. Which of the following is arranged in increasing order of ionic mobility? (a) I < Br⁻ < Cl⁻ < F⁻ (b) F⁻ < Cl⁻ < Br⁻ < I⁻ (c) F⁻ < I⁻ < Cl⁻ < Br⁻ (d) F⁻ < Cl⁻ < I⁻ < Br⁻ 148. HNO₃(aq) is titrated with NaOH(aq) conductometrically, graphical representation of the titration as: (a) (b) 8 Vol of NaOH Vol of NaOH (c) (d) Vol of NaOH Vol of NaOH — PAGE 23 —

Problem 73**[Narendra Avasthi Q149]**

Which of the following plots will be obtained for a conductometric titration of strong acid against a weak base? (a) (b) Vol of alkaline Vol of alkaline uonjos Vol of alkaline Vol of alkaline solution

Problem 74**[Narendra Avasthi Q150]**

Conductometric titration curve of a equimolar mixture of a HCl and HCN with NaOH(aq) is : (a) Vol of NaOH Vol of NaOH Conductance (c) (d) Vol of NaOH Vol of NaOH — PAGE 24 — ELECTROCHEMISTRY Level 2 1. In the Hall process, aluminum is produced by the electrolysis of molten Al_2O_3 by the following reactions. How many second would it take to produce enough aluminum by the Hall process to make a case of 24 cans of aluminum soft-drink, if each can uses 5.0 g of Al, a current of 9650 amp is employed, and the current efficiency of the cell is 90.0% (a) 203.2 (b) 148.14 (c) 333 (d) 6.17 2. 108 g fairly concentrate solution of AgNO_3 is electrolyzed using 0.1 F of electricity. The weight of resulting solution is: (a) 94 g (b) 11.6 g (c) 96.4g (d) None of these

Problem 75**[Narendra Avasthi Q3]**

The electrolysis of acetate solution produces ethane according to reaction : $2\text{CH}_3\text{COO}^- \rightarrow \text{C}_2\text{H}_6(\text{g}) + 2\text{CO}_2(\text{g}) + 2\text{e}^-$ The current efficiency of the process is 80% and 740 torr, if the current of 0.5 amp is passed through the solution for 96.45 min? (a) 6.0 L (b) 0.60 L (c) 1.365 L (d) 0.91 L 4. A layer of chromium metal 0.25 mm thick is to be plated on an auto bumper with a total area of 0.32 m^2 from a solution containing CrO_2^- . What current flow is required for this electroplating if the bumper is to be plated in 60 s ? The density of chromium metal is 7.20 g/cm^3 . (a) 4.9 /times 103A (b) 1.78 x103A (c) 5.3x104A (d) 10.69 /times 10 /circ A

Problem 76**[Narendra Avasthi Q5]**

100 mL of 0.05 M $\text{CuSO}_4(\text{aq})$ solution was electrolyzed using inert electrodes by passing current till the pH of the resulting solution was 2. The solution after electrolysis was neutralized and then treated with excess KI and formed I_2 titrated with 0.04 M $\text{Na}_2\text{S}_2\text{O}_3$. Calculate the required volume (in mL) of $\text{Na}_2\text{S}_2\text{O}_3$: (a) 112.5 mL (b) 100 mL (c) 125 mL (d) None of these

Problem 77**[Narendra Avasthi Q6]**

If the equilibrium constant for the reaction $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$ is 103 at certain temperature then what is the E° for the reaction, $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$ 2.303RT Given: $F = 0.066$ (a) 1.230 V (b) -0.858V (c) -0.80V (d) -0.8274 V 7. A fuel cell develops an electrical potential from the combustion of butane at 1 bar and 298K $\text{C}_4\text{H}_{10}(\text{g}) + 6.5\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 5\text{H}_2\text{O}(\text{l})$; $\Delta_r G^\circ = -2746 \text{ kJ/mol}$ What is E° of a cell? (a) 4.74 V (b) 0.547V (c) 4.37 V (d) 1.09 V

Problem 78**[Narendra Avasthi Q8]**

The cell $\text{Pt}|\text{H}_2(\text{g}, 0.1 \text{ bar})|\text{H}^+(\text{aq}), \text{pH} = \text{X}||\text{Cl}^-(\text{aq}, 1 \text{ M})|\text{Hg}_2\text{Cl}_2|\text{Hg}|\text{Pt}$, has e.m.f. of 0.5755 V at 25 $^\circ\text{C}$. The SOP of calomel electrode is -0.28V, then pH of solution will be : (a) 11 (b) 4.5 (c) 5.5 (d) None of these' — PAGE 25 — 412

Problem 79**[Narendra Avasthi Q9]**

For a cell reaction $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$ $\Delta_r H^\circ_{298} = -0.32 \text{ kJ/K}$. What is the value of $\Delta_r H^\circ_{298}(\text{H}_2\text{O}, \text{l})$? Given: $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\text{l})$; $E^\circ = 1.23 \text{ V}$ (a) -285.07 kJ/mol (b) -570.14 kJ/mol (c) 285.07 kJ/mol (d) None of these 10. What is the potential of an electrode which originally contained 0.1 M NO_3^- and 0.4 M H^+ and which has been treated by 80% at 1 bar? Given: $\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{NO} + 2\text{H}_2\text{O}$; $E^\circ = 0.96 \text{ V}$; $\log 2 = 0.3$ (a) 0.84V (b) 1.08 V (c) 1.23 V (d) 1.36 V

Problem 80**[Narendra Avasthi Q11]**

The standard reduction potential of normal calomel electrode and reduction potential of saturated calomel electrodes are 0.27 and 0.33 volt respectively. What is the concentration of Cl in saturated solution of KCl? (a) 0.1 M (b) 0.01 M (c) 0.001 M (d) None

12. Determine the potential of the following cell: $\text{Pt}[\text{H}(\text{g}, 0.1 \text{ bar})|\text{H}^+(\text{aq}, 10^{-4} \text{ M})|\text{MnO}_4^-(\text{aq}, 0.1 \text{ M}), \text{Mn}^{2+}(\text{aq}, 0.01 \text{ M}), \text{H}^+(\text{aq}, 0.01 \text{ M})|\text{Pt}]$
 Given: $E^\circ_{\text{MnO}_4^-/\text{Mn}^{2+}} = 1.51 \text{ V}$ (a) 1.54 V (b) 1.48 V (c) 1.84 V (d) none of these

Problem 81**[Narendra Avasthi Q13]**

Copper reduces NO into NO and NO₂ depending upon concentration of HNO₃ in solution. Assuming $[\text{Cu}^{2+}] = 0.1 \text{ M}$, and $P_{\text{NO}} = P_{\text{NO}_2} = 10^{-4} \text{ bar}$. At which concentration of HNO₃, thermodynamic tendency for reduction of NO into NO and NO₂ by copper is same? $E_{\text{NO}/\text{NO}_2} = +0.96 \text{ V}$, (a) 101.23 M (b) 100.56 M (c) 100.66 M (d) 100.12 M

Problem 82**[Narendra Avasthi Q14]**

For the cell, $\text{Pt}|\text{Cl}_2(\text{g}, 0.4 \text{ bar})|\text{Cl}^-(\text{aq}, 0.1 \text{ M})|\text{ICl}(\text{aq}, 0.01 \text{ M})|\text{Cl}_2(\text{g}, 0.2 \text{ bar})|\text{Pt}$ The measured potential at 298 K is: (a) 0.051 V (b) -0.051 V (c) 0.102 V (d) 0.0255 V

Problem 83**[Narendra Avasthi Q15]**

The chlorate ion can disproportionate in basic solution according to reaction, $2\text{ClO}_3^- \rightarrow \text{ClO}_4^- + \text{ClO}_2^-$ What is the equilibrium concentration of perchlorate ions from a solution initially at 0.1 M in chlorate ions at 298 K? Given: $E^\circ_{\text{ClO}_4^-/\text{ClO}_3^-} = 0.36 \text{ V}$ and $E^\circ_{\text{ClO}_3^-/\text{ClO}_2^-} = 0.33 \text{ V}$ at 298 K (a) 0.019 M (b) 0.024 M (c) 0.1 M (d) 0.19 M — PAGE 26 — 413

16. A cell diagram shown below contains one litre of buffer solution of HA ($\text{p}K_a = 4$) and NaA in both compartment. What is the cell e.m.f.? $\text{H}_2(\text{g}):1 \text{ bar} | \text{H}^+(\text{aq}):1 \text{ bar} || \text{HA}(\text{aq}, 0.1 \text{ M}) | \text{HA}(\text{aq}, 1 \text{ M})$ and $\text{NaA}(\text{aq}, 1 \text{ M}) | \text{NaA}(\text{aq}, 1 \text{ M})$ (a) 0.03 V (b) 0.06 V (c) -0.06 V (d) None of these

Problem 84**[Narendra Avasthi Q17]**

Given the cell: $\text{Cd}(\text{s})|\text{Cd}(\text{OH})_2(\text{s})|\text{NaOH}(\text{aq}, 0.01 \text{ M})|\text{H}_2(\text{g}, 1 \text{ bar})|\text{Pt}(\text{s})$ (a) 0.1 (b) 10⁻¹³ (c) 10⁻¹⁵ (d) None of these

18. Calculate the e.m.f. (in V) of the cell: $\text{Pt}|\text{H}_2(\text{g})|\text{BOH}(\text{aq})|\text{HA}(\text{aq})|\text{H}^+(\text{g})|\text{Pt}$; 0.1 bar 1 M 0.1 M 1 bar Given: $K_a(\text{HA}) = 10^{-7}$, $K_b(\text{BOH}) = 10^{-5}$ (a) 0.39 V (b) 0.36 V (c) 0.93 V (d) None of these

Problem 85**[Narendra Avasthi Q19]**

Calculate the potential of a half cell having reaction: $\text{Ag}_2\text{S}(\text{s}) + 2\text{e}^- \rightarrow 2\text{Ag}(\text{s}) + \text{S}^{2-}(\text{aq})$ in a solution buffered at pH=3 and which is also saturated with 0.1 M $\text{HS}^-(\text{aq})$: [Given: $K_{\text{sp}}(\text{Ag}_2\text{S}) = 10^{-49}$, $K_a(\text{HS}^-) = 10^{-14}$] (a) 1.18 (b) 0.39 (c) -0.19 V (d) none of these

Problem 86**[Narendra Avasthi Q20]**

The conductivity of 0.1 N NaOH solution is 0.022 S cm⁻¹. When equal volume of 0.1 N HCl solution is added, the conductivity of resultant solution decreases to 0.0055 S cm⁻¹. The equivalent conductivity in S cm² eq⁻¹ of NaCl solution is: (a) 0.0055 (b) 0.11 (c) 110 (d) None of these

21. In above question after formation of NaCl, further 0.1 N HCl is added, the volume of which is double to that of the first portion added, the conductivity increases to 0.018 S cm⁻¹. The value of $\Lambda_{\text{eq}}(\text{HCl})$ is [assume no change in conductivity of NaCl (aq)]: (a) 330 S cm² eq⁻¹ (b) 305 S cm² eq⁻¹ (c) 415 S cm² eq⁻¹ (d) 360 S cm² eq⁻¹

Problem 87**[Narendra Avasthi Q22]**

Given the following molar conductivities at 25°C : HCl , $426\text{ cm}^2\text{mol}^{-1}$; NaCl , $126\text{ cm}^2\text{mol}^{-1}$; NaC (sodium crotonate), $83\text{ cm}^2\text{mol}^{-1}$. What is the ionization constant of crotonic acid? If the conductivity of a 0.001 M crotonic acid solution is $3.83 \times 10^{-5}\text{ cm}^{-1}$? (a) 10^{-5} (b) 1.11×10^{-5} (c) 1.11×10^4 (d) 0.01 — PAGE 27 — PROBLEMS IN CHEMISTRY

Problem 88**[Narendra Avasthi Q23]**

Equivalent conductivity of BaCl_2 , H_2SO_4 and HCl , are x , X_2 and $x\text{ Scm}^{-1}$ at infinite dilution. If conductivity of saturated BaSO_4 solution is $x\text{ Scm}^{-1}$, then K_{sp} of BaSO_4 is : $500x \times 10^6 x^2$ (a) (b) $(x_1 + x_2 - 2x)^3$ (c) 2.5×10^2 (d) $(x_1 + x_2 - x_3)^2$ $(x_1 + x_2 - x_3)^2$

Problem 89**[Narendra Avasthi Q24]**

The conductivity of 0.001 M Na_2SO_4 solution is $2.6 \times 10^4\text{ Scm}^{-1}$ and increases to $7.0 \times 10^4\text{ Scm}^{-1}$, when the solution is saturated with CaSO_4 . The molar conductivities of Na^+ and Ca^{2+} are 50 and $120\text{ Scm}^2\text{mol}^{-1}$, respectively. Neglect conductivity of used water. What is the solubility product of CaSO_4 ? (a) 4×10^{-6} (b) 1.57×10^{-3} (c) 4×10^{-4} (d) 2.46×10^{-6}

Problem 90**[Narendra Avasthi Q25]**

The ionization constant of a weak acid is 1.6×10^{-5} and the molar conductivity at infinite dilution is $380 \times 10^{-4}\text{ S m}^2\text{mol}^{-1}$. If the cell constant is 0.01 m^{-1} then conductance of 0.01 M acid solution is: (a) $1.52 \times 10^{-5}\text{ s}$ (b) 1.52 S (c) $1.52 \times 10^{-3}\text{ s}$ (d) $1.52 \times 10^{-4}\text{ s}$ 26. Three electrolytic cells X, Y, Z containing solution of NaCl , AgNO_3 and CuSO_4 respectively are connected in series combination. During electrolysis 21.6 gm of silver deposit at cathode in cell Y. Which is incorrect statement. (a) 6.35 gm copper deposit at cathode in cell Z (b) 2.24 litre Cl_2 liberated at 1 atm and 273 K at cathode in cell X (c) 2.24 litre O_2 liberated at 1 atm and 273 K at anode in cell Y (d) 2.24 litre H_2 liberated at 1 atm and 273 K at cathode in cell X moles of H_2SO_4 formed is : (a) 0.25 (b) 0.50 (c) 0.75 (d) 1.00

Problem 91**[Narendra Avasthi Q28]**

$\text{Zn(s)} \mid \text{Zn(CN)}_2 (0.5\text{M}), \text{CN}^- (0.01) \parallel \text{Cu(NH)}_2^{2+} (0.5\text{M}), \text{NH}_3 (1\text{M}) \mid \text{Cu(s)}$ Given: K of $\text{Zn(CN)}_2 = 10^{16}$; K of $\text{Cu(NH)}_2 = 10^{12}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34\text{V}$, $2.303RT = 0.06$ The emf of above cell is : (a) 1.22 V (b) 1.10 V (c) 0.98 V (d) None of these — PAGE 28 — 415 ELEOROCERSTRY Level 3 PASSOE Porous barrier $\text{Zn(s)} \mid \text{Zn}^{2+} + \text{NaNO}_3 \parallel \text{Cu}^{2+} \mid \text{Cu(s)}$ (1M)(1M)(1M)) A. Galvanic cell consist of three compartment as shown in figure. The first compartment contain $\text{ZnSO}_4 (1\text{ M})$ and the second compartment contain $\text{CuSO}_4 (1\text{ M})$. The mid compartment contain $\text{NaNO}_3 (1\text{ M})$. Each compartment contain 1 L solution:

Problem 92**[Narendra Avasthi Q1]**

The concentration of Zn^{2+} in first compartment after passage of 0.1 F charge will be : (a) 1 M (b) 1.05 M (c) 1.025 M (d) 0.5 M

Problem 93**[Narendra Avasthi Q2]**

The concentration of NO in mid compartment after passage of 0.1 F of charge will be : (a) 0.95 M (b) 0.90 M (c) 0.975 M (d) 1.05 M

Problem 94**[Narendra Avasthi Q3]**

The concentration of SO_4^{2-} ion in II compartment will be : (a) 1.05 M (b) 1.025 M (c) 0.95 M (d) 0.975 M .

POSSAGE 2 The cell potential (E_{cell}) of a reaction is related as $-\Delta G = nF E_{\text{cell}}$, where $-\Delta G$ represents max. useful electrical work n -no. of moles of electrons exchanged during the reaction for reversible cell reaction $dG = -nF dE$ at constant pressure $dG = -nF dE$ at constant pressure. $\Delta G = -nF E_{\text{cell}}$ (1) $\Delta G = -nF E_{\text{cell}}$ + $T \left(\frac{d\Delta G}{dT} \right)$... (2) $\left(\frac{dE_{\text{cell}}}{dT} \right)$ is known as temperature coefficient of the e.m.f of the cell $\frac{dE_{\text{cell}}}{dT}$ — PAGE 29 — 416 .

1. The temperature coefficient of the e.m.f. of cell, is given by: $\frac{dE_{\text{cell}}}{dT} = \frac{\Delta H}{nF T^2}$ (a) F (b) S (c) AS (d) $-\frac{nFE_{\text{cell}}}{nFT^2}$ 2. At 300 K, ΔG for the reaction $\text{Zn(s)} + \text{AgCl(s)} \rightarrow \text{ZnCl}_2(\text{aq}) + 2\text{Ag(s)}$ is -218 kJ/mol while the e.m.f. of the cell was 1.015 V $\frac{dE_{\text{cell}}}{dT}$ of the cell is: (a) $4.2 \times 10^{-4} \text{ V K}^{-1}$ (b) $3.81 \times 10^{-4} \text{ V K}^{-1}$ (c) 0.11 V K^{-1} (d) $7.62 \times 10^{-4} \text{ V K}^{-1}$ 3. Calculate ΔS for the given cell reaction in Q.no.2: (a) -73.53 J/Kmol (b) 83.53 J/Kmol (c) 100 J/Kmol (d) none of these

PASSAGE 3 Molar conductivity (Λ_m) is defined as conducting power of the ions produced by 1 mole of an electrolyte in a solution. $\Lambda_m = \frac{\kappa}{C}$ where κ is conductivity (in $\text{Scm}^2\text{mol}^{-1}$) and C is molar concentration (in mol/cm^3) The molar conductivity of 0.04 M solution of MgCl_2 is $200 \text{ Scm}^2\text{mol}^{-1}$ at 298 K . A cell with electrodes that are 2.0 cm^2 in surface area and 0.50 cm apart is filled with MgCl_2 solution.

1. Conductance of MgCl_2 solution is: (a) $8 \times 10^{-3} \text{ S}$ (b) 32 S (c) 0.032 S (d) None of these' 2. How much current will flow when the potential difference between the two electrodes is 5.0 V ? (a) 156.25 A (b) 0.16 A (c) 160 A (d) None of these

PASSAGE 4 In a hydrogen oxygen fuel cell, electricity produced. In this process $\text{H}_2(\text{g})$ is oxidised at anode and $\text{O}_2(\text{g})$ reduced at cathode. Given : Cathode $\text{O}_2(\text{g}) + 2\text{H}_2\text{O(l)} + 4e^- \rightarrow 4\text{OH}^-(\text{aq})$ Anode $\text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \rightarrow 2\text{H}_2\text{O(l)} + 2e^-$ 4.48 litre H_2 at 1 atm and 273 K oxidised in 9650 sec . 1. The current produced is (in amp) : (a) 1 amp (b) 2 amp (c) 4 amp (d) 8 amp

Problem 95**[Narendra Avasthi Q2]**

The mass of water produced is : (a) 7.2 gm (b) 3.6 gm (c) 1.8 gm (d) 0.9 gm

Problem 96**[Narendra Avasthi Q3]**

If current produced in fuel cell, use for the deposition of Cu^{+2} in 1 L , 2 M $\text{CuSO}_4(\text{aq})$ solution for 241.25 sec using Pt electrode. The pH of solution after electrolysis is : (a) 1 (b) 2 (c) 3 (d) 4 — PAGE 30 — 417

ELECTROCHEMISTRY IPASSAGE A saturated solution in AgX ($K_{\text{sp}} = 3 \times 10^{-12}$) and AgY ($K_{\text{sp}} = 10^{-12}$) has conductivity $0.4 \times 10^{-6} \text{ Q-cm}^{-1}$. Given : Limiting molar conductivity of $\text{Ag}^+ = 60 \text{ Q-cm}^2\text{mol}^{-1}$ Limiting molar conductivity of $\text{X}^- = 90 \text{ Q-cm}^2\text{mol}^{-1}$

Problem 97**[Narendra Avasthi Q1]**

The conductivity of Y^- is (in Q-cm^{-1}): (a) 1.45×10^{-7} (b) 1.45×10^{-5} (c) 1.45×10^{-9} (d) None of these

2. The limiting molar conductivity of Y^- is (in $\text{Q-cm}^2\text{mol}^{-1}$) : (b) 2900 (c) 2.90 (d) None of these

(a) 290 ONE OR MORE ANSWERS IS/ARE CORRECT 1. If the e.m.f of a galvanic cell is negative, it implies that : (a) the cell reaction is spontaneous . (b) the cell reaction is non-spontaneous (c) the cell reaction is exothermic (d) the cell is working in reverse direction

Problem 98**[Narendra Avasthi Q2]**

Select right statement(s) about electrolysis : (b) $-\Delta G$ is positive for chemical process during electrolysis (c) Cations and anions are moved toward the anode and cathode respectively (d) Over voltage is generally associated with evolution of O_2 gas

Problem 99**[Narendra Avasthi Q3]**

If the half-cell reaction $A + e^- \rightarrow A^-$ has a large negative reduction potentials, it follows that:
 (a) A is readily reduced (b) A is readily oxidised (c) A^- is readily reduced (d) A is readily oxidised
 4. Which of the following statement is correct? $E^\circ = -0.0V$ $H^+ | H_2$ (a) Cu^{2+} ions can be reduced by $H_2(g)$ (b) Cu can be oxidized by H^+ (c) Sn^{2+} ions can be reduced by H_2 (d) Sn can be oxidized by Cu^{2+}

Problem 100**[Narendra Avasthi Q5]**

The oxidation potential of hydrogen half-cell will be negative if : (a) $p(H_2) = 1 \text{ atm}$ and $[H^+] = 1 \text{ M}$
 (b) $p(H_2) = 1 \text{ atm}$ and $[H^+] = 2 \text{ M}$ (c) $p(H_2) = 0.2 \text{ atm}$ and $[H^+] = 1 \text{ M}$ (d) $p(H_2) = 0.2 \text{ atm}$ and $[H^+] = 0.2 \text{ M}$
 6. Which of the following arrangement will produce oxygen at anode during electrolysis?
 (a) Dilute H_2SO_4 with Pt electrodes (b) Fused $NaOH$ with inert electrodes — PAGE 31 — 418
 PROBLEMS IN CHEMISTRY (c) Dilute H_2SO_4 with Cu electrodes (d) Concentrate aq. $NaCl$ with Pt electrodes

Problem 101**[Narendra Avasthi Q7]**

When an aqueous concentrate solution of lithium chloride is electrolysed using inert electrodes : (a) Cl_2 is liberated at the anode (b) Li is deposited at the cathode (c) as the current flows, pH of the solution around the cathode remains constant (d) as the current flows, pH of the solution around the cathode increases

Problem 102**[Narendra Avasthi Q8]**

Oxygen and hydrogen gas are produced at the anode and cathode during the electrolysis of fairly concentrate aqueous solution of : (a) K_2SO_4 (b) $AgNO_3$ (c) H_2SO_4 (d) $NaOH$

Problem 103**[Narendra Avasthi Q9]**

During the purification of copper by electrolysis : (a) the anode used is made of copper ore (b) pure copper is deposited on the cathode (c) the impurities such as Ag, Au present in solution as ions (d) concentration of $CuSO_4$ solution remains constant during dissolution of Cu

Problem 104**[Narendra Avasthi Q10]**

When a lead storage battery is discharged : (a) SO_2 is evolved (b) lead sulphate is produced at both electrodes (c) sulphuric acid is consumed (d) water is formed
 11. Which of the following is characteristic of the cathode in a voltaic cell? (a) It may gain weight during reaction (b) Electrons flow to it through the external circuit (c) It is where oxidation occurs (d) It received electrons from ions in solution

Problem 105**[Narendra Avasthi Q12]**

In an electrochemical process, a salt bridge is used : (a) to maintain electrical neutrality in each solution (b) to complete the external circuit so that current can flow for long time (c) to mix the solution of anodic and cathodic compartment (d) to supply voltage

Problem 106**[Narendra Avasthi Q13]**

For a reaction in a galvanic cell the value of $-AG^\circ$ at certain temperature is not necessarily equal to: (a) nFE (b) $RT \ln K$ (c) $T \Delta S^\circ$ (d) zero

Problem 107**[Narendra Avasthi Q14]**

Standard electrode potential of two half-reactions are given below: $\text{Fe}^{2+}/\text{Fe} = -0.44\text{V}$ $\text{Fe}^{3+}/\text{Fe}^{2+} = +0.77\text{V}$ If Fe^{2+} , Fe^{3+} and Fe are kept together : (a) the concentration of Fe^{3+} increases (b) the concentration of Fe^{2+} decreases (c) the mass of Fe increases (d) the concentration of Fe^{2+} decreases — PAGE 32 — 419 ELECTROCHEMISTRY

Problem 108**[Narendra Avasthi Q15]**

Which of the following statements are correct regarding to galvanic cell? (a) A reaction is spontaneous from left to right if $E_{\text{cell}} > 0$ (b) A reaction occurs from right to left if $E_{\text{cell}} < 0$ (c) If the system is at equilibrium no reaction occurs (d) E_{cell} is temperature-independent

Problem 109**[Narendra Avasthi Q16]**

Which of the following are concentration cells? (a) $\text{Cd} | \text{Cd}^{2+} || \text{Hg}^{2+} | \text{Hg}$ (b) $\text{Pt} | \text{H}_2(\text{g}) | \text{HCl} | \text{H}_2(\text{g}) | \text{Pt}$ (c) $\text{Ag} | \text{AgCl} | \text{Cl}^-(\text{aq}) || \text{Br}^-(\text{aq}) | \text{AgBr} | \text{Ag}$ (d) $\text{Ag} | \text{AgCl} | \text{Cl}^-(\text{aq}) || \text{Br}^-(\text{aq}) | \text{AgBr} | \text{Ag}$

Problem 110**[Narendra Avasthi Q17]**

In electrolyte concentration cell : (a) the electrode material and the solution in both half-cells are composed of the same substances (b) only the concentrations of solutions of the same substances are different (c) $E_{\text{cell}} = 0$ (d) the Nernst equation reduces to $E_{\text{cell}} = \frac{RT}{nF} \ln \frac{C_2}{C_1}$

Problem 111**[Narendra Avasthi Q18]**

The standard electrode potential of a metal-metal ion ($\text{Ag} | \text{Ag}^+$) and metal-sparingly soluble salt anion ($\text{Ag} | \text{AgCl} | \text{Cl}^-$) are related as : (a) $E^\circ_{\text{Ag} | \text{AgCl} | \text{Cl}^-} = E^\circ_{\text{Ag} | \text{Ag}^+} - \frac{RT}{F} \ln K_{\text{sp}}$ (b) $E^\circ_{\text{Ag} | \text{AgCl} | \text{Cl}^-} = E^\circ_{\text{Ag} | \text{Ag}^+} + \frac{RT}{F} \ln K_{\text{sp}}$ (c) $E^\circ_{\text{Ag} | \text{AgCl} | \text{Cl}^-} = E^\circ_{\text{Ag} | \text{Ag}^+} - \frac{RT}{F} \ln [\text{Cl}^-]$ (d) $E^\circ_{\text{Ag} | \text{AgCl} | \text{Cl}^-} = E^\circ_{\text{Ag} | \text{Ag}^+} + \frac{RT}{F} \ln [\text{Cl}^-]$

Problem 112**[Narendra Avasthi Q19]**

Which of the following units is correctly matched? (a) SI units of conductivity is S m^{-1} (b) SI units of molar conductivity is $\text{S cm}^2 \text{mol}^{-1}$ (c) SI unit of conductance is S (d) All of these 20. Which of the following statements is/are correct? (a) The conductance of one cm^2 (or 1 unit) of a solution is called specific conductance (b) Specific conductance increases while molar conductivity decreases on progressive dilution (c) The limiting equivalent conductivity of weak electrolyte cannot be determined exactly by extrapolation of the plot of Λ_{eq} against $\sqrt{C}$ (d) The conductance of metals is due to the movement of free electrons

Problem 113**[Narendra Avasthi Q21]**

Which is/are correct statement ? (a) No corrosion takes place in vacuum (b) corrosion protecting by electroplating (c) During rusting $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ formed (d) In presence of electrolyte, corrosion takes place with greater rate. 22. A dilute solution of KCl was placed between two Pt electrode 10 cm apart across which a potential difference of 10 volt was applied. Which is/are correct statement (Given: molar conductivity of K^+ at infinite dilution is $96.5 \text{ Scm}^2 \text{ mol}^{-1}$). — PAGE 33 — 420

Which is/are correct statement? (a) Ionic mobility of K^+ is $10^{-3} \text{ cm}^2 \text{ sec}^{-1} \text{ volt}^{-1}$. (b) The speed of K^+ is $10^{-3} \text{ cm sec}^{-1}$. (c) Distance travelled by K^+ in $5 \times 10^2 \text{ sec}$ is 5 cm. (d) The potential gradient is 1.0 volt cm^{-1} . (S) (H) (D) HOOHWIO (H) OHNWIO (H) (S): A 1 bar 1 bar pK, $(\text{NH}_4\text{OH}) = 5$; pK, $(\text{CHCOOH}) = 5$; $2.303RT = 0.06 \text{ F}$ Volume of $0.1 \text{ M NH}_4\text{OH}$ in anode half cell = 100 mL, Volume of 0.1 M CHCOOH in cathode half cell = 100 mL Which is/are correct statement? (a) The emf of given cell is 0.48 V. (b) The emf of given cell is 0.36 V when 50 mL, 0.1 M NaOH added to cathode compartment (c) The emf of given cell is 0.36 V when 50 mL 0.1 M HCl added to anode compartment (d) The emf of given cell is 0.192 V when 100 mL 0.1 M NaOH added to anode compartment

Problem 114**[Narendra Avasthi Q24]**

Given : $\text{Pt(s)} | \text{Cl}_2(\text{g}) | \text{Cl}^-(\text{C}_1) | \text{Cl}^-(\text{C}_2) | \text{Cl}_2(\text{g}) | \text{Pt(s)}$ P atm P_2 atm Identify in which of following condition working of cell takes place: (a) $\text{C}_1 > \text{C}_2$ and $\text{P} = \text{P}_2$ (b) $\text{P}_2 > \text{P}$ and $\text{C}_1 = \text{C}_2$ (c) $\text{C}_1 < \text{C}_2$ and $\text{P} = \text{P}_2$ (d) $\text{P}_2 < \text{P}$ and $\text{C}_1 = \text{C}_2$ 25. 1000 mL $1 \text{ M CuSO}_4(\text{aq})$ is electrolysed by 9.65 amp current for 100 sec using Pt-electrode. Which is/are correct statement? (a) Blue colour intensity decreases during electrolysis. (b) Blue colour intensity remains constant if Cu-electrode used. (c) pH of solution is 8 after electrolysis. (d) 28 mL of CH_4 at 1 atm and 273 K required to its combustion by O_2 , liberated during electrolysis. MATCH THE COLUMN Column-I and Column-II contains four entries each. Entries of Column-I are to be matched with some entries of Column-II. One or more than one entries of Column-I may have the matching with the same entries of Column-II

Problem 115**[Narendra Avasthi Q1]**

Column-I (A) Dilute solution of HCl (P) O_2 evolved at anode (B) Dilute solution of NaCl (Q) H_2 evolved at cathode (C) Concentrate solution of NaCl (R) Cl_2 evolved at anode (D) Fairly concentrate solution of AgNO_3 (S) Ag deposition at cathode — PAGE 34 — 421 ELECTROCHEMISTRY

Column-I (A) If SOP of substance is exist between (P) Oxidation of substance is not possible -1.23 to -0.81 V (B) If SOP of substance is exist between (Q) Oxidation possible only in acidic medium -0.81 V to -0.40 V (R) Oxidation possible in any medium (C) If SOP is less than -1.23 V (D) If SOP is greater than -0.40 V (S) Oxidation easily takes place

Column-I (SRP) 3 (P) 0.54 (A) $\text{F}_2 + 2\text{e}^- = 2\text{F}^-$ (Q) 1.09 (B) $\text{Cl}_2 + 2\text{e}^- = 2\text{Cl}^-$ (R) 1.36 (C) $\text{Br}_2 + 2\text{e}^- = 2\text{Br}^-$ (D) $\text{I}_2 + 2\text{e}^- = 2\text{I}^-$ (S) 2.87

Column-I (A) $\text{Pt} | \text{Fe}^{3+}, \text{Fe}^{2+} || \text{Pt}$ (P) Metal-metal ion half-cell (B) $\text{Pt} | \text{H}_2 | \text{H}^+$ (Q) Gas-gas ion half-cell (C) $\text{Pt} | \text{Hg} | \text{Hg}_2^{2+}$ (R) Oxidation-reduction half-cell (D) $\text{Pb} | \text{PbSO}_4 | \text{SO}_4^{2-}$ (S) Metal sparing soluble salt half-cell

Column-I (Property) Column-II (Unit)

(P) $\text{Sm}^{-1} \text{ cm}^{-1}$ (A) Conductance (Q) $\text{s}^{-1} \text{ m}^{-1}$ (B) Conductivity (R) $\text{Sm}^2 \text{ mol}^{-1}$ (C) Molar conductivity (S) S (D) Resistivity

Column-I (Ion) Column-II (Molar Conductivity)

H^+ (P) 350 (V) Na^+ (Q) 50 (B) Li^+ (R) 39 (C) Cs^+ (S) 77

Column-II Column-I Galvanic cell (P) Used in space craft (V) (Q) No transformation of electrical energy (B) Electrolytic cell into chemical energy — PAGE 35 — 422 PROBLEMS IN CHEMISTRY (C) Dead battery (R) Cell reaction is spontaneous (D) Fuel cell (S) Cell reaction is non-spontaneous

ASSERTION-REASON TYPE QUESTIONS

Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Examine the statements carefully and mark the correct answer according to the instructions :- given below : (A) If both the statements are TRUE and STATEMENT-2 is the correct explanation of STATEMENT-1 (B) If both the statements are TRUE but STATEMENT-2 is NOT the correct explanation of STATEMENT-1 (C) If STATEMENT-1 is TRUE and STATEMENT-2 is FALSE (D) If STATEMENT-1 is FALSE and STATEMENT-2 is TRUE

Problem 116**[Narendra Avasthi Q1]**

STATEMENT-1 : E and n is negative for electrolytic cell. STATEMENT-2: ΔG is +ve for electrolytic cell. 2.STATEMENT-1:When 2 faraday of electricity is passed through 0.1 M $\text{H}_2\text{SO}_4(\text{aq})$, 11.2 litre O_2 evolved at STP STATEMENT-2:Molecular weight of oxygen is 32.

Problem 117**[Narendra Avasthi Q3]**

STATEMENT-1: Copper is dissolved at anode and deposited at cathode when Cu electrodes are used and electrolyte is 1 M $\text{CuSO}_4(\text{aq})$ solution. STATEMENT-2:SOP of Cu is less than SOP of water and SRP of Cu is greater than SRP of water. 4.STATEMENT-1:1 coulomb electricity deposits 1 g-equivalent of a substance. STATEMENT-2:1 faraday is charge on 1 mole of electricity. 5.STATEMENT-1:If SRP of substance is -0.3V, it's reduction is possible at cathode. STATEMENT-2:Reduction potential of water exist between 0 to -0.8274V at 25°C. 6.STATEMENT-1:If SRP of substance is -0.5V then reduction of substance is possible only in basic medium. STATEMENT-2:SRP of water is -0.8274V and at reduction potential is zero at pH=7 7.STATEMENT-1:The voltage of mercury cell remains constant for longer period of time. STATEMENT-2:It is because net cell reaction does not involve ions. 8.STATEMENT-1:Lead storage battery is a galvanic cell without salt bridge. STATEMENT-2:A secondary cell is rechargeable cell. 9.STATEMENT-1:The SRP of three metallic ions A,B,C are -0.3,-0.5,0.8 volt respectively, so oxidizing power of ions is $\text{C} > \text{A} > \text{B}$. STATEMENT-2:Higher the SRP higher the oxidizing power. 10.STATEMENT-1:If SOP of substance is less than -1.23 V and over voltage = 0V, then it's oxidation in it's aqueous solution is not possible at 298 K. STATEMENT-2:Standard reduction potential(SRP) of water is +1.23 V. 11.STATEMENT-1:We cannot add the electrode potential in order to get electrode potential of third electrode if no. of moles of electrons exchanged are not same. — PAGE 36 — 423 ELECTROCHEMISTRY STATEMENT-2 : Electrode potential is an extensive property.

Problem 118**[Narendra Avasthi Q12]**

STATEMENT-1: E and $n = 0$ for a chloride ion concentration cell. $R_{\text{th}} [\text{Cl}^-]_{\text{LHS}} nF^\circ [\text{Cl}^-]_{\text{RHS}}$ dE_{cell} 13.STATEMENT-1: $\Delta G > 0$ for a cell reaction then ΔS is positive. IP E STATEMENT-2: $\Delta S = nFT$

Problem 119**[Narendra Avasthi Q14]**

STATEMENT-1: Molar conductivity increases with decrease in concentration for weak electrolytes. STATEMENT-2: No. of ions increases and no. of ions per unit volume decreases due to dilution. and strong electrolytes. STATEMENT-2:No. of ions per unit volume linearly decreases in both electrolytes. SUBJECTIVE PROBLEMS 1.How many faradays are required for reduction of 1mol CH_3NO_2 into CH_3NH_2 ? $\text{O}_2 + 2\text{e}^- \rightarrow 2\text{OH}^-$? 2.What is the equivalent weight of O in the following reaction, $\text{HO} + \dots$ 3.The amount of electricity which releases 2.0g of gold from a gold salt is same as that which dissolves 0.967g of copper anode during the electrolysis of copper sulphate solution. What is the oxidation number of gold in the gold state? (At. wt. of Cu = 63.5; Au = 197) 4.When a molten salt was electrolysed for 5 min with 9.65A current, 0.72g of the metal was deposited. Calculate the Eq.wt. of metal. 5.During the electrolysis of a concentrated brine solution. Calculate the moles of chlorine gas produced by the passage of 4F electricity. 6.Calculate the cell potential (in V) if $\Delta G^\circ = -96.5 \text{ kJ/mol}$ and $n = 1$. 7.If K for the reaction $\text{Cu}^{2+}(\text{aq}) + \text{Sn}^{2+}(\text{aq}) \rightarrow \text{Sn}^{4+}(\text{aq}) + \text{Cu}(\text{s})$ at 25°C is represented as $y \times 10^9$ then find the value of y. (Given : $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34 \text{ V}$) 8.If ΔG° for the half cell $\text{MnO}|\text{MnO}_2$ in an acid solution is xF then find the value of x. (Given : $E^\circ = 1.67 \text{ V}$) 9.If the equilibrium constant for the reaction $\text{Cd}^{2+}(\text{aq}) + 4\text{NH}_3(\text{aq}) \rightarrow \text{Cd}(\text{NH}_3)_4^{2+}(\text{aq})$ is 10^x then find the value of x. — PAGE 37 — 426 PROBLEMS IN CHEMISTRY Passage-1:

Problem 120**[Narendra Avasthi Q2]**

(a) One or More Answers is/are correct (a,b,d) (d)

Problem 121

[Narendra Avasthi Q10]

20.(a,c,d). 21.(a,b,c,d)

Problem 122

[Narendra Avasthi Q25]

(a,b,d) Match the Column 1.A/toP, Q; B/toP, Q; C/toQ,R; D/toP, S 2.A-/toQ; B/toR; C/toP; D/toS, R 3.A/toS; B/toR; C/toQ; D/toP 4.A/toR; B/toQ; C/toP; D/toS 5.A/toS; B/toP; D/toQ 6.A/toP; C/toR; D/toS 7.A/toQ,R; B/toS; C/toQ; D/toP,Q,R Assertion-Reason Type Questions

Problem 123

[Narendra Avasthi Q15]

(C) Subjective Problems

Problem 124

[Narendra Avasthi Q15]

7 — PAGE 38 — Hints and Solutions Level i..

Problem 125

[Narendra Avasthi Q32]

(d) Initial moles of $\text{Cr}^{3+} = 0.25 \times 0.2 = 0.05$; 6.(c) $1F = 96500C$, final moles of $\text{Cr}^{3+} = 0.25 \times 0.1 = 0.025$ 6.023/1023 moles of Cr^{3+} reduced, $1C = 6.24/1018 \times 96500 \times 0.05 = 0.025 \Rightarrow 0.025$;

Problem 126

[Narendra Avasthi Q10]

(c) $0.55 \times 100/60 \text{ M} \times 3 = 96500 \times 96500 \text{ M} = 48.25 \text{ g/mol}$ $t = 75 \text{ sec}$ **Problem 127**

[Narendra Avasthi Q13]

(c) $W \propto I \times t$ $0.195 \times 60/60 \times I$ 34.(a) $W \propto I \times t$ $0.838 \text{ M} \times n = E$ $96500 \times 195 \times 96500 \times 184 \times n \times 96500$ $I = 21.44 \times 40/60/1.0$

Problem 128

[Narendra Avasthi Q15]

(d) When 1 mole of e passed, wt. of substance 96500 deposited $/\text{ton} = 6 \times 38$.(d) no.of eq = no.of F = $1000 \times 6.023/1023 \times 8 = 65.3$ $-/3.34 \text{ g} = \text{eq. wt}$ $122.5 \times 1.81/1023 \text{ W}$ $1930 \times 0.75 \times 40$.(a) $W = 0.105 \text{ gm}$. atomic wt. of metal = $3.34 \times 3 = 10.03 \text{ g}$ 7 96500 8

Problem 129

[Narendra Avasthi Q41]

(a) equivalents of Cl_2 produced 16.(c) 1 mole MnO_4 lose $\times 3 = 10$ mole $1000/9.65/3600$ e; so total charge required = $2 \times 10 = 20 \text{ F}$ $96500 = 360$ moles of $\text{Cl} = 180$ 28.(b) Let t sec be used; No. of faradays = $4 \times t \times 96500/180/0.0821/300$ No. of moles of $\text{H}_2\text{O} = 4$, no. of equivalents P 1.8 of $\text{H}_2\text{O} = 4/2 = 2463 \text{ L}$ $4 \times t$

Problem 130

[Narendra Avasthi Q42]

(c) No. of equivalent of H_2 produced = Eq. of O_2 $96500 = 4/2 \times 30/193/60$ $/t$ $t = 1.93 \times 10^5$ 96500 29.(c) only Au^{3+} , Ag^{+} and Cu^{2+} will deposit at = 3.6 cathode. vol. of O_2 and H_2 produced are Li will not deposit at cathode because SRP of = $3.6/5.6 + 3.6 \times 11.2$ water is -0.8274 V so after Cu^{2+} ; H_2 will = 60.48 litre evolve at cathode.

Problem 134

[Narendra Avasthi Q15]

(a) $\text{HO(D)} + \text{ClO(aq)} \rightarrow \text{HClO(aq)} + 2\text{H}^+(\text{aq}) + 2\text{e}^-$ $2\text{H}^+(\text{aq}) + \text{ClO(aq)} + 2\text{e}^- \rightarrow \text{HClO(aq)} + \text{HO(D)}$
 $2\text{ClO(aq)} \rightarrow \text{HClO}_2(\text{aq}) + \text{ClO(aq)}$ RT E_{cell} = 0.33 - 0.36 = -0.03; $\ln K = \frac{2F}{RT} \Delta E = \frac{2 \times 96485}{8.314 \times 298} \times (-0.03) = -7.16$ or $K = 0.12$
 $2\text{ClOClO} + \text{ClO}_2 \rightleftharpoons 2\text{H}^+ + 2\text{ClO}^-$ $1.36 \times 10^{-1} = 0.1 - 2x$ $(0.1 - 2x)^2 = 10^{-1.16} = 0.069$
 $1.9 \times 10^{-2} = 0.06 [\text{H}^+]_{\text{RHS}}$

Problem 135

[Narendra Avasthi Q16]

(b) E_{cell} = 1 [H]_{HS} J₀ E_{cell} = 0.06 [(pH)_{LHS} - (pH)_{RHS}] (pH)_{HS} = pK_a + log $\frac{[\text{A}^-]}{[\text{HA}]}$ 0.1 (pH)_{RHS}
 = 4 E_{cell} = 0.06 (5-4) = + 0.06 V

Problem 136

[Narendra Avasthi Q17]

(c) 0.06 [Cd²⁺] 10 E_{cell} 2 [H]⁺? E_{cell} = 0, [Cd²⁺][OH⁻]² E_{ll} = 0.03 log $\frac{[\text{Cd}^{2+}]}{[\text{OH}^-]^2}$ > K₂ 0.39 log = 13 K₂ 0.03
 K_p = 10¹³ $\times (10^{-14})^2 = 10^{-15}$ [H⁺]_{RHS} PHUS

Problem 137

[Narendra Avasthi Q18]

(a) 0.06 E_{cell} 2 [H₂SPHJRHS] $\gg [\text{OH}^-] = \sqrt{K_{\text{a}} \times \text{C}}$; [OH⁻] = 10⁻³ or [H⁺]_{HS} = 10⁻¹¹ [H⁺]_{RHS}
 $= \sqrt{K_{\text{a}} \times \text{C}} = 10^{-3}$ 0.06 (10⁻³)² / 10⁻¹¹ E_{cell} 801 $\times 10^{-1} = 0.39 \text{ V}$ 2 (10⁻¹¹)² — PAGE 42 — ELECTROCHEMISTRY 433

Problem 138

[Narendra Avasthi Q19]

(c) HS $\rightarrow$ 2H⁺ + S²⁻ 23. (c) A(BaSO₄) = 2A(BaSO) [H⁺]² [S²⁻] K_a, K_{a2} A_{eq}(BaSO₄) = A_q(Ba²⁺) + A_q(SO₄²⁻) [H₂S] = A_q(BaCl) + A_q(HSO₄⁻) - A_q(HCl) = 10⁻²¹ / 10⁻¹⁴ = (10⁻³)² $\times$ 10⁻¹⁴ A_{eq}(BaSO) = $\frac{[\text{Ba}^{2+}][\text{SO}_4^{2-}]}{[\text{HS}][\text{H}^+]}$ = 10⁻¹⁶ m = 2(x + x² - x) 0.06 [S²⁻] for sparingly soluble salt Ag⁺ Ag log 2 K_{sp} = K / 1000
 10⁻¹⁶ M = 0.80 - 0.06 log or M = $\frac{x}{2 \times 10^{-49} \times (x + x^2 - x^3)} \times 1000$ 0.06 $\times 33 = -0.19 \text{ V}$ 500x = 0.80 - 2(x + x² - x)

Problem 139

[Narendra Avasthi Q20]

(c) Normality of resultant solution 2.5 / 10⁻² 0.1 $\times \text{V} = 0.05 \text{ N}$ (x + x² - x³) V_xV

Problem 140

[Narendra Avasthi Q24]

(a) Conductivity of Na₂SO₄ = 2.6 / 10⁻⁴ 0.0055 / $\text{cm}^2 \text{eq}^{-1}$ $\text{cm}^2 \text{eq}^{-1} \text{Am}(\text{NaSO}_4) = 1000 / 2.6 / 10^{-4} \times 0.05 = 0.001$

Problem 141

[Narendra Avasthi Q21]

(b) Resultant conc. of NaCl = 0.1 $\times \text{V}$ 0.025N = 260 S $\text{cm}^2 \text{mol}^{-1}$ 4V m(SO₂) = A_m(Na₂SO) - 2m(Na) and HCl 0.1 / 2V = 0.05 N 4V = 260 - 2 / 50 = 160 S $\text{cm}^2 \text{mol}^{-1}$ K_{total} = K_{NaCl} + K_{HCl} N(NaCl). A_{eq}(NaCl) Conductivity of CaSO₄ solution K_{solution} 1000 = 7 / 10⁻² - 2.6 / 10⁻⁴ N(HCl). A_{eq}(HCl) = 4.4 $\times 10^{-4}$ S $\text{cm}^2 \text{mol}^{-1}$ 1000 A_m(CaSO) = m(Ca²⁺) + m(SO₂) 0.025 / 110 0.05. A_{eq}(HCl) = 120 + 160 0.018 = 1000 1000 = 280 S $\text{cm}^2 \text{mol}^{-1}$ 18 - 2.75 Solubility C 1000 / K 1000 / 4.4 / 10⁻⁴ eg(HCl) = 305 S $\text{cm}^2 \text{mol}^{-1}$ 0.05 A_m 280

Problem 142**[Narendra Avasthi Q22]**

(b) The molar conductivity of the dissociated 1.57×10^{-3} M form of crotonic acid is $\kappa_{sp} = [\text{Ca}^{2+}][\text{SO}_4^{2-}]_{\text{total}} \Lambda_m(\text{HC}) = \Lambda_m(\text{HCl}) + \Lambda_m(\text{NaC}) - \Lambda_m(\text{NaCl}) = (0.00157)(0.00157 + 0.001) = (426 + 83 - 126) \text{ Q-cm}^2 \text{mol}^{-1} = 4.0 \times 10^{-6} \text{ M}^2 = 383 \text{ Q}/\Omega\text{-cm}^2 \text{mol}^{-1}$ 25. (b) $\kappa = C/\alpha^2$ $0.01x/\alpha^2$ $1/\alpha$ 1.6×10^{-5} = The molar conductivity of HC, $1/\alpha$ $1.6 \times 10^{-5} / 1000 / 0.04$ $\text{C} \cdot 0.001$ $0.01 = 38.3 \text{ Q-cm}^2 \text{mol}^{-1}$ $\Lambda_m = 0.04 / 380 / 10^{-4}$ The degree of dissociation, $\Lambda_m / \Lambda_m(\text{HC})$ $(38.3 \text{ Q-cm}^2 \text{mol}^{-1}) = 15.2 / 10^{-4} \text{ S m}^2 \text{mol}^{-1}$ $(383 \text{ Q-cm}^2 \text{mol}^{-1})$ 0.1 (DH) $V \cdot \kappa = \Lambda_m x C = 15.2 / 10^{-4} / 102 / 10^{-2}$ C/α^2 $(10^{-1})(0.1)^2$ $(:1 \text{ m}^2 = 1000 \text{ litre})$ $1 - 0.1$ $1.11 / 10^{-5} \text{ k} = G \cdot G^*$ and $G^* = 0.01 \text{ m}^{-1}$, — PAGE 43 — 434 PROBLEMS IN CHEMISTRY where $G^* / \Delta H = nF / A$ (dT) $1.52 / 10^{-2} - 218 / 1000 = 2 / 96500 / 300$ $dE / G = 1.52 \text{ S} \cdot 0.01$ $27. (a) x_H = x_O + x_{H_2SO_4}$ $-2 / 96500 / 1.015$ 22.4 $4 / 8.4$ $2x + 2 / n$ $-3.81 \times 10^{-4} \text{ V} \cdot K^{-1}$ 22.4 $2 = 1.5 + 2n$ $= / n$ $= 0.25$ 3. (a) $\Delta S = nF / E = 2 / 96500 / -3.81 / 10^{-4} \text{ dT}$ Level 3 = $-75.53 \text{ J/mol} \cdot K$ Passage-1 Passage-3 1. (c), 2. (a), 3. (d)

Problem 143**[Narendra Avasthi Q2]**

(b) $\kappa = \Lambda_m \cdot C = m \cdot M$ Equivalent of Zn^{2+} produced = 0.1 or 1000 moles of Zn^{2+} 0.1 0.05 $200 / 0.04$ $\kappa = 8 / 102 \text{ S cm}^{-1}$ 2000 +ve charge increases in first compartment so 0.50 due to interaction and maintain electrical $8 / 10^{-3}$ Zn^{2+} 2 neutrality move toward I compartment and NO move towards first or $G = 0.032 \text{ S}$; $V = IR$ I compartment. Solution is always electrically G neutral so charge of 1 Zn^{2+} is neutralized by $I = 5 \times 0.032 = 0.16 \text{ A}$ 2 NO_3^- . $[\text{Zn}^{2+}]$ in first compartment Subjective Problems $0.05 = 1 + 1.025 \text{ M}$ $14. A(\text{HA}) = A(\text{HCl}) + A(\text{NaA}) - A(\text{NaCl})$ 2 Concentration second = $425 + 100 - 125 = 400 \text{ S cm}^2 \text{mol}^{-1}$ Jo NO_3 in $\text{pH} = 4$, $[\text{H}^+] = 10^{-4} = 1/\alpha C$ compartment = $1 - 0.05 = 0.95 \text{ M}$ $200 = 0.5$; In third compartment moles of Cu^{2+} 400 reduced $(\text{Ca}) / \alpha$ $10^{-4} (0.5)$ $K_a = 10^{-4}$; $\text{p}K_a = 4$ $0.05 = 0.025$ $(1/\alpha)$ $(1/\alpha)^2$ $15. E_{2a} / \text{AgCl} / \text{Ag} = E_{\text{Ag} / \text{Ag}}$ 0.0591 $\log K_{sp}$ Relatively -ve charge increased so SO_2 and 1 Na^+ move toward opposite direction to $0.209 = 0.80 + 0.0591 \log K_{sp}$ maintain electrical neutrality 1 $[\text{SO}_2]_{\text{remaining}} = 0.025 > 0.975 \text{ M}$ $K_{sp} = 10^{-1}$; Let solubility of AgCl in $2 \cdot 0.01 \text{ M}$ solution is x Passage-2 $10^{-10} = x(x + 0.01)$ $d(\Delta G)$

Problem 144**[Narendra Avasthi Q1]**

(b) From equation 1 and 2, $\Delta S : x = 10^{-8} \text{ dT}$ $d(-nFE) / dE / \Delta S$. Moles of AgCl dissolved in $\Delta S = dT / dT$ or dT / nF $10 \text{ L} = 10^{-8} / 10 = 10^{-7}$ 2. (b) From equation 2;

Chapter 11

Pearson Advanced Electrochemistry: Complete Exercise Bank

Problem 1

[Pearson Advanced Electrochemistry Q12]

A metal having negative reduction of three metallic cations, X, Y and Z are potential, when dipped in the solution of +0.52, -3.03 and -1.18 V, respectively. its own ions, has a tendency to The order of reducing power of the (a) remain as metal atoms corresponding metals is (b) become electrically positive (a) $Y > Z > X$ (c) become electrically negative (b) $X > Y > Z$ (d) be deposited from the solution (c) $Z > Y > X$ (d) $Z > X > Y$ 13. The calomel electrode is reversible with 9. A gas 'X' at 1 atm is bubbled through respect to a solution containing a mixture of 1 M (a) Hg^{+} -Y and 1 M -Z at 25/°C. If the reduction (b) H^{+} potential of $Z > Y > X$, then, (c) Hg^{2+} (a) Y will oxidize X and not Z (d) Cr (b) Y will oxidize Z and not X 14. Which one of the following does not get (c) Y will oxidize both X and Z oxidized by bromine water? (d) Y will reduce both X and Z (a) Fe^{2+} to Fe^{3+} + 10. S Standard electrode potential data are (b) Cut to Cu^{2+} useful for understanding the suitability (c) Mn^{2+} to MnO_4 of an oxidant in a redox titration. Some (d) Sn^{2+} to Sn^{4+} half-cell reactions and their standard potentials are given below:

Problem 2

[Pearson Advanced Electrochemistry Q15]

In salt bridge, normally KCl is used because $MnO(aq) + 8H^{+}(aq) + 5e^{-} \rightarrow Mn^{2+}(aq) + 4H_2O(l)$ $E^{\circ} = 1.51$ V (a) it is a strong electrolyte. (b) it is good conductor of electricity. $CrO_2(aq) + 14H^{+}(aq) + 6e^{-} \rightarrow 2Cr^{+}(aq) + 7H_2O(l)$ $E^{\circ} = 1.38$ V ionic mobility. $Fe^{2+}(aq) + e^{-} \rightarrow Fe^{+}(aq)$ $E^{\circ} = 0.77$ V (d) it is an ionic compound. $Cl_2(g) + 2e^{-} \rightarrow 2Cl^{-}(aq)$ $E^{\circ} = 1.40$ V

Problem 3

[Pearson Advanced Electrochemistry Q16]

By how much would the oxidizing power of MnO/Mn^{2+} couple change if the H^{+} Identify the only incorrect statement ions concentration is decreased 100 times regarding the quantitative estimation of at 25/°C? aqueous $Fe(NO)$ (a) increases by 189 mV (a) MnO can be used in aqueous HCl (b) decreases by 189 mV (b) $Cr_2O_7^{2-}$ can be used in aqueous HCl (c) will increase by 19 mV (c) MnO can be used in aqueous H_2SO_4 (d) will decrease by 19 mV (d) $Cr_2O_7^{2-}$ can be used in aqueous H_2SO_4 17. The solution of $CuSO_4$, in which copper

Problem 4

[Pearson Advanced Electrochemistry Q11]

The decreasing order of standard electrode rod is immersed, is diluted to 10 times. potential of Mg, K, Ba and Ca, is The reduction electrode potential (a) K, Ca, Ba, Mg (a) increases by 0.0295 V (b) Ba, Ca, K, Mg (b) decreases by 0.0295 V (c) Ca, Mg, K, Ba (c) increases by 0.059 V (d) Mg, Ca, Ba, K (d) decreases by 0.059 V — PAGE 3 — Electrochemistry 8.3

Problem 5**[Pearson Advanced Electrochemistry Q18]**

The standard reduction potential of $2\text{H}^+/\text{H}_2$ oxygen in acidic solution is +1.23 V. What of Li^+/Li , Ba^{2+}/Ba , Na^+/Na and Mg^{2+}/Mg is the standard reduction potential of are -3.05, -2.73, -2.71 and -2.37 V, oxygen in basic solution? respectively. Which is the strongest (a) +0.404 V (b) -0.404 V reducing agent? (c) +2.056 V (d) -2.056 V (a) Li (b) Ba 19. The standard reduction potentials of (c) Na (d) Mg Cu^{2+}/Cu and $\text{Cu}^{2+}/\text{Cu}^+$ are 0.337 V Some standard electrode potentials are and 0.153 V, respectively. The standard given: electrode potential of Cu^+/Cu half-cell is $\text{Fe}^{2+} + 2\text{e}^-/\text{Fe}$; $E^\circ = -0.440$ V (a) 0.184 V $\text{Fe}^{3+} + 3\text{e}^-/\text{Fe}$; $E^\circ = -0.036$ V (b) 0.827 V (c) 0.521 V The standard electrode potential for: (d) 0.490 V $\text{Fe}^{3+} + \text{e}^-/\text{Fe}^{2+}$, is

Problem 6**[Pearson Advanced Electrochemistry Q20]**

The electrode potential of hydrogen (a) -0.476 V (b) -0.404 V (c) +0.988 V electrode in neutral solution and 298 K is (d) +0.772 V (a) -0.413 V (b) zero

Problem 7**[Pearson Advanced Electrochemistry Q26]**

The dissociation constant for CH_3COOH (c) -0.826 V (d) +0.413 V is 1.8×10^{-4} at 298 K. The electrode

Problem 8**[Pearson Advanced Electrochemistry Q21]**

Electrode potential will be more for potential for the half-cell: $\text{Pt}|\text{H}_2$ hydrogen electrode at pH (at the same (1 bar) 0.5 M- CH_3COOH , at 298 K temperature) is ($\log 2 = 0.3$; $\log 3 = 0.48$; $2.303RT/F = 0.06$) (a) 4 (c) 2 (d) 5 (a) -0.3024 V (b) -0.1512 V 22. Saturated solution of KNO_3 is used to (c) +0.3024 V make salt bridge because (d) +0.1512 V (a) velocity of K^+ is greater than that of NO_3^- : At 25°C, the solubility product of CuCl_2 (b) velocity of NO_3^- is greater than that of K^+ The value of K_{sp} is ($\log 2 = 0.3$; (c) velocities of both K^+ and NO_3^- are $2.303 RT/F = 0.06$) nearly the same (d) KNO_3 is highly soluble in water (a) -0.274 V (b) -0.402 V (c) +0.53 V (d) +0.402 V 23. The standard reduction potentials of $\text{Pt}|\text{Cr}, \text{O}_2$, Cr^{3+}/Cr ; $\text{Pt}|\text{MnO}_2, \text{Mn}^{2+}$;

Problem 9**[Pearson Advanced Electrochemistry Q28]**

The standard reduction potential of $\text{Pt}|\text{Ce}^{4+}, \text{Ce}^{3+}$ in the presence of acid are oxygen in acidic solution is 1.23 V (O_2 1.33 V, 1.51 V and 1.61 V, respectively, at $4\text{H}^+ + 4\text{e}^-/\text{O}_2$). The standard 25°C. The decreasing order of oxidizing reduction potential of oxygen in basic power is solution is ($2.303RT/F = 0.06$) (a) $\text{CrO}_2^- > \text{MnO}_2 > \text{Ce}^{4+}$ (a) -1.23 V (b) $\text{MnO}_2 > \text{CrO}_2^- > \text{Ce}^{4+}$ (b) -0.39 V (c) $\text{Ce}^{4+} > \text{MnO}_2 > \text{CrO}_2^-$ (c) +0.39 V (d) $\text{MnO}_2 > \text{Ce}^{4+} > \text{CrO}_2^-$ (d) +2.07 V — PAGE 4 — 8.4 Chapter 8 29. The standard potentials of $\text{MnO}_2/\text{Mn}^{2+}$ 30. The following reactions represent the and $\text{MnO}_2/\text{Mn}^{2+}$ electrodes in acid reduction of IO_3^- ion into I^- ion in acidic solution are 1.51 and 1.23 V, respectively. and basic medium. Predict in which medium Standard electrode potential for IO_3^- ion will act as a better oxidizing agent? the electrode, $\text{MnO}_4^-/\text{MnO}_2$ in acid $\text{IO}_3^- + 6\text{H}^+ + 6\text{e}^-/\text{I}^- + 3\text{H}_2\text{O}$; solution is $E^\circ = +0.907$ V (a) +1.697 V $\text{IO}_3^- + 3\text{H}_2\text{O} + 6\text{e}^-/\text{I}^- + 6\text{OH}^-$; (b) +5.09 V $E^\circ = +0.260$ V (c) +0.28 V (a) Acid medium (b) Basic medium (d) +1.37 V (c) Equally in both (d) Not predictable Galvanic Cell 31. The correct cell diagram for the following

Problem 10**[Pearson Advanced Electrochemistry Q34]**

The cell reaction for the given cell is reaction and E° for the cell is spontaneous if Pt, Cl (P, atm) | Cl⁻ | 2AgBr(s) + H₂(g) / 2Ag(s) + 2H⁺ + 2Br⁻ (P, atm), Pt E° AgBr/Ag = +0.10 V. (a) $P > P^\circ$ (a) (Pt) H₂ | H⁺ | Br⁻ | AgBr | Br (Pt); (b) $P < P^\circ$ E° = 0.10V (c) $P = P^\circ$ (b) (Pt) H₂ | H⁺ | Br⁻ | AgBr | Br (Pt); (d) $P = 1$ atm E° = -0.10V (c) (Pt) Br₂ | AgBr | Br⁻ | H⁺ | H₂ (Pt); Which one of the following statements E° = 0.10V is incorrect regarding an electrochemical (d) (Pt) Br₂ | AgBr | Br⁻ | H⁺ | H₂ (Pt); E° = -0.10V cell? (a) The electrode on which oxidation takes place is called anode. measurement of EMF of a half-cell using a reference electrode and a salt bridge, when (b) Anode is the negative pole. the salt bridge is removed, the voltage (c) The direction of the current is same (a) remains the same as that of the direction of flow of (b) increases to maximum electrons. (c) decreases half the value (d) The flow of current is partly due to (d) drops to zero flow of electrons and partly due to 33. After some time, the voltage of an flow of ions. electrochemical cell becomes zero. This is because 36. When an electric current is drawn from a (a) their electrode potential becomes galvanic cell zero. (a) EMF suddenly increases. (b) their reduction potential become (b) EMF gradually increases and attains equal but have opposite sign. a maximum value. (c) their reduction potential become equal and have the same sign. (c) EMF decreases and finally falls to (d) the ions of the electrolyte in the salt zero. bridge stop moving. (d) EMF must remain constant. — PAGE 5 — Electrochemistry 8.5 37. Identification of anode and cathode in an 44. Use of lithium metal as an electrode in electrochemical cell is made by the use of high energy density batteries is due to (a) Galvanometer (a) lithium is the lightest element. (b) Salt bridge (b) lithium has the highest oxidation (c) Voltmeter potential. (d) Potentiometer (c) lithium is quite reactive. 38. For the cell Zn | Zn²⁺ || Cu²⁺ | Cu, if the (d) lithium does not corrode readily. concentration of both, Zn²⁺ and Cu²⁺ ions 45. A depolarizer used in dry cell batteries is are doubled, the EMF of the cell (a) NH₄Cl (b) MnO₂ (a) doubles (c) KOH (d) Na₂PO₄ (b) reduces to half 46. Which is correct about fuel cells? (c) remains same (d) becomes zero (a) Cell continuously run as long as fuels are supplied. 39. The standard EMF of a galvanic cell can (b) These are more efficient and free from pollution. (a) the size of the electrode (c) These are used to provide power and (b) the pH of the solution drinking water to astronauts in space (c) the amount of metal in the anode programme. (d) the E° values of the half-cells (d) All of these 47. When a lead storage battery is discharged

Problem 11**[Pearson Advanced Electrochemistry Q40]**

The value of equilibrium constant for a feasible cell reaction must be (a) SO₂ is evolved (b) lead sulphate is consumed (a) < 1 (b) Zero (c) lead is formed (c) = 1 (d) > 1 (d) sulphuric acid is consumed 41. If the cell reaction is spontaneous then 48. For a cell reaction involving two- (a) ΔG° = -ve electron change, the standard EMF of (b) E° = -ve the cell is found to be 0.295 V at 25°C. (c) E° = +ve The equilibrium constant of the reaction (d) ΔG° = -ve at 25°C will be (a) 10^{10} (b) 10^{-10} 42. Which of the following pair of metals, (c) 10^{-2} (d) 10^{10} when coupled, will give maximum EMF 49. The E° for Ag(s) | AgI (satd) || Ag⁺ for a voltaic cell? (0.10 M) | Ag (s) is +0.413 V. What is the (a) Fe and Cu value of K of AgI? (b) Pb and Au (c) Cu and Au (a) 10^{-8} (b) 10^{-7} (c) 10^{-14} (d) 10^{-16} (d) Ca and Cu 50. Assuming that hydrogen behaves as an 43. When a lead storage battery is charged ideal gas, what is the EMF of the cell at (a) PbO dissolves. 25°C if P = 600 mm and P = 420 mm: (b) the lead electrode becomes coated Pt | H₂ (P) | HCl | H₂ (P) | Pt? [Given: with lead sulphate. $2.303RT/F = 0.06$, $\log 7 = 0.85$] (c) sulphuric acid is regenerated. (a) -0.0045 V (b) -0.0V (d) the amount of acid decreases. (c) +0.0045 V (d) +0.0015 V — PAGE 6 — 8.6 Chapter 8 51. For the electrochemical cell, M | M⁺ || I⁻ | I₂ For a reaction A(s) + 2B⁺ / 2A²⁺ + B(s); K_c = X | X, $E^\circ_M = 0.44$ V and $E^\circ_{X/X} = 0.33$ V. has been found to be 10. The EMF of From this data, one can deduce that the cell is (a) M + X / 2M⁺ + X is the spontaneous (a) 0.354V (b) 0.708 V reaction (c) 0.534V (d) 0.453 V (b) M + X / 2M⁺ + X is the spontaneous

Problem 12**[Pearson Advanced Electrochemistry Q58]**

An electrochemical cell is set up as reaction follows: (c) $E_c = 0.77$ V (d) $E = -0.77$ V $\text{Pt}|\text{H}(\text{s}) (1 \text{ atm})||0.001 \text{ M HCl} || 0.1 \text{ M HA} |\text{H}(\text{s}) (1 \text{ atm}) | \text{Pt}$

Problem 13**[Pearson Advanced Electrochemistry Q52]**

The EMF of the cell: $\text{Zn}|\text{Zn}^{2+}(0.01 \text{ M})||$ EMF of this cell is zero because ($pK_a \text{ Fe}^{2+} (0.001 \text{ M})|\text{Fe}$ at 298 K is 0.2905 V, of $\text{HA} = 5$) then the value of equilibrium constant (a) molar concentrations of acids are for the cell reaction is different 0.32 0.32 (a) 0.0295 (b) 100.0295 (b) Temperature is constant 0.26 (c) pH of two solutions are same 0.32 (c) 100.0295 (d) 100.0591 (d) both are standard hydrogen electrodes

Problem 14**[Pearson Advanced Electrochemistry Q59]**

.The 53. For the cell reaction: $4\text{Br}^- + \text{O}_2 + 4\text{H}^+ \rightarrow 2\text{H}_2(\text{g}) + 2\text{AgCl}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{Cl}^-(\text{aq}) + 2\text{Ag}(\text{s})$ occurs in $2\text{Br}^- + 2\text{H}^+$; $E^\circ_{\text{cell}} = 0.18$ V. The value of the galvanic cell ($\log K$) at 298 K is $[2.303RT/F = 0.06]$ (a) $\text{Ag}|\text{AgCl}(\text{s})|\text{KCl}(\text{aq})|\text{AgNO}_3(\text{aq})|$ (a) 12 (b) 6 (c) 18 (d) 3 Ag (b) $\text{Pt}|\text{H}_2(\text{g})|\text{HCl}(\text{aq})|\text{AgNO}_3(\text{aq})|\text{Ag}$ 54. The standard EMF for the cell reaction: (c) $\text{Pt}|\text{H}_2(\text{g})|\text{HCl}(\text{aq})|\text{AgCl}(\text{s})|\text{Ag}$ $\text{Zn}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu}(\text{s})$ is (d) $\text{Pt}|\text{H}_2(\text{g})|\text{KCl}(\text{aq})|\text{AgCl}(\text{s})|\text{Ag}$ 1.10 volts at 25°C. The EMF of the cell 60. The standard reduction potentials in reaction when 0.1 M Cu^{2+} and 0.1 M Zn^{2+} acidic conditions are 0.77V and 0.53 V, solutions are used at 25°C is respectively, for $\text{Fe}^{2+}|\text{Fe}$ and $\text{I}^-|\text{I}_2$ (a) 1.10 v (b) 1.041 V couples. The equilibrium constant for (c) -1.10 V (d) -1.041 V the reaction: $2\text{Fe}^{2+} + \text{I}_2 \rightarrow 2\text{Fe}^{3+} + 2\text{I}^-$ is 55. For the cell: $\text{Ni}|\text{Ni}^{2+}||\text{Cu}^{2+}|\text{Cu}$; ($2.303RT/F = 0.06$) $E^\circ_{\text{cell}} = 0.77$ V. By which of the following (a) 2 times 108 (b) 108 (c) 10^8 (d) 10^{-8} activity, E will increase? (a) On decreasing $[\text{Ni}^{2+}]$ 61. The following electrochemical cell has (b) On decreasing $[\text{Cu}^{2+}]$ been set up: $\text{Pt}(\text{s})|\text{Fe}^{2+}, \text{Fe}^{3+}(a=1)||\text{Ce}^{4+}, \text{Ce}^{3+}(a=1)|\text{Pt}(\text{s})$; $E^\circ_{\text{cell}}(\text{Fe}^{3+}/\text{Fe}^{2+}) = 0.77$ V; (d) On increasing mass of Cu electrode $E^\circ_{\text{cell}}(\text{Ce}^{4+}/\text{Ce}^{3+}) = 1.61$ V. If an ammeter is connected between the two platinum 56. The standard EMF of a Daniel cell at 298 K, predict the direction of flow of K^+ ions. When the concentration of ZnSO_4 is 1.0 M and that of CuSO_4 is 0.01 M, the decrease with time? EMF becomes E , at 298 K. The correct (a) Ce electrode to Fe electrode, decrease relationship between E , and E° is (b) Ce electrode to Fe electrode, increase (a) $E = E^\circ$ (b) $E = 0$ (c) Fe electrode to Ce electrode, decrease (c) $E > E^\circ$ (d) $E < E^\circ$ (d) Fe electrode to Ce electrode, increase — PAGE 7 — Electrochemistry 8.7 62. For the reaction: $\text{H}_2(1 \text{ bar}) + 2\text{AgCl}(\text{s})$

Problem 15**[Pearson Advanced Electrochemistry Q64]**

The standard potentials of OCl^-/Cl^- and $2\text{Ag}(\text{S}) + 2\text{H}^+ (0.1 \text{ M}) + 2\text{Cl}^- (0.1 \text{ M})$; Cl^-/Cl_2 are 0.94 V and 1.36 V, respectively. $\Delta G^\circ = -48,250 \text{ J}$ at 25°C . The EMF of cell The E° value of OCl^-/Cl_2 , will be in which the given reaction takes place is (a) 3.24 V (b) -0.42 V (a) 0.25 V (c) -2.30 V (d) 0.52 V (b) 0.37 V 65. From the following E° values for the (c) 0.13 V half-cells: (d) 0.049 V (i) $\text{D} \rightarrow \text{D}^{2+} + 2\text{e}^-$; $E^\circ = -1.5 \text{ V}$ 63. A cell contains two hydrogen electrodes. The negative electrode is in contact (ii) $\text{B} + \text{e}^- \rightarrow \text{B}^-$; $E^\circ = -0.5 \text{ V}$ with a solution of 10^{-4} M hydrogen (i) $\text{A}^{2+} \rightarrow \text{A} + 2\text{e}^-$; $E^\circ = 1.5 \text{ V}$ ions. The EMF of the cell is 0.118 V at (iv) $\text{C}^{2+} + \text{e}^- \rightarrow \text{C}^+$; $E^\circ = +0.5 \text{ V}$ 25°C . The concentration of hydrogen ions at the positive electrode is Which combination of two half-cells (a) 10^{-4} M would result in a cell with largest (b) 10^{-2} M potential? (c) 10^{-4} M (a) i and ii (b) i and iv (d) 10^{-8} M (c) ii and iv (d) i and iv Electrolysis 66. Which of the following statements does 69. If mercury is used as a cathode during the not differentiate between electrochemical electrolysis of an aqueous NaCl solution, cell and electrolytic cell? the ions discharged at cathode are (a) Spontaneous or non-spontaneous (a) H^+ nature of the chemical process. (b) Na^+ (b) Chemical reactions occurring at the (c) OH^- electrodes (d) Cl^- (c) Positive and negative nature of anode. (d) EMF measurement. 67. The electrode through which electrons 70. Electrochemical equivalent is more for enter the electrolytic solution is (a) Hydrogen (a) cathode (b) Silver (b) anode (c) Copper (c) may be anode or cathode (d) Zinc (d) both, anode and cathode Which process occurs in the electrolysis 71. The electrolytic bath used in gold plating of an aqueous solution of nickel chloride of copper articles contains at nickel anode? (a) Molten gold (a) $\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^-$ (b) Copper sulphate(aq) (b) $\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}$ (c) $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ (c) AuCl_4^- (aq) (d) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ (d) $\text{AuCl}_4^- + \text{NaCN}$ (aq) — PAGE 8 — 8.8 Chapter 8

Problem 16**[Pearson Advanced Electrochemistry Q72]**

Copper can be deposited from acidified 77. A certain current liberated 0.50 g of copper sulphate and alkaline cuprous hydrogen in 2 h. How many grams of cyanide. If the same current is passed for copper can be liberated by the same a definite time current flowing for the same time in a (a) the amount of copper deposited from copper sulphate solution? ($\text{Cu} = 63.5$) acidic copper sulphate will be higher (a) 12.7 g (b) 15.88 g (b) the amount of copper deposited (c) 31.75 g (d) 63.5 g from alkaline cuprous cyanide will be 78. A current of 3.7 A is passed for 6 h higher (c) the same amount of copper will be between nickel electrodes in 0.501 of 2 M solution of $\text{Ni}(\text{NO})$. The molarity of deposited Ni^{2+} at the end of electrolysis is (d) copper will not deposit in either case (a) 1.172 M (b) 0.172 M 73. In the electrolytic cell, flow of electrons is (c) 0.586 M (d) 2 M from 79. The current efficiency of an (a) Cathode to anode in solution electrodeposition of copper metal in (b) Cathode to anode through external which 9.8 g of copper is deposited by a supply current of 3 A for 10000 s, from aqueous (c) Cathode to anode through internal copper sulphate solution, is about supply (a) 60/ (b) 99/ (d) Anode to cathode through internal (c) 92/ (d) 75/ supply 80. On passing electricity through dilute 74. Electrolytic cell is used to convert H_2SO_4 solution, the mass of substances (a) Chemical energy to electrical energy liberated at the cathode and anode are in (b) Electrical energy to chemical energy the ratio of (c) Chemical 1 energy to mechanical (a) 1:8 (b) 8:1 energy (c) 1:32 (d) 1:16 (d) Electrical energy to mechanical 81. The electrochemical equivalents of two energy substances are E_1 and E_2 . The current that must pass to deposit the same amount at 75. Faraday's law of electrolysis fails when the cathodes in the same time must be in (a) temperature is increased the ratio of (b) inert electrodes are used (a) $E_1:E_2$ (b) $E_2:E_1$ (c) a mixture of electrolytes is used (c) $(E_1 - E_2):E_1$ (d) $E_1:(E_1 - E_2)$ (d) in none of these cases 82. T The same quantity of electricity is passed 76. Using same quantity of current, which through one molar solution of H_2SO_4 among Na, Mg and Al is deposited more and one molar solution of HCl . The (by mass) during electrolysis of their amount of hydrogen evolved from H_2SO_4 molten salts? as compared to that from HCl is (a) Na (a) the same (b) Mg (b) twice as such (c) Al (c) one half as such (d) All in same (d) dependent on size of electrode — PAGE 9 — Electrochemistry 8.9 83. In the electrolysis of acidified AgNO_3 ; series. The mass ratio of iron deposited at solution using Pt-electrodes, the anode cathodes in the two cells will be reaction is (a) 3:1 (b) 2:3 (a) $2\text{NO}/\text{to } 2\text{NO}_2 + \text{O}_2 + 2\text{e}^-$ (c) 1:1 (d) 3:2 (b) $\text{NO}/\text{to } \text{NO} + \text{O}_2 + \text{e}^-$ 88. A galvanic cell is set up from a zinc bar 2 (c) $2\text{H}^+/\text{to } 4\text{H}^+ + \text{O}_2 + 4\text{e}^-$ weighing 100 g and 1.0 L of 1.0 M copper (d) $\text{Pt}/\text{to } \text{Pt}^{2+} + 3\text{e}^-$ sulphate solution. How long would the cell run if it is assumed to deliver a steady

Problem 17**[Pearson Advanced Electrochemistry Q84]**

Two platinum electrodes were immersed in a solution of CuSO_4 and an electric current of 1.0 A was passed through the solution. (a) 53.6 h (b) 26.8 h (c) 81.97 h (d) 40.99 h After some time, it was found that the colour of CuSO_4 disappeared with the evolution of a gas at the electrode. The colourless solution absorbs 6×10^4 electrons. The number of equivalents of the ion is (a) platinum sulphate (a) 0.10 (b) 0.01 (b) copper sulphate (c) 0.001 (d) 0.0001 (c) copper hydroxide 90 . In the lead storage battery, the anode reaction is $\text{Pb(s)} + \text{HSO}_4^- + \text{H}^+ \rightarrow \text{PbSO}_4(\text{s}) + \text{H}_2\text{O} + 2\text{e}^-$. How many grams of Pb will be used up to deliver 1 A in being electrolysed using inert electrodes. for 100 h ? ($\text{Pb} = 208$) The values of standard electrode potential (a) 776 g (b) 388 g are: $\text{Ag}|\text{Ag}^+ = 0.80 \text{ V}$, $\text{Hg}_2^{2+}|\text{Hg} = 0.79 \text{ V}$, (c) 194 g (d) 0.1 g $\text{Cu}^{2+}|\text{Cu} = 0.34 \text{ V}$, $\text{Mg}^{2+}|\text{Mg} = -2.37 \text{ V}$. With increasing voltage, the sequence of deposition of metals on the cathode will be (a) Ag, Hg, Cu, Mg (b) Mg, Cu, Hg, Ag (c) Ag, Mg, Cu (d) Cu, Hg, Ag (a) 0.96 g (b) 60 g (c) 86.1 g (d) 60.96 g During the electrolysis of an aqueous salt solution, the pH in the space near the cathode will be (a) 59.04 (b) 60.96 (c) 59.04 (d) 60.96 The quantity of electricity required for one of the electrode was increased and the reduction of $1 \text{ mole of FeO to Fe}$ is the other one was decreased. The salt (a) 1 F (b) 0.33 F (c) 3 F (d) 6 F (a) NaCl (very dilute) (b) ZnCl_2 (c) NaCl (Conc.) (d) $\text{Cu(NO}_3)_2$ Three faradays of electricity is passed through molten Al_2O_3 , aqueous solutions (a) $\text{Cu(NO}_3)_2$ (b) CuSO_4 (c) NaCl (d) $\text{Cu(NO}_3)_2$ The amounts of Al , Cu and Na deposited at the cathodes will be in the molar ratio of acidified ferrous chloride and another (a) $1:2:3$ (b) $3:2:1$ (c) $1:1.5:3$ (d) $6:3:2$ — PAGE 10 — 8.10 Chapter 8 94. Electrolysis of a solution of HSO_4^- ions produces SO_2 . Assuming a Daniel cell for 96.5 min , the cathode will efficiency, what current should be gain in weight by ($\text{Cu} = 63.5$, $\text{Zn} = 65.4$) employed to achieve a production rate of (a) 1.905 g (b) 1.962 g (c) 3.81 g (d) 3.924 g (a) 71.5 A (b) 35.7 A (c) 53.0 A (d) 143 A at the rate of $2.8 \text{ ml per second}$ during (a) 53.0 A (b) 143 A (c) 53.0 A (d) 143 A The number of Faradays required to electrolysis of acidulated water is produce 1 g-atom of Mg from MgCl_2 (a) 48.25 A/s (b) 1 (c) 2 (d) 24.12 A/s (a) 0.5 (b) 4 (c) 96.5 A/s

Problem 18**[Pearson Advanced Electrochemistry Q96]**

Passage of $96,500 \text{ coulomb}$ of electricity (a) 0.0048 A/s liberates L of O_2 at 273°C and 103 atm . Sodium amalgam is prepared by electrolysis of aqueous NaCl using 10 g (a) 5.6 (b) 16.8 mercury as cathode. How many Faraday (c) 22.4 (d) 11.2 of electricity is required to prepare 18.7 g of Na-amalgam , by weight, with a current efficiency of 50% of AgSO_4 and platinum electrodes. A (a) 0.1 F (b) 0.2 F current is passed until 1.6 g of O has (c) 0.05 F (d) 0.16 F been liberated at anode. The amount of silver deposited at cathode would be (a) 108.0 g (b) 1.6 g (c) 10.8 g (d) 21.6 g element A (atomic mass = 112) and element B (atomic mass = 27) form chlorides. Solutions of these chlorides are electrolysed separately and it is found that when the same quantity of electricity is passed through the electrolyte, 3.0 g of A was deposited, 5.6 g of B was deposited. The electro-valency of the valency of B is 3 . The valency of A is metal cation in the electrolyte is (a) 1 (b) 2 (c) $+4$ (d) $+3$ (a) 1 (b) 2 (c) $+2$ (d) -4 99. How many electrons flow when a current of

Problem 19**[Pearson Advanced Electrochemistry Q105]**

The same current was passed successively 5 A is passed through a solution for 200 s? through solution of zinc-ammonium sulphate and nickel-ammonium sulphate (a) 6.022/10²³ rendered alkaline with ammonia. The (b) 6.24/10²¹ weights of zinc and nickel deposited in (c) 6.024/10²¹ a certain time were found to be 22.89 g (d) 6.022/10²⁰ and 20.55 g, respectively. Given that the

Problem 20**[Pearson Advanced Electrochemistry Q100]**

The current of 9.65 A flowing for 10 min chemical equivalent weight of zinc is 32.7, deposits 3.0 g of a metal. The equivalent what is the chemical equivalent weight of the metal is nickel? (a) 10 (b) 30 (a) 58.71 (b) 29.36 (c) 50 (d) 96.5 (c) 14.39 (d) 36.42 — PAGE 11 — Electrochemistry 8.11 Conductance 106. Which of the following solutions have

Problem 21**[Pearson Advanced Electrochemistry Q112]**

The specific conductance of AgCl highest resistance? solution in water was determined to be (a) 1N-NaCl 1.8/10⁻³ Q-cm- at 298 K. The molar (b) 0.05N-NaCl conductances at infinite dilution, of Ag⁺ (c) 2N-NaCl and Cl are 67.9 and 82.1 Q-cm²mol⁻¹, (d) 0.1N-NaCl respectively. What is the solubility of AgCl in water? 107. Variation of molar conductance of an (a) 1.2/10⁻⁸ M electrolytic solution with temperature is (b) 1.44/10⁻¹ M that it (a) increases with increase of temperature (c) 1.2/10⁻⁶ M (b) decreases increase of (d) 1.44/10⁻¹⁶ M with temperature 113. Equivalence conductance at infinite (c) first increases then decreases dilution of NHCl, NaOH and NaCl (d) is not affected by temperature are 129.8, 217.4 and 108.9 Q-cm²mol⁻¹ 108. Which pure substance will not conduct mol⁻¹, respectively. If the equivalent electricity? conductance of 0.01 N solution of (a) Molten NaCl NHOH is 9.532 Q-cm²mol⁻¹, then the (b) Molten KOH degree of dissociation of NHOH at this (c) Liquefied HCl temperature is (d) Liquid Hg (a) 0.04/(b) 2.1/109. The correct order of molar conductance (c) 4.0/(d) 44.7/at infinite dilution of LiCl, NaCl and KCl is

Problem 22**[Pearson Advanced Electrochemistry Q114]**

The resistance of 1 M - CH₃COOH (a) LiCl > NaCl > KCl solution is 250 Q, when measured in a (b) KCl > NaCl > LiCl cell of cell constant 125 m⁻¹. The molar (c) NaCl > KCl > LiCl conductivity, in Q-cm²mol⁻¹ is (d) LiCl > KCl > NaCl (a) 5.0/10⁻⁴ (b) 500

Problem 23**[Pearson Advanced Electrochemistry Q110]**

The molar conductance of a strong (c) 2/10⁻³ (d) 200 electrolyte at infinite dilution 115. How does the electrical conductivity of (a) tends to a finite value, which is above 20 ml of 0.2 M-MgSO₄ change when that at higher concentration 0.5 M -Ba(OH)₂, solution is gradually (b) tends to a finite value, which is below added in it, to excess? that at higher concentration (c) tends to zero (a) decreases continuously (b) increases continuously (d) tends to a finite value, which is equal (c) increases and then decreases to that at high concentration (d) decreases and then increases

Problem 24**[Pearson Advanced Electrochemistry Q111]**

The best conductor of electricity is a 0.1 M solution of

Problem 25**[Pearson Advanced Electrochemistry Q116]**

The equivalent conductivity (in $\Omega^{-1}\text{cm}^2\text{eq}^{-1}$) of 1.0 M H_2SO_4 solution of (b) Sulphuric acid specific conductance $2.6 \times 10^{-4} \text{ cm}^{-1}$, is (c) Acetic acid (a) 1.3×10^2 (b) 6.5×10^1 (d) Propanoic acid (c) 1.3×10^{-1} (d) 2.6×10^2 — PAGE 12 — 8.12 Chapter 8 117. The molar conductance of a 0.01 M CH_3COOH solution of acetic acid was found to be and 0.015 M NaOH solutions are $16.30 \Omega^{-1}\text{cm}^2\text{mol}^{-1}$ at 25°C . The ionic mixed together. What would be the molar conductances of hydrogen and acetate ions conductivity of mixture if conductivity at infinite dilution are 349.8 and $40.9 \Omega^{-1}\text{cm}^2\text{mol}^{-1}$ of CH_3COONa is $6.3 \times 10^{-4} \text{ S cm}^{-1}$, respectively, at the same (a) $0.84 \text{ S cm}^2\text{mol}^{-1}$ temperature. What percentage of acetic (b) $8.4 \text{ S cm}^2\text{mol}^{-1}$ acid is dissociated at this concentration? (c) $84 \text{ S cm}^2\text{mol}^{-1}$ (a) 0.04172/(b) 4.172/(d) 42 $\text{S cm}^2\text{mol}^{-1}$ (c) 41.72/(d) 0.4172/

Problem 26**[Pearson Advanced Electrochemistry Q123]**

Calculate Λ_m (in $\Omega^{-1}\text{cm}^2\text{mol}^{-1}$) for SrCl_2 118. The distance between two electrodes of a cell at 25°C , from the following data: cell is 2.5 cm and area of each electrode is 5 cm^2 . The cell constant is 0.25 M^{-1} . (a) 0.5 m^{-1} (b) $12.5 \text{ cm}^2 / \Delta$ (in $\Omega^{-1}\text{cm}^2\text{mol}^{-1}$) 260 250 (c) 2.0 cm (d) 50 m^{-1} (a) 270 (b) 265

Problem 27**[Pearson Advanced Electrochemistry Q119]**

The molar conductivity of NH_4Cl , OH^- (c) 240 (d) 275 and Cl^- at infinite dilution is 150, 200 and $75 \Omega^{-1}\text{cm}^2\text{mol}^{-1}$, respectively. If the

Problem 28**[Pearson Advanced Electrochemistry Q124]**

The resistance of a solution A is 50Ω and molar conductivity of a $0.01 \text{ M-NH}_4\text{OH}$ that of solution B is 100Ω , both solutions solution is $22 \Omega^{-1}\text{cm}^2\text{mol}^{-1}$, then its being taken in the same conductivity cell. degree of dissociation is If equal volumes of solution A and B (a) 0.146 (b) 0.063 are mixed, what will be the resistance of (c) 0.080 (d) 0.293 the mixture using the same cell? Assume that there is no increase in the degree of

Problem 29**[Pearson Advanced Electrochemistry Q120]**

Calculate the ionic product of water at dissociation of A and B on mixing. 25°C from the following data: (a) 150Ω Conductivity of water = $5.5 \times 10^{-6} \text{ S cm}^{-1}$ (b) 75Ω /times 10-mho m^{-1} (c) 33.33Ω $\kappa = 0.035 \text{ mho m}^2\text{mol}^{-1}$ (d) 66.67Ω /textasciicircum()/circumH = $0.020 \text{ mho m}^2\text{mol}^{-1}$

Problem 30**[Pearson Advanced Electrochemistry Q125]**

In a conductivity cell, the two platinum (a) $2 \times 10^{-14} \text{ M}^2$ electrodes, each of area 10 cm^2 are fixed (b) $1 \times 10^{-2} \text{ M}^2$ 1.5 cm apart. The cell contained 0.05 N (c) $1 \times 10^{-8} \text{ M}^2$ solution of a salt. If the two electrodes are (d) $1 \times 10^{-14} \text{ M}^2$ just half dipped into the solution which has a resistance of 50Ω , the equivalent 121. Calculate K_a of acetic acid if its 0.05 M conductance of the salt solution, in $\Omega^{-1}\text{cm}^2\text{mol}^{-1}$ solution has molar conductivity of $7.814 \text{ cm}^2\text{eq}^{-1}$, is $1 \times 10^{-10} \Omega^{-1}\text{cm}^2\text{mol}^{-1}$ at 25°C . Given: Λ° for (a) 120 $\text{CH}_3\text{COOH} = 3.907 \times 10^{-2} \Omega^{-1}\text{cm}^2\text{mol}^{-1}$ (b) 60 (a) $2 \times 10^{-10} \text{ s}$ (b) $1.8 \times 10^{-10} \text{ s}$ (c) 240 (c) 4×10^{-4} (d) 0.02 (d) 3000 — PAGE 13 — Electrochemistry 8.13 Answer Keys-Exercise I Electrode Potential

Problem 31**[Pearson Advanced Electrochemistry Q99]**

(b) 100. (c) 101. (a) 102. (a) 103. (b) 104. (b) 105. (b) Conductance 106. (b) 107. (a) 108. (c) 109. (b) 110. (a) 111. (b) 112. (c) 113. (c) 114. (a) 115. (d)

Problem 32**[Pearson Advanced Electrochemistry Q116]**

(a) 117. (b) 118. (d) 119. (c) 120. (b) 121. (a) 122. (c) 123. (a) 124. (d) 125. (a) — PAGE 14 — 8.14 Chapter 8 EXERCISE II (JEE ADVANCED) Section A (Only one correct) 1. A student made the following observations 4. The standard reduction potential for the in the laboratory: reactions: $\text{Ag}^+ + e^- \rightarrow \text{Ag}$ and $\text{Ag}(\text{NH}_3)_2^+ + e^- \rightarrow \text{Ag} + 2\text{NH}_3$, are +0.79 V and +0.37 V, (I) Clean copper metal did not react respectively. From these values and the with 1 M-Pb(NO₃) solution. Nernst equation, what should be K, for the (II) Clean lead metal dissolves in 1 M Ag(NH₃)⁺ ion? [Given: $2.303RT/F = 0.06$] AgNO₃ solution and crystals of (a) 1.0×10^{-7} Ag metal appeared. (b) 1.0×10^{-2} (III) Clean silver metal did not react with (c) 2.15×10^{-19} 1 M-Cu(NO₃) solution. (d) 4.64×10^{-20} The order of decreasing reducing 5. The overall formation constant for the character of the three metals is reaction of 6 mole of CN⁻ with cobalt (I) (a) Cu > Pb > Ag is 1×10^7 . What is the formation constant (b) Cu > Ag > Pb for the reaction of 6 moles of CN⁻ with (c) Pb > Cu > Ag cobalt (III)? Given that (d) Pb > Ag > Cu $\text{Co}(\text{CN})_2 + e^- \rightarrow \text{Co}(\text{CN})^-$; 2. We have an oxidation-reduction $E^\circ = -0.83$ V system: $[\text{Fe}(\text{CN})_6]^{2-} + e^- = [\text{Fe}(\text{CN})_6]^-$; $\text{Co}^{2+} + e^- \rightarrow \text{Co}^+$; $E^\circ = +1.81$ V $E^\circ = +0.36$ V. The ratio of concentrations of oxidized and reduced form at which and $2.303RT/F = 0.06$. the potential of the system becomes (a) 1.0×10^6 3 (b) 1.0×10^2 0.24 V, is [Given: $2.303 RT/F = 0.06$] (c) 1.0×10^{-25} (d) 1.0×10^{-63} (a) 2:1 (b) 1:2 (c) 1:20 (d) 1:100 6. For the process: $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$; $\log[\text{Cu}^{2+}]$ vs. E graph is shown in the figure, where 3. The standard reduction potential for the OA = 0.34 V. The electrode potential of process: $[\text{Co}(\text{H}_2\text{O})_6]^{2+} + e^- \rightarrow [\text{Co}(\text{H}_2\text{O})_6]^+$ the half-cell of Cu|Cu²⁺ (0.1 M) will be is 1.8 V. The standard reduction potential $[2.303RT/F = 0.06]$ for the process: $[\text{Co}(\text{NH}_3)_6]^{2+} + e^- \rightarrow [\text{Co}(\text{NH}_3)_6]^+$ is 0.1 V. Which of the complex, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ or $[\text{Co}(\text{NH}_3)_6]^{2+}$ can be oxidized to the corresponding cobalt (III) complex, by oxygen, in basic A medium, under standard condition? (a) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ $\log[\text{Cu}^{2+}]$ (b) $[\text{Co}(\text{NH}_3)_6]^{2+}$ (c) both (a) -0.31 V (b) +0.31 V (d) none of these (c) -0.37 V (d) +0.37 V — PAGE 15 — Electrochemistry 8.15 7. Tell which of the following statements 12. The EMF of the cell: Hg(l)|HgCl₂(s), accurately describes the effect of adding CN⁻ KCl sol. (1.0 N)|Quinohydrone|Pt, is to the cathode of a cell with a cell reaction: $\text{Cd} + 2\text{Ag}^+ \rightarrow 2\text{Ag} + \text{Cd}^{2+}$, $E^\circ = 1.2$ V quinohydrone solution, the potential of (a) E increases because Cd(CN)₂ forms the normal calomel electrode is 0.279 V (b) E decreases because Cd(CN)₂ forms and E for the quinohydrone electrode (c) E increases because Ag(CN)₂⁻ forms is 0.699 V, both at the same temperature. (d) E decreases because Ag(CN)₂⁻ forms $[2.303RT/F = 0.06]$ (a) 3.5 (b) 7.0 8. For an electrochemical cell Zn|Zn²⁺ (C) (c) 1.85 M Cu²⁺ (C M)| Cu, the decrease in (d) -3.5 free energy at a given temperature is a 13. The potential (EMF) of a cell consisting of an anode of silver in 0.10 M-AgNO₃ (a) In C (b) In C solution and a cathode of Pt immersed in (c) In CC, (d) In C/C in a solution of 1.6 M - Cr₂O₇²⁻, 9. Select the correct option if it is known $E^\circ = 0.80$ V and $E^\circ_{\text{Co}^{2+}/\text{Co}} = 1.33$ V that $K(\text{AgCl}) > K(\text{AgBr}) > K(\text{AgI})$ (a) $E^\circ_{\text{AgI}/\text{Ag}} = 0.46$ V (b) 0.60 V (c) 0.53 V (d) -0.17 V (b) $E^\circ_{\text{AgI}/\text{Ag}} = 0.53$ V 14. The EMF of the cell: Zn-Hg (C, M)| Zn²⁺(aq)| Zn-Hg (CM) at 25°C, if the (d) $E^\circ_{\text{AgI}/\text{Ag}} = 0.53$ V concentrations of the zinc amalgams are: C, = 10 g per 100 g of mercury, C₂ = 1 g 10. The EMF of a galvanic cell composed of two hydrogen electrodes is 272 mV. (a) 0.059 V (b) 0.0295 V What is the pH of the solution in which (c) 0.59 V (d) 0.295 V the anode is immersed if the cathode is in contact with a solution of pH=3?

Problem 33**[Pearson Advanced Electrochemistry Q15]**

What is the equilibrium constant of the reaction: $2\text{Fe}^{2+} + \text{Au}^+ \rightarrow 2\text{Fe}^{3+} + \text{Au}$? (a) 3 (b) 6.7 Given $E^\circ_{\text{Au}^+/\text{Au}} = 1.68$ V, $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 1.50$ V, (c) 7.6 (d) 1.6 $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.75$ V and $2.303RT/F = 0.06$. 11. At what pH does the potential (EMF) for the disproportionation of chlorine (a) 1×10^{22} change from a negative value to a positive (b) 1×10^{22} value, assuming 1.0 bar pressure and (c) 1×10^{-1} 1.0 M concentration for all species except (d) 1×10^{-72} hydrogen ion? Given:

Problem 34**[Pearson Advanced Electrochemistry Q16]**

What is the EMF of the cell: $\text{Pt}, \text{H}_2 \text{ Cl} + 2\text{e}^- / \text{to} 2\text{Cl}^-; E^\circ = 1.36 \text{ V (1 atm)} | \text{CH}_3\text{COOH (0.1 M)} | \text{H}^+ \text{ (0.01 M)} | \text{H}_2\text{O} | \text{Pt}$? Given: $E^\circ = 1.63 \text{ V}$. $K_{\text{for CH}_3\text{COOH}} = 1.8 \times 10^{-5}$, $K_{\text{for NHOH}} = 1.8 \times 10^{-2}$, $2.303RT/F = 0.06$, $[2.303RT/F = 0.06] \log 1.8 = 0.25$ (a) 4.5 (b) 1.5 (a) 0.465 V (b) -0.465 V (c) 2.25 (d) 9.0 (c) -0.2325 V (d) -0.93 V — PAGE 16 — 8.16 Chapter 8 17. Two electrochemical cells are assembled -0.018 V, respectively, at 25°C . If the in which the following reactions occur: potential of normal calomel electrode is $\text{V}^{2+} + \text{VO}_2 + 2\text{H}^+ / \text{to} 2\text{V} + \text{H}_2\text{O}$; $E^\circ = -0.28 \text{ V}$, the EMF of the Daniel cell is = 0.616 V (a) 1.065 V (b) 1.101 V $\text{V}^{3+} + \text{Ag}^+ + \text{H}_2\text{O} / \text{to} \text{VO}_2 + \text{Ag(s)} + 2\text{H}^+$; (c) 0.803 V (d) 0.262 V $E^\circ_{\text{cal}} = 0.439 \text{ V}$ 22. A hydrogen electrode placed in a buffer If $E^\circ_{\text{H}^+/\text{H}_2} = 0.799 \text{ V}$, what is E° solution of CH_3COONa and CH_3COOH in the ratio's $x:y$ and $y:x$ has electrode (a) -0.256 V (b) +0.256 V potential values E , and E , volts, (c) +1.854 V (d) -1.854 V respectively, at 25°C . The pK_a value of 18. The EMF of cell: $\text{H(g)} | \text{Buffer} | \text{Normal acetic acid is calomel electrode, is } 0.70 \text{ V at } 25^\circ\text{C}$, when (a) $(E_+ - E_-) / 0.118$ barometric pressure is 760 mm. What is (b) $(E_+ - E_-) / 0.118$ the pH of the buffer solution? $E^\circ_{\text{cal}} = 0.439 \text{ V}$ (c) $-(E_+ + E_-) / 0.118 = 0.28 \text{ V}$. $[2.303RT/F = 0.06]$ (d) $(E_+ - E_-) / 0.118$ (a) 3.5 23. The standard reduction potential at 25°C (b) 7.0 of the reaction $2\text{H}_2\text{O} + 2\text{e}^- = \text{H}_2 + 2\text{OH}^-$ (c) tending to zero is -0.84 V. Calculate equilibrium constant (d) tending to 14.0 for the reaction: $2\text{H}_2\text{O} = \text{H}_2 + \text{O}_2$ at 25°C . $(2.303RT/F = 0.06)$ 19. What is the solubility product of a saturated solution of Ag_2CrO_4 in water (a) 10^{-14} (b) 10^{-14} at 298 K if the EMF of the cell: $\text{Ag} | \text{Ag}^+ \text{ (c) } 10^2 \text{ (d) } 10^{-2} \text{ (satd. Ag}_2\text{CrO}_4) | \text{Ag}^+ \text{ (0.1 M)} | \text{Ag}$ is 0.162

Problem 35**[Pearson Advanced Electrochemistry Q24]**

The voltage of the cell given below is V at 298 K? $[2.303RT/F = 0.06, \log 2 = 0.3]$ 0.61 V. (a) 2.0×10^{-4} (b) 3.2×10^{-4} (c) 8.0×10^{-12} (d) 4.0×10^{-12} $\text{Pt} | \text{H}_2 (1 \text{ bar}) | \text{NaHSO}_3 (0.4 \text{ M}), \text{Na}_2\text{SO}_3 (6.4 \times 10^{-2} \text{ M}) | \text{H}^+ | \text{Zn}^{2+} (0.4 \text{ M}) | \text{Zn}$ 20. A $\text{Ti} | \text{TiCl}_2$ couple was prepared by saturating If $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$, Calculate K_{a2} of 0.10 M H_2SO_4 with TiBr and allowing H_2SO_4 ($2.303RT/F = 0.06$) Ti^+ ions form the insoluble bromide to equilibrate. This couple was observed to (a) 3.2×10^{-4} (b) 3.2×10^{-2} have a potential -0.444 V with respect (c) 3.2×10^{-3} (d) 6.4×10^{-7} to $\text{Pb}^{2+} | \text{Pb}$ couple in which Pb^{2+} was 25. B Estimate the cell potential of a Daniel cell 0.10 M. What is the K_{sp} of TiBr . [Given: having 1.0 M Zn^{2+} and originally having $E^\circ_{\text{Pb}^{2+}/\text{Pb}} = -0.126 \text{ V}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = -0.336 \text{ V}$, 1.0 M Cu^{2+} after sufficient ammonia has $\log 2.5 = 0.4$, $2.303 RT/F = 0.06$] been added to the cathode compartment (a) 4.0×10^{-6} (b) 2.5×10^{-4} to make the NH_3 concentration 2.0 M. Given: $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ V}$, $K_{\text{for Cu(NH}_3)_4^{2+}} = 1 \times 10^{12}$. 21. The EMFs of the cell obtained by (a) 1.10 V combining separately Zn and Cu (b) 0.704 V electrodes of a Daniel cell with normal (c) 0.396 V calomel electrodes are 1.083 V and (d) 1.496 V — PAGE 17 — Electrochemistry 8.17 26. The minimum mass of NaOH required measured between the electrodes at 25°C to be added in RHS to consume all the H^+ present in RHS of the cell of EMF, = $10^{-4} + 0.70 \text{ V}$ at 25°C , before its use. (a) 0.92 V (b) 0.32 V $\text{Zn} | \text{Zn}^{2+} (0.01 \text{ M}) | \text{HCl (V=500 ml)} | \text{H}^+ (1 \text{ bar}) | \text{Pt}$ 31. Although aluminium is above hydrogen Given: $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$; $2.303RT/F$ in the electrochemical series, it is stable in = 0.06 air and water. Why? (a) 2.0 g (b) 4.0 g (a) Aluminium is non-reactive metal. (c) 0.2 g (d) 0.4 g (b) A layer of oxide is formed at metal surface, which prevents further reaction. 27. The EMF of the cell: Ag, AgCl in 0.1 M - KCl satd. NH_4NO_3 0.1 M - AgNO_3 (c) The reaction is thermodynamically unfavourable, i.e., ΔG is not negative. ΔG is 0.42 V at 25°C . 0.1 M - KCl is 50% (d) Kinetically, the reaction has very low dissociation and 0.1 M - AgNO_3 is 40% dissociated. The solubility product of activation energy and hence do not AgCl is ($2.303RT/F = 0.06$) occur. (a) 1.0×10^{-10} 32. When silver chloride is dissolved in a (b) 2.0×10^{-9} large excess of ammonia, practically all (c) 1.0×10^{-9} silver ion can be assumed to exist in form (d) 2.0×10^{-10} of a single ionic species $[\text{Ag(NH}_3)_2]^+$. Compute the values of x and y using the

Problem 36**[Pearson Advanced Electrochemistry Q28]**

On the basis of information available two following cells: from the reaction Cell I: $\text{Ag} \mid 4.0 \times 10^{-4} \text{ M} - \text{AgCl}, 1 \text{ M}$ $\text{Al} + 3\text{O}_2 \rightarrow 2\text{AlO}$; $\Delta G = -965 \text{ kJ/mol}$ - NH, II $4 \times 10^{-2} \text{ M} - \text{AgCl}, 1 \text{ M}$ of O_2 - NH | Ag; $E_{\text{cell}} = 0.118 \text{ V}$ at 298 K. The minimum EMF required to carry out Cell III: $\text{Ag} \mid 3 \times 10^{-2} \text{ M} - \text{AgCl}, 1 \text{ M}$ electrolysis of AlO is - NH, II $3 \times 10^{-4} \text{ M} - \text{AgCl}, 0.1 \text{ M}$ (a) 0.833 V (b) 2.5 V - NH | Ag; $E_{\text{cell}} = 0.118 \text{ V}$ at 298 K. (c) 5.0 V (d) 1.67 V (a) $x=1, y=2$ (b) $x=1, y=4$ 29. The theoretical efficiency of a hypothetical (c) $x=2, y=4$ cell is about 84% (d) $x=1, y=6$ following reaction $\text{A(s)} + \text{B}_2(\text{aq}) \rightarrow \text{A}_2(\text{aq}) + \text{B(s)}$; 33. When metallic copper is shaken with a $\Delta H = -285 \text{ kJ}$, solution of a copper salt, the reaction $\text{Cu} + \text{Cu}^{2+} + 2\text{Cu}^+$ proceeds. When then, the standard EMF of the cell is equilibrium is established at 298 K, (a) 1.10 V (b) 1.24 V $[\text{Cu}^{2+}/[\text{Cu}^+] = 1.667 \times 10^{-3} \text{ M}]$. If the (c) 2.48 V (d) 2.20 V standard potential of the Cu^{2+}/Cu half-

Problem 37**[Pearson Advanced Electrochemistry Q30]**

To a beaker containing 0.1 M - HCl, a cell is $+0.3376 \text{ V}$, what is the standard little pure solid AgCl is added. Both potential of $\text{Cu} \mid \text{Cu}$ half-cell? (Given: a silver and a hydrogen electrode (PH, $2.303RT/F = 0.06$, $\log 2 = 0.3$, $\log 3 = 0.48$) = 1.0 bar) are then placed in the solution. (a) -0.3732 V (b) 0.6752 V What is the approximate value of EMF (c) 0.5242 V (d) 0.151 V — PAGE 18 — 8.18/cdotChapter8

Problem 38**[Pearson Advanced Electrochemistry Q34]**

Zinc granules are added in excess to a Use the following data to calculate ΔG 500 ml of 1.0 M nickel nitrate solution for the above reaction. at 25°C until the equilibrium is reached. $\text{K}[\text{Au}(\text{CN})]$ If the standard reduction potential of $\text{X Zn} \mid \text{Zn}$ and $\text{Ni} \mid \text{Ni}$ are -0.75 V and $\text{O}_2 + 2\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$; $E^\circ = +0.41 \text{ V}$ -0.24 V, respectively, the concentration $\text{Au}^{2+} + 3\text{e}^- \rightarrow \text{Au}$; $E^\circ = +1.50 \text{ V}$ of Ni^{2+} in solution at equilibrium is $\text{Au}^{2+} + 2\text{e}^- \rightarrow \text{Au}^+$; $E^\circ = +1.40 \text{ V}$ ($2.303RT/F = 0.06$) (a) $-RT \ln X + 1.29 \text{ F}$ (a) $1.0 \times 10^{-2} \text{ M}$ (b) $-RT \ln X - 1.29 \text{ F}$ (b) $1.0 \times 10^{-2} \text{ M}$ (c) $+RT \ln X + 2.11 \text{ F}$ (c) $5 \times 10^{-1} \text{ M}$ (d) $-RT \ln X - 2.11 \text{ F}$ (d) $2 \times 10^{-1} \text{ M}$ 35. The EMF for the cell: $\text{Ag(s)} \mid \text{AgCl(s)} \mid$

Problem 39**[Pearson Advanced Electrochemistry Q39]**

F₂ gas cannot be obtained by the KCl(0.2 M)|| KBr (0.001 M)|AgBr(s) | electrolysis of any aqueous fluoride salt Ag(s) at 25°C is $K(\text{AgCl}) = 2.0 \times 10^{-10}$; because $K(\text{AgBr}) = 4.0 \times 10^{-13}$, $2.303RT/F = 0.06$, (a) F⁻ is the strongest oxidizing agent $\log 2 = 0.3$ (b) Fe easily combines with water (a) 0.024 V (c) F⁻, readily combines with the (b) -0.024 V electrodes (c) -0.24 V (d) F⁻ can never be oxidized (d) -0.012 V 40. A volume of 100 ml of a buffer of 1 M 36. Two students use same stock solution of NH₃, and 1 M -NH₄⁺ is placed in two ZnSO₄ but different solutions of CuSO₄ half-cells connected by a salt bridge. A The EMF of one cell is 0.03 V higher current of 1.5 A is passed through the cell than the other. The concentration of for 20 min. If electrolysis of water takes CuSO₄ in the cell with higher EMF value place only and the electrode reactions are: is 0.5 M. The concentration of CuSO₄ in the other cell is $(2.303RT/F = 0.06)$ Right: $2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^- \rightarrow 4\text{OH}^-$ (a) 0.05 M (b) 5.0 M and Left: $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + \text{O}_2 + 4\text{e}^-$, (c) 0.5 M (d) 0.005 M then, the pH of the 37. Two weak acid solutions HA, and HA (a) right electrode will increase each with the same concentration and (b) left electrode will increase having pK_a values 3 and 5 are placed in (c) both electrode will increase contact with hydrogen electrodes (1 atm, (d) both electrode will decrease 25°C) and are interconnected through a 41. When molten ICl₃ is electrolysed using salt bridge. EMF of the cell is platinum electrodes (a) 0.0295 V (b) 0.118 V (c) 0.0885 V (d) 0.059 V (a) I₂ is evolved at cathode and Cl₂ at anode 38. Consider the reaction of extraction of (b) Cl₂ is evolved at cathode and I at gold from its ore: anode $\text{Au(s)} + 2\text{CN}^- (\text{aq}) \rightarrow \text{Au(CN)}_2^- (\text{aq}) + 2\text{e}^-$ (c) I₂ is evolved at cathode and both I₂ O₂(g) + 4 and Cl₂ at anode $\text{HO(I)} \rightarrow \text{Au(CN)}_2^- (\text{aq}) + \text{OH}^- (\text{aq})$ (d) electrolysis does not take place — PAGE 19 — Electrochemistry 8.19 42. When electric current is passed through 47. Which of the following substances: Na, a cell having an electrolyte, positive ions Hg, S, Pt and graphite can be used as move towards the cathode and negative electrodes in electrolytic cells having ions towards the anode. If the cathode is aqueous solution? pulled out of the solution, then the (a) Na, Pt and graphite (a) positive and negative ions will move (b) Na and Hg towards the anode (c) Hg, Pt and graphite (b) positive ions will start moving (d) Na and S towards the anode, the negative ions

Problem 40**[Pearson Advanced Electrochemistry Q48]**

3. An ammeter and a copper voltameter will stop moving are connected in series in an electric (c) negative ions will continue to move towards the anode, the positive ions circuit through which a constant direct current flows. The ammeter shows 0.5 A. will start moving randomly If 0.635 g of Cu is deposited in 0.965 h, (d) positive and negative ions will start what is the percentage error of ammeter? moving randomly (Cu = 63.5)

Problem 41**[Pearson Advanced Electrochemistry Q43]**

When 10⁻⁶ M -HCl is electrolysed (a) 5 / (b) 10 / (a) O₂ is produced at the anode / (b) H₂ is produced at the anode 49. The same current is passed through (c) Cl₂ is produced at the anode acidulated water and stannous chloride (d) Cl₂ and O₂ are produced at the anode solution. What volume of dry detonating gas at 0°C and 1 atm is evolved from 44. A dilute aqueous solution of Na₂SO₄ is electrolysed using platinum electrodes. water, when 1.20 g of tin is deposited The products at the anode and cathode from the other solution? (Sn = 120) are, respectively, (a) 112 ml (b) 336 ml (a) O₂, H₂ (c) 224 ml (d) 672 ml (b) S, O₂-, Na 50. The preparation of LiOH by the (c) O₂, Na electrolysis of a 35 / (d) S, O₂, H using a platinum anode led to a current efficiency of 80 / 45. In the electrolysis of a fused salt, the mass was formed by the passage of 2.5 A for of substance deposited on an electrode 4825 s? will not depend on (a) 1.92 g (b) 2.40 g (a) temperature of bath (c) 0.96 g (d) 0.672 g (b) current intensity (c) time of electrolysis

Problem 42**[Pearson Advanced Electrochemistry Q51]**

Assuming that copper contains only (d)electrochemical equivalent of ions iron, silver and gold as impurities. After passage of 12.4 A for 4825 s, the 46. When an electric current is passed through mass of anode decreased by 20.00 g an aqueous solution of the following, the and the cathode increased by 19.05 g. concentrations of cation as well as anion The percentages of iron and copper in will not change for (assume constant the original sample are, respectively, volume of the electrolytic solution) (Cu=63.5, Fe=56) (a) CsCl (a) 4.75/(b) KNO₃, (b) 2.8/(c) AgNO₃; (c) 95.25/(d) HCl (d) 95.25/— PAGE 20 — 8.20 Chapter 8

Problem 43**[Pearson Advanced Electrochemistry Q52]**

After electrolysis of an aqueous sodium 57. Most of the copper used to make wire has chloride solution, it was found that the been electrically refined by depositing it solution is being neutralized by 60 ml N from copper salts solution (divalent) on -HCl solution. During the same period to a cathode. What is the cost of electrical of electrolysis, 3.18 g of copper was energy required per kg of copper if the deposited in a copper voltameter in series. cost of electricity is Rs 4.00 per kWh What is the percentage of the theoretical and the cell operates at 0.33 V? The yield of sodium hydroxide obtained? electrochemical equivalent of copper is (Cu=63.6) 0.00033 g/coulomb. (a) 40/(b) 60/(a) Rs 11.11 (b) Rs 5.55 (c) 30/(d) 80/(c) Rs 2.22 (d) Rs 1.11 53. A lead storage battery has initially 200g

Problem 44**[Pearson Advanced Electrochemistry Q58]**

Electrolysis of an acetate solution of lead and 200 g of PbO₂ plus excess produces ethane according to the H₂SO₄. Theoretically, how long could this reaction: cell deliver a current of 10.0 A, without 2CH₃COO⁻ to CH₄(g) + 2CO₂(g) + 2e⁻ recharging, if it were possible to operate it so that the reaction goes to completion? What total volume of ethane and CO₂ (Pb = 208) would be produced at 0/°C and 1 atm, if a current of 0.5 A is passed through the (a) 16083.33 s (b) 8041.67 s solution for 482.5 min? Assume current (c) 44.68 s (d) 18557.7 s efficiency 80/54. A volume of 100 ml of 0.6 N-CuSO₄ (a) 1.344 L (b) 2.688 L solution is electrolysed between two (c) 4.032 L (d) 1.792 L platinum electrodes till the concentration

Problem 45**[Pearson Advanced Electrochemistry Q59]**

To perform an analysis of a mixture of in the residual liquid is 0.1 N, when a steady current of 5.0 A is used. How long metal ions by electro-deposition, the second metal to be deposited must not should the current be passed to get the above change? being plating out until the concentration ratio of the second to the first is about 10/circ. (a) 965 s (b) 96500 s What must be the minimum difference (c) 1930 s (d) 482.5 s in standard potential of the two metals 55. A constant current flowed for 2h through which form dipositive ions in order for a potassium iodide solution, oxidizing such an analysis to be feasible? the iodide ion to iodine. At the end of the (a) 0.177 V (b) 0.354 V experiment, the iodine was titrated with (c) 0.708V (d) 0.088V 72 ml of 1.0 M-Na₂SO₃ solution. What

Problem 46**[Pearson Advanced Electrochemistry Q60]**

During the electrolysis of 0.1 M-CuSO₄ was the average rate of current flow, in solution using copper electrodes, a ampere? depletion of Cu²⁺ occurs near the cathode (a) 0.965 A (b) 1.93 A with a corresponding excess near the (c) 0.483 A (d) 0.0965 A anode, owing to inefficient stirring of 56. Anthracene, C₁₄H₁₀, can be oxidized to the solution. If the local concentration anthraquinone, CHO. What weight of Cu²⁺ near the anode and cathode are, anthraquinone can be produced by the respectively, 0.12 M and 0.08 M, the passage of a current of 1 A of 40 min if back EMF developed at 298 K is (log 1.5 the current efficiency is 96.5% = 0.18, $2.303RT/F = 0.06$) (a) 0.862 g (b) 0.832 g (a) 0.33 V (b) 5.4 mV (c) 0.624 g (d) 0.738 g (c) 2.7 mV (d) 10.8 mV — PAGE 21 — Electrochemistry 8.21 61. The following galvanic cell: The first reaction is an electrolytic Zn|Zn(NO₃)(100 ml, 1 M)|Cu(NO₃) reaction and second is steam distillation. What amount of current would have to be (100 ml, 1 M)| Cu used in first reaction to produce enough was operated as an electrolytic cell as Cu intermediate to yield 102 g pure H₂O per as the anode and Zn as the cathode. A hour. Assume current efficiency 50% / current of 0.4825 A was passed for 10 h (a) 643.33 A and then the cell was allowed to function (b) 321.67 A as galvanic cell. What would be the final (c) 160.83 A EMF of the cell at 25°C? Assume that the only electrode reactions occurring (d) 1286.67 A were those involving Cu|Cu²⁺ and Zn|Zn²⁺. Given: $E^\circ_{Cu^{2+}/Cu} = +0.34$ V, $E^\circ_{Zn^{2+}/Zn} = -0.76$ V, $2.303RT/F = 0.06$, log 1.9 = 0.28) a solution containing arsenious acid, (a) 1.10 V (b) 1.0616 V HAsO₂, KI and a small amount of starch (c) 1.1384 V (d) 1.1768 V is electrolysed. The electrolysis produces free I⁻ from I₂ and the I immediately

Problem 47**[Pearson Advanced Electrochemistry Q62]**

At the Nangal fertilizer plant in Punjab, oxidizes the arsenious acid to hydrogen hydrogen is produced by the electrolysis arsenate ion, HAsO₄²⁻ of water. The hydrogen is used for the production of ammonia and nitric acid $I + HAsO_4 + HO \rightarrow 2I + HAsO_2$ (by the oxidation of ammonia). If the + 4H⁺ average production of ammonium nitrate is 1152 kg/day, the daily consumption of When the oxidation of arsenic is complete, the free iodine combines with electricity (in A / day) is the starch to give a deep blue colour. If (a) 96,500 (b) 48,250 during a particular run, it takes 96.5 s (c) 32,166.67 (d) 24,125 for a current of 10.0 mA to give an end

Problem 48**[Pearson Advanced Electrochemistry Q63]**

Lactic acid, HCH₂COOH, produced in 1 g point (indicated by the blue colour), sample of muscle tissue was titrated how many g of arsenic are present in the using phenolphthalein as indicator solution. (As = 75) against OH⁻ ions which were obtained by (a) 6.3×10^{-4} g the electrolysis of water. As soon as OH⁻ (b) 1.26×10^{-3} g ions are produced, they react with lactic (c) 3.15×10^{-3} g acid and at complete neutralization, (d) 6.3×10^{-3} g immediately a pink colour is noticed. If electrolysis was made for 1158 s using 50.0 mA current to reach the end point,

Problem 49**[Pearson Advanced Electrochemistry Q66]**

Electrolysis of a solution of MnSO_4 in aqueous sulphuric acid is a method what was the percentage of lactic acid in for the preparation of MnO as per the muscle tissue? reaction: $\text{Mn}^{2+}(\text{aq}) + 2\text{H}_2\text{O} \rightarrow \text{MnO}_2(\text{s})$ (a) 5.4/(b) 2.7/+ $2\text{H}(\text{aq}) + \text{H}(\text{g})$. Passing a current of (c) 10.8/(d) 0.054/19.3 A for 2 h gives only 52.2 g of MnO_2 . HO_2 can be prepared by successive The current efficiency is (Mn=55) reactions: (a) 8.33/ $2\text{NHHSO}_4 \rightarrow \text{H}_2\text{O} + (\text{NH})\text{SO}$ (b) 83.33/ $(\text{NH})\text{SO} + 2\text{H}_2\text{O} \rightarrow 2\text{NHHSO}_4$ (c) 41.67/+ H_2O_2 (d) 100/— PAGE 22 — 8.22 Chapter 8 67. Copper sulphate solution (250 ml) was $\text{Pb} + \text{SO}_2 \rightarrow \text{PbSO}_4 + 2\text{e}^-$ electrolysed using a platinum anode and $\text{PbO}_2 + 4\text{H}^+ + \text{SO}_2 + 2\text{e}^- \rightarrow \text{PbSO}_4$ a copper cathode. A constant current + $2\text{H}_2\text{O}$ of 2 mA was passed for 19.3 min. It was found that after electrolysis the (a) 965 (b) 482.5 absorbance of the solution was reduced (c) 1930 (d) 241.25 to 50/71. A cell whose resistance, when filled with concentration of copper sulphate in the solution to begin with. 0.1 M-KCl is 200 Q, is measured to be 6400 Q, when filled with 0.003 M -NaCl (a) 9.6/ $\times 10^{-10}$ M solution. What is the molar conductance (b) 4.8/ $\times 10^{-10}$ M of NaCl solution, in $\text{Q}^{-1}\text{cm}^2\text{mol}^{-1}$ if the (c) 2.4/ $\times 10^{-10}$ s M molar conductance of 0.1 M -KCl is (d) 1.2/ $\times 10^{-10}$ M $120 \text{ Q}^{-1}\text{cm}^2\text{eq}^{-1}$? 68. (Calculate the quantity of electricity that (a) 41.67 (b) 250 would be required to reduce 12.3 g of (c) 125 (d) 375 nitrobenzene to aniline if the current efficiency for the process is 50/72. At 18/ $^\circ\text{C}$, the ionic mobilities of NH_4^+ and potential drop across the cell is 3.0V, how ClO^- ions are 6.6×10^{-4} and 5.4×10^{-8} much energy will be consumed? $\text{m}^2\text{volt}^{-1}\text{s}^{-1}$ at infinite dilution. What is (a) 0.3 F, 86.85 kJ the molar conductance of ammonium chromate solution in $\text{Q}^{-1}\text{m}^2\text{mol}^{-1}$? (b) 0.6 F, 173.7 kJ (c) 1.2 F, 347.4 kJ (a) 0.01795 (b) 0.01158 (d) 0.6 F, 173.7 kJ (c) 1.2/ $\times 10^{-8}$ (d) 1.86/ $\times 10^{-7}$ 69. Perdisulphuric acid, $\text{H}_2\text{S}_2\text{O}_8$ can be

Problem 50**[Pearson Advanced Electrochemistry Q73]**

At 25/ $^\circ\text{C}$, the molar conductance at prepared by electrolytic oxidation of infinite dilution for HCl solution is H_2SO_4 , oxygen and hydrogen gases are by $4.25 \times 10^{-2} \text{ Q}^{-1}\text{m}^2\text{mol}^{-1}$, while its specific products. In such an electrolysis, 4.54 L conductance is $382.5 \text{ Q}^{-1}\text{m}^{-1}$. If the degree of H_2 and 2.27 L of O_2 were generated at of dissociation is 90/STP. What is the mass of H_2SO_4 formed? solution is (a) 0 (a) 0.9 M (b) 77.6 g (b) 1.0 M (c) 38.8 g (c) 0.1 M (d) 19.4 g (d) 1.1 M

Problem 51**[Pearson Advanced Electrochemistry Q70]**

During the discharge of a lead storage 74. A big irregular shaped vessel contained battery, the density of sulphuric acid fell water, conductivity of which was 2.56 from 1.5 to 1.1 g/ml. Sulphuric acid of $\times 10^{-3} \text{ s}^{-1}\text{m}^{-1}$. 585 g of NaCl was then density 1.5 g/ml is 40/added to the water and conductivity after and that of density 1.1 g/ml is 10/the addition of NaCl, was found to be 3.06 by weight. The battery holds 3.6 L of the $\times 10^{-3} \text{ s}^{-1}\text{m}^{-1}$. The molar conductivity of acid and the volume remained practically NaCl at this concentration is 1.5×10^{-2} constant during the discharge. Calculate $\text{S}^{-1}\text{m}^2\text{mol}^{-1}$. The capacity of vessel if it is the number of ampere-hours which fulfilled with water, is the battery should have been used. The (a) 3/ $\times 10^4$ 1 (b) 301 electrode reactions are: (c) 3/ $\times 10^8$ 1 (d) 3/ $\times 10^{10}$ 1 — PAGE 23 — Electrochemistry 8.23 75. The molar conductivity of 0.10 M 78. B Resistance of 0.2 M solution of solution of MgCl_2 is 100 mho $\text{cm}^2\text{mol}^{-1}$, an electrolyte is 50 Q. The specific at 25/ $^\circ\text{C}$. A cell with electrodes that are conductance of the solution is 1.4 S m^{-1} . 1.50 cm^2 in surface area and 0.50 cm The resistance of 0.5 M solution of the apart is filled with 0.10 M - MgCl_2 same electrolyte is 280 Q. The molar solution. How much current will flow conductivity of 0.5 M solution of the when the potential difference between the electrolyte (in S $\text{m}^2\text{mol}^{-1}$) is electrode-sis 5 volts? (a) 5/ $\times 10^{-3}$ (b) 5/ $\times 10^3$ (a) 0.03 A (b) 3.0 A (c) 5/ $\times 10^2$ (d) 5/ $\times 10^{-4}$ (c) 0.15 A (d) 15 A

Problem 52**[Pearson Advanced Electrochemistry Q79]**

The conductivity of a saturated solution of NaCl is $3.75 \times 10^{-8} \text{ Q}^{-1} \text{ cm}^{-1}$. For Na^+ , the value of λ° is $106 \text{ cm}^2 \text{ mol}^{-1} \text{ Q}^{-1}$. The speed of Na^+ ion in the solution is $3.75 \times 10^{-8} \text{ Q}^{-1} \text{ cm}^{-1}$. If the limiting molar conductivity of Ag^+ and Cl^- are $135 \text{ cm}^2 \text{ mol}^{-1} \text{ Q}^{-1}$ and $77 \text{ cm}^2 \text{ mol}^{-1} \text{ Q}^{-1}$, respectively, the limiting molar conductivity of AgCl is (a) $212 \text{ cm}^2 \text{ mol}^{-1} \text{ Q}^{-1}$ (b) $135 \text{ cm}^2 \text{ mol}^{-1} \text{ Q}^{-1}$ (c) $77 \text{ cm}^2 \text{ mol}^{-1} \text{ Q}^{-1}$ (d) $270 \text{ cm}^2 \text{ mol}^{-1} \text{ Q}^{-1}$.

The conductivity of a saturated solution of AgCl at 298 K was found to be $3.40 \times 10^{-6} \text{ Q}^{-1} \text{ cm}^{-1}$; the conductivity of water is $1.2 \times 10^{-6} \text{ Q}^{-1} \text{ cm}^{-1}$. The conductivity of saturated solution used to make up the solution was $1.60 \times 10^{-6} \text{ Q}^{-1} \text{ cm}^{-1}$. Determine the solubility limiting equivalent conductivities of AgCl in water in mole per litre at 298 K . BaCl_2 , K_2PO_4 and KCl are 160 , 140 and $100 \text{ Q}^{-1} \text{ cm}^2 \text{ eq}^{-1}$, respectively. The equivalent conductivity of AgCl at infinite dilution is $150.0 \text{ Q}^{-1} \text{ cm}^2 \text{ eq}^{-1}$. The solubility product of $\text{Ba}_3(\text{PO}_4)_2$ is (a) 1.44×10^{-10} (b) 1.2×10^{-5} (c) 1.08×10^{-23} (d) 3.33×10^{-25} .

Section B (One or More than one Correct)

1. Which of the following statement(s) is/are correct? (a) The net chemical change in a galvanic and electrolytic cell is always redox reactions. (b) Spontaneous or non-spontaneous (c) In a galvanic cell made of cobalt and cadmium electrodes, cobalt electrode acts as anode. (d) Chemical reactions occurring at the electrodes. (e) Standard potential increases with concentration of the anode electrolyte. (f) Dependence on Faraday's law. (g) Calomel electrode is a reference electrode having 0.00 volt potential.

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3. Consider the cell: $\text{Ag(s)} | \text{AgCl(s)} | \text{KCl (0.1 M)} || \text{Hg}_2\text{Cl}_2(\text{s}) | \text{Hg(l)}$. The function(s) of salt bridge in a cell is/are (a) It maintains standard electrode potential of cell constant which of Cl^- ions. (b) It completes the electrical circuit. (c) It separates both the solutions from each other. (d) It maintains the electrical neutrality of Cl^- ions. (e) It is independent of amounts of AgCl of both electrolytic solutions. and Hg_2Cl_2 .

9. The reactions taking place in the dry cell are: (a) $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ (b) $\text{MnO}_2 + 2\text{NH}_4^+ + 2\text{e}^- \rightarrow \text{Mn(OH)}_2 + 2\text{NH}_3$ (c) $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ (d) $\text{MnO}_2 + 2\text{NH}_4^+ + 2\text{e}^- \rightarrow \text{Mn(OH)}_2 + 2\text{NH}_3$

4. The passage of electricity in the Daniel cell are: (a) $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ (b) $\text{MnO}_2 + 2\text{NH}_4^+ + 2\text{e}^- \rightarrow \text{Mn(OH)}_2 + 2\text{NH}_3$ (c) $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ (d) $\text{MnO}_2 + 2\text{NH}_4^+ + 2\text{e}^- \rightarrow \text{Mn(OH)}_2 + 2\text{NH}_3$

5. From an electrolyte, one mole of electron chemical reactions occurring in the cell will deposit at cathode (a) 2.943 g Zn (b) 7.83 g MnO_2 (c) 63.5 g of Cu (d) 24 g of Mg (e) 1.62 g NH_4^+ (f) 3.915 g MnO_2 (g) 11.5 g of Na (h) 9.0 g of Al

Problem 53**[Pearson Advanced Electrochemistry Q10]**

EMF of the cell: $\text{Cd(s)}|\text{CdCl}_2\text{(aq)}||\text{AgCl(s)}|\text{Ag(s)}$ is +0.70 V at 0°C correct? and +0.60 V at 50°C. If ΔH° and ΔS° (a) The conductivity of molten NaCl is temperature independent, then the due to movement of Na^+ and Cl^- ions. correct information(s) regarding the cell (b) Solid NaCl is good conductor of electricity. (a) $\Delta G^\circ = -115.8 \text{ kJ}$ at 50°C (c) Molten sodium is a good conductor (b) $\Delta G^\circ = 135.1 \text{ kJ}$ at 0°C because of mobile electrons. (c) $\Delta S^\circ = -386 \text{ J/K}$ (d) Resistivity is reciprocal of molar conductivity of electrolyte. 11. Two litre solution of a buffer mixture of H_2PO_4^- and HPO_4^{2-} ions are 60 and 80 Q-1. H_2PO_4^- is placed in two compartments, respectively. The correct detail(s) (one litre in each) of an electrolytic cell. regarding Al(SO)₄ is/are The platinum electrodes are inserted in (a) the molar conductance is 140 each compartment and 1.25 A current is Q-'cm²mol⁻¹. passed for 965 min. Assuming electrolysis (b) the equivalent conductance is 140 of only water at each compartment, Q-' cm² eq⁻¹. what will be pH in each compartment (c) the molar conductance : is 840 after passage of above charge? (pK_a for Q-'cm²mol⁻¹. $\text{H}_2\text{PO}_4^- = 2.15, \log 7 = 0.85$) (d) the molar conductance is 23.33 (a) Anode: 3.00 (b) Cathode: 3.00 Q-'cm²mol⁻¹. (c) Anode: 1.30 (d) Cathode: 1.30 — PAGE 25 — Electrochemistry 8.25 12. For the reduction of NO_3^- ion in an acidic solution, the salt bridge aqueous solution, E/circ is +0.96 V. Values of (a) does not participate chemically in the E/circ for some metal ions are given below: cell reaction. $\text{V}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{V(s)}$ E/circ = -1.19 V (b) stops the diffusion of ions from one electrode to another. $\text{Fe}^{2+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Fe(s)}$ E/circ = -0.04 V (c) is necessary for the occurrence of the cell reaction. $\text{Au}^{2+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Au(s)}$ E/circ = +1.40 V (d) ensures mixing of the two electrolytic $\text{Hg}_2^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Hg(l)}$ E/circ = +0.86 V solutions.

Problem 54**[Pearson Advanced Electrochemistry Q15]**

A lead storage cell is discharged which The pair(s) of metal that is(are) oxidized causes the H_2SO_4 electrolyte to change by H_2O in aqueous solution is(are) from a concentration of 40% (a) V and Hg (density = 1.260 g/ml) to 28% (b) Hg and Fe The original volume of electrolyte was (c) Fe and Au 1 L. Identify the correct statement(s): (d) Fe and V (a) The overall cell reaction is: $\text{Pb(s)} + \text{H}_2\text{O(l)} \rightarrow \text{PbO}_2\text{(s)} + \text{H}_2\text{(g)}$ (b) A total of 2.0 moles of H_2SO_4 is (a) Molar conductivity reacted. (c) The total charge released from anode (b) Electromotive force of the cell is $1.93 \times 10^5 \text{ coulomb}$. (c) Resistance (d) The mass of electrolytic solution has (d) Heat capacity decreased. Section C (Comprehensions) Comprehension 1 Suppose that the S.H.E. was arbitrarily assigned a value of 1.00 V for $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$. What would this do to the observed voltage under standard condition for each of the following: (Given: E°Zn = -0.76 V, E°Cu = +0.34 V)

Problem 55**[Pearson Advanced Electrochemistry Q1]**

3. Zn-Cu Cell (a) +0.34 V (b) -0.76 V (a) +1.10 V (c) +1.76 V (d) -0.34 V (b) -1.10 V

Problem 56**[Pearson Advanced Electrochemistry Q2]**

$E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}$. The solubility product of AgI is 6.4×10^{-10} at 298 K. (2.303RT/F = 0.06) 4. The potential of Ag⁺/Ag electrode in a 0.04 M AgI saturated solution at 298 K is (a) -0.314 V (b) +0.314 V (c) -0.088 V (d) +0.172 V 5. The standard reduction potential of I⁻/AgI|Ag electrode at 298 K is (a) -0.314 V (b) +0.314 V (c) -0.172 V (d) +0.172 V Comprehension I The Edison storage cell is represented as Fe(s)|FeO(s)|KOH(aq)|NiO(s)|Ni(s). The half-cell reactions are $\text{NiO(s)} + \text{H}_2\text{(l)} + 2\text{e}^- \rightleftharpoons \text{Ni(s)} + 2\text{OH}^-$; $E^\circ = +0.40 \text{ V}$ $\text{FeO(s)} + \text{H}_2\text{(l)} + 2\text{e}^- \rightleftharpoons \text{Fe(s)} + 2\text{OH}^-$; $E^\circ = -0.87 \text{ V}$ 7. What is the cell reaction? 9. How does cell EMF depend on increasing (a) NiO(s) + Fe(s) $\rightarrow$ 2NiO(s) + FeO(s) the concentration of KOH? (b) 2NiO(s) + FeO(s) $\rightarrow$ NiO(s) + Fe(s) (a) increases (c) NiO(s) + FeO(s) $\rightarrow$ 2NiO(s) + Fe(s) (b) decreases + O₂ (c) remains unaffected (d) None of these (d) none of these 8. What is the standard cell EMF? 10. What is the maximum amount of (a) 1.27 V electrical energy that can be obtained (b) 0.47 V from one mole of NiO? (c) -1.27 V (a) 2.54 J (b) 245.11 kJ (d) -0.47 V (c) 122.56 kJ (d) 61.28 kJ Comprehension IV The cell potential for the unbalanced chemical reaction: $\text{Hg(aq)} + \text{NO}_3^-(\text{aq}) + 3\text{H}^+(\text{aq}) \rightarrow 2\text{Hg}_2^{2+}(\text{aq}) + \text{HNO}_3(\text{aq}) + 4\text{H}_2\text{O(l)}$ is measured under standard conditions in the electrochemical cell shown in the accompanying diagram. $E^\circ_{\text{cell}} = 0.02 \text{ V}$ dish A dish B anode cathode — PAGE 27 — Electrochemistry 8.27 11. In which dish, the solution is acidic?

Problem 57**[Pearson Advanced Electrochemistry Q13]**

How long will it take to produce 0.10 mole (a) Dish A (b) Dish B of HNO_2 by this reaction if a current

of (c) Both (d) None 10 A passes through the cell? 12. How many moles of electrons pass (a) 965 s

(b) 193 s through the circuit when 0.60 mole of Hg_2^{2+} (c) 1930 s and 0.30 mole of HNO_2 are produced

in the cell that contains 0.50 mole of Hg_2^{2+} (d) 482.5 s and 0.40 mole of NO , at the beginning

of the reaction? (a) 0.30 (b) 0.60 (c) 0.15 (d) 1.20 Comprehension V A fuel cell is the device to

convert the energy of a fuel into electrical energy without the use of a heat engine, where the fuel is

burnt directly. Such conversions are possible because the combustion reactions are essentially redox

reactions and highly exothermic as well as highly exergonic. Electrical energy can be obtained indefi-

nately from a fuel cell as long as the outside supply of fuel is maintained. In hydrogen-oxygen fuel

cell, the following reaction takes place: Anode reaction: $2\text{H}(\text{g}) + 4\text{OH}(\text{aq}) \rightarrow 4\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$ Cathode reac-

tion: $\text{O}(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^- \rightarrow 4\text{OH}(\text{aq})$ Overall reaction: $2\text{H}(\text{g}) + \text{O}(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$ The overall reaction has

a value of $\Delta H^\circ = -285.8 \text{ kJ/mol}$ and $\Delta G^\circ = -237.39 \text{ kJ/mol}$ at 25°C . 14. What is the standard EMF of the cell?

17. What is the approximate value of ΔS° for (a) 0.615 V (b) 1.23 V the fuel cell reaction at 25°C ? (c) 2.46 V (d) 0.74 V (a) -0.1624 JK^{-1} 15. What volume

of gaseous H_2 (at STP), (b) -162.4 JK^{-1} (c) $+162.4 \text{ JK}^{-1}$ when combined with excess O_2 in the fuel

(d) $+0.1624 \text{ JK}^{-1}$ cell, is needed to produce 23.739 kJ of useful work under standard conditions?

18. The theoretical efficiency of the fuel cell is (a) 2.27 L (b) 4.54 L given by (c) 1.13 L (d) 2.24 L (a)

83.06% (b) 100% 16. Suppose the concentration of hydroxide ion in the cell is doubled at 298 K. The (c)

67.53% (d) 97.88% cell voltage will be (a) reduced by half (b) increased by a factor of 2 (c) increased by

a factor of 4 (d) unchanged — PAGE 28 — 8.28 Chapter 8 Comprehension VI Billions of dollars are

spent each year to replace or prevent the corrosion and subsequent destruction of government property

such as ships, bridges, and metallic piping. Corrosion may result in decreased susceptibility to corrosion.

One part of the hull acts as the anode, while another acts as the cathode, the iron hull itself connects

the two parts, completing the circuit. Part of the metal hull begins oxidizing to Fe^{2+} in the presence of

H_2O and O_2 , the reduction reaction proceeds with an $E^\circ = 1.229 \text{ V}$. It is there that Fe^{2+} ions migrate from

the anode and are further oxidized as $2\text{Fe}^{2+}(\text{aq}) + \text{O}(\text{g}) + (6 + x)\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{Fe}(\text{OH})_x(\text{s}) + 4\text{H}^+(\text{aq})$

2 The $\text{Fe}(\text{OH})_x$ rust, formed, is only a small part of the problem. Pitting, or loss of solid metal, weakens the struc-

ture of the hull and may lead to serious damage and destruction of the hull. Using a sacrificial

anode often successfully prevents corrosion. This metal, more easily oxidized than iron. Although the

sacrificial anode eventually corrodes as well, the cost of replacing it is far less than that of replacing the

iron hull. Half-reaction Potential $\text{O} + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$ $E^\circ = 1.229 \text{ V}$ $\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$ $E^\circ = 0.771$

V $\text{Cu}^+ + \text{e}^- \rightarrow \text{Cu}$ $E^\circ = 0.521 \text{ V}$ $\text{O} + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$ $E^\circ = 0.401 \text{ V}$ $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ $E^\circ = 0.342$

V $\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$ $E^\circ = -0.126 \text{ V}$ $\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$ $E^\circ = -0.447 \text{ V}$ $\text{Mn}^{2+} + 2\text{e}^- \rightarrow \text{Mn}$ $E^\circ = -$

1.185 V $\text{Mg} + 2\text{e}^- \rightarrow \text{Mg}$ $E^\circ = -2.372 \text{ V}$ 19. Which of the following reactions may How many grams

of $\text{Fe}(\text{s})$ are lost in occur at the anode of the iron hull? 1.0 h? (a) $\text{Fe}^{2+}(\text{aq}) \rightarrow \text{Fe}^{2+}(\text{aq}) + \text{e}^-$ (a) 0.261 g

(b) 0.522 g (b) $\text{Fe}(\text{s}) \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$ (c) 1.044 g (d) 0.783 g (c) $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$

23. A ship that has sunk to the bottom of _____ $7 + (3)2\text{H} + (\text{be})\text{HO} \rightarrow (\text{DO})2\text{H}7(\text{p})$ the

ocean may still exhibit corrosion. One 20. What is the standard voltage generated by differen-

ce in the corrosion reaction between the corrosion reaction on the iron hull of ship above water and one

that has sunk of a ship? the bottom of the ocean maybe that (a) -1.676 V (b) -0.782 V (a) the

oxidation reaction of the sunken (c) 1.782 V (d) 1.676 V ship does not include H_2O as a reactant.

21. Which of the following materials could (b) the oxidation reaction of the sunken not serve

as a sacrificial anode for lead? ship does not include O_2 as a reactant. (a) Copper (b) Iron (c) the

reduction reaction of the sunken (c) Magnesium (d) Manganese ship does not include H_2O as a reactant.

22. An ammeter reading shows a current of (d) the reduction reaction of the sunken 0.50 A running

through the iron hull ship does not include O_2 as a reactant. — PAGE 29 — Electrochemistry 8.29

Comprehension VII A current of 15.0 A is employed to plate nickel in a NiSO_4 solution. Both Ni

and H_2 are formed at the cathode. The current efficiency with respect to formation of Ni is 60% $= 8.9$

g/ml . ($\text{Ni} = 58.7$) 24. How much of nickel is plated on the 26. What volume of H_2 at 0°C and 1 atm

is cathode per hour? formed per hour? (a) 16.43 g (b) 32.85 g (a) 6.271 (b) 3.761 (c) 19.7 g

(d) 9.85 g (c) 2.51 (d) 5.011 25. What is the thickness of the plating if the 27. At the end of the

electrolysis, how many cathode consists of a sheet of metal 4.0 cm, grams of the gaseous product

appear at which is to be coated on both sides? the anode? (a) 13.8 mm (b) 27.6 mm (a) 4.48

g (b) 1.79 g (c) 6.9 mm (d) 23.0 mm (c) 2.69 g (d) 7.46 g Comprehension VIII Tollen's reagent

is used for the detection of aldehyde when a solution of AgNO_3 is added to glucose with NaOH

then gluconic acid is formed. $\text{A}80 = 40 + \text{CHO} + \text{HO} \rightarrow \text{CHO}, (\text{Gluconic acid}) + 2\text{H}^+ + 2\text{e}^-; E^\circ_{\text{ox}} =$

-0.05 V $\text{Ag}(\text{NH}_3)_2^+ + \text{e}^- \rightarrow \text{Ag}(\text{s}) + 2\text{NH}_3$ $E^\circ_{\text{red}} = 0.337 \text{ V}$ [$U_{\text{S}} = 2.303 \text{ RT} E^\circ = -0.0592 \text{ V}$ and $E^\circ_{\text{red}} = 0.337 \text{ V}$ at 298 K]

Problem 58**[Pearson Advanced Electrochemistry Q53]**

Which of the following graph truly VNaOH represents the titration of CH₃COOH (c) solution against NHOH solution? (a) Cond. Cond. VNaOH (d) VNHOH (b) Cond. Cond. VNaOH-52. Which of the following graph truly VNHOH represents the titration of AgNO₃ solution against KCl solution? (c) (a) Cond. Cond. VNHOH VKCl (b) (d) Cond. Cond. VKCl — /to VNHOH — PAGE 35 — Electrochemistry 8.35 Section D (Assertion-Reason) The following questions consist of two Statement II: Metals having negative statements. Mark reduction potential have large hydration (a) If both statements are CORRECT, energy. and Statement II is the CORRECT 6. Statement I: Specific conductance explanation of Statement I. decreases with dilution while molar (b) If both statements are CORRECT, and conductance increases. Statement II is NOT the CORRECT Statement II: On dilution, number of explanation of Statement I. ions per unit volume decreases but total (c) If Statement I is CORRECT, but number of ions increases considerably. Statement II is INCORRECT. (d) If Statement I is INCORRECT, but 7. Statement I: When acidified zinc sulphate Statement II is CORRECT. solution is electrolysed between zinc Statement I: Electrolysis of CuCl₂(aq) electrodes, it is zinc that is deposited at

Problem 59**[Pearson Advanced Electrochemistry Q1]**

the cathode and evolution of hydrogen gives 1 mole of Cu and 1 mole of Cl₂ by the passage of suitable charge. gas does not take place. Statement II: The electrode potential Statement II: Equal equivalents of Cu and Cl₂ are formed during the passage of zinc becomes more negative than hydrogen as the overvoltage for the same charge. hydrogen evolution on zinc is quite large. Statement I: Zinc metal can displace Ag metal from a solution containing the 8. Statement I: The conductivity of complex [Ag(CN)₂]. solutions of different electrolytes in the is greater than same solvent and at a given temperature is same. Statement II: The conductivity depends

Problem 60**[Pearson Advanced Electrochemistry Q3]**

Statement I: In electrolysis, the quantity on the charge and size of the ions, the of electricity needed for depositing 1 mole concentrations of ions and ease with of silver is different from that required for which the ions move under potential 1 mole copper. gradient. Statement II: The atomic masses of silver 9. Statement I: Sodium ions are discharged and copper are different. in preference to hydrogen ions at a Statement II: In an electrochemical cell, mercury cathode.

Problem 61**[Pearson Advanced Electrochemistry Q4]**

anode and cathode are, respectively, Statement II: Na⁺ is a strong reducing negative and positive electrode. agent in comparison to H⁺ ion. Statement II: At anode, oxidation takes 10. S Statement I: The conductivity of metals place and at cathode reduction takes decreases while that of electrolytic place. solution increases with increase in 5. Statement I: A metal having negative temperature. reduction potential when dipped in the Statement II: Electrons in metals are very solution of its own ions has a tendency to tightly held by the nucleus and are not pass into the solution. free to move. — PAGE 36 — 8.36 / Chapter 8 11. Statement I: An electrochemical cell can Statement II: A salt bridge contains be set up only when the redox reaction is concentrated solution of an inert spontaneous. electrolyte like KCl, KNO₃, K₂SO₄, etc., Statement II: A reaction is spontaneous in agar-agar. if free energy change at constant 14. Statement I: During electrolysis of temperature and pressure is negative. aqueous sodium acetate solution, the

Problem 62**[Pearson Advanced Electrochemistry Q12]**

Statement I: In the Daniel cell, if molar ratio of gases formed at cathode concentrations of Cu^{2+} and Zn^{2+} ions and anode is 1:3. are doubled, the EMF of cell does not change. Statement II: Acetate ion discharges at anode and H^+ ion, at cathode. Statement II: If the concentration of ions 15. Statement I: Electrode potential of any in contact with the metal is doubled, the electrode will change on changing any of electrode potential will be doubled. its intensive properties. 13. Statement I: KCl , NaCl , NHCl , etc., Statement II: Any intensive property of cannot be used in the salt bridge of a cell system changes on changing any of its containing silver. properties, whether intensive or extensive. Section E (Column Match) 1. Match Column I with Column II Column I Column II (Electrolysis product using inert electrodes) (A) Dilute solution of HCl (P) O_2 evolved at anode (B) Dilute solution of NaCl (Q) H_2 evolved at cathode (C) Concentrated solution of NaCl (R) Cl_2 evolved at anode (D) AgNO_3 solution (S) Ag deposited at cathode

Problem 63**[Pearson Advanced Electrochemistry Q2]**

The variation in conductivity of these reactions is given in List II. Match List I with List II. List I List II (A) $\text{CH}_3\text{COOH} + \text{NaOH}$ (P) Conductivity decreases and then increases (B) $\text{KI} (0.1 \text{ M}) + \text{AgNO}_3 (0.01 \text{ M})$ (Q) Conductivity decreases and then does not change much (C) $\text{CH}_3\text{COOH} + \text{KOH}$ (R) Conductivity increases and then does not change much (D) $\text{NaOH} + \text{HI}$ (S) Conductivity does not change much and then increases — PAGE 37 — Electrochemistry / 8.37 3. The standard reduction potential data at 25°C is given below: $E^\circ(\text{Fe}^{2+}/\text{Fe}) = +0.77 \text{ V}$ $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) = -0.44 \text{ V}$ $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$ $E^\circ(\text{Cu}^+/\text{Cu}) = +0.52 \text{ V}$ $E^\circ[\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}] = +1.23 \text{ V}$ $E^\circ[\text{O}_2(\text{g}) + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-] = +0.40 \text{ V}$ $E^\circ(\text{Cr}^{2+}/\text{Cr}) = -0.74 \text{ V}$ $E^\circ(\text{Cr}^{3+}/\text{Cr}^{2+}) = -0.91 \text{ V}$ Match E° of the redox pair in List I with the values given in List II. List I List II (A) $E^\circ(\text{Fe}^{2+}/\text{Fe})$ (P) -0.18 V (B) $E^\circ(4\text{H}^+/\text{O}_2 + 4\text{H}^+ + 4\text{OH}^-)$ (Q) -0.4 V (C) $E^\circ(\text{Cu}^{2+}/\text{Cu})$ (R) -0.04 V (D) $E^\circ(\text{Cr}^{2+}/\text{Cr})$ (S) -0.83 V

Problem 64**[Pearson Advanced Electrochemistry Q4]**

Match the following Column I Column II (A) Concentration cell (P) $\text{Ag} | \text{AgCl} | \text{Cl}^- || \text{Ag}^+ | \text{Ag}$ (B) Spontaneous cell reaction (Q) $\text{Ag} | \text{AgCl} | \text{Cl}^- || \text{Br}^- | \text{AgBr} | \text{Ag}$ (C) Non-spontaneous cell reaction (R) $\text{Ag} | \text{Ag}^+ (0.1 \text{ M}) || \text{Ag}^+ (1.0 \text{ M}) | \text{Ag}$ (S) $\text{Ag} | \text{AgCl} | \text{Cl}^- (0.1 \text{ M}) || \text{Cl}^- (1.0 \text{ M}) | \text{AgCl} | \text{Ag}$ Section F (Subjective) Single-digit Integer Type 1. Percentage of aniline hydrochloride potential for the above electrode becomes hydrolysed in its $\text{M}/40$ solution at 25°C 0.31 V , is $(2.303RT/F = 0.06)$ 3. The reduction potential at 25°C for $2.303RT/F = 0.06$ 2. The standard reduction potential for -0.44 V and $E^\circ = -0.04 \text{ V}$, the ratio

Problem 65**[Pearson Advanced Electrochemistry Q2]**

$\text{Fe} | \text{Fe}^{2+} | \text{Cu}^{2+} | \text{Cu}$ electrode is $+0.34 \text{ V}$. The solubility of molar concentrations of Fe^{2+} to Fe^{3+} product of $\text{Cu}(\text{OH})_2$ is 1.0×10^{-19} . The ions in solution is $(2.303RT/F = 0.06)$, pH of solution at which the reduction $\log 5 = 0.7$ — PAGE 38 — 8.38 Chapter 8 4. If $[\text{Fe}^{2+}]$ at equilibrium, when potassium

Problem 66**[Pearson Advanced Electrochemistry Q10]**

A test for complete removal of Cu^{2+} iodide is added to a solution of Fe^{3+} ions from a solution of Cu^{2+} is to add initially at 0.50 M until $[\text{I}^-] = 1.0 \text{ M}$, $\text{NH}_3(\text{aq})$. A blue colour signifies the is $x \times 10^{-4} \text{ M}$, the value of x is (Given formation of complex $[\text{Cu}(\text{NH}_3)_4]^{2+} = 0.53 \text{ V}$, having $K = 1.1 \times 10^3$ and thus confirms $2.303RT/F = 0.06$) I | I the presence of Cu^{2+} in solution. 250 ml of 0.1 M CuSO_4 is electrolysed by 5. If it is desired to construct the following passing a current of 5 A for 1351 s. galvanic cell: After passage of this charge, sufficient quantity of NH_3 is added to electrolysed $\text{Ag(s)} | \text{Ag}^+(\text{saturated AgI}) || \text{Ag}^+(\text{solution maintaining } [\text{NH}_3] = 0.10 \text{ M. (saturated AgCl, } x \times 10^{-4} \text{ M Cl)} | \text{Ag(s)}$ If $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is detectable up to its to have $E_{\text{cell}} = 0.102 \text{ V}$, what should be concentration as low as $1 \times 10^{-4} \text{ M}$, the value of x to get $[\text{Cl}^-]$, which must would a blue colour be shown by the be present in the cathodic half-cell to electrolysed solution on addition of achieve the desired EMF. Given K of NH_3 . Mark "1", if the answer is 'yes' and AgCl and AgI are 1.8×10^{-1} and 8.1×10^{-1} mark "2", if the answer is 'no'. $x \times 10^{-4}$, respectively. ($2.303RT/F = 0.06$) 6. An alloy of lead (valency = 2) - thallium 11. By passing a certain amount of charge (valency = 1) containing 70% through NaCl solution, 9.08 L of chlorine Cl_2 , by weight, can be electroplated onto a gas were liberated at STP. When the same cathode from a perchloric acid solution. amount of charge is passed through a How many hours would be required to nitrate solution of metal M, 52.8 g of the deposit 5.0 g of this alloy at a current of metal was deposited. If the specific heat 1.10 A? ($\text{Pb} = 208$, $\text{Tl} = 204$) of metal is 0.032 Cal/°C-g, the valency of metal is 7. Iridium was plated from a solution containing IrCl_3 for 2.0 h with a current

Problem 67**[Pearson Advanced Electrochemistry Q12]**

The electrode reactions for charging of a of 0.075 A. The Iridium deposited on the lead storage battery are: cathode weighed 0.36 g. If the oxidation state of Ir in IrCl_3 is x , then the value of $\text{PbSO}_4 + 2e^- \rightarrow \text{Pb} + \text{SO}_4^{2-}$ ($x + y$) is ($\text{Ir} = 192$) $\text{H}^+ + \text{OS} + 209 \rightarrow \text{H}_2 + \text{OS} + 9$ 8. The electrolysis of cold sodium chloride + $2e^-$ solution produces sodium hypochlorite by reacting NaOH and Cl_2 , thoroughly The electrolyte in the battery is an aqueous How long (in days) will a cell operate to solution of sulphuric acid. Before recharging, produce 10 L of 7.45% the specific gravity of the liquid was found of NaClO if the cell current is 2.5 A? Assume that the density of solution is to be 1.10 (16/1.0 g/ml. 965 charging for h, the specific gravity 9.9. A volume of 500 ml of 0.1 M CuSO_4 of the liquid was found to be 1.42 (40% solution is electrolysed for 5 min at a HSO by weight). If the battery contained current of 0.161 A. If Cu is produced at 2 L of the liquid and the volume remains one electrode and oxygen at the other, the constant during charge, the average current approximate pH of the final solution is (in A) used for charging the battery is — PAGE 39 — Electrochemistry 8.39 13. A dilute solution of KCl was placed 14. In the refining of silver by electrolytic between two platinum electrodes, 12 cm method, what will be the mass of 72.8 g apart, across which a potential of 1.93 silver anode (60 volts was applied. How far (in cm) would 9.65 A current is passed for 1 h? ($\text{Ag} = 108$) the K^+ ion move in 20 h at 25°C? Ionic

Problem 68**[Pearson Advanced Electrochemistry Q15]**

The pH of 100 L of concentrated KCl conductance of K^+ ion at infinite dilution at 25/°C is $7.5 \times 10^{-2} \text{ mho m}^2 \text{ mol}^{-1}$. solution after the electrolysis for 10 s using 9.65 A at 298 K is Four-digit Integer Type 1. The specific conductivity of a saturated 5. An alloy weighing 2.70 mg of Pb-Ag was solution of AgCl is $2.80 \times 10 \text{ mho m}^{-1}$ dissolved in desired amount of HNO and volume was made 250 ml. A silver electrode was dipped in solution and $E_{\text{cell}} = 7.81 \text{ mV}$, the of the cell: solubility of silver chloride (in order of 10^{-g1}) at 25/°C, is Pt, H(1 bar) | H⁺(1 M) | Ag⁺ | Ag. 2. The electrode potential (in millivolts) of was 0.50 V at 298 K. The percentage of 2Ag(s) + S₂(aq) to Ag₂S(s) + 2e⁻ in a lead in alloy is (Given: $E_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}$, solution buffered at pH = 3 and which $\text{Ag} = 108$, $2.303 RT/F = 0.06$) is also saturated with 0.1 M -HS, is (Given: for HS, $K_a = 10^{-7}$; $K_a = 10^{-13}$ 6. Given $E_{\text{Fe}^{2+}(\text{cyt b})/\text{Fe}^{2+}} = 0.08 \text{ V}$ for Fe₂(cyt b) | Fe²⁺ K of AgS = 4×10^{-5} , $E_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}$, (cyt b) couple and $E_{\text{Fe}^{2+}(\text{cyt c})/\text{Fe}^{2+}(\text{cyt c})} = 0.20 \text{ V}$ for log 2 = 0.3, $2.303 RT/F = 0.06$) Fe₂(cyt c) | Fe₂(cyt c), couple, where $E_{\text{Fe}^{2+}(\text{cyt c})/\text{Fe}^{2+}(\text{cyt c})}$ represents the standard state reduction 3. An excess of liquid mercury is added to potentials at pH = 7.0 at 25/°C and cyt an acidified solution of 10⁻² M - Fe³⁺. is an abbreviation for cytochromes. The It is found that 10/value of K for the reaction equilibrium at 25/°C. The value of $E_{\text{Hg}_2^{2+}(\text{Hg})/\text{Hg}}$ (in mV), assuming that the only reaction Fe₂(cyt c) + Fe²⁺ (cyt b) = Fe₂(cyt c) that occurs is: 2Hg + 2Fe²⁺ to Hg₂^{2+} + c) + Fe²⁺ (cyt b) + 2Fe³⁺, is (Given: $E_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.7724 \text{ V}$, is ($2.303 RT/F = 0.06$) log 2 = 0.3, log 3 = 0.48, $2.303 RT/F = 0.06$) 7. Calculate potential (in mV) of the cell: 4. Determine potential (in mV) of the cell: Cu | Mn(s) | MnCl₂(0.001 M), HCl(0.01 M) | Pt | Fe₂, Fe₃CrO₂, Cr₂, H⁺ | Pt in M) O₂(0.25 bar) Pt | Cu. Given, which $[\text{Fe}^{2+}] = 0.75 \text{ M}$, $[\text{Fe}^{3+}] = 0.75 \text{ M}$, $E_{\text{Fe}^{3+}/\text{Fe}^{2+}} = -1.85 \text{ V}$ for the Mn²⁺ | Mn couple $[\text{CrO}_2] = 2 \text{ M}$, $[\text{Cr}_2^{3+}] = 4 \text{ M}$ and $[\text{H}^+] = 1.229 \text{ V}$ for the O₂ | HO, H couple. = 1 M. Given: ($2.303 RT/F = 0.06$, log 2 = 0.3) Fe₂ + e⁻ to Fe₂; $E_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.77 \text{ V}$ 8. The temperature coefficient of the cell: Zn | 14 H⁺ + 6e⁻ + CrO₂ to 2Cr³⁺ + 7HO; | ZnCl₂, | AgCl(s) | Ag is + 0.001 V/K. The $E_{\text{cell}} = 1.35 \text{ V}$ entropy change (in J/K) accompanying $2.303 RT/F = 0.06$, log 2 = 0.3, log 3 = 0.48 this reaction, at 298 K is — PAGE 40 — 8.40 Chapter 8 9. For the reaction: 4Al(s) + 3O₂(g) + 6HO(l) solution of Au(NO₃); and a current of +4OH(aq) = 4Al(OH)(aq); $E_a = 2.4 \text{ A}$ is applied. The time (in seconds) = 2.5 V. If G for HO(l) and OH(aq) are required for the electroplating to be -280.0 and -156.25 kJ/mol, respectively, completed, assuming that the layer of the magnitude of ΔG for Al(OH)₃(aq) gold builds up evenly, is (Au = 197) (in kJ/mol) is

Problem 69**[Pearson Advanced Electrochemistry Q13]**

We have taken a saturated solution 10. In a Zn-MnO₂ Cell, the anode is made of AgBr. K_{sp} of AgBr is 12×10^{-14} . If up of Zn and cathode of carbon rod 10-moles of AgNO₃ is added to 1 litre of surrounded by a mixture of MnO₂ this solution, find conductivity (specific Carbon, NHCl and ZnCl₂ in aqueous conductance) of this solution in terms of base. If 8.7 g MnO₃ is present in cathodic 10-7 S/cm units. Given (Ag^+) = 6×10^{-3} compartment, how many days the dry Sm²⁺, A(Br) = 8×10^{-3} Sm²⁺, cell will continue to give a current of $i_{\text{cell}}(\text{NO}) = 7 \times 10^{-3} \text{ S/cm}^2 \text{ mol}^{-1}$. 3.99 $\times 10^{-3} \text{ A}$?

Problem 70**[Pearson Advanced Electrochemistry Q14]**

On passing electricity through 11. An amount of 19 g of molten SnCl₂ nitrobenzene solution, it is converted into is electrolysed for sometime. Inert azobenzene. The mass of azobenzene (in electrodes are used. 1.19 g of tin is mg) produced, if the same quantity of deposited at the cathode. No substance electricity produces oxygen just sufficient is lost during electrolysis. If the ratio to burn 96 g of fullerene (C), is of the masses of SnCl₂ and SnCl₄ after 15. The conductivity of saturated solution electrolysis is 261: x, the value of x is (Sn = 119) of sparingly soluble salt, Ba₃(PO₄), is $1.2 \times 10^{-10} \text{ } \Omega \text{-cm}$. The limiting

Problem 71**[Pearson Advanced Electrochemistry Q12]**

An object whose surface area is 80 cm^2 is equivalent conductances of BaCl_2 , K_2PO_4 to be plated with an even layer of gold and KCl are 160, 140 and $100 \text{ } \Omega \text{ cm}^2$. $8.0 \times 10^4 \text{ cm}$ thick. The density of gold eq', respectively. The k of $\text{Ba}(\text{PO}_4)_2$ is 19.7 g/cm^2 . The object is placed in a (in the order of 10^{-25}) is Answer Keys-Exercise II Section A (only one Correct)

Problem 72**[Pearson Advanced Electrochemistry Q80]**

(b) Section B (One or More than one Correct)

Problem 73**[Pearson Advanced Electrochemistry Q15]**

(a), (b), (c), (d) — PAGE 41 — Electrochemistry / $\cdot 8.41$ Section C Comprehension I Comprehension IX

Problem 74**[Pearson Advanced Electrochemistry Q34]**

(c) Comprehension II Comprehension X

Problem 75**[Pearson Advanced Electrochemistry Q38]**

(c) Comprehension III Comprehension XI

Problem 76**[Pearson Advanced Electrochemistry Q41]**

(a) Comprehension IV Comprehension XII

Problem 77**[Pearson Advanced Electrochemistry Q43]**

(d) Comprehension V Comprehension XI

Problem 78**[Pearson Advanced Electrochemistry Q44]**

(b) 45. (c) Comprehension VI Comprehension XIV

Problem 79**[Pearson Advanced Electrochemistry Q47]**

(b) Comprehension VII Comprehension XV

Problem 80**[Pearson Advanced Electrochemistry Q30]**

(d) Section D (Assertion-Reason)

Problem 81**[Pearson Advanced Electrochemistry Q15]**

(c) Section E (Column Match)

Problem 82**[Pearson Advanced Electrochemistry Q1]**

A/toP, Q; B/toP, Q; C/toQ, R; D/toP, S

Problem 83**[Pearson Advanced Electrochemistry Q2]**

A/toR; B/toS; C/toQ; D/toP 3. A/toR; B/toS; C/toP; D/toQ

Problem 84

[Pearson Advanced Electrochemistry Q4]

A/toR,S;B/toP,R; C/toQ,S Section F(Subjective) Single-digitIntegerType

Problem 85

[Pearson Advanced Electrochemistry Q15]

(9) Four-digit Integer Type

Problem 86

[Pearson Advanced Electrochemistry Q5]

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Chapter 12

Cengage DPP (3.1 to 3.5): Complete Question Bank

Problem 1

[Cengage DPP Q9]

The equilibrium constant of acetic acid in an aqueous solution of concentration c is given by K_a . Why on dilution the K_a of CH_3COOH increases drastically, while that of CH_3COONa increases gradually? (a) $K_a = \frac{c}{\sqrt{c}}$ (b) $K_a = \frac{1}{\sqrt{c}}$ (c) $K_a = \frac{1}{c}$ (d) $K_a = \frac{1}{c^2}$

4. A 100 watt, 110 volt incandescent lamp is connected in series with an electrolytic cell containing cadmium sulphate solution. What mass of cadmium will be deposited (a) 1.0 g (b) 2.0 g (c) 3.0 g (d) 4.0 g by the flowing for 10 hours?

Problem 2

[Cengage DPP Q5]

The resistance of a solution A is 50Ω and that of

Problem 3

[Cengage DPP Q10]

The molar conductivity of 0.05 M solution of an electrolyte A is $100 \text{ S cm}^2 \text{ mol}^{-1}$, both solutions being taken in the electrolyte is $200 \text{ S cm}^2 \text{ mol}^{-1}$. The resistance offered same conductivity cell. If equal volumes of solutions A and B are mixed, what will be the resistance of the mixture be about using the same cell? (Assume that there is no increase in the degree of dissociation of A and B on mixing). (a) 11.11 Ω (b) 22.22 Ω (c) 33.33 Ω (d) 44.44 Ω

6. The equivalent conductivity of acetic acid at infinite

Problem 4**[Cengage DPP Q11]**

If $\kappa_m(\text{Kt}) = 73.5 \text{ Q}^{-1} \text{ cm}^2 \text{ mol}^{-1}$, $\kappa_m(\text{At}) = 189 \text{ Q}^{-1}$ dilution is $387 \text{ S cm}^2 \text{ eq}^{-1}$. At the same temperature, 0.001 M solution of acetic acid, it is $55 \text{ S cm}^2 \text{ eq}^{-1}$. What κ_m and $(\text{SO}_2) = 160 \text{ Q}^{-1} \text{ cm}^2 \text{ mol}^{-1}$, the molar is the degree of dissociation of 0.1 N acetic acid? Assume conductivity of potash alum will be about $1/\alpha = 1$. (a) $1165 \text{ Q}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ 7. Calculate the dissociation constant of water at 25°C from (b) $422.5 \text{ Q}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ the following data: (c) $582.5 \text{ Q}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ Conductivity of $\text{H}_2\text{O} = 5.8 \times 10^{-8} \text{ S cm}^{-1}$, $\kappa_{\text{g}^+} = 350.0$ (d) $145.6 \text{ Q}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and $2\text{OH}^- = 198.0 \text{ S cm}^2 \text{ eq}^{-1}$ — PAGE 2 — 3.2 Physical Chemistry 12. The molar conductivity of a weak electrolyte HA at infinite dilution can be determined by (c) (p) (a) extrapolating $\Delta \kappa_m$ versus c plot to $c \rightarrow 0$. (b) extrapolating $\Delta \kappa_m$ versus $\sqrt{c}$ plot to $c \rightarrow 0$. (c) using Kohlrausch law with the data on Multiple Correct Answers Type A (HCl), A (NaA) and A (NaCl) (d) using Kohlrausch law with the data on. 18. Which one is a correct statement? (a) Conductance is an additive property A(HA) and A(NaA). (b) Conductance order: $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Rb}^+$ 13. When a certain conductivity cell was filled with 0.01 M (c) Molar conductivity of NH in liquid NH is higher solution of KCl, it had a resistance of 160 Q at 25°C , and when filled with 0.005 M NaOH, it had a resistance of than in H_2O 190 Q . If specific resistance of KCl solution is $700 \text{ Q}^{-1} \text{ cm}$, (d) At infinite dilution all the interionic attraction specific conductance ($\text{Q}^{-1} \text{ cm}^{-1}$) solution is vanishes (a) 0.00120 (b) 0.00170 19. An ion travels 0.4 cm in 1 minute under the influence (c) 0.00180 (d) 0.00190 2 volt potential difference when electrodes are 3 cm apart 14. Variation of resistance with increase in cell constant gives Which one are correct? graph of the type (a) Speed of ion $= 6.67 \times 10^{-3} \text{ cm s}^{-1}$ (b) Ionic mobility $= 1 \times 10^{-6} \text{ m}^2 \text{ volt}^{-1} \text{ sec}^{-1}$ R R (c) Ionic mobility $= 1.0 \times 10^{-3} \text{ cm}^2 \text{ volt}^{-1} \text{ sec}^{-1}$ (d) Ionic mobility $= 4.44 \times 10^{-3} \text{ cm}^2 \text{ volt}^{-1} \text{ sec}^{-1}$ (a) (b) 20. Which ion(s) has/have exceptionally higher A values? (a) H^+ (b) K^+ [Cell constant] (c) OH^- (d) NH_2 21. For which, the formula $1/\alpha =$. (a) CH_3COOH (b) HCl R (c) HClO_4 (d) CH_3NH_2 (c) R (d) 22. The factor(s) which influence(s) the conductance solutions is. (a) Solute-solute interaction (b) Solute-solvent interaction (c) Temperature 15. The molar conductivity of LiNO_3 at infinite dilution is (d) Solvent-solvent interaction $105 \text{ S cm}^2 \text{ mol}^{-1}$. If 0.02 M solution shows conductance 23. Which of the following graph incorrectly represents κ_m of $0.158 \text{ } \Omega^{-1} \text{ cm}^{-1}$ using cell of cell constant 1 cm^{-1} . The variation of molar conductance ($\Delta \kappa_m$) with dilution for concentration of Li^+ ion in g litre^{-1} is: strong electrolyte? (a) 0.01106 (b) 0.1106 (c) 11.06 (d) 0.2212

Problem 5**[Cengage DPP Q16]**

The absolute velocity of Ag^+ and NO_3^- are $5.7 \times \sqrt{c}$ (e) (b) $\Delta \kappa_m$ $10^{-4} \text{ cm sec}^{-1}$ and $6.9 \times 10^{-4} \text{ cm sec}^{-1}$ respectively. Assuming complete dissociation of AgNO_3 , calculate the Dilution- molar conductivity of AgNO_3 at infinite dilution. (a) $121.59 \text{ S cm}^2 \text{ mol}^{-1}$ (b) $141.59 \text{ S cm}^2 \text{ mol}^{-1}$ (c) $131.59 \text{ S cm}^2 \text{ mol}^{-1}$ (d) none of these (c) $\Delta \kappa_m$ (d) $\Delta \kappa_m$ and the conductivity of the solution was measured. The plots of conductance A vs volume of $\text{AgNO}_3(\text{aq})$ is: Dilution— Dilution—/to a) v (q)

Problem 6**[Cengage DPP Q24]**

— PAGE 3 — Electrochemistry 3.3 Comprehension Type Above plot represents the variation of molar conductance against $\sqrt{C}$ (where C = molar concentration of the electrolyte). The molar conductance of NaCl varies with the concentration as electrolyte). Select the incorrect options among the shown in the following table: following: Molar conductance in (a). Both I and II are for strong electrolyte Molar concentration (b) Both I and II are for weak electrolyte $\Delta \kappa_m$ ($^{-1}$) $\text{cm}^2 \text{ mol}^{-1}$ of NaCl (c) I is for strong electrolyte and II for weak electrolyte 107.4×10^{-4} (d) I is for weak electrolyte and II for strong electrolyte 97.9×10^{-4} Matching Column Type 87 16×10^{-4}

Problem 7**[Cengage DPP Q25]**

Match the physical term in Column I with the unit in Column II. When a certain conductivity cell (C) was filled with 25×10^{-4} M NaCl solution, the resistance of the cell was found to be 1000 Ω . At infinite dilution ion conductances of Cl^- and SO_4^{2-} are 80 $\text{cm}^2 \text{mol}^{-1}$ and 160 $\text{cm}^2 \text{mol}^{-1}$ respectively. (a) Conductance (p) siemen $\text{m}^2 \text{mol}^{-1}$ 28. What is the molar conductance of NaCl at infinite dilution? (b) Resistivity (q) m^{-1} (a) 147 $\text{cm}^2 \text{mol}^{-1}$ (c) Conductivity (1) volt per ampere (b) 107 $\text{cm}^2 \text{mol}^{-1}$ (d) Molar conductivity (s) siemen (c) 127 $\text{cm}^2 \text{mol}^{-1}$ Cell constant (cm^{-1}) siemen m^{-1} (d) 157 $\text{cm}^2 \text{mol}^{-1}$ (e) 29. What is the cell constant of the conductivity cell (C)? (f) Resistance (u) cm^{-1} (a) 0.385 cm^{-1} (b) 3.85 cm^{-1} (c) 38.5 cm^{-1} (d) 0.1925 cm^{-1} Integer Type 30. If the cell (C) is filled with 5×10^{-3} N Na_2SO_4 , the observed resistance was 400 Ω . What is the molar conductivity of acetic acid at infinite dilution conductance of Na_2SO_4 is 390.7 and for 0.01 M acetic acid is 3.907 S $\text{cm}^2 \text{mol}^{-1}$ (a) 19.25 $\text{cm}^2 \text{mol}^{-1}$ What is the pH solution? (b) 96.25 $\text{cm}^2 \text{mol}^{-1}$ (c) 385 $\text{cm}^2 \text{mol}^{-1}$ The specific conductivity of N/50 solution of KCl at 298 K (c) 385 $\text{cm}^2 \text{mol}^{-1}$ is 0.002501 S cm^{-1} . If the resistance of the same solution (d) 192.4 $\text{cm}^2 \text{mol}^{-1}$ placed in the cell is 2000 Ω , what is the cell constant? Answers Key Subjective Type Matching Column Type

Problem 8**[Cengage DPP Q4]**

19.06 g 5. 66.67 $\text{cm}^2 \text{mol}^{-1}$ (1 $\rightarrow$ J) (b $\rightarrow$) (d $\rightarrow$ p) (1 $\rightarrow$ o) (n $\rightarrow$ q) (s $\rightarrow$ r) 6.1.5/7.2 $\times 10^{-16}$ mol/litre Integer Type Single Correct Answer Type

Problem 9**[Cengage DPP Q12]**

(c) Comprehension Type

Problem 10**[Cengage DPP Q30]**

(d) Multiple Correct Answers Type

Problem 11**[Cengage DPP Q24]**

(a, b, d) — PAGE 4 — DPP Electrolysis and Faraday's Laws 9. Which of the following gives the charge of an electron? Subjective Type (a) 1 F/1 NA (b) 1F (c) 1F/1N (d) 1N/1F

Problem 12**[Cengage DPP Q1]**

An aqueous solution of copper sulphate is electrolysed using platinum electrodes in one case and copper electrodes in another. In the electrolysis of copper(II) chloride solution, the mass of cathode is increased by 3.15 g. At copper anode, we will in another case. Will the products of electrolysis be same or different? Give reason. have

Problem 13**[Cengage DPP Q2]**

State the products of electrolysis obtained on the cathode (a) the liberation of 1120 mL of Cl_2 at STP and the anode in the following cases: (b) the liberation of 560 mL of O_2 at STP (i) A dilute solution of H_2SO_4 with platinum electrodes. (c) the loss of 3.15 g of Cu (i) An aqueous solution of AgNO_3 with silver electrodes. (d) the gain of 3.15 g of Cu 3. A current of 2 ampere is passed for one hour between nickel 11. One equivalent of Na metal is formed from electrolysis electrodes in 0.5 L of 2 M Ni(NO)_3 solution. What will be the molarity of fused NaCl. No. of moles of Al from the fused NaAlF_4 be the molarity of the solution at the end of electrolysis? with the same current passed is: 4. (i) On electrolysis of an aqueous solution NaCl, why H_2 (a) 1 (b) 3 (c) $1/3$ (d) 2 and not Na is liberated at the cathode? (i) One faraday of electricity deposits one mole of Na from the molten salt but mol of Al from an 12. Which of the following statements are correct? aluminium salt. Why? 3 (a) Sodium can be obtained by the electrolysis of aqueous solution of NaCl using Pt electrodes. Single Correct Answer Type (b) The current carrying ions in an electrolytic cell are not necessarily discharged at the electrodes. 5. In electrolysis of an aqueous solution of sodium sulphate, (c) The mass of a substance deposited on the cathode 2.4 L of oxygen at STP was liberated at anode. The volume of hydrogen at STP liberated at cathode would be: (d) In the purification of a metal by electrolytic method (a) 1.2 L (b) 2.4 L the impure metal is made cathode. (c) 2.6 L (d) 4.8 L 13. Which of the following statements are not correct? 6. What mass of silver could be plated out on a serving tray by (a) The reaction involving reduction of water in electrolysis of a solution containing silver in +1 oxidation $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$ state for a period of 8 hours at a current of 8.46 amperes? (b) The species having a least reduction potential is What is the area of tray if the thickness of the silver plating preferentially reduced at the cathode. is 0.00254 cm? Density of silver is 10.5 g/cm³. (c) Water is reduced in preference to the cations having (a) 11200 cm² (b) 10204.7 cm² more positive potential than -0.828 V. (c) 10300 cm² (d) 9237.4 cm² (d) In the electrolysis of a solution containing SO_3^{2-} anion

Problem 14**[Cengage DPP Q7]**

A certain electricity deposited 0.54 g of Ag from AgNO_3 only, the anodic reaction is $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$ solution. What volume of hydrogen will the same quantity

Problem 15**[Cengage DPP Q14]**

Which of the following statements are not correct? of electricity liberate at 27°C and 728 mm Hg pressure? (a) The electrolysis of a solution containing Cl^- anion (a) 62.48 mL (b) 60.00 mL produces Cl_2 at anode but O_2 is also liberated as the (c) 64.28 mL (d) 66.5 mL solution is diluted. 8. The time taken to decompose electrolytically 18 g of water (b) The quantitative relationship between electricity and by 2 A current is about chemical change were established by Faraday. (a) 24.8 h (b) 26.8 h (c) When an aqueous solution of sodium fluoride (c) 28.8 h (d) 30.8 h electrolysed, the gas liberated at the anode is F_2 (d) In an electrolytic refining of an impure metal, the pure metal is obtained at anode. — PAGE 5 — Electrochemistry 3.5 19. A current strength of 96.5 A is passed for 10 sec through

Problem 16**[Cengage DPP Q15]**

Three moles of electrons are passed through three solutions 1 L of solution of 0.1M aqueous-
 CuSO₄, calculate the pH in succession containing AgNO₃, CuSO₄ and AuCl₃, of the solution. re-
 spectively. The molar ratio of amounts of cations reduced at cathode will be Comprehension Type 1.11
 (a) 1:2:3 (b) 123 Electrolysis is the process in which electrical energy is converted (c) 3:2:1 (d) 6:3:2
 to chemical energy. In electrolytic cell, oxidation takes place at 16. On electrolysis, in which of the
 following, O₂ would be anode and reduction at cathode. Electrode process depends on the liberated
 at the anode? electrode taken for electrolysis. Amount of substance liberated (a) dilute H₂SO₄ with Pt
 electrodes at an electrode is directly proportional to the amount of charge (b) aqueous AgNO₃ solution
 with Pt electrodes passed through it. The mass of substance liberated at electrode (c) dilute H₂SO₄
 with Cu electrodes is calculated using the following relation: (d) aqueous NaOH with a Fe cathode and
 a Pt anode It $E = \frac{m}{Z}$ $m = 96500$ Matching Column Type Here, E represents the equivalent mass and 96500
 C is called the Faraday constant. Faraday (96500 C) is the charge of 1 mol 17. Match the Column I
 with Column II: electron, i.e., 6.023×10^{23} electrons; it is used to liberate one Column I Column II
 gram equivalent of the substance. (Electrolysis) (pH and products at 20. The platinum electrodes
 were immersed in a solution of anode and cathode) cupric sulphate (CuSO₄) and electric current is
 passed Electrolysis of 100 L (p) pH = 12.3 (a) through the solution. After some time, it was observed that
 aqueous solution Anode = Ethane (g) the colour of copper sulphate disappeared with evolution + CO (g)
 CH₃COOH by passing of a gas at the electrode. The colourless solution contains: 2 F of electricity
 Cathode = H₂ (g) (a) platinum sulphate (b) copper nitrate (c) copper sulphate (d) sulphuric acid
 (b) Electrolysis of 10 L (b) pH = 13.0 aqueous solution Anode = H₂ (g) 21. The passage of current
 liberates H₂ at cathode and Cl₂ at HCOOH by passing + CO (g) anode. The solution is: 1 F of electricity
 Cathode = H₂ (g) (a) copper chloride in water (b) NaCl in water (c) Electrolysis of 10 L (1) pH = 7.0
 (c) mercuric chloride in water aqueous solution Anode = O₂ (g) Cathode = H₂ (g) (d) AuCl₃ in water
 K₂SO₄ by passing 1 F

Problem 17**[Cengage DPP Q22]**

On electrolysis of dilute sulphuric acid using platinum of electricity electrodes, the product obtained
 at the anode will be: (d) Electrolysis of 10 L (s) pH = 1.0 (a) hydrogen (b) oxygen aqueous solution
 Anode = O₂ (g) (c) hydrogen sulphite (d) sulphur oxide CuF₂ by passing 1 F Cathode = Cu (g) 23. How
 many faradays are required to reduce 1 mol BrO₃⁻; to of electricity Br⁻? (a) 3 (b) 5 (c) 6 (d) 4
 24. Calculate the volume of gas liberated at the anode at S.T.P. Integer Type during the electrolysis
 of a CuSO₄ solution by a current of 1 A passed for 16 min and 5 sec.

Problem 18**[Cengage DPP Q18]**

How many grams of Cl₂ (g) can be produced by the (a) 224 mL (b) 56 mL electrolysis of molten
 NaCl with a current of 5.5 A for 25 (c) 112 mL (d) 448 mL min? [Atomic weight of Cl = 35.5 amu]
 Answers Key Matching Column Type Single Correct Answer Type

Problem 19**[Cengage DPP Q17]**

(a) / to p), (b) / to q), (c) / to r), (d) / to s) 5. (d) 10. (c)

Problem 20**[Cengage DPP Q11]**

(c) Integer Type Multiple Correct Answers Type

Problem 21**[Cengage DPP Q14]**

(c, d) Comprehension Type

Problem 22**[Cengage DPP Q24]**

(b) — PAGE 6 — 3.6 Physical Chemistry DPP 3.3 Fuel Cell and Batteries (a) $\text{Pb}^{2+} + 2\text{e}^- / \text{to Pb}$ Subjective Type (b) $\text{Pb} / \text{to Pb}^{2+} + 2\text{e}^-$ (c) $\text{Pb}^{2+} + \text{SO}_4^{2-} / \text{to PbSO}_4$ 1. (i) What advantage do the fuel cells have over primary (d) $\text{PbSO}_4 + 2\text{H}_2\text{O} / \text{to PbO}_2 + 4\text{H}^+ + \text{SO}_4^{2-} + 2\text{e}^-$ and secondary batteries? (i) Write the cell reaction of a lead storage battery when it is discharged. How does the density of the electrolyte supply change when the battery is discharged? (a) power for heat and light

Problem 23**[Cengage DPP Q2]**

(i) What is the role of ZnCl_2 in a dry cell? (b) power for pressure (c) oxygen its life? (d) none of these 3. Which type of cells are rechargeable? 11. In a H-O cell, combustion of hydrogen occurs due to: 4. During the discharge of a lead storage battery, density of (a) remove adsorbed oxygen from electrode surface H_2SO_4 fell from 1.294 g/mL to 1.139 g/mL. Sulphuric acid (b) create potential difference between the two electrodes of density 1.294 g/mL is 39/(c) produce high purity water density 1.139 g/mL is 20/(d) generate heat holes 3.5 litres of the acid and volume remains practically constant during discharge. Calculate ampere-hour of Multiple Correct Answers Type which the battery must have been used. The charging and discharging reactions are:

Problem 24**[Cengage DPP Q12]**

Which is/are correct about fuel cells? $\text{Pb} + \text{SO}_4^{2-} / \text{to PbSO}_4 + 2\text{e}^-$ (charging) (a) Cells continuously run as long as fuels are supplied (b) These are more efficient and free from pollution $\text{PbO}_2 + 4\text{H}^+ + \text{SO}_4^{2-} + 2\text{e}^- / \text{to PbSO}_4 + 2\text{H}_2\text{O}$ (discharging) (c) These are used to provide power and drinking water 5. In a fuel cell H_2 and O_2 react to produce electricity. In the H_2 and O_2 reaction process H_2 is oxidised at the anode and O_2 is reduced at (d) None of these the cathode. If 67.2 litres of H_2 at NTP reacts in 15 minutes,

Problem 25**[Cengage DPP Q13]**

During the discharge of lead storage battery: what is the average current produced? If the entire current (a) PbO_2 is reduced is used for electro-deposition of Cu from Cu^{2+} , how many (b) Pb is oxidised of Cu are deposited? (c) specific gravity of H_2SO_4 gives extent of discharging (d) equivalent mass of H_2SO_4 is 98

Problem 26**[Cengage DPP Q14]**

Which of the following statements are correct about Ni-Cd Single Correct Answer Type cell 6. When a lead storage battery is discharged, then: (a). It consists of a cadmium electrode (as anode) and a (a) SO_2 is evolved metal grid containing nickel (IV) oxide (as cathode) (b) lead is formed immersed in KOH solution. (c) lead sulphate is consumed (b) At cathode: $\text{NiO}(\text{s}) + 2\text{H}_2\text{O} + 2\text{e}^- = \text{Ni}(\text{OH})_2(\text{s}) + 2\text{OH}^-$ (d) sulphuric acid is consumed 2OH^- (aq) 7. A depolarizer used in dry cell batteries is: (c) At anode: $\text{Cd}(\text{s}) + 2\text{OH}^- (\text{aq}) = \text{Cd}(\text{OH})_2(\text{s}) + 2\text{e}^-$ (a) ammonium chloride (b) manganese dioxide (d) The potential of each Ni-Cd cell is approximately 2.6 (c) potassium hydroxide (d) sodium phosphate V. 8. An example of a simple fuel cell is:

Problem 27**[Cengage DPP Q15]**

During the charging of lead storage battery, which reactions (a) lead storage battery (b) H_2 - O_2 cell don't occur at anode are: (c) Daniell cell (d) Leclanche cell (a) $\text{Pb}^{2+} + \text{SO}_4^{2-} / \text{to PbSO}_4$

Problem 28**[Cengage DPP Q9]**

Which of the following reactions occurs at cathode during charging of storage battery?
 (b) $\text{PbSO}_4 + \text{H}_2\text{O} \rightarrow \text{PbO}_2 + \text{SO}_2 + 2\text{H}^+$ — PAGE 7 — Electrochemistry 3.7 Comprehension Type
 (c) $\text{Pb} \rightarrow \text{Pb}^{2+} + 2\text{e}^-$ (d) $\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$ A lead storage battery consists of a lead anode and a grid of

Problem 29**[Cengage DPP Q16]**

Fuel cell involves following reaction(s): (at cathode) lead packed with lead dioxide as the cathode. The electrolyte (a) $\text{O}(\text{g}) + 2\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\text{l})$ (at anode) taken is 39% (b) $\text{O}(\text{g}) + 2\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\text{l})$ (aq.) The battery holds 35 L of the acid. During the discharge of the (at anode) (c) $2\text{H}(\text{g}) + 4\text{OH}^-(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$ battery, the density of H_2SO_4 falls from 1.294 g mL⁻¹ to 1.139 g mL⁻¹ (at cathode) (d) $2\text{H}(\text{g}) + 4\text{OH}^-(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$ mL-I which is 20%

Problem 30**[Cengage DPP Q19]**

The reaction occurring at the anode during charging is Matching Column Type
 (a) $\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$ (b) $\text{Pb}^{2+} + \text{SO}_4^{2-} \rightarrow \text{PbSO}_4$ 17. Match the following: (c) $\text{Pb} \rightarrow \text{Pb}^{2+} + 2\text{e}^-$ Column II (d) $\text{PbSO}_4 + 2\text{H}_2\text{O} \rightarrow 2\text{PbO}_2 + 4\text{H}^+ + \text{SO}_4^{2-} + 2\text{e}^-$ Column I (p) Cell reaction $2\text{H} + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ 20. Moles of sulphuric acid lost during discharge is (a) Leclanche (a) 9.88 (b) 8.88 cell (c) 7.88 (d) 6.88 (b) Ni-Cd cell (b) Does not involve any ion in 21. Molarity of the solution after the discharge is solution and is used in hearing (a) 8.136 (b) 4.068 aids. (c) 2.32 (d) 1.16 (c) Fuel cell (r) Rechargeable 22. The amount of charge in coulombs used up by the battery Mercury cell (s) Reaction at anode, is nearly (d) (a) 954180 (b) 477090 ++uzuz (c) 95418 (d) 47709 23. The number of ampere-hour for which the battery must Integer Type have been used is (a) 2650.5 (b) 265.05

Problem 31**[Cengage DPP Q18]**

The emf of lead storage battery is V. (c) 26.505 (d) 2.6505 Answers Key Matching Column Type Subjective Type

Problem 32**[Cengage DPP Q17]**

(a /tos), (b /tor), (c /top), (d /toq) 4.265.04 5. 643.3 amp, 190.5 g Integer Type Single Correct Answer Type 9. (a)

Problem 33**[Cengage DPP Q11]**

(b) Comprehension Type

Problem 34**[Cengage DPP Q23]**

(b) Multiple Correct Answers Type

Problem 35**[Cengage DPP Q16]**

(a, c) — PAGE 8 — DPP Construction and Working of a Cell, Electrochemical Series and Its Applications Which is the strongest reducing agent? SubjectiveType (a) Zn(s) (b) Cr(s) (c) H(g) (d) Fe²⁺(aq) 1. Predict whether zinc and silver react with 1M sulphuric acid. The following facts are available: acid to give hydrogen or not. Given that the standard $2X + Y_2 \rightarrow 2Y + X_2$; reduction potentials of zinc and silver are - 0.76 volt and 2W- + Y₂ / to 2Y- + X₂; reduction potentials of zinc and silver are - 0.76 volt and 2W- + Y₂ / to No reaction; 0.80 volt respectively. 2Z- + X₂ 2X- + Z₂ 2. Iodine (I) and bromine (Br) are added to a solution containing iodide (I-) and bromide (Br-) ions. What of the following statements is correct? reaction would occur if the concentration of each species (a) FO is 1 M? The electrode potentials for the reactions are: W- / W E = 0.54V, E Br/Br- = 1.08V (b) 1/ W / W₂ 3. (i) Which of the following oxides is reduced by hydrogen? (c) W / W MgO, CuO and NaO (ii) Which of the following oxides will decompose on W / W₂ Z / Z₂ heating? Ag₂X at 1 atm is bubbled through a solution containing ZnO, CuO, MgO and Ag₂O a mixture of 1M Y and 1M Z at 25 / circ C. If the reduction (ii) The value of E for electrode reactions, potential of Z > Y > X, then (a) Y will oxidise X and not Z + 2e are 0.444, -0.337 and 0.763 volts respectively. (b) Y will oxidise Z and not X State which of these metals can replace the other two (c) Y will oxidise both X and Z from the solution of their salts? (d) Y will reduce both X and Z 4. Determine range of E / circ values for this reaction 8. A large quantity of cobalt metal was dropped into a solution of X₂ + 2e- / to X(s) for given conditions: containing Ag⁺, Fe³⁺ and Cu²⁺, all at unit concentration. E / circ (Ag⁺ / Ag) = 0.799V, E / circ (Fe³⁺, Fe²⁺ / Pt) = 0.77V, E (Cu²⁺ (a) If the metal X dissolves in HNO₃ but not in HCl it can Cu = 0.34V, E / circ (Fe³⁺ / Fe) = -0.04V, E / circ (Cu²⁺, Cu⁺ / Pt) displace Ag⁺ ion but not Cu²⁺ ion. 0.153 V and E / circ (Co²⁺ / Co) = 0.28V, then the solution would (b) If the metal X is HCl acid producing H(g) but does not displace either Zn or Fe²⁺ Given: (a) Ag, Fe and Cu (b) Ag, Fe and Cu⁺ (c) Fe²⁺, Cu and Co²⁺ (d) Fe²⁺, Cu, Ag and Co⁺ = 0.8V, 0.44V, E = 0.34V, C²⁺ / C 9. Given : E (Sn²⁺, Sn⁴⁺ / Pt) = 0.15 V, E / circ (Hg₂ Hg⁺ / Pt) -0.76V 0.92 V and E / circ (Pb²⁺, H⁺ / PbO₂) = 1.45 V. Based on this data = 0.96V NONO Zn²⁺ / Zn which of the following statements is correct? (a) Sn⁴⁺ is a stronger oxidizing agent than Pb²⁺ Single Correct Answer Type (b) Sn²⁺ is a stronger reducing agent than Hg²⁺* (c) Hg²⁺ is a stronger oxidizing agent than Pb²⁺ 5. The standard potential at 298 K for the following half (d) Pb²⁺ is a stronger reducing agent than Sn²⁺ reactions are given against each: 10. Given below are some standard reduction potentials: Zn²⁺(aq) + 2e- Zn(s) 0.762 V Sn²⁺ + 2e- = -0.14V Hg²⁺ + 2e- = 2Hg E = 0.79 2H⁺(aq) + 2e- Cr(s) 0.000V Which of the following statements is correct? Cr³⁺(aq) + 3e- Cr(s) 0.740 V (a) Hg can reduce Sn²⁺ to Sn Fe³⁺(aq) + 2Fe²⁺(aq) 0.770V (b) Sn²⁺ can oxidize Hg to Hg²⁺ — PAGE 9 — Electrochemistry 3.9

Problem 36**[Cengage DPP Q19]**

In which of the following, salt bridge is not needed? (c) Hg²⁺ can oxidize Sn to Sn²⁺ (a) Pb | PbSO₄(s) | H₂SO₄ | PbO₂(s) | Pb (d) Between Hg₂ and Sn²⁺, Sn²⁺ is stronger oxidising (b) Cd | CdO(s) | KOH(aq) | NiO₂(s) | Ni agent than Hg₂⁺. (c) Fe(s) | FeO(s) | KOH(aq), NiO(s) | Ni 11. Given that E values of Ag⁺ / Ag, K⁺ / K, Mg²⁺ / Mg and Cr³⁺ / (d) Zn | ZnSO₄ | CuSO₄ | Cu Cr are 0.80V, -2.90V, -2.37V, and -0.74V, respectively. Which of the following orders regarding the reducing Matching Column Type power of metals is correct? (a) Ag > Cr > Mg > K (b) Ag < Cr < Mg < K 20. Match the items of Column I and Column II on the basis (c) Ag > Cr > K > Mg (d) Cr > Ag > Mg > K of data given below:

Problem 37**[Cengage DPP Q12]**

Indicator electrode is: $E^\circ = 2.87\text{V}$, $E = -3.5\text{V}$ (b) Calomel electrode F_2/F^- Li/Li^+ (a) SHE (c) Ag/AgCl electrode (d) Quinhydrone electrode $E^\circ = 1.4\text{V}$, $E = 1.09\text{V}$ Au^{3+}/Au Br/Br^- 13. When an aqueous solution of lithium chloride is electrolysed using graphite electrodes: Column I Column II (a) pH of the resulting solution increases (b) pH of the resulting solution decreases (a) F_2 (P) Metal is the strongest reducing agent (c) as the current flows, pH of the solution around the (b) Li (q) Metal ion which is the weakest cathode increases oxidising agent (d) none of the above 14. The calomel electrode is reversible with respect to: (c) Au^{3+} (r) Non-metal which is the best oxidising (a) Hg_2^{2+} (b) H^+ (c) Hg_2^{2+} (d) Cl^- agent (d) Br (s) Unreactive metal Multiple Correct Answers Type (e) Au () Anion that can be oxidised by Au^{3+} 15. Which of the following is/are true for standard hydrogen (f) Lit (u) Anion which is the weakest reducing electrode? agent (a) Temperature is maintained at 25°C (g) F^- (v) Metal ion which is an oxidising agent (b) Hydrogen ion concentration is 1M (c) Pressure of hydrogen gas is maintained at 1atm sphere (d) It contains platinum metal which is used to adsorb Integer Type hydrogen gas,

Problem 38**[Cengage DPP Q16]**

The electrode potential of a glass electrode does not depend

Problem 39**[Cengage DPP Q21]**

If E° , E_2 and E are standard oxidation potentials for : uodn (a) concentration of chloride ions The value of n is n (b) concentration of hydrogen ions

Problem 40**[Cengage DPP Q22]**

The no. of cells which may be constructed with different (c) concentration of KCl solution (d) all of these E° values for the reaction: $\text{Fe} + 2\text{Fe}^{3+} \rightarrow 3\text{Fe}^{2+}$. 17. A KI solution containing starch, turns blue on the addition of chlorine. Which of the following statements explain(s) Comprehension Type this? (a) The standard reduction potential of Cl_2 is more than The standard reduction potentials of some half reactions are that of L_2 given as below: (b) The standard oxidation potential of Cl_2 is more than $\text{Cu}^{2+} + e^- \rightarrow \text{Cu}^+$; $E^\circ = 0.15\text{V}$ that of $\text{I}_2 + 2e^- \rightarrow 2\text{I}^-$; $E^\circ = 0.5\text{V}$ (c) The product formed when Cl_2 combines with starch is blue in colour $\text{Ag}^+ + e^- \rightarrow \text{Ag}$; $E^\circ = 0.799\text{V}$ (d) The product formed when I_2 combines with starch is The decrease in free energy ($-\Delta G$) is responsible for the blue in colour production of equivalent amount of electrical work. Thus,

Problem 41**[Cengage DPP Q18]**

Which of the following statements is/are correct? (a) A metal in its highest oxidation state acts as an oxidant $-\Delta G = nFE$

Problem 42**[Cengage DPP Q23]**

The E° for $\text{CuI}/\text{Cu}^{2+}$ in volt is: (b) In the reaction, $\text{F}_2 + \text{O}_2 \rightarrow \text{OF}_2$, oxygen is an (a) 0.35 (b) -0.35 (c) $+0.325$ (d) -0.325 oxidant (c) The oxidant number of Ni in $\text{Ni}(\text{CO})_4$ is zero

Problem 43**[Cengage DPP Q24]**

The E° for $\text{CuI}/\text{Cu}^{2+} + \text{I}^- + \text{I}_2$ in volt is: (d) Copper metal can be oxidized by Zn^{2+} ions (a) $+0.449$ (b) $+0.474$ (c) -0.449 (d) -0.474 — PAGE 10 — 3.10 Physical Chemistry (a) 1,3 (b) 2,3

Problem 44**[Cengage DPP Q25]**

The E_{cell} for $\text{Cu}|\text{Cu}^{2+}||\text{Ag}^+|\text{Ag}$ is: (c) 1, 4 (d) 2, 4 1M 0.1M 27 . The m.f. of Cl^- for spontaneous cell reaction having two (a) 0.415V (b) 0.374V electrodes as $\text{Ag}|\text{Ag}^+$ and $\text{Cu}|\text{Cu}^{2+}$ containing 1M Cu^{2+} (c) 0.15V (d) -0.374V and 1M Ag^+ ions after the passage of 9.65 ampere current 26 . In the cell $\text{Ag}|\text{Ag}^+||\text{Cu}^{2+}|\text{Cu}$ containing 1M Cu^{2+} and 1M for 1 hour will: Ag^+ ions. 9.65 ampere current is passed for 1 hour. Select (a) increase by 0.010V (b) decrease by 0.010V the correct statement. (c) increase by 0.020V (d) decrease by 0.020V (1) Ag will oxidise to Ag^+ and new $[\text{Ag}^+] = 1.36\text{M}$ (2) Ag^+ will reduce to Ag and new $[\text{Ag}^+] = 0.64\text{M}$ 28 . The total change in free energy during the cell reaction in (3) Cu^{2+} will reduce to Cu and new $[\text{Cu}^{2+}] = 0.82\text{M}$ $\text{Cu}|\text{Cu}^{2+}||\text{Ag}^+|\text{Ag}$ would be: (4) Cu will oxidise to Cu^{2+} and new $[\text{Cu}^{2+}] = 0.82\text{M}$ (a) -80.1kJ (b) -40.05kJ (c) -31.36kJ (d) -62.72kJ Answers Key SingleCorrectAnswerType IntegerType

Problem 45**[Cengage DPP Q14]**

(d) ComprehensionType Multiple CorrectAnswersType

Problem 46**[Cengage DPP Q19]**

(a, b, c) MatchingColumnType (b/leftrarrow)(s/leftrarrow)(1/leftrarrowP)(A/leftrarrow)(d/leftrarrowq)(1/leftrarrowg/rov,u) — PAGE 11 — Electrochemistry 3.11 DPP 3.5 Nernst Equation and Its Applications Subjective Type (C) (P) Ecell 1. How will the pH of brine (aq. NaCl solution) be affected Ecell when it is electrolysed? 2. From the standard cell potentials given below, calculate $\log [\text{Mg}^{2+}]$ the indicated cell potential E_{cell} . $\log [\text{Zn}^{2+}]$ $[\text{Zn}^{2+}]$ Br(l) 8. The overall cell reaction of Daniell cell is: E_{cell} $\text{An(s)} + \text{Cu}^{2+} + \text{Zn}^{2+} + \text{Cu(s)}$; $E_{\text{cell}} = +1.10\text{V}$ 3. Write the Nernst equation for the cell reaction in the Daniell cell. How will the E_{cell} be affected when concentration of $[\text{Zn}^{2+}]$ Zn^{2+} ions is increased? $[\text{Cu}^{2+}]$ (i) How can the reduction potential of an electrode be increased? When $\log Q$ is plotted on X-axis and E_{cell} on Y-axis, then which of the following will correctly represent the Explain why? variation? (ii) What is the free energy change for (a) galvanic cell (b) electrolytic cell? 5. An excess of solid AgCl is added to a 0.1M solution of (a) Brions at 298K . Calculate the concentrations of Cl^- and 1.10V Br ions at equilibrium. Given: $E_{\text{cell}}(\text{Cl}^-|\text{Ag}) = 0.222\text{V}$ and $E(\text{Br}|\text{Ag}|\text{Br}|\text{Ag}) = 0.095\text{V}$. 6. Find the solubility product of a saturated solution of Ag_2CrO_4 in water at 298K if the emf of the cell $\text{Ag}|\text{Ag}^+(\text{satd. Ag}_2\text{CrO}_4\text{ soln.})||\text{Ag}^+(0.1\text{M})|\text{Ag}$ is 0.164V at (b) 1.10V 298K . SingleCorrectAnswerType 7. In the electrochemical cell, $\text{Mg(s)}|\text{Mg}^{2+}(\text{aq})||\text{Zn}^{2+}(\text{aq})|\text{Zn(s)}$; $E_{\text{cell}} = +3.13\text{V}$ (c) 1.10V The correct plot of E_{cell} versus $\log [\text{Mg}^{2+}]$ will be represented as: $[\text{Zn}^{2+}]$ (a) Ecell (b) Ecell (d) $\log [\text{Mg}^{2+}]$ 1.10V $[\text{Zn}^{2+}]$ $\log [\text{Mg}^{2+}]$ $[\text{Zn}^{2+}]$ — PAGE 12 — 3.12 Physical Chemistry Ag (s) | AgI (saturated solution) || $\text{Ag}_2\text{C}_2\text{O}_4$ (saturated 9. An alloy of Pb-Ag weighing 1.08g was dissolved in dilute solution) $\text{I}|\text{Ag(s)}|\text{HNO}_3$ and the volume made to 100mL . A silver electrode $[\text{Ag}^+]$ AgI was dipped in the solution and the emf of the cell set-up, (a) $E_{\text{cell}} = 0.0592$ $\log \text{Pt(s)}|\text{H(g)}|\text{H}^+(1\text{M})|\text{Ag}^+(\text{aq})|\text{Ag(s)}$ was 0.62V . If $E_{\text{cell}} [\text{Ag}^+]\text{Ag}_2\text{C}_2\text{O}_4$ is 0.80V , what is the percentage of Ag in the alloy? (At 25°C , $RT/F = 0.06$) $[\text{Ag}^+]\text{Ag}_2\text{C}_2\text{O}_4$ (b) 2.50 (b) $E_{\text{cell}} = 0.0592$ $\log$ (a) 25 $[\text{Ag}^+]\text{AgI}$ (c) 10 (d) 50 $K_{\text{sp}}(\text{Ag}_2\text{C}_2\text{O}_4)$ 10. Determine the solubility and solubility product of (c) $E_{\text{cell}} = 0.0592$ $\log \text{Co}[\text{Fe}(\text{CN})_6]$ in water at 25°C from the following data: $K_{\text{sp}}(\text{AgI})$ Conductivity of saturated solution of $\text{Co}[\text{Fe}(\text{CN})_6] = 2.06 \times 10^{-6} \text{ } \Omega^{-1} \text{cm}^{-1}$ and that of water $= 4.1 \times 10^{-7} \text{ } \Omega^{-1} \text{cm}^{-1}$. The ionic molar conductivities of Co^{2+} and (d) $E_{\text{cell}} = 0.0592$ $\log [\text{Fe}(\text{CN})_6]$ are 86 and $444 \text{ } \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$ respectively. $[K_{\text{sp}}(\text{AgI})]/2$ (a) 7.69×10^{-17} (b) 2.67×10^{-6} (c) 9.68×10^{-6} (d) 8.23×10^{-17}

Problem 47**[Cengage DPP Q16]**Consider the cell: $\text{Ag}|\text{AgCl(s)}|\text{KCl(1M)}||\text{Hg}_2\text{Cl}_2(\text{s})|$

Problem 48**[Cengage DPP Q11]**

The standard reduction potential for the following two Hg (l) | I Pt. The cell potential: reactions are given: (a) increases on increasing concentration of Cl^- ions $\text{AgCl} + e^- \rightarrow \text{Ag(s)} + \text{Cl}^-(\text{aq})$; $E^\circ = 0.22\text{V}$ (i) (b) decreases on decreasing concentration of Cl^- ions $\text{Ag}^+(\text{aq}) + e^- \rightarrow \text{Ag(s)}$ $E^\circ = 0.80\text{V}$ (i) (c) is independent of concentration of Cl^- ions The solubility product of AgCl under standard condition (d) is independent of amounts of AgCl (s) and HgCl (s) will be: (a) $1.613 \times 10^{-5}\text{M}^2$ (b) $1.535 \times 10^{-8}\text{M}^2$ 7. The temperature coefficient of cell is $\Delta E/\Delta T$ Then: (c) $3.213 \times 10^{-10}\text{M}^2/\text{K}$ (d) $1.535 \times 10^{-10}\text{M}^2/\text{K}$

Problem 49**[Cengage DPP Q12]**

Calculate the cell potential of following cell: $\text{DE} | \text{Pt(s)} | \text{H(g)} (0.1\text{ bar}) | \text{BOH} (0.1\text{ M aq}) | \text{HA} (0.1\text{ M}) | \text{H}^+(\text{g}) (1\text{ bar}) | \text{Pt}$ (b) When $3E^\circ$ Given: $K_a(\text{HA}) = 10^{-7}$, $K_b(\text{BOH}) = 10^{-5}$ ΔH° and it is endothermic ΔH° (a) 0.40V (b) 0.36V reaction (c) 0.93V (d) 0.63V

Problem 50**[Cengage DPP Q13]**

For a cell reaction involving two electron charges, the (c) When ΔE° standard emf of the cell is 0.295V at 25°C . The equilibrium constant of the reaction at 25°C will be: (a) 29.5×10^{-2} (b) 10 (c) 1×10^{10} (d) 2.95×10^{10} (d) When $\Delta E^\circ = 0$, then $\Delta H^\circ > nFE^\circ$ and it is endothermic Multiple Choice Type reaction 14. A sample of water has a hardness expressed as 77.5 ppm . In the following electrochemical cell: Ca^{2+} . This sample is passed through an ion exchange column and the Ca^{2+} is replaced by H^+ . Select the correct pH (x atm) H^+ ($\text{pH} = y$) Zn^{2+} ($z\text{ M}$) Zn state-ment(s) $E_{\text{cell}} = E^\circ_{\text{cell}}$. This will be possible when (a) pH of the water after it has been so treated is 2.4 (b) Every Ca^{2+} ion is replaced by one H^+ ion (a) $[\text{Zn}^{2+}] = [\text{H}^+] = 1\text{ M}$ and $\text{pH} = 1$ atm (c) Every Ca^{2+} ion is replaced by two H^+ ions (b) $[\text{Zn}^{2+}] = 0.01$, $[\text{H}^+] = 0.1\text{ M}$ and $\text{pH} = 1$ atm (d) pH of the solution remains unchanged when $100 = \text{Hardness}$ $\text{pH} = 10$ (3) 15. Which of the following expression(s) represent the voltage of cell at 298K ? (d) $[\text{Zn}^{2+}] = [\text{H}^+] = 0.1\text{ M}$ and $\text{pH} = 0.1\text{ atm}$ — PAGE 13 — Electrochemistry 3.13 Matching Column Type 19. Match the Column I with Column II: Column II (Potential of overall process) Column I (Combination of half-cell reactions) (p) $E^\circ = +1.56\text{V}$ $E^\circ = -0.61\text{V}$ (a) $6\text{OH}^- + \text{Br}_2 \rightarrow 3\text{H}_2\text{O} + 6\text{e}^-$; $E^\circ = -0.76\text{V}$ $7\text{OH}^- + \text{O}_2 \rightarrow 4\text{H}_2\text{O} + 4\text{e}^-$ (b) $E^\circ = -0.535\text{V}$ $E^\circ = 0.45\text{V}$ (b) $\text{HSO}_4^- + 4\text{H}^+ + 4\text{e}^- \rightarrow \text{S} + 3\text{H}_2\text{O}$; $E^\circ = 0.17\text{V}$ $\text{SO}_2 + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{HSO}_3^- + \text{H}^+$; $E^\circ = ?$ $\text{SO}_2 + 8\text{H}^+ + 6\text{e}^- \rightarrow \text{S} + 4\text{H}_2\text{O}$; $E^\circ = +1.45\text{V}$ (c) $E^\circ = +0.36\text{V}$ (c) $\text{ClO}^- + 6\text{H}^+ + 6\text{e}^- \rightarrow \text{Cl}^- + 3\text{H}_2\text{O}$; $E^\circ = +1.36\text{V}$ $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$; $E^\circ = ?$ $2\text{ClO}^- + 6\text{H}^+ + 5\text{e}^- \rightarrow \text{Cl}^- + 3\text{H}_2\text{O}$; $E^\circ = ?$ $2\text{ClO}^- + 6\text{H}^+ + 5\text{e}^- \rightarrow \text{Cl}^- + 3\text{H}_2\text{O}$; $E^\circ = -0.76\text{V}$ (s) $E^\circ = +1.47\text{V}$ (d) $\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn(s)}$; $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag(s)}$; $E^\circ = +0.80\text{V}$ $\text{Zn(s)} + 2\text{Ag}^+ \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{Ag(s)}$

Problem 51**[Cengage DPP Q22]**

For the following reaction, Integer Type

Problem 52**[Cengage DPP Q20]**

The voltage of the cell: Determine the value of $\ln K_{\text{eq}}$ $\text{Pb} | \text{PbSO}_4 | \text{Na}_2\text{SO}_4 (0.1\text{M}) | \text{O}^2-$ (salt) H_2 Hg , $\text{SO}_4^{2-} | \text{Hg}$ (a) 58.38 (b) 46.2 (c) 28.3 (d) 66.13 is $+0.9647$ at 25°C . The temperature coefficient is $1.74 \times 10^{-3}\text{V/K}$. With the increase in concentration of NH_3 , pH increases 10V/K . Calculate the value of ΔS° to 11 . Consider there is no change in concentration of

Problem 53**[Cengage DPP Q21]**

K of $\text{Cu}(\text{OH})$ is 1×10^{-19} . If reduction potential of $\text{Cu}^{2+}/\text{CH}_2\text{O}$ and CHO , on addition of NH_3 , the Cu couple is 0.1335 V in a solution and E° for Cu^{2+}/Cu is 0.34 V , the pH of solution is _____.

Comprehension Type (b) E_{ox} decreases by 0.65 V from E° (c) E_{red} increases by 0.65 V from E° (d) E_{red} decreases by 0.65 V from E°

Tollen's reagent is used for the detection of aldehyde when a solution of AgNO_3 is added to glucose with NH_4OH , then gluconic acid is formed. 24. Ammonia is always added in this reaction. Which of the following must be incorrect?

(a) Ag^+ easily forms complex with Ag^+ (b) $\text{Ag}(\text{NH}_3)_2^+$ is a strong oxidising reagent than Ag^+ . (c) In absence of NH_3 , silver salt of gluconic acid is formed. (d) NH_3 has affected the standard reduction potential of glucose/gluconic acid electrode.

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3.14 Physical Chemistry Answers Key Subjective Type Matching Column Type

Problem 54**[Cengage DPP Q2]**

0.522 V 5.0993 M , 0.0007 M

Problem 55**[Cengage DPP Q19]**

(a) $6.2 \times 10^{-12} \text{ M}$ (b) $2.9 \times 10^{-12} \text{ M}$ (c) $2.9 \times 10^{-12} \text{ M}$ (d) $6.2 \times 10^{-12} \text{ M}$

Integer Type Single Correct Answer Type

Problem 56**[Cengage DPP Q13]**

(c) Comprehension Type Multiple Correct Answers Type

Problem 57**[Cengage DPP Q24]**

(d) 14. (a, c) 15. (b, d) 16. (c, d) 17. (a, b, c)

Cengage Physical Chemistry: Comprehensive Chapter Exercises

From the following half-cell reactions and their standard () electrode potentials, what is the smallest possible standard e.m.f. for a spontaneous reaction?

Which of the following statements is correct about Galvanic cell. (1) It converts chemical energy into electrical energy. (2) It converts electrical energy into chemical energy. (3) It converts metal from its free state to the combined state. (4) It converts electrolyte into individual ions.

Which graph correctly correlates Ee as a function of cancestrioes fer dhe cell

Determine which substance is the better reducing agent in (3) +0.56 (4) +0.28 2xs) +2Ag (eg)—
 $\text{Zn (ag)} + 2\text{Ag(s)}, E=1.56\text{V}$ (I) HPO_4^{2-} (3) 109 (4) IO_3^- (2) PO_4^{3-} (4) E° , J-axis: leg; 9,
 Consider the following half-cell reactions and associated standard half-cell potentials and determine
 the maximum A9S'I voltage that can be obtained by combination resulting in (I) 1.56 V (2) AuBe ,
 (e) +)r sesseoud ssceuruds $\text{Eu(ad)} + \text{Ee(ag)} E^\circ = -0.43\text{V}$ $\text{Au(s)} + 4\text{Br}^- (\text{aq})$; and $E^\circ = 0.56\text{V}$ -1.0 1.0
 $\text{Al}^{3+} (\text{aq}) + 3\text{e}^- \rightarrow \text{Al(s)} E^\circ = -1.66\text{V}$ $\text{I}_2 (\text{s}) + 2\text{e}^- \rightarrow 2\text{I}^- (\text{aq}) E^\circ = 0.54\text{V}$ $\text{I}_2 (\text{s}) + 2\text{OH}^- (\text{aq}) \rightarrow 2\text{IO}_3^- (\text{aq}) + \text{H}_2\text{O(l)} E^\circ = 0.72\text{V}$ (2) + 1.54 $E^\circ = +0.49\text{V}$ -1.0 1.0 1.0 0 1.0

A solution coextruding H^* and D^* jets is in equilibrium with (3) + 1.00 (4) + 1.35

Give the standard electrode potential, what is the free energy of formation, and the partial pressures of H_2 and D_2 gases at 25 °C if the partial pressures of H_2 and D_2 are 1.0 atm, find the ratio of $[\text{D}]/[\text{H}]$; (b) $\text{H}_2 + \text{OHP} + (\text{g}) \rightleftharpoons \text{H}_2\text{O} + \text{H}^+ + \text{OH}^-$ (A E000 - x(p + SH*a) + 4r N₂, -0.23V Bioia) + 2H⁺(aq) 3r Bis) + H₂O(l), (1) 1.23 (3) 0.11 (4) 1.0 (2) 1.12 and = +0.32 V IIL The and 25 fr de following reactioe is 0.55. Calculane (t) +0.55 (2) +0.34 AG ia k : () +1.88 (4) + 0.99 4B0 () + 3N₂,* 4Bis) + 3N₂(g) + 4L,O(I) HClO (1) 637 (2) 424 + 7H⁺ BClO(g) + H⁺(aq) 2r Gives Cr(aq) CT (x) + H₂O(l)

Problem 7**[Cengage Chapter Exercises P85 Q12]**

A large quantity of cobalt metal was dropped into a solution of Cr^{3+} (aq) r , $= 0.41 \text{ V}$ containing Ag^+ and Fe^{2+} , all at its standard state. $E^\circ(\text{Ag}^+/\text{Ag}) = 0.799 \text{ V}$, $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V}$.

Problem 8**[Cengage Chapter Exercises P86 Q21]**

In an electrolytic cell, a common fews $E^\circ(\text{CC}) = -0.4 \text{ V}$ (1) From cathode to anode in outer circuit (Fe^{2+}) $= -0.04 \text{ V}$

Problem 9**[Cengage Chapter Exercises P86 Q3]**

From cathode to anode inside the cell (2) From anode to cathode inside the cell pur A ($S_{10} = (4/40) \cdot 2 \text{ E}$ ($\text{Co}/\text{Ce}) = 0.28 \text{ V}$, then the solution would be (4) None of the above (1) Ag^+/Fe and C (2) Ag^+/Fe and C (3) Fe , C and Co (4) Fe , Cu . Ag^+/Mg 13, Gives: $E^\circ(\text{Sa}, \text{Sn}^{2+}) = -0.15 \text{ V}$, (2) Can be made to occur in an electrolytic cell $E^\circ(\text{Hg} - \text{Hg}_2^{2+}) = 0.92 \text{ V}$ and (3) Can occur in acidic medium only $E^\circ(\text{Pb}^{2+}, \text{H}^+/\text{PtO}_2) = 1.45 \text{ V}$: Based on this data, which of (4) Can occur in basic medium only (1) Sn^{2+} is a stronger reducing agent than Pb (2) Mg is a stronger reducing agent than Fe (3) Fe is a stronger oxidizing agent than Pb (4) Pb is a stronger oxidizing agent than Fe (3) $E^\circ(\text{Sa}, \text{Sn}^{2+}) > 0$; $E^\circ(\text{Fe}^{2+}/\text{Fe}) < 0$; Quotient $Q < K$, is galvanic cell? (2) Se^{2+} in a drier medium than Hg^{2+} (2) Hg^{2+} is a stronger oxidizing agent than Pb (2) $E^\circ(\text{Mg}^{2+}/\text{Mg}) < 0$, $Q < K$, (4) Pb is a stronger oxidizing agent than Se (3) $E^\circ(\text{Sa}, \text{Sn}^{2+}) > 0$; $E^\circ(\text{Fe}^{2+}/\text{Fe}) < 0$, $Q > K$

Problem 10**[Cengage Chapter Exercises P86 Q14]**

Indicate electrode: (4) $E^\circ(\text{Ag}^+/\text{Ag}) > 0$, $Q < K$ (1) SHE (3) Ag electrode (2) Calomel electrode 1, $+2r$ 21; (4) Quinhydrone electrode $E^\circ = 0.54 \text{ V}$ (1) $E^\circ(\text{Zn}^{2+}/\text{Zn}) < E^\circ(\text{Fe}^{2+}/\text{Fe})$ of iron because

Problem 11**[Cengage Chapter Exercises P86 Q15]**

Given that: $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$ (2) $E^\circ(\text{Zn}^{2+}/\text{Zn}) > E^\circ(\text{Fe}^{2+}/\text{Fe})$, $E^\circ(\text{Zn}^{2+}/\text{Zn}) > E^\circ(\text{Fe}^{2+}/\text{Fe})$ (3) of $\text{Zn} = E^\circ(\text{Fe}^{2+}/\text{Fe})$ (1) 1 ions will be able to reduce bromine.

Problem 12**[Cengage Chapter Exercises P86 Q25]**

The standard potential of a hydrogen electrode is 0 V (4) Zn is cheaper than iron (2) Br^- ions will be able to reduce iodine. G) Iodine will be able to reduce bromine. and P , $= 1 \text{ atm}$ is (4) Bromine will be able to reduce iodine ions. $E^\circ(\text{Mg}^{2+}/\text{Mg}) = -2.37 \text{ V}$, $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$, and $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V}$ (1) 0.39 V (2) 0.00 V (3) 0.65 V (4) 0.65 V (f)

Problem 13**[Cengage Chapter Exercises P86 Q16]**

As an electrochemical cell stops working after some time because is correct? 0.76 V , and 0.44 V , respectively. Which of the following (1) Electrode potential of both the electrodes becomes zero. (2) Electrode potential of both the electrodes becomes equal. (1) Mg oxidizes Fe (3) Zn reduces Mg^{2+} (2) Zn oxidizes Fe (3) One of the electrodes is eaten away. 17, The solution of CuSO_4 in which copper rod is immersed (4) Zn reduces Fe^{2+} 17, Which of the following statements is correct for a galvanic cell (4) The reaction starts proceeding in opposite direction. diluted to 10 times. The reduction electrode potential cell? (1) Increases by 30 mV (1) Reduction occurs at cathode (3) Increases by 59 mV (2) (2) Oxidation occurs at anode

Problem 14**[Cengage Chapter Exercises P86 Q28]**

Deduce from the following values of half cells $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$ (3) Electrodes from anode to cathode combination of two half-cells would result in $E^\circ(\text{cell}) = 0.44 \text{ V}$ (4) All the statements are correct the largest potential?

Problem 15**[Cengage Chapter Exercises P86 Q18]**

Given $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34 \text{ V}$, $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$. Which metal will corrode fastest? (1) Ag (2) Cu (3) Zn (4) None of these

Problem 16**[Cengage Chapter Exercises P86 Q19]**

Red hot carbon will remove oxygen from the oxides XO (3) Cu (4) Zn (1) Pb, IV D, and YO but not from ZO. Y will remove oxygen from XO. (2) IL, III (3) ML, IV (4) I, II Use this evidence to deduce the order of activity of the three metals X, Y, and Z, putting the most reactive first. Z'XA(0 XX'Z(2D ZXX(1) If Fe, Fe and Fe solid are kept together, m

Problem 17**[Cengage Chapter Exercises P86 Q20]**

Among Na, Hg, S, P, and graphite which can be used as (4) Z, X, Y (1) Fe increases electrodes in electrolytic cell having aqueous solutions? (2) Fe decreases (1) Na and S (2) Hg, Pt, and S (3) Fe, Feb remains unchanged (3) Na, Hg, and S (4) Hg, Pt, and graphite (4) Fe* decreases

Problem 18**[Cengage Chapter Exercises P87 Q42]**

The EMF of a concentration cell consisting of two zinc (I) (-36 - 0.439) V electrodes, one dipping in M/4 solution of ZnSO₄ and the other in M/16 solution of the same salt at 25 is (1) (-0.6 - 0.409) V (2) [3(0.36) + 2(0.439)] V (3) [X-4.36]2(-0.439)] V (4) J. A certain electrode has standard (reduction potential) of . The standard electrode potential of a metal ion (Ag I Ag) 0.34 V. The potential when measured against a normal calomel electrode (with electrode potential = 0.28 V) is (1) 0.104 (2) 0.664 (3) 0.332 (4) 0.218 RT F ln K,

Problem 19**[Cengage Chapter Exercises P87 Q44]**

The potential of a hydrogen electrode in a solution with pH = 5.25 is (1) $E = E^\circ - \frac{RT}{n} \ln a_{\text{H}^+}$, (I) +0.59 (2) 0.59 (3) -0.295 (4) 0.295

Problem 20**[Cengage Chapter Exercises P87 Q45]**

The potential of a hydrogen electrode in a solution with (3) $E = E^\circ - \frac{RT}{n} \ln [CT] pOH = 4 = 25$ is K (1) 0.059 (2) 6500 (3) 6500 (4) 0.59 RT Ke

Problem 21**[Cengage Chapter Exercises P87 Q46]**

Consider the electrode $\text{Ag}|\text{AgCl(s)}, \text{Cl}^- (0.1 \text{ M})$, i.e., silver electrode in contact with 0.1 M KCl solution saturated with AgCl. If it is combined with the electrode $\text{Ag}|\text{Ag}^+ / \text{percent}$. The standard reduction potential at 25 for the reaction, $(\text{K}, \text{efAgCl} = 10^{-25}) (0.1 \text{ M})$ to form a complete cell, the EMF would be 28,0 + 2r vt H, + 20H is 0.8277 volt The equilibrium constant for the reaction (1) 0.799 (2) 0.636 (3) 650 (4) 0.472 O, St / No. $\text{HO} + \text{OH} = \text{OH}^-$ 47, The standard EMF of decinormal calomel electrode is (1) 0.298 V (2) 0.327 V (3) 0.327 V (4) 0.298 V 0.268 V. The EMF is

Problem 22**[Cengage Chapter Exercises P87 Q31]**

What would be the magnitude of the EMF of the following (I) 2 (2) 19- (3) 18-11 (4) 194 48, The standard EMF of quinhydrone electrode is 0.699 V. The cell EMF of the quinhydrone electrode dipped in a solution with [pH = 10] P(H₂) pH = 10 is H(H₂) 0.1 M sodium 0.1 M sodium (1) 0.109 V (2) 0.109 V acetate forate

Problem 23**[Cengage Chapter Exercises P87 Q49]**

A hydride ion is oxidized to sodium (3) 1.289 V A 681() 25 7 The ionization constant of acetic acid, and, 10⁻, while that of formic acid, K, 10. (1) 8.0295 V (2) 0.059 V (3) 0.059 V (4) 8295 V Electrode potential values, respectively, are 25. The pK_a value of acetic acid is (1) (6, + E) (2) 2 0.059 E + E the salt bridge is removed, the voltage is 2 0.059 (3) E-E (4) (6 + E) (1) Does it change 2 0.059 0.059

Problem 24**[Cengage Chapter Exercises P88 Q50]**

Given standard 0.59, For the half cell $\text{Fe} + \text{r} = \text{.} = -0.440 \text{ V}$ $g = 0.036 \text{ V}$ $\text{OH Fe} + 2\text{r} = \text{Fe}$: (1) 0.476 V (2) -0.404 V (3) 0.404 V (4) 0.772 V The standard $\text{Fe} + \text{r} = \text{Fe}$ is

Problem 25**[Cengage Chapter Exercises P88 Q51]**

For the cell $\text{Zn} | \text{Zr Cr (Ce)}, \text{the standard cell voltage is } 1.10 \text{ V}$: When a cell in these mgas HX was prepared in the lab, the measured cell voltage was At plf 2, the electrode potential is 0.96 V. One possible explanation for the observed voltage is (1) 1.36 V (2) 1.30 V A 2F1 () (1) The worst 200 mol of Zs^* but only 1.00 mol of C (4) 126 V (2) The Zn electrode had twice the surface of the Cu electrode: $\text{Co} | \text{CoCl}_2, \text{M}, 1 \text{HCl}, \text{M}, \text{P}$ (3) $\text{Te} | \text{Zn}$ was larger than the $[\text{Cv}]$ electrode. (4) The volume of the Zn solution was larger than the 200 mL. volume of the Ce^* solution. (3) In case of 102 N 0.50 0.9 N 10/6 uog as a use (2)

Problem 26**[Cengage Chapter Exercises P88 Q52]**

$\text{SG} = \text{SA Ta Sand G} - - 7 \text{ aG}$ 6L. $\text{Ag} | \text{Ag} (1 \text{ M}) [\text{Ag} [\text{Ag} (4) \text{Increase M, from } 0.91 \text{ to } 0.59 \text{ M. T (I) S (2) xF S (3) sFE (4) nFEt } 0.5 \text{ F electricity is used LS (anode and 1 F of elect)} 1 \text{ L xleie } 1 \text{ L solution}$

Problem 27**[Cengage Chapter Exercises P88 Q53]**

2s $\text{CV}() = \text{C} + \text{Za(aq)}$. the RHS (cathode) is first passed making bile Reaction quixtiestis Q $[\text{Zn}]$ cell 29 KEMF the $[\text{Cu}]$ Variation of E, with Q will be E/ wi 1.1591 V when (1) increased 16 - L, = 21, *0 (1) (2) Decreased 2) $[2e]/[C] = 0.0$ (3) No change 10 - L; o1/1., az]0 (4) Time is also required (4) $[2e]/[0] - 1$

Problem 28**[Cengage Chapter Exercises P88 Q62]**

The cell EMF is independent of the concentration (ft 54, $\text{Cv} + \text{Cs} = 1$, vol species of the cell in (1) Fe 1 $\text{FeO(x)} | \text{KOH(aq)} | \text{NiO(o)} (\text{Ni}, 0,) \text{Cr} + \text{Cv}^*$, E (vol) will be (2) R (L,) IHC (CI) P (1jx, 2x, (2)x, *2, G)x, (4) 2s, 3 (3) Z|Z 1Cv |C

Problem 29**[Cengage Chapter Exercises P88 Q55]**

Zs Zs/ percent s, 1Zslc,) Zn For @is cell. G is negative if:

Problem 30**[Cengage Chapter Exercises P88 Q63]**

Given the standard potential 2C (4) $\text{Hg}_2\text{Hg}; \text{C}_3, \text{KCl AgNO}_3, \text{Ag} (, *, (c, >, 0)c >$,

Problem 31**[Cengage Chapter Exercises P88 Q56]**

P(H, (P, IH (1 M) I0, (e, IP cell reaction will be (4) Nonc Ms0, ; Mt ; see thermodynamic if The standard potential of MsO , — Ms is (1) ; (2, > (4)p, 1 atm (1) -0.5 V (2) -246 V (3) -123 V (0.12 V 51, P(C,) $[\text{HCl } 1.1 \text{ M}] \text{Cl}_1$, (e, cell reaction will

Problem 46

[Cengage Chapter Exercises P92 Q118]

of different half cells are given as: (1) $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^-$, $P = 0.34 \text{ V}$, $E = -0.76 \text{ V}$. (4) $\text{B}_2\text{O}_3 + 6\text{H}^+ + 12\text{e}^- \rightarrow 2\text{B} + 3\text{H}_2\text{O}$, $E = 0.80 \text{ V}$. $E = 2.37 \text{ V}$.

Problem 47

[Cengage Chapter Exercises P92 Q126]

In a passing 3 F of electricity through three electrolytic cells connected in series containing Ag⁺, Cu²⁺, and Al³⁺ in which cell is most negative? respectively. The molar ratio in which the three metals (1) Zn | Zn²⁺ (1 M) | Mg (1 M) are liberated at the electrodes is (2) Zn | Zn²⁺ (1 M) | Ag (1 M) | Ag (1 M) 1 : 2 : 3 (2) 2 : 3 : 1 (3) Ca | Co (1 M) | Ag⁺ (1 M) | Ag (3) 6 : 3 : 2 (4) 3 : 4 : 2 127, During electrolysis of molten calcium hydride, the byproduct (4) Ag | Ag⁺ (1 M) | Mg (1 M) | Mg is produced at 119, For the reaction: (1) Cathode $\text{A} + 2\text{B} \rightarrow 2\text{B} + \text{A}$, (2) Anode E of the given redox reaction is: (4) H₂, produced reacts with oxygen to form (3) Hydrogen is not liberated at all $\text{A}^{2+} + 2\text{e}^- \rightarrow \text{A}$

Problem 48

[Cengage Chapter Exercises P92 Q128]

Which of the following is anodic reaction? $\text{S}^{2-} + 8\text{H}^+ \rightarrow \text{S} + 4\text{H}_2\text{O}$, $\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{SO}_2 + 2\text{H}_2\text{O}$ (1) $\text{x} + 2\text{y}$ (2) $\text{x} + \text{y}/2$ (3) $\text{x} - \text{y}$ (4) y (2) $\text{H} + 1/2\text{H}_2$

Problem 49

[Cengage Chapter Exercises P92 Q120]

Excess of solid AgCl is added to a 0.1 M solution of Br⁻ (3) $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$ for half cell: (4) None of these $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$; $E = 0.095 \text{ V}$ $\text{AgCl} + \text{e}^- \rightarrow \text{Ag} + \text{Cl}^-$; $E = 0.222 \text{ V}$ ions to Cr is 9 tungsten atoms [08], $\log_{10} 4$ (1) 96500 C (3) 3 96500 C (2) 2 96500 C (4) 6 96500 C [Gives: $\text{Antilog}(2.152) = 142$] (1) 0.0317 M (2) 0.013 M (3) 0.99 M (4) 0.099 M (1) 96500 C (3) 9650 C (2) 2 96500 C (4) 96.50 C

Problem 50

[Cengage Chapter Exercises P93 Q142]

Which of the following is a strong electrolyte? per 0.5 mole of Na_2CO_3 (1) $\text{Ca}(\text{NO}_3)_2$, (2) HCN (1) 1/2 (3) 3 (4) 1.5 (3) H_2SO_4 , (4) NH_4OH H₂SO₄ can be oxidized by a current 1 mole of B_2O_3 to O_2 of 2 ampere hours? (119) 000 C (2) 1.93 10 c (1) 0.746 L (2) 7.46 L (4) 19.3 [0 C] (1) 1.482 L (4) 0.373 L 144, 500 mL of 1 N solution of CuCl_2 was electrolyzed with a constant current, I (1) 1150, (2) 0, (3) 0, (4) 11, the remaining CuCl_2 solution? (1) 0.85 (2) 8.15 (3) 9.16 (4) 1.9 dattdes 145, the strength of 0.4 M CaCl_2 solution, is evolved at anode and O_2 at cathode (1) 1.2 (2) 1.0 (3) 0.9 (4) 9.8 1N, is produced at anode 146, ionic strength of 0.2 M Na_2PO_4 is Cl^- is retained at cathode (1) 2.8 (2) 1.2 (3) 1. (4) 1.3 (40, is produced at cathode and H_2 at anode 147, ionic strength of a solution made by mixing the volumes

Problem 51

[Cengage Chapter Exercises P93 Q8]

During electrolysis of fused NaCl , with reaction excess of 0.01 M NaCl and 0.02 M AlCl_3 , is E in (1) 9.965 (2) 0.1 (3) 0.0325 (4) 0.0216 (1) Chloride ions are oxidized 2 Chloride ions are reduced

Problem 52

[Cengage Chapter Exercises P93 Q148]

Marschall's acid is prepared by the electrolytic oxidation of H_2SO_4 as

Problem 53**[Cengage Chapter Exercises P93 Q3]**

Sodium ions are oxidized $2\text{H}_2\text{SO}_4 \rightarrow \text{B}_2\text{S}_3\text{O}_6 + 2\text{H}_2 + 2\text{SO}_2$. Sodium ions are reduced D/ percent. Two platinum electrodes were immersed in a solution. Oxygen and hydrogen are byproducts. In such electrolysis 224 L of H_2 and 0.56 L of O_2 were produced = STP. The effluent of CeSO_4 and electric current was passed through the weight of $\text{H}_2\text{S}_2\text{O}_8$ formed is solution. After some time, it was found that color of CuSO_4 (1) 9.7 g (2) 19.4 g (3) 14.55 g (4) 29.1 g disappeared with evolution of gas at the electrode. The colorless solution contains 149, The volume of gases evolved at STP by passing 0.1 A of (1) sulphuric acid (2) Copper hydroxide consumed 965 s, formed an aqueous solution of sodium / Copper sulphate (4) Sulphuric acid acetate [71, Wm conductivity is passed through a solution of AlCl_3 and 15L. The mass of products evolved in Q. 149 is 965 (1) 11 (2) 1W 22 (1) (4) +4.8 mL 3.5 g of acid, the number of Farad (1) 9.5 (2) 1.9 (3) 1.5 (4) 2.0 (1) 9.06 g (2) 0.6 g (3) 6.0 g 15L. The volume of gases evolved at STP by passing 0.2 A of (4) 60 g 2 find the ratio of cathodic to anodic current for 95 s through an aqueous solution of sodium) of Cs^+ - 61.5 e (1) 11 (2) +22 (1) 1; 2.9 (2) 3.17 (3) 63.5 (4) 127.8

Problem 54**[Cengage Chapter Exercises P93 Q152]**

The products at the cathode and anode on electrolysis (4) 44.5 mL Metal which is monovalent. The mass of the metal of aqueous sodium succinate are (1) Anode : $\text{H}_2\text{SO}_4 + \text{CO}_2$, Cathode : H_2 ,

Problem 55**[Cengage Chapter Exercises P93 Q1]**

10 (2) 5 (3) 39 (4) 96.5 (2) Anode : H_2 + O_2 , Cathode : H_2 , (3) Anode : H_2 + B_2 , Cathode : Na my opposite half cell is 153, A current strength of 1.0 A is used for 96.5 s. Brought (4) Anode : H_2 + H_2 Cathode : Na (1) 12.7 g (2) 15.9 g (3) 31.8 g (4) 63.5 g the final solution with respect to O_2 ions is 190 st of a solution of O_2 M KCL. The concentration of C_2M is 0.01 M (1) 0.001 M (2) 0.01 M (3) 0.001 M (4) 0.001 M (5) 0.01 M

Problem 56**[Cengage Chapter Exercises P93 Q154]**

Is Q 153 pH of the solution will be (1) 9.095 M (2) 0.05 M (3) 0.01 M (4) 0.001 M (5) 0.01 M (6) 0.001 M (7) 0.01 M (8) 0.001 M (9) 0.01 M (10) 0.001 M (11) 0.01 M (12) 0.001 M (13) 0.01 M (14) 0.001 M (15) 0.01 M (16) 0.001 M (17) 0.01 M (18) 0.001 M (19) 0.01 M (20) 0.001 M (21) 0.01 M (22) 0.001 M (23) 0.01 M (24) 0.001 M (25) 0.01 M (26) 0.001 M (27) 0.01 M (28) 0.001 M (29) 0.01 M (30) 0.001 M (31) 0.01 M (32) 0.001 M (33) 0.01 M (34) 0.001 M (35) 0.01 M (36) 0.001 M (37) 0.01 M (38) 0.001 M (39) 0.01 M (40) 0.001 M (41) 0.01 M (42) 0.001 M (43) 0.01 M (44) 0.001 M (45) 0.01 M (46) 0.001 M (47) 0.01 M (48) 0.001 M (49) 0.01 M (50) 0.001 M (51) 0.01 M (52) 0.001 M (53) 0.01 M (54) 0.001 M (55) 0.01 M (56) 0.001 M (57) 0.01 M (58) 0.001 M (59) 0.01 M (60) 0.001 M (61) 0.01 M (62) 0.001 M (63) 0.01 M (64) 0.001 M (65) 0.01 M (66) 0.001 M (67) 0.01 M (68) 0.001 M (69) 0.01 M (70) 0.001 M (71) 0.01 M (72) 0.001 M (73) 0.01 M (74) 0.001 M (75) 0.01 M (76) 0.001 M (77) 0.01 M (78) 0.001 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Problem 60**[Cengage Chapter Exercises P94 Q162]**

Adil us stfsimideis elrly (1) 0.01 N (2) 0.01 M (3) 0.002 M (4) 0.62 M e prducts at fe atode and caflode are (1) O, H, (2) F, Na (3) O, Na (4) F, H, made using Pt electrodes? 163, Copper can be deposited from acidified copper sulphate and (1) pH of solution increases. (2) pH of solution decreases. (3) The amount of copper deposited from acidic copper

Problem 61**[Cengage Chapter Exercises P94 Q2]**

The amount of copper deposited from a solution of copper sulfate is 1.0 g. The amount of copper deposited from a solution of copper sulfate is 1.0 g.

Problem 62**[Cengage Chapter Exercises P94 Q172]**

Select the weight of silver. (1) 1.0 g (2) 1.5 g (3) 2.0 g (4) 2.5 g. The same amount of copper will be deposited (4) Nost of the acid is consumed. 144, Silver is removed electrolytically from 200 mL of a 0.1 N solution of AgNO_3 by a current of 0.1 A. How long will it take to remove half of the silver from the solution?

Problem 63**[Cengage Chapter Exercises P95 Q188]**

0.224 L H_2 is formed at the cathode of a cell at (1) 2.2 V (2) 1.5 V (3) 1.0 V (4) 0.5 V. The amount of copper deposited at the cathode of the other cell. During the electrolysis of AgNO_3 , the volume of O_2 evolved electrolytic cell arranged in series? STP. The usage of 2 A of current for 965 s is (1) 0.24 g (2) 24 g (3) 0.43 g (4) 4.8 g (5) 112 L (6) 11.2 L (7) 22.4 L (8) 224 L. A certain amount of charge is used to produce H_2 and O_2 (g) were evolved at STP: 1 mol of CaCl_2 by a current of 25 mA for 60 s will be. What is the magnitude of charge passed during electrolysis? (1) 4.68×10^4 C (2) 4.68×10^5 C (3) 4.68×10^6 C (4) 4.68×10^7 C. 1 L of 1 M CuSO_4 solution is electrolyzed using Pt cathode and Pt anode. After passing 2 F of electricity, the number of moles of gases obtained at cathode, anode, and total will be (1) 9 (2) 10 (3) 11 (4) 12 (5) 13 (6) 14 (7) 15 (8) 16 (9) 17 (10) 18. How many Faradays are required to reduce 1 mole of BO_2 to B? (1) 2 (2) 3 (3) 4 (4) 5 (5) 6 (6) 7 (7) 8 (8) 9 (9) 10 (10) 11. During electrolysis of acidified water, O_2 gas is evolved (1) 6 (2) 5 (3) 4 (4) 3 (5) 2 (6) 1 (7) 0. To produce O_2 gas at the anode at the rate of 1 L per minute, the current required is (1) 1 A (2) 2 A (3) 3 A (4) 4 A (5) 5 A (6) 6 A (7) 7 A (8) 8 A (9) 9 A (10) 10 A.

Problem 64**[Cengage Chapter Exercises P95 Q192]**

Which of the following gives solution molarity (1) 3.86 A (2) 2.24 A (3) 9.64 A (4) 42.24 A. After electrolysis? (1) K_2SO_4 (2) NaCl (3) AgNO_3 (4) CuSO_4 . The amount of current passed through moles of NaCl 0.01 and the amount of Al deposited in the electrolytic cell is 9 W (1) 0.9 g (2) 1.8 g (3) 2.7 g (4) 3.6 g (5) 4.5 g (6) 5.4 g (7) 6.3 g (8) 7.2 g (9) 8.1 g (10) 9.0 g.

Problem 65**[Cengage Chapter Exercises P95 Q194]**

Two platinum electrodes were immersed in a solution of CuSO_4 , and a direct current was passed through the solution. After some time, it was found that the color of CuSO_4 had disappeared with the evolution of O_2 at the anode. (1) The concentration of Cu^{2+} ions in the solution is (1) 0.1 M (2) 0.2 M (3) 0.3 M (4) 0.4 M (5) 0.5 M (6) 0.6 M (7) 0.7 M (8) 0.8 M (9) 0.9 M (10) 1.0 M. The amount of Al deposited in the electrolytic cell is 9 W (1) 0.9 g (2) 1.8 g (3) 2.7 g (4) 3.6 g (5) 4.5 g (6) 5.4 g (7) 6.3 g (8) 7.2 g (9) 8.1 g (10) 9.0 g.

Problem 66

[Cengage Chapter Exercises P96 Q28]

A 0.1 M solution of acetic acid is 250 Ω cm. The resistance of a 0.1 N solution of acetic acid is 250 Ω cm. The conductivity of an electrolyte A, B is when measured in a cell of cell constant 1.15 cm. The (1) $\times 10^{-9}$, (2) $\times 10^{-8}$, (3) $\times 10^{-7}$, (4) $\times 10^{-6}$.

Problem 67

[Cengage Chapter Exercises P96 Q1]

The equivalent conductance of acetic acid is (3) 200 Ω^{-1} cm² eq⁻¹ (1) 46.0 (2) 9.2 (3) 18.4 (4) 2.3. The values of Λ , for NH₄Cl, NaOH, and NaCl are.

Problem 68

[Cengage Chapter Exercises P96 Q2]

The equivalent conductances of BaCl₂ and CaCl₂ are 63.5 Ω^{-1} cm² eq⁻¹ and 149.74, 248.1, and 126.4 Ω^{-1} cm² eq⁻¹ at infinite dilution. The equivalent conductance (in Ω^{-1} cm² eq⁻¹) of (1) 371.44 value of Λ , NH₄OH is BaCl₂, at infinite dilution will be dilution. The equivalent conductance (in Ω^{-1} cm² eq⁻¹) of (2) 271.44 (3) 203.0 (3) 279.0 (4) 101.15 (3) 71.44. The equivalent conductance of an electrolyte MA at

Problem 69

[Cengage Chapter Exercises P96 Q210]

The increase in the molar conductivity of HCl via dilution (4) It cannot be calculated from the data given. The increase in the molar conductivity of HCl is (more than the correct answer) is due to (2) Λ^∞ (1) Increase in the self ionization of water (3) Λ^∞ (2) Hydrolysis of HCl (3) Decrease in the self ionization of water (4) Λ^∞ , Λ^∞ , Λ^∞ (4) Decrease in the interionic forces

Problem 70

[Cengage Chapter Exercises P96 Q282]

Which of the following has the highest ionic mobility? (1) Li⁺ (2) F⁻ (3) SO₄²⁻ (4) Co^{2+} . Callate the solubility product of Co₃(PO₄)₂ at 25 $^\circ\text{C}$ (2) 10^{-28} (3) 10^{-26} (4) 10^{-24} .

Problem 71

[Cengage Chapter Exercises P96 Q203]

Λ^∞ of BaCl₂, H₂SO₄, and HCl are 120, 80, and 430 Ω^{-1} cm² eq⁻¹, respectively. The Λ^∞ of H₂SO₄ is (1) 80 (2) 3 (3) 4 (4) 444.9 Ω^{-1} cm² eq⁻¹, respectively (3) $\times 10^{-3}$, (2) $\times 10^{-2}$, (4) $\times 10^{-1}$, (1) $\times 10^{-1}$.

Problem 72

[Cengage Chapter Exercises P96 Q204]

Λ^∞ of BaCl₂, H₂SO₄, and HCl are 120, 80, and 430 Ω^{-1} cm² eq⁻¹, respectively. The Λ^∞ of H₂SO₄ is (1) 80 (2) 3 (3) 4 (4) 444.9 Ω^{-1} cm² eq⁻¹, respectively (3) $\times 10^{-3}$, (2) $\times 10^{-2}$, (4) $\times 10^{-1}$, (1) $\times 10^{-1}$.

Problem 73

[Cengage Chapter Exercises P96 Q295]

Which of the following statements is correct? (4) None of the above. The Van't Hoff factor for 10% NaCl solution is 39/35 (Λ^∞ = 350 Ω^{-1} cm² eq⁻¹ and Λ = 4 $\times 10^3$ Ω^{-1} cm² eq⁻¹). The acid with dilution is due to

Problem 74

[Cengage Chapter Exercises P97 Q222]

For a dilute solution of a strong electrolyte, the variation of increase with dilution. molar conductivity with concentration is given by (1) $\Lambda^\infty - \Lambda^\infty \sqrt{c}$ and (2) $\Lambda^\infty = \Lambda^\infty / \sqrt{c}$ (3) HCl is strongly acidic (3) $\Lambda^\infty - \Lambda^\infty + \Lambda^\infty c$ (4) $\Lambda^\infty = \Lambda^\infty + \Lambda^\infty c$

Problem 75

[Cengage Chapter Exercises P97 Q223]

A cell constant is generally found by measuring the conductivity of aqueous solution of KCl. The cell constant is

Problem 76**[Cengage Chapter Exercises P97 Q224]**

A solution of sodium sulphate was electrolyzed using some (1) BaCl, (2) KCl (3) NaCl (4) MgCl, (I) +4 (2) +3 (3) +2 (4) +1 inert electrode. The products at the electrodes are 15maphecs-setil fach letde 225, The resistance of 1 N solution of acetic acid is 250 ohm, (1) O, H, (2) O, Na (3) O, 50, (4) O, S, O, is 0.75 cm. What is the cell constant? (I) 1.125 (2) 0.5 cm (3) 2.0 cm⁻¹ (4) 0.2 cm equivalent conductance (in ohm⁻¹ cm² eq⁻¹) of 1 N acetic when measured is a cell of cell constant 1.15 cm. The Λ^∞ of acetic acid is 0.0110 S cm² mol⁻¹. The molar conductivity of acetic acid is (1) 4.6 (2) 9.2 (3) 18.4 (4) 0.023 (1) 8.055 cm² mol⁻¹ / No. (2) 550 S cm² mol⁻¹

Problem 77**[Cengage Chapter Exercises P97 Q226]**

Given the ionic equivalent conductivities for the following ions: (1) 9.22 S cm² mol⁻¹ (4) 220 S cm² mol⁻¹ Λ^∞ of H₂SO₄ is 73.5 S cm² mol⁻¹ eq⁻¹ AP 149 cm² mol⁻¹ eq⁻¹ If the specific conductance of 1 M H₂SO₄ solution is 26×10^{-3} S cm⁻¹, then the equivalent conductivity of H₂SO₄ is 85.8 cm² mol⁻¹ eq⁻¹ AP 149 cm² mol⁻¹ eq⁻¹ The Λ^∞ of K₂SO₄ is (1) 1.3 $\times 10^3$ S cm² mol⁻¹ (2) 1.6 $\times 10^3$ S cm² mol⁻¹ (3) 368.2 (4) 108.52 (1) 13 S cm² mol⁻¹ (4) 1.3 $\times 10^3$ S cm² mol⁻¹ 227, Gives the ionic conductance of COO⁻, K⁺, and Na⁺ are 18, The highest electrical conductivity of the following aqueous solutions is of 74, 50, and 73 cm² ohm⁻¹ eq⁻¹, respectively. The equivalent (1) 0.1 M acetic acid conductance at infinite dilution of the salt COONa (2) 0.1 M ethanoic acid COOK (3) 0.1 M formic acid (1) 197 cm² ohm⁻¹ eq⁻¹ (2) 172 cm² ohm⁻¹ eq⁻¹ (4) 0.1 M diformic acid (3) 135.5 cm² ohm⁻¹ eq⁻¹ (4) 160.5 cm² ohm⁻¹ eq⁻¹ (1) The conductance of 1 cm of a material is called specific

Problem 78**[Cengage Chapter Exercises P97 Q228]**

The variation of equivalent conductance of strong electrolyte conductance with concentration is correctly shown in the figure conductance decreases on progressive dilution. (1) A (2) B of the plot of Λ against concentration. (3) A (4) B of dilution.

Problem 79**[Cengage Chapter Exercises P98 Q236]**

Rusting of iron is (4) All of the above are true. End point End point

Problem 80**[Cengage Chapter Exercises P98 Q230]**

20 mL of KOH solution was titrated with 0.20 M H₂SO₄, (2) A photochemical process (1) A decomposition process solution in a conductivity cell The data obtained were plotted to give the graph shown below. (3) An electrochemical cell (4) A redox process 80- 237, The cell reaction of a cell is (1) $\text{O}_2 + 4\text{H}^+ \rightarrow 2\text{H}_2\text{O}$ (2) $\text{O}_2 + 4\text{H}^+ \rightarrow 2\text{H}_2\text{O}$ (3) $\text{O}_2 + 4\text{H}^+ \rightarrow 2\text{H}_2\text{O}$ (4) $\text{O}_2 + 4\text{H}^+ \rightarrow 2\text{H}_2\text{O}$ 30 238, The cathode reaction during the charging of a lead-acid (800 + (3) + (4)) Tiolurme of acid added (mL) The concentration of the KOH solution was better leads to the (1) 0.30 mol L⁻¹ (2) 0.15 mol L⁻¹ (1) Formation of PbSO₄ 0.12 mol L⁻¹ (4) 0.075 mol L⁻¹ (2) Reduction of Pb²⁺ to Pb 231, Λ^∞ of acetic acid is 4.55 $\times 10^{-3}$ S cm² mol⁻¹ for a 0.001 M solution. The reciprocal (3) Formation of PbO of the degree of dissociation of acetic acid, if Λ^∞ of acetic acid is 400 S cm² mol⁻¹ 239, Which of the following cells is reversible (4) Deposition of Pb at the anode

Problem 87**[Cengage Chapter Exercises P99 Q257]**

A graph was plotted between molar conductivity of various $x + 7 \text{ K } 188 (1) 322 \text{ kJ mol}^{-1}$ 3152 kJ mol^{-1} (2) 161 kJ mol^{-1} electrolytes (NaCl, HCl and NH_4OH) and C (in mol L^{-1}) (4) 76 g mol^{-1} Coerret set is: cell K and of the cell are (1) $9.55 > 10$ percent, 1.096 V (2) $9.55, 1.096 \text{ V}$ (3) $1.023 \times 10^{-2}, 2.85 \text{ V}$ (4) $5 > 10, 0.55 \text{ V}$ (tyC and OH), (2) Ni(OH)_2 ; (1) I percent C), $\text{II(ICH}_2\text{OH)}$ G) OH (4) Nosr of thee (2) HCl , INaCD , $\text{II(N}_2\text{OH)}$ (3) $\text{I(NH}_4\text{OH)}$, IB(NaCD) , III(HCl) 25 percent, IINO , b itrated win NaOH(aq) conductometrically. (4) $\text{I(NH}_4\text{OH)}$, IB(HT) , $(\text{NGCl} + r \times \text{O}, \text{HO(I)})$ A $(\text{Zn} + \text{Ie (a)} + 2r \text{ Feos } 0.44 \text{ V}$ gtaptical rpeesenlatioe of te tieaties is Cfovlase aGP for he set proces. (1) 322 kJ mol^{-1} (2) 152 kJ mol^{-1} (1) -76 kJ mol^{-1} (4) 161 kJ mol^{-1} (Z WOP SJPPR Wfof NaOB

Problem 88**[Cengage Chapter Exercises P100 Q260]**

Coodscetmetric titice csre ef a oqrineler mishure ef $x(\text{HON})$ (NOH PRT IOH (I) (2) R. R. Vil of alalne Volefskslist (3) (4) soltion (3) 266 AgNO_3 , aqasadn aqskCse Wl ef aloline sltios Vil of allkaline of oondectunce A vs volume of AgNO_3 (aq) ic:

Problem 89**[Cengage Chapter Exercises P100 Q261]**

A layer of cheomtiem mxtal 0.25 mm tidk is so he plated-on an ao umper w atol a ef 0.32 m fma slatie oomtaining CrO_2 ? Wat cumet fow is neuired fer fs (1) A (2) A elptsg if de bmper is to b plad is 60 s ? The detsity of chromium metal s 7.20 g cm^{-3} (1) 4.9 MA (3) $5.3 \times 10^{-4} \text{ A}$ (2) $1.78 \times 10^{-4} \text{ A}$

Problem 90**[Cengage Chapter Exercises P100 Q262]**

Calcslatc the potential of a half cell having rcactios: (4) $10.69 \times 10^{-4} \text{ A Ag}_2\text{S(s)} + 2r \text{ 2Ag(s)} + \text{S (aq)}$ in a solution (3) A (4) A buffend pl = 3 and wtich is abo sluralcd with $9.1 \text{ M H}_2\text{S(aq)}$: [Given: $K_{\text{sp}}(\text{Ag}_2\text{S}) = 10^{-49}$, x , $X \times 10^{-21} \text{ 267}$, In the electrochemical cell. (1) 1.18 (2) 0.19 V (3) 0.19 V (4) 1.18 V (aq) | $\text{Zn(s)} | \text{Zn}^{2+}(\text{aq})$ (f = +3.1X

Problem 91**[Cengage Chapter Exercises P100 Q263]**

Eeuivalent conductivity of BaCl_2 , H_2SO_4 , and HCl , are x (3) 0.19 V (4) None of these The correct plot of E_{cell} versus $\log [\text{Mg}^{2+}]$ s, and a, Sm eq al inie dtio. If cdcitivity of represesled as: $[\text{Zn}^{2+}]$ snemed BaSO_4 soios is $x \text{ Sema}$, hes X of BaSO_4 is

Problem 92**[Cengage Chapter Exercises P101 Q4]**

A gas X at 1 atm is bebed trough a solution containing Mg^{2+} a rtixture of 1 M Y and 1 M Z at 25°C . If the reduction $[\text{Zn}]$ potestial is $Z > Y > X$, fher (1) Y will exifize X cely g aepxo 1 X (2) (4) Z will reduce both X and Y A poe x anpo gn 2 () (3)

Problem 93**[Cengage Chapter Exercises P101 Q5]**

Which of the following charges will increase the EMF of Be cell $\text{Co(s)} | \text{CoCl}_2(\text{M}) || \text{HCl}(\text{M}) | \text{H}_2(\text{g})$ P? (1) Increase the volume of CoCl_2 solution from 190 mL to (2) Increase M , from 0.01 M to 0.50 M (3) decrease the pressure of the $\text{H}_2(\text{g})$ from 1.0 to 2.0 atm . 41

Problem 94**[Cengage Chapter Exercises P101 Q6]**

Gives: $NOSOI$ N 100 ng nr aeue () Bo $[Mg^*]$ $[Zn^{2+}]$ 0.80 V. E° 2.37 V; Multiple Correct Answers
 Type Which of the following statements is incorrect? Electrochemical Cell (I) $AgNO_3$, can be stored in copper vessel. (2) $Cu(NO_3)_2$, can be stored in magnesium vessel. $Zn(s) | Zn^{2+} (0.024 M) || Zn^{2+} (0.480 M) | Zn(s)$ (3) $CaCl_2$, can be stored in silver vessel which of the following statements is correct? (4) $HgCl_2$, can be stored in copper vessel. (1) The EMF of the cell at 25 is nearly +0.039 V. 7, 100 mL of buffer of 1 M $NH_3(aq)$ and 1 M $NH_4^+(aq)$ are

Problem 95**[Cengage Chapter Exercises P101 Q2]**

The EMF of the cell at 25 °C is nearly -0.039 V placed in two compartments of a voltaic cell separately. (3) If water is added in LHE, so that the $[Zn^{2+}]$ is reduced. If electrolysis of water takes place, then a current of 1.5 A is passed through both cells for 20 min. 9.012 M, the cell voltage increases. (4) If water is added in LHE, so that the $[Zn^{2+}]$ is reduced (1) pH of LHE half cell will increase to 9.912 M, the cell voltage decreases. 2 I9 (a) JP 38 J Hd (2)

Problem 96**[Cengage Chapter Exercises P101 Q1]**

Given (3) pH of both half cell will increase Oxidation: $H_2O \rightarrow O_2 + 2H^+ + 2e^-$ -0.69 V, (4) pH of both half cell will decrease at $-y + sd = -2.87$ V;

Problem 97**[Cengage Chapter Exercises P101 Q8]**

For the following electrochemical cell: $Zn | Zn^{2+} (1.0 \times 10^{-2} M) || P$

Problem 98**[Cengage Chapter Exercises P101 Q210]**

1, + 2 - -8.54 V; E_a F This will be when (1) H_2O , behaves as an oxidant for Fe (1) $[Zn^{2+}]$ $[H^+] = 1 M$ and, Au , = 1 atm (2) H_2O , behaves as a reductant for Fe (2) $[Zn^{2+}] = 0.91 M$, $[H^+] = 0.1 M$, and $P_{H_2} = 1 atm$ (3) $[Zn^{2+}] = 1 M$, $(H^+) = 0.1 M$, and $P_{H_2} = 1 atm$ (4) H_2O , behaves as a reductant for Fe (4) None of the above

Problem 99**[Cengage Chapter Exercises P102 Q18]**

Which of the following statements is correct reaction (1) 17 percent in kJ at 25 is (2) 2H.7(3) 123.5 (4) 167.6 (1) F_2 is the strongest oxidizing agent. IL. In which of the following cells, EMF is greater than and? (2) Li is the strongest reducing agent. (1) $P, H_2(g) | H^+ (H = 5) | H^+ (H = 3) | Br_2(l), Pt$ (3) Li^+ is the weakest oxidizing agent. (2) $Zn(s) | Zn^{2+} (0.2 M) || Cr^{3+} (0.1 M) | Cr(s)$ (3) $Cr(s) | Cr^{3+} (0.1 M) || Cr^{3+} (9.2 M) | Cr(s)$ 19, For $I_2 + 2e^- \rightleftharpoons 2I^-$, standard reduction potential (4) F_2 has a highest reduction potential. (4) $P, I_2(g) | H^+ (H = 4) | H^+ (pH = 6) | H_2(g) | Pt$

Problem 100**[Cengage Chapter Exercises P102 Q12]**

If $A + B \rightleftharpoons C + D$; $K = X$, $\Delta E = V$ $2A + 2B \rightleftharpoons 2C + 2D$; $K = K$, and $9 = V$ reactions is (are) spontaneous? (1) $e = (2) K = A$, (3) $a = 2b$ (4) $b = a^*$ (1) $Br_2 + 200 \rightarrow 2Br + 1, I_2$. Select the wrong relationship(s) (2) $Fe + Br_2 \rightleftharpoons Fe^{2+} + 2Br^-$ (1) $S = (3) Fe + 1, Fe^{2+} + 219 \text{ kJ}$ (4) $1, + 2Br + 21 + Br, H + nEF$

Problem 101**[Cengage Chapter Exercises P102 Q20]**

Consider the cell: $P(H_2, atm) | H^+ (, M) || H^+ (x, M) | B_2(l, atm)$

[Cengage Chapter Exercises P104 Q46]

Iron can be prevented from rusting by (1) CH₃COONa (1) Coating in moist air (3) NaCl (2) KNO₃, of cathodic protection. 4L, A solution containing Na⁺, NO₃⁻, Cl⁻, and SO₄²⁻ ions, all (4) LiF at unit concentrations, is electrolysed between a silver anode and a platinum cathode. As the current is passed through the cell anodic protection. (1) pH of the cathodic solution (-) Oxygen is the major product at anode cathodic protection. (3) Nickel is deposited at cathode (1) Basic copper carbonate forms (4) Chlorine is the major product at anode Conductance, specific, equivalent and molar conductance changes as follows (1) (2) Copper sulphide

[Cengage Chapter Exercises P104 Q42]

Which of the following statements is correct? (1) The cell constant of an electrolytic cell is measured as (4) Copper oxide the product: after being lia. to (1) Al_2O_3 (2) Al_2S_3 (3) Ag_2CO_3 , (4) Al_2O_3 ,

[Cengage Chapter Exercises P105 Q57]

In electrolyte concentration cell: (1) The electrode material and the solution in both half-cells are the same substances (2) Only the concentration of solutions of the same substances is different (3) The overall reaction is $\text{Zn} + 2\text{H}^+ \rightarrow \text{Zn}^{2+} + 2\text{H}_2$ (4) The rate of reaction is 0.0591 V at 25°C . Which is correct statement? (1) No corrosion takes place in vacuum (2) Mercury cell (3) During rusting $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ is formed (4) In presence of electrolyte, corrosion takes place with CO_2 atmosphere. 59, 1000 mL 1M $\text{CuSO}_4(\text{aq})$ is electrolysed by 9.65 A current greater than 1 produces Fe (I) oxide. (4) It is an electrochemical process. for 100 sec using Pt-electrode. Which is correct statement?

[Cengage Chapter Exercises P105 Q9]

Which of the following statements is incorrect? (1) Rate is Fe₂O₃ (1) Blue colour intensity decreases during electrolysis. (2) Zn-Cu cell is called Daniell cell. (3) Blue colour intensity remains constant if Cu-electrode is immersed in solution. (4) Impure metals undergo corrosion faster than pure metals. (5) pH of solution is 8 after electrolysis. (6) 28 mL of H₂, at 1 atm and 273 K required for its complete combustion by O₂, liberated during electrolysis. (7) Standard potential of hydrogen half-cell will be 0V, As ion travels 0.4 cm in 1 minute under the influence of negative electrode. (8) 2 vol potestial after 10 minutes are 3 m apart. (9) NI [gH] per sec (=) Which one are correct? (10) In per cent (=) (g (1) Speed of ion = 6.67×10^{-10} cm s⁻¹) (2) RK1 Ipr e0 g (3) Iomic miliy 1.9×10^{-10} <cn volt se (2) Iosie mobiliey 1×10^{-10} m volt- sec- W co gH pr 0d ((4) Iomic mohilty - 4.44×10^{-10} cmr voltr / sec aodc during electrolysis ?

[Cengage Chapter Exercises P105 Q61]

Wich of the fwing a incerrel nrests e 1),Dite H,5O, wih h detods variation of mola conductanv (/) with dilution fof a (2)Feed NafI with inet dectrdes steng dectulytc ? l,Dilte H,SO, wih Cs clstrsdes (4)Ctd aq N1 wihPh dcds (1) 121 Dilation* Dilutioo - 2)C4,(Mg)ICeP (Mg). Cd F, P; (k)

[Cengage Chapter Exercises P105 Q4]

'2 [C Dilutios - Dilution *

Problem 122**[Cengage Chapter Exercises P106 Q4]**

Calculate ΔG of the reaction: (4) 1 s for weak electrolyte and II for strong electrolyte $\text{Ag}(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow$ (s) 0 u s p a p o p e t j p t g a p e p a y 9 Give: a 109 kJ/mol - standard ΔG° , $\Delta^\circ = -129 \text{ kJ/mol}$ (1) Coefficient of chloride ions $\Delta^\circ = 77 \text{ kJ/mol}$ (2) Coefficient of hydrogen ions $\Delta^\circ = 97 \text{ kJ/mol}$ (3) Coefficient of KCl solution (3) 57 kJ/mol - (4) 97 kJ/mol - (4). All of these

Problem 123**[Cengage Chapter Exercises P106 Q64]**

A sample of water has a hardness expressed as 77.5 ppm

Problem 124**[Cengage Chapter Exercises P106 Q5]**

of the reaction above in Q. 4. is Cr. This sample is passed through a ion exchange (1) 0.59 V (2) 0.59 V volume and de Ca is replaced by H Select the correct (3) 0.295 V A 56Z() statement(s) 6, K, of AgCl is (1) p f f e d e w a t e r a f t e r i t h a s b e n s o t r e a t e d i s 2.4 (1) 1g-1 (2) 10-12 (3) 10-11 (2) Every Ca^{2+} ion is replaced by one H^+ ion (4) 20- (3) Every Ca^{2+} ion is replaced by two H^+ ions 7, 6.537 10- g of metallic Zn was added to 10 mL - (4) pH of the solution remains unchanged saturated solution of AgCl . Calculate $\log [\text{Zn}^{2+}]$

Problem 125**[Cengage Chapter Exercises P106 Q65]**

Is the electrolysis of a 40-L CuSO_4 solution, there are two $[\text{Ag}^+]$ L C + 2r possible reactions at anode: K/ of AgCl = 10-, atomic weight of Zn = 65.37 E 2, 0 4* + 4r - 0, A current of 1.07 A is passed for 2 hours. The loss in the . Calculate the number of moles of Ag formed after 592 (1) (2) 13.24 (3) 53 (4) 106 mass of Cu at anode was 1.27 g - , now I s = n p o w l a x s e n y) Q. 7. go on arts (ssas le;*oggs 2 Jp 9p/ percent (1) 10 (1) 4:08 mol of electrons are gained through the solution Paragraph 3 (2) 10-5 (3) 10* (4) * 2) 224 nL of O_2 is liberated at STP at anode. during entire electrolysis. (3) Fraction of current in the production of C^{2+} ions (4) pH drops to 3. = 6.5. Linked Comprehensive Type Paragraph 1 is neutralized by OH^- , i.e., anodic reaction. -4 W t6e pt 24 IK 100 leeseo uogajps moerbe vy was timed via a concentration of NaOH = 30 , s0 H, = $2\text{H}^+ + 2$ sies foed uig ition a $\text{Fe}(\text{OH})$, ad $\text{Fe}(\text{OH})$, oly. dat changs ia volumes wet negligible. Ating at he sew oHt $\text{HOC} + \text{H}_2\text{O} + 20\text{H}^+ + 2\text{H}_2\text{O} + \text{Aafolk}$, O, gets reduced to OH^-

Problem 126**[Cengage Chapter Exercises P107 Q14]**

Finally, the I compartment contains (1) 1 (4) II, III, IV (2) 1, II I5. Finally the II compartment contains (1) 1, I, (2) He.V () 0.41 V (4) 0.83 V (I) 1 (2) 1, / No. (1) I, (4) I , , IV 80JK- (2) -160 JK- (4) 1600 1 K- IK. The initial concentration of Co in I, II and III compartment is (1) 0.105, 0.005, 0.0 M (2) 0.005 , 0.105 , 0.9 M (3) 0.105 , 0.0, 0.005 M (4) 0.6, 0.605, 0.195 M oL d e g e ill be

Problem 127**[Cengage Chapter Exercises P107 Q17]**

The final concentration of NO_2 in I, II, and III compartment is (Rabwel ly lal (1) 0.109, 0.210 , 0.0900 M (2) 0.210 , 0.100 , 0.0900 M Choely fetr of? (3) 0.0900 , 0.210, 0.100 M I(4) 0.0900 , 0.100 , 0.210 M japg ipm llsaepr

Problem 128**[Cengage Chapter Exercises P107 Q18]**

The final concentration of K⁺ in I, II, and III compartment is (1) 0.090, 0.9, 0.0100 M (2) 0.0, 0.0160 , 0.090 M IAsdelia L velic cell in which continuous supply of fels are (3) 0.0, 0.090 , 0.0100 M (4) 0.100 , 0.090 , 0.0 M set / No. asedr perfem oxidertee.

Problem 129**[Cengage Chapter Exercises P107 Q19]**

The final concentration of Ag^+ in I, II and III compartments is 0.01 M, 0.001 M, 0.08 M (4) 0.0 M in all compartments (2) 0.9, 0.08, 0.9 M are used as an anode. (3) 0.08, 0.0, 0.0 M III On the other hand involves the reaction of H_2O , the cell is a Matrix Match Type

Anode: $2\text{H}_2 + 4\text{OH}^- \rightarrow 4\text{H}_2\text{O} + 4\text{e}^-$ This section contains questions each with two columns I and II. Match the items given in column I with the item in column II. Cathode: $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$ The electrolyte is NaOH , the cell is 75% percent complete. Column I (1) L (2) L, II, IV Electric unit Single unit (3) L, IV (4) 1, II, III a. Vol-ampere and joule second-1 p. Coulomb per hour 4. The cell is composed of two electrodes separated by. Ampere-second and joule per volt-1. The number of moles of electrons in 5.0 L of 0.10 M AgNO_3 ; in 5.0 L of 0.10 M KNO_3 , the 'Volt ampere' and 'joule ampere' second-1 are the same as the cell is carried equally by the negative and positive ions. Ag electrode in 5.0 L of 0.10 M AgNO_3 , assuming the current is 0.1 A for 1 hour. Watt ampere-seconds-1 Ampere

Problem 130**[Cengage Chapter Exercises P107 Q2]**

Column 1 Column II Properties Units MnO_4^- ON Ne ON Nee a. Cell constant $\text{M}^{-1}\text{cm}^{-1}$ [0.10 M Ag b. Molar conductance $\text{cm}^2\text{mol}^{-1}$ Equivalent conductance $\text{cm}^2\text{mol}^{-1}$ SAL compartment I/ No. 5.0 L Specific conductance cm^{-1} Conductance mho Gnes $\text{C} + 2\text{r}$ $\text{Ce}^{4+} = 0.28 \text{ V}$ L mhe cs^{-1} Resistance Ω .

Problem 131**[Cengage Chapter Exercises P108 Q6]**

Glass electrode [Buffer solution $\text{pH} = 6$ | Standard calomel electrode $E^\circ = 0.1 \text{ V}$ $\text{AS}^\circ = 0.069 \text{ V}$

Problem 132**[Cengage Chapter Exercises P108 Q5]**

Column 1 Electrolysis Column II pH and products Electrolysis of 100 L aqueous solution of CH_3COOK by at anode and cathode passing 2 F of electricity $\text{pH} = 12.3$ Anode = Ethane(g) - CO_2 (g) Electrolysis of 10 L aqueous solution of HCOOK by passing Cathode = H_2 (g) b. 1 F of electricity. $\text{pH} = 13.0$ Anode = H_2 (g) + CO_2 (g) Electrolysis of 10 L aqueous solution of K_2SO_4 by passing Cathode = H_2 (g) 1 F of electricity. $\text{pH} = 7.0$ Cathode = H_2 (g) Anode = O_2 (g) d Electrolysis of 10 L aqueous solution of CaF_2 by passing 1 F of electricity $\text{pH} = 1.0$ Anode = O_2 (g) Cathode = Cu (g) Column 1 Cell and EMF Column II PHJ1 m | H^+ (10 M) H^+ (10 M) H^+ (1 M) R Characteristics a) 6.177 V P. Spontaneous b. H^+ (1 M) P (1e M) 1 F (10 M) 1 F (2 atm) $R_g = 0.699 \text{ V}$ Nos-spontaneous Hg , Hg_2Cl_2 (s) KCl (saturated solution) Ir ($\text{H}^+ = 10$) 1 Q. QH, 1 R $\text{O}_2 = 0.7 \text{ V}$ $\text{H}^+ = 0.13$, $\text{Egc} = 0.24 \text{ f}$. Exergonic $(\text{OH}^- + \text{Z} = \text{H}_2 + \text{O})$ $g = 0.7 \text{ V}$, $\text{Egc} = 0.582 \text{ V}$ Endergonic

Chapter 14

Essential Physical Chemistry: Complete Numerical Bank

Problem 1

[Essential Physical Chemistry p.42 Q1]

Define or explain the following terms : (a) Arrhenius theory (b) Transport number (c) Hittorf's method (d) Moving boundary method (e) Kohlrausch's law (f) Degree of dissociation

Problem 2

[Essential Physical Chemistry p.42 Q2]

(a) Explain ionic conductance, transport number of an ion, and absolute ionic velocity. (b) The absolute velocity of Ag^+ is $0.00057 \text{ cm sec}^{-1}$ and of the NO_3^- is $0.00069 \text{ cm sec}^{-1}$. Assuming complete dissociation, calculate the specific conductivity of 0.01 M solution of silver nitrate. Answer. (b) 0.00012 mhos

Problem 3

[Essential Physical Chemistry p.42 Q3]

Describe Hittorf's method for the experimental determination of the transference number. The speed ratio of silver and nitrate ions in AgNO_3 electrolysed between silver electrodes was found to be 0.89 . Calculate the transference number of silver and nitrate ions. Answer. $\text{Ag}^+ = 0.4708$; $\text{NO}_3^- = 0.5292$

Problem 4

[Essential Physical Chemistry p.42 Q4]

(a) What do you understand by the transport number of an ion? Does it vary? (b) In a moving boundary experiment with 0.1 N KCl using 0.65 N LiCl as indicator solution, a constant current of 0.006 amp was passed for 1900 secs and the boundary was observed to move through 5 cm in a tube of 0.1142 cm^2 cross-section. Calculate the transport number of K^+ and Cl^- ions. Answer. (b) $\text{K}^+ = 0.4833$; $\text{Cl}^- = 0.5167$

Problem 5

[Essential Physical Chemistry p.42 Q5]

(a) What is meant by transport number of ions? Describe Hittorf's method for the determination of transport number of silver ions. (b) A dilute solution of silver nitrate was electrolysed using platinum electrodes. After passing small

Problem 6

[Essential Physical Chemistry p.42 Q1]

Conductance measurements are done to check end points.

Problem 7

[Essential Physical Chemistry p.42 Q2]

Titration can be carried out even with coloured solution.

Problem 8

[Essential Physical Chemistry p.42 Q3]

Accurate results are obtained.

Problem 9

[Essential Physical Chemistry p.42 Q4]

End point is determined graphically.

Problem 10

[Essential Physical Chemistry p.42 Q5]

In case of polybasic acids conductometric titrations can be used.

Problem 11

[Essential Physical Chemistry p.42 Q6]

These are successful even in weak acids and strong bases.

Problem 12

[Essential Physical Chemistry p.42 Q1]

Volume measurements are done to check end points.

Problem 13

[Essential Physical Chemistry p.42 Q2]

These titrations fail in coloured solutions as suitable indicators are not available some times.

Problem 14

[Essential Physical Chemistry p.42 Q3]

Results are not so accurate.

Problem 15

[Essential Physical Chemistry p.42 Q4]

End point is determined by change in colour of indicator.

Problem 16

[Essential Physical Chemistry p.42 Q5]

In case of polybasic acids volumetric titrations do not give correct end points.

Problem 17

[Essential Physical Chemistry p.42 Q6]

These are not successful in weak acids and strong bases.

Problem 18

[Essential Physical Chemistry p.43 Q6]

(a) Explain Kohlrausch's law of ionic mobility. How does it help in determining the equivalent conductivity of infinite dilution of weak electrolyte? (b) Describe the Hittorf's method for determining the transport number of Ag^+ and NO_3^- ions in solution of silver nitrate.

Problem 19

[Essential Physical Chemistry p.43 Q7]

The specific conductance of a saturated solution of AgCl is $1.55 \times 10^{-6} \text{ } \Omega^{-1} \text{ cm}^{-1}$ at 25°C . The ion conductance of Ag^+ Cl^- are 61.94 and $76.34 \text{ } \Omega^{-1} \text{ cm}^2$ respectively. Calculate the solubility of AgCl in gram equivalent per litre at 25°C . Neglect the specific conductance of water. Answer. $1.6148 \times 10^{-3} \text{ g lit}^{-1}$

Problem 20**[Essential Physical Chemistry p.43 Q8]**

In a moving boundary experiment with 0.1 N KCl solution, the boundary moved 4.94 cm during 67 minutes when a current of 5.21 A was used. The cross-sectional area of the tube was 0.230 cm². Calculate the transport number of K⁺ ions. Answer. 0.523

Problem 21**[Essential Physical Chemistry p.43 Q9]**

What do you understand by the terms: transport number and ionic mobility of an ion? How are these related to each other? Write their units. (Punjabi BSc, 2000)

Problem 22**[Essential Physical Chemistry p.43 Q10]**

What do you understand by transport number? Give one method for the determination of it. (Punjabi BSc, 2000)

Problem 23**[Essential Physical Chemistry p.43 Q11]**

Describe Hittorf's method for determining the transport number of Ag⁺ and NO₃⁻ ions in a solution of AgNO₃ when the electrodes are made up of silver. (Jiwaji BSc, 2000)

Problem 24**[Essential Physical Chemistry p.43 Q12]**

Give the applications of transport number. (Himachal Pradesh BSc, 2000)

Problem 25**[Essential Physical Chemistry p.43 Q13]**

State Kohlrausch's law. (Madurai BSc, 2000)

Problem 26**[Essential Physical Chemistry p.43 Q14]**

Write short notes on : (a) Transport number (b) Kohlrausch law of independent migration of ions (Delhi BSc, 2000)

Problem 27**[Essential Physical Chemistry p.43 Q15]**

(a) How is solubility of sparingly soluble salt is determined by conductance measurement? (b) Discuss the application of Kohlrausch's law. (Lucknow BSc, 2001)

Problem 28**[Essential Physical Chemistry p.43 Q16]**

If the conductance at infinite dilution of NaCl, HCl and CH₃COONa are 125.45 $\times 10^{-4}$, 426.1 $\times 10^{-4}$ and 91.0 $\times 10^{-4}$ cm² eqvt⁻¹ respectively, calculate equivalent conductance of acetic acid at infinite dilution. Answer. 390.7 $\times 10^{-4}$ Ω^{-1} cm² eqvt⁻¹ (Guru Nanak Dev BSc, 2002)

Problem 29**[Essential Physical Chemistry p.43 Q17]**

Define Transport number. Describe briefly the principle of experimental determination of Transport number by Hittorf's method. (Panjab BSc, 2002)

Problem 30**[Essential Physical Chemistry p.43 Q18]**

(a) State the principle of conductometric titrations. Draw the titration curves for (i) Weak acid with a strong base (ii) Na_2SO_4 solution with BaCl_2 solution (b) Calculate the ionisation constant of water if the specific conductivity of water is $5.51 \times 10^{-8} \text{ } \Omega^{-1} \text{ cm}^{-1}$ and ionic conductance of H^+ and OH^- ions at the same temperature are 349.8 and 196.5 respectively. Answer. 1.013×10^{-14} (Arunachal BSc, 2002)

Problem 31**[Essential Physical Chemistry p.43 Q19]**

How does Kohlrausch's law helps in determining the dissociation constants of a weak acid by conductance measurements? (Vidyasagar BSc, 2002)

Problem 32**[Essential Physical Chemistry p.43 Q20]**

Derive the expression for the determination of Transport number by Hittorf's method when electrodes are not attacked? (Arunachal BSc, 2002)

Problem 33**[Essential Physical Chemistry p.43 Q21]**

Explain the factors on which the transport number depends. (MD Rohtak BSc, 2002)

Problem 34**[Essential Physical Chemistry p.43 Q22]**

Explain why the transport number of cadmium ion in cadmium iodide solution at high concentration may become zero or negative also. (Arunachal BSc, 2002)

Problem 35**[Essential Physical Chemistry p.43 Q23]**

Explain how is Kohlrausch's law helpful in determining the equivalent conductance of weak electrolyte at infinite dilution. (Jammu BSc, 2002)

Problem 36**[Essential Physical Chemistry p.44 Q24]**

Write short notes on : (i) Activity and mean ionic activity of an electrolyte. (ii) Ionic atmosphere (Aligarh BSc, 2002)

Problem 37**[Essential Physical Chemistry p.44 Q25]**

Discuss the effect of concentration on transference number. (Kalyani BSc, 2003)

Problem 38**[Essential Physical Chemistry p.44 Q26]**

During the electrolysis of a solution of potassium chloride between platinum electrodes, 0.0137 g of chloride was lost from the anodic compartment and 0.0857 g of silver was deposited in a silver coulometer connected in series with the cell. Find at the transport number of K^+ and Cl^- ions. Answer. 0.4854; 0.5146 (Delhi BSc, 2003)

Problem 39**[Essential Physical Chemistry p.44 Q27]**

(a) Explain the term 'ionic mobility'. Describe Hittorf's theoretical device to show that although most of the ions differ largely in their mobilities, but their equivalent amounts are discharged on electrolysis at appropriate electrodes. (b) In moving boundary experiment with 0.100 N KCl using 0.065 N LiCl as indicator solution, a constant current of 0.005893 amp was passed for 2130 seconds. The boundary was observed to move through 5.6 cm in a tube of 0.1142 sq cm cross-section. Calculate the transport number of K^+ and Cl^- ions. Answer. 0.492; 0.508 (Rajasthan BSc, 2003)

Problem 40**[Essential Physical Chemistry p.44 Q28]**

What is meant by limiting equivalent conductance of weak electrolyte? How do you determine it by the help of Kohlrausch's law? (Sambalpur BSc, 2003)

Problem 41**[Essential Physical Chemistry p.44 Q29]**

Explain the use of Kohlrausch's law to determine the solubility of a sparingly soluble salt. (Nagpur BSc, 2003)

Problem 42**[Essential Physical Chemistry p.44 Q30]**

(a) Define transport number. (b) A solution of silver nitrate containing 0.51 gm of the salt in 60.40 gm of the solution was electrolysed under similar electrodes. After electrolysis 64 gm of the anode solution was found to contain 0.56 gm of silver nitrate. A current of 0.04 amp was passed for 2950 seconds. Calculate the transference number of nitrate ion. Answer. (b) 0.520 (Delhi BSc, 2004)

Problem 43**[Essential Physical Chemistry p.44 Q31]**

(a) How is Kohlrausch's law helpful in determining the solubility of a sparingly soluble salt in water? (b) How can you find the transference number of an ion by Hittorf's method? (Madurai BSc, 2004)

Problem 44**[Essential Physical Chemistry p.44 Q32]**

A solution of HCl was electrolysed in a cell between two platinum electrodes. The cathode compartment contained 0.354 g of chloride ions before electrolysis and 0.32 g after electrolysis. A silver coulometer in series had a deposit of 0.532 g of silver after passing the same amount of current. Calculate the transport number of H^+ and Cl^- ions. Answer. 0.84 ; 0.160 (Rajasthan BSc, 2005)

Problem 45**[Essential Physical Chemistry p.44 Q33]**

In the Hittorf cell using silver electrodes and $AgNO_3$ as an electrolyte, a certain amount of current was passing which deposited 9.886×10^{-4} g equivalent of silver in the coulometer. The anode compartment had the composition 28.235 g of H_2O and 0.099 g of $AgNO_3$ before electrolysis and 28.435 g of water and 0.1874 g of $AgNO_3$ after electrolysis. Calculate the transport number of Ag^+ and NO_3^- ions. Answer. 0.48 and 0.52 (Kalyani BSc, 2005)

Problem 46**[Essential Physical Chemistry p.44 Q34]**

In a moving boundary experiment with 1 N KCl solution using $CaCl_2$ as indicator, a current of 0.0115 A was passed for half an hour and the boundary moved through a volume of 0.106 ml. Calculate the transport number of K^+ ion. Answer. 0.494 (Vikram BSc, 2006)

Problem 47**[Essential Physical Chemistry p.44 Q35]**

A conductivity cell whose cell constant is 2 cm^{-1} , is filled with 0.1 M acetic acid solution. Its resistance is found to be $3765 \text{ } \Omega$. Calculate the degree of dissociation of acetic acid. Answer. 0.0136 (Delhi BSc, 2006)

Problem 48**[Essential Physical Chemistry p.44 Q36]**

In a moving boundary method, a current of 30 mA was passed for 80 seconds . During this time the boundary of a solution of HCl containing 0.01 mol dm^{-3} moved 17.0 cm towards the cathode. Calculate the transport number of H^+ and Cl^- ions. (The cross-sectional area of the glass tube is $1.0 \times 10^{-5} \text{ m}^2$). Answer. 0.68 ; 0.32 (Mysore BSc, 2006)

Problem 49**[Essential Physical Chemistry p.45 Q1]**

According to Arrhenius theory an electrolyte when dissolved in water gives two types of (a) charged particles (b) molecules (c) ion pairs (d) fundamental particles Answer. (a)

Problem 50**[Essential Physical Chemistry p.45 Q2]**

For an electrolyte AB , the ionisation constant is given by the expression (a) $K = \frac{[\text{A}][\text{B}]}{[\text{AB}]}$ (b) $K = \frac{[\text{A}][\text{B}]}{[\text{AB}] + [\text{A}][\text{B}]}$ (c) $K = \frac{[\text{A}][\text{B}]}{[\text{AB}] + [\text{A}][\text{B}] + [\text{A}][\text{B}]}$ (d) $K = \frac{[\text{A}][\text{B}]}{[\text{AB}] + [\text{A}][\text{B}]}$ Answer. (d)

Problem 51**[Essential Physical Chemistry p.45 Q3]**

The conductivity of an electrolyte is due to the (a) presence of ions in the electrolyte (b) free movement of ions in the solution (c) reunion of ions in the solution (d) release of heat energy due to ionisation Answer. (b)

Problem 52**[Essential Physical Chemistry p.45 Q4]**

The electrical conductivity of an electrolyte depends upon (a) the number of molecules in the electrolyte (b) the number of ions present in the electrolyte (c) the number of ions present in the solution (d) the number of molecules of the solvent Answer. (c)

Problem 53**[Essential Physical Chemistry p.45 Q5]**

On passing electrical current through an electrolyte solution (a) cations move towards anode (b) anions move towards cathode (c) cations move towards cathode and anions towards anode (d) both cations and anions move in same direction Answer. (c)

Problem 54**[Essential Physical Chemistry p.45 Q6]**

On passing electrical current through an electrolyte solution, the cations (a) move towards cathode with the speed equal to that of anions towards anode (b) move with faster speed than that of anions (c) move with different speed as compared to that of anions (d) move with slower speed than that of anions Answer. (c)

Problem 55**[Essential Physical Chemistry p.45 Q7]**

The Hittorf's rule states that (a) the loss of concentration around any electrode is proportional to the speed of the ions moving towards it (b) the loss of concentration around any electrode is proportional to the speed of the ions moving away from it (c) the loss of concentrations around both the electrodes is proportional to the sum of speed of cations and anions (d) none of the above Answer. (b)

Problem 56

[Essential Physical Chemistry p.45 Q8]

The Hittorf's rule can be represented by the expression

Problem 57

[Essential Physical Chemistry p.46 Q9]

The fraction of the total current carried by the cation or anion is termed as (a) fractional number (b) speed number (c) carrier number (d) transport number Answer. (d)

Problem 58

[Essential Physical Chemistry p.46 Q10]

If v_+ is the speed of cation and v_- that of anion, the transport number of cation is given by (a) $\frac{v_+}{v_+ + v_-}$ (b) $\frac{v_-}{v_+ + v_-}$ (c) $\frac{v_+}{v_+ - v_-}$ (d) $\frac{v_-}{v_+ - v_-}$ Answer. (c)

Problem 59

[Essential Physical Chemistry p.46 Q11]

If v_+ is the speed of cation and v_- that of anion, then the speed ratio is given by (a) $\frac{v_+}{v_-}$ (b) $\frac{v_-}{v_+}$ (c) $\frac{v_+}{v_-} + 1$ (d) $\frac{v_-}{v_+} + 1$ Answer. (d)

Problem 60

[Essential Physical Chemistry p.46 Q12]

The sum of the transport number of cation and anion is equal to (a) 1 (b) 0 (c) 0.5 (d) *propto* Answer. (a)

Problem 61

[Essential Physical Chemistry p.46 Q13]

In Hittorf method for determination of transport numbers we make use of a (a) H-tube (b) V-tube (c) U-tube (d) L-tube Answer. (c)

Problem 62

[Essential Physical Chemistry p.46 Q14]

While determining transport number by Hittorf method, _____ electrodes are used when electrodes are attackable (a) silver (b) copper (c) mercury (d) platinum Answer. (d)

Problem 63

[Essential Physical Chemistry p.46 Q15]

In the moving boundary method for the determination of transport number, the formula used is (where n is the number of faradays of current passed) (a) $\frac{Q}{nF}$ (b) $\frac{2Q}{nF}$ (c) $\frac{Q}{2nF}$ (d) $\frac{2Q}{nF}$ Answer. (a)

Problem 64

[Essential Physical Chemistry p.46 Q16]

One Faraday is equal to (a) 96,500 coulombs (b) 9650 coulombs

Problem 65

[Essential Physical Chemistry p.47 Q17]

The statement of Kohlrausch's law is (a) the equivalent conductance of an electrolyte at infinite dilution is equal to the product of equivalent conductances of the component ions (b) the equivalent conductance of an electrolyte at infinite dilution is equal to the difference of equivalent conductances of the component ions (c) the equivalent conductance of an electrolyte at infinite dilution is equal to the sum of equivalent conductances of the component ions (d) none of the above Answer. (c)

Problem 66**[Essential Physical Chemistry p.47 Q18]**

Kohlrausch's law can be expressed as (a) $\lambda_{\alpha} = \lambda_{\infty} - \lambda_{\alpha}$ (b) $\lambda_{\alpha} = \lambda_{\infty} - \lambda_{\alpha}$ (c) $\lambda_{\alpha} = \lambda_{\infty} + \lambda_{\alpha}$ (d) $\lambda_{\alpha} = \lambda_{\infty} \times \lambda_{\alpha}$ Answer. (c)

Problem 67**[Essential Physical Chemistry p.47 Q19]**

Kohlrausch's law can be used to determine (a) λ_{α} for weak electrolytes (b) absolute ionic mobilities (c) solubility of sparingly soluble salts (d) all of these Answer. (d)

Problem 68**[Essential Physical Chemistry p.47 Q20]**

If λ_{α} is the equivalent conductance at infinite dilution and λ_{α} the equivalent conductance of the electrolyte at the dilution v , the degree of dissociation is given by (a) $v \lambda_{\alpha}$ (b) $v \lambda_{\alpha} / \lambda_{\alpha}$ (c) $v \lambda_{\alpha} / \lambda_{\alpha}$ (d) $v \lambda_{\alpha} / \lambda_{\alpha}$ Answer. (b)

Problem 69**[Essential Physical Chemistry p.47 Q21]**

When a strong acid is titrated against a strong base the end point is the point of (a) zero conductance (b) maximum conductance (c) minimum conductance (d) none of these Answer. (c)

Problem 70**[Essential Physical Chemistry p.47 Q22]**

During titration of a weak acid against a weak base, there is a sharp increase in _____ at the end point (a) conductivity (b) equivalent conductance (c) specific conductance (d) none of these Answer. (a)

Problem 71**[Essential Physical Chemistry p.47 Q23]**

If the equivalent conductance of a certain solution of acetic acid is $39.07 \text{ cm}^2 \text{ eqvt}^{-1}$. If λ_{α} of CH_3COOH is 390.7, the degree of dissociation of acetic acid is (a) 0.1 (b) 0.2 (c) 0.5 (d) 0.75 Answer. (a)

Problem 72**[Essential Physical Chemistry p.47 Q24]**

The equivalent conductance at infinite dilution of NaCl , HCl and CH_3COONa at 25°C are 126.0, 426.0 and 91.0 cm^2 respectively. The equivalent conductance of acetic acid at infinite dilution at 25°C will be (a) 643.0 (b) 517.0 (c) 217.0 (d) 391.0 Answer. (d)

Problem 73**[Essential Physical Chemistry p.47 Q25]**

The equivalent conductance of decinormal solution of acetic acid at 298 K is 80 and at infinite dilution 400 cm^2 . The percentage dissociation of acetic acid is

Problem 74**[Essential Physical Chemistry p.48 Q26]**

If the transport number of K^+ is 0.492 in KCl solution. The transport number of Cl^- ion will be (a) 0.984 (b) 0.492 (c) 0.508 (d) 0.016 Answer. (c)

Problem 75**[Essential Physical Chemistry p.48 Q27]**

In a binary electrolyte AB of certain concentration, the transport number of cation is 0.4. The transport number of the anion will be (a) 0.2 (b) 0.4 (c) 0.6 (d) 0.8 Answer. (c)

Problem 76**[Essential Physical Chemistry p.48 Q28]**

The equivalent conductance at 18°C of a normal solution of KCl is 98.2 and at infinite dilution at the same temperature is 131. The degree of dissociation of KCl will be (a) $131/98.2$ (b) $98.2/131$ (c) $98.2/131 + 98.2$ (d) $131/131 + 98.2$ Answer. (b)

Problem 77**[Essential Physical Chemistry p.48 Q29]**

If the degree of dissociation of weak electrolyte at a certain temperature is 0.1, the percentage of the molecules undissociated will be (a) 10/(b) 20/(c) 50/(d) 90/Answer. (d)

Problem 78**[Essential Physical Chemistry p.48 Q30]**

For strong electrolytes, the degree of dissociation is (a) nearly equal to one (b) nearly equal to zero (c) nearly equal to infinity (d) nearly equal to 0.5 Answer. (a)

Problem 79**[Essential Physical Chemistry p.48 Q31]**

The equivalent conductance at infinite dilution of NH_4Cl , NaOH and NaCl at 18°C are respectively, 129.8, 227.4 and $108.9 \text{ } \Omega^{-1} \text{ cm}^2 \text{ g eqvt}^{-1}$. The equivalent conductance of NH_4OH at infinite dilution at 18°C will be (a) 128.8 (b) 108.9 (c) 227.4 (d) 238.3 Answer. (d)

Problem 80**[Essential Physical Chemistry p.48 Q32]**

At infinite dilution the conductance of NH_4Br , KOH and KBr at 298 K are 151, 271 and $151 \text{ } \Omega^{-1} \text{ cm}^2 \text{ g eqvt}^{-1}$ respectively. If the equivalent conductance of NH_4OH at 0.01 N concentration at 298 K is $27.1 \text{ } \Omega^{-1} \text{ cm}^2 \text{ g eqvt}^{-1}$, the degree of dissociation of NH_4OH would be (a) 0.1 (b) 0.01 (c) 0.001 (d) 1.0 Answer. (a)

Problem 81**[Essential Physical Chemistry p.48 Q33]**

Λ° for $\text{Ba}(\text{OH})_2$, BaCl_2 and NH_4Cl are 228, 120 and $129 \text{ } \Omega^{-1} \text{ cm}^2 \text{ g eqvt}^{-1}$. The value of Λ° for NH_4OH (a) $108 \text{ } \Omega^{-1} \text{ cm}^2 \text{ g eqvt}^{-1}$ (b) $249 \text{ } \Omega^{-1} \text{ cm}^2 \text{ g eqvt}^{-1}$ (c) $348 \text{ } \Omega^{-1} \text{ cm}^2 \text{ g eqvt}^{-1}$ (d) $183 \text{ } \Omega^{-1} \text{ cm}^2 \text{ g eqvt}^{-1}$ Answer. (d)

Problem 82**[Essential Physical Chemistry p.48 Q34]**

The ionic mobility of an ion is given by the relation (a) Λ/α (b) $\Lambda/\alpha \div 96500$ (c) $\Lambda/\alpha \times 96500$ (d) $\Lambda/\alpha + 96500$ Answer. (a)